Plasma Physics

Christopher Albert, TU Graz

SI units throughout, with ε_0 and μ_0; CGS forms are collected in the appendix.

1. Introduction

A plasma is a many-particle system whose long-range electromagnetic fields make the particles respond collectively. This chapter establishes the vocabulary through examples, characteristic lengths and times, and the models used to describe different observations. The next two chapters develop Debye shielding and plasma oscillations as separate physical responses. The scale hierarchy follows standard graduate treatments [1–3].

1.1 Plasma as a collective state

A plasma is recognized by its response: a local charge or current must influence particles beyond the nearest collision partner, while the system still contains enough particles for a smooth field description.

Learning objectives

- distinguish a plasma state from a temperature threshold
- identify collective electromagnetic phenomena in a physical example
- select a model level from the relevant length and time scales

Unit ledger

The temperature T_s below is a thermodynamic temperature in kelvin, so $k_B T_s$ is an energy in J. A symbol without a unit is explicitly marked dimensionless.

Working definition of a plasma — Definition 1

A plasma is a collection of charged species whose collective electromagnetic response is important on the scale of interest. It need not be fully ionized, and there is no universal temperature at which matter abruptly becomes a plasma. Ionization, collisionality, boundaries, and the observation scale all matter.

Three examples show why the observation matters as much as the name of the plasma:

- In a glow discharge, energetic electrons sustain ionization while much of the gas remains neutral. Collisions with neutrals and particle collection at surfaces influence the electrical response.
- In a magnetic fusion device, the field confines a hot plasma. Particle orbit sizes and collective fluid motion matter on different length and time scales, so one model need not describe both.
- In the solar wind, a dilute plasma expands through a magnetic field. Infrequent collisions allow velocity distributions to retain structure that a density and a single temperature cannot specify.

Other examples include the solar corona, planetary magnetospheres, lightning, arcs, and processing plasmas. These settings span many orders of magnitude in density and temperature; the table below lists typical values.

The charge density is the species sum

$$\rho_q = \sum_s q_s n_s$$

The charge density ρ_q is in C m^{-3}. The sum runs over all charged species.

Charge and current generate electromagnetic fields, which in turn act on the particles. The self-consistent equations are given in the particle–field model.

Quasineutrality Definition 2

On a macroscopic scale, a simple electron–positive-ion plasma is quasineutral when its net charge density is small compared with the individual species charge densities. This is an ordering statement, not an exact pointwise identity.

$$\rho_q = e\left(\sum_i Z_i n_i - n_e\right) \approx 0, \quad n_e \approx \sum_i Z_i n_i$$

e is the positive elementary charge in C and Z_i is the integer ion charge state. For a singly ionized hydrogen plasma, the condition reduces to $n_e \approx n_i$. Charge-separated regions of Debye-scale thickness and boundary sheaths are controlled departures from this bulk ordering.

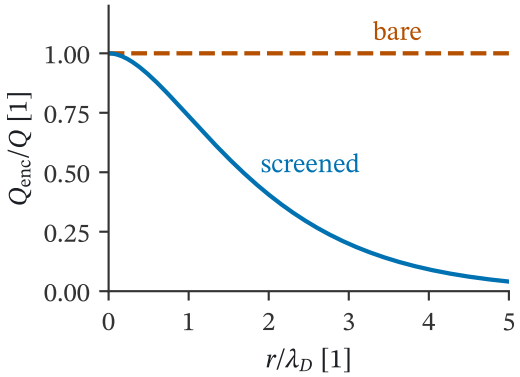
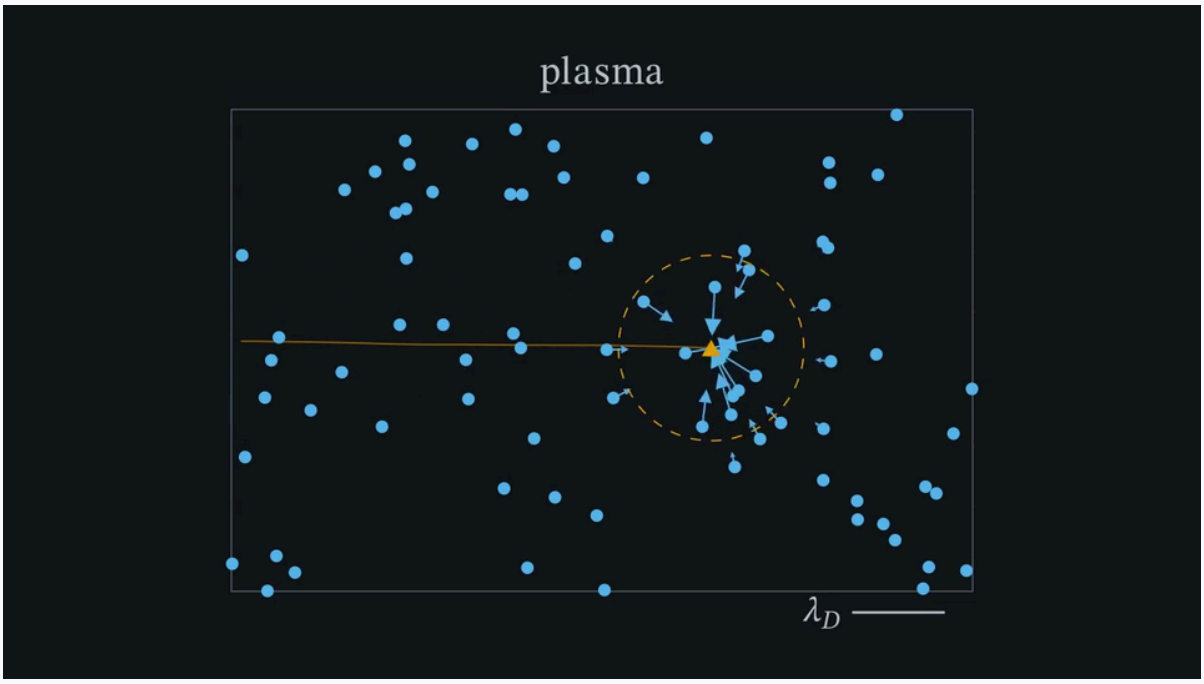


Figure 1: Net charge inside radius r around a point charge Q: bare (dashed) and Debye-screened (solid), $Q_{\mathrm{enc}}/Q = (1 + r/\lambda_D)e^{-r/\lambda_D}$ from Gauss's law and the screened potential derived in the Debye-shielding chapter. Radius in units of λ_D.

A plasma differs from a neutral gas in the range of its response. A charge imbalance can launch an electric field, a current can launch a magnetic perturbation, and the resulting fields can move many particles before local collisions erase the correlation.



Watch animation

Range of the interaction. Neutral gas: hard spheres of equal mass that interact only at contact. Plasma: a positive test charge and resting electrons with the screened force derived from $-e^{-r/\lambda_D}/r$; arrow length grows with the force and saturates near the charge. Lengths in units of λ_D; the dashed circle has radius λ_D. The screened force is prescribed.

Figure 2: Range of the interaction. Neutral gas: hard spheres of equal mass that interact only at contact. Plasma: a positive test charge and resting electrons with the screened force derived from $-e^{-r/\lambda_D}/r$; arrow length grows with the force and saturates near the charge. Lengths in units of λ_D; the dashed circle has radius λ_D. The screened force is prescribed.

Order-of-magnitude examples

The density column is n_e in m^{-3}, and the energy column is $k_B T_e$ in electron-volts. A single named object can occupy more than one row as its local state changes.

Setting	n_e	$k_B T_e$	Characteristic emphasis
Solar wind	10^6–10^7	1–100	dilute, weakly collisional, magnetized
Ionosphere	10^{10}–10^{12}	0.03–0.3	partially ionized and collisional
Glow discharge	10^{15}–10^{18}	1–10	weak ionization and boundaries
Solar corona	10^{14}–10^{16}	10^2–10^3	hot, magnetized, nearly fully ionized
Fusion plasma	10^{19}–10^{21}	10^3–$2 \cdot 10^4$	hot, confined, collective

The characteristic responses introduced by this script are Debye shielding, electron plasma oscillations, collective waves, gyromotion and guiding-center drifts, instabilities, and boundary sheaths. These are different limits of the same coupled particle–field system.

An ideal plasma, as used here, contains enough weakly interacting particles for smooth collective fields to be useful. Bulk quasineutrality also requires observing lengths larger than the charge-separation scale. The collective validity criteria quantify these conditions. Very dense plasmas can require quantum statistics; the degeneracy criterion distinguishes that regime from the classical description used in this course.

Summary

Summary 3

A plasma is identified by its scale-dependent collective electromagnetic response. Bulk quasineutrality, weak coupling, and the validity of classical statistics are separate conditions, each checked at the scale of interest.

Exam questions

(b) Describe some typical plasmas in nature and technology (Fig. 1.3).

(c) What are characteristic phenomena in plasmas?

(g) List features of an ideal plasma.

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 4

- What is the bulk meaning of quasineutrality?
- What do weak coupling and quasineutrality assert, respectively?
- Why can a weakly ionized gas still behave as a plasma?
- What is lost when moving from the particle–field model to a fluid model?

Answers

What is the bulk meaning of quasineutrality?
On scales large compared with the charge-separation layer, the signed species charge densities nearly cancel: $\rho_q \approx 0$. This does not forbid Debye-scale charge separation or a sheath.

What do weak coupling and quasineutrality assert, respectively?
Weak coupling means that a typical pair interaction is small compared with thermal energy. Quasineutrality compares the net charge density with the individual species charge densities on a stated macroscopic scale.

Why can a weakly ionized gas still behave as a plasma?
Its charged component can sustain collective electromagnetic responses on the observation scale even when most particles are neutral.

What is lost when moving from the particle–field model to a fluid model?
Velocity-space and particle-level detail are averaged into moments, and a closure assumption supplies the unresolved higher moments.

1.2 Speed, energy, and temperature

What does temperature measure for a plasma particle, and why is it not a universal switch into the plasma state? Start with the kinetic-energy scale, then keep the equilibrium and non-equilibrium meanings of temperature distinct.

Learning objectives

- relate particle speed, kinetic energy, and the thermal energy $k_B T_s$
- use one stated thermal-speed convention consistently
- explain why different species can have different temperatures

Unit ledger

When temperature is quoted in electron-volts, the intended quantity is the energy $k_B T_s$, not the thermodynamic temperature symbol by itself.

For a particle of species s, the kinetic energy associated with speed v is

$$\varepsilon_{\mathrm{kin,s}} = \frac{m_s v^2}{2}, \quad \varepsilon_{\mathrm{th,s}} = k_B T_s$$

The first quantity depends on the individual particle speed. The second is the thermal energy scale of an equilibrium population; both are in J. A single particle need not have the population's average kinetic energy.

Temperature measures the spread of random velocities around the bulk motion. Heating a population and accelerating it as a whole are therefore different changes. Throughout this course the thermal-speed convention is

$$v_{\mathrm{th,s}} = \sqrt{\frac{2k_B T_s}{m_s}}$$

$v_{\mathrm{th,s}}$ is in m s^{-1}. It is a characteristic width, not the mean speed of the particles.

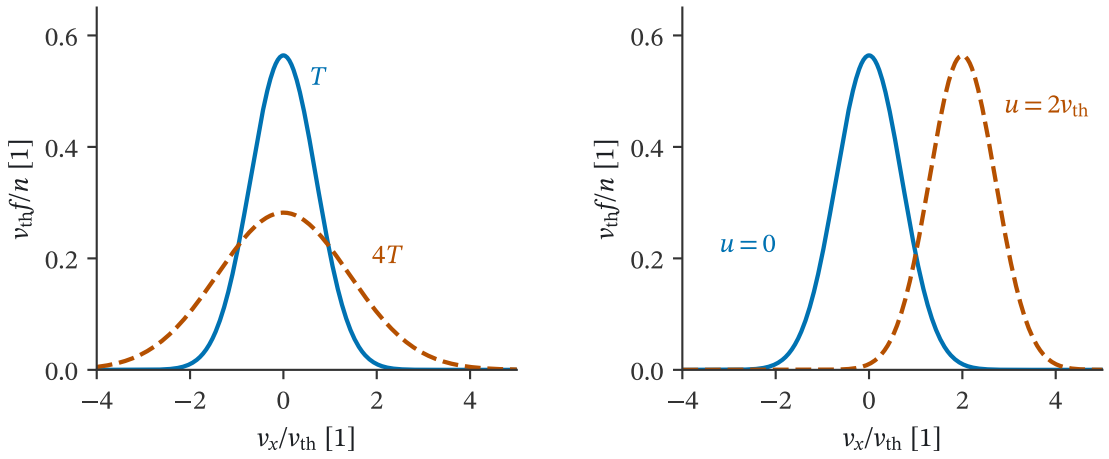


Figure 3: One velocity component of a Maxwellian with the same density. Left: heating from T to $4T$ doubles the width v_{th} at fixed bulk velocity. Right: acceleration to $u = 2v_{\mathrm{th}}$ shifts the distribution at fixed width. Velocity in units of the initial v_{th}, f in units of n/v_{th}.

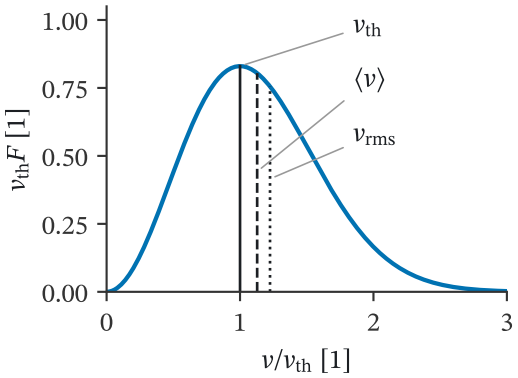


Figure 4: Speed distribution $F(v)$ of an isotropic Maxwellian, $\int_0^\infty F \, \mathrm{d}v = 1$. The thermal speed v_{th} is the most probable speed; the mean speed is $2v_{\mathrm{th}}/\sqrt{\pi}$ and the root-mean-square speed $\sqrt{3/2}v_{\mathrm{th}}$. Speed in units of v_{th}.

Distribution functions and phase space develops the Maxwellian, its normalization, and the connection between temperature and mean random kinetic energy.

The conversion $1\ \mathrm{eV} = 1.602176634 \cdot 10^{-19}\ \mathrm{J}$ corresponds to $1\ \mathrm{eV}/k_B \approx 1.1605 \cdot 10^4\ \mathrm{K}$. Thus a statement such as $k_B T_e = 10\ \mathrm{eV}$ identifies an energy scale of roughly $1.16 \cdot 10^5\ \mathrm{K}$, while the electron and ion temperatures may still differ. A temperature also does not specify the total heat content: that depends on the number of particles and the volume.

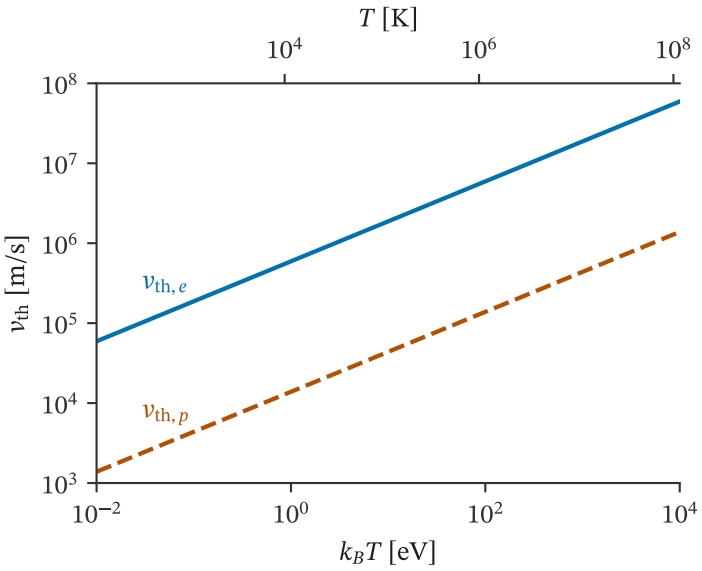


Figure 5: Thermal speed $v_{\mathrm{th}} = \sqrt{2k_B T/m}$ of electrons (solid) and protons (dashed) against $k_B T$ in eV; top axis: T in K, $1\ \mathrm{eV}/k_B \approx 1.16 \cdot 10^4\ \mathrm{K}$. Both lines have slope 1/2; their ratio is $\sqrt{m_p/m_e} \approx 43$.

Nonrelativistic.

Electron and ion temperatures can differ when energy exchange between species is slow. Outside thermal equilibrium, one temperature does not describe beams or unequal velocity spreads in different directions. Equilibrium and kinetic model limits explains when a thermal description is adequate.

There is consequently no universal temperature at which matter abruptly enters the plasma state. Ionization, the energy source, collisionality, and the scale of the electromagnetic response decide whether an ionized gas behaves as a plasma.

Summary

Summary 5

Speed sets the single-particle kinetic energy, while temperature sets the width and average energy of an equilibrium distribution. The script uses $v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$ and treats electron and ion temperatures as separate quantities unless an equilibration assumption is stated.

Exam questions

(a) At which temperature does matter enter the plasma state?

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 6

- What quantity does a quoted plasma temperature in electron-volts represent?
- How does the chosen thermal-speed convention relate to $k_B T_s$?
- Why can a plasma have separate electron and ion temperatures?
- Why is there no universal temperature threshold for the plasma state?

Answers

What quantity does a quoted plasma temperature in electron-volts represent?
It represents the thermal energy $k_B T_s$ expressed in electron-volts. The thermodynamic temperature T_s can be recovered using the conversion to kelvin.

How does the chosen thermal-speed convention relate to $k_B T_s$?
$v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$. At fixed mass, multiplying thermal energy by four doubles this characteristic speed.

Why can a plasma have separate electron and ion temperatures?
Interspecies energy exchange can be slower than equilibration within each species, so each species can have its own approximately Maxwellian distribution for the time scale being considered.

Why is there no universal temperature threshold for the plasma state?
Plasma behavior depends on ionization, collisionality, boundaries, and whether collective electromagnetic response matters on the chosen scales, not on temperature alone.

1.3 Characteristic scales and ordering

A model must resolve the response being studied. Compare the size and duration of the observation with the distances and times over which particles and fields respond.

Learning objectives

- distinguish screening, collective oscillation, and gyromotion
- explain why length and time scales affect model selection
- locate the quantitative treatment of a relevant response

Unit ledger
Comparing two lengths or two times produces a dimensionless ratio; both must refer to the same physical problem.

A displaced electron population overshoots equilibrium because electrons have inertia. The charge separation supplies a restoring force. Plasma oscillations derives this motion and its natural response time.

Screening describes how mobile charges rearrange around an electric disturbance. Far enough from a localized source, its influence is reduced. Debye shielding derives the screening length and the assumptions behind bulk quasineutrality.

In a magnetic field, individual particles turn around field lines. Their orbit size and period determine whether this motion must be followed or can be averaged. Uniform-field gyromotion defines these scales, with numerical examples in Comparing orbit and collective scales.

Collisions introduce a further distance: how far a particle travels before its motion changes appreciably through scattering. Collisions in gases and plasmas relates this mean free path to the collision rate.

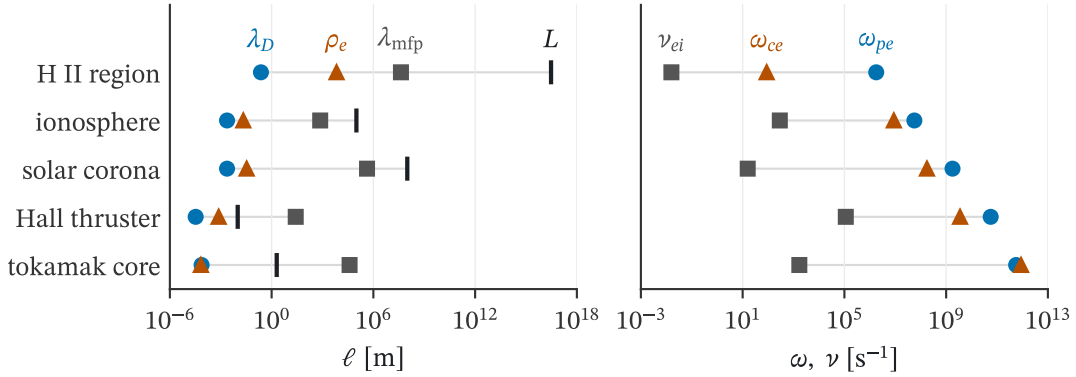


Figure 6: Characteristic lengths (left) and rates (right) of five example plasmas: Debye length λ_D, electron thermal gyroradius $\rho_e = v_{\mathrm{th},e}/\omega_{ce}$, Coulomb mean free path $\lambda_{\mathrm{mfp}} = \langle v \rangle / \nu_{ei}$, system size L; electron plasma and cyclotron frequencies ω_{pe}, ω_{ce} and electron–ion collision frequency ν_{ei}. Order-of-magnitude n_e, $k_B T_e$, B and L; the ionosphere's dominant electron–neutral collisions are not included.

For example, an instrument that averages over many tiny orbits may not need a model of each turn. An experiment that resolves a single orbit does. Likewise, bulk neutrality can be useful far from a wall while failing in the thin charge-separated layer next to it. Each approximation must match the observation; one small scale does not justify every simplification.

Summary

Summary 7

Screening, oscillation, gyromotion, and collisions describe different responses. Comparing their scales with the observation tells us which processes to resolve and which might be averaged.

Knowledge check

Knowledge check 8

- How do shielding and plasma oscillations differ?
- When might averaging over gyromotion be useful?
- Why can bulk quasineutrality fail close to a wall?
- Does rapid screening imply frequent collisions?

Answers

How do shielding and plasma oscillations differ?

Shielding concerns an equilibrium rearrangement of charge; plasma oscillations concern the time-dependent restoring motion of displaced charges.

When might averaging over gyromotion be useful?

When observations and fields vary over distances and times much larger than an orbit and its period, and the omitted orbit detail does not control the response.

Why can bulk quasineutrality fail close to a wall?

The wall can create a thin charge-separated layer. Resolving that layer requires keeping the charge imbalance.

Does rapid screening imply frequent collisions?

No. Collective field response and collisional scattering have distinct scales that must be compared separately with the observation.

1.4 From microscopic particles to a model

Which description should be used for a given plasma problem? Keep the variables needed by the observable, then justify every average, closure, and ordering that removes detail.

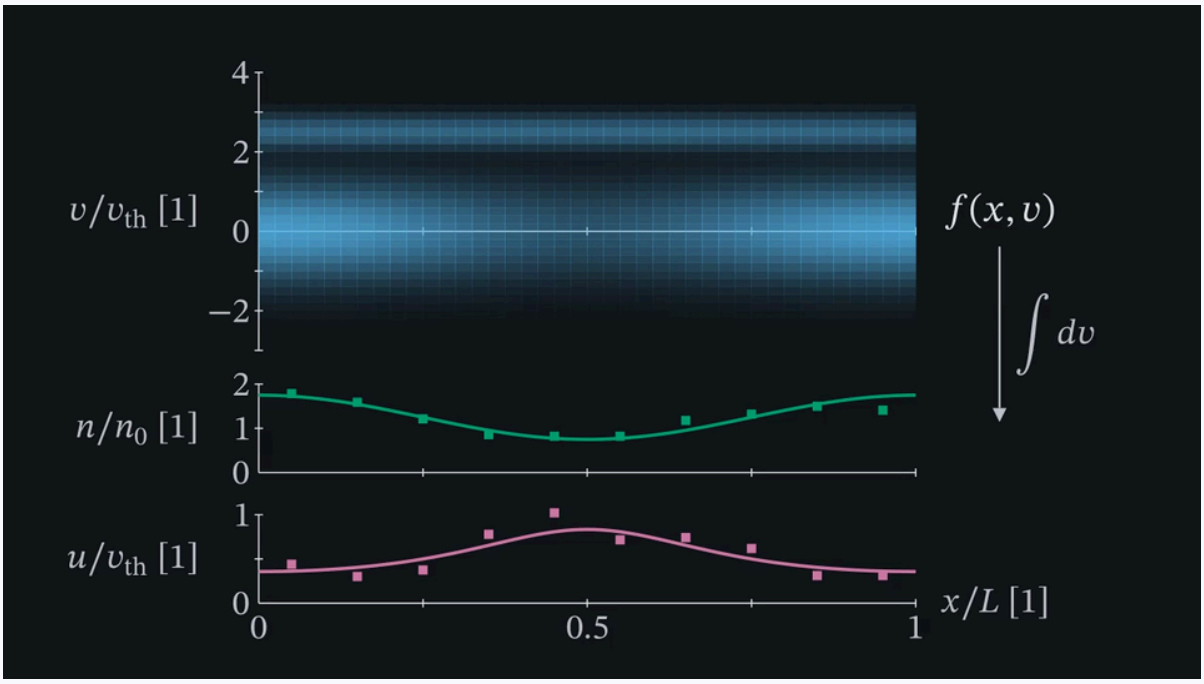
Learning objectives

- map particle, kinetic, multiple-fluid, and MHD descriptions to retained variables
- explain what is lost when taking moments or combining species
- connect equilibrium distributions to the next kinetic description

The most complete classical description follows all charged particles together with the fields they generate. A single-particle approximation instead prescribes the fields; the Lorentz-force model explains what can be learned from that approximation.

Kinetic theory replaces the list of particles with a distribution over positions and velocities. It retains beams and directional structure that a density and temperature cannot describe. The Boltzmann and Vlasov equations evolve this distribution under forces and collisions.

Fluid models keep averages such as density, bulk velocity, and pressure. The continuity equation illustrates how their conservation laws follow from kinetic theory. Discarding the rest of the distribution requires a closure assumption. Combining species into a single conducting fluid leads to magnetohydrodynamics, when its scale assumptions hold.



Watch animation

Particles, distribution function and moments in one dimension: a density-modulated Maxwellian at rest plus a uniform beam at $v = 2.5v_{\mathrm{th}}$. Integration over v gives $n(x)$ and $u(x)$; squares are estimates from the plotted particles. Position in units of the period L, velocity in units of v_{th}, density in units of the mean background density n_0.

Figure 7: Particles, distribution function and moments in one dimension: a density-modulated Maxwellian at rest plus a uniform beam at $v = 2.5v_{\mathrm{th}}$. Integration over v gives $n(x)$ and $u(x)$; squares are estimates from the plotted particles. Position in units of the period L, velocity in units of v_{th}, density in units of the mean background density n_0.

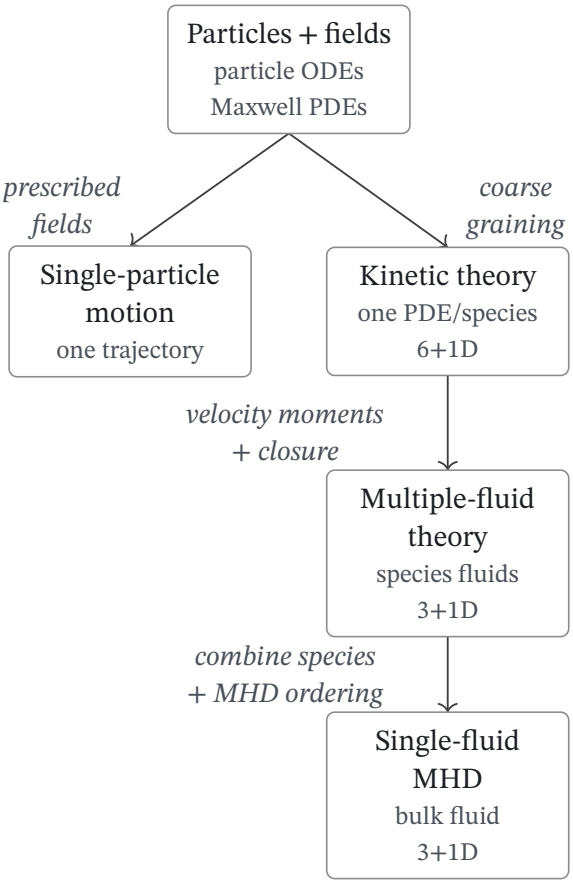


Figure 8: Two reductions of the particle–field description. Prescribing the fields gives a test-particle model; retaining collective feedback leads to kinetic and fluid descriptions, with an additional assumption at each step. 6+1D: three position and three velocity coordinates and time; 3+1D: position and time. At 100 grid points per coordinate a 3D grid has 10^6 points and a 6D grid 10^{12}: the curse of dimensionality.

Map of plasma models

These four descriptions are a small part of a larger landscape. The map below places 65 plasma models between the first principles at the top, the classical Maxwell–Lorentz many-body system and the many-electron Schrödinger equation, and the fluid, wave and equilibrium models at the bottom; each arrow is a reduction with its own small parameters and assumptions.

This course follows the highlighted route. From the quantum branch it takes only thermal equilibrium (Saha ionization, the Fermi energy). It reaches the Vlasov equation by coarse graining instead of deriving it from the many-body dynamics, and it leaves the many-body hierarchy, the collision operators and drift kinetics to the Kinetic Theory course. The interactive map shows the equations, assumptions and verification status of every model and reduction.

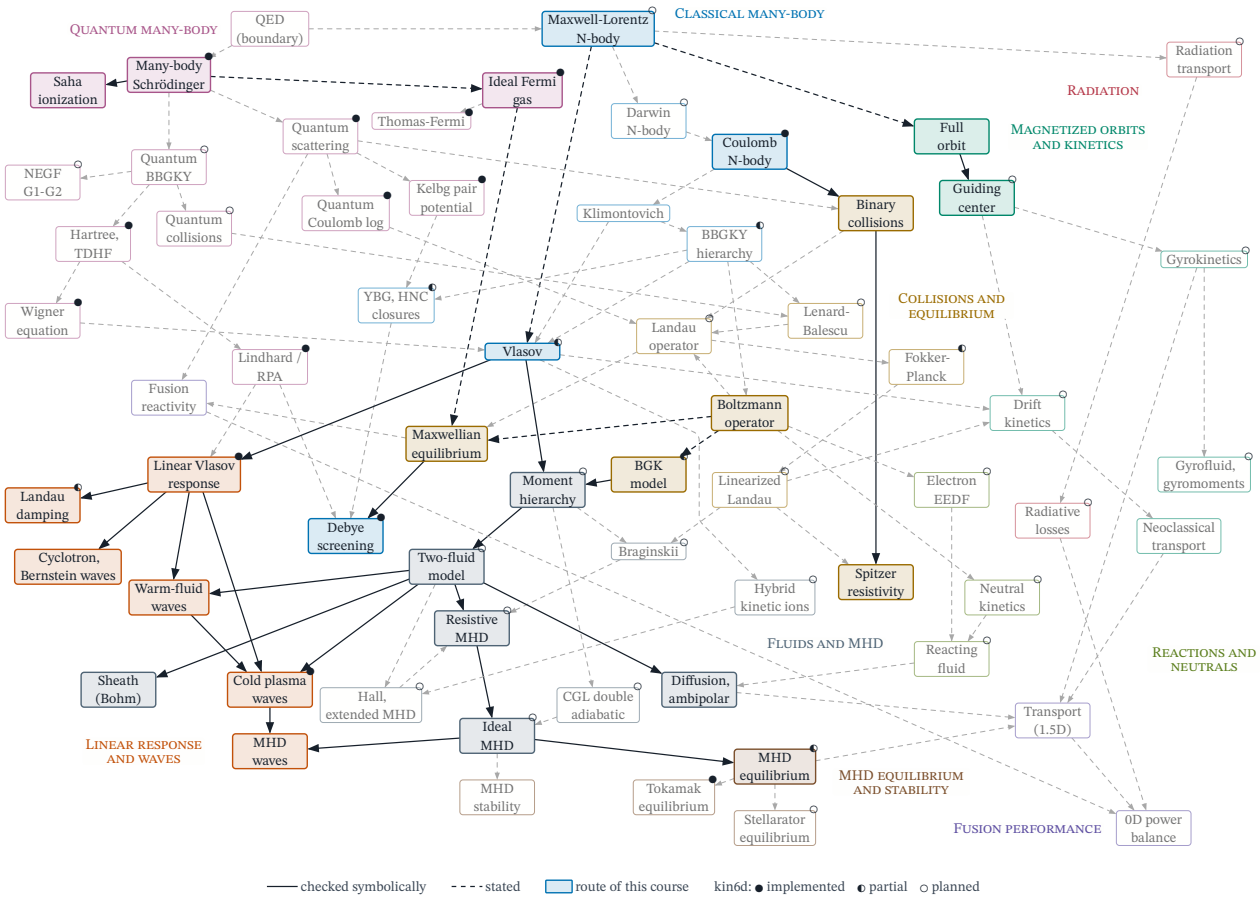


Figure 9: Map of plasma models, from first principles at the top to reduced models at the bottom. Highlighted boxes and arrows form the route of this course; solid arrows are reductions whose algebra is checked in the SymPy derivations of the script, dashed arrows are stated only. Small circles mark the state of the research code kin6d: filled implemented, half filled partially implemented, open planned. Open the interactive map for the equations, assumptions and evidence of every model and reduction.

Equilibrium is a model statement Interpretation 9

A Maxwell–Boltzmann distribution is an equilibrium candidate when the relevant collisions establish a thermal state. A collisionless plasma can retain non-Maxwellian structure, beams, or anisotropy. Calling a state equilibrium therefore requires specifying which dynamics and time scale have equilibrated.

Summary Summary 10

Model reduction is an ordered argument about retained variables, averaging, and closure. The distribution function connects the full particle picture to moments and fluid equations, while equilibrium is always relative to a specified equilibration process.

Exam questions

(k) Describe the most complete plasma model? (All particles interacting with electromagnetic forces)

(l) What kind of models can be used for plasma and how are they related?

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 11

- What variable is removed when passing from a kinetic description to a fluid density?
- Why can a collisionless plasma be non-Maxwellian?
- Why does a fluid description need a closure assumption?
- State a model-selection question that must be answered before using MHD.

Answers

What variable is removed when passing from a kinetic description to a fluid density?
Velocity dependence is integrated out. The density is the zeroth velocity moment, while information about distribution shape is no longer retained directly.

Why can a collisionless plasma be non-Maxwellian?
Without collisions that efficiently mix phase space, beams, anisotropy, and other structures can persist on the observation time.

Why does a fluid description need a closure assumption?
It retains only a few averages. A closure supplies a model for the discarded distribution information that still affects their evolution.

State a model-selection question that must be answered before using MHD.
One must justify combining species into one fluid and closing the pressure or energy response at the length and time scales of interest.

2. Temperature, entropy, and thermal ionization

Ionization costs energy, yet a hydrogen gas can become substantially ionized far below the temperature corresponding to the binding energy of one atom. Counting accessible states explains this balance. It also derives the Boltzmann electron response used for Debye shielding in Chapter 4. The statistical framework follows [4]; thermal ionization follows [5].

2.1 Entropy as counted microstates

A macroscopic description specifies totals, such as energy and particle number, without specifying each particle's microscopic state. How many microscopic arrangements realize the same totals? Their number is the multiplicity W. A microstate is one fully specified microscopic arrangement; a macrostate is the collection of constraints that those arrangements share.

Learning objectives
- count independent binary choices and distinguish multiplication from addition
- explain why the logarithm converts multiplicity into additive entropy

Unit ledger
Multiplicity W is a dimensionless count. Entropy S is in $\mathrm{J\,K^{-1}}$; Boltzmann's constant k_B is in $\mathrm{J\,K^{-1}}$.

Each cell in a simple counting model can be open or filled. A configuration specifies the state of every cell. For a independent two-state cells, the next cell doubles the possibilities:

$$W_a = 2^a, \quad W_1 = 2, \quad W_2 = 4, \quad W_3 = 8.$$

Combining two independent systems A and B pairs every state of A with every state of B. Thus the combined count is $W_A W_B$, not $W_A + W_B$. Define entropy by taking the natural logarithm, denoted ln:

$$S = k_B \ln W, \quad S_{A+B} = S_A + S_B.$$

The logarithm is dimensionless because its argument is a count. Multiplying by k_B fixes the physical entropy scale. This counting definition assumes equal statistical weight for accessible microstates of an isolated system.

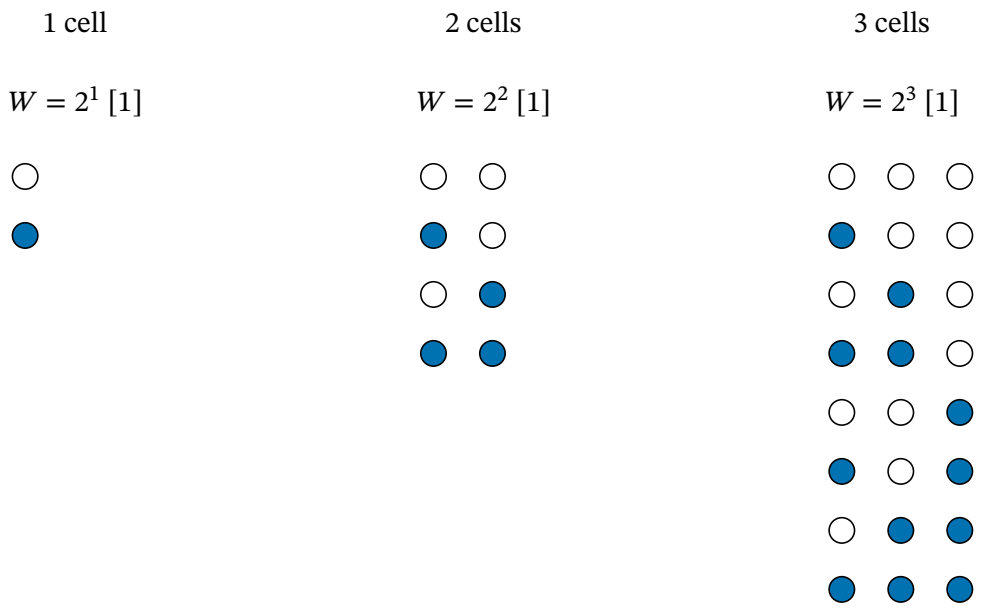


Figure 10: Rows enumerate microstates. Open and filled circles distinguish the two choices without relying on color. The counts W have unit [1].

Interaction energy or shared constraints can prevent factorization; then independence must be justified before adding subsystem entropies.

Summary

Summary 12

Independent possibilities multiply; their logarithms add. Entropy records the accessible microscopic freedom behind a macroscopic description.

Knowledge check

Knowledge check 13

- Two independent cells have three states each. What is their multiplicity?
- Why does adding one binary cell to two cells yield eight rather than six states?
- How much entropy does an independent binary cell add?
- If two cells must always have equal states, may their individual counts be multiplied?

Answers

Two independent cells have three states each. What is their multiplicity?
Nine, a dimensionless count: each of the three first choices pairs with three second choices.
Why does adding one binary cell to two cells yield eight rather than six states?
Each of the four existing states receives two alternatives, so the count is multiplied by two.
How much entropy does an independent binary cell add?
$k_B \ln 2$, in $\mathrm{J\,K^{-1}}$, because it multiplies the total multiplicity by two.
If two cells must always have equal states, may their individual counts be multiplied?
No. The constraint leaves only the two equal-state pairs, so the choices are correlated.

2.2 Temperature

Two boxes exchange a small energy while their combined energy remains fixed. The favored transfer is the one that opens more combined microstates. Temperature measures the entropy gained per unit energy, rather than the total energy in a box.

Learning objectives

- define temperature through an entropy derivative with its fixed constraints
- infer the direction of energy exchange and the condition for equilibrium

Unit ledger

Pressure p is in Pa. Chemical potential μ is in J per particle; total momentum $\boldsymbol{P}$ is in $\mathrm{kg\,m\,s^{-1}}$. Its conjugate bulk velocity $\boldsymbol{u}$ is in $\mathrm{m\,s^{-1}}$.

Let E be total energy, V volume, N particle number, and $\boldsymbol{P}$ total momentum. At fixed V, N, and $\boldsymbol{P}$, define absolute temperature T, measured in K, by

$$\frac{1}{T} = \frac{\partial S}{\partial E}, \quad \mathrm{d}S = \frac{\mathrm{d}E}{T}.$$

The derivative holds V, N, and $\boldsymbol{P}$ fixed. Its unit is $\mathrm{K^{-1}}$.

If A receives energy $\mathrm{d}E_A$ from B, then $\mathrm{d}E_B = -\,\mathrm{d}E_A$ and

$$\mathrm{d}S_{A+B} = \left(\frac{1}{T_A} - \frac{1}{T_B}\right)\mathrm{d}E_A.$$

At positive temperatures, energy flows from hotter to colder: taking $T_B > T_A$ makes the coefficient positive. At the maximum of total entropy, the first-order change vanishes for either transfer direction, so $T_A = T_B$. A stable maximum also requires negative curvature along the transfer direction.

Thermodynamic differential Definition 14

Pressure p is the mechanical force per area. Chemical potential μ measures the energy cost of adding a particle at fixed entropy, volume, and momentum. With bulk velocity $\boldsymbol{u}$ conjugate to momentum,

$$\mathrm{d}E = T\,\mathrm{d}S - p\,\mathrm{d}V + \mu\,\mathrm{d}N + \boldsymbol{u}\cdot\mathrm{d}\boldsymbol{P}.$$

In a mixture, replace $\mu\,\mathrm{d}N$ by the sum over species.

The ideal translational gas considered here has an unbounded kinetic-energy spectrum and increasing entropy with energy, so $T > 0$. Negative absolute temperatures require a bounded accessible energy spectrum and do not describe the Maxwellian plasma model.

Derivation: entropy maximum for two boxes

Write $S_{\mathrm{tot}(E_A)} = S_{A(E_A)} + S_{B(E_{\mathrm{tot}}-E_A)}$ with fixed total energy E_{tot}. The chain rule gives

$$\frac{\mathrm{d}S_{\mathrm{tot}}}{\mathrm{d}E_A} = \frac{\partial S_A}{\partial E_A} - \frac{\partial S_B}{\partial E_B} = \frac{1}{T_A} - \frac{1}{T_B}.$$

Setting this derivative to zero gives equality of temperatures. For positive heat capacities $C_A = \frac{\mathrm{d}E_A}{\mathrm{d}T_A}$ and $C_B = \frac{\mathrm{d}E_B}{\mathrm{d}T_B}$, each in $\mathrm{J\,K^{-1}}$,

$$\frac{\mathrm{d}^2 S_{\mathrm{tot}}}{\mathrm{d}E_A^2} = -\frac{1}{T_A^2 C_A} - \frac{1}{T_B^2 C_B} < 0.$$

Thus the stationary state is a maximum under energy exchange.

Summary

Summary 15

Inverse temperature is the entropy slope with respect to energy. Equality of those slopes stops net energy exchange at thermal equilibrium.

Knowledge check

Knowledge check 16

- Which constraints belong to the entropy definition of temperature?
- A colder box receives a small energy from a hotter box. What happens to total entropy?
- Does equal energy in two boxes imply equal temperature?
- Why are negative absolute temperatures excluded in this gas model?

Answers
Which constraints belong to the entropy definition of temperature?
Volume, particle number, and total momentum stay fixed while energy changes.
A colder box receives a small energy from a hotter box. What happens to total entropy?
It increases: the colder box gains more entropy than the hotter box loses per joule.
Does equal energy in two boxes imply equal temperature?
No. Different particle numbers, volumes, or spectra can give different entropy slopes at the same energy.
Why are negative absolute temperatures excluded in this gas model?
Its kinetic-energy spectrum has no upper bound; the bounded-spectrum condition for a negative-temperature equilibrium is absent.

2.3 The Boltzmann factor and the Maxwellian

A small system can take energy from a much larger reservoir. A high-energy state of the small system leaves less energy, and fewer possibilities, for the reservoir. This loss supplies the Boltzmann factor.

Learning objectives
- derive the weight of a subsystem microstate from a reservoir entropy expansion
- separate conserved bulk momentum from thermal motion and normalize a Maxwellian

Unit ledger
Velocity probability density F_s is in $\mathrm{s}^3\,\mathrm{m}^{-3}$.

Let P_a be the dimensionless probability of a particular subsystem microstate a with energy E_a. The reservoir, labelled rest, has energy $E_{\mathrm{tot}} - E_a$. At fixed reservoir volume and particle number,

$$P_a \propto W_{\mathrm{rest}(E_{\mathrm{tot}}-E_a)} = \exp\left(\frac{S_{\mathrm{rest}(E_{\mathrm{tot}}-E_a)}}{k_B}\right).$$

A large reservoir changes temperature negligibly on exchanging E_a. Expanding its entropy to first order gives

$$S_{\mathrm{rest}(E_{\mathrm{tot}}-E_a)} \approx S_{\mathrm{rest}(E_{\mathrm{tot}})} - \frac{E_a}{T}, \quad P_a = \frac{\exp\left(-\frac{E_a}{k_B T}\right)}{Z}.$$

The dimensionless partition sum $Z = \sum_a \exp\left(-\frac{E_a}{k_B T}\right)$ normalizes the probabilities. For an energy interval, also count the subsystem states in that interval; the Boltzmann factor alone is a weight per microstate, not an energy histogram. For N equal-mass particles of mass m, conserved momentum sets the mean velocity $\boldsymbol{u} = \frac{\boldsymbol{P}}{Nm}$. Their kinetic energy separates as

$$\sum_{j=1}^{N} \frac{m\boldsymbol{v}_j^2}{2} = \frac{\boldsymbol{P}^2}{2Nm} + \sum_{j=1}^{N} \frac{m(\boldsymbol{v}_j - \boldsymbol{u})^2}{2}.$$

The bulk term is fixed. The fluctuating velocity relative to $\boldsymbol{u}$ therefore receives the thermal weight. For species s, define F_s as a velocity probability density with $\int F_s \mathrm{d}^3 v = 1$:

$$F_{s(\boldsymbol{v})} = \frac{1}{\pi^{\frac{3}{2}} v_{\mathrm{th},s}^3} \exp\left(-\frac{(\boldsymbol{v} - \boldsymbol{u}_s)^2}{v_{\mathrm{th},s}^2}\right), \quad v_{\mathrm{th},s} = \sqrt{\frac{2k_B T_s}{m_s}}.$$

F_s has unit $\mathrm{s}^3\,\mathrm{m}^{-3}$. Multiplying by the density n_s gives the phase-space distribution at a homogeneous point. This is exactly the §1.2 thermal-speed convention: the one-component variance is $k_B \frac{T_s}{m_s}$, and the most probable three-dimensional speed relative to the flow is $v_{\mathrm{th},s}$.

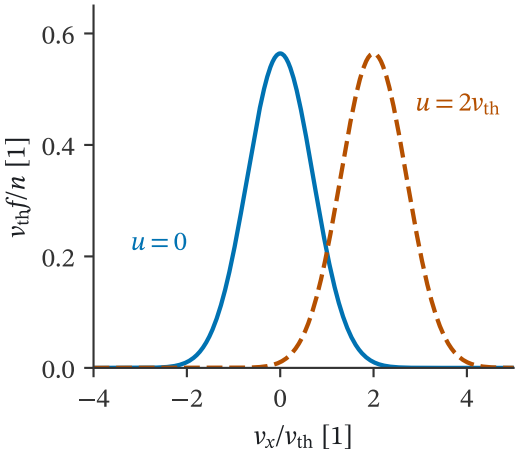


Figure 11: Changing mean flow shifts a Maxwellian; heating changes its width. The velocity normalization uses the thermal speed defined in §1.2.

A Maxwellian requires thermal equilibration. Beams, anisotropy, or different electron and ion temperatures need a kinetic description beyond one common equilibrium temperature. The reservoir expansion fails if its temperature changes appreciably during the exchange.

Summary Summary 17

The reservoir's loss of states gives $\exp\left(-\frac{E}{k_B T}\right)$. At fixed bulk momentum, the relevant kinetic energy is measured relative to the mean flow.

Knowledge check — Knowledge check 18

- Why does a subsystem energy cost reduce its probability?
- Is the probability of an energy interval just its Boltzmann factor?
- A flow doubles while the random velocity variance stays fixed. Does the temperature change?
- When is the reservoir entropy expansion unreliable?

Answers

Why does a subsystem energy cost reduce its probability?

It removes energy from the reservoir, reducing reservoir entropy by approximately $\frac{E}{T}$.

Is the probability of an energy interval just its Boltzmann factor?

No. Multiply the microstate weight by the number of subsystem states in that interval.

A flow doubles while the random velocity variance stays fixed. Does the temperature change?

No. Temperature measures the velocity spread relative to the mean flow, whose kinetic energy is separately conserved.

When is the reservoir entropy expansion unreliable?

When exchanged energy changes the reservoir temperature substantially, so higher-order entropy terms matter.

2.4 The Saha equation

Removing an electron from an atom costs ionization energy but gives that electron a vast set of free translational states. Counting the change in possibilities determines the equilibrium composition without requiring a typical particle's thermal energy to equal the binding energy.

Learning objectives

- derive the add and remove factors for indistinguishable dilute particles
- derive Saha balance by canceling heavy-particle translational factors
- identify electron freedom, internal states, and the reservoir energy penalty

Unit ledger

Planck's constant h is in J s; thermal de Broglie wavelength $\lambda_{\mathrm{th},s}$ is in m. Ionization energy χ is in J or eV. Box side L and volume V are in m and m^3. Momenta p_s and $p_{\mathrm{th},s}$ are in $\mathrm{kg\,m\,s^{-1}}$. Sharp-cutoff wavelength λ_s is in m; density n_s is in $\mathrm{m^{-3}}$. Counts M_s, N_s, and g_s have unit [1].

Dilute equilibrium mixture — Assumptions 19

Neutral atoms n, singly ionized ions i, and free electrons e share one temperature and volume. Local thermodynamic equilibrium means that collisions establish thermal velocities and reversible ionization and recombination balance locally. Quantum occupation is dilute: $n_s \frac{\lambda_{\mathrm{th},s}^3}{g_s} \ll 1$. Interactions do not shift the isolated atom's binding energy. Only one ionization stage is retained.

First estimate the number of one-particle patterns in a box. A wave reflects from the walls and forms a standing mode. In one direction its wavelength $\lambda_s = \frac{h}{p_s}$ fits an integer number of half-waves across side L. Independent directions give about one distinguishable pattern per spatial cell of volume λ_s^3. For momentum scale p_s, the sharp-cutoff estimate is

$$\lambda_s = \frac{h}{p_s}, \quad M_s \approx g_s \frac{V}{\lambda_s^3}, \quad \frac{M_s}{N_s} \approx \frac{g_s}{n_s \lambda_s^3}.$$

Geometry and boundary conditions change only order-one factors in this estimate. The internal degeneracy g_s counts states with the same energy; $N_s = n_s V$ is the number of particles of species s.

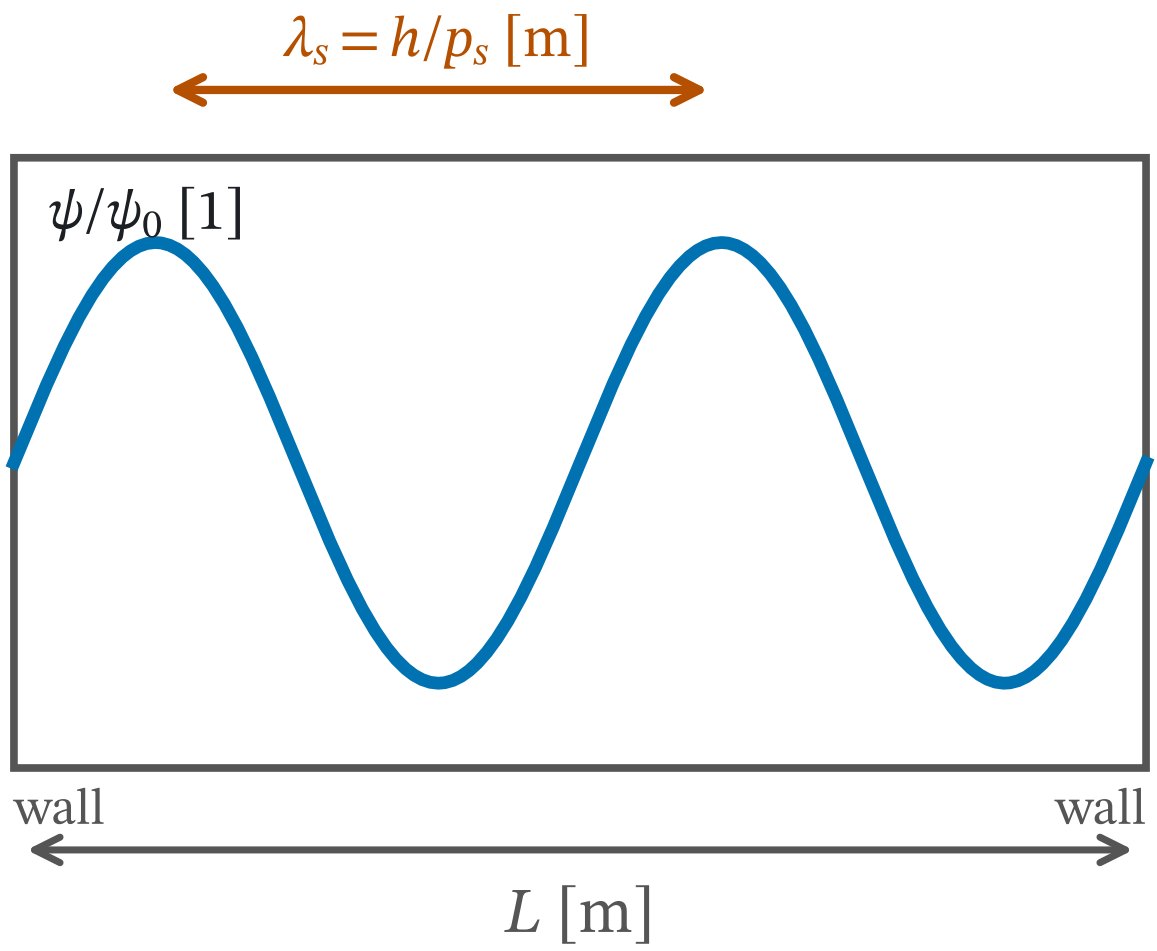


Figure 12: A one-dimensional section of a box mode; independent standing-wave choices in the other two directions give the volume estimate $\frac{V}{\lambda_s^3}$. The plotted amplitude is normalized and labelled [1]; L and λ_s are lengths in m.

Let $\boldsymbol{p}$ be a particle's momentum vector and $p = |\boldsymbol{p}|$ its magnitude. Thermal particles do not stop at a sharp momentum boundary. Momentum states carry a Maxwell–Boltzmann weight $\exp\left(-\frac{\boldsymbol{p}^2}{2m_s k_B T}\right)$. The characteristic thermal momentum is $p_{\mathrm{th},s} = \sqrt{2m_s k_B T}$, so the one-wavelength estimate would use $\lambda_s = \frac{h}{p_{\mathrm{th},s}}$.

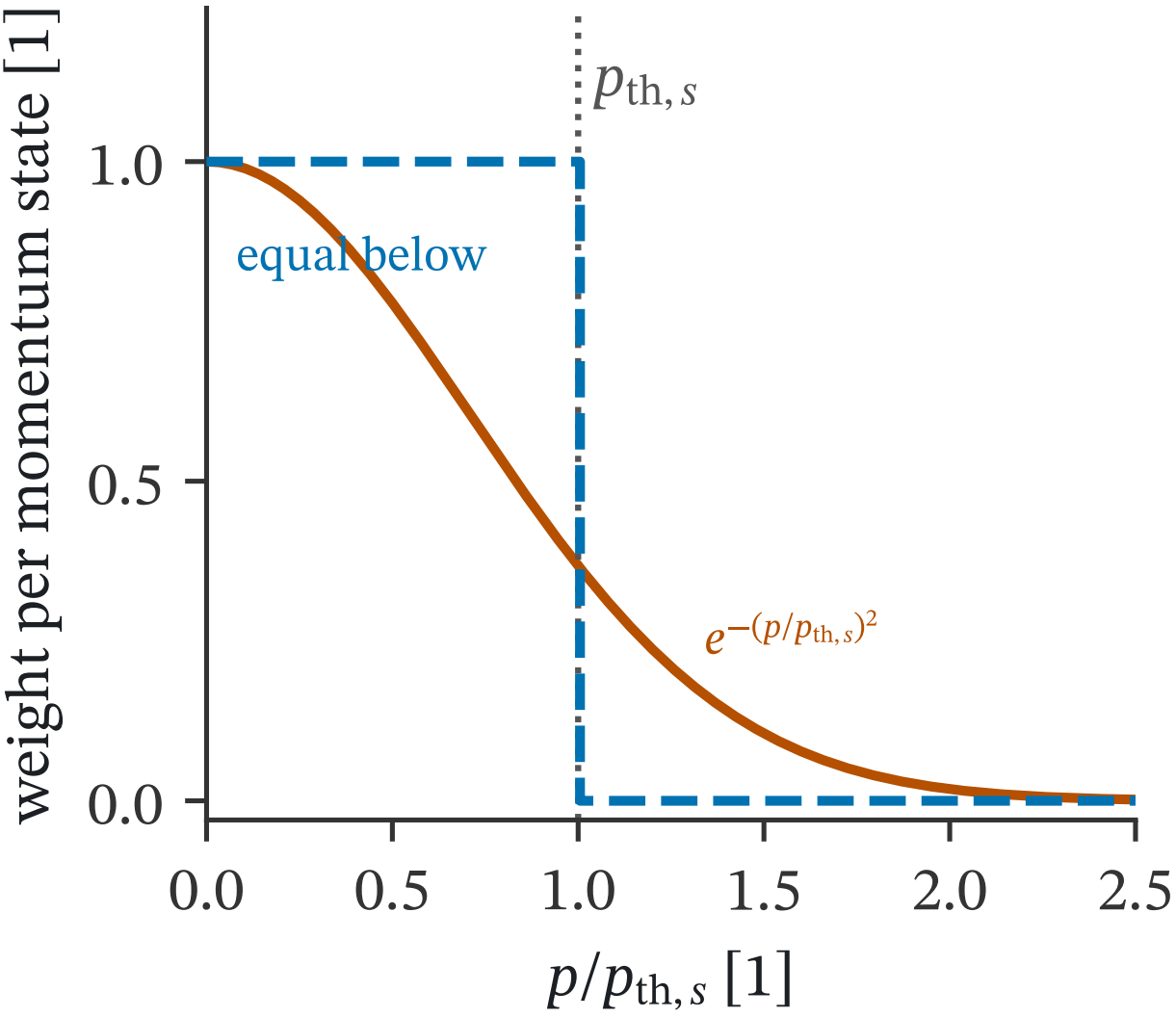


Figure 13: The cutoff counts all states below $p_{\mathrm{th},s}$ equally. The Maxwellian assigns weight $\exp\left(-\left(\frac{p}{p_{\mathrm{th},s}}\right)^2\right)$ per state, with a tail beyond the marked thermal momentum. Both plotted axes are normalized and labelled [1].

The exact effective slot count averages over these weights. Each quantum state occupies phase-space volume h^3:

$$\int \exp\left(-\frac{\boldsymbol{p}^2}{2m_s k_B T}\right) \mathrm{d}^3 p = (2\pi m_s k_B T)^{\frac{3}{2}}.$$

Thus the Gaussian integral replaces the sharp momentum boundary by

$$\lambda_{\mathrm{th},s} = \frac{h}{\sqrt{2\pi m_s k_B T}} = \frac{h}{\sqrt{\pi} m_s v_{\mathrm{th},s}} = \frac{\lambda_s}{\sqrt{\pi}}.$$

The wavelength changes by the order-one factor $\sqrt{\pi}$; the corresponding momentum-space volume changes by $\pi^{\frac{3}{2}}$. The thermal slot count and its count per particle are

$$M_s = \frac{g_s V}{\lambda_{\mathrm{th},s}^3}, \quad \frac{M_s}{N_s} = \frac{g_s}{n_s \lambda_{\mathrm{th},s}^3}.$$

Exact-to-sharp-cutoff slot-volume ratio:

$$\frac{M_{s(T)}}{g_s \frac{V}{\lambda_s^3}} = \pi^{\frac{3}{2}}.$$

The full Gaussian momentum integral is the precise replacement for the de Broglie-cell estimate; the estimate captures the scale of the available translational states.

Derivation: thermally weighted slots
A quantum state occupies phase-space volume h^3. With momentum $\boldsymbol{p}$,

$$M_s = \frac{g_s V}{h^3} \int \exp\left(-\frac{\boldsymbol{p}^2}{2m_s k_B T}\right) \mathrm{d}^3 p.$$

Each Cartesian Gaussian integral is $\sqrt{2\pi m_s k_B T}$. Their product gives $(2\pi m_s k_B T)^{\frac{3}{2}}$ and hence $g_s \frac{V}{\lambda_{\mathrm{th},s}^3}$.

For N_s indistinguishable particles, count each slot assignment once, rather than once per particle permutation. In the dilute limit,

$$W_{s(N_s)} \approx \frac{M_s^{N_s}}{N_{s!}}.$$

Adding and removing one particle respectively give

$$\frac{W_{s(N_s+1)}}{W_{s(N_s)}} = \frac{M_s}{N_s + 1} \approx \frac{M_s}{N_s}.$$

$$\frac{W_{s(N_s-1)}}{W_{s(N_s)}} = \frac{N_s}{M_s}.$$

The approximation $N_s + 1 \approx N_s$ also assumes macroscopic populations. Counts factor across species in this ideal-mixture model. For one reaction $\mathrm{A} \Longleftrightarrow \mathrm{A}^+ + e^-$, remove a neutral and add an ion and an electron. The reservoir pays energy χ, multiplying its possibilities by $\exp\left(-\frac{\chi}{k_B T}\right)$. With $n_s = \frac{N_s}{V}$, the complete ratio is

$$R = \frac{W_{\text{after}}}{W_{\text{before}}} = \frac{n_n \lambda_{\mathrm{th},n}^3}{g_n} \times \frac{g_i}{n_i \lambda_{\mathrm{th},i}^3} \times \frac{g_e}{n_e \lambda_{\mathrm{th},e}^3} \times \exp\left(-\frac{\chi}{k_B T}\right).$$

Every factor and R are dimensionless. If $R > 1$, further ionization increases entropy; if $R < 1$, recombination does. At the composition maximum, $R = 1$. Since electron mass is small compared with atom or ion mass, $m_n \approx m_i$ and $\lambda_{\mathrm{th},n} \approx \lambda_{\mathrm{th},i}$. The wavelength and internal-state part of the neutral–ion factor cancels:

$$\left(\frac{\lambda_{\mathrm{th},n}}{\lambda_{\mathrm{th},i}}\right)^3 \left(\frac{g_i}{g_n}\right) \approx 1.$$

The remaining $\frac{n_n}{n_i}$ is a composition factor, not part of that cancellation. This uses the lecture approximation $g_i = g_n = 1$. Electron spin gives $g_e = 2$:

$$R \approx \frac{n_n}{n_i n_e} \frac{2 g_i}{g_n \lambda_{\mathrm{th},e}^3} \exp\left(-\frac{\chi}{k_B T}\right).$$

Saha composition balance Governing law 20

Setting $R = 1$ gives

$$\frac{n_i n_e}{n_n} = \underbrace{\frac{g_e}{\lambda_{\mathrm{th},e}^3}}_{\text{free-electron states}} \underbrace{\frac{g_i}{g_n}}_{\text{internal states}} \underbrace{\exp\left(-\frac{\chi}{k_B T}\right)}_{\text{energy penalty}}.$$

With $g_i = g_n = 1$ and $g_e = 2$, substituting the thermal wavelength gives

$$\frac{n_i n_e}{n_n} = 2\left(2\pi m_e k_B \frac{T}{h^2}\right)^{\frac{3}{2}} \exp\left(-\frac{\chi}{k_B T}\right).$$

The left side and electron-state factor have unit m^{-3}. The other factors are dimensionless. Internal quantum states add order-one factors; we ignore them here by setting $g_i = g_n = 1$. This is a lecture simplification, not a detailed hydrogen partition-function calculation. The newly liberated free electron is the new translational freedom: its huge number of states compensates the energy penalty. Saha equilibrium [5] does not describe every ionized gas; chemical detailed balance must also hold.

Summary Summary 21

One ionization removes a neutral, adds an ion and electron, and costs reservoir energy. Heavy-particle translational factors cancel; free-electron states provide the large entropy gain.

Knowledge check Knowledge check 22

- Why does adding a particle give $\frac{M_s}{N_s+1}$ rather than M_s?
- What does the sharp-cutoff slot estimate count?
- How does $\lambda_s = \frac{h}{p_{\mathrm{th},s}}$ compare with $\lambda_{\mathrm{th},s}$?
- Can this Saha model describe ionization by a nonthermal beam?

Answers

Why does adding a particle give $\frac{M_s}{N_s+1}$ rather than M_s?

Indistinguishability gives the factorial ratio $\frac{N_s!}{N_s+1} \neq \frac{1}{N_s+1}$.

What does the sharp-cutoff slot estimate count?

About $g_s \frac{V}{\lambda_s^3}$ standing-wave patterns, one per de Broglie spatial cell up to geometric factors.

How does $\lambda_s = \frac{h}{p_{\mathrm{th},s}}$ compare with $\lambda_{\mathrm{th},s}$?

$\lambda_s = \sqrt{\pi}\lambda_{\mathrm{th},s}$; the Gaussian momentum integral gives the exact thermal wavelength.

Can this Saha model describe ionization by a nonthermal beam?

No. Beam ionization depends on the electron distribution and reaction kinetics; a reservoir weight is insufficient.

2.5 How much thermal ionization?

A composition equation becomes predictive when the total inventory is specified. For single-stage hydrogen, neutrality and nucleus conservation reduce three densities to one ionization fraction.

Learning objectives

- compute a hydrogen ionization fraction at given temperature and nucleus density
- explain the logarithmic thermal threshold and its relation to binding energy
- select Saha or a kinetic rate model from the equilibration conditions

Unit ledger

Ionization fraction x and Saha parameter A are dimensionless. Nucleus density $n = n_i + n_n$ is in m^{-3}. Plot temperature is $k_B T$ in eV, with a second axis showing T in K.

Use $\chi = 13.6\,\mathrm{eV}$, $g_i = g_n = 1$, and neutrality $n_e = n_i$. Density n counts nuclei, not electrons plus ions plus neutrals. Define $x = \frac{n_i}{n}$, so $n_i = n_e = xn$ and $n_n = (1-x)n$. Then

$$\frac{x^2}{1-x} = A(T,n), \quad A(T,n) = \frac{2}{n\lambda_{\mathrm{th},e}^3} \exp\!\left(-\frac{\chi}{k_B T}\right).$$

The physical root lies between zero and one:

$$x = 2\frac{A}{A + \sqrt{A^2 + 4A}} = \frac{2}{1 + \sqrt{1 + \frac{4}{A}}}.$$

For $A \ll 1$, $x \approx \sqrt{A}$; for $A \gg 1$, $1 - x \approx \frac{1}{A}$.

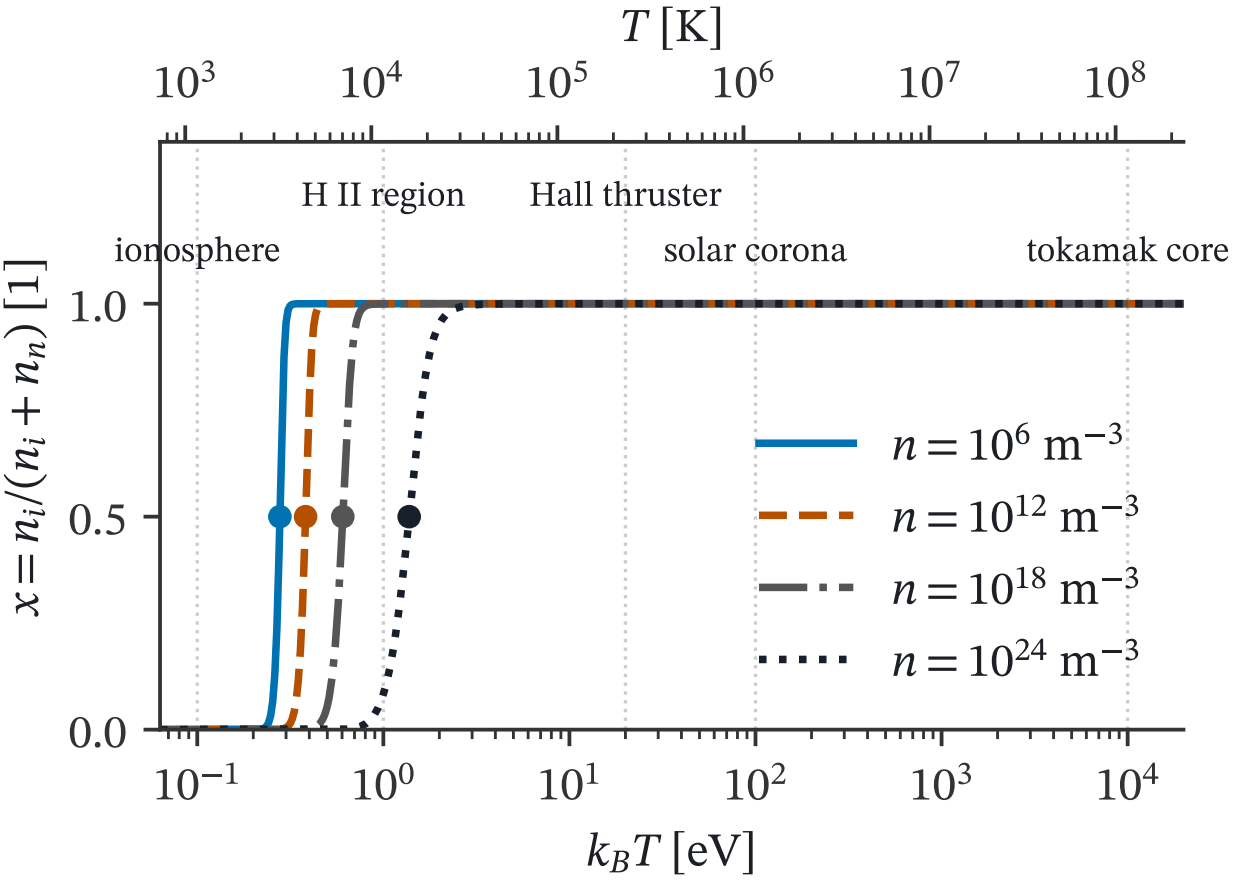


Figure 14: Single-stage equilibrium hydrogen with $g_i = g_n = 1$. Markers show half ionization. Example temperatures provide context, not predictions for actual compositions; several examples are outside local thermodynamic equilibrium.

At $x = \frac{1}{2}$, the left side is $\frac{1}{2}$, not one. The exact condition is

$$k_B T_{\frac{1}{2}} = \frac{\chi}{\ln\left(\frac{4}{n\lambda_{\mathrm{th},e}\left(T_{\frac{1}{2}}\right)^3}\right)}.$$

Iterate because the wavelength depends on temperature. Replacing 4 by 2 gives a rough threshold at equal entropy and energy factors. That crossing actually has $x = \frac{\sqrt{5}-1}{2} \approx 0.618$ instead of half ionization.

n (1 m^{-3})	$T_{\frac{1}{2}}$ (K)	$k_B T_{\frac{1}{2}}$ (eV)	$\frac{\chi}{k_B T_{\frac{1}{2}}}$ [1]
$1 \cdot 10^6$ m^{-3}	3226 K	0.278 eV	48.93
$1 \cdot 10^{12}$ m^{-3}	4435 K	0.382 eV	35.59
$1 \cdot 10^{18}$ m^{-3}	7026 K	0.605 eV	22.46
$1 \cdot 10^{24}$ m^{-3}	15976 K	1.377 eV	9.88
$1 \cdot 10^{26}$ m^{-3}	26230 K	2.26 eV	6.02

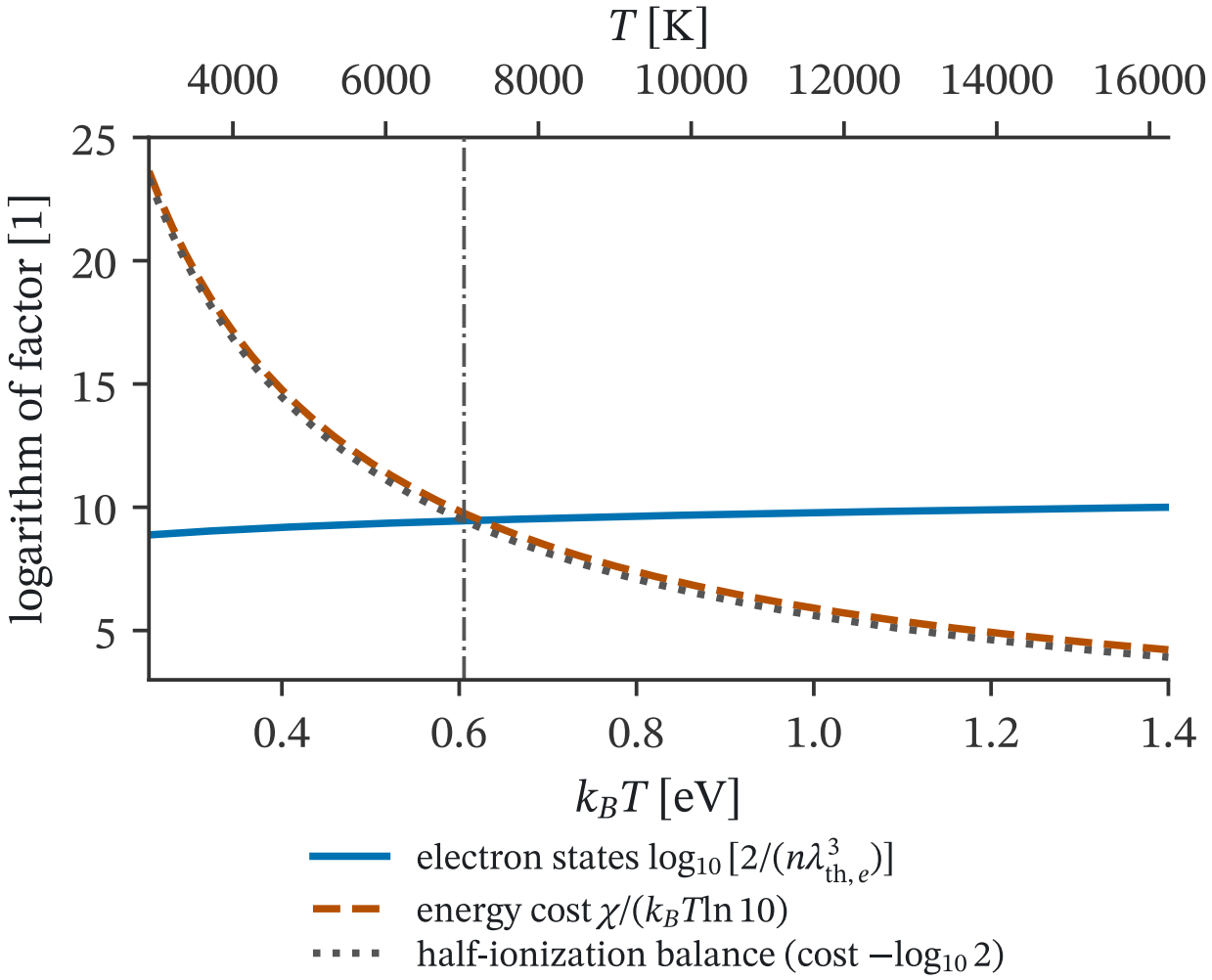


Figure 15: Competing factors at $n = 1 \cdot 10^{18}\,\mathrm{m}^{-3}$. Their unshifted equality gives $A = 1$. The dotted cost includes the factor $\frac{1}{2}$ needed for exact half ionization, whose temperature is marked vertically.

Constants follow [6]. Here $\frac{\chi}{k_B} \approx 157\,821\,\mathrm{K}$. Over the table's densities the exact logarithm falls from 48.93 to 6.02; half-ionization temperature rises from 3226 K to 26230 K. At $n = 1 \cdot 10^{18}\,\mathrm{m}^{-3}$, the logarithm is 22.46 and $k_B T_{\frac{1}{2}} = 0.605\,\mathrm{eV}$. Density enters the logarithm, so many orders of magnitude in density produce a much smaller relative threshold shift. The familiar $T \sim 1 \cdot 10^4\,\mathrm{K}$ captures the order of magnitude at intermediate densities. It is neither a universal half-ionization temperature nor a definition of plasma. A dense, detached divertor provides a test of equilibrium among highly excited levels. In the Alcator C-Mod tokamak, the measured populations of excited deuterium atoms agree with Saha–Boltzmann predictions within a factor of 1.5 for principal quantum number $p \geq 5$, at $k_B T_e = 1.2\,\mathrm{eV} \approx 1.4 \cdot 10^4\,\mathrm{K}$ and $n_e = n_i = 8.8 \cdot 10^{20}\,\mathrm{m}^{-3}$ [7]. Here p labels the bound level (dimensionless), and $n_{\mathrm{D}}(p)$ is its neutral-deuterium population density in m^{-3}. This partial equilibrium of excited levels does not imply Saha balance of ground-state atoms throughout the divertor.

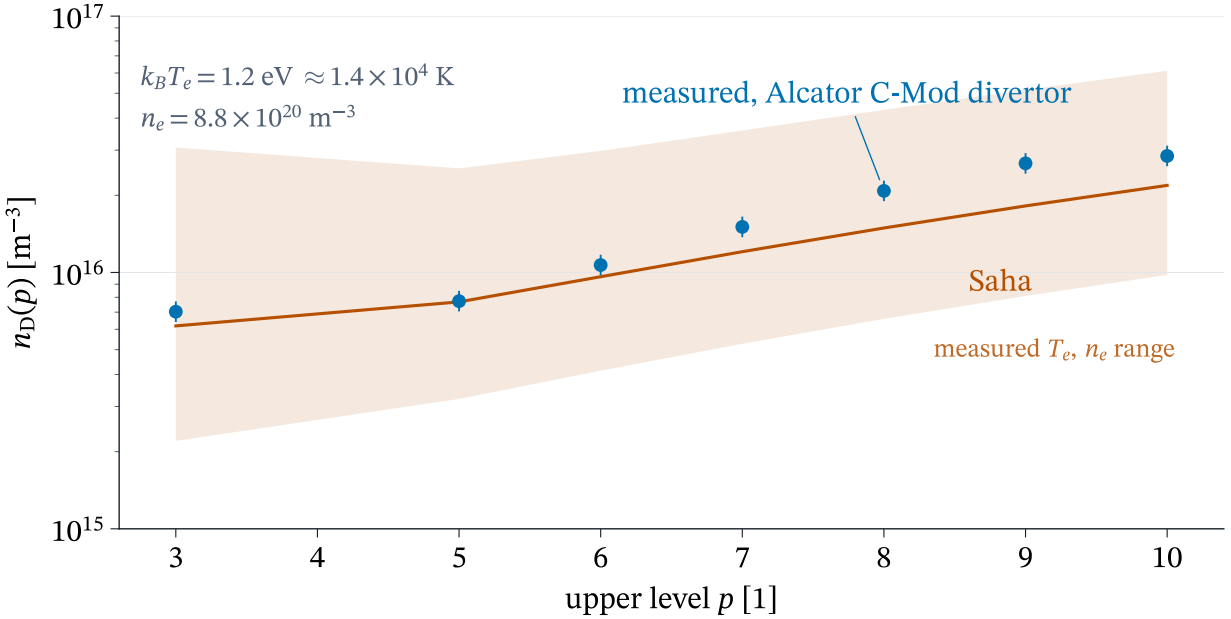


Figure 16: Saha–Boltzmann populations compared with redrawn data from Lumma, Terry, and Lipschultz, *Physics of Plasmas* **4**, 2555 (1997), doi:10.1063/1.872234, Fig. 7b. Bars show digitization uncertainty only (±0.04 dex), not experimental confidence intervals. The band spans the measured $k_B T_e = 0.8$ eV–1.5 eV and ±20% n_e ranges, with quasineutrality $n_i = n_e$.

Validity limits Interpretation 23

Saha requires local thermodynamic equilibrium and the appropriate reverse reactions. In dilute plasmas where radiative escape dominates recombination, such as the solar corona, tokamak edge, and many laboratory discharges, use coronal or collisional-radiative rate balance. External photons and nonthermal electrons ionize gases below a thermal threshold. At very high density, interactions lower or erase bound-state thresholds (pressure ionization), invalidating ideal dilute slots. High-density rows illustrate the ideal-model trend, without guaranteeing validity for a real sample. See [2,3].

Summary Summary 24

Saha plus nucleus conservation predicts $x(T, n)$. Electron-state freedom lowers the threshold by a logarithm, supporting a qualified 10^4 K thermal rule; non-equilibrium ionization needs rate models.

Knowledge check

- What densities follow from $x = \frac{1}{4}$ at nucleus density n?
- What is A at half ionization?
- Why can half ionization occur with $k_B T \ll \chi$?
- Recombination photons escape a dilute plasma. Is a Maxwellian enough to justify Saha?

Answers

What densities follow from $x = \frac{1}{4}$ at nucleus density n?

$n_i = n_e = \frac{n}{4}$ and $n_n = 3\frac{n}{4}$, each in m^{-3}.

What is A at half ionization?

$A = \frac{1}{2}$, dimensionless, from $\frac{\left(\frac{1}{2}\right)^2}{1-\frac{1}{2}}$.

Why can half ionization occur with $k_B T \ll \chi$?

Free-electron states compensate the Boltzmann penalty; binding energy is divided by a large logarithm.

Recombination photons escape a dilute plasma. Is a Maxwellian enough to justify Saha?

No. Chemical detailed balance is also necessary; escaping radiation can require rate balance even with thermal electrons.

3. Plasma oscillations

A plasma is neutral on average, but its electrons are light and mobile. If a slab of electrons is shifted rigidly against the ions, the separated charge pulls it back, and electron inertia carries it past equilibrium into an oscillation. This cold slab model is the first rigid charge-separation model of the course and supplies the plasma-frequency scale used throughout, following [2,3].

3.1 Electron plasma oscillations

What happens when all electrons in a region are displaced together, faster than thermal motion can rearrange them? Gauss's law gives the restoring electric field, and that field produces an electron plasma oscillation.

Learning objectives

- derive the cold electron plasma frequency from a displacement model
- separate the restoring force from the equilibrium charge balance
- state how temperature, collisions, and magnetic fields change the model

Unit ledger

The electron displacement ξ is in m. Ions are singly charged and initially stationary.

Cold, unmagnetized, collisionless response — Assumptions 26

Displace a uniform electron slab by a small amount ξ relative to fixed ions. Ignore pressure, collisions, magnetic fields, and density variation within the slab. The model isolates the electrostatic restoring force and describes small-amplitude motion about equilibrium.

A displacement creates two oppositely charged boundary sheets. The sheet charge magnitude is $en_0|\xi|$, so the electric field between them is

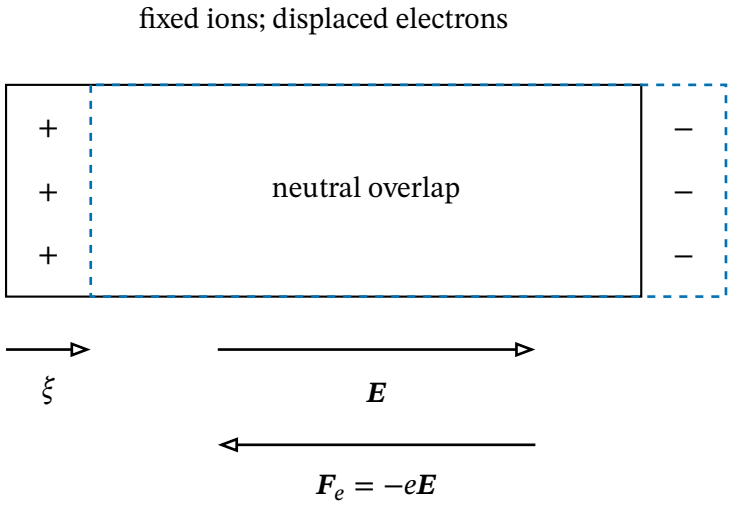


Figure 17: The charge signs and arrow directions show the restoring force. This schematic shows geometry, not a numerical length scale.

$$\boldsymbol{E} = \frac{e n_0 \xi}{\varepsilon_0} \boldsymbol{e}_x$$

$\boldsymbol{e}_x$ is a fixed unit vector and ξ is the signed electron displacement along it. The field reverses when ξ changes sign and the electron force points opposite to it. The field is in $\mathrm{V\,m^{-1}}$ when n_0 is in $\mathrm{m^{-3}}$ and ξ in m.

The electron force is opposite to the displacement:

$$m_e \frac{\mathrm{d}^2 \xi}{\mathrm{d}t^2} = -eE = -\frac{n_0 e^2}{\varepsilon_0} \xi$$

The force is in N. The negative sign is the restoring sign for the electron charge $-e$.

Dividing by m_e gives the harmonic-oscillator equation

$$\frac{\mathrm{d}^2 \xi}{\mathrm{d}t^2} + \omega_{p,e}^2 \xi = 0, \quad \omega_{p,e} = \sqrt{\frac{n_0 e^2}{\varepsilon_0 m_e}}$$

$\omega_{p,e}$ is an angular frequency in $\mathrm{s^{-1}}$. The cold model has no damping and no thermal dispersive correction.

Frequency and length scales

For a homogeneous species s with number density n_s in $\mathrm{m^{-3}}$, charge q_s in C, and mass m_s in kg, the plasma frequency is

$$\omega_{p,s} = \sqrt{\frac{n_s q_s^2}{\varepsilon_0 m_s}}$$

$\omega_{p,s}$ is in $\mathrm{s^{-1}}$ as an angular frequency. The corresponding ordinary frequency is $f_{p,s} = \omega_{p,s}/(2\pi)$ in hertz.

If the ions move as well, the electron and ion slabs oscillate against each other. Ions of charge Ze and density $n_i = n_e/Z$ shift by ξ_i, the electrons by ξ_e, and the field depends on the relative shift $\xi = \xi_e - \xi_i$. Both species feel it, so the frequencies add:

$$\omega_p^2 = \omega_{p,e}^2 + \omega_{p,i}^2 = \frac{n_e e^2}{\varepsilon_0 m_{\mathrm{eff}}}, \quad \frac{1}{m_{\mathrm{eff}}} = \frac{1}{m_e} + \frac{Z}{m_i}$$

m_{eff} is in kg. Since $m_i \gg m_e$, $m_{\mathrm{eff}} \approx m_e$: the ion correction to ω_p^2 is of relative size $Z m_e/m_i$.

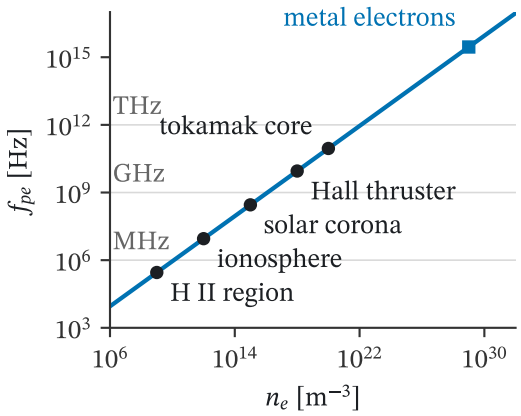


Figure 18: Electron plasma frequency $f_{p,e} = \omega_{p,e}/(2\pi)$ against electron density, with the five example plasmas of the density–temperature map. $f_{p,e}$ grows as $\sqrt{n_e}$: radio for space plasmas, microwave for laboratory and fusion plasmas.

The species inertial length $d_s = c/\omega_{p,s}$ compares a light-transit time with the plasma-response time. Here c is in $\mathrm{m\,s^{-1}}$ and d_s is in m. This electromagnetic scale differs from the thermal screening length derived in Debye shielding.

The inertial length also decides how an electromagnetic wave meets a plasma. A transverse wave of angular frequency ω in a cold, unmagnetized plasma obeys $\omega^2 = \omega_{p,e}^2 + c^2k^2$, with wavenumber k in $\mathrm{m^{-1}}$, derived in electromagnetic dispersion. Below $\omega_{p,e}$ the electrons follow the field and screen it within the skin depth $c/\sqrt{\omega_{p,e}^2 - \omega^2}$, which tends to d_e for $\omega \ll \omega_{p,e}$; the wave is reflected. Above $\omega_{p,e}$ the electron bulk can no longer follow, and the wave is transmitted with a longer wavelength than in vacuum.

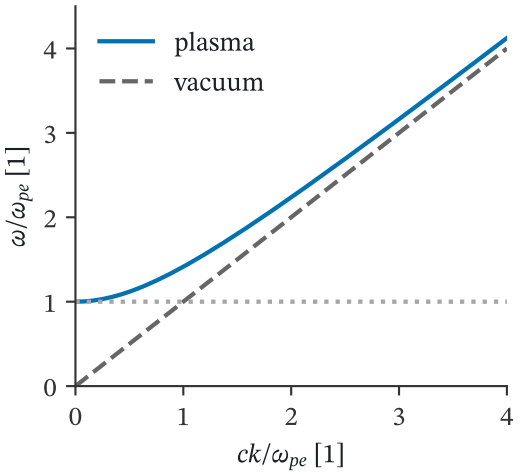
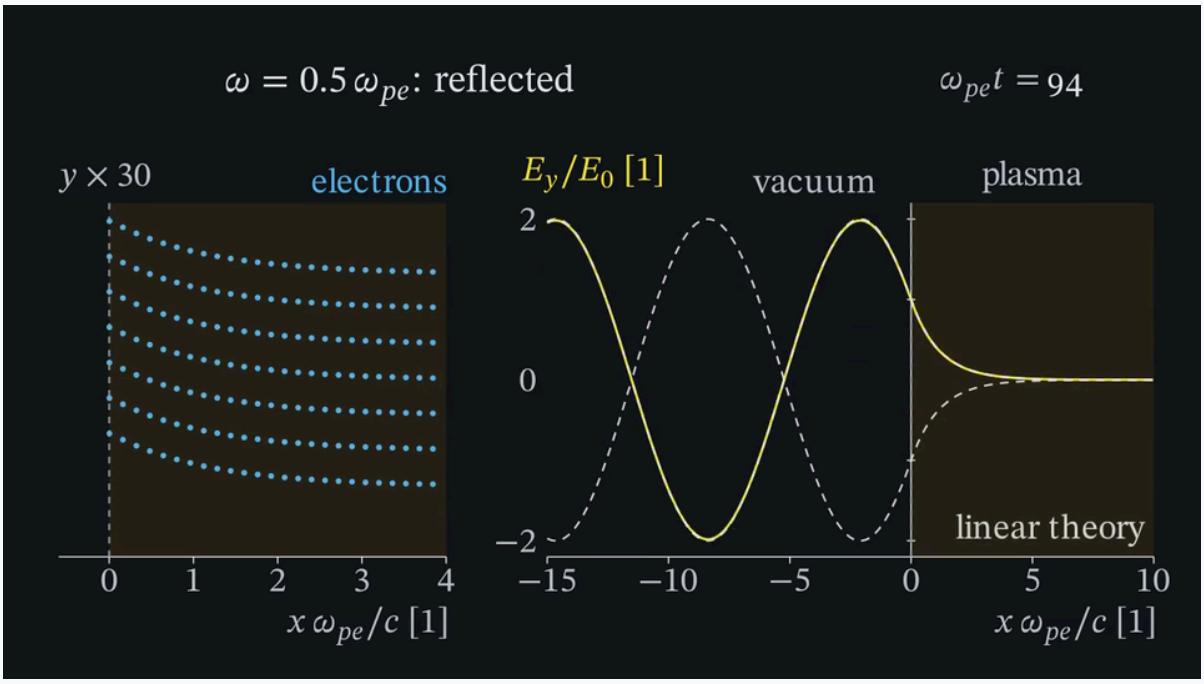


Figure 19: Transverse waves in a cold, unmagnetized plasma, $\omega^2 = \omega_{p,e}^2 + c^2k^2$ (solid), against the vacuum line $\omega = ck$ (dashed). The plasma branch starts at the cutoff $\omega = \omega_{p,e}$ (dotted): lower frequencies have no real k and are reflected. Frequency in units of $\omega_{p,e}$, wavenumber in units of $\omega_{p,e}/c$.



Watch animation

An electromagnetic wave at a plasma edge, computed with kin6d: a one-dimensional Maxwell–Lorentz run with cold electron macroparticles, fixed ions and absorbing boundaries. At $\omega = 0.5\omega_{p,e}$ the measured reflection is $|R| = 0.9996$ and the skin depth $1.155c/\omega_{p,e}$; at $\omega = 2\omega_{p,e}$ the reflection is $R = 0.072$ and the transmitted wavelength $3.628c/\omega_{p,e}$, each within 0.1 % of linear theory. Positions are $x\omega_{p,e}/c$ and time $\omega_{p,e}t$, with unit [1]; the electron displacement is magnified for visibility.

Figure 20: An electromagnetic wave at a plasma edge, computed with kin6d: a one-dimensional Maxwell–Lorentz run with cold electron macroparticles, fixed ions and absorbing boundaries. At $\omega = 0.5\omega_{p,e}$ the measured reflection is $|R| = 0.9996$ and the skin depth $1.155c/\omega_{p,e}$; at $\omega = 2\omega_{p,e}$ the reflection is $R = 0.072$ and the transmitted wavelength $3.628c/\omega_{p,e}$, each within 0.1 % of linear theory. Positions are $x\omega_{p,e}/c$ and time $\omega_{p,e}t$, with unit [1]; the electron displacement is magnified for visibility.

Derivation: the displacement oscillator

Step: Create the charge-separation field

At equilibrium, positive ion and negative electron charge densities cancel. Shift the electron slab by ξ. The overlap region remains nearly neutral, while the two boundary layers carry sheet charges of magnitude $en_0|\xi|$.

Gauss's law for two infinite sheets gives the uniform internal field

$$E = \frac{en_0\xi}{\varepsilon_0}.$$

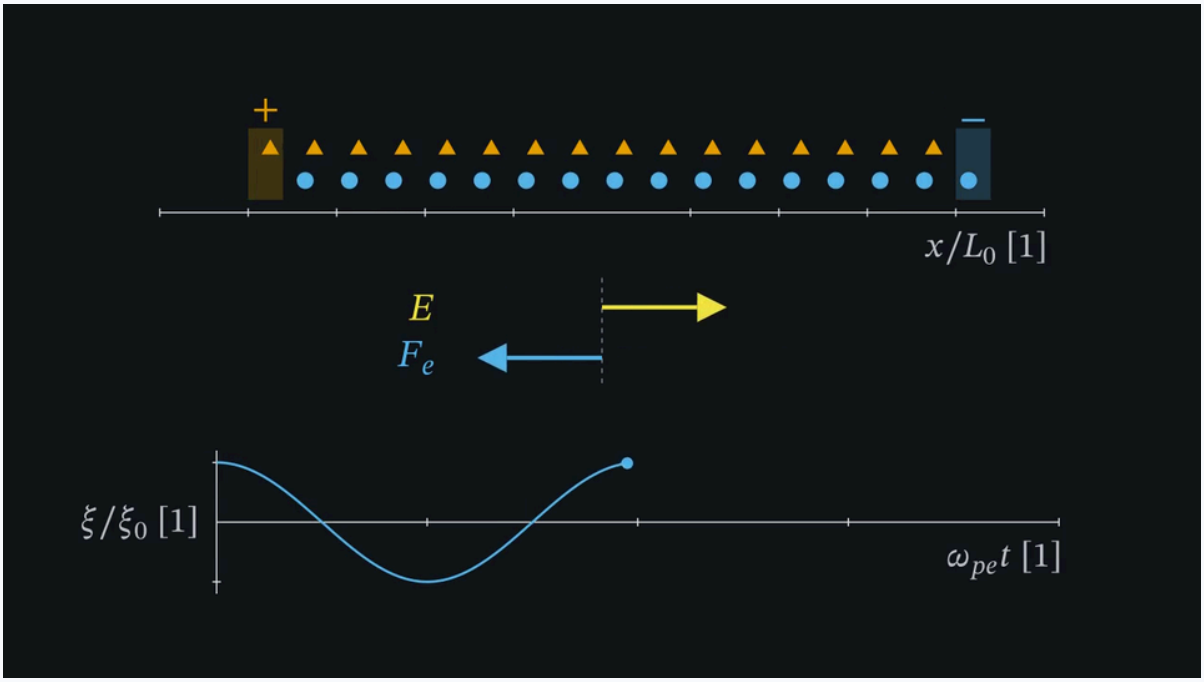
Step: Identify the oscillator

An electron feels $F = -eE$, so

$$m_e\frac{\mathrm{d}^2\xi}{\mathrm{d}t^2} = -\frac{n_0e^2}{\varepsilon_0}\xi.$$

The coefficient of ξ has units of s^{-2} and identifies the square of the plasma frequency. Within the small-amplitude model,

$$\xi(t) = \xi_0 \cos\left(\omega_{p,e} t + \delta\right).$$



Watch animation

Cold electron plasma oscillation: the charge-separation field produces a restoring electron force. The animation uses ξ/ξ_0 and $\tau = \omega_{p,e}t$, both with unit [1]. The trace ξ/ξ_0 is a numerical solution of the cold slab equation computed with kin6d, enclosed by a certified error band thinner than the drawn line. The position scale L_0 is a quarter of the slab half-width a, with $\xi_0/a = 0.1$, so $\xi_0/L_0 = 0.4$. The field scale is $E_0 = en_0\xi_0/\varepsilon_0$; field and electron force are shown as E/E_0 and $F_e/(eE_0)$, respectively, both with unit [1].

Figure 21: Cold electron plasma oscillation: the charge-separation field produces a restoring electron force. The animation uses ξ/ξ_0 and $\tau = \omega_{p,e}t$, both with unit [1]. The trace ξ/ξ_0 is a numerical solution of the cold slab equation computed with kin6d, enclosed by a certified error band thinner than the drawn line. The position scale L_0 is a quarter of the slab half-width a, with $\xi_0/a = 0.1$, so $\xi_0/L_0 = 0.4$. The field scale is $E_0 = en_0\xi_0/\varepsilon_0$; field and electron force are shown as E/E_0 and $F_e/(eE_0)$, respectively, both with unit [1].

Limits and extensions Interpretation 27

Finite temperature adds pressure and produces a wavenumber-dependent Langmuir-wave frequency. Collisions damp the motion. A background magnetic field couples the displacement to gyromotion. Those effects belong to the kinetic and wave chapters, while the oscillator here provides the local response scale.

Summary

Summary 28

A small collective electron displacement creates a restoring electric field. In the cold, collisionless, unmagnetized limit its natural angular frequency is $\omega_{p,e}$. Pressure, collisions, and magnetic fields are controlled extensions of this baseline model.

Exam questions

(f) Describe and derive electron plasma oscillations.

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 29

- What supplies the restoring force in the cold plasma oscillator?
- Why does the cold model produce a single local frequency rather than a dispersive frequency?
- How would electron collisions change the oscillator qualitatively?
- What is the dimensional status of $t\omega_{p,e}$ in the animation?

Answers

What supplies the restoring force in the cold plasma oscillator?
The charge separation between displaced electrons and stationary ions creates an electric field, and the electron charge makes the force oppose the displacement.

Why does the cold model produce a single local frequency rather than a dispersive frequency?
Pressure and spatial gradients were omitted, so the restoring coefficient contains density and particle constants but no wavenumber.

How would electron collisions change the oscillator qualitatively?
They introduce momentum loss and therefore damp the oscillation. The undamped harmonic equation is no longer complete.

What is the dimensional status of $t\omega_{p,e}$ in the animation?
It is dimensionless because time in seconds is multiplied by an angular frequency in s^{-1}.

4. Debye shielding

The previous chapter displaced a rigid electron slab and found the plasma frequency. A second rigid model, an electron-free sphere, estimates how large a charge-separated region thermal electrons can sustain. Letting the electrons respond with the Boltzmann factor of Chapter 2 then gives Debye shielding: the equilibrium response defines the Debye length and explains the reduction of the potential around a localized charge. The treatment follows [2,3].

4.1 Plasma parameter

How large a region can thermal motion charge up, and does a single neighbour matter there? Comparing electrostatic with thermal energy, first for an emptied sphere and then for one neighbour, gives the Debye length and the coupling parameter that decide whether a plasma is ideal.

Learning objectives

- estimate the largest charge-separated region from the thermal energy
- relate the coupling parameter Γ_s to the Debye number
- decide from N_D, Γ_s and λ_D/L whether a collective, quasineutral description applies

Unit ledger

The Debye length λ_D and the mean spacing a are in m.

A first estimate of the size of a charge-separated region can be obtained before solving the shielding profile. Let N denote a number density in a uniformly charged spherical region of radius R, with charge magnitude e per particle. Gauss's law gives the boundary potential scale

$$\Phi(R) = \frac{NeR^2}{3\varepsilon_0}, \quad |e\Phi(R)| \approx k_B T_e \Rightarrow R \approx \sqrt{\frac{3\varepsilon_0 k_B T_e}{Ne^2}}$$

N is a number density in m^{-3}, R is in m, and Φ is in V. The numerical factor depends on the assumed charge profile; the robust result is the scaling R proportional to $\sqrt{\varepsilon_0 k_B T_e/(Ne^2)}$. For $N = n_0$ this is of order the Debye length λ_D. An electron at the thermal speed $v_{\mathrm{th,e}}$ crosses λ_D in a time $\lambda_D/v_{\mathrm{th,e}}$, which the relation to the plasma frequency below shows to be of order $1/\omega_{p,e}$: charge separation on this scale is undone on the plasma-oscillation time.

Derivation: potential scale of a uniformly charge-separated sphere

Step: Compute the enclosed charge and field

The enclosed charge at radius r is

$$Q(r) = \frac{4\pi}{3} Ner^3.$$

Applying Gauss's flux law to a sphere gives

$$E(r)4\pi r^2 = \frac{Q(r)}{\varepsilon_0},$$

and hence

$$E(r) = \frac{Ner}{3\varepsilon_0}.$$

Step: Estimate the boundary potential

Measured relative to infinity, the potential at the boundary of the uniformly charged sphere is

$$\Phi(R) = \frac{Q(R)}{4\pi\varepsilon_0 R} = \frac{NeR^2}{3\varepsilon_0}.$$

A thermal particle can cross or substantially rearrange the region when $|e\Phi(R)|$ is comparable to $k_B T_e$. Solving that balance gives the displayed estimate. If $N = n_0$, then

$$R \approx \sqrt{3}\lambda_D.$$

The order-one factor depends on the charge geometry.

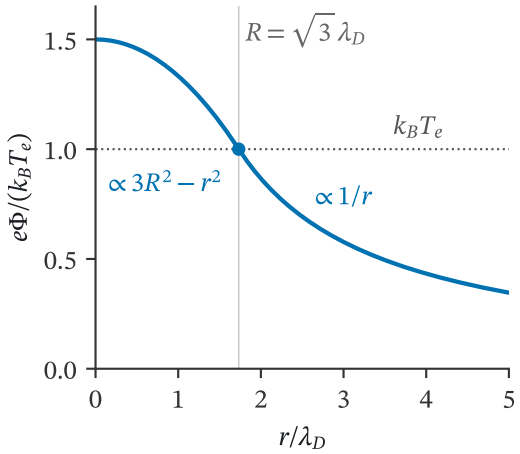
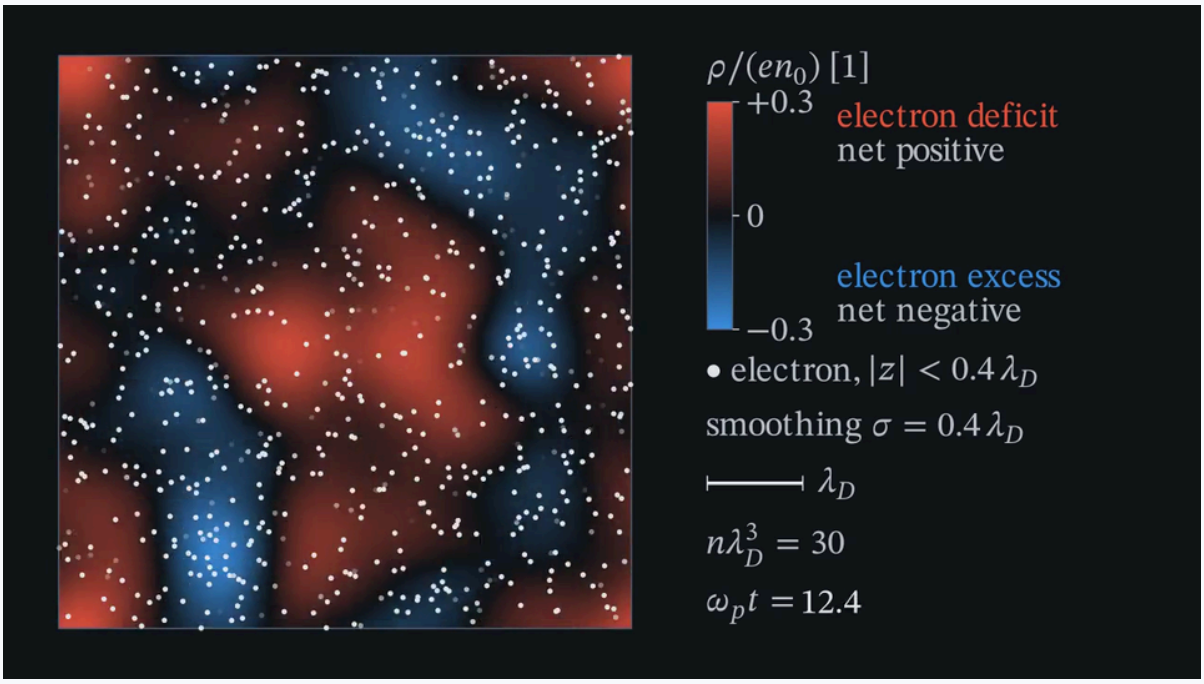


Figure 22: Rigid charge-separation estimate: all electrons removed from a sphere of radius $R = \sqrt{3}\lambda_D$ in a plasma of density n_0, no plasma response. The energy $e\Phi$ needed to move an electron from radius r to infinity follows from Gauss's law: quadratic inside, Coulomb $1/r$ outside. The last electron at $r = R$ costs $k_B T_e$ (dot), so thermal motion sustains separated regions up to this size. Radius in units of λ_D, energy in units of $k_B T_e$.

Thermal motion opens such regions continually, without any external charge: wherever electrons happen to crowd or thin out, a net charge appears and is undone again. The estimate fixes their largest size, $R \approx \sqrt{3}\lambda_D$ with the Debye length $\lambda_D = \sqrt{\varepsilon_0 k_B T_e/(n_0 e^2)}$.



Watch animation

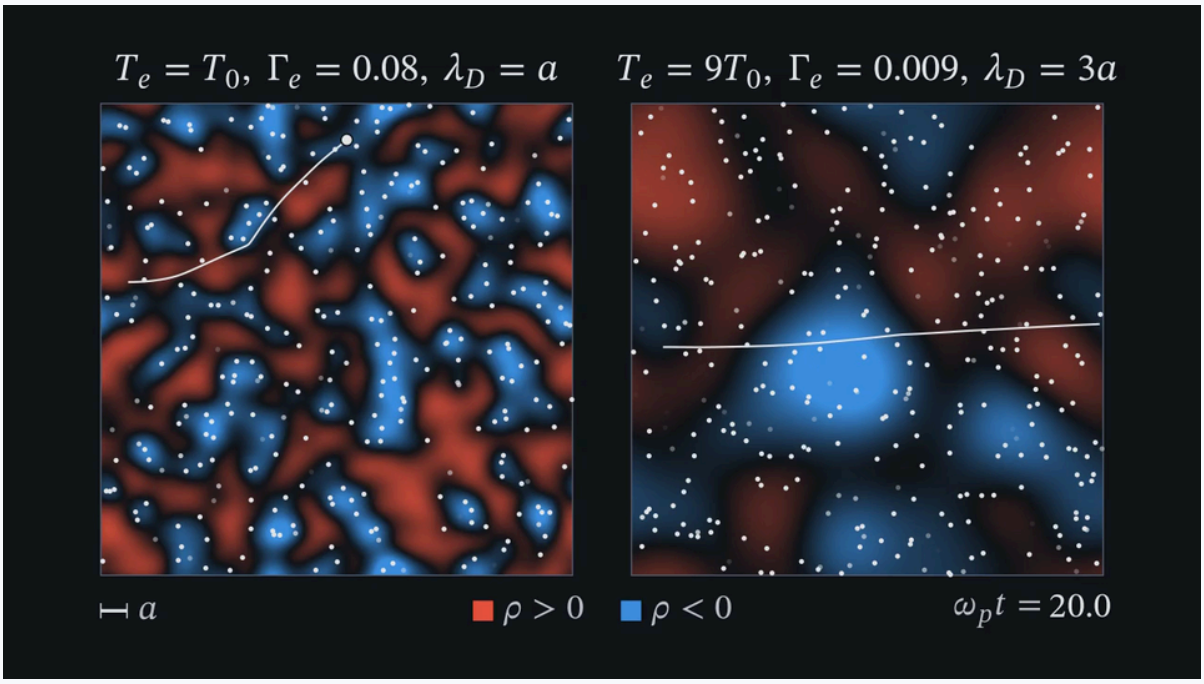
Thermal charge fluctuations, computed with kin6d. 6480 electrons of a classical one-component plasma in a periodic cube of edge $6\lambda_D$ with $n_0\lambda_D^3 = 30$, unsoftened Coulomb force, no test charge. Colour: net charge density $\rho/(en_0)$ [1] in the plane $z = 0$, smoothed with a Gaussian of width $0.4\lambda_D$; red marks an electron deficit, blue an excess. Dots: electrons with $|z| < 0.4\lambda_D$. Time is $\omega_{p,e}t$ [1].

Figure 23: Thermal charge fluctuations, computed with kin6d. 6480 electrons of a classical one-component plasma in a periodic cube of edge $6\lambda_D$ with $n_0\lambda_D^3 = 30$, unsoftened Coulomb force, no test charge. Colour: net charge density $\rho/(en_0)$ [1] in the plane $z = 0$, smoothed with a Gaussian of width $0.4\lambda_D$; red marks an electron deficit, blue an excess. Dots: electrons with $|z| < 0.4\lambda_D$. Time is $\omega_{p,e}t$ [1].

The number of particles inside that size decides how the plasma behaves. If the sphere holds about one particle, $\lambda_D \sim a$ with mean spacing $a = n_0^{-1/3}$, then a single neighbour already carries an energy of order $k_B T_e$: encounters are strong and the particles are correlated. If it holds many, $\lambda_D \gg a$, the energy $k_B T_e$ is shared among many charges, and each neighbour contributes only the fraction

$$\Gamma_s = \frac{q_s^2}{4\pi\varepsilon_0 a_s k_B T_s}, \quad \Gamma_e = \frac{a^2}{4\pi\lambda_D^2} \ll 1.$$

Dimensionless. The coupling parameter Γ_s compares the Coulomb energy of two particles of species s, charge q_s, at the mean spacing $a_s = n_s^{-1/3}$ with their thermal energy $k_B T_s$. With $\Gamma_s \ll 1$ each particle moves nearly freely and feels the smooth field of many charges. This weak coupling defines the ideal plasma used in this script: it justifies the ideal-gas pressure $p_s = n_s k_B T_s$ and a mean potential Φ.



Watch animation

Weak and strong coupling at equal density, computed with kin6d. 5832 electrons of a classical one-component plasma in a periodic cube of edge $18a$, unsoftened Coulomb force, no test charge, at $T_e = T_0$ ($\Gamma_e = 0.08$, $\lambda_D = a$, left) and $T_e = 9T_0$ ($\Gamma_e = 0.009$, $\lambda_D = 3a$, right), with $\lambda_D/a = (4\pi\Gamma_e)^{-1/2}$. Colour: net charge density in the plane $z = 0$, smoothed with a Gaussian of width $\lambda_D/2$, red for an electron deficit, blue for an excess; each panel is scaled to its own fluctuation level, about five times larger at strong coupling. Dots: electrons with $|z| < a/2$. After $\omega_{p,e}t = 12$ the electrons are frozen and one of them is launched at $v_{\mathrm{th,e}} = \sqrt{2k_BT_e/m_e}$ through the frozen charges and the background; its three-dimensional path is drawn in projection. Below each panel: its potential energy $U = -e\Phi$, with Φ the electrostatic potential of the frozen electrons and the background, plotted as $\Delta U = U - U_0$ relative to the launch value U_0 in units of k_BT_e [1], on the same scale in both panels. Since $U + m_e v^2/2$ is conserved and the launch kinetic energy is k_BT_e, $\Delta U/(k_BT_e)$ is the fraction of that energy lost. Time is $\omega_{p,e}t$ [1].

Figure 24: Weak and strong coupling at equal density, computed with kin6d. 5832 electrons of a classical one-component plasma in a periodic cube of edge $18a$, unsoftened Coulomb force, no test charge, at $T_e = T_0$ ($\Gamma_e = 0.08$, $\lambda_D = a$, left) and $T_e = 9T_0$ ($\Gamma_e = 0.009$, $\lambda_D = 3a$, right), with $\lambda_D/a = (4\pi\Gamma_e)^{-1/2}$. Colour: net charge density in the plane $z = 0$, smoothed with a Gaussian of width $\lambda_D/2$, red for an electron deficit, blue for an excess; each panel is scaled to its own fluctuation level, about five times larger at strong coupling. Dots: electrons with $|z| < a/2$. After $\omega_{p,e}t = 12$ the electrons are frozen and one of them is launched at $v_{\mathrm{th,e}} = \sqrt{2k_BT_e/m_e}$ through the frozen charges and the background; its three-dimensional path is drawn in projection. Below each panel: its potential energy $U = -e\Phi$, with Φ the electrostatic potential of the frozen electrons and the background, plotted as $\Delta U = U - U_0$ relative to the launch value U_0 in units of k_BT_e [1], on the same scale in both panels. Since $U + m_e v^2/2$ is conserved and the launch kinetic energy is k_BT_e, $\Delta U/(k_BT_e)$ is the fraction of that energy lost. Time is $\omega_{p,e}t$ [1].

The same condition counts particles: the Debye number, the number of electrons within a sphere of radius λ_D, must be large,

$$N_D = \frac{4\pi}{3} n_e \lambda_D^3 \gg 1$$

Dimensionless. N_D counts electrons inside a Debye sphere. The symbol $\gg 1$ states an ordering assumption.

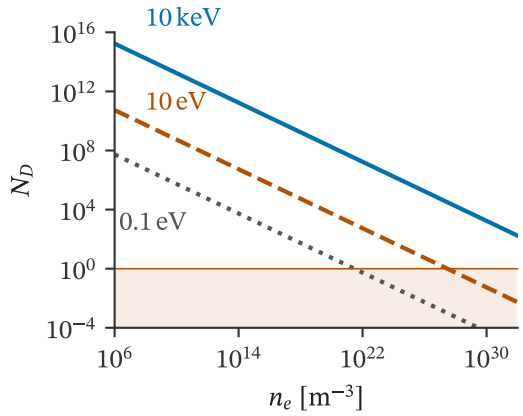


Figure 25: Debye number $N_D = (4\pi/3) n_e \lambda_D^3$ against electron density n_e at $k_B T_e = 10$ keV (solid), 10 eV (dashed) and 0.1 eV (dotted). $N_D \propto T_e^{3/2} n_e^{-1/2}$: dense, cold plasmas reach the shaded region $N_D < 1$, where the collective description fails.

Compare the Debye length with the system or gradient length L in m. The dimensionless ratio $\lambda_D / L \ll 1$ supports bulk quasineutrality away from boundaries; it does not establish a collisional fluid closure. Collision scales are treated in Collisions in gases and plasmas.

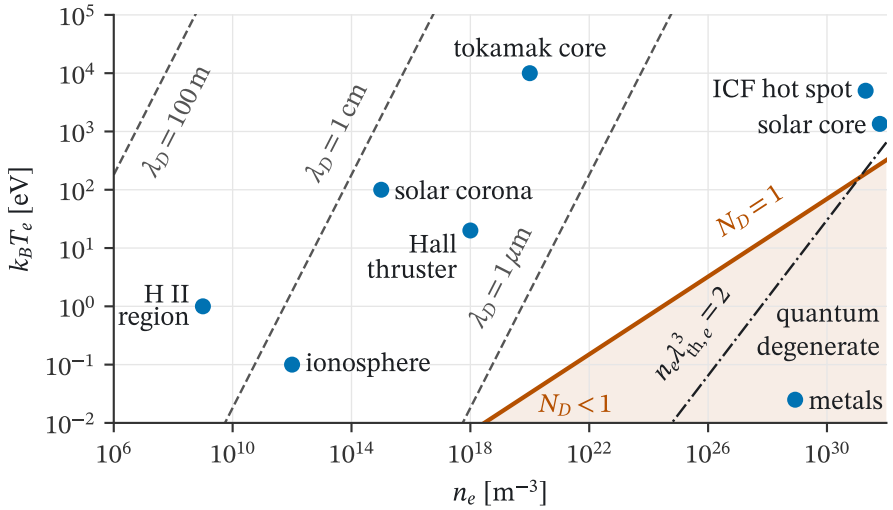


Figure 26: Where the collective ordering holds. Dashed: $\lambda_D = L$, i.e. $k_B T_e = e^2 n_e L^2 / \varepsilon_0$, for three system sizes L; plasmas to the right of a line are quasineutral on that scale. Solid: $N_D = 1$, i.e. $k_B T_e = \left(e^2/\varepsilon_0\right)(3/(4\pi))^{2/3} n_e^{1/3}$; the shaded side has $N_D < 1$. Dots: typical parameters of five example plasmas (order of magnitude).

Summary Summary 30

Thermal energy can separate charge only over about a Debye length. The coupling parameter Γ_s compares the Coulomb energy of neighbours at the mean spacing with $k_B T_s$; an ideal plasma has $\Gamma_s \ll 1$. The Debye number $N_D = (4\pi/3) n_e \lambda_D^3$ states the same condition, since $\Gamma_e N_D^{2/3} = (36\pi)^{-1/3} \approx 0.21$ independent of n_e and T_e. Bulk quasineutrality additionally requires $\lambda_D / L \ll 1$ for the system or gradient length L.

Exam questions

(d) A spherical region of complete charge separation has a potential $\Phi(r) = \left(N\frac{q_e}{\varepsilon_0}\right)r^2$ at its boundary.

What size can such a region roughly have in a thermal plasma of temperature T (potential energy = thermal energy)?

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 31

- How does N_D scale with the electron density n_e at fixed T_e, and with T_e at fixed n_e? Which corner of the density–temperature plane violates $N_D \gg 1$?
- Show that $\Gamma_e N_D^{2/3}$ contains neither n_e nor T_e, using $a_e = n_e^{-1/3}$ and $\lambda_D^2 = \varepsilon_0 k_B T_e/(n_e e^2)$.
- A plasma has $N_D \gg 1$ and $\Gamma_e \ll 1$, but its size L is comparable to λ_D. Which condition fails, and which modeling assumption must be dropped?
- The line $\lambda_D = L$ in the density–temperature plane is $k_B T_e = e^2 n_e L^2/\varepsilon_0$. On which side is the bulk quasineutral, and how does the line move when L doubles?

Answers

How does N_D scale with the electron density n_e at fixed T_e, and with T_e at fixed n_e? Which corner of the density–temperature plane violates $N_D \gg 1$?

Since λ_D is proportional to $\sqrt{T_e/n_e}$, N_D is proportional to $n_e\lambda_D^3$ and hence to $T_e^{3/2}n_e^{-1/2}$. Dense, cold plasmas reach $N_D \approx 1$, the shaded high-density, low-temperature region of the regime map.

Show that $\Gamma_e N_D^{2/3}$ contains neither n_e nor T_e, using $a_e = n_e^{-1/3}$ and $\lambda_D^2 = \varepsilon_0 k_B T_e/(n_e e^2)$.

$$\Gamma_e = \frac{e^2}{4\pi\varepsilon_0 k_B T_e} n_e^{\frac{1}{3}} \quad \text{and} \quad N_D^{\frac{2}{3}} = \left(\frac{4\pi}{3}\right)^{\frac{2}{3}} n_e^{\frac{2}{3}} \frac{\varepsilon_0 k_B T_e}{n_e e^2}.$$

In the product, e^2, ε_0 and $k_B T_e$ cancel, and the density factors combine to n_e/n_e, leaving the pure number $(36\pi)^{-1/3} \approx 0.21$. A large N_D therefore implies $\Gamma_e \ll 1$, and conversely.

A plasma has $N_D \gg 1$ and $\Gamma_e \ll 1$, but its size L is comparable to λ_D. Which condition fails, and which modeling assumption must be dropped?

The ordering $\lambda_D/L \ll 1$ fails. Screening layers then span the whole system, and bulk quasineutrality cannot be assumed. The collective field remains smooth because $N_D \gg 1$ still holds.

The line $\lambda_D = L$ in the density–temperature plane is $k_B T_e = e^2 n_e L^2/\varepsilon_0$. On which side is the bulk quasineutral, and how does the line move when L doubles?

Quasineutral bulk requires $\lambda_D < L$, which is $k_B T_e < e^2 n_e L^2/\varepsilon_0$, the side of higher density at given temperature. Doubling L multiplies the temperature on the line by four at each density, so the quasineutral region grows.

4.2 Debye shielding

A test charge disturbs the surrounding plasma. How far does its electrostatic influence extend once mobile electrons rearrange? In the weak-potential, equilibrium limit, the answer is the Debye length.

Learning objectives

- state the equilibrium and ordering assumptions behind Debye shielding
- connect the Boltzmann response to the screened Poisson equation
- interpret the Debye length as a collective response scale

Linearized Boltzmann response — Assumptions 32

Take a uniform, stationary ion background with $n_i = n_0$, mobile electrons at temperature T_e, no magnetic force in the equilibrium response, and a weak potential satisfying $|e\Phi| \ll k_B T_e$. The ions are treated as fixed on the electron response time.

The electron equilibrium density follows from the electrostatic potential energy $-e\Phi$:

$$n_e = n_0 \exp\left(\frac{e\Phi}{k_B T_e}\right) \approx n_0\left(1 + \frac{e\Phi}{k_B T_e}\right)$$

n_e and n_0 are in m^{-3}, and $e\Phi$ and $k_B T_e$ are energies in J. The approximation is dimensionless and requires $|e\Phi|/(k_B T_e) \ll 1$.

The net charge density is therefore

$$\rho_q = en_i - en_e \approx -\frac{e^2 n_0}{k_B T_e}\Phi$$

The sign expresses the restoring response of electrons to a positive potential perturbation.

Poisson's equation is

$$\nabla^2 \Phi = -\frac{\rho_q}{\varepsilon_0}$$

This equation defines the electrostatic field convention used here, with vacuum permittivity ε_0 in $\mathrm{F\,m}^{-1}$.

Substitution gives the screened equation

$$\nabla^2 \Phi - \frac{\Phi}{\lambda_D^2} = 0, \quad \lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_0 e^2}}$$

λ_D is in m. The equation is valid outside the localized source and within the linearized, static response model.

The Debye length also connects to the electron plasma frequency $\omega_{p,e} = \sqrt{n_0 e^2/(\varepsilon_0 m_e)}$. With the electron thermal speed $v_{\mathrm{th,e}} = \sqrt{2k_B T_e/m_e}$, in $\mathrm{m\,s}^{-1}$,

$$\lambda_D = \frac{v_{\mathrm{th,e}}}{\sqrt{2}\omega_{p,e}}, \quad \omega_{p,e} = \frac{v_{\mathrm{th,e}}}{\sqrt{2}\lambda_D}$$

A thermal electron crosses the screening length in about one inverse plasma frequency. The factor $\sqrt{2}$ follows from the convention $v_{\mathrm{th,e}} = \sqrt{2k_B T_e/m_e}$. If a source defines thermal speed as $\sqrt{k_B T_e/m_e}$, the same relation is written without that factor.

Derivation: from particle response to shielding

Step: Linearize the electron response

For a positive test potential, the electron potential energy is $-e\Phi$. The equilibrium Boltzmann factor is therefore

$$\exp\left(\frac{-(-e\Phi)}{k_B T_e}\right) = \exp\left(\frac{e\Phi}{k_B T_e}\right).$$

When $|e\Phi|/(k_B T_e) \ll 1$, expand it to first order:

$$n_e \approx n_0\left(1 + \frac{e\Phi}{k_B T_e}\right).$$

With immobile ions, the charge density becomes

$$\rho_q = en_0 - en_e \approx -\frac{e^2 n_0 \Phi}{k_B T_e}.$$

Step: Insert the response into Poisson's equation

Define the Debye coefficient by

$$\lambda_D^{-2} = \frac{n_0 e^2}{\varepsilon_0 k_B T_e}.$$

Inserting the charge response into the Poisson equation gives

$$\nabla^2\Phi - \frac{\Phi}{\lambda_D^2} = 0.$$

In spherical symmetry, the decaying source solution has the form

$$\Phi(r) \text{ proportional to } \frac{\exp\left(-\frac{r}{\lambda_D}\right)}{r}.$$

The exponential factor is the shielding; $1/r$ is the unscreened geometric spreading of a point source.

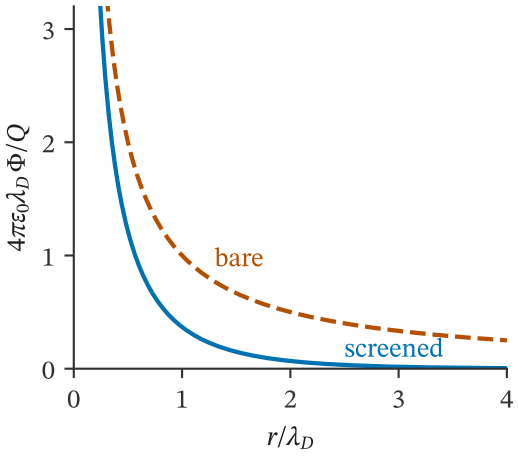


Figure 27: Bare (dashed) and Debye-screened (solid) potential of a point charge, $Q/(4\pi\varepsilon_0 r)$ and $Qe^{-r/\lambda_D}/(4\pi\varepsilon_0 r)$, from the expressions derived above. Radius in units of λ_D, potential in units of $Q/(4\pi\varepsilon_0\lambda_D)$. Their ratio is e^{-r/λ_D}; at $r = \lambda_D$ the screened potential is $1/e$ of the bare one.

The two spherical pictures of this section answer different questions. The charge-separated sphere is a rigid estimate: electrons are removed and the plasma does not respond, so its potential is the bare Coulomb result. The screened point charge is the plasma response itself: mobile electrons rearrange

until the Coulomb tail is cut off beyond λ_D. An extended source much smaller than λ_D is screened like a point charge of the same total charge.

The linearization $|e\Phi| \ll k_B T_e$ fails close to the charge, where the bare potential is large. Solving the full Boltzmann response numerically shows how small that region is when $N_D \gg 1$: outside it the screened potential above is accurate.

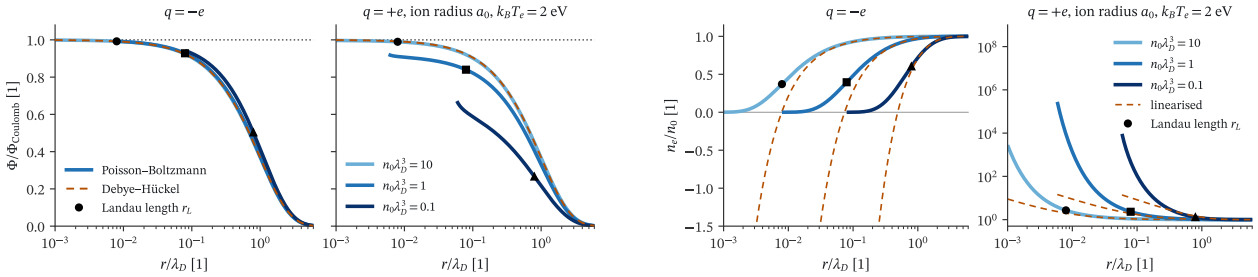


Figure 28: Screening of a positive point charge without linearization. Solid: numerical solution of the Poisson–Boltzmann equation $\nabla^2\Phi = (en_0/\varepsilon_0)\left(e^{e\Phi/(k_B T_e)} - 1\right)$ for fixed ions; dashed: the Debye–Hückel solution derived above; dotted: bare Coulomb. Left: potential $e\Phi/(k_B T_e)$; right: excess electron density $n_e/n_0 - 1$. The strength $\Lambda = qe/(4\pi\varepsilon_0 \lambda_D k_B T_e)$ sets where $e\Phi = k_B T_e$ (markers): at $r \approx \Lambda\lambda_D$. For a unit charge $q = e$, $\Lambda = 1/\left(4\pi n_0 \lambda_D^3\right) = 1/(3N_D)$, so $\Lambda = 10^{-4}, 10^{-3}, 10^{-2}$ correspond to $N_D \approx 3300$, 330, 33. Only inside that small radius is the density far from linear; beyond one Debye length the two potentials differ by about 0.3 % for $\Lambda = 10^{-2}$ and less for weaker charges. The charge is a small sphere of radius $0.1\Lambda\lambda_D$, because a positive point charge would collapse the Boltzmann electron density. Radius in units of λ_D.

In Fourier space, with $\hat{f}(\boldsymbol{k}) = \int f(\boldsymbol{r}) e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} \mathrm{d}^3 r$, the screened equation for a test charge $Q\delta(\boldsymbol{r})$ becomes algebraic:

$$\left(k^2 + \frac{1}{\lambda_D^2}\right)\hat{\Phi}(k) = \frac{Q}{\varepsilon_0}, \quad \hat{\Phi}(k) = \frac{Q}{\varepsilon_0 k^2 \varepsilon(k)}, \quad \varepsilon(k) = 1 + \frac{1}{k^2\lambda_D^2}$$

k is the wavenumber in m^{-1}; $\varepsilon(k)$ is the static dielectric function of the plasma, dimensionless. Without the plasma, $\varepsilon = 1$ and $\hat{\Phi} = Q/\left(\varepsilon_0 k^2\right)$ is the bare Coulomb potential.

For an imposed charge pattern of wavelength $2\pi/k$, the electrons add an induced charge of the same shape, and $1/\varepsilon(k) = k^2/\left(k^2 + 1/\lambda_D^2\right)$ is the net charge over the imposed charge, equally the potential over its vacuum value. Patterns much longer than λ_D are cancelled almost completely: the electrons have room to pile up in the troughs, and $\varepsilon \to \infty$ as $k \to 0$. The induced charge $\hat{\rho}_{\mathrm{ind}} = -Q/\left(1 + k^2\lambda_D^2\right)$ tends to $-Q$, so the cloud neutralises the test charge exactly. Patterns much shorter than λ_D survive, $\varepsilon \to 1$: thermal motion smears the electrons out before they can follow.

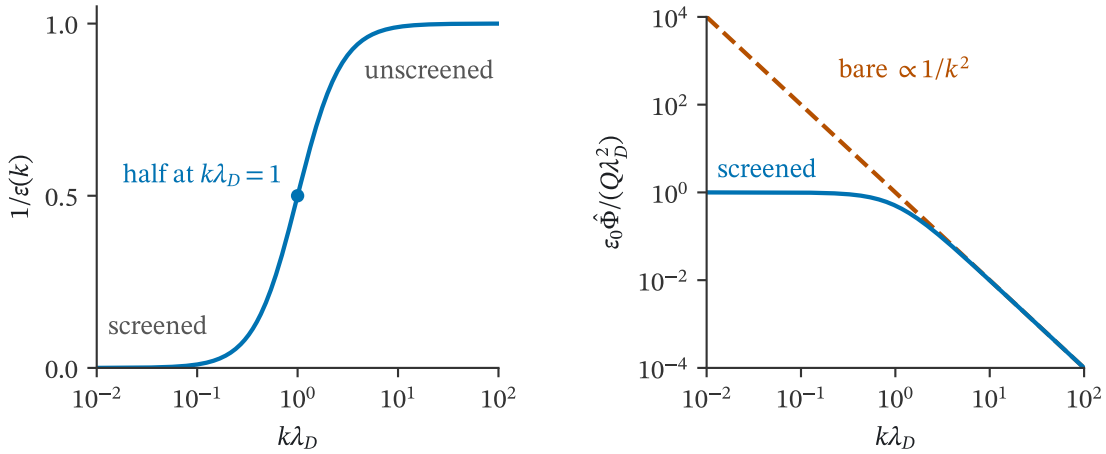
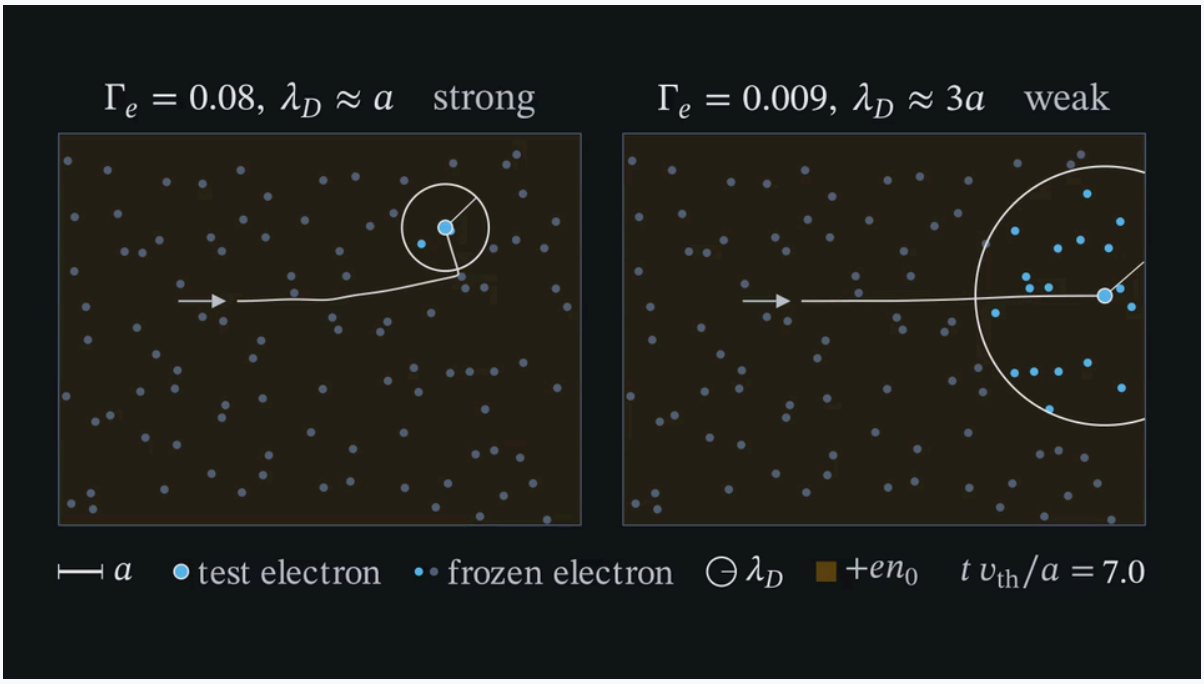


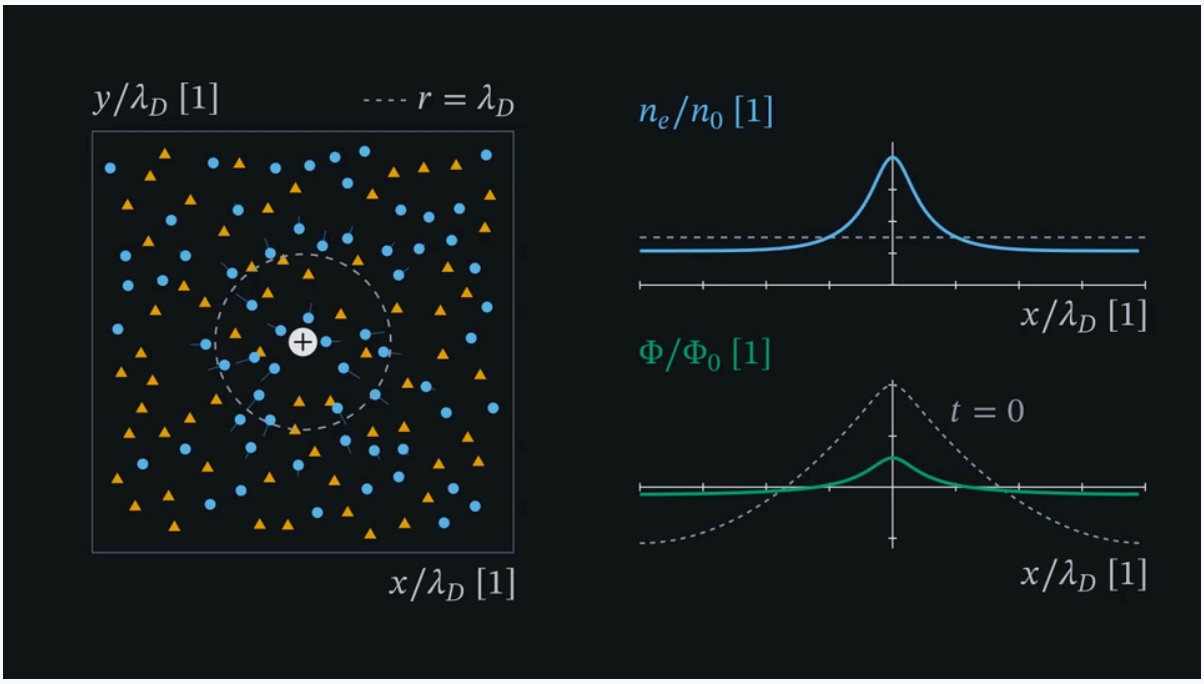
Figure 29: Debye screening in Fourier space. Left: $1/\varepsilon(k) = k^2/\left(k^2 + 1/\lambda_D^2\right)$, the net charge over the imposed charge for a pattern $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$; it is one half at $k\lambda_D = 1$. Right: bare $Q/\left(\varepsilon_0 k^2\right)$ (dashed) and screened $Q/\left(\varepsilon_0\left(k^2 + 1/\lambda_D^2\right)\right)$ (solid) potential of a point charge, in units of $Q\lambda_D^2/\varepsilon_0$. Wavenumber in units of $1/\lambda_D$.



Watch animation

Strong and weak coupling with the Debye circle. A test electron crosses frozen electrons on a uniform neutralizing background, at $\Gamma_e = 0.08$, $\lambda_D \approx a$ (left) and $\Gamma_e = 0.009$, $\lambda_D \approx 3a$ (right), with $\lambda_D/a = (4\pi\Gamma_e)^{-1/2}$; configuration, launch point and clock are identical. The circle of radius λ_D moves with the test electron; the drawn electrons (the slab $0 < z < a$) inside it are blue, about $\pi\lambda_D^2/a^2$ of them: few at strong, many at weak coupling. The frozen charges do not move and do not screen. Exact three-dimensional Coulomb forces; the paths stay in the drawn plane by mirror symmetry of the frozen configuration. Lengths are in units of a and time is $tv_{\mathrm{th,e}}/a$ with $v_{\mathrm{th,e}} = \sqrt{2k_B T_e/m_e}$, all with unit [1].

Figure 30: Strong and weak coupling with the Debye circle. A test electron crosses frozen electrons on a uniform neutralizing background, at $\Gamma_e = 0.08$, $\lambda_D \approx a$ (left) and $\Gamma_e = 0.009$, $\lambda_D \approx 3a$ (right), with $\lambda_D/a = (4\pi\Gamma_e)^{-1/2}$; configuration, launch point and clock are identical. The circle of radius λ_D moves with the test electron; the drawn electrons (the slab $0 < z < a$) inside it are blue, about $\pi\lambda_D^2/a^2$ of them: few at strong, many at weak coupling. The frozen charges do not move and do not screen. Exact three-dimensional Coulomb forces; the paths stay in the drawn plane by mirror symmetry of the frozen configuration. Lengths are in units of a and time is $tv_{\mathrm{th,e}}/a$ with $v_{\mathrm{th,e}} = \sqrt{2k_B T_e/m_e}$, all with unit [1].



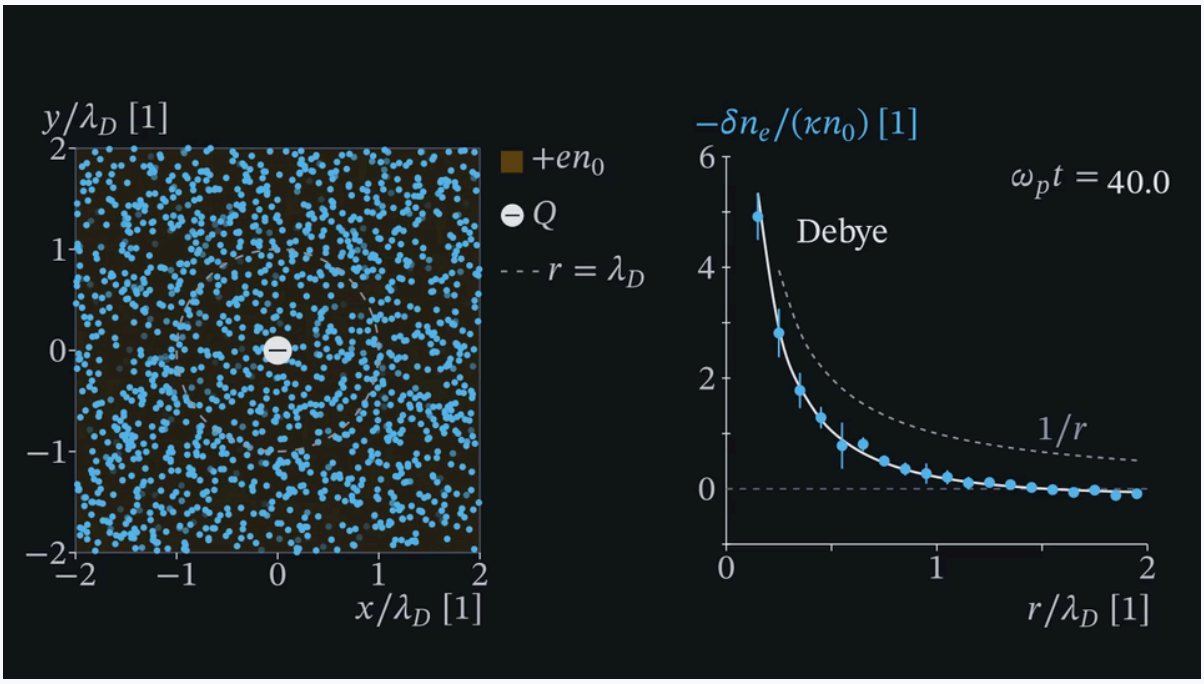
Watch animation

Two views of shielding: prescribed electron markers move toward a positive source in the left panel. The right panel shows a separate normalized one-dimensional slab calculation, in which an electron excess reduces the potential. Ions remain fixed; the slab calculation uses a periodic box with compensating background charge.

Figure 31: Two views of shielding: prescribed electron markers move toward a positive source in the left panel. The right panel shows a separate normalized one-dimensional slab calculation, in which an electron excess reduces the potential. Ions remain fixed; the slab calculation uses a periodic box with compensating background charge.

The two panels illustrate related but distinct models. On the left, prescribed two-dimensional marker trajectories show inward rearrangement; those markers do not generate the plotted field. On the right, density and potential come from one coupled one-dimensional drift–diffusion and Poisson calculation in a periodic box with compensating background charge. That computed electron excess reduces the source field. The independently drawn marker cloud must not be read as a self-consistent two-dimensional simulation of the right-hand profiles.

The next animation removes the prescription: the electrons move under their own Coulomb forces. The test charge $Q = -4\pi\kappa n_0 \lambda_D^3 e$ has the electrons' sign, so the linearized Boltzmann response and the screened potential derived above predict an electron deficit $-\delta n_e/(\kappa n_0) = (\lambda_D/r)e^{-r/\lambda_D}$, valid where $r \gg \kappa\lambda_D$. Here $\delta n_e = n_e - n_0$ and $\kappa = 0.05$ is dimensionless. Time is measured in units of $1/\omega_{p,e}$, with the electron plasma frequency $\omega_{p,e} = \sqrt{n_0 e^2/(\varepsilon_0 m_e)}$ of Chapter 3 and the electron mass m_e.



Watch animation

Debye shielding from particle dynamics, computed with kin6d. A classical electron one-component plasma (mobile electrons, uniform neutralizing background in place of immobile ions) fills a periodic cube of edge $4\lambda_D$ with $n_0\lambda_D^3 = 100$, that is $N_D \approx 420$ and 6400 electrons, interacting by the unsoftened Coulomb force. A fixed test charge of the electrons' sign sits at the centre. Left: one run, electrons in the slab $|z| < \lambda_D/2$ around the charge; a single snapshot shows no visible hole, because beyond $r \approx \lambda_D/2$ the deficit is only a few percent of n_0, smaller than the counting fluctuations. Right: the deficit $-\delta n_e/(\kappa n_0)$ averaged over time up to $\omega_{p,e}t = 40$ and over eight independent runs. It follows the linear Debye reference for the periodic cube (solid) and lies far below the unscreened response $1/r$ (dashed). Positions are x/λ_D, y/λ_D, r/λ_D and time $\omega_{p,e}t$, all with unit [1]. The bars are statistical 95 % intervals, not exact error bounds.

Figure 32: Debye shielding from particle dynamics, computed with kin6d. A classical electron one-component plasma (mobile electrons, uniform neutralizing background in place of immobile ions) fills a periodic cube of edge $4\lambda_D$ with $n_0\lambda_D^3 = 100$, that is $N_D \approx 420$ and 6400 electrons, interacting by the unsoftened Coulomb force. A fixed test charge of the electrons' sign sits at the centre. Left: one run, electrons in the slab $|z| < \lambda_D/2$ around the charge; a single snapshot shows no visible hole, because beyond $r \approx \lambda_D/2$ the deficit is only a few percent of n_0, smaller than the counting fluctuations. Right: the deficit $-\delta n_e/(\kappa n_0)$ averaged over time up to $\omega_{p,e}t = 40$ and over eight independent runs. It follows the linear Debye reference for the periodic cube (solid) and lies far below the unscreened response $1/r$ (dashed). Positions are x/λ_D, y/λ_D, r/λ_D and time $\omega_{p,e}t$, all with unit [1]. The bars are statistical 95 % intervals, not exact error bounds.

The screening cloud of the derivation is therefore an average, not a picture of individual electrons. The Boltzmann density describes the mean occupation of a shell around the charge, and with hundreds of electrons per Debye sphere this mean emerges only after averaging over many configurations. The same requirement, $N_D \gg 1$, returns as the plasma parameter below.

Scope of the Debye length

Interpretation 33

The Debye length is the distance over which this particular equilibrium response reduces a localized electrostatic field. Strong potentials, rapid time dependence, boundaries, insufficient Debye number, and kinetic non-equilibrium require a more complete model.

Summary

Summary 34

Debye shielding is a linear equilibrium response: the Boltzmann electron density modifies Poisson's equation and introduces the length λ_D. The derivation depends on weak potential, mobile electrons, and a stationary ion background.

Exam questions

(e) Describe and give an overview of the derivation for Debye shielding.

(i) Estimate the plasma frequency ω_p from thermal velocity v_t and Debye length λ_D.

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

Knowledge check 35

- Why does a positive electrostatic potential increase the equilibrium electron density in the Boltzmann response?
- Which approximation turns the exponential Boltzmann response into a linear screening equation?
- What happens to the screened potential at distances much larger than λ_D in this model?
- Give one situation in which the linear Debye-shielding derivation should not be used without modification.

Answers

Why does a positive electrostatic potential increase the equilibrium electron density in the Boltzmann response?

An electron has charge $-e$, so its potential energy is $-e\Phi$. A positive Φ lowers that energy and produces the factor $\exp(e\Phi/(k_B T_e))$.

Which approximation turns the exponential Boltzmann response into a linear screening equation?

The weak-potential ordering $|e\Phi| \ll k_B T_e$ permits a first-order expansion of the exponential.

What happens to the screened potential at distances much larger than λ_D in this model?

The exponential factor suppresses it, so the localized electrostatic influence is small compared with the unscreened $1/r$ field.

Give one situation in which the linear Debye-shielding derivation should not be used without modification.

A potential comparable to or larger than $k_B T_e/e$, a time-dependent kinetic response, or a boundary within the shielding region violates the stated assumptions.

5. Single-particle motion

The single-particle model prescribes electromagnetic fields and follows one charged particle through them. It supplies the orbit geometry that guiding-center, kinetic, and fluid descriptions later average or close.

5.1 The Lorentz force and uniform-field gyromotion

What can an electric or magnetic field do to a charged particle? Splitting the Lorentz force into its electric and magnetic parts separates energy change from bending, before any approximation is introduced.

Learning objectives

- separate the energy-changing and trajectory-bending parts of the force
- derive the circular orbit in a uniform magnetic field
- use the charge sign to determine the sense of gyration

Lorentz force — Governing law 36

For prescribed fields, a particle obeys

$$m_\alpha \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = q_\alpha(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

q_α and m_α are the charge and mass of a particle of species α, for example $\alpha = e$ for electrons and $\alpha = i$ for ions. Multiplying each field term by q_α gives its contribution to the force, in N. Dividing the total force by m_α gives the particle acceleration.

The instantaneous power supplied by the fields is

$$\frac{\mathrm{d}\frac{m_\alpha v^2}{2}}{\mathrm{d}t} = q_\alpha \boldsymbol{E} \cdot \boldsymbol{v}$$

The kinetic energy is in J. The magnetic term does no work because $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0$.

In a uniform magnetic field with $\boldsymbol{E} = 0$, choose $\boldsymbol{B} = B\boldsymbol{e}_z$. The perpendicular velocity components satisfy

$$\frac{\mathrm{d}v_x}{\mathrm{d}t} = \omega_c v_y, \quad \frac{\mathrm{d}v_y}{\mathrm{d}t} = -\omega_c v_x, \quad \omega_c = \frac{q_\alpha B}{m_\alpha}$$

The gyrofrequency ω_c is a signed angular frequency in s^{-1}: positive for ions, negative for electrons. The parallel velocity is constant, so the full orbit is a helix unless $v_\parallel = 0$.

Derivation: circular motion and the gyroradius

Step: Balance magnetic bending and inertia

The magnetic force is perpendicular to $\boldsymbol{v}_\perp$, so it supplies centripetal acceleration without changing $v_\perp$. Equating magnitudes,

$$\frac{m_\alpha v_\perp^2}{\rho_L} = |q_\alpha| v_\perp B,$$

gives the Larmor radius

$$\rho_L = \frac{m_\alpha v_\perp}{|q_\alpha| B} = \frac{v_\perp}{|\omega_c|}.$$

Step: Decouple the component equations

Differentiating the first component equation once more and inserting the second gives

$$\frac{\mathrm{d}^2 v_x}{\mathrm{d}t^2} = \omega_c \frac{\mathrm{d}v_y}{\mathrm{d}t} = -\omega_c^2 v_x,$$

a harmonic oscillator at the angular frequency $|\omega_c|$; v_y obeys the same equation.

Step: Integrate the component motion

Solving the perpendicular component equations gives, for the chosen coordinate orientation,

$$v_x = v_\perp \cos(\omega_c t + \phi), \quad v_y = -v_\perp \sin(\omega_c t + \phi),$$

where the gyrophase ϕ at $t = 0$ is set by the initial velocity. Integrating once more gives

$$x = x_0 + \frac{v_\perp}{\omega_c} \sin(\omega_c t + \phi), \quad y = y_0 + \frac{v_\perp}{\omega_c} \cos(\omega_c t + \phi),$$

a circle of radius $v_\perp/|\omega_c| = \rho_L$ about the constant guiding-center position (x_0, y_0). Reversing q_α reverses the sign of ω_c and therefore reverses the sense of rotation, while the radius is unchanged. With $v_z = v_\parallel$ constant, $z = z_0 + v_\parallel t$ and the orbit is a helix.

The guiding center is the center of the circular orbit in the perpendicular plane. Its radius is the Larmor radius or gyroradius $\rho_L = v_\perp/|\omega_c|$, where $|\omega_c|$ is the positive gyrofrequency. Thus a faster perpendicular particle has a larger orbit, while a stronger field bends the same particle more tightly.

For species α, write the signed frequency as $\omega_{c\alpha} = q_\alpha B/m_\alpha$, so that $\omega_{ci} > 0$ and $\omega_{ce} < 0$; both are angular frequencies in s^{-1}. A thermal orbit estimate uses $v_{\perp,\alpha} = v_{\mathrm{th},\alpha} = \sqrt{2k_B T_\alpha/m_\alpha}$ in $\mathrm{m\,s}^{-1}$, giving $\rho_{L\alpha} = v_{\mathrm{th},\alpha}/|\omega_{c\alpha}|$ in m.

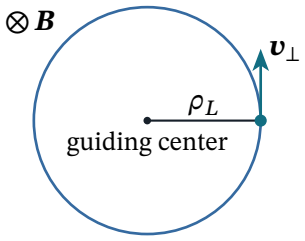
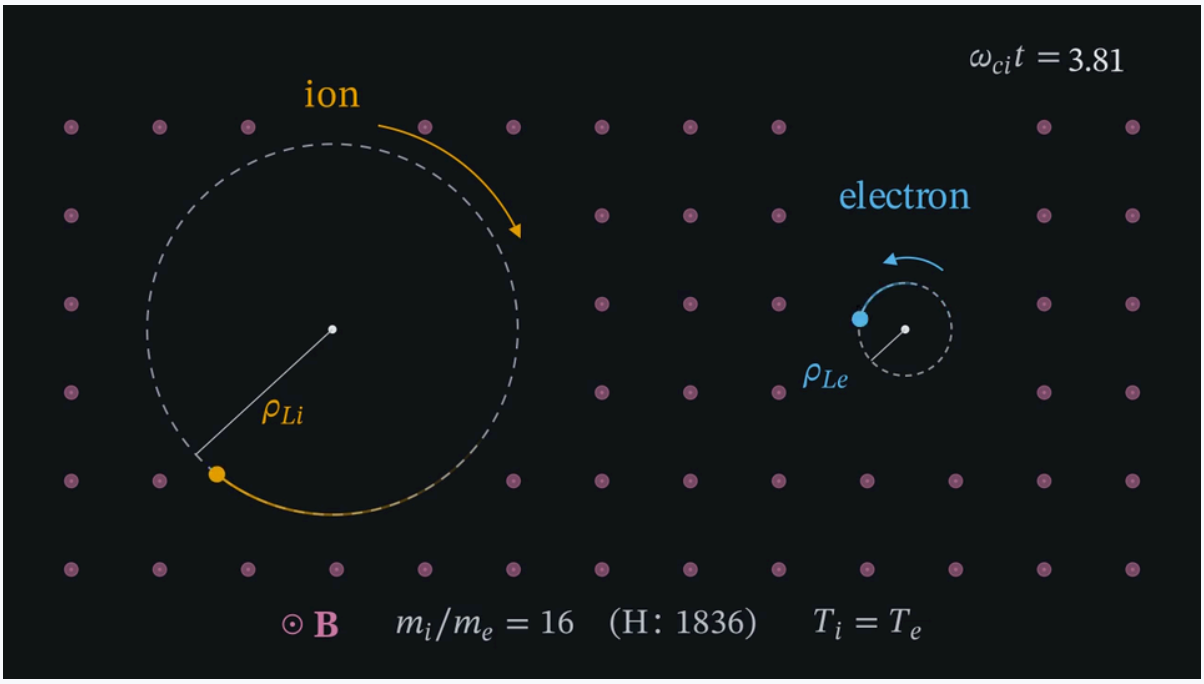


Figure 33: Geometry of uniform-field gyromotion for a positive charge ($q_\alpha > 0$), with $\boldsymbol{B}$ into the page (⊗). The radius is the Larmor radius ρ_L (a length in m), and the tangent arrow represents the perpendicular velocity $\boldsymbol{v}_\perp$; a negative charge circles the other way.

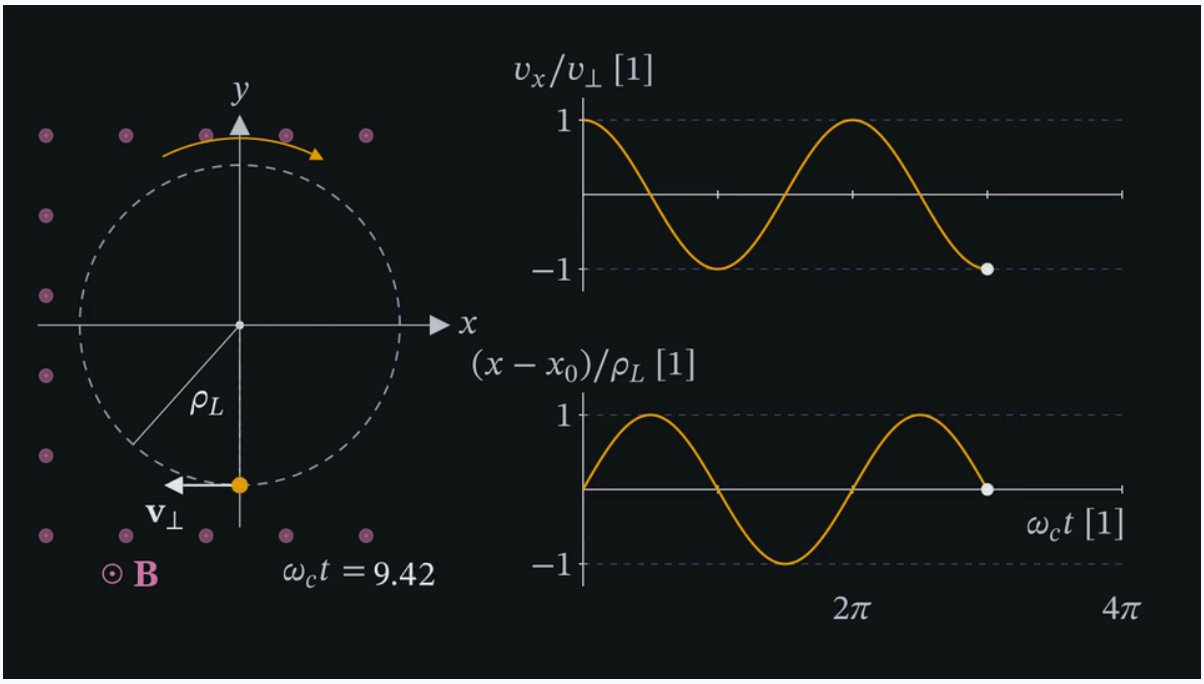
Viewed with $\boldsymbol{B}$ pointing toward the reader, ions ($\omega_c > 0$) gyrate clockwise and electrons ($\omega_c < 0$) counterclockwise. In both cases the current of the orbiting charge produces a magnetic moment opposite to $\boldsymbol{B}$: gyrating particles weaken the applied field, so a plasma is diamagnetic.



Watch animation

Ion and electron gyration in a uniform $\boldsymbol{B}$ out of the page, at equal temperature. A reduced mass ratio $m_i/m_e = 16$ (hydrogen: 1836) keeps both orbits visible: $\rho_{Li}/\rho_{Le} = \sqrt{m_i/m_e} = 4$ and $|\omega_{ce}|/\omega_{ci} = 16$. Time is $\omega_{ci}t$ [1].

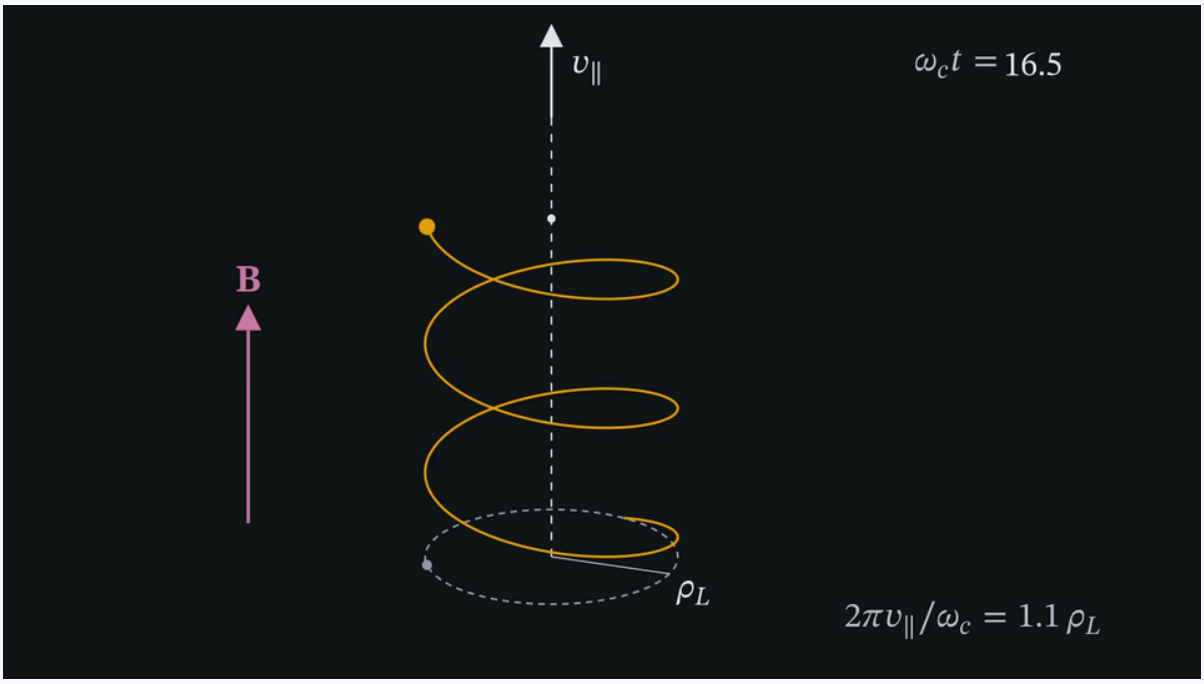
Figure 34: Ion and electron gyration in a uniform $\boldsymbol{B}$ out of the page, at equal temperature. A reduced mass ratio $m_i/m_e = 16$ (hydrogen: 1836) keeps both orbits visible: $\rho_{Li}/\rho_{Le} = \sqrt{m_i/m_e} = 4$ and $|\omega_{ce}|/\omega_{ci} = 16$. Time is $\omega_{ci}t$ [1].



Watch animation

Velocity and position components of a gyrating ion ($q_\alpha > 0$, gyrophase $\phi = 0$): $v_x/v_\perp = \cos\omega_c t$ and $(x - x_0)/\rho_L = \sin\omega_c t$, both [1], over two gyroperiods.

Figure 35: Velocity and position components of a gyrating ion ($q_\alpha > 0$, gyrophase $\phi = 0$): $v_x/v_\perp = \cos\omega_c t$ and $(x - x_0)/\rho_L = \sin\omega_c t$, both [1], over two gyroperiods.



Watch animation
Helical orbit with $v_\parallel \neq 0$ along $\boldsymbol{B} = B\boldsymbol{e}_z$: the pitch is $2\pi v_\parallel/\omega_c = 1.1\rho_L$; time is $\omega_c t$ [1].

Figure 36: Helical orbit with $v_\parallel \neq 0$ along $\boldsymbol{B} = B\boldsymbol{e}_z$: the pitch is $2\pi v_\parallel/\omega_c = 1.1\rho_L$; time is $\omega_c t$ [1].

Energy versus orbit geometry — Interpretation 37

An electric field can change the particle's kinetic energy because it can have a component along the velocity. A magnetic field can change the direction of the velocity but not its speed in this single-particle model. This distinction remains useful when fields are later solved self-consistently.

Comparing orbit and collective scales

The following estimates compare orbit sizes with the Debye length and plasma frequency. Each measures a different response; a small screening length alone does not justify averaging over a particle orbit.

Rechenbeispiel — Rechenbeispiel 38

Context: use a hydrogen plasma with $n_e = 1 \cdot 10^{20}\,\mathrm{m}^{-3}$, $T_e = 1 \cdot 10^6\,\mathrm{K}$, $T_i = 1 \cdot 10^6\,\mathrm{K}$, and $B = 1\,\mathrm{T}$. Assume singly charged ions, $n_i = n_e$, and use $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m}^{-1}$, and $k_B = 1.381 \cdot 10^{-23}\,\mathrm{J\,K}^{-1}$. Take the perpendicular speed to be the thermal speed, $v_{\perp,\alpha} = v_{\mathrm{th},\alpha} = \sqrt{2k_B T_\alpha/m_\alpha}$, for each species.

Target: report λ_D, $\omega_{p,e}$, ρ_{Le}, and ρ_{Li}.

Numerical result: $\lambda_D = 6.9 \cdot 10^{-6}\,\mathrm{m}$, $\omega_{p,e} = 5.6 \cdot 10^{11}\,\mathrm{s}^{-1}$, $\rho_{Le} = 3.1 \cdot 10^{-5}\,\mathrm{m}$, and $\rho_{Li} = 1.3 \cdot 10^{-3}\,\mathrm{m}$. These values are dimensional results.

Rechenbeispiel

Rechenbeispiel 39

Context: use a representative thermonuclear hydrogen plasma with $n_e = 1 \cdot 10^{20}\,\mathrm{m}^{-3}$, $T_e = 1 \cdot 10^8\,\mathrm{K}$, $T_i = 1 \cdot 10^8\,\mathrm{K}$, and $B = 5\,\mathrm{T}$. Assume singly charged ions, $n_i = n_e$. Use $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m}^{-1}$, and $k_B = 1.381 \cdot 10^{-23}\,\mathrm{J\,K}^{-1}$.

Take the perpendicular speed to be the thermal speed, $v_{\perp,\alpha} = v_{\mathrm{th},\alpha} = \sqrt{2k_B T_\alpha / m_\alpha}$, for each species.

Target: report λ_D, $\omega_{p,e}$, $|\omega_{ce}|$, ω_{ci}, ρ_{Le}, and ρ_{Li}.

Numerical result: $k_B T_e \approx 8.62\,\mathrm{keV}$, $\lambda_D = 6.9 \cdot 10^{-5}\,\mathrm{m}$, $\omega_{p,e} = 5.6 \cdot 10^{11}\,\mathrm{s}^{-1}$, $|\omega_{ce}| = 8.8 \cdot 10^{11}\,\mathrm{s}^{-1}$, $\omega_{ci} = 4.8 \cdot 10^{8}\,\mathrm{s}^{-1}$, $\rho_{Le} = 6.3 \cdot 10^{-5}\,\mathrm{m}$, and $\rho_{Li} = 2.7 \cdot 10^{-3}\,\mathrm{m}$. These are rough dimensional values for a hot confined plasma.

Summary

Summary 40

The Lorentz force separates energy transfer by $\boldsymbol{E}$ from magnetic bending. In a uniform $\boldsymbol{B}$ field the perpendicular motion is circular with radius $\rho_L = v_\perp / |\omega_c|$, while the parallel motion is uniform.

Exam questions

(a) What are the main differences between the electric and magnetic force acting on a charged particle? How does a background magnetic field B change the effect of an electric field on particle orbits if it is parallel or perpendicular to B?

(b) Compute the gyroradius by balancing centrifugal force and Lorentz force. Draw the direction of the gyration for ions/electrons if the B field points inside the paper plane.

Verbatim from Plasma Physics Exam.pdf, p. 2

Exam questions

(j) Give rough values of particle density, temperature, plasma frequency, gyrofrequency (electrons, ions) and Debye length and gyroradius (thermal electrons, ion) in a thermonuclear plasma.

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

- Which part of the Lorentz force changes a particle's kinetic energy?
- How does the gyroradius scale with particle mass at fixed $v_\perp$ and B?
- What changes when the sign of q_α changes in a uniform magnetic field?
- What limiting orbit results when $v_\perp$ tends to zero?

Answers

Which part of the Lorentz force changes a particle's kinetic energy?
Only $q_\alpha \boldsymbol{E}$ can do work. The magnetic force is perpendicular to the instantaneous velocity and has zero power.

How does the gyroradius scale with particle mass at fixed $v_\perp$ and B?
It is proportional to m_α, because $\rho_L = m_\alpha v_\perp/(|q_\alpha|B)$.

What changes when the sign of q_α changes in a uniform magnetic field?
The signed gyrofrequency and the sense of rotation change. The gyroradius magnitude and kinetic energy do not.

What limiting orbit results when $v_\perp$ tends to zero?
The gyroradius tends to zero and only uniform motion parallel to the magnetic field remains in the ideal uniform-field model.

5.2 Homogeneous-force drift and the $E \times B$ drift

A slowly moving orbit center can be found without solving the rapid gyromotion. What velocity makes a constant force balance the magnetic Lorentz force in a uniform field?

Learning objectives

- derive the $E \times B$ drift from a gyration-plus-drift ansatz
- generalize it to the drift velocity for a homogeneous force perpendicular to $\boldsymbol{B}$
- identify which particle properties cancel from the common drift

Unit ledger

A drift ratio such as $v_D/v_\perp$ is dimensionless only after both speeds use the same reference state.

Resolve the electric field into components parallel and perpendicular to the magnetic field, with $\boldsymbol{b} = \boldsymbol{B}/B$:

$$\boldsymbol{E} = E_\parallel \boldsymbol{b} + \boldsymbol{E}_\perp, \quad m_\alpha \frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = q_\alpha E_\parallel$$

$E_\parallel$ is the scalar component along $\boldsymbol{B}$. It accelerates a particle along the field. The perpendicular component participates in gyromotion and, when static and homogeneous, in the common $E \times B$ drift derived below.

For uniform, static $\boldsymbol{E}$ and $\boldsymbol{B}$, try a gyration carried along by a constant drift:

$$\boldsymbol{v} = v_\parallel \boldsymbol{b} + \boldsymbol{v}_\perp(t) + \boldsymbol{v}_D$$

All velocities are in $\mathrm{m\,s^{-1}}$. $\boldsymbol{v}_\perp(t)$ is the gyration of the previous section, $m_\alpha \frac{\mathrm{d}\boldsymbol{v}_\perp}{\mathrm{d}t} = q_\alpha \boldsymbol{v}_\perp \times \boldsymbol{B}$, and $\boldsymbol{v}_D$ is a constant drift velocity perpendicular to $\boldsymbol{B}$.

Insert the ansatz into the Lorentz force. The gyration terms cancel on both sides, $\mathrm{d}\boldsymbol{v}_D/\mathrm{d}t = 0$, and the parallel part was treated above, so the perpendicular remainder is

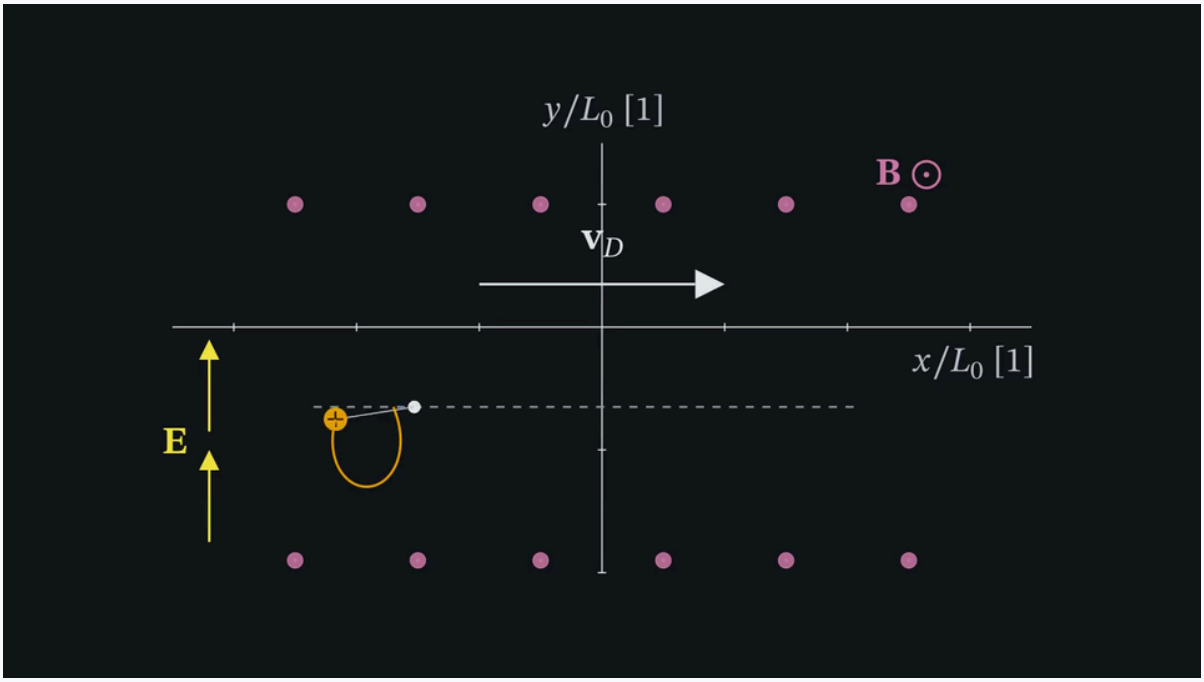
$$\boldsymbol{0} = \boldsymbol{E}_\perp + \boldsymbol{v}_D \times \boldsymbol{B}.$$

Crossing with $\boldsymbol{B}$ and using $(\boldsymbol{v}_D \times \boldsymbol{B}) \times \boldsymbol{B} = -B^2 \boldsymbol{v}_D$ gives

$$\boldsymbol{v}_D = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2}$$

With E in $\mathrm{V\,m^{-1}}$ and B in T, E/B is directly a speed in $\mathrm{m\,s^{-1}}$. The common $E \times B$ drift is independent of species mass and charge sign.

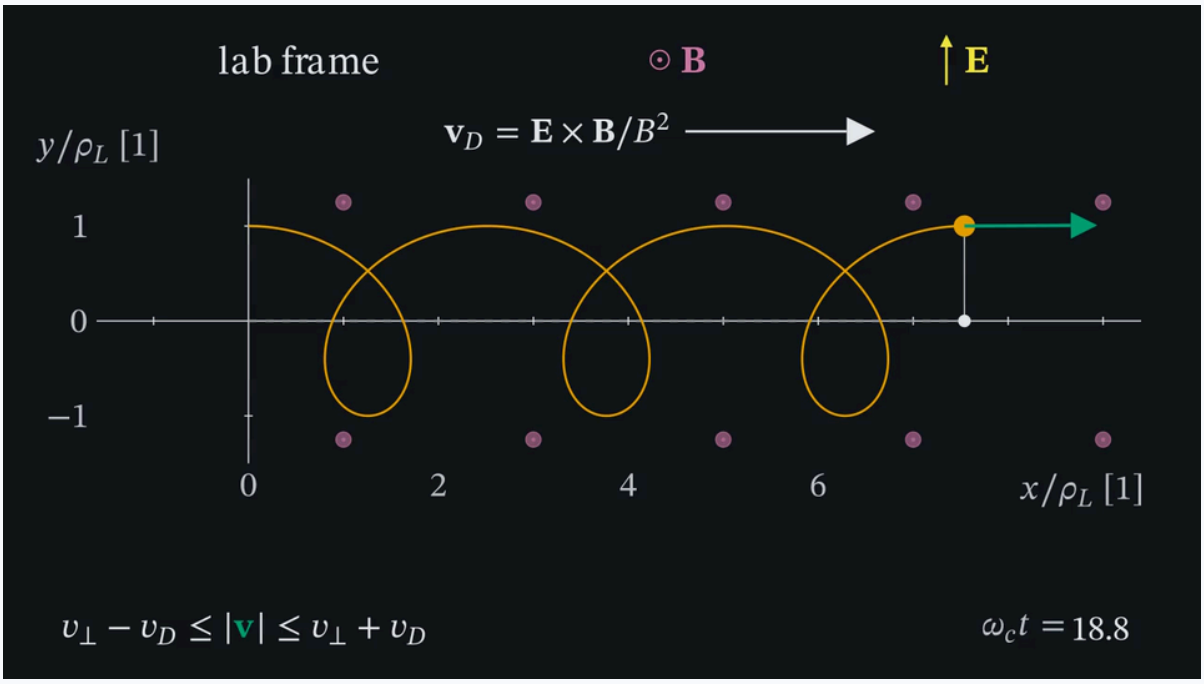
The remainder equation also states what the drift means. In a frame moving with $\boldsymbol{v}_D$, the perpendicular electric field is $\boldsymbol{E}'_\perp = \boldsymbol{E}_\perp + \boldsymbol{v}_D \times \boldsymbol{B} = \boldsymbol{0}$, so there the particle only gyrates. In the laboratory the same circle is carried along at $\boldsymbol{v}_D$. For $E/B \ll c$ this split is exact, not an average over the orbit.



Watch animation

Gyromotion plus the $E \times B$ drift. With reference length L_0 and time t_0, the trajectory uses $\omega_c t_0 = 2$, $\rho_L/L_0 = 0.65$, and $v_D t_0/L_0 = 0.55$, all dimensionless.

Figure 37: Gyromotion plus the $E \times B$ drift. With reference length L_0 and time t_0, the trajectory uses $\omega_c t_0 = 2$, $\rho_L/L_0 = 0.65$, and $v_D t_0/L_0 = 0.55$, all dimensionless.



Watch animation

The $E \times B$ drift as a change of frame, computed from the full Lorentz force for an ion with $E/B = 0.4 v_\perp$ over three turns. Laboratory frame: a trochoid whose fitted drift speed equals E/B. Frame moving with $\boldsymbol{v}_D$: $\boldsymbol{E}' = \boldsymbol{E} + \boldsymbol{v}_D \times \boldsymbol{B} = 0$ and the orbit is a circle of radius ρ_L. Lengths in ρ_L and time $\omega_c t$, both [1].

Figure 38: The $E \times B$ drift as a change of frame, computed from the full Lorentz force for an ion with $E/B = 0.4 v_\perp$ over three turns. Laboratory frame: a trochoid whose fitted drift speed equals E/B. Frame moving with $\boldsymbol{v}_D$: $\boldsymbol{E}' = \boldsymbol{E} + \boldsymbol{v}_D \times \boldsymbol{B} = 0$ and the orbit is a circle of radius ρ_L. Lengths in ρ_L and time $\omega_c t$, both [1].

Uniform crossed-field ordering Assumptions 42

Take uniform, time-independent $\boldsymbol{B}$ and a constant force with $\boldsymbol{F} \cdot \boldsymbol{B} = 0$. The force may be electric, gravitational, or another prescribed force on the particle. Averaging over gyromotion leaves a constant orbit-center velocity. The pressure-gradient force used later in fluid theory acts on a population; it is not an additional microscopic force on an isolated particle.

The same argument holds with $q_\alpha \boldsymbol{E}_\perp$ replaced by any such force. The force balance for a constant orbit-center velocity is

$$\boldsymbol{F} + q_\alpha (\boldsymbol{v}_D \times \boldsymbol{B}) = 0$$

This is a vector force balance in N. The drift has no acceleration because it describes the constant orbit-center translation.

Crossing with $\boldsymbol{B}$ and using $\boldsymbol{v}_D \cdot \boldsymbol{B} = 0$ gives

$$\boldsymbol{v}_D = \frac{\boldsymbol{F} \times \boldsymbol{B}}{q_\alpha B^2}$$

The result has velocity units, $\mathrm{m\,s^{-1}}$. It is valid for a homogeneous perpendicular force and a uniform magnetic field. For $\boldsymbol{F} = q_\alpha \boldsymbol{E}$ the charge cancels and the $E \times B$ drift is recovered.

Derivation: the force drift as a cross product

Step: Start from the orbit-center force balance

A constant drift velocity satisfies

$$\boldsymbol{F} + q_\alpha(\boldsymbol{v}_D \times \boldsymbol{B}) = \boldsymbol{0}.$$

The drift is perpendicular to $\boldsymbol{B}$, so crossing the equation with $\boldsymbol{B}$ uses

$$(\boldsymbol{v}_D \times \boldsymbol{B}) \times \boldsymbol{B} = -\boldsymbol{v}_D B^2.$$

Step: Solve for the drift

The cross-product equation becomes

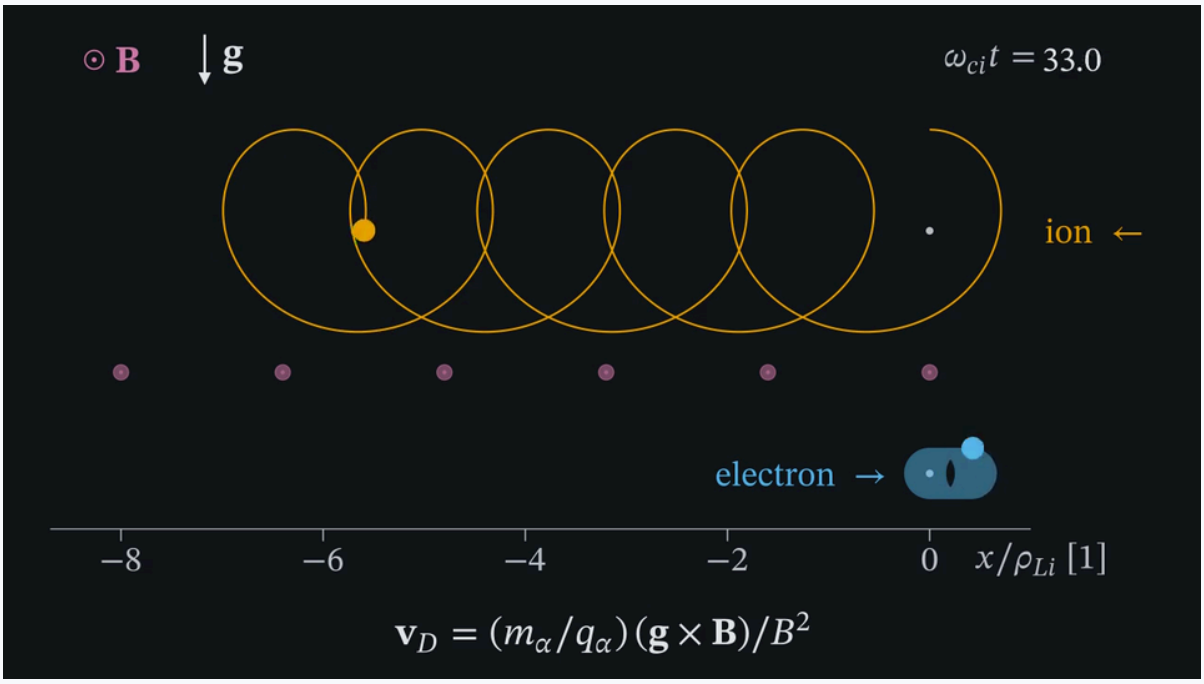
$$\boldsymbol{v}_D = \frac{\boldsymbol{F} \times \boldsymbol{B}}{q_\alpha B^2}.$$

For an electric force, $\boldsymbol{F} = q_\alpha \boldsymbol{E}$, so both the charge magnitude and its sign cancel. Direct substitution into the original force balance checks the sign of the resulting $E \times B$ drift.

Gravity, $\boldsymbol{F} = m_\alpha \boldsymbol{g}$ with the gravitational acceleration $\boldsymbol{g}$ in $\mathrm{m\,s^{-2}}$, does not contain the charge:

$$\boldsymbol{v}_D = \frac{m_\alpha}{q_\alpha} \frac{\boldsymbol{g} \times \boldsymbol{B}}{B^2}$$

Ions and electrons drift in opposite directions, the electrons slower by m_e/m_i. In a hydrogen plasma with $n_i = n_e = n$ in $\mathrm{m^{-3}}$ the drifts carry the current density $\boldsymbol{j} = \sum_\alpha n_\alpha q_\alpha \boldsymbol{v}_D = n(m_i + m_e)(\boldsymbol{g} \times \boldsymbol{B})/B^2$ in $\mathrm{A\,m^{-2}}$.



Watch animation

Gravity drift of an ion and an electron, computed from the full Lorentz force at a reduced mass ratio $m_i/m_e = 16$ (hydrogen: 1836) and equal temperatures. The measured drifts agree with $(m_\alpha/q_\alpha)(\boldsymbol{g}\times\boldsymbol{B})/B^2$ to better than 0.01 %: opposite directions, electron drift 16 times smaller. Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

Figure 39: Gravity drift of an ion and an electron, computed from the full Lorentz force at a reduced mass ratio $m_i/m_e = 16$ (hydrogen: 1836) and equal temperatures. The measured drifts agree with $(m_\alpha/q_\alpha)(\boldsymbol{g}\times\boldsymbol{B})/B^2$ to better than 0.01 %: opposite directions, electron drift 16 times smaller. Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

Summary

A homogeneous perpendicular force produces a drift $\boldsymbol{v}_D = \boldsymbol{F}\times\boldsymbol{B}/(q_\alpha B^2)$. For an electric force the charge cancels, so all magnetized species share the same $E\times B$ drift; in the frame moving with it the electric field vanishes.

Exam questions

(c) Derive the drift velocity in a magnetized plasma for a general homogeneous force.

(d) What is the $E\times B$ drift?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check — Knowledge check 44

- Why does the $E \times B$ drift not depend on charge sign?
- What component of a constant force is not represented by the perpendicular force drift?
- If $\boldsymbol{E}$ is parallel to $\boldsymbol{B}$, what happens to the $E \times B$ drift?
- What is the physical meaning of averaging over gyromotion in the drift derivation?

Answers

Why does the $E \times B$ drift not depend on charge sign?
The electric force is $q_\alpha \boldsymbol{E}$, and the q_α cancels the $1/q_\alpha$ in the general force-drift expression.

What component of a constant force is not represented by the perpendicular force drift?
A component parallel to $\boldsymbol{B}$ accelerates the particle along the field and cannot be balanced by a magnetic force.

If $\boldsymbol{E}$ is parallel to $\boldsymbol{B}$, what happens to the $E \times B$ drift?
The cross product vanishes. The electric field instead accelerates particles along the field, with a response that depends on charge and mass.

What is the physical meaning of averaging over gyromotion in the drift derivation?
It removes the rapid circular oscillation and retains the slower translation of the orbit center.

5.3 Guiding-center approximation and magnetic moment

Real plasmas rarely have perfectly uniform fields. How can a circular orbit remain a useful local picture when the field changes across the orbit? The guiding-center approximation separates the fast gyration from the slow field variation.

Learning objectives

- state the scale ordering behind guiding-center motion
- decompose position and velocity into guiding-center and gyromotion parts
- define the magnetic moment and identify its adiabatic character

Unit ledger

Let L_B denote the magnetic-field variation length in m and $|\omega_c|^{-1}$ the characteristic gyration time in seconds; the gyroperiod is $2\pi/|\omega_c|$. The ordering parameters $\varepsilon = \rho_L/L_B$ and $\omega_{\mathrm{slow}}/|\omega_c|$ are dimensionless. The magnetic moment defined below is in $\mathrm{J\,T^{-1}}$, equivalently $\mathrm{A\,m^2}$.

Slowly varying fields — Assumptions 45

Assume $\rho_L/L_B \ll 1$, field variation during one gyroperiod is small, and the fields are smooth enough for a local orbit expansion. Collisions, wave-particle resonances, and abrupt boundaries can invalidate the averaging.

Write the particle position as

$$\boldsymbol{r}(t) = \boldsymbol{R}(t) + \boldsymbol{\rho}(t), \quad |\boldsymbol{\rho}| \approx \rho_L$$

$\boldsymbol{R}$ is the guiding-center position and $\boldsymbol{\rho}$ is the fast gyration vector. Both positions and ρ_L are in m.

The velocity has the corresponding decomposition

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}t} + \frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}, \quad \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}t} = v_{\parallel}\boldsymbol{b} + \boldsymbol{v}_D, \quad \frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t} \approx \boldsymbol{v}_{\perp}$$

Every velocity is in $\mathrm{m\,s^{-1}}$. The first equality follows from the position split. The guiding center moves along the field at $v_{\parallel}$ and across it at the slow drift $\boldsymbol{v}_D$; the fast gyration velocity $\boldsymbol{v}_{\perp}$ averages to zero over one turn.

For perpendicular gyromotion, define the magnetic moment

$$\mu = \frac{m_\alpha v_\perp^2}{2B}$$

μ is an energy divided by magnetic field, in $\mathrm{J\,T^{-1}}$. It is an adiabatic invariant under the stated slow-variation ordering: it is constant to leading order in ε as the particle samples a slowly changing field. Here “adiabatic” refers to the separation of orbit and field-variation scales, not to thermodynamic heat exchange.

Derivation: magnetic moment, and its invariance from the mirror force

For this energy argument, take a static magnetic configuration with no electric work at the retained order. Keep leading parallel motion along the field when evaluating the sampled change in B.

Step: Compute the magnetic moment of one orbit

For one circular orbit, the gyroperiod and orbit area are

$$T_\mathrm{c} = \frac{2\pi}{|\omega_c|}, \quad S = \pi\rho_L^2.$$

The magnitude of the orbit current is

$$I_\mathrm{gyro} = \frac{|q_\alpha|}{T_\mathrm{c}} = \frac{|q_\alpha||\omega_c|}{2\pi}.$$

The magnetic dipole moment is current times area:

$$\mu = I_\mathrm{gyro} S = \frac{|q_\alpha||\omega_c|\rho_L^2}{2} = \frac{m_\alpha v_\perp^2}{2B}.$$

Here μ is the positive scalar magnitude; the gyration produces a diamagnetic vector dipole.

Step: Use the averaged mirror force

Let s measure distance along a field line. The averaged mirror force is

$$F_\parallel = -\mu\frac{\partial B}{\partial s}, \quad m_\alpha\frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = -\mu\frac{\partial B}{\partial s}.$$

Since $\mathrm{d}s/\mathrm{d}t = v_\parallel$, the particle samples the field according to

$$\frac{\mathrm{d}B}{\mathrm{d}t} = v_\parallel\frac{\partial B}{\partial s}.$$

Therefore the parallel kinetic energy changes as

$$\frac{\mathrm{d}\frac{m_\alpha v_\parallel^2}{2}}{\mathrm{d}t} = -\mu v_\parallel\frac{\partial B}{\partial s}.$$

Step: Show adiabatic conservation

The magnetic part changes according to

$$\frac{\mathrm{d}\mu B}{\mathrm{d}t} = B\frac{\mathrm{d}\mu}{\mathrm{d}t} + \mu v_{\parallel}\frac{\partial B}{\partial s}.$$

Adding the two balances gives

$$\frac{\mathrm{d}\frac{m_\alpha v_\parallel^2}{2} + \mu B}{\mathrm{d}t} = B\frac{\mathrm{d}\mu}{\mathrm{d}t}.$$

In this static, leading-order guiding-center approximation, the total guiding-center energy is conserved, so $\mathrm{d}\mu/\mathrm{d}t \approx 0$. The ordering requires the field to vary little over one gyroperiod and one gyroradius.

This static mirror argument transfers energy between parallel and perpendicular motion. In a slowly time-varying magnetic field, μ can remain invariant while an induced electric field supplies energy: the increase of μB is then betatron acceleration, and total kinetic energy need not be conserved. Faraday's law gives this second route directly.

Derivation: μ from Faraday's law
Take a uniform field $\boldsymbol{B} = B(t)\boldsymbol{e}_z$ that changes slowly, $|\partial B/\partial t| \ll |\omega_c|B$, with no parallel electric field and an orbit centred on the symmetry axis, so that the guiding center does not drift.

Step: Work done in one turn

The induced electric field obeys $\boldsymbol{\nabla} \times (\boldsymbol{E}) = -\partial \boldsymbol{B}/\partial t$. Over one turn it changes the perpendicular energy by

$$\delta\left(\frac{m_\alpha v_\perp^2}{2}\right) = q_\alpha \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l},$$

with the line integral taken along the orbit in the direction of motion.

Step: Apply Stokes' theorem

Ions run around $\boldsymbol{B}$ in the left-handed sense and electrons in the right-handed sense, so for both signs of the charge

$$q_\alpha \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = |q_\alpha|\pi\rho_L^2\frac{\partial B}{\partial t}.$$

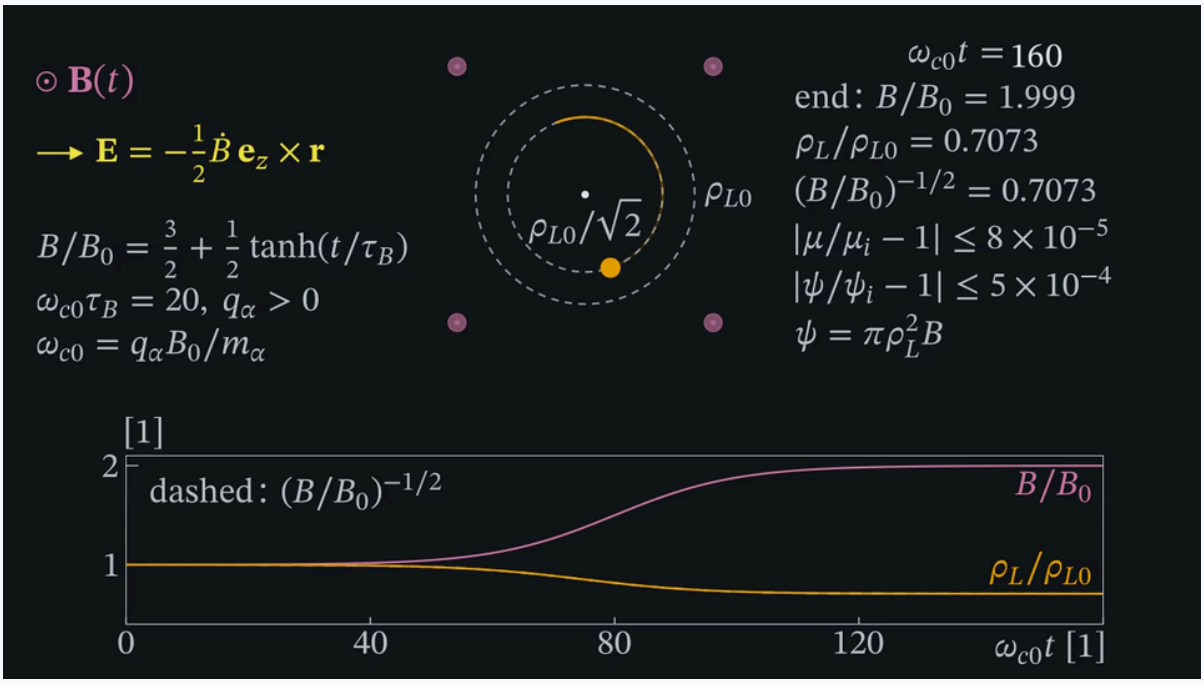
Step: Compare with the field change

One turn lasts $2\pi/|\omega_c|$, during which the field changes by $\delta B = \partial B/\partial t 2\pi/|\omega_c|$. With $\rho_L = m_\alpha v_\perp/(|q_\alpha|B)$ the energy gain becomes

$$\delta\left(\frac{m_\alpha v_\perp^2}{2}\right) = \frac{m_\alpha v_\perp^2}{2B}\delta B = \mu\delta B.$$

To first order in δB the magnetic moment then changes by

$$\delta\mu = \frac{\delta\left(m_\alpha v_\perp^2/2\right)}{B} - \mu\frac{\delta B}{B} = 0.$$



Watch animation

An ion in a uniform field that grows smoothly from B_0 to $2B_0$ over about twenty gyrations, computed from the full Lorentz force with the induced electric field of Faraday's law. The orbit shrinks as $\rho_L \propto B^{-1/2}$, while μ and the flux $\pi\rho_L^2 B$ through the orbit stay constant to better than 0.05 %. Lengths in the initial ρ_L and time $\omega_{c0}t$, both [1], with $\omega_{c0} = q_\alpha B_0/m_\alpha$.

Figure 40: An ion in a uniform field that grows smoothly from B_0 to $2B_0$ over about twenty gyrations, computed from the full Lorentz force with the induced electric field of Faraday's law. The orbit shrinks as $\rho_L \propto B^{-1/2}$, while μ and the flux $\pi\rho_L^2 B$ through the orbit stay constant to better than 0.05 %. Lengths in the initial ρ_L and time $\omega_{c0}t$, both [1], with $\omega_{c0} = q_\alpha B_0/m_\alpha$.

"Constant to leading order" can be tested. The figure below doubles the field at different rates and records how much μ has changed once the field is constant again.

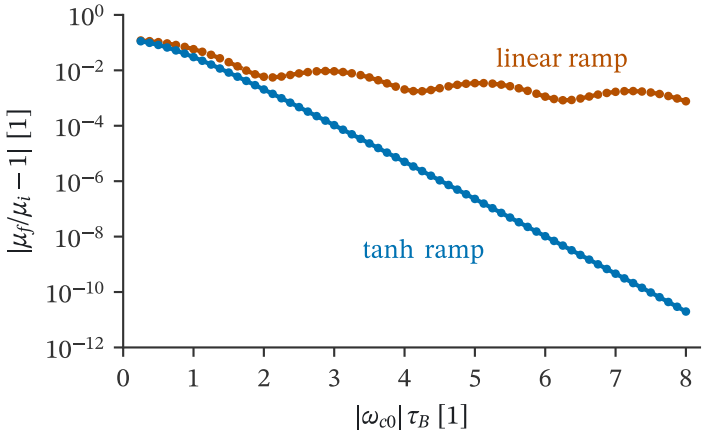


Figure 41: Change of μ when a uniform field doubles, $B : B_0 \to 2B_0$, from the full orbit with the induced field $\boldsymbol{E} = -(1/2)(\partial B/\partial t)\boldsymbol{e}_z \times \boldsymbol{r}$, started on a circle about the axis. Smooth ramp $B/B_0 = 3/2 + \tanh(t/\tau_B)/2$: the change falls exponentially with the ramp time τ_B, to $2 \cdot 10^{-11}$ at $|\omega_{c0}|\tau_B = 8$. Linear ramp of the same largest slope, with kinks at $t = \pm\tau_B$: the change falls only algebraically, to $8 \cdot 10^{-4}$ at $|\omega_{c0}|\tau_B = 8$. μ is an adiabatic invariant, not an exact constant, and its error is set by the sharpest feature of the field's time dependence. Ramp time in units of $1/|\omega_{c0}|$, $\omega_{c0} = q_\alpha B_0/m_\alpha$.

Rechenbeispiel Rechenbeispiel 46

A proton has $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$ and charge $q_i = e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. Use an initial field $B_0 = 0.0100\,\mathrm{T}$, a final field $B_1 = 0.0400\,\mathrm{T}$, and $v_{\perp,0} = 1.00 \cdot 10^5\,\mathrm{m\,s^{-1}}$. Assume that the field changes adiabatically, μ is conserved, and there is no electrostatic energy exchange.

Target: report $v_{\perp,1}$, ρ_{L0}, and ρ_{L1}.

Numerical result: $v_{\perp,1} = 2.00 \cdot 10^5\,\mathrm{m\,s^{-1}}$, $\rho_{L0} = 0.104\,\mathrm{m}$, and $\rho_{L1} = 0.0522\,\mathrm{m}$.

The invariance has a Hamiltonian form. For a periodic motion whose parameters change slowly compared with its period, the action $J = \oint \boldsymbol{p} \cdot \mathrm{d}\boldsymbol{l}$ over one period is an adiabatic invariant. For the gyration, J is proportional to μ, and the guiding center moves with the Hamiltonian

$$H = \frac{m_\alpha v_\parallel^2}{2} + \mu B(\boldsymbol{R}) + q_\alpha \Phi(\boldsymbol{R})$$

H is an energy in J and Φ the electrostatic potential in V, both evaluated at the guiding center $\boldsymbol{R}$. The term μB acts as a potential energy for the guiding center. Its parallel gradient gives the mirror force, its perpendicular gradient the grad-B drift of the next section.

Derivation: μ as the action of the gyration

Step: Canonical momentum on the orbit

With the vector potential $\boldsymbol{A} = \boldsymbol{B} \times \boldsymbol{r}/2$ of the uniform field, $\boldsymbol{r}$ measured from the orbit center, the canonical momentum across $\boldsymbol{B}$ is $\boldsymbol{p}_\perp = m_\alpha \boldsymbol{v}_\perp + q_\alpha \boldsymbol{A}$, in $\mathrm{kg\,m\,s^{-1}}$.

Step: Integrate over one turn

Along the orbit in the direction of motion, the kinetic part gives $2\pi m_\alpha v_\perp \rho_L$. The vector-potential part is q_α times the magnetic flux through the orbit, counted with the sense of rotation; since ions circle $\boldsymbol{B}$ left-handed and electrons right-handed, it is $-\pi|q_\alpha|B\rho_L^2$ for both. Hence

$$J = 2\pi m_\alpha v_\perp \rho_L - \pi |q_\alpha| B \rho_L^2 = \frac{2\pi m_\alpha}{|q_\alpha|}\mu = |q_\alpha| \pi \rho_L^2 B.$$

The action is proportional to μ and to the flux $\pi\rho_L^2 B$ through the orbit.

Step: Gyrophase as an ignorable coordinate

H does not depend on the gyrophase ϕ, so its conjugate momentum $p_\phi = (m_\alpha/q_\alpha)\mu$ is constant. Along the field line, $-\partial H/\partial s$ gives

$$m_\alpha \frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = -\mu \frac{\partial B}{\partial s} - q_\alpha \frac{\partial \Phi}{\partial s}.$$

Within the guiding-center model μ is exactly conserved. The particle's μ differs from the guiding-center value by terms of order $\varepsilon = \rho_L/L_B$, so for the particle it is constant to leading order in ε: an adiabatic invariant.

Fast and slow variables Interpretation 47

The particle position contains a rapidly rotating vector $\boldsymbol{\rho}$ and a slowly evolving center $\boldsymbol{R}$. Guiding-center theory does not erase the physics of the fast orbit. It replaces its detailed phase by averaged quantities such as μ, parallel momentum, and drift velocities.

Summary Summary 48

Guiding-center theory requires small orbit size and slow field variation. The decomposition $\boldsymbol{r} = \boldsymbol{R} + \boldsymbol{\rho}$ separates gyromotion from center motion, and $\mu = m_\alpha v_\perp^2/(2B)$ is conserved to leading order in $\varepsilon = \rho_L/L_B$: by the mirror force and energy conservation, by Faraday's law, and as the action of the gyration.

Exam questions

(e) How are particle orbits approximated in a magnetized plasma and how is the particle motion decomposed and averaged in a not fully homogeneous plasma? What are the conditions that such an approximation is applicable?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 49

- What is the small parameter that compares a gyroradius with magnetic-field structure?
- Which part of the orbit is averaged out in guiding-center theory?
- What happens to μ if B increases adiabatically while it remains invariant?
- Give one process that can break magnetic-moment conservation.

Answers

What is the small parameter that compares a gyroradius with magnetic-field structure?

$\varepsilon = \rho_L/L_B$. Guiding-center averaging requires this dimensionless ratio to be much smaller than one.

Which part of the orbit is averaged out in guiding-center theory?

The fast gyrophase dependence of $\boldsymbol{\rho}$ is averaged, while the guiding-center position and slow invariants are retained.

What happens to μ if B increases adiabatically while it remains invariant?

The perpendicular kinetic energy $m_\alpha v_\perp^2/2 = \mu B$ increases in proportion to B, so $v_\perp$ increases as $\sqrt{B}$.

Give one process that can break magnetic-moment conservation.

A field variation on the gyroradius or gyroperiod scale, a collision, a sharp boundary, or a resonant wave can change the gyrophase coherently.

5.4 Nonuniform fields: grad-B, curvature, and mirrors

A nonuniform magnetic field exerts an averaged force on the guiding center. How do field gradients and curved field lines turn the local circular orbit into a drift or a mirror reflection?

Learning objectives

- derive the grad-B drift from the magnetic-moment force
- identify the curvature drift as a response to parallel inertia
- explain magnetic-mirror reflection using the adiabatic invariant

Adiabatic guiding-center force

Assumptions 50

Use the same $\rho_L/L_B \ll 1$ ordering and neglect rapid changes of μ. The magnetic-moment force is the effective guiding-center force

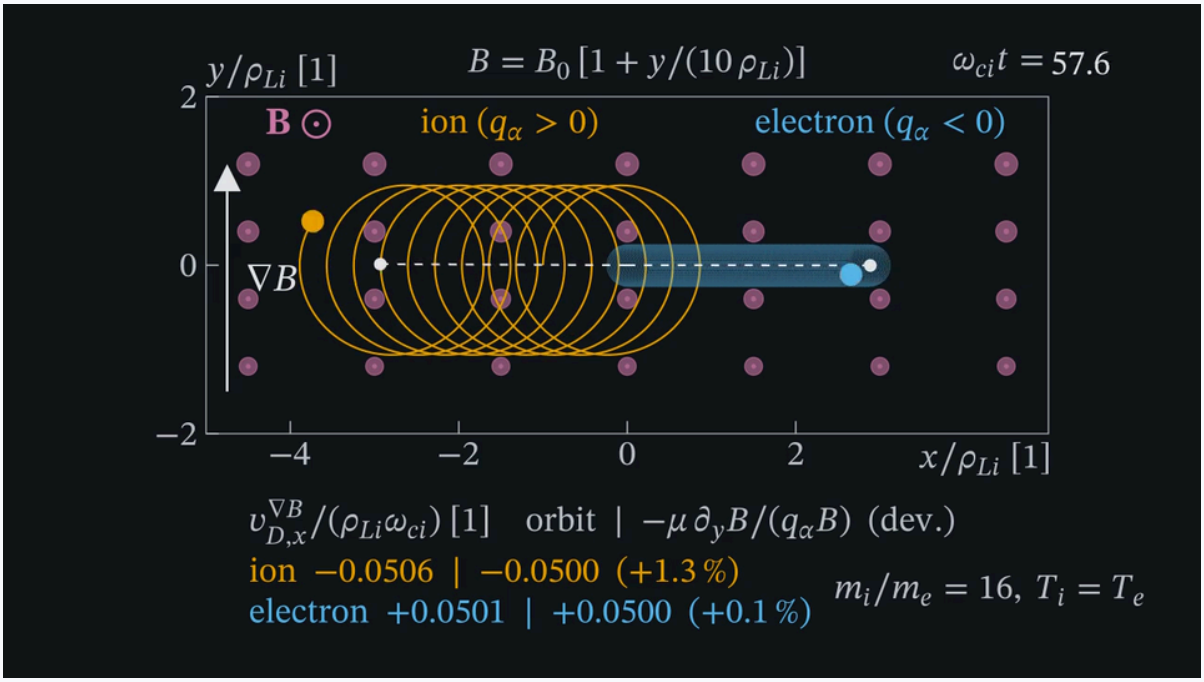
$$\boldsymbol{F}_\mu = -\mu \boldsymbol{\nabla}(B)$$

$\boldsymbol{F}_\mu$ is in N when μ is in J T^{-1} and $\boldsymbol{\nabla}(B)$ in T m^{-1}. Its component along $\boldsymbol{b}$ is the mirror force, derived below from $\boldsymbol{\nabla}\cdot(\boldsymbol{B}) = 0$; its perpendicular part drives the grad-B drift.

Because the field varies little across an orbit, the local perpendicular part of this force, $\boldsymbol{F}_{\mu,\perp} = -\mu \boldsymbol{\nabla}_\perp B$, can be inserted into the homogeneous-force drift:

$$\boldsymbol{v}_D^{\nabla B} = \frac{\boldsymbol{F}_{\mu,\perp} \times \boldsymbol{B}}{q_\alpha B^2} = \frac{\mu}{q_\alpha} \frac{\boldsymbol{B} \times \boldsymbol{\nabla}(B)}{B^2}$$

The drift is perpendicular to both $\boldsymbol{B}$ and the field gradient. Its direction changes with the sign of q_α.



Watch animation

Grad-B drift from the full Lorentz force in $\boldsymbol{B} = B_0(1 + y/L)\boldsymbol{e}_z$ with $L = 10\rho_{Li}$, reduced mass ratio $m_i/m_e = 16$ and equal temperatures. The orbit curves more tightly where B is stronger, so ion and electron drift along x in opposite directions. The measured drift speeds agree with $\mu|\boldsymbol{\nabla}B|/(|q_\alpha|B)$ within 1.3 % (ion) and 0.1 % (electron). Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

Figure 42: Grad-B drift from the full Lorentz force in $\boldsymbol{B} = B_0(1 + y/L)\boldsymbol{e}_z$ with $L = 10\rho_{Li}$, reduced mass ratio $m_i/m_e = 16$ and equal temperatures. The orbit curves more tightly where B is stronger, so ion and electron drift along x in opposite directions. The measured drift speeds agree with $\mu|\boldsymbol{\nabla}B|/(|q_\alpha|B)$ within 1.3 % (ion) and 0.1 % (electron). Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

A curved field line also pushes the guiding center sideways. Let R_c be the local radius of curvature in m and $\boldsymbol{e}_R$ the unit vector normal to the field line, pointing away from the centre of curvature. Along the line, $\boldsymbol{b} = \boldsymbol{B}/B$ turns as

$$(\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{b} = -\frac{\boldsymbol{e}_R}{R_c}$$

$(\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{b}$ is the rate of change of $\boldsymbol{b}$ per unit length along the field line, in m^{-1}. It points toward the centre of curvature.

A guiding center streaming at $v_\parallel$ along the curved line therefore has the centripetal acceleration $v_\parallel^2(\boldsymbol{b}\cdot\boldsymbol{\nabla})\boldsymbol{b} = -(v_\parallel^2/R_c)\boldsymbol{e}_R$. The magnetic force must supply it. In the frame of the guiding center this appears as the centrifugal force

$$\boldsymbol{F}_{\mathrm{curv}} = \frac{m_\alpha v_\parallel^2}{R_c}\boldsymbol{e}_R$$

The effective centrifugal force is in N and points away from the centre of curvature.

Applying the same force-drift map gives

$$\boldsymbol{v}_D^{\mathrm{curv}} = \frac{\boldsymbol{F}_{\mathrm{curv}} \times \boldsymbol{B}}{q_\alpha B^2} = \frac{m_\alpha v_\parallel^2}{q_\alpha B R_c} \boldsymbol{e}_R \times \boldsymbol{b}$$

The curvature drift is in $\mathrm{m\,s^{-1}}$. The sign of q_α determines its direction, and it assumes the same adiabatic ordering as the grad-B drift. With the signed $\rho_\parallel = v_\parallel/\omega_c$ and the angular frequency $\Omega_\parallel = v_\parallel/R_c$ at which the guiding center circles the centre of curvature, $\boldsymbol{v}_D^{\mathrm{curv}} = \rho_\parallel \Omega_\parallel \boldsymbol{e}_R \times \boldsymbol{b}$.

Derivation: curvature drift from the guiding-center acceleration

Step: Curvature of a circular field line

Take a field line that is a circle of radius R_c about the z axis. With $R = \sqrt{x^2 + y^2}$, its unit vectors are $\boldsymbol{b} = \big(-y\boldsymbol{e}_x + x\boldsymbol{e}_y\big)/R$ and $\boldsymbol{e}_R = \big(x\boldsymbol{e}_x + y\boldsymbol{e}_y\big)/R$. Differentiating $\boldsymbol{b}$ along itself gives

$$(\boldsymbol{b} \cdot \boldsymbol{\nabla})\boldsymbol{b} = -\frac{x\boldsymbol{e}_x + y\boldsymbol{e}_y}{R^2} = -\frac{\boldsymbol{e}_R}{R_c}$$

on the line $R = R_c$. A smooth field line is locally such a circle.

Step: Acceleration of the guiding center

The guiding center moves with $v_\parallel \boldsymbol{b}$, and $\boldsymbol{b}$ changes along its path at the rate $v_\parallel (\boldsymbol{b} \cdot \boldsymbol{\nabla})\boldsymbol{b}$. For constant $v_\parallel$ in a static field,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(v_\parallel \boldsymbol{b}\big) = v_\parallel^2 (\boldsymbol{b} \cdot \boldsymbol{\nabla})\boldsymbol{b} = -\frac{v_\parallel^2}{R_c}\boldsymbol{e}_R.$$

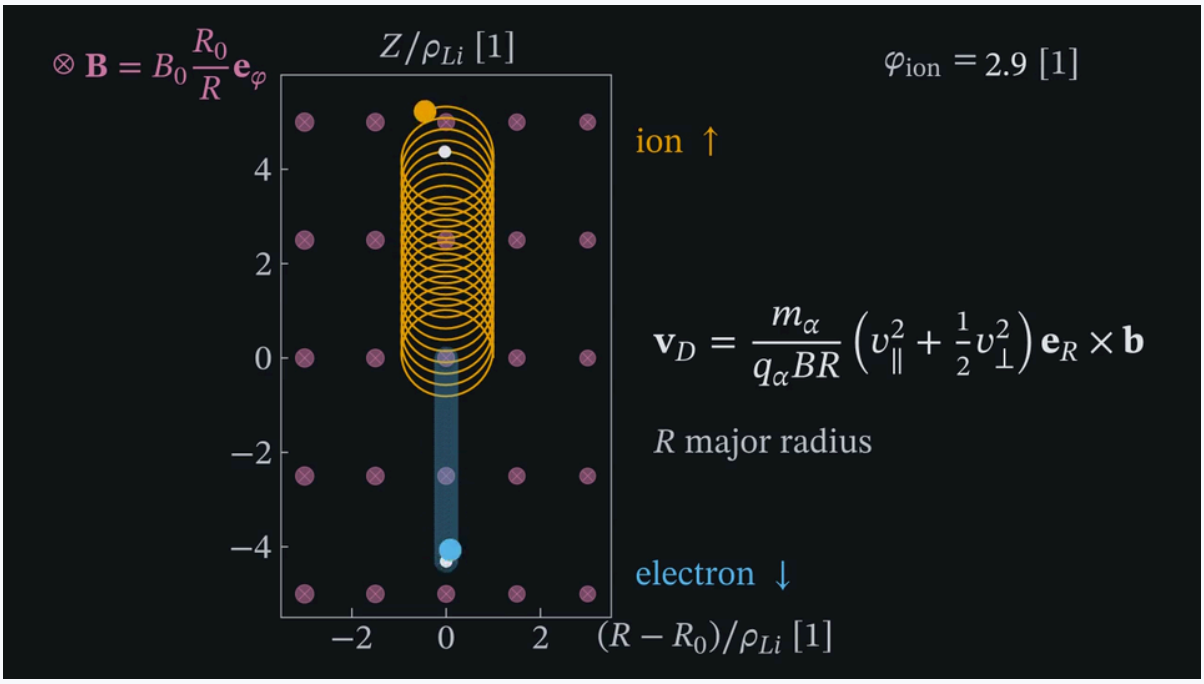
Step: Insert the centrifugal force into the force drift

The force $\boldsymbol{F}_{\mathrm{curv}} = \big(m_\alpha v_\parallel^2/R_c\big)\boldsymbol{e}_R$ in $\boldsymbol{v}_D = (\boldsymbol{F} \times \boldsymbol{B})/\big(q_\alpha B^2\big)$ gives $\boldsymbol{v}_D^{\mathrm{curv}} = m_\alpha v_\parallel^2 (\boldsymbol{e}_R \times \boldsymbol{b})/(q_\alpha B R_c)$. At $(R_c, 0, 0)$, $\boldsymbol{e}_R = \boldsymbol{e}_x$ and $\boldsymbol{b} = \boldsymbol{e}_y$, so $\boldsymbol{e}_R \times \boldsymbol{b} = \boldsymbol{e}_z$: ions drift along $+z$, electrons along $-z$.

In a vacuum toroidal field, $B \propto 1/R$, so $\boldsymbol{\nabla} B = -(B/R_c)\boldsymbol{e}_R$ and the grad-B drift points the same way. The two drifts add to

$$\boldsymbol{v}_D^{\mathrm{curv}} + \boldsymbol{v}_D^{\nabla B} = \frac{m_\alpha}{q_\alpha B R_c}\left(v_\parallel^2 + \frac{v_\perp^2}{2}\right)\boldsymbol{e}_R \times \boldsymbol{b}$$

In $\mathrm{m\,s^{-1}}$. Both terms are vertical in a torus and opposite for ions and electrons, so they separate charge.



Watch animation

Curvature and grad-B drift from the full Lorentz force in the vacuum toroidal field $B \propto 1/R$, seen in the poloidal plane. The ion drifts up and the electron down, a charge separation that a purely toroidal field cannot hold. The measured vertical drift speeds agree with $(m_\alpha/(q_\alpha B R_c))\big(v_\parallel^2 + v_\perp^2/2\big)$ within 1.6 % (ion) and 0.4 % (electron). Lengths in ρ_{Li} [1].

Figure 43: Curvature and grad-B drift from the full Lorentz force in the vacuum toroidal field $B \propto 1/R$, seen in the poloidal plane. The ion drifts up and the electron down, a charge separation that a purely toroidal field cannot hold. The measured vertical drift speeds agree with $(m_\alpha/(q_\alpha B R_c))\big(v_\parallel^2 + v_\perp^2/2\big)$ within 1.6 % (ion) and 0.4 % (electron). Lengths in ρ_{Li} [1].

Derivation: grad- B drift

Step: Identify the magnetic-moment force

The guiding-center force from the adiabatic magnetic moment is

$$\boldsymbol{F}_\mu = -\mu \boldsymbol{\nabla}(B) = -\boldsymbol{\nabla}(\mu B),$$

with μ held constant. Its part along $\boldsymbol{b}$ is the mirror force; only the perpendicular part enters the drift.

Step: Apply the general force-drift formula

A perpendicular homogeneous force drifts at

$$\boldsymbol{v}_D = \frac{\boldsymbol{F} \times \boldsymbol{B}}{q_\alpha B^2}.$$

Substitute the magnetic-moment force and use

$$-\boldsymbol{\nabla}(B) \times \boldsymbol{B} = \boldsymbol{B} \times \boldsymbol{\nabla}(B).$$

The grad-B drift is therefore

$$\boldsymbol{v}_D^{\nabla B} = \frac{\mu}{q_\alpha}\frac{\boldsymbol{B}\times\boldsymbol{\nabla}(B)}{B^2}.$$

Reversing q_α while holding $\boldsymbol{B}$ fixed checks the direction of the drift.

Magnetic mirrors Interpretation 51

If a particle moves into a region of increasing B while μ is approximately conserved, its perpendicular energy μB increases. With nearly constant total kinetic energy, parallel energy decreases. The parallel velocity can reach zero, after which the particle reverses and is reflected without a collision. The pitch angle determines whether the particle reaches the high-field region. The pitch angle is the angle between the velocity and the local magnetic field.

Let s measure distance along a field line. The magnetic-moment force has a parallel component

$$m_\alpha \frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = F_\parallel = -\mu\frac{\partial B}{\partial s}$$

The derivative $\partial B/\partial s$ is taken along the field-line coordinate s. A positive field gradient therefore opposes motion into the stronger-field region. This is the one-dimensional mirror force in the adiabatic model.

Derivation: mirror force from $\boldsymbol{\nabla}\cdot(\boldsymbol{B}) = 0$

Step: Radial field of a converging field

Take an axisymmetric field whose axial component $B_z(z)$ varies slowly along the axis, in cylinder coordinates r, θ, z about the axis. Near the axis, $\boldsymbol{\nabla}\cdot(\boldsymbol{B}) = 0$ reads $(1/r)\frac{\partial}{\partial r}(rB_r) = -\partial B_z/\partial z$, so

$$B_r = -\frac{r}{2}\frac{\partial B_z}{\partial z}.$$

Where the field lines converge, B_r points toward the axis.

Step: Average the axial Lorentz force over one turn

Put the orbit on the axis at $r = \rho_L$. With $B_\theta = 0$, the axial force is $q_\alpha(\boldsymbol{v}\times\boldsymbol{B})_z = -q_\alpha v_\theta B_r$. Ions circle $\boldsymbol{B}$ left-handed, $v_\theta = -v_\perp$; electrons right-handed, $v_\theta = +v_\perp$. For both signs of the charge,

$$q_\alpha(\boldsymbol{v}\times\boldsymbol{B})_z = |q_\alpha| v_\perp B_r = -\frac{|q_\alpha| v_\perp \rho_L}{2}\frac{\partial B_z}{\partial z},$$

the same at every gyrophase, so it is also the gyro-average.

Step: Identify the magnetic moment

With $\rho_L = m_\alpha v_\perp/(|q_\alpha|B)$, $|q_\alpha| v_\perp \rho_L/2 = m_\alpha v_\perp^2/(2B) = \mu$. On the axis $B = B_z$ and $s = z$, which gives the mirror force $F_\parallel = -\mu\partial B/\partial s$. The same average in a field whose strength varies across the orbit gives the perpendicular part, so that $\boldsymbol{F}_\mu = -\mu\boldsymbol{\nabla}(B)$.

In a static magnetic field with no electrostatic potential, the effective parallel energy is

$$K = \frac{m_\alpha v_\parallel^2}{2} + \mu B(s) = \text{const.}$$

In this one-dimensional description, $\mu B(s)$ acts as a potential for parallel motion, although it is physically the perpendicular kinetic energy. A particle reflects when this term grows to equal K, leaving no parallel kinetic energy. Particles with too little perpendicular energy reach the maximum field without reflecting; they form the loss cone.

Derivation: mirror reflection and the loss cone

Step: Show conservation of parallel plus magnetic energy

Along a field line, the chain rule gives

$$\frac{\mathrm{d}B(s(t))}{\mathrm{d}t} = \frac{\mathrm{d}s}{\mathrm{d}t}\frac{\partial B}{\partial s} = v_\parallel \frac{\partial B}{\partial s}.$$

The parallel kinetic energy therefore obeys

$$\frac{\mathrm{d}\frac{m_\alpha v_\parallel^2}{2}}{\mathrm{d}t} = m_\alpha v_\parallel \frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = -\mu v_\parallel \frac{\partial B}{\partial s}.$$

If $\mathrm{d}\mu/\mathrm{d}t = 0$, the magnetic energy changes as

$$\frac{\mathrm{d}\mu B}{\mathrm{d}t} = \mu v_\parallel \frac{\partial B}{\partial s}.$$

The two terms cancel, proving conservation of

$$K = \frac{m_\alpha v_\parallel^2}{2} + \mu B.$$

Step: Find the mirror point

At the initial point, use

$$v_{\perp,0} = v_0 \sin\vartheta_0, \quad v_{\parallel,0} = v_0 \cos\vartheta_0.$$

At the mirror point B_m, $v_\parallel$ is zero. Conservation of K and μ gives

$$\mu B_m = \frac{m_\alpha v_0^2}{2}, \quad \mu B_0 = \frac{m_\alpha v_0^2 \sin^2\vartheta_0}{2}.$$

Hence

$$\frac{B_m}{B_0} = \frac{1}{\sin^2\vartheta_0}.$$

If the largest available field is $B_{\max}$, reflection occurs when

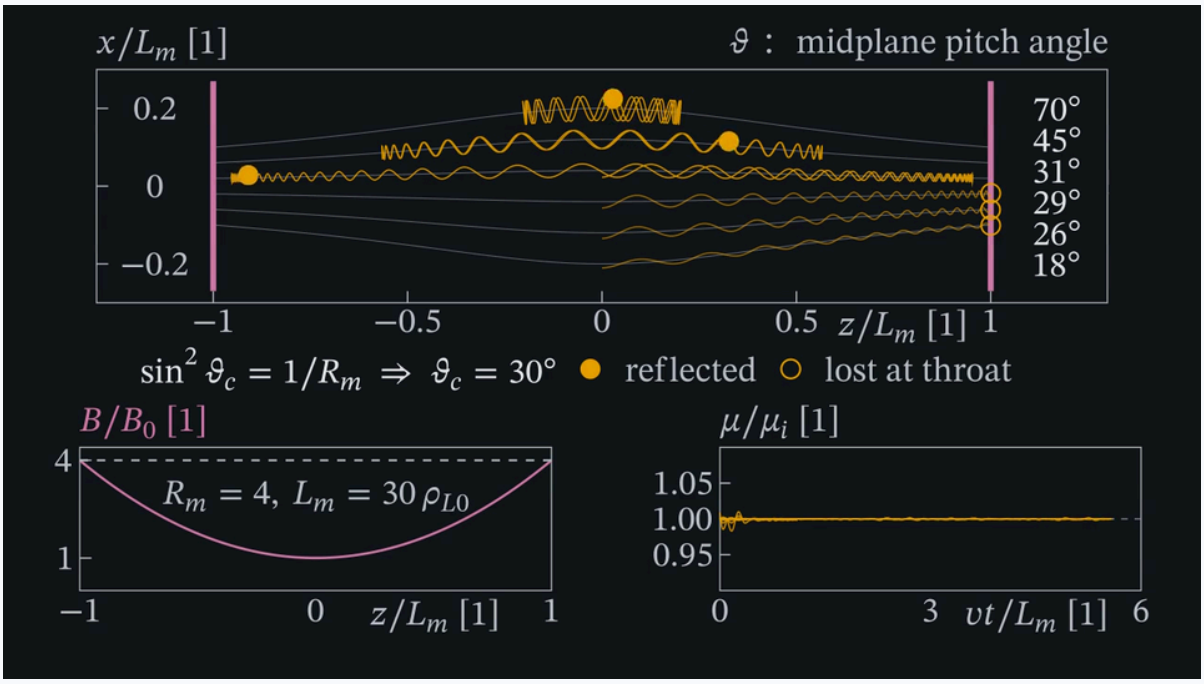
$$\sin^2\vartheta_0 \geq \frac{B_0}{B_{\max}}.$$

The complementary range is the loss cone: particles with $\sin^2\vartheta_0 < B_0/B_{\max}$ pass through the mirror and escape.

The boundary of the loss cone depends only on the mirror ratio $R_m = B_{\max}/B_0$:

$$\sin^2\vartheta_c = \frac{1}{R_m}$$

ϑ_c is the loss-cone angle and R_m is dimensionless. Particles with $\vartheta_0 < \vartheta_c$, or $\vartheta_0 > \pi - \vartheta_c$ when moving the other way, leave through the mirror throat. The angle does not depend on the particle's speed, charge or mass.



Watch animation

Six ions of equal speed in a mirror with $R_m = 4$, computed from the full Lorentz force. Ions with $\sin^2\vartheta_0 < 1/R_m$, inside the 30° loss cone, leave through the throat; the others are reflected. μ stays constant within 0.3 % for the reflected and 1.6 % for the escaping ions. Positions in units of the mirror half-length L_m [1].

Figure 44: Six ions of equal speed in a mirror with $R_m = 4$, computed from the full Lorentz force. Ions with $\sin^2\vartheta_0 < 1/R_m$, inside the 30° loss cone, leave through the throat; the others are reflected. μ stays constant within 0.3 % for the reflected and 1.6 % for the escaping ions. Positions in units of the mirror half-length L_m [1].

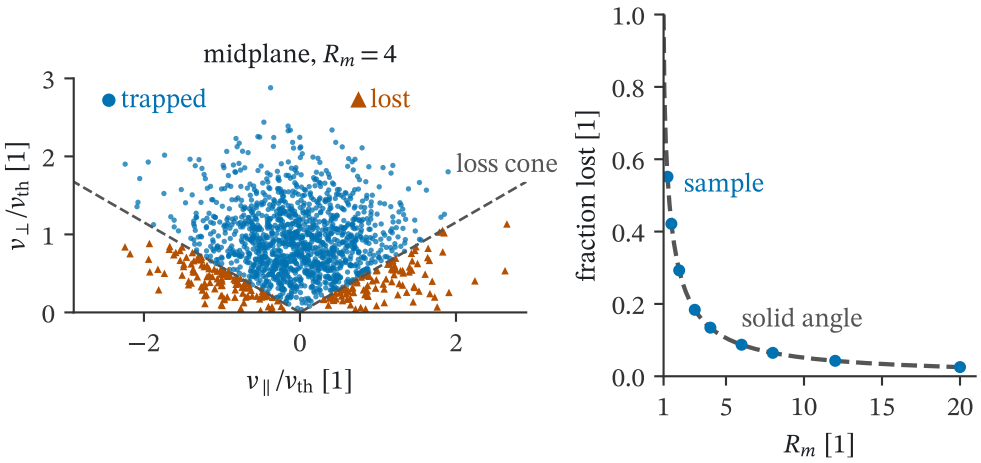


Figure 45: Loss cone in velocity space. Left: an isotropic Maxwellian sample at the midplane of a mirror with $R_m = 4$. Each particle is classified by following $v_\parallel^2(z) = v^2 - v_{\perp 0}^2 B(z)/B_0$ to the throat in the field $B(z)/B_0 = 1 + (R_m - 1)(z/L_m)^2$, without using the loss-cone formula; the lost particles (triangles) lie inside the dashed lines $\sin^2 \vartheta_c = 1/R_m$. Right: lost fraction of an isotropic distribution against R_m, sample (dots) and the solid-angle fraction $1 - \sqrt{1 - 1/R_m}$ of the two cones. Velocities in units of $v_{\mathrm{th}} = \sqrt{2k_B T/m_\alpha}$.

Rechenbeispiel Rechenbeispiel 52

A particle starts in a minimum field $B_0 = 0.0100\,\mathrm{T}$ and sees a maximum field $B_{\max} = 0.0500\,\mathrm{T}$. Determine the critical pitch angle ϑ_c separating reflected particles from the loss cone. Here $0 \le \vartheta_0 \le \pi/2$ is the acute pitch angle to the direction of approach, in a static field with no electric work.

Numerical result: $\vartheta_c = 26.6°$. Particles with $\vartheta_0 \ge \vartheta_c$ reflect in this ideal adiabatic model.

Drift inventory Interpretation 53

The single-particle drifts developed here are the common $E \times B$ drift, the general homogeneous-force drift, the grad-B drift, and the curvature drift. The first is species independent; the latter two reverse direction with charge sign. A mirror reflection is not itself a transverse drift: it is parallel slowing and reversal caused by the effective potential μB.

Summary Summary 54

Gradients and curvature create effective guiding-center forces. The force drift turns $-\mu \nabla(B)$ into grad- B drift, while parallel inertia in a curved field produces curvature drift. Magnetic mirrors follow from conserving μ while the field strength changes.

Exam questions

(f) How is the magnetic moment μ defined and how does the grad-B-drift follow from $F = -\mu \nabla(B)$?

(g) Draw and explain a magnetic mirror.

(h) What is the curvature drift?

(i) What drifts do you know in magnetized plasmas?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 55

- Why is the grad-B drift charge-sign dependent while the $E \times B$ drift is not?
- What energy conversion occurs as a mirrored particle approaches stronger B?
- What is the common geometric direction of grad- B drift?
- What approximation must be checked before classifying a drift with guiding-center theory?

Answers

Why is the grad-B drift charge-sign dependent while the $E \times B$ drift is not?
The magnetic-moment force does not contain a factor of q_α, so the force-drift formula retains $1/q_\alpha$. The electric force contains q_α and cancels it.

What energy conversion occurs as a mirrored particle approaches stronger B?
Perpendicular energy μB increases while parallel kinetic energy decreases if total energy is conserved.

What is the common geometric direction of grad- B drift?
It is perpendicular to the local field and to $\nabla(B)$, with its sign set by the charge.

What approximation must be checked before classifying a drift with guiding-center theory?
The field and force must vary slowly across a gyroradius and a gyroperiod, so the relevant dimensionless ordering parameters must be small.

5.5 Polarization drift in a slowly varying electric field

The $E \times B$ drift is an instantaneous force balance. What happens when the perpendicular electric field changes slowly in time? The orbit center acquires an inertial correction so that its drift can follow the changing field.

Learning objectives

- state the temporal ordering for the polarization-drift approximation
- separate the static $E \times B$ drift from the inertial correction
- derive the polarization drift in a uniform magnetic field
- interpret the species-summed polarization current

Unit ledger

The drive frequency ω_d and ω_c are in s^{-1}. The ordering ratio $\omega_\mathrm{d}/|\omega_c|$ is normalized and therefore carries unit [1].

Slowly varying perpendicular electric field

Assumptions 56

Take a uniform, time-independent $\boldsymbol{B} = B\boldsymbol{b}$ and a spatially uniform perpendicular electric field whose characteristic drive frequency satisfies $\omega_\mathrm{d}/|\omega_c| \ll 1$. The field changes slowly compared with one gyroperiod, and the correction to the $E \times B$ drift remains small.

Extend the $E \times B$ ansatz by a slow inertial correction to the drift:

$$\boldsymbol{v} = v_\parallel \boldsymbol{b} + \boldsymbol{v}_D + \boldsymbol{v}_D^{\mathrm{pol}} + \boldsymbol{v}_\perp$$

The leading force balance is the familiar common drift

$$\boldsymbol{v}_D = \frac{\boldsymbol{E}_\perp \times \boldsymbol{B}}{B^2}$$

The leading drift is in $\mathrm{m\,s^{-1}}$ and follows the instantaneous electric field. As in the guiding-center split, $\boldsymbol{v}_\perp$ is the rapidly rotating gyration velocity.

The next-order orbit-center response is the polarization drift:

$$\boldsymbol{v}_D^{\mathrm{pol}} = \frac{m_\alpha}{q_\alpha B^2}\frac{\partial \boldsymbol{E}_\perp}{\partial t}$$

The factor $1/q_\alpha$ makes the polarization drift point in opposite directions for positive and negative charges. Unlike the $E \times B$ drift, it depends on particle mass.

Derivation: inertial correction to the $E \times B$ drift

Step: Separate the leading drift

Insert the split into the Lorentz force for a uniform magnetic field. As for the $E \times B$ ansatz, the gyration obeys $m_\alpha \mathrm{d}\boldsymbol{v}_\perp/\mathrm{d}t = q_\alpha \boldsymbol{v}_\perp \times \boldsymbol{B}$ and cancels, leaving for the perpendicular drifts

$$m_\alpha \frac{\partial}{\partial t}\left(\boldsymbol{v}_D + \boldsymbol{v}_D^{\mathrm{pol}}\right) = q_\alpha\left(\boldsymbol{E}_\perp + \left(\boldsymbol{v}_D + \boldsymbol{v}_D^{\mathrm{pol}}\right) \times \boldsymbol{B}\right).$$

The $E \times B$ drift is defined by

$$\boldsymbol{E}_\perp + \boldsymbol{v}_D \times \boldsymbol{B} = \boldsymbol{0}.$$

Keeping the leading inertial term gives

$$m_\alpha \frac{\partial \boldsymbol{v}_D}{\partial t} = q_\alpha\left(\boldsymbol{v}_D^{\mathrm{pol}} \times \boldsymbol{B}\right).$$

Step: Invert the magnetic operator

Cross with $\boldsymbol{B}$. Because the correction is perpendicular to the field,

$$\left(\boldsymbol{v}_D^{\mathrm{pol}} \times \boldsymbol{B}\right) \times \boldsymbol{B} = -B^2 \boldsymbol{v}_D^{\mathrm{pol}}.$$

Thus

$$\boldsymbol{v}_D^{\mathrm{pol}} = \frac{m_\alpha}{q_\alpha B^2}\left(\boldsymbol{B} \times \frac{\partial \boldsymbol{v}_D}{\partial t}\right).$$

Since $\boldsymbol{B}$ is constant,

$$\frac{\partial \boldsymbol{v}_D}{\partial t} = \frac{\frac{\partial \boldsymbol{E}_\perp}{\partial t} \times \boldsymbol{B}}{B^2}.$$

Use $\boldsymbol{B} \times (\boldsymbol{A} \times \boldsymbol{B}) = B^2 \boldsymbol{A}$ for $\boldsymbol{A} \cdot \boldsymbol{B} = 0$. The correction is therefore

$$\boldsymbol{v}_D^{\mathrm{pol}} = \frac{m_\alpha}{q_\alpha B^2}\frac{\partial \boldsymbol{E}_\perp}{\partial t}.$$

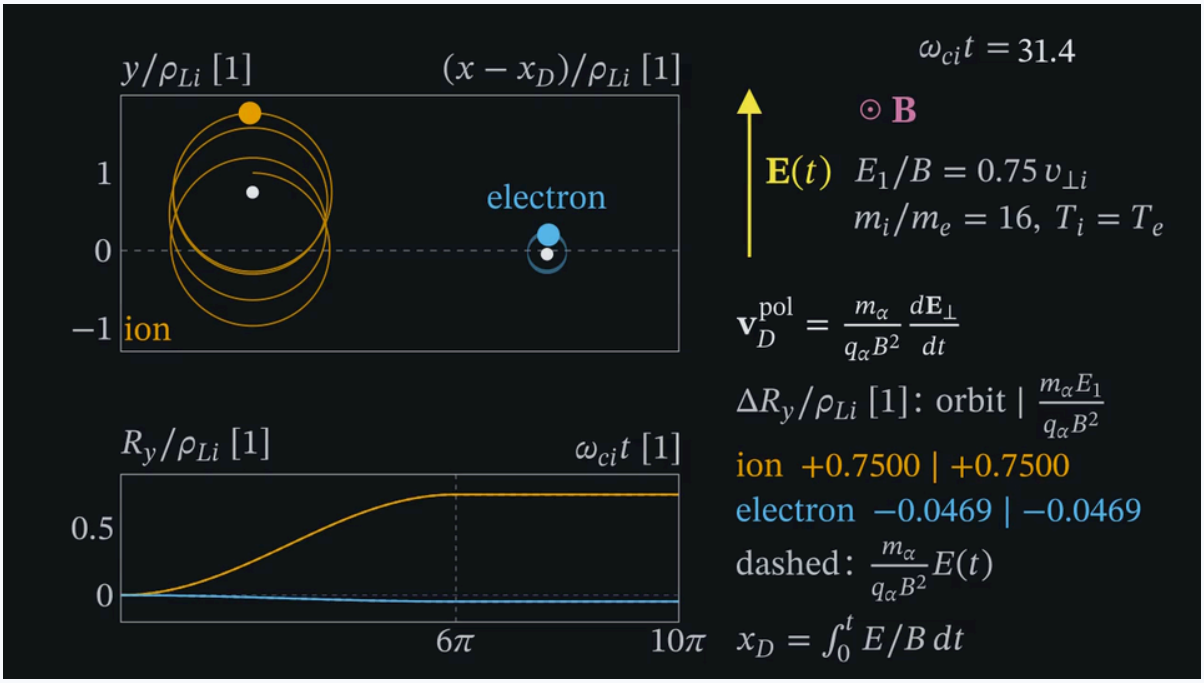
Step: State the ordering

The discarded term $m_\alpha \partial \boldsymbol{v}_D^{\mathrm{pol}}/\partial t$ is smaller by $\omega_{\mathrm{d}}/|\omega_c|$. The polarization drift is therefore valid when the electric field varies slowly compared with the gyrofrequency.

Summing over species gives a polarization current density

$$\boldsymbol{j}_{\mathrm{pol}} = \sum_\alpha n_\alpha q_\alpha \boldsymbol{v}_{D,\alpha}^{\mathrm{pol}} = \frac{1}{B^2}\left(\sum_\alpha n_\alpha m_\alpha\right)\frac{\partial \boldsymbol{E}_\perp}{\partial t}$$

The current density is in $\mathrm{A\,m^{-2}}$ and n_α is the density of species α in $\mathrm{m^{-3}}$. The charge signs cancel in the species sum, so the mass density weights the polarization current; in an electron–ion plasma the ion contribution is usually larger.



Watch animation

Polarization drift, computed from the full Lorentz force while $\boldsymbol{E}_\perp$ rises smoothly to E_1 over three ion gyroperiods, with $E_1/B = 0.75 v_\perp$ of the ion and $m_i/m_e = 16$. With the common $E \times B$ displacement removed, the ion guiding center moves along $\boldsymbol{E}$ by $m_i E_1/(eB^2)$ and the electron 16 times less the other way, the time integral of $\boldsymbol{v}_D^{\mathrm{pol}}$. Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

Figure 46: Polarization drift, computed from the full Lorentz force while $\boldsymbol{E}_\perp$ rises smoothly to E_1 over three ion gyroperiods, with $E_1/B = 0.75 v_\perp$ of the ion and $m_i/m_e = 16$. With the common $E \times B$ displacement removed, the ion guiding center moves along $\boldsymbol{E}$ by $m_i E_1/(eB^2)$ and the electron 16 times less the other way, the time integral of $\boldsymbol{v}_D^{\mathrm{pol}}$. Lengths in ρ_{Li} and time $\omega_{ci}t$, both [1].

Rechenbeispiel

Rechenbeispiel 57

Consider an electron in a uniform field $B = 0.0100\,\mathrm{T}$. Use $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $q_e = -1.602 \cdot 10^{-19}\,\mathrm{C}$, and a perpendicular drive $\boldsymbol{E}_\perp(t) = 3.00 \cdot 10^4\,\mathrm{V\,m^{-1}}\cos(\omega_\mathrm{d} t)\boldsymbol{e}_x$ with $\omega_\mathrm{d} = 1.00 \cdot 10^5\,\mathrm{s^{-1}}$. Evaluate the polarization-drift amplitude and the ordering ratio $\omega_\mathrm{d}/|\omega_{ce}|$.

Assumptions: uniform fields, collisionless motion, and the slowly varying-field approximation.

Numerical result: $\left|v_{D,e}^{\mathrm{pol}}\right| = 171\,\mathrm{m\,s^{-1}}$ and $\omega_\mathrm{d}/|\omega_{ce}| = 5.69 \cdot 10^{-5}$ [1].

Polarization is an inertial response

Interpretation 58

The $E \times B$ drift transports both species together, whereas the polarization drift separates their orbit centers when the electric field changes. A growing electric field therefore produces a current even though the leading $E \times B$ drift carries no net current in a quasineutral pair of species. The approximation fails when the field varies on the gyrofrequency scale, where the full cyclotron response is needed.

Summary

Summary 59

For $\omega_\mathrm{d} \ll |\omega_c|$, a changing perpendicular electric field adds $\boldsymbol{v}_D^{\mathrm{pol}} = \left(m_\alpha/(q_\alpha B^2)\right)\partial \boldsymbol{E}_\perp/\partial t$ to the common $E \times B$ drift. This correction is mass dependent and reverses with charge, so it contributes to the polarization current.

Knowledge check

Knowledge check 60

- Which ordering makes the polarization-drift expansion valid?
- Why does the polarization drift depend on mass while the $E \times B$ drift does not?
- What is the direction of the electron polarization drift when $\boldsymbol{E}_\perp$ grows in the $\boldsymbol{e}_x$ direction?
- What physical effect is missed if only the $E \times B$ drift is retained?

Answers

Which ordering makes the polarization-drift expansion valid?

The electric-field drive must be slow compared with gyromotion: $\omega_\mathrm{d}/|\omega_c| \ll 1$, with a uniform field over the orbit and a small inertial correction.

Why does the polarization drift depend on mass while the $E \times B$ drift does not?

The $E \times B$ drift is an instantaneous electric–magnetic force balance. The polarization drift is the velocity needed to supply the inertia $m_\alpha \partial \boldsymbol{v}_D/\partial t$, so its coefficient contains m_α/q_α.

What is the direction of the electron polarization drift when $\boldsymbol{E}_\perp$ grows in the $\boldsymbol{e}_x$ direction?

It points in the negative $\boldsymbol{e}_x$ direction because the electron charge is negative, while the positive-ion response points in the positive $\boldsymbol{e}_x$ direction.

What physical effect is missed if only the $E \times B$ drift is retained?

The model misses the species-opposite inertial response and the associated polarization current generated by a time-varying electric field.

5.6 Cyclotron resonance and circular polarization

The slow polarization approximation has a sharp boundary. How does a rotating electric field exchange energy with a particle when its rotation matches the particle's gyrofrequency? This matching condition is cyclotron resonance.

Learning objectives

- represent perpendicular motion with circular complex amplitudes
- identify the circular polarization that couples to a given charge sign
- locate the cyclotron-resonance denominator
- state why an ideal collisionless resonance cannot predict unlimited physical energy

Unit ledger

The angular frequency ω, signed gyrofrequency $\omega_c = q_\alpha B/m_\alpha$, and detuning are in s^{-1}. The resonance condition $\omega/|\omega_c| = 1$ is normalized and carries unit [1].

Uniform harmonic drive — Assumptions 61

Use a uniform, static $\boldsymbol{B} = B\boldsymbol{e}_z$ and a small transverse field with time dependence $\exp(-i\omega t)$. Neglect collisions, field gradients, radiation reaction, and relativistic corrections. The physical field is the real part of the complex representation.

Circular complex amplitudes — Definition 62

In the perpendicular plane define the clockwise combinations $v_{\mathrm{cw}} = v_x + iv_y$ and $E_{\mathrm{cw}} = E_x + iE_y$, and the counterclockwise combinations $v_{\mathrm{ccw}} = v_x - iv_y$ and $E_{\mathrm{ccw}} = E_x - iE_y$. For $q_\alpha > 0$, free gyromotion has v_{cw} proportional to $\exp(-i\omega_c t)$, so the clockwise field is the resonant polarization.

Circular cyclotron response — Governing law 63

The perpendicular Lorentz equation becomes

$$\frac{\partial v_{\mathrm{cw}}}{\partial t} + i\omega_c v_{\mathrm{cw}} = \left(\frac{q_\alpha}{m_\alpha}\right) E_{\mathrm{cw}}$$

For a harmonic drive $E_{\mathrm{cw}} = \tilde{E}_{\mathrm{cw}} \exp(-i\omega t)$, the response amplitude is

$$\tilde{v}_{\mathrm{cw}} = \frac{q_\alpha \tilde{E}_{\mathrm{cw}}}{i m_\alpha (\omega_c - \omega)}.$$

The denominator becomes small when the drive rotation matches the signed gyrofrequency.

Derivation: the resonant circular response

Step: Combine the transverse equations

With $\boldsymbol{B} = B\boldsymbol{e}_z$, the component equations are

$$\frac{\partial v_x}{\partial t} = \left(\frac{q_\alpha}{m_\alpha}\right) E_x + \omega_c v_y, \qquad \frac{\partial v_y}{\partial t} = \left(\frac{q_\alpha}{m_\alpha}\right) E_y - \omega_c v_x.$$

Define $v_{\mathrm{cw}} = v_x + iv_y$ and $E_{\mathrm{cw}} = E_x + iE_y$. Adding i times the second equation to the first gives

$$\frac{\partial v_{\mathrm{cw}}}{\partial t} + i\omega_c v_{\mathrm{cw}} = \left(\frac{q_\alpha}{m_\alpha}\right) E_{\mathrm{cw}}.$$

Step: Insert a harmonic drive

Use

$$E_{\mathrm{cw}}(t) = \tilde{E}_{\mathrm{cw}} \exp(-i\omega t), \quad v_{\mathrm{cw}}(t) = \tilde{v}_{\mathrm{cw}} \exp(-i\omega t).$$

The forced amplitude satisfies

$$-i\omega\tilde{v}_{\mathrm{cw}} + i\omega_c\tilde{v}_{\mathrm{cw}} = \left(\frac{q_\alpha}{m_\alpha}\right)\tilde{E}_{\mathrm{cw}},$$

so

$$\tilde{v}_{\mathrm{cw}} = \frac{q_\alpha \tilde{E}_{\mathrm{cw}}}{im_\alpha(\omega_c - \omega)}.$$

The homogeneous solution is $v_{\mathrm{cw},0} \exp(-i\omega_c t)$. The forced response therefore resonates at $\omega = \omega_c$ for this polarization.

Step: Reverse the circular polarization

With $v_{\mathrm{ccw}} = v_x - iv_y$ and $E_{\mathrm{ccw}} = E_x - iE_y$,

$$\frac{\partial v_{\mathrm{ccw}}}{\partial t} - i\omega_c v_{\mathrm{ccw}} = \left(\frac{q_\alpha}{m_\alpha}\right) E_{\mathrm{ccw}}.$$

Its denominator is proportional to $\omega + \omega_c$. For positive drive frequency, one circular polarization resonates at $\omega = |\omega_c|$; the matching handedness is determined by the sign of q_α.

A linearly polarized field is the sum of two counter-rotating circular fields. Only the component with the matching handedness resonates, while the other component remains off resonance. In a real plasma, collisions, finite pulse duration, spatial inhomogeneity, and nonlinear effects limit the energy transfer and broaden or shift the ideal resonance.

Rechenbeispiel

Rechenbeispiel 64

In a uniform field $B = 0.0100\,\mathrm{T}$, determine the positive resonant angular frequencies for an electron and a proton. Use $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, and $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$. For each species, identify the circular polarization that couples to the positive-frequency drive.

Assumptions: collisionless, nonrelativistic, uniform magnetic field and ideal harmonic forcing.

Numerical result: $\omega_{\mathrm{res,e}} = 1.76 \cdot 10^{9}\,\mathrm{s}^{-1}$ and $\omega_{\mathrm{res,i}} = 9.58 \cdot 10^{5}\,\mathrm{s}^{-1}$. The resonant polarization has the handedness of the species' free gyromotion.

Resonance is a model boundary Interpretation 65

When the drive frequency is much smaller than the gyrofrequency, the response can be expanded into the common $E \times B$ drift and the small polarization correction. Near $\omega = |\omega_c|$, that ordering fails: the drive remains phase coherent with the orbit and transfers energy over many gyroperiods. A finite collision rate or a finite interaction time must then be included before predicting an amplitude or absorbed power.

Summary Summary 66

Circular decomposition exposes the cyclotron denominator. A positive frequency resonates with the circular polarization matching the free gyromotion, at $\omega = |\omega_c|$. The collisionless singularity marks the limit of the slow-drift approximation.

Knowledge check Knowledge check 67

- Why is circular polarization useful for diagnosing cyclotron resonance?
- How does changing the sign of q_α affect the resonant polarization?
- What does the factor $1/(\omega_c - \omega)$ predict as the drive approaches resonance?
- Why can a linearly polarized wave still heat a charged particle at cyclotron resonance?

Answers

Why is circular polarization useful for diagnosing cyclotron resonance?
It separates a rotating field into the two handednesses. Only the component rotating with the free gyromotion has the small resonant denominator.

How does changing the sign of q_α affect the resonant polarization?
It reverses the signed gyrofrequency and therefore selects the opposite circular handedness for a positive-frequency drive; the positive resonance remains at $|\omega_c|$.

What does the factor $1/(\omega_c - \omega)$ predict as the drive approaches resonance?
The ideal forced-response amplitude grows as the detuning tends to zero. The divergence signals missing broadening or saturation physics, such as collisions, finite pulse duration, or nonlinear motion.

Why can a linearly polarized wave still heat a charged particle at cyclotron resonance?
A linear polarization is the sum of two circular components. The component with matching handedness supplies the resonant rotating force, while the opposite component is nonresonant.

6. Kinetic theory of plasmas

Single-particle theory follows individual orbits. Kinetic theory instead describes how many particles of each species occupy each range of positions and velocities. One distribution function per species retains velocity-space structure, such as beams and unequal spreads in different directions, that a few fluid variables cannot fully describe.

Distribution function in phase space — Key idea 68

The species distribution $f_s(t, \boldsymbol{r}, \boldsymbol{v})$ is a density in phase space. Its velocity moments give density, bulk velocity, pressure, and higher transport variables. The kinetic equation tells this density how streaming, electromagnetic forces, and collisions change it.

6.1 Collisions in gases and plasmas

When does a particle retain the memory of its previous flight? The answer is set by the distance between effective collisions and by the time over which the plasma is being observed.

Learning objectives

- define mean free path and collision frequency from a collision probability
- distinguish short-range neutral collisions from long-range Coulomb scattering
- explain the Coulomb logarithm and identify the limits of a collisional model

Unit ledger

A neutral collision cross section σ is in m^2, mean free path is in m, and collision frequency is in s^{-1}. The dimensionless parameters K_n and $\ln\Lambda$ are explicitly marked as such.

Binary collision picture — Definition 69

For a projectile of species a moving through a background of species b, let σ_{ab} be the effective cross section for the chosen collision criterion. In a distance $d\ell$, the probability of a collision is $dP_{ab} = n_b \sigma_{ab} d\ell$. This local statement assumes an uncorrelated background and a cross section that is defined for the relative velocity under consideration.

If the background is uniform, the survival probability $S(\ell)$ after a flight of length ℓ obeys

$$\frac{\mathrm{d}S}{\mathrm{d}\ell} = -n_b \sigma_{ab} S, \quad S(0) = 1$$

The coefficient $n_b\sigma_{ab}$ has units m^{-1}. The equation describes loss from the uncollided population, not loss of particles from the plasma.

Therefore the mean free path and collision frequency are

$$\lambda_{\mathrm{mfp},ab} = \frac{1}{n_b \sigma_{ab}}, \quad \nu_{ab} = n_b \sigma_{ab} v_\mathrm{rel}, \quad \lambda_{\mathrm{mfp},ab} = \frac{v_\mathrm{rel}}{\nu_{ab}}$$

The mean free path is in m and the collision frequency is in s^{-1}. The relation uses the same relative speed in both quantities.

Derivation: exponential flights and the mean free path

Step: Write the survival equation

During a path segment $d\ell$, the probability S of having no collision changes by

$$dS = -Sn_b\sigma_{ab}d\ell.$$

Dividing by $d\ell$ gives the differential equation for survival.

Step: Find the first-collision distribution

Integrating from 0 to ℓ gives

$$S(\ell) = \exp(-n_b\sigma_{ab}\ell).$$

The probability density for the first collision is the loss of survival probability per unit length:

$$p(\ell) = n_b\sigma_{ab}\exp(-n_b\sigma_{ab}\ell).$$

Step: Calculate the mean free path and collision rate

The mean path length is

$$\int_0^\infty \ell\, p(\ell)\,\mathrm{d}\ell = \frac{1}{n_b\sigma_{ab}}.$$

A particle traveling at v_{rel} samples this length in a mean time $\lambda_{\mathrm{mfp}}/v_{\mathrm{rel}}$. The inverse time is therefore

$$\nu_{ab} = \frac{v_{\mathrm{rel}}}{\lambda_{\mathrm{mfp}}}.$$

The exponential model is valid when successive encounters can be treated as independent and the background properties are approximately constant over one mean free path.

In a neutral gas, the interaction range is often short compared with the macroscopic system size. A geometric cross section can then summarize the encounter. A plasma has a different collision geometry. Two charged particles interact through a Coulomb field, so many weak deflections can accumulate even when no single encounter is a large-angle event.

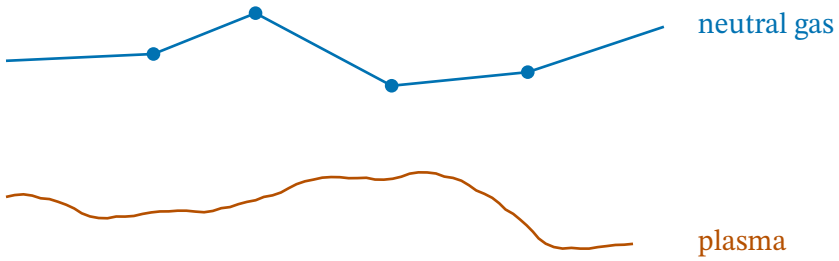


Figure 47: Neutral gas (top): straight flights between a few hard, large-angle collisions (dots). Plasma (bottom): many small-angle Coulomb deflections whose accumulated effect turns the particle by a comparable angle.

Small-angle Coulomb scattering — Assumptions 70

For a weakly coupled, fully ionized plasma, treat the dominant encounters as many small deflections. The distant part of the interaction is limited by collective screening, while the close part is limited by strong deflection or quantum diffraction. The resulting collision estimate is a transport frequency.

The impact parameter b is the perpendicular separation of the incoming, undeflected relative trajectory from the scattering center. For relative speed v_{rel} and reduced mass m_r, the classical impact parameter for a ninety-degree deflection is

$$m_r v_{\mathrm{rel}}^2 b_{90} = \frac{|q_a q_b|}{4\pi\varepsilon_0}, \quad b_{90} = \frac{|q_a q_b|}{4\pi\varepsilon_0 m_r v_{\mathrm{rel}}^2}, \quad m_r = \frac{m_a m_b}{m_a + m_b}$$

The potential energy of two charges is $q_a q_b/(4\pi\varepsilon_0 r)$ in J, so b_{90} is in m.

The small-angle contributions add a logarithmic weight over impact parameters. With $b_{\max}$ set by collective screening and $b_{\min}$ set by the larger of strong deflection and quantum diffraction,

$$b_{\max} \approx \lambda_D, \quad b_{\min} \approx \max(b_{90}, b_{\mathrm{qm}}), \quad \ln\Lambda = \ln\left(\frac{b_{\max}}{b_{\min}}\right)$$

$\ln\Lambda$ is dimensionless. In a classical plasma, b_{qm} may be omitted when the de Broglie wavelength is much smaller than b_{90}. The precise cutoffs depend on the collision operator and velocity average. This is the cutoff-logarithm convention, denoted $\ln\Lambda_{\mathrm{cut}}$ in Chapter 10; that chapter separately defines the plasma parameter $\Lambda = n_e\lambda_D^3$ for its cited approximate collision rate.

A representative scaling for the deflection frequency of a test species a in a background b is

$$\nu_{ab} \text{ scales as } \frac{n_b q_a^2 q_b^2 \ln\Lambda}{\varepsilon_0^2 m_a^2 v_a^3}$$

This is a scaling relation. Numerical coefficients and the replacement of v_a by a relative-velocity average depend on the selected collision operator. The v_a^{-3} dependence makes slow particles especially sensitive to Coulomb scattering.

Interaction range in gases and plasmas — Interpretation 71

Gases and plasmas differ in the interaction range. Neutral collisions are often summarized by a finite cross section. Coulomb interactions are long range, and collective screening plus cumulative small-angle deflections determine the transport rate. A plasma can therefore be collisionless on a chosen time scale even though every charged particle feels electromagnetic forces continuously.

The comparison with a macroscopic length L is summarized by the Knudsen number and, for a process with time scale τ, by a collisionality parameter:

$$K_{\mathrm{n}} = \frac{\lambda_{\mathrm{mfp}}}{L}, \quad \nu\tau$$

Both K_{n} and $\nu\tau$ are dimensionless. $K_{\mathrm{n}} \ll 1$ means many collisions occur over a macroscopic flight, while $\nu\tau \ll 1$ means the process is effectively collisionless over the time τ.

Rechenbeispiel

Rechenbeispiel 72

Consider a weakly ionized gas with $n_b = 1 \cdot 10^{18}\,\mathrm{m}^{-3}$, $\sigma_{ab} = 1 \cdot 10^{-19}\,\mathrm{m}^2$, and $v_{\mathrm{rel}} = 1 \cdot 10^{6}\,\mathrm{m\,s}^{-1}$. Use the neutral-collision model above. Determine the mean free path and collision frequency.

Numerical result: $\lambda_{\mathrm{mfp}} = 10\,\mathrm{m}$ and $\nu_{ab} = 1 \cdot 10^{5}\,\mathrm{s}^{-1}$.

Summary

Summary 73

A mean free path follows from the survival probability of an uncollided particle. In a gas, a finite cross section often controls it directly. In a plasma, long-range Coulomb deflections produce a Coulomb logarithm and a velocity-dependent transport frequency. The ratios K_{n} and $\nu\tau$ decide whether collisions must be retained for a particular process.

Exam questions

(a) What's the difference between a gas and a plasma in terms of collisions?

(b) What's the mean free path and how is it related to the collision frequency?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 74

- What probability law gives the mean free path in a uniform background?
- Why is the Coulomb logarithm dimensionless?
- What does $K_{\mathrm{n}} \approx 1$ indicate?
- Why can a plasma be collisionless while particles still interact?

Answers

What probability law gives the mean free path in a uniform background?
The first-collision distance is exponentially distributed, $p(\ell) = n_b\sigma \exp(-n_b\sigma\ell)$, with mean $\lambda_{\mathrm{mfp}} = 1/(n_b\sigma)$.

Why is the Coulomb logarithm dimensionless?
It is the logarithm of the ratio of two impact-parameter lengths, $\ln\Lambda = \ln(b_{\max}/b_{\min})$.

What does $K_{\mathrm{n}} \approx 1$ indicate?
The mean free path and the macroscopic scale are comparable. A local fluid closure is then questionable and kinetic information is generally needed.

Why can a plasma be collisionless while particles still interact?
Collisionless means $\nu\tau \ll 1$ for the process under study. Electromagnetic forces can remain strong even when discrete collisional deflections are too infrequent to affect that process.

6.2 Distribution functions and phase space

A single particle has a position and a velocity. How can an ensemble with an enormous number of particles be represented without storing every trajectory? A distribution function assigns density to the corresponding phase-space cells.

Learning objectives

- define a species distribution and its normalization
- distinguish phase-space density from a normalized probability density
- compute density, bulk velocity, and pressure as velocity moments
- identify the parameters and meaning of a shifted Maxwellian

Unit ledger

The phase-space measure $\mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$ has units $\mathrm{m}^6\,\mathrm{s}^{-3}$. The distribution f_s therefore has units $\mathrm{s}^3\,\mathrm{m}^{-6}$ so that its integral over a phase-space cell gives a particle count.

Species distribution function Definition 75

For species s, $f_s(t,\boldsymbol{r},\boldsymbol{v})$ is defined by $dN_s = f_s\mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$. It is the number density in six-dimensional phase space: three position coordinates and three velocity coordinates. Dividing f_s by the total species particle number in a chosen domain gives a probability density on that domain; its integral over the domain is one. At a fixed position, a different normalization, f_s/n_s, gives the local velocity probability density.

For any single-particle property $g(t,\boldsymbol{r},\boldsymbol{v})$, the local velocity average is

$$\langle g\rangle_s = \frac{1}{n_s}\int_{\mathbb{R}^3} g(t,\boldsymbol{r},\boldsymbol{v})f_s(t,\boldsymbol{r},\boldsymbol{v})\mathrm{d}^3\boldsymbol{v}$$

The average is local in position. Setting $g = 1$ gives one, while setting $g = \boldsymbol{v}$ gives the bulk velocity $\boldsymbol{u}_s$. The factor f_s/n_s is a velocity-space probability density with units $\mathrm{s}^3\,\mathrm{m}^{-3}$.

At a fixed position, the velocity integral gives the number density:

$$n_s(t,\boldsymbol{r}) = \int f_s(t,\boldsymbol{r},\boldsymbol{v})\mathrm{d}^3\boldsymbol{v}$$

The velocity integral is over all of $\mathbb{R}^3$. The result is a number density in m^{-3}. The position dependence remains because the integration removes velocity information only.

The first two useful velocity moments are the bulk velocity and the central pressure tensor:

$$\boldsymbol{u}_s = \frac{1}{n_s}\int \boldsymbol{v}f_s\mathrm{d}^3\boldsymbol{v}, \quad \boldsymbol{w} = \boldsymbol{v} - \boldsymbol{u}_s, \quad \boldsymbol{P}_s = m_s\int \boldsymbol{w}\boldsymbol{w}f_s\mathrm{d}^3\boldsymbol{v}$$

$\boldsymbol{w}$ is the random velocity measured in the local bulk frame. The dyadic product $\boldsymbol{w}\boldsymbol{w}$ has components w_iw_j: each pairs two components of the random velocity. $\boldsymbol{P}_s$ is a momentum-flux or pressure tensor in Pa (force per area). Its scalar isotropic part is introduced in the moments chapter.

A local equilibrium distribution with uniform temperature T_s and drift velocity $\boldsymbol{u}_s$ is the shifted Maxwellian

$$f_{M,s} = n_s\left(\frac{m_s}{2\pi k_B T_s}\right)^{\frac{3}{2}}\exp\left(-\frac{m_s|\boldsymbol{v}-\boldsymbol{u}_s|^2}{2k_BT_s}\right)$$

Temperature is expressed through the thermal energy k_BT_s in J. The one-dimensional standard deviation is $\sqrt{k_BT_s/m_s}$, while the parameter $v_{\mathrm{th},s} = \sqrt{2k_BT_s/m_s}$ is the thermal-speed convention from Chapter 1. The plot below shows a one-component velocity distribution, whose centered peak is at zero velocity. It is not the distribution of speed magnitudes, which includes a velocity-space volume factor.

Derivation: normalization and central moments of a Maxwellian

Step: Shift to random velocity

Set

$$\boldsymbol{c} = \boldsymbol{v} - \boldsymbol{u}_s, \quad a = \frac{m_s}{2k_B T_s}.$$

In these variables the three-dimensional Gaussian factorizes:

$$\int \exp\left(-a|\boldsymbol{c}|^2\right) \mathrm{d}^3\boldsymbol{c} = \left(\int \exp\left(-ac_x^2\right) \mathrm{d}c_x\right)^3 = \left(\sqrt{\frac{\pi}{a}}\right)^3.$$

Step: Normalize the distribution

The prefactor in $f_{M,s}$ is chosen so that the velocity integral is

$$\int f_{M,s} \mathrm{d}^3\boldsymbol{v} = n_s.$$

Thus n_s is the zeroth moment, independent of the choice of bulk velocity.

Step: Evaluate the first and second central moments

Symmetry gives

$$\int \boldsymbol{c} f_{M,s} \mathrm{d}^3\boldsymbol{v} = \boldsymbol{0},$$

so the first raw moment is $n_s \boldsymbol{u}_s$. Each Cartesian component has variance

$$\frac{1}{2a} = \frac{k_B T_s}{m_s}.$$

Therefore the pressure tensor is isotropic:

$$\boldsymbol{P}_s = m_s n_s \left(\frac{k_B T_s}{m_s}\right) \boldsymbol{I} = n_s k_B T_s \boldsymbol{I}.$$

The derivation assumes a smooth Maxwellian with finite temperature; a cold delta distribution has the same density and flow moments but zero central pressure.

Mean random kinetic energy $3k_B T_s/2$

Step: Sum the independent velocity components

Each Cartesian component of a Maxwellian has variance

$$\left\langle \left(v_j - u_{s,j}\right)^2 \right\rangle = \frac{k_B T_s}{m_s}.$$

The three components are independent, so their kinetic-energy average is

$$\left\langle \frac{m_s (\boldsymbol{v} - \boldsymbol{u}_s)^2}{2} \right\rangle = \frac{m_s}{2} \sum_j \left\langle \left(v_j - u_{s,j}\right)^2 \right\rangle = \frac{3k_B T_s}{2}.$$

Step: Interpret the thermal-speed convention

The convention

$$v_{\mathrm{th},s} = \sqrt{\frac{2k_B T_s}{m_s}}$$

defines a width parameter of the distribution. It is not the mean particle speed.

Thermodynamic temperature characterizes equilibrium. Away from equilibrium, a kinetic temperature can still be defined by the second central velocity moment, $3k_B T_s = m_s\left\langle |\boldsymbol{v} - \boldsymbol{u}_s|^2 \right\rangle$. It measures random kinetic energy but does not specify the whole distribution. If the velocity spread depends on direction, a single scalar temperature does not capture that anisotropy; a pressure tensor is needed. A species that has reached thermal equilibrium through collisions can be described by the Maxwellian above, whereas a collisionless species may retain beams or other non-Maxwellian structure.

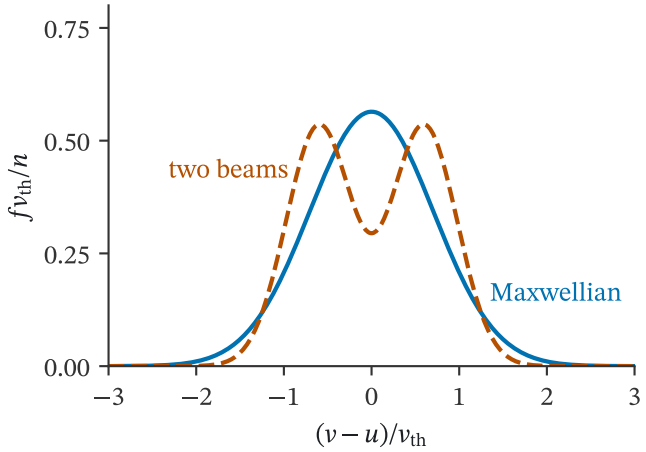


Figure 48: Same n, $\boldsymbol{u}$ and T, different f: a Maxwellian (solid) and two beams at $(v-u)/v_{\mathrm{th}} = \pm 0.6$ (dashed) whose width is chosen to give the same variance $k_B T/m$. One velocity component, $v_{\mathrm{th}} = \sqrt{2k_B T/m}$; f in units of n/v_{th}. The moments first differ at fourth order.

Information retained by low-order moments Interpretation 76

The density keeps the total number at a position. The bulk velocity keeps the first velocity average. The pressure tensor keeps the covariance of random velocities. Distinct distributions can share these moments, so a finite set of fluid variables cannot reproduce every kinetic feature. Closure assumptions enter when the retained moments are evolved.

Summary Summary 77

The distribution function is a phase-space density. Integrating it over velocity gives density, weighting it once gives bulk flow, and weighting it twice about the bulk flow gives pressure. The Maxwellian is the normalized equilibrium reference state for elastic collisions at fixed density, flow velocity, and temperature.

Exam questions

(c) What's the meaning of the distribution function?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 78

- What does $f_s \mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$ count?
- Why does a shifted Maxwellian have zero first central moment?
- Which moment distinguishes a hot isotropic plasma from a cold fluid?
- Can two different distributions have the same density and bulk velocity?

Answers

What does $f_s \mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$ count?

It gives the number of species- s particles in the specified six-dimensional phase-space cell.

Why does a shifted Maxwellian have zero first central moment?

After shifting to $\boldsymbol{w} = \boldsymbol{v} - \boldsymbol{u}_s$, the Gaussian is even in every velocity component, so the integral of the odd vector $\boldsymbol{w}$ vanishes.

Which moment distinguishes a hot isotropic plasma from a cold fluid?

The second central moment, or pressure tensor, is nonzero for a finite-temperature plasma and vanishes in the cold-fluid idealization.

Can two different distributions have the same density and bulk velocity?

Yes. Density and bulk velocity are only the zeroth and first moments. The distributions may differ in temperature, anisotropy, or non-Maxwellian tails.

6.3 Convective derivatives and phase-space transport

A distribution changes because particles cross phase-space cell faces and because collisions redistribute them. The Eulerian description measures this change at a fixed point. The Lagrangian description follows a moving phase-space trajectory.

Learning objectives

- write the chain rule along a phase-space characteristic
- interpret Eulerian and Lagrangian views of the same evolution
- identify phase-space fluxes in position and velocity
- explain why collisionless Hamiltonian flow preserves phase-space volume

Unit ledger

For the phase-space characteristic $\boldsymbol{z} = (\boldsymbol{r}, \boldsymbol{v})$, the two blocks of its velocity are $\boldsymbol{V}_z = (\boldsymbol{v}, \boldsymbol{a})$. We also use the normalized variables $\xi = \boldsymbol{r}/L_0$, $\eta = \boldsymbol{v}/v_0$, and $\tau = tv_0/L_0$, all explicitly dimensionless.

Convective derivative Definition 79

For a scalar $g(t, \boldsymbol{r}, \boldsymbol{v})$, the derivative along a particle trajectory is

$$\frac{\mathrm{D}g_s}{\mathrm{D}t} = \frac{\partial g_s}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(g_s) + \boldsymbol{a}_s \cdot \frac{\partial g_s}{\partial \boldsymbol{v}}$$

. It follows the particle through both real space and velocity space.

A characteristic is the phase-space curve defined by

$$\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \boldsymbol{v}, \quad \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \boldsymbol{a}_s, \quad \boldsymbol{z} = (\boldsymbol{r}, \boldsymbol{v})$$

Position has units m, velocity has units m s^{-1}, and acceleration has units m s^{-2}. The phase-space coordinate is a bookkeeping pair with mixed units, so its normalized version is useful for visualizations.

For any phase-space density, the conservative balance has the form

$$\frac{\partial f}{\partial t} + \boldsymbol{\nabla} \cdot (f \boldsymbol{V}_z) = S$$

Here $\boldsymbol{\nabla} \cdot$ is the divergence in the full phase space and S is a source density. Expanding the flux gives separate spatial and velocity-space contributions. A collision operator is one possible source term in this balance.

The normalized free-streaming equation used in the animation is

$$\frac{\partial f}{\partial \tau} + \boldsymbol{\eta} \cdot \boldsymbol{\nabla}(f) = 0$$

Both $\boldsymbol{\xi} = \boldsymbol{r}/L_0$ and $\boldsymbol{\eta} = \boldsymbol{v}/v_0$ are dimensionless. The time coordinate is $\tau = t/\tau_0$ with $\tau_0 = L_0/v_0$; restore positions, velocities, and times with L_0, v_0, and τ_0. In the one-dimensional visualization, the vector equation reduces to $\partial f/\partial \tau + \eta \partial f/\partial \xi = 0$ with $\xi = x/L_0$ and $\eta = v/v_0$. The physical kinetic equation is three-dimensional in both position and velocity.

Derivation: the Eulerian chain rule

Step: Apply the multivariable chain rule

Let $g(t, \boldsymbol{r}, \boldsymbol{v})$ be evaluated on a trajectory $\boldsymbol{r} = \boldsymbol{r}(t)$ and $\boldsymbol{v} = \boldsymbol{v}(t)$. A small time step gives

$$dg = \frac{\partial g}{\partial t} dt + \boldsymbol{\nabla}(g) \cdot d\boldsymbol{r} + \frac{\partial g}{\partial \boldsymbol{v}} \cdot d\boldsymbol{v}.$$

Step: Insert the characteristic velocities

Divide by dt and use

$$\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \boldsymbol{v}, \quad \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \boldsymbol{a}_s.$$

The derivative along the characteristic is then

$$\frac{\mathrm{d}g}{\mathrm{d}t} = \frac{\partial g}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(g) + \boldsymbol{a}_s \cdot \frac{\partial g}{\partial \boldsymbol{v}}.$$

Step: Distinguish the two viewpoints

At a fixed phase-space point, only the partial derivative $\partial g/\partial t$ is measured. Along a characteristic, the spatial and velocity-space advection terms contribute as well. The two descriptions are equivalent because they evaluate the same scalar field in different ways.

Derivation: phase-space flux balance

Step: Account for the four phase-space contributions

Take a small cell $d^3\boldsymbol{r}d^3\boldsymbol{v}$. Its particle count changes by temporal accumulation, spatial flux through the position faces, velocity-space flux through the velocity faces, and collisions.

Step: Write the two fluxes

The spatial flux is $f\boldsymbol{v}$ and the velocity-space flux is $f\boldsymbol{a}_s$. Dividing the cell balance by its volume and taking the cell-size limit gives

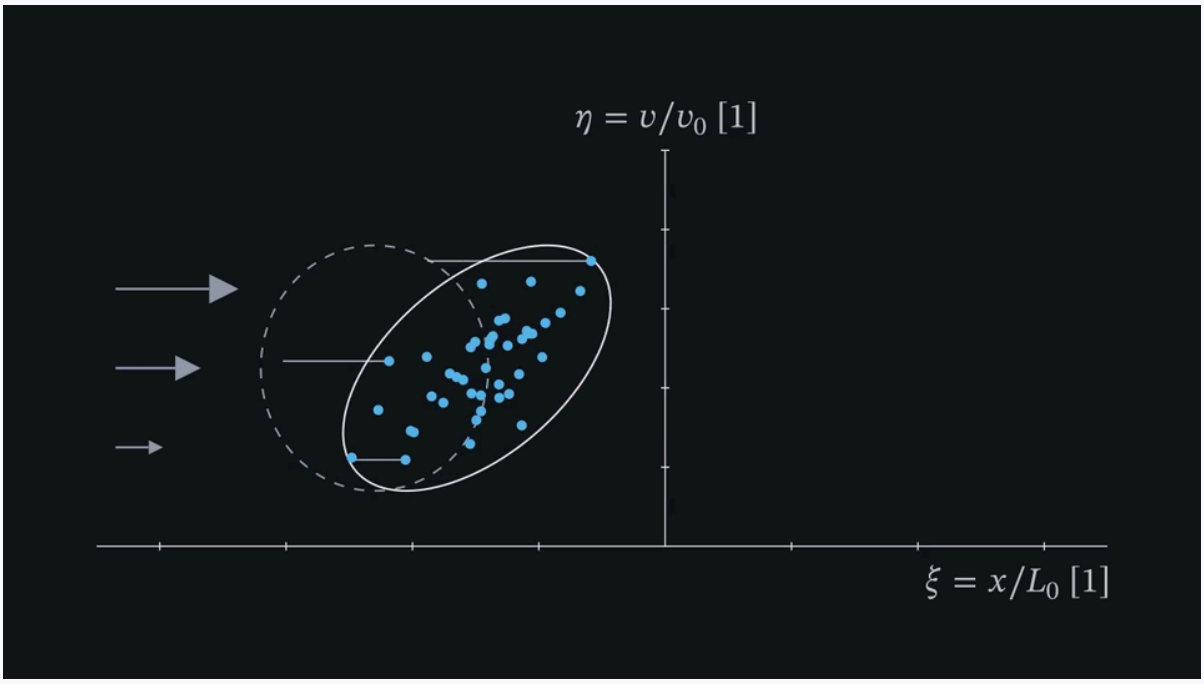
$$\frac{\partial f}{\partial t} + \boldsymbol{\nabla} \cdot (f\boldsymbol{v}) + \boldsymbol{\nabla} \cdot (f\boldsymbol{a}_s) = S.$$

Step: Combine position and velocity space

Define the phase-space velocity

$$\boldsymbol{V}_z = (\boldsymbol{v}, \boldsymbol{a}_s).$$

The two divergence terms are then the compact conservative flux of particles through phase space. This form is valid provided the source S represents all collisions, sources, and sinks not included in the characteristic flow.



Watch animation

Free streaming in one spatial and one velocity dimension. The visualization uses normalized variables and a deterministic sample of a distribution. Arrow lengths are proportional to $\eta = v/v_0$ and represent displacement over the same normalized time interval.

Figure 49: Free streaming in one spatial and one velocity dimension. The visualization uses normalized variables and a deterministic sample of a distribution. Arrow lengths are proportional to $\eta = v/v_0$ and represent displacement over the same normalized time interval.

Eulerian and Lagrangian pictures Interpretation 80

The Eulerian picture asks how the distribution at one fixed phase-space location changes. The Lagrangian picture follows a characteristic through phase space. In the free-streaming example, a particle keeps its velocity and moves horizontally in the position–velocity plane. The distribution at a fixed point can change as particles pass, even while its value along each characteristic remains constant.

Summary Summary 81

A convective derivative follows a particle through position and velocity. A conservative balance counts particles entering and leaving phase-space cells. Relating the two forms requires accounting for any compression of the characteristic flow. The free-streaming animation isolates spatial transport: velocities stay fixed and the cloud shears because faster particles travel farther. Acceleration would also move particles through velocity-space cell faces.

Exam questions

(d) Sketch how the distribution function changes over time due to fluxes in phase-space (also collisions, see Fig. 3.3)

(e) What is a convective derivative in terms of Euler and Lagrange picture?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 82

- What does the spatial part of a phase-space flux transport?
- What does the velocity-space flux transport?
- When is the convective derivative of a distribution zero?
- Why does free streaming shear a phase-space cloud?

Answers

What does the spatial part of a phase-space flux transport?
The flux $f\boldsymbol{v}$ transports particles across position-space cell faces at their physical velocity.

What does the velocity-space flux transport?
The flux $f\boldsymbol{a}$ transports particles across velocity-space cell faces as forces change their velocities.

When is the convective derivative of a distribution zero?
It is zero without collisions or sources when the phase-space flow is incompressible, as for Lorentz motion. In general,

$$\frac{\mathrm{D}f}{\mathrm{D}t} = -f(\boldsymbol{\nabla}\cdot_{\boldsymbol{r}}(\boldsymbol{v}) + \boldsymbol{\nabla}\cdot_{\boldsymbol{v}}(\boldsymbol{a}))$$

along a source-free characteristic; velocity-dependent drag can compress phase space.

Why does free streaming shear a phase-space cloud?
Each sample keeps its velocity but advances by a distance proportional to that velocity. The faster part therefore moves farther in position during the same normalized time.

6.4 The Boltzmann and Vlasov equations

The phase-space balance becomes a plasma kinetic equation after the force and collision terms are specified. The conservative and convective forms carry the same particle balance when the phase-space flow is incompressible.

Learning objectives

- write the Boltzmann equation for a species
- transform its conservative form into convective form
- state the phase-space incompressibility condition for Lorentz motion
- distinguish the collisional Boltzmann equation from the collisionless Vlasov equation

Unit ledger

The collision operator has units of f_s per second, so it is written as s^{-1} times a distribution density.

From discrete particles to a distribution

Complete classical particle–field model

At the most complete classical level, every particle trajectory and both electromagnetic fields are evolved self-consistently. In each particle's force, its singular self-field is excluded or regularized; radiation reaction is neglected in this nonrelativistic model:

$$\frac{\mathrm{d}\boldsymbol{r}_a(t)}{\mathrm{d}t} = \boldsymbol{v}_a(t)$$

$$m_a \frac{\mathrm{d}\boldsymbol{v}_a(t)}{\mathrm{d}t} = q_a[\boldsymbol{E}(\boldsymbol{r}_a(t), t) + \boldsymbol{v}_a(t) \times \boldsymbol{B}(\boldsymbol{r}_a(t), t)]$$

$$\rho_q(\boldsymbol{r}, t) = \sum_a q_a \delta(\boldsymbol{r} - \boldsymbol{r}_a(t)), \quad \boldsymbol{j}(\boldsymbol{r}, t) = \sum_a q_a \boldsymbol{v}_a(t) \delta(\boldsymbol{r} - \boldsymbol{r}_a(t))$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{E}) = \frac{\rho_q}{\varepsilon_0}, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$$

$$\boldsymbol{\nabla} \times (\boldsymbol{E}) = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \boldsymbol{\nabla} \times (\boldsymbol{B}) = \mu_0 \boldsymbol{j} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$$

The discrete particle sources are coarse-grained when one passes to a kinetic distribution or to fluid moments. This is the sense in which the model is complete: all charged particles interact through the shared electromagnetic fields, subject to the classical and nonrelativistic assumptions stated here.

The electromagnetic acceleration follows from the Lorentz force:

$$\boldsymbol{a}_s = \frac{q_s}{m_s}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

The acceleration is in $\mathrm{m\,s^{-2}}$. The magnetic term changes the direction of velocity and the electric term can change its magnitude.

Let $C_s[f]$ denote the velocity-space redistribution caused by collisions. The conservative kinetic equation is

$$\frac{\partial f_s}{\partial t} + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) = C_s[f]$$

The first divergence acts on position at fixed velocity; the second acts on velocity at fixed position. This convention also applies to the expanded fluxes below. The equation is a number balance in phase space. The source $C_s[f]$ conserves the particle number of species s for ordinary binary collisions.

Expanding the two fluxes gives

$$\boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) = \boldsymbol{v} \cdot \boldsymbol{\nabla}(f_s) + f_s \boldsymbol{\nabla} \cdot (\boldsymbol{v}),$$

$$\boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) = \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} + f_s \boldsymbol{\nabla} \cdot (\boldsymbol{a}_s).$$

Position and velocity are independent phase-space coordinates, so the spatial divergence of $\boldsymbol{v}$ vanishes. The Lorentz acceleration does depend on velocity, but its velocity-space divergence vanishes: the electric part translates velocities and the magnetic part rotates them without compressing velocity-space volume. Thus

$$\boldsymbol{\nabla} \cdot (\boldsymbol{v}) = 0, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{a}_s) = 0.$$

These are the incompressibility conditions for the six-dimensional characteristic flow. They are statements about phase-space volume, not about the spatial compression of a fluid density.

The convective form is therefore

$$\frac{\mathrm{D}f_s}{\mathrm{D}t} = \frac{\partial f_s}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(f_s) + \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} = C_s[f]$$

When the collision operator is neglected, this becomes the Vlasov equation:

$$\frac{\partial f_s}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(f_s) + \frac{q_s}{m_s}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} = 0$$

The Vlasov equation is collisionless, not force-free. Self-consistent electromagnetic fields can still accelerate particles and bend their trajectories. The omitted collisions must be small on the process time scale.

Derivation: conservative form to convective form

Step: Start from the conservative equation

Use

$$\frac{\partial f_s}{\partial t} + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) = C_s[f].$$

Step: Apply the product rule

Expanding both flux divergences gives

$$\frac{\partial f_s}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(f_s) + \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} + f_s \cdot [\boldsymbol{\nabla} \cdot (\boldsymbol{v}) + \boldsymbol{\nabla} \cdot (\boldsymbol{a}_s)] = C_s[f].$$

Step: Use incompressibility of Lorentz characteristics

For Lorentz motion, $\boldsymbol{\nabla} \cdot (\boldsymbol{v}) = 0$ because $\boldsymbol{v}$ is an independent velocity coordinate in the spatial divergence. Also $\boldsymbol{\nabla} \cdot (\boldsymbol{a}_s) = 0$: the electric acceleration is velocity independent, while the magnetic acceleration is linear in $\boldsymbol{v}$ with an antisymmetric cross-product matrix of zero trace.

Removing the vanishing bracket gives the convective form. The equivalence relies on the zero divergence of the phase-space characteristic flow.

Derivation: particle-number conservation

Step: Integrate over velocity space

Integrate the conservative equation over all velocity space. Assume f_s and the velocity-space flux vanish as $|\boldsymbol{v}|$ tends to infinity. The velocity-space divergence becomes a boundary term and vanishes.

Step: Use the collision invariant

Number-conserving binary collisions obey

$$\int C_s[f] \mathrm{d}^3\boldsymbol{v} = 0.$$

Step: Identify the fluid density and flux

The remaining velocity integrals are

$$\int f_s \mathrm{d}^3\boldsymbol{v} = n_s, \quad \int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} = n_s \boldsymbol{u}_s.$$

Therefore

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0.$$

This species continuity equation is the first bridge from kinetic theory to fluid theory. It assumes that no particles are created or removed by the selected collision model.

Collisional redistribution in velocity space

Interpretation 83

Binary elastic collisions redistribute particles in velocity space. They can relax a non-equilibrium distribution toward a Maxwellian while conserving species particle number. Momentum and energy conservation are imposed on the combined interacting species, not necessarily on each species separately.

Summary

Summary 84

The Boltzmann equation is a phase-space conservation law with a collision operator. Lorentz characteristics are incompressible in phase space, so the conservative equation and the convective equation are equivalent. The Vlasov equation is the collisionless limit and still contains the full electromagnetic acceleration.

Exam questions

(f) Write down and explain the convective and conservative variant of the plasma kinetic (Boltzmann) equation. Why can one transform one into the other?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 85

- What condition makes the two kinetic forms equivalent?
- What is the Vlasov approximation?
- Why must $C_s[f]$ integrate to zero over velocity for ordinary collisions?
- What is the first fluid equation obtained by integrating the kinetic equation?

Answers

What condition makes the two kinetic forms equivalent?

The phase-space characteristic flow must have zero divergence, $\boldsymbol{\nabla} \cdot (\boldsymbol{v}) + \boldsymbol{\nabla} \cdot (\boldsymbol{a}) = 0$. Lorentz motion satisfies this condition.

What is the Vlasov approximation?

It neglects the collisional redistribution term on the chosen time and length scales while retaining self-consistent electromagnetic forces.

Why must $C_s[f]$ integrate to zero over velocity for ordinary collisions?

Collisions rearrange the velocities of species- s particles, so they do not create or destroy the number of particles of that species.

What is the first fluid equation obtained by integrating the kinetic equation?

The species continuity equation, $\partial n_s/\partial t + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0$, provided the velocity-space boundary flux vanishes and collisions conserve particle number.

6.5 Maxwellian equilibrium and kinetic model limits

The Maxwellian is a reference state. Its use is justified by collisional relaxation and the absence of unresolved sources that sustain non-equilibrium structure.

Learning objectives

- state the conditions under which a Maxwellian is a collisional equilibrium
- derive the density response to a static electrostatic potential
- select between kinetic and fluid descriptions using scale orderings
- connect the kinetic description to the moment hierarchy of the next chapter

Unit ledger

The ratios $\nu\tau$, K_n, and λ_D/L are dimensionless.

Static isothermal equilibrium

Assumptions 86

Take a time-independent electrostatic potential, uniform temperature, no bulk flow, and a collision operator whose stationary state is the Maxwellian. For full multispecies collisional equilibrium the temperature is common to all species; distinct species temperatures describe only self-collision equilibrium or an approximation neglecting interspecies energy exchange. The equilibrium is local in position and isotropic in the local rest frame.

Let $n_{0,s}$ be the species density where the reference potential is $\Phi = 0$. A species in this equilibrium has the energy-dependent distribution

$$f_{\mathrm{eq,s}}(\boldsymbol{r},\boldsymbol{v}) = n_{0,s}\left(\frac{m_s}{2\pi k_B T_s}\right)^{\frac{3}{2}} \exp\left(-\frac{\frac{m_s v^2}{2} + q_s\Phi(\boldsymbol{r})}{k_B T_s}\right)$$

Integrating over velocity gives the Boltzmann density response:

$$n_s(\boldsymbol{r}) = n_{0,s}\exp\left(-\frac{q_s\Phi(\boldsymbol{r})}{k_B T_s}\right)$$

The exponent is dimensionless because both $q_s\Phi$ and $k_B T_s$ are energies in J. For electrons, $q_e < 0$, so a positive potential increases the electron density in this convention.

Derivation: Maxwellian equilibrium in a potential

Step: Use the single-particle invariant

In a static electrostatic field, the single-particle energy is

$$H_s = \frac{m_s v^2}{2} + q_s\Phi(\boldsymbol{r}).$$

A stationary collisional equilibrium depends on velocity through this invariant and has the Maxwellian form

$$f_{\mathrm{eq,s}} = A_s \exp\left(-\frac{H_s}{k_B T_s}\right).$$

Step: Choose the normalization

Choose

$$A_s = n_{0,s}\left(\frac{m_s}{2\pi k_B T_s}\right)^{\frac{3}{2}}.$$

At $\Phi = 0$, the velocity integral is then $n_{0,s}$.

Step: Separate velocity and position

Factor the exponential into a velocity part and a position part. The normalized Gaussian integrates to one, leaving

$$n_s(\boldsymbol{r}) = n_{0,s}\exp\left(-\frac{q_s\Phi}{k_B T_s}\right).$$

The result assumes a static potential, a spatially uniform temperature, and an equilibrium that is Maxwellian in the conserved single-particle energy.

Collision frequency is a model selector Key idea 87

For a process with characteristic time τ and length L, use $\nu\tau$ and $K_{\mathrm{n}} = \lambda_{\mathrm{mfp}}/L$ together. A small $\nu\tau$ permits collisionless kinetic behavior. A small K_{n} supports local fluid moments. These criteria answer different questions and can select hybrid models in which some directions or species are collisional while others are kinetic.

The kinetic description remains necessary when the distribution contains velocity-space beams, anisotropic tails, resonant structure, or boundary layers whose width is comparable to a mean free path. A fluid model becomes efficient when a small set of moments evolves slowly and a closure can represent the discarded velocity dependence. The next chapter performs the first velocity moments and makes the closure issue explicit.

Departures from equilibrium Interpretation 88

Equilibrium gives a useful reference distribution and a way to test a collision model. Perturbations, boundaries, waves, and weakly collisional transport can carry information in the parts of f_s that a Maxwellian does not contain.

Classical statistics and quantum degeneracy

Classical statistics also have a validity boundary. For electrons, define the degeneracy parameter $\theta_e = k_B T_e / E_{F,e}$, with the nonrelativistic Fermi energy

$$E_{F,e} = \left(\frac{\hbar^2}{2m_e}\right)\left(3\pi^2 n_e\right)^{\frac{2}{3}}$$

$E_{F,e}$ and $k_B T_e$ are energies in J, and θ_e is dimensionless. The classical limit has $\theta_e \gg 1$; $\theta_e \leq 1$ signals quantum degeneracy. White-dwarf interiors and dense laser-compressed matter are representative settings.

Summary Summary 89

Collisions drive elastic systems toward Maxwellian velocity distributions when sources and boundaries do not sustain non-equilibrium structure. A static electrostatic potential changes the density through the Boltzmann factor. The ratios $\nu\tau$ and K_{n} determine which kinetic information a reduced model may safely discard.

Exam questions

(h) What is a quantum degenerate plasma and where does it appear?

(m) What is a Boltzmann distribution and how is it related to equilibrium states?

Verbatim from Plasma Physics Exam.pdf, p. 1

Knowledge check

- What makes the Maxwellian a stationary collisional state?
- Why is $q_s\Phi/(k_B T_s)$ dimensionless?
- Which ordering supports a local fluid closure?
- Give one example of kinetic information a Maxwellian cannot represent.

Answers

What makes the Maxwellian a stationary collisional state?

The elastic self-collision operator vanishes on a local Maxwellian. Full multispecies collisional equilibrium also requires common mean flow and temperature; otherwise interspecies collisions exchange momentum or energy even between Maxwellian species.

Why is $q_s\Phi/(k_B T_s)$ dimensionless?

Both the numerator and denominator are energies in J, so their ratio has no units.

Which ordering supports a local fluid closure?

A small Knudsen number, $K_{\mathrm{n}} = \lambda_{\mathrm{mfp}}/L \ll 1$, supports frequent local collisions over the macroscopic scale. A closure is still required for the moment hierarchy.

Give one example of kinetic information a Maxwellian cannot represent.

A beam, loss-cone, anisotropic tail, or resonant velocity-space structure requires information beyond density, flow, and temperature.

7. Moments of the Boltzmann equation

Many plasma problems concern density, flow, or energy transport rather than the full velocity distribution. Velocity moments turn the kinetic equation from Chapter 6 into evolution equations for these fluid fields. The reduction remains incomplete: the transport equation for one moment usually introduces a higher moment, so a finite model needs a closure.

Moment models and closure Key idea 91

A moment model is obtained by integrating the kinetic equation against a chosen weight, keeping the resulting conservation laws, and then stating how the first omitted moment is represented. The closure is part of the model and must be visible.

7.1 Velocity moments and fluid variables

Which macroscopic quantities can be computed from a one-particle distribution? Start with the number of particles in a velocity-space element, then use increasingly informative velocity weights.

Learning objectives

- define number, mass, charge, and current densities as moments of f_s
- compute the species fluid velocity from the first moment
- construct the pressure tensor from the random velocity
- identify which information is discarded at each moment order

Unit ledger

Mass density is in $\mathrm{kg\,m^{-3}}$, momentum density is in $\mathrm{kg\,m^{-2}\,s^{-1}}$, and pressure is in $\mathrm{Pa} = \mathrm{N\,m^{-2}} = \mathrm{kg\,m^{-1}\,s^{-2}}$. The symbols $\boldsymbol{u}_s$, $\boldsymbol{P}_s$, and $\boldsymbol{q}_s$ below denote velocity, pressure tensor, and heat-flux vector, respectively. The bold heat-flux symbol $\boldsymbol{q}_s$ is distinct from the scalar particle charge q_s.

Distribution and velocity moment Definition 92

For species s, the phase-space distribution is defined by $dN_s = f_s(t, \boldsymbol{r}, \boldsymbol{v})\mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$. For a velocity weight $A(\boldsymbol{v})$, its local velocity moment is

$$\int A(\boldsymbol{v}) f_s \mathrm{d}^3\boldsymbol{v}.$$

The zeroth moment counts particles, while the weight determines which physical average is retained.

At fixed position, integration over velocity gives the species number density:

$$n_s = \int_{\mathbb{R}^3} f_s \mathrm{d}^3\boldsymbol{v}$$

The corresponding mass density, charge density, and current density are

$$\rho_s = m_s n_s, \quad \rho_q = \sum_s q_s n_s, \quad \boldsymbol{j}_s = q_s \int_{\mathbb{R}^3} \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}$$

$$\boldsymbol{j} = \sum_s \boldsymbol{j}_s$$

The charge density ρ_q is in C m^{-3}. The current density $\boldsymbol{j}$ is the charge-weighted particle flux. Because both are sums over species, an electrically neutral plasma can have $\rho_q \approx 0$ while carrying a nonzero current.

The first velocity moment is the particle flux. Dividing it by number density defines the species fluid velocity:

$$\boldsymbol{u}_s = \left(\frac{1}{n_s}\right) \int_{\mathbb{R}^3} \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}, \quad \boldsymbol{j}_s = q_s n_s \boldsymbol{u}_s$$

Introduce the random velocity relative to this local mean,

$$\boldsymbol{w}_s = \boldsymbol{v} - \boldsymbol{u}_s, \quad \int_{\mathbb{R}^3} \boldsymbol{w}_s f_s \mathrm{d}^3\boldsymbol{v} = \boldsymbol{0}$$

The second central moment is the pressure tensor:

$$\boldsymbol{P}_s = m_s \int_{\mathbb{R}^3} \boldsymbol{w}_s \boldsymbol{w}_s f_s \mathrm{d}^3\boldsymbol{v}$$

The product $\boldsymbol{w}_s \boldsymbol{w}_s$ is dyadic, so $\boldsymbol{P}_s$ is a rank-two tensor. Its diagonal entries are normal momentum fluxes and its off-diagonal entries are shear momentum fluxes. In an isotropic state the tensor reduces to a scalar pressure times the identity, but isotropy is an additional assumption, not a consequence of taking a moment.

"Raw" moments use the laboratory velocity $\boldsymbol{v}$; "central" moments use the velocity relative to the local flow. The raw second moment therefore contains both directed flow and random motion:

$$\boldsymbol{M}_s = m_s \int_{\mathbb{R}^3} \boldsymbol{v}\boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} = \boldsymbol{P}_s + \rho_s \boldsymbol{u}_s \boldsymbol{u}_s$$

Derivation: central and raw second moments

Step: Split velocity into bulk and random parts

Insert

$$\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{w}_s$$

into the raw second moment. The dyadic expansion is

$$\boldsymbol{v}\boldsymbol{v} = \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{u}_s \boldsymbol{w}_s + \boldsymbol{w}_s \boldsymbol{u}_s + \boldsymbol{w}_s \boldsymbol{w}_s.$$

Step: Use the definition of the local mean

By construction,

$$\int \boldsymbol{w}_s f_s \mathrm{d}^3\boldsymbol{v} = \boldsymbol{0}.$$

The two cross terms therefore vanish. The bulk term gives $n_s \boldsymbol{u}_s \boldsymbol{u}_s$, which becomes $\rho_s \boldsymbol{u}_s \boldsymbol{u}_s$ after multiplication by m_s.

Step: Identify the pressure tensor

The remaining central term is the pressure tensor:

$$\boldsymbol{P}_s = m_s \int \boldsymbol{w}_s \boldsymbol{w}_s f_s \mathrm{d}^3 \boldsymbol{v}.$$

Consequently,

$$\boldsymbol{M}_s = \boldsymbol{P}_s + \rho_s \boldsymbol{u}_s \boldsymbol{u}_s.$$

Both terms are momentum fluxes with units $\mathrm{Pa} = \mathrm{kg\,m^{-1}\,s^{-2}}$. The central tensor vanishes for a cold delta-like velocity distribution.

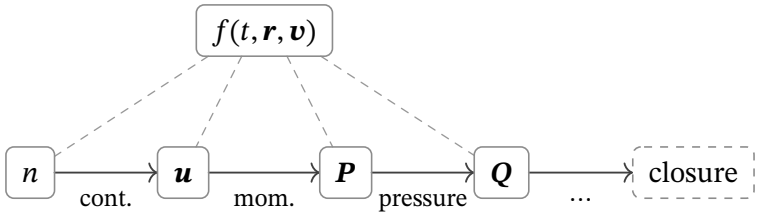
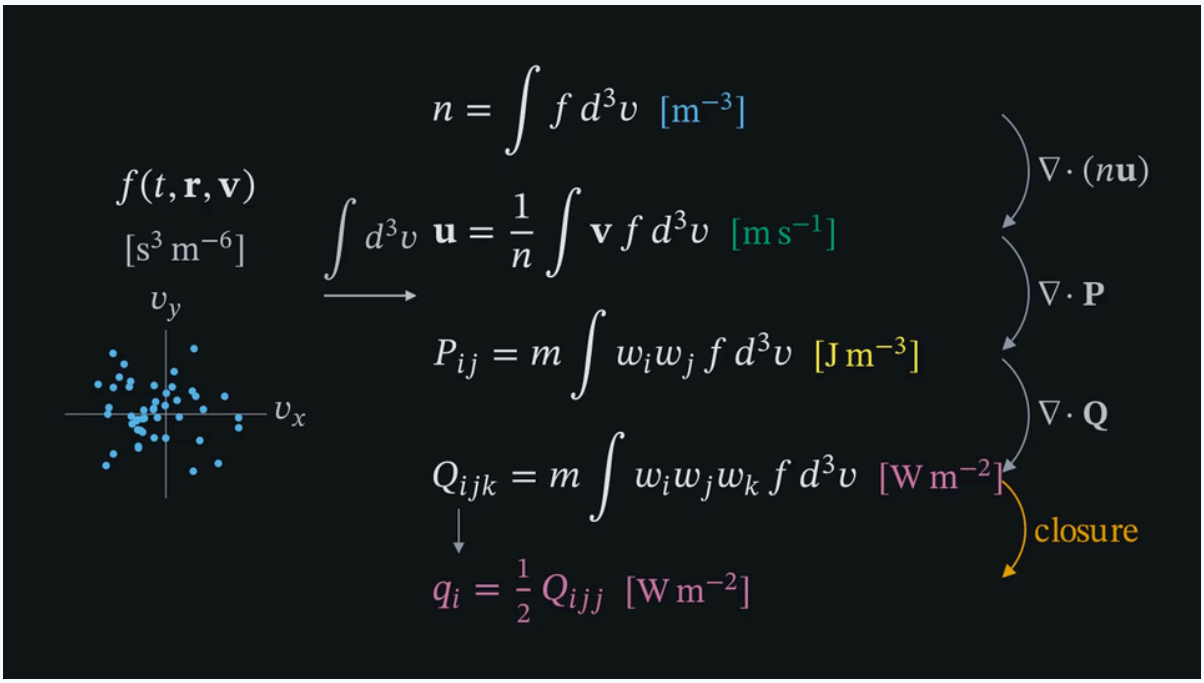


Figure 50: Each moment equation contains the next moment. The equation for n contains $\boldsymbol{u}$, the one for $\boldsymbol{u}$ contains $\boldsymbol{P}$, and the one for $\boldsymbol{P}$ contains the third central tensor $\boldsymbol{Q}$, whose contraction is the heat flux $\boldsymbol{q}$. A fluid model cuts the chain with a closure. Dashed lines: moments of f.



Watch animation

Moment hierarchy from kinetic information to fluid fields. The animation labels dimensional quantities. The full third central tensor Q_{ijk} contains more information than heat flux $q_i = (1/2)\sum_j Q_{ijj}$.

Figure 51: Moment hierarchy from kinetic information to fluid fields. The animation labels dimensional quantities. The full third central tensor Q_{ijk} contains more information than heat flux $q_i = (1/2)\sum_j Q_{ijj}$.

Information discarded at each moment order — Interpretation 93

The density forgets how particles are distributed in velocity. The flow velocity restores the first directional average. The pressure tensor retains the covariance of random motion, including anisotropy and shear. Higher moments retain asymmetric tails and energy transport. Two distributions can therefore have the same n_s, $\boldsymbol{u}_s$, and $\boldsymbol{P}_s$ while carrying different heat fluxes or kinetic instabilities.

Summary — Summary 94

Integrating f_s over velocity gives number density. Multiplying by mass or charge gives mass or charge density, and weighting once gives the fluid velocity and current. The central second moment is the pressure tensor. These are exact definitions. Replacing the tensor or the higher moments by simpler forms is a later closure assumption.

Exam questions

(a) How are particle density, mass density and charge density computed from a given (one-particle) distribution function f(t,x,v)?

(b) How are species fluid velocity and pressure tensor computed starting from a kinetic description?

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check

Knowledge check 95

- Which velocity weight produces the number density?
- Why can a neutral plasma carry current?
- What is removed when the pressure tensor is replaced by a scalar pressure?
- How would a cold distribution appear in the central second moment?

Answers

Which velocity weight produces the number density?

The zeroth weight, $A = 1$, gives

$$n_s = \int f_s \mathrm{d}^3\boldsymbol{v}$$

in m^{-3}.

Why can a neutral plasma carry current?

Neutrality constrains the charge-weighted density $\rho_q = \sum_s q_s n_s$, whereas current depends on the charge-weighted flow $\boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$. Opposite species flows can cancel charge while adding current.

What is removed when the pressure tensor is replaced by a scalar pressure?

Directional information is removed. The replacement keeps only

$$p_s = \frac{\sum_i (P_s)_{ii}}{3}$$

and discards anisotropic normal stresses and off-diagonal shear stresses.

How would a cold distribution appear in the central second moment?

If every particle has the local velocity $\boldsymbol{u}_s$, then $\boldsymbol{w}_s = \boldsymbol{0}$ and $\boldsymbol{P}_s = \boldsymbol{0}$. Finite spread in velocity produces nonzero pressure.

7.2 Zeroth moment and continuity

A fluid density changes only when particles cross a spatial boundary or when the model creates or removes particles. The zeroth velocity moment of the kinetic equation turns that statement into a local continuity law.

Learning objectives

- integrate the kinetic equation over all velocity space
- identify the vanishing velocity-space and collision contributions
- derive species number, mass, and charge continuity equations
- rewrite continuity in Eulerian and fluid-following forms

Unit ledger
The continuity equation balances a density in m^{-3} per s with the divergence of a flux in $\mathrm{m}^{-2}\,\mathrm{s}^{-1}$. The same balance multiplied by mass or charge gives mass or charge continuity.

Velocity-space boundary and particle conservation — Assumptions 96

Assume f_s and the velocity-space flux vanish sufficiently rapidly as $|\boldsymbol{v}| \to \infty$. Assume ordinary collisions do not create or destroy particles of species s, so

$$\int C_s[f]\mathrm{d}^3\boldsymbol{v} = 0.$$

Ionization, recombination, or an imposed source would add a nonzero right-hand side and must be stated separately.

Start from the conservative kinetic equation for the species. As in Chapter 6, the streaming divergence acts on position and the acceleration divergence acts on velocity:

$$\frac{\partial f_s}{\partial t} + \boldsymbol{\nabla}\cdot(f_s\boldsymbol{v}) + \boldsymbol{\nabla}\cdot(f_s\boldsymbol{a}_s) = C_s[f]$$

Integrate it over all velocity space. The result is

$$\int_{\mathbb{R}^3}\frac{\partial f_s}{\partial t}\mathrm{d}^3\boldsymbol{v} + \int_{\mathbb{R}^3}\boldsymbol{\nabla}\cdot(f_s\boldsymbol{v})\mathrm{d}^3\boldsymbol{v} + \int_{\mathbb{R}^3}\boldsymbol{\nabla}\cdot(f_s\boldsymbol{a}_s)\mathrm{d}^3\boldsymbol{v} = \int_{\mathbb{R}^3} C_s[f]\mathrm{d}^3\boldsymbol{v}$$

The time derivative passes through the velocity integral, and the spatial divergence does too because velocity is an independent integration variable:

$$\int_{\mathbb{R}^3}\frac{\partial f_s}{\partial t}\mathrm{d}^3\boldsymbol{v} = \frac{\partial n_s}{\partial t}, \quad \int_{\mathbb{R}^3}\boldsymbol{\nabla}\cdot(f_s\boldsymbol{v})\mathrm{d}^3\boldsymbol{v} = \boldsymbol{\nabla}\cdot(n_s\boldsymbol{u}_s)$$

The velocity-space term is a surface flux at infinity and vanishes under the stated boundary assumption. The collision integral vanishes by particle conservation. Thus the species continuity equation is

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla}\cdot(n_s\boldsymbol{u}_s) = 0$$

Multiplying by the constant species mass gives mass continuity:

$$\frac{\partial \rho_s}{\partial t} + \boldsymbol{\nabla}\cdot(\rho_s\boldsymbol{u}_s) = 0$$

Multiplying by charge and summing over species gives total charge continuity:

$$\frac{\partial \rho_q}{\partial t} + \boldsymbol{\nabla}\cdot(\boldsymbol{j}) = 0$$

The flux in the number equation is $n_s\boldsymbol{u}_s$ in $\mathrm{m}^{-2}\,\mathrm{s}^{-1}$. A source, sink, or ionization model would appear as S_s on the right-hand side. The source-free equation is local, so it applies even when the total number in a finite region changes by flux through its boundary.

Derivation: every term in the zeroth moment

Step: Integrate the temporal term

Integrate the kinetic equation over a velocity-space domain V_v and then let its boundary tend to infinity:

$$\int_{V_v} \frac{\partial f_s}{\partial t} \mathrm{d}^3\boldsymbol{v} = \frac{\partial \int_{V_v} f_s \mathrm{d}^3\boldsymbol{v}}{\partial t} \to \frac{\partial n_s}{\partial t}.$$

Step: Identify the spatial particle flux

Commute the spatial divergence with the velocity integral:

$$\int_{V_v} \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) \mathrm{d}^3\boldsymbol{v} = \boldsymbol{\nabla} \cdot \left(\int_{V_v} \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} \right) = \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s).$$

Step: Discard velocity-space and collision sources

The velocity-space divergence becomes a boundary integral:

$$\int_{V_v} \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) \mathrm{d}^3\boldsymbol{v} = \int_{\partial V_v} f_s \boldsymbol{a}_s \cdot d\boldsymbol{S}_v.$$

It vanishes when the distribution decays faster than the surface measure grows. Number-conserving collisions obey

$$\int_{\mathbb{R}^3} C_s[f] \mathrm{d}^3\boldsymbol{v} = 0.$$

Step: Assemble the continuity equation

Substituting the four terms yields

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0.$$

Integrating this local equation over a finite spatial volume gives a rate of change equal to the negative outward particle flux. The local and integral forms are therefore the same conservation law.

Rechenbeispiel

Rechenbeispiel 97

Context: a uniform singly ionized particle stream has $n_s = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, normal speed $u_s = 2.0 \cdot 10^5\,\mathrm{m\,s}^{-1}$, and reaches a planar collector of area $A = 1.0 \cdot 10^{-4}\,\mathrm{m}^2$.

Assumptions: steady source-free flow with the velocity normal to the collector.

Target: report the particle flux Γ_s and the collection rate $\dot{N}_s$.

Numerical result: $\Gamma_s = 2.0 \cdot 10^{21}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$ and $\dot{N}_s = 2.0 \cdot 10^{17}\,\mathrm{s}^{-1}$.

Eulerian reading of continuity Interpretation 98

The term $\partial n_s/\partial t$ measures accumulation at a fixed position. The divergence $\nabla \cdot (n_s \boldsymbol{u}_s)$ measures net outward particle flux. If the flow converges, the divergence is negative and the density increases. This is the divergence of particle flux, which depends on both density and velocity. Even an incompressible velocity field can transport a density gradient past a fixed observer.

Summary Summary 99

The zeroth moment of a number-conserving kinetic equation is continuity. Velocity-space force transport contributes only a boundary flux, and ordinary collisions rearrange velocities without changing species number. Multiplication by mass or charge produces the corresponding continuity laws. Any source process must be added explicitly.

Exam questions

(c) How do fluid equations follow from the Boltzmann equation? Perform the derivation for the continuity equation.

Verbatim from Plasma Physics Exam.pdf, p. 2

Knowledge check Knowledge check 100

- Which kinetic term becomes the particle flux in continuity?
- Why does the Lorentz force not create particles in the zeroth moment?
- What changes in continuity if ionization is included?
- What does a negative divergence of particle flux imply locally?

Answers

Which kinetic term becomes the particle flux in continuity?

The spatial streaming term becomes

$$\nabla \cdot \left(\int \boldsymbol{v} f_s \mathrm{d}^3 \boldsymbol{v} \right) = \nabla \cdot (n_s \boldsymbol{u}_s).$$

Why does the Lorentz force not create particles in the zeroth moment?

It transports the distribution through velocity space. Its divergence integrates to a boundary flux, which is zero when the distribution vanishes at infinite speed.

What changes in continuity if ionization is included?

A source term appears, for example $\partial n_s/\partial t + \nabla \cdot (n_s \boldsymbol{u}_s) = S_s$. The source has the same density-per-time units as the left-hand side.

What does a negative divergence of particle flux imply locally?

With no source, $\partial n_s/\partial t = -\nabla \cdot (n_s \boldsymbol{u}_s)$, so a negative flux divergence means positive local density accumulation.

7.3 First moment and momentum transport

Number continuity tracks how many particles occupy a region. Momentum transport additionally records how directed motion and random motion move across its boundary and how electromagnetic forces change the momentum.

Learning objectives

- derive the momentum equation by weighting the kinetic equation with $m_s\boldsymbol{v}$
- separate directed momentum flux from the pressure tensor
- evaluate the Lorentz force moment
- interpret the collision moment as interspecies momentum exchange

Unit ledger

The divergence of a pressure tensor and an electromagnetic force density are in $\mathrm{N\,m^{-3}} = \mathrm{kg\,m^{-2}\,s^{-2}}$. The Lorentz acceleration is $\boldsymbol{a}_s = (q_s/m_s)(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$.

Fields and velocity-space boundary — Assumptions 101

The prescribed electromagnetic fields depend on t and $\boldsymbol{r}$, not on the velocity integration variable. The distribution and weighted velocity-space flux vanish at $|\boldsymbol{v}| \to \infty$. Collisions may exchange momentum between species, so their first moment is retained as a source until species are summed.

Multiply the conservative kinetic equation by $m_s\boldsymbol{v}$ and integrate over velocity. The result is

$$m_s \int \boldsymbol{v} \frac{\partial f_s}{\partial t} \mathrm{d}^3\boldsymbol{v} + m_s \int \boldsymbol{v} \boldsymbol{\nabla} \cdot (\boldsymbol{v} f_s) \mathrm{d}^3\boldsymbol{v} + m_s \int \boldsymbol{v} \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) \mathrm{d}^3\boldsymbol{v} = m_s \int \boldsymbol{v} C_s[f] \mathrm{d}^3\boldsymbol{v}$$

Define the collision momentum source

$$\boldsymbol{R}_s = m_s \int_{\mathbb{R}^3} \boldsymbol{v} C_s[f] \mathrm{d}^3\boldsymbol{v}$$

The time term is the time derivative of momentum density:

$$m_s \int \boldsymbol{v} \frac{\partial f_s}{\partial t} \mathrm{d}^3\boldsymbol{v} = \frac{\partial \rho_s \boldsymbol{u}_s}{\partial t}$$

For the spatial term, move the divergence outside the velocity integral:

$$m_s \int \boldsymbol{v} \boldsymbol{\nabla} \cdot (\boldsymbol{v} f_s) \mathrm{d}^3\boldsymbol{v} = \boldsymbol{\nabla} \cdot \left(m_s \int \boldsymbol{v}\boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} \right) = \boldsymbol{\nabla} \cdot (\boldsymbol{M}_s)$$

Use the decomposition of the raw second moment from the first section:

$$\boldsymbol{\nabla} \cdot (\boldsymbol{M}_s) = \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s)$$

Integration by parts transfers the velocity derivative from the force flux to the weight $m_s\boldsymbol{v}$; the assumed boundary decay removes the surface term. In components, with repeated Cartesian indices summed and δ_{ij} equal to one for $i = j$ and zero otherwise,

$$m_s \int v_i \frac{\partial (a_{s,j} f_s)}{\partial v_j} \mathrm{d}^3\boldsymbol{v} = -m_s \int \delta_{ij} a_{s,j} f_s \mathrm{d}^3\boldsymbol{v} = -m_s \int a_{s,i} f_s \mathrm{d}^3\boldsymbol{v}$$

Therefore the force contribution on the right-hand side of the momentum balance is

$$m_s \int \boldsymbol{a}_s f_s \mathrm{d}^3\boldsymbol{v} = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B})$$

The species momentum equation is consequently

$$\frac{\partial \rho_s \boldsymbol{u}_s}{\partial t} + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) + \boldsymbol{R}_s$$

The pressure force density is $-\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s)$. In the fluid-following form, obtained with species continuity, the equation reads

$$\rho_s \left(\frac{\partial \boldsymbol{u}_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(\boldsymbol{u}_s) \right) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) - \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) + \boldsymbol{R}_s.$$

Every term has force-density units in $\mathrm{N\,m^{-3}}$. The acceleration now follows a fluid element moving at $\boldsymbol{u}_s$, not an individual particle moving at $\boldsymbol{v}$. Pressure accounts for the momentum transport caused by their velocity difference.

Derivation: the first moment term by term

Step: Evaluate the time and streaming terms

Commute the time derivative with the velocity integral and use

$$m_s \int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} = \rho_s \boldsymbol{u}_s.$$

For spatial streaming, the component identity is

$$m_s \int v_i \frac{\partial (v_j f_s)}{\partial x_j} \mathrm{d}^3\boldsymbol{v} = \frac{\partial \left(m_s \int v_i v_j f_s \mathrm{d}^3\boldsymbol{v} \right)}{\partial x_j}.$$

This is the divergence of the raw second-moment tensor.

Step: Integrate the force term by parts

For each component,

$$m_s \int v_i \frac{\partial (a_{s,j} f_s)}{\partial v_j} \mathrm{d}^3\boldsymbol{v} = m_s \int_{\text{boundary}} v_i a_{s,j} f_s dS_j - m_s \int \frac{\partial v_i}{\partial v_j} a_{s,j} f_s \mathrm{d}^3\boldsymbol{v}.$$

The surface term vanishes and $\partial v_i / \partial v_j = \delta_{ij}$. The force term is therefore

$$-m_s \int a_{s,i} f_s \mathrm{d}^3\boldsymbol{v}$$

on the left.

Step: Insert the Lorentz acceleration

Move the force term to the right. Substitution of the Lorentz acceleration gives

$$q_s \boldsymbol{E} \int f_s \mathrm{d}^3\boldsymbol{v} + q_s \left(\int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} \right) \times \boldsymbol{B} = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}).$$

Step: Split the momentum flux and use continuity

Insert $\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{w}_s$ into the raw second moment. The terms linear in $\boldsymbol{w}_s$ vanish, leaving

$$\boldsymbol{M}_s = \rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s.$$

Finally use

$$\frac{\partial \rho_s \boldsymbol{u}_s}{\partial t} + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s \boldsymbol{u}_s) = \rho_s \left(\frac{\partial \boldsymbol{u}_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(\boldsymbol{u}_s) \right) + \boldsymbol{u}_s \left(\frac{\partial \rho_s}{\partial t} + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s) \right).$$

The final bracket is zero by mass continuity, so the conservative equation becomes the species material-acceleration equation.

Rechenbeispiel Rechenbeispiel 102

Context: a uniform singly ionized hydrogen plasma has $n_i = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $q_i = e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, and an electric field $E = 6.00 \cdot 10^{4}\,\mathrm{V\,m}^{-1}$ along x.

Assumptions: neglect pressure gradients and magnetic forces, and use the electrostatic force density $f_{E,x} = n_i q_i E_x$.

Target: report the x-directed electric force density.

Numerical result: $f_{E,x} = 96.1\,\mathrm{N\,m}^{-3}$, equivalently $96.1\,\mathrm{kg\,m}^{-2}\,\mathrm{s}^{-2}$.

Pressure is a momentum flux Interpretation 103

Particles with random velocities carry momentum across a surface, and the imbalance of that flux produces $-\boldsymbol{\nabla} \cdot (\boldsymbol{P})$ in the momentum equation. Anisotropic velocity spread therefore produces a force that cannot generally be represented by the gradient of one scalar.

Summary Summary 104

Weighting the kinetic equation with $m_s \boldsymbol{v}$ gives momentum density, raw momentum flux, Lorentz force density, and the first collision moment. The raw flux splits exactly into directed transport and pressure tensor. The collision term is not automatically zero for one species because collisions can transfer momentum between species.

Knowledge check

- Which moment produces the pressure tensor in the momentum flux?
- Why does the magnetic part of the Lorentz force do no work but still affect momentum?
- When can the collision momentum sources cancel?
- What approximation turns the pressure force into a scalar gradient?

Answers

Which moment produces the pressure tensor in the momentum flux?
The raw second velocity moment produces $\boldsymbol{M}_s = \boldsymbol{P}_s + \rho_s \boldsymbol{u}_s \boldsymbol{u}_s$. The central part is the random momentum flux.

Why does the magnetic part of the Lorentz force do no work but still affect momentum?
It is perpendicular to the instantaneous velocity, so it does no kinetic-energy work. It is nevertheless a directional force and appears in the momentum source $q_s n_s(\boldsymbol{u}_s \times \boldsymbol{B})$.

When can the collision momentum sources cancel?
For internal binary collisions, summing over all participating species cancels equal and opposite momentum exchange. A single species equation generally retains $\boldsymbol{R}_s$.

What approximation turns the pressure force into a scalar gradient?
Assume isotropy, $\boldsymbol{P}_s = p_s \boldsymbol{I}$. Then $\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) = \boldsymbol{\nabla}(p_s)$. Without isotropy the full tensor divergence is required.

7.4 Second moment: energy and heat transport

Contracting the second moment to a scalar gives kinetic-energy density. Some of that energy is in the bulk flow and some is in random motion. Their combined flux contains transport by the flow, pressure work, and heat carried relative to the moving fluid.

Learning objectives

- derive the total kinetic-energy equation from the second moment
- decompose energy density and energy flux into bulk, pressure, and heat parts
- relate pressure-tensor divergence to scalar-pressure gradient under isotropy
- identify the next moment that prevents an exact finite fluid hierarchy

Unit ledger

Energy density is in $\mathrm{J\,m^{-3}} = \mathrm{kg\,m^{-1}\,s^{-2}}$, and energy flux is in $\mathrm{W\,m^{-2}} = \mathrm{kg\,s^{-3}}$. Heat flux $\boldsymbol{q}_s$ is an energy flux. The scalar pressure p_s has the same units as $\boldsymbol{P}_s$. The ratio $\gamma = 5/3$ below is dimensionless and applies to a three-dimensional monatomic closure.

Finite energy and decaying velocity tails

Assume the distribution and all weighted velocity-space boundary fluxes vanish at infinity. The electromagnetic fields are independent of velocity. The energy collision moment may remain nonzero for one species because collisions can exchange thermal energy, even though total energy is conserved for a closed set of interacting species.

Define the kinetic-energy density and its bulk and internal parts:

$$E_{\mathrm{kin},s} = \left(\frac{m_s}{2}\right)\int_{\mathbb{R}^3} |\boldsymbol{v}|^2 f_s \mathrm{d}^3\boldsymbol{v}, \quad \varepsilon_s = \frac{\sum_i (P_s)_{ii}}{2}, \quad E_{\mathrm{kin},s} = \frac{\rho_s |\boldsymbol{u}_s|^2}{2} + \varepsilon_s$$

Randomly moving particles also carry kinetic energy relative to the flow. Weighting their random velocity by that energy gives the heat-flux vector, a contraction of the mass-weighted third central tensor:

$$\boldsymbol{q}_s = \left(\frac{m_s}{2}\right)\int_{\mathbb{R}^3} |\boldsymbol{w}_s|^2 \boldsymbol{w}_s f_s \mathrm{d}^3\boldsymbol{v}$$

and the energy collision source is

$$Q_s = \left(\frac{m_s}{2}\right)\int_{\mathbb{R}^3} |\boldsymbol{v}|^2 C_s[f] \mathrm{d}^3\boldsymbol{v}$$

Weight the kinetic equation by $m_s|\boldsymbol{v}|^2/2$. The time term is $\partial E_{\mathrm{kin},s}/\partial t$. The spatial term is the divergence of the raw energy flux

$$\boldsymbol{F}_s = \left(\frac{m_s}{2}\right)\int_{\mathbb{R}^3} |\boldsymbol{v}|^2 \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}$$

Integration by parts of the force term gives

$$\left(\frac{m_s}{2}\right)\int |\boldsymbol{v}|^2 \boldsymbol{\nabla}\cdot(f_s \boldsymbol{a}_s)\mathrm{d}^3\boldsymbol{v} = -m_s \int \boldsymbol{v}\cdot\boldsymbol{a}_s f_s \mathrm{d}^3\boldsymbol{v}$$

For Lorentz acceleration, $\boldsymbol{v}\cdot(\boldsymbol{v}\times\boldsymbol{B}) = 0$, so the force moment is electric work only. The conservative energy equation is

$$\frac{\partial E_{\mathrm{kin},s}}{\partial t} + \boldsymbol{\nabla}\cdot(\boldsymbol{F}_s) = q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E} + Q_s$$

Decompose the energy flux with $\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{w}_s$:

$$\boldsymbol{F}_s = E_{\mathrm{kin},s}\boldsymbol{u}_s + \boldsymbol{P}_s\cdot\boldsymbol{u}_s + \boldsymbol{q}_s$$

Thus the total kinetic-energy equation is

$$\frac{\partial E_{\mathrm{kin},s}}{\partial t} + \boldsymbol{\nabla}\cdot\left(E_{\mathrm{kin},s}\boldsymbol{u}_s + \boldsymbol{P}_s\cdot\boldsymbol{u}_s + \boldsymbol{q}_s\right) = q_s n_s \boldsymbol{u}_s\cdot\boldsymbol{E} + Q_s$$

The magnetic field does no direct work because it is perpendicular to $\boldsymbol{v}$. The pressure work term is contained in $\boldsymbol{\nabla}\cdot(\boldsymbol{P}\cdot\boldsymbol{u})$. The heat flux is independent information: it transports random energy relative to the local bulk frame and is not fixed by n_s, $\boldsymbol{u}_s$, and $\boldsymbol{P}_s$ alone.

The pressure tensor has scalar pressure

$$p_s = \frac{\sum_i (P_s)_{ii}}{3}$$

as its isotropic part. If the distribution is isotropic in the local fluid frame, then

$$\boldsymbol{P}_s = p_s \boldsymbol{I}, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) = \boldsymbol{\nabla}(p_s)$$

In components, however,

$$(\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s))_i = \sum_j \frac{\partial (P_s)_{ij}}{\partial x_j}$$

so the scalar gradient is insufficient when anisotropy or shear is present.

Derivation: energy density and energy flux decomposition

Step: Weight the kinetic equation by particle energy

Multiply the kinetic equation by $m_s v^2/2$ and integrate. The time derivative gives $\partial E_{\mathrm{kin},s}/\partial t$. The spatial streaming term gives

$$\boldsymbol{\nabla} \cdot \left(\left(\frac{m_s}{2} \right) \int v^2 \boldsymbol{v} f_s \mathrm{d}^3 \boldsymbol{v} \right).$$

Step: Evaluate the force work

Integrating the force term by parts in velocity gives

$$\left(\frac{m_s}{2} \right) \int v^2 \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) \mathrm{d}^3 \boldsymbol{v} = -\left(\frac{m_s}{2} \right) \int \frac{\partial v^2}{\partial \boldsymbol{v}} \cdot \boldsymbol{a}_s f_s \mathrm{d}^3 \boldsymbol{v} = -m_s \int \boldsymbol{v} \cdot \boldsymbol{a}_s f_s \mathrm{d}^3 \boldsymbol{v}.$$

The magnetic force does no work because $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0$. The remaining electric term is

$$-q_s \boldsymbol{E} \cdot \int \boldsymbol{v} f_s \mathrm{d}^3 \boldsymbol{v}$$

on the left. Move it to the right and use

$$\int \boldsymbol{v} f_s \mathrm{d}^3 \boldsymbol{v} = n_s \boldsymbol{u}_s.$$

Step: Split the energy flux

Use

$$|\boldsymbol{v}|^2 = |\boldsymbol{u}_s|^2 + 2\boldsymbol{u}_s \cdot \boldsymbol{w}_s + |\boldsymbol{w}_s|^2, \quad \boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{w}_s.$$

The terms separate into bulk kinetic energy transported by $\boldsymbol{u}_s$, internal energy transported by $\boldsymbol{u}_s$, the pressure-work cross term

$$m_s \int (\boldsymbol{u}_s \cdot \boldsymbol{w}_s) \boldsymbol{w}_s f_s \mathrm{d}^3 \boldsymbol{v} = \boldsymbol{P}_s \cdot \boldsymbol{u}_s,$$

and the third central moment $\boldsymbol{q}_s$. Terms with one unpaired $\boldsymbol{w}_s$ vanish by the definition of the local mean. The total energy flux is

$$\boldsymbol{F}_s = E_{\mathrm{kin},s}\boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_s.$$

Step: Check the pressure divergence

In Cartesian components,

$$\left(\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s)\right)_i = \sum_j \frac{\partial (P_s)_{ij}}{\partial x_j}.$$

If $P_{ij} = p_s \delta_{ij}$, the sum reduces to $\partial p_s / \partial x_i$, which is $\boldsymbol{\nabla}(p_s)$. A scalar pressure is therefore a special isotropic closure.

Derivation: internal-energy balance

Step: Form the bulk kinetic-energy balance

Dot the momentum equation with $\boldsymbol{u}_s$ and use mass continuity. The bulk kinetic-energy balance is

$$\frac{\partial \frac{\rho_s |\boldsymbol{u}_s|^2}{2}}{\partial t} + \boldsymbol{\nabla} \cdot \left(\frac{\rho_s |\boldsymbol{u}_s|^2 \boldsymbol{u}_s}{2} \right) + \boldsymbol{u}_s \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) = q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E} + \boldsymbol{u}_s \cdot \boldsymbol{R}_s.$$

Step: Rewrite the pressure work

The colon denotes a double contraction: multiply corresponding tensor components and sum both indices. Use the product identity

$$\boldsymbol{u}_s \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) = \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s \cdot \boldsymbol{u}_s) - \boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s).$$

Step: Subtract from the total energy balance

Subtract the bulk equation from the total energy equation. The electric work cancels, while collisional momentum transfer contributes to the internal-energy source:

$$\frac{\partial \varepsilon_s}{\partial t} + \boldsymbol{\nabla} \cdot (\varepsilon_s \boldsymbol{u}_s + \boldsymbol{q}_s) + \boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) = Q_s - \boldsymbol{u}_s \cdot \boldsymbol{R}_s.$$

Step: Apply the isotropic adiabatic closure

For an isotropic three-dimensional pressure tensor,

$$\varepsilon_s = \frac{3 p_s}{2}, \quad \boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) = p_s \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s).$$

With no heat flux and no net collisional internal heating, this becomes

$$\frac{\partial p_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(p_s) + \frac{5 p_s}{3} \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s) = 0.$$

This is the adiabatic pressure law used by one common warm-fluid closure; retaining anisotropy or heat flux requires the unclosed tensor equation.

Rechenbeispiel

Rechenbeispiel 107

Context: an isotropic hydrogen-ion population has $n_i = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, $k_B T_i = 1.602 \cdot 10^{-18}\,\mathrm{J}$ (10 eV), and bulk speed $u_i = 1.0 \cdot 10^{5}\,\mathrm{m\,s}^{-1}$.

Assumptions: Maxwellian random motion, isotropic pressure, and the three-dimensional kinetic-energy definitions used above.

Target: report the internal energy density ε_i, bulk kinetic-energy density, and total kinetic-energy density $E_{\mathrm{kin},i}$.

Numerical result: $\varepsilon_i = 0.0240\,\mathrm{J\,m}^{-3}$, $E_{\mathrm{bulk},i} = 0.0836\,\mathrm{J\,m}^{-3}$, and $E_{\mathrm{kin},i} = 0.108\,\mathrm{J\,m}^{-3}$.

Open moment hierarchy

Interpretation 108

The continuity equation needs the first moment. The momentum equation needs the second moment. The scalar energy equation needs the contracted third central moment $\boldsymbol{q}_s$; evolving a general anisotropic pressure tensor requires the full third central tensor. Its evolution introduces fourth moments. Exact kinetic information therefore generates an infinite moment hierarchy. A finite fluid system exists only after a constitutive closure.

Summary

Summary 109

The second moment gives energy density and an energy flux made of bulk energy transport, pressure work, and heat flux. The magnetic field does no kinetic-energy work, while the electric field supplies power. Scalar pressure is equivalent to the pressure tensor only under isotropy. The third central moment is the first omitted variable in a pressure-based fluid model.

Exam questions

(d) Give the key steps of the derivation of momentum and heat transport equation.

(e) How are pressure tensor and scalar pressure and their divergence/gradient related?

Verbatim from Plasma Physics Exam.pdf, p. 3

Knowledge check

Knowledge check 110

- Which field supplies power in the species energy equation?
- What physical transport does $\boldsymbol{q}_s$ represent?
- When may $\nabla(p_s)$ replace $\nabla \cdot (\boldsymbol{P}_s)$?
- Why is the energy equation not the end of the exact hierarchy?

Answers

Which field supplies power in the species energy equation?

The electric field supplies $q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E}$. The magnetic force is perpendicular to the particle velocity and does no direct work.

What physical transport does $\boldsymbol{q}_s$ represent?

It is the energy flux carried by random motion in the local bulk frame,

$$\left(\frac{m_s}{2}\right) \int |\boldsymbol{w}_s|^2 \boldsymbol{w}_s f_s \mathrm{d}^3\boldsymbol{v}.$$

When may $\nabla(p_s)$ replace $\nabla \cdot (\boldsymbol{P}_s)$?

Only when $\boldsymbol{P}_s = p_s \boldsymbol{I}$ is isotropic. Anisotropic or sheared distributions require the full tensor divergence.

Why is the energy equation not the end of the exact hierarchy?

Its flux contains the third central moment $\boldsymbol{q}_s$. The transport equation for that heat flux introduces a fourth moment, so a finite model needs closure.

7.5 Cold and warm closures

A fluid description becomes predictive only after it specifies the first omitted moment. Cold and warm plasma models make different choices about pressure and energy transport, while collisional models choose how velocity space relaxes toward equilibrium.

Learning objectives

- state what is retained in cold and warm plasma models
- write an empirical BGK collision model and identify its moment sources
- distinguish species exchange from total conservation
- close the hierarchy with an equation of state or a polytropic law

Unit ledger

The Maxwellian reference $f_{M,s}$ has the units of f_s, $\mathrm{s}^3\,\mathrm{m}^{-6}$. The collision operator has units of distribution per time, $\mathrm{s}^2\,\mathrm{m}^{-6}$. Closure parameters such as γ, the Knudsen number, and $\nu_s \tau$ are dimensionless.

Closure

Definition 111

A closure is a constitutive relation or distributional assumption that expresses an unretained moment in terms of retained fields. It supplies information that cannot be recovered from the retained moment equations alone.

The hierarchy may be written schematically as

$$f_s \to (n_s, \boldsymbol{u}_s, \boldsymbol{P}_s) \to \boldsymbol{q}_s \to \text{fourth moment} \to \ldots$$

The cold-plasma closure sets random momentum transport to zero:

$$\boldsymbol{P}_s = \boldsymbol{0}, \quad \boldsymbol{q}_s = \boldsymbol{0}$$

It is the formal zero-temperature limit in which all particles of a species share one local velocity. It can be useful when thermal speed is small compared with the flow or wave speed, but it cannot describe pressure balance, thermal conduction, or pressure-driven waves.

A warm isotropic model retains a scalar pressure:

$$\boldsymbol{P}_s = p_s \boldsymbol{I}, \quad \varepsilon_s = \frac{3p_s}{2}$$

and must supply one additional relation. Examples include an isothermal equation of state and an adiabatic or polytropic law:

$$p_s = n_s k_B T_s \quad (T_s \text{ prescribed}), \quad p_s n_s^{-\gamma} = \text{constant along a fluid path}$$

For a source-free, heat-flux-free three-dimensional monatomic fluid, the second relation is equivalently

$$\frac{\partial p_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(p_s) + \gamma p_s \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s) = 0, \quad \gamma = \frac{5}{3}$$

This pressure law closes the model without evolving heat flux as an independent variable. It describes compression with no heat transport or internal heating under the stated isotropic assumptions. An isothermal closure instead holds temperature fixed, so compression requires energy exchange to maintain that temperature.

BGK relaxation model Definition 112

The Bhatnagar–Gross–Krook (BGK) model replaces the detailed collision integral by relaxation toward a local Maxwellian: $C_{\mathrm{BGK}}[f_s] = -\nu_s(f_s - f_{M,s})$. Here ν_s is a model collision frequency and $f_{M,s}$ is chosen to share selected local moments with f_s. The model is useful for explaining relaxation and for constructing reduced calculations, but its conservation properties depend on which moments are matched.

If $f_{M,s}$ has the same density as f_s, then

$$\int C_{\mathrm{BGK}}[f_s] \mathrm{d}^3\boldsymbol{v} = 0.$$

If it also has the same flow and energy moments, the corresponding momentum and energy collision moments vanish for that single-species model. In a multispecies plasma, physical interspecies collisions may still exchange momentum and energy between species even though the total exchange sums to zero.

Derivation: where the collision term remains

Step: Define the collision moments

For a general collision operator, the number, momentum, and energy sources are respectively

$$S_{\mathrm{N},s} = \int C_s[f]\mathrm{d}^3\boldsymbol{v}, \quad \boldsymbol{R}_s = m_s \int \boldsymbol{v} C_s[f]\mathrm{d}^3\boldsymbol{v}, \quad Q_s = \left(\frac{m_s}{2}\right)\int v^2 C_s[f]\mathrm{d}^3\boldsymbol{v}.$$

Ordinary elastic collisions within one species have $S_{\mathrm{N},s} = 0$. Interspecies collisions can give nonzero $\boldsymbol{R}_s$ and Q_s for each species.

Step: Apply total conservation

For an isolated elastic system, summing over all collision partners gives

$$\sum_s \boldsymbol{R}_s = \boldsymbol{0}, \quad \sum_s Q_s = 0.$$

Individual species equations may still contain exchange terms; only the total balance removes them.

Step: Inspect the BGK example

For BGK relaxation, substitute

$$C_{\mathrm{BGK}}[f_s] = -\nu_s \cdot (f_s - f_{M,s}).$$

Matching the density makes the zeroth moment zero. Matching density, flow, and temperature additionally makes the first and second collision moments zero. If a background is fixed or only number is matched, the remaining moments represent momentum or energy exchange with that background.

The collision term therefore disappears from a moment equation only after a conservation or closure condition justifies that step.

Cold and warm model assumptions

Key idea 113

"Warm" means that a pressure or temperature scale is retained, not that the distribution must be exactly Maxwellian. "Cold" means that pressure forces are neglected in the selected equations, not that electromagnetic forces or all kinetic effects vanish. The validity of either model is a scale-ordering statement.

Rechenbeispiel

Rechenbeispiel 114

Context: a warm isotropic closure follows a polytropic law with $\gamma = 5/3$. The density changes from n_0 to $n_1 = 8n_0$.

Assumptions: source-free adiabatic compression with $pn^{-\gamma}$ constant along the fluid path.

Target: report the pressure ratio p_1/p_0 and temperature ratio T_1/T_0.

Numerical result: $p_1/p_0 = 32$ and $T_1/T_0 = 4$; both ratios are dimensionless.

Choosing a closure Interpretation 115

Use a cold closure when thermal pressure is asymptotically small for the phenomenon of interest. Use an isotropic warm closure when collisions or another isotropization mechanism erase directional pressure differences and an equation of state is credible. Gyrophase mixing only removes dependence on the angle around $\boldsymbol{B}$: a gyrotropic distribution can still have $p_{\parallel} \neq p_{\perp}$. Retain both pressures, a general tensor, or return to kinetic theory when anisotropy, heat flux, resonances, or boundary layers are dynamically important.

Summary Summary 116

The moment equations are exact until a closure is introduced. Cold models set pressure transport to zero. Warm models retain pressure and then choose an equation of state, energy law, or pressure tensor model. BGK is an empirical collision operator whose moments vanish only when its reference state is matched appropriately. Species exchange can remain nonzero even when total momentum and energy are conserved.

Exam questions

(f) How can one empirically model the collision term? In which equations does it stay or disappear and why?

(g) What is the difference between cold and warm plasma model?

(h) How can one close the hierarchy of fluid equations to avoid requiring a heat transport law?

Verbatim from Plasma Physics Exam.pdf, p. 3

Knowledge check Knowledge check 117

- What does the cold closure remove from the momentum equation?
- Which BGK matching condition guarantees number conservation?
- Why can a collision term vanish in total momentum but remain in a species equation?
- Give one closure that avoids a separate heat-transport law and state its assumption.

Answers

What does the cold closure remove from the momentum equation?
It sets $\boldsymbol{P}_s = \mathbf{0}$, so the pressure force $-\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s)$ is absent. It does not remove the Lorentz force.

Which BGK matching condition guarantees number conservation?
Choose $f_{M,s}$ with the same density as f_s. Then the velocity integral of C_{BGK} is zero.

Why can a collision term vanish in total momentum but remain in a species equation?
Interspecies collisions transfer equal and opposite momentum. The species sources are nonzero, but their sum is zero for an isolated elastic system.

Give one closure that avoids a separate heat-transport law and state its assumption.
A polytropic law $p_s n_s^{-\gamma} =$ constant along a fluid path closes pressure using density. It assumes a prescribed effective energy response, such as adiabatic compression when $\gamma = 5/3$.

8. Multiple-fluid theory of plasmas

A plasma can contain several interpenetrating fluids at the same position. Electrons and ions feel the same fields but have different masses, charges, temperatures, and collision rates. Keeping their equations separate makes those differences explicit before a one-fluid approximation is chosen.

The species-resolved model — Key idea 118

A multiple-fluid model is obtained by taking moments of one kinetic equation for each species. The electric and magnetic fields couple the species, while pressure and collision terms can remain species-specific. Summing the species equations gives total-fluid balances, but further assumptions are needed to close a one-fluid model. Quasi-neutrality alone does not supply those assumptions.

8.1 Species-resolved plasma variables

When is one bulk velocity insufficient? It is insufficient whenever electrons and ions carry different momentum or respond differently to a force. The first step is therefore to define one set of fluid variables for each species.

Learning objectives

- define the species variables needed by a two-fluid model
- distinguish quasi-neutrality from equality of species velocities
- identify the common electromagnetic coupling and species-specific terms
- trace the information retained when kinetic distributions become fluids

Hydrogen two-fluid variables — Definition 119

Use $s \in \{e, i\}$ for electrons and singly charged ions. Their charges are $q_e = -e$ and $q_i = +e$, with masses $m_e = 9.109 \cdot 10^{-31}$ kg and $m_i = 1.673 \cdot 10^{-27}$ kg. For each species, n_s is number density, $\boldsymbol{u}_s$ is mean velocity, $\boldsymbol{P}_s$ is the pressure tensor, and $\boldsymbol{R}_s$ is the collisional momentum source.

The species moments from the kinetic description are

$$\rho_s = m_s n_s, \quad \rho_q = \sum_s q_s n_s, \quad \boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$$

The mass density ρ_s is in kg m^{-3}, charge density ρ_q is in C m^{-3}, and $\boldsymbol{j}$ is the total current density in A m^{-2}. The charge e is measured in C. In a quasi-neutral hydrogen plasma, $n_e \approx n_i$, but generally $\boldsymbol{u}_e \neq \boldsymbol{u}_i$; the difference is precisely what permits a current.

Derivation: species moments retain separate fluids

Step: Take moments species by species

Begin with a distribution $f_s(t, \boldsymbol{r}, \boldsymbol{v})$ for every species. Its zeroth and first velocity moments are

$$n_s = \int f_s \mathrm{d}^3\boldsymbol{v}, \quad n_s \boldsymbol{u}_s = \int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}.$$

Multiplication by m_s or q_s gives the species mass and current contributions.

Step: Form total variables only after the species moments

Because the moments are taken before summing over s, electron and ion velocities remain independent fields. Only after those fields have been defined may one form

$$\rho = \sum_s \rho_s, \quad \boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s.$$

A mass-weighted bulk velocity is another derived quantity.

Step: Separate quasi-neutrality from current

Quasi-neutrality constrains only the zeroth charge moment:

$$\rho_q = \sum_s q_s n_s \approx 0.$$

It does not set the first moments equal. For hydrogen,

$$\rho_q \approx e(n_i - n_e), \quad \boldsymbol{j} = e n_i \boldsymbol{u}_i - e n_e \boldsymbol{u}_e.$$

The current can therefore remain finite even when the charge density is small.

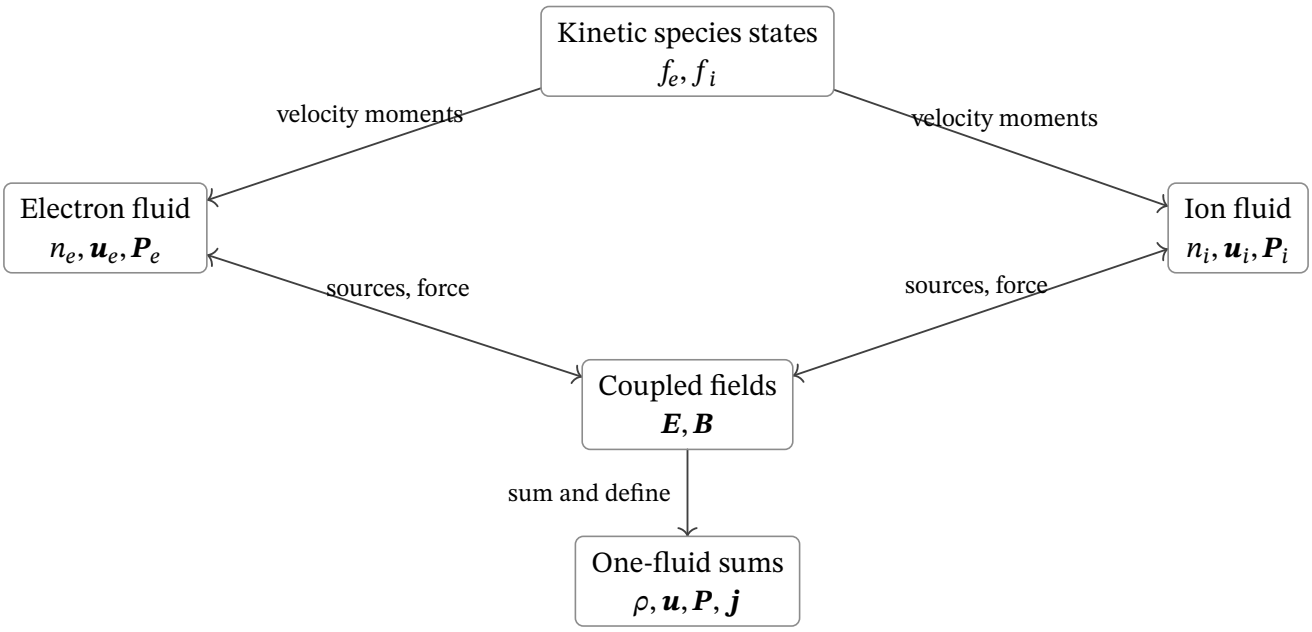


Figure 52: Species-resolved moments retain the relative motion of electrons and ions. Each fluid supplies charge and current to Maxwell's equations and feels the Lorentz force of the shared field. A one-fluid description follows only after the sums and relative-flow terms are defined.

Quasi-neutrality is not one-fluid motion Interpretation 120

The net charge density on bulk scales can be small while electron and ion flows differ substantially. Quasi-neutrality removes a large charge-density imbalance from the bulk ordering; it does not remove species momentum, pressure, or current.

Summary

Summary 121

A two-fluid description assigns density, velocity, pressure, and source terms to each species. Quasi-neutrality constrains the charge-weighted densities, whereas current depends on charge-weighted velocities. The electromagnetic field is common to all species, but the response to it is species-dependent through q_s and m_s.

Knowledge check

Knowledge check 122

- Which species variables are retained by a two-fluid model?
- What does quasi-neutrality constrain?
- Why can a quasi-neutral plasma carry a current?
- What is the physical role of the magnetic field in the species equations?

Answers

Which species variables are retained by a two-fluid model?

At minimum, each species has n_s, $\boldsymbol{u}_s$, and a pressure model $\boldsymbol{P}_s$, together with fields and collision or source terms.

What does quasi-neutrality constrain?

It constrains the charge density $\rho_q = \sum_s q_s n_s$ to be small in the adopted ordering. It does not require $\boldsymbol{u}_e = \boldsymbol{u}_i$.

Why can a quasi-neutral plasma carry a current?

Because the current is $\boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$. Equal and opposite charge densities can be accompanied by different species velocities.

What is the physical role of the magnetic field in the species equations?

It couples to each species through $q_s n_s(\boldsymbol{u}_s \times \boldsymbol{B})$. The force is perpendicular to the instantaneous velocity and changes direction of motion without directly doing work.

8.2 Complete two-fluid equations

What must be evolved once the species variables are defined? Each species needs particle conservation, momentum transport, and an energy or pressure closure. Maxwell's equations provide the shared fields and the coupling between the species currents and charge density.

Learning objectives

- write the conservative continuity, momentum, and energy equations
- identify the Lorentz force and collision moments in each species equation
- state the Maxwell equations that close the field coupling
- separate exact moment equations from pressure and collision closures

Unit ledger

The Lorentz force density is in $\mathrm{N\,m^{-3}}$, because $q_s n_s \boldsymbol{E}$ and $q_s n_s(\boldsymbol{u}_s \times \boldsymbol{B})$ have that unit. The pressure divergence and $\boldsymbol{R}_s$ are also force densities.

Use fixed species labels e and i, no ionization or recombination source, and a smooth distribution that vanishes at large velocity. The collision moments are left explicit: $\boldsymbol{R}_s$ transfers momentum and Q_s transfers energy. For isolated elastic interspecies collisions, $\sum_s \boldsymbol{R}_s = \boldsymbol{0}$ and $\sum_s Q_s = 0$.

The species continuity equation is

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0$$

In conservative momentum form,

$$\frac{\partial \rho_s \boldsymbol{u}_s}{\partial t} + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) + \boldsymbol{R}_s$$

Equivalently, after using continuity, the left-hand side can be written as the species material derivative:

$$\rho_s \left(\frac{\partial \boldsymbol{u}_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(\boldsymbol{u}_s) \right) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) - \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) + \boldsymbol{R}_s$$

The momentum density is in $\mathrm{kg\,m^{-2}\,s^{-1}}$. Every term in the material equation is a force density in $\mathrm{N\,m^{-3}}$. The tensor divergence becomes $\boldsymbol{\nabla}(p_s)$ only for an isotropic pressure tensor.

As in Chapter 7, $\boldsymbol{w}_s = \boldsymbol{v} - \boldsymbol{u}_s$ is the random velocity relative to the species flow. Define the total species kinetic-energy density and heat-flux vector by

$$E_{\mathrm{kin},s} = \frac{\rho_s \boldsymbol{u}_s^2}{2} + \varepsilon_s, \quad \varepsilon_s = \frac{m_s \int \boldsymbol{w}_s^2 f_s \mathrm{d}^3 \boldsymbol{v}}{2}$$

$$\boldsymbol{q}_{h,s} = \frac{m_s \int \boldsymbol{w}_s^2 \boldsymbol{w}_s f_s \mathrm{d}^3 \boldsymbol{v}}{2}$$

The subscript h labels heat flux, distinguishing this vector from particle charge q_s. The energy $E_{\mathrm{kin},s}$ contains bulk and random kinetic energy; electromagnetic field energy is separate.

The corresponding energy balance is

$$\frac{\partial E_{\mathrm{kin},s}}{\partial t} + \boldsymbol{\nabla} \cdot \left(E_{\mathrm{kin},s} \boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_{h,s} \right) = q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E} + Q_s$$

The field equations are

$$\boldsymbol{\nabla} \cdot (\boldsymbol{E}) = \frac{\rho_q}{\varepsilon_0}, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$$

$$\boldsymbol{\nabla} \times (\boldsymbol{E}) = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \boldsymbol{\nabla} \times (\boldsymbol{B}) = \mu_0 \boldsymbol{j} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$$

Derivation: from the kinetic equation to the fluid balances

Step: Start with the conservative kinetic equation

Use

$$\frac{\partial f_s}{\partial t} + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) + \boldsymbol{\nabla} \cdot (f_s \boldsymbol{a}_s) = C_s[f],$$

with

$$\boldsymbol{a}_s = \frac{q_s}{m_s}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}).$$

Step: Take the zeroth moment

Integrating over velocity gives $\partial n_s/\partial t$ from the time term and $\boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s)$ from spatial streaming. The velocity divergence is a vanishing surface term, and particle-conserving collisions have zero zeroth moment. This gives the species continuity equation.

Step: Take the first moment

Multiply by $m_s \boldsymbol{v}$ and integrate. In components, spatial streaming gives

$$m_s \int v_i \frac{\partial (v_j f_s)}{\partial x_j} \mathrm{d}^3\boldsymbol{v} = \frac{\partial \left(m_s \int v_i v_j f_s \mathrm{d}^3\boldsymbol{v}\right)}{\partial x_j}.$$

Writing $\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{w}_s$ decomposes the raw tensor into

$$\rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s.$$

Step: Evaluate the Lorentz and collision moments

Integration by parts in velocity gives

$$m_s \int v_i \frac{\partial (a_{s,j} f_s)}{\partial v_j} \mathrm{d}^3\boldsymbol{v} = -m_s \int a_{s,i} f_s \mathrm{d}^3\boldsymbol{v}.$$

Moving this term to the right gives

$$m_s \int \boldsymbol{a}_s f_s \mathrm{d}^3\boldsymbol{v} = q_s n_s \boldsymbol{E} + q_s n_s (\boldsymbol{u}_s \times \boldsymbol{B}).$$

The remaining collision moment is

$$\boldsymbol{R}_s = m_s \int \boldsymbol{v} C_s[f] \mathrm{d}^3\boldsymbol{v}.$$

Step: Take the energy moment

Multiply by $m_s v^2/2$. After integration by parts, the force work is

$$-m_s \int \boldsymbol{v} \cdot \boldsymbol{a}_s f_s \mathrm{d}^3\boldsymbol{v}.$$

The magnetic contribution vanishes because $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0$, while the electric contribution is $q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E}$.

Splitting

$$v^2 = \boldsymbol{u}_s^2 + 2\boldsymbol{u}_s \cdot \boldsymbol{w}_s + \boldsymbol{w}_s^2$$

separates directed kinetic energy, internal energy, pressure work, and heat flux. The collision energy moment is

$$Q_s = \left(\frac{m_s}{2}\right) \int v^2 C_s[f] \mathrm{d}^3\boldsymbol{v}.$$

No pressure equation has been assumed. Replacing $\boldsymbol{P}_s$ by a scalar or $\boldsymbol{q}_{h,s}$ by a constitutive law is a separate closure choice.

Rechenbeispiel Rechenbeispiel 124

Context: a quasi-neutral hydrogen plasma has $n_e = n_i = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, ion speed $u_{i,x} = 2.0 \cdot 10^5\,\mathrm{m\,s}^{-1}$, electron speed $u_{e,x} = 1.5 \cdot 10^5\,\mathrm{m\,s}^{-1}$, and $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$.

Assumptions: singly charged species, equal densities, and one-dimensional flows along x.

Target: report the charge density and the total x-directed current density.

Numerical result: $\rho_q = 0\,\mathrm{C\,m}^{-3}$ and $j_x = 80.1\,\mathrm{A\,m}^{-2}$.

Electromagnetic and collisional coupling Interpretation 125

The electron and ion equations are separate, but they share $\boldsymbol{E}$ and $\boldsymbol{B}$. The fields depend on the total charge and current, while collisions can transfer momentum and energy between species. The coupling is therefore both electromagnetic and collisional.

Summary Summary 126

The complete two-fluid model consists of one continuity, momentum, and energy balance for each species, Maxwell's equations, and constitutive choices for pressure and collisions. The moment balances are exact under the stated boundary assumptions. Cold, isotropic, collisional, and collisionless models arise only after additional terms are ordered or closed.

Exam questions

(a) Write down the two-fluid equations of a hydrogen plasma.

Verbatim from Plasma Physics Exam.pdf, p. 3

Knowledge check

Knowledge check 127

- Which term couples a species momentum equation to the electromagnetic field?
- When can the pressure-tensor divergence be written as a pressure gradient?
- Why does the magnetic force not appear in the species energy source?
- What must be specified in addition to the exact moment balances to obtain a usable fluid model?

Answers

Which term couples a species momentum equation to the electromagnetic field?
The Lorentz force density $q_s n_s(\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B})$ couples the species momentum to the common fields.

When can the pressure-tensor divergence be written as a pressure gradient?
When the pressure tensor is isotropic, $\boldsymbol{P}_s = p_s \boldsymbol{I}$, so that $\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) = \boldsymbol{\nabla}(p_s)$. Isotropy is an additional model assumption.

Why does the magnetic force not appear in the species energy source?
Because $\boldsymbol{u}_s \cdot (\boldsymbol{u}_s \times \boldsymbol{B}) = 0$. The magnetic force changes direction but does no direct work; the electric field contributes $q_s n_s \boldsymbol{u}_s \cdot \boldsymbol{E}$.

What must be specified in addition to the exact moment balances to obtain a usable fluid model?
One must choose pressure and heat-flux closures and specify collision or source models. Those choices state which kinetic information has been discarded.

8.3 Perpendicular drift balance

How does a pressure gradient move a magnetized species? On scales longer than its gyroradius, the perpendicular momentum equation can balance the pressure force against the Lorentz force. Solving that local balance reveals a common electric drift and a species-dependent diamagnetic drift.

Learning objectives

- state the ordering behind a local perpendicular force balance
- derive the electric and pressure-gradient drift velocities
- track the sign of the diamagnetic drift for electrons and ions
- identify the terms that invalidate the low-inertia drift ordering

Local drift ordering

Assumptions 128

Assume a locally uniform magnetic field, an isotropic pressure tensor, and a perpendicular scale much larger than the gyroradius. Neglect the perpendicular inertial, collisional, and time-derivative terms for the leading balance.

Let $\boldsymbol{b} = \boldsymbol{B}/B$ and project the species momentum balance perpendicular to the field:

$$0 \approx q_s n_s \left(\boldsymbol{E}_\perp + \boldsymbol{u}_{s,\perp} \times \boldsymbol{B}\right) - \boldsymbol{\nabla}_\perp p_s$$

Here $\boldsymbol{E}_\perp = \boldsymbol{E} - \boldsymbol{b}(\boldsymbol{b} \cdot \boldsymbol{E})$ and $\boldsymbol{\nabla}_\perp = \boldsymbol{\nabla} - \boldsymbol{b}(\boldsymbol{b} \cdot \boldsymbol{\nabla})$ are perpendicular projections. Solving for the perpendicular velocity gives

$$\boldsymbol{u}_{s,\perp} = \boldsymbol{u}_{E\times B} + \boldsymbol{u}_{*,s}$$

$$\boldsymbol{u}_{E\times B} = \frac{\boldsymbol{E}\times\boldsymbol{B}}{B^2}$$

$$\boldsymbol{u}_{*,s} = \frac{\boldsymbol{B}\times\boldsymbol{\nabla}(p_s)}{q_s n_s B^2}$$

Both velocities are in m s^{-1}. The electric drift is independent of species, while the diamagnetic drift changes sign with q_s and depends on the species pressure gradient. The star labels the diamagnetic contribution; it is not a complex conjugate. If the pressure is uniform, $\boldsymbol{u}_{*,s} = \boldsymbol{0}$.

Derivation: solve the perpendicular force balance

Step: Write the perpendicular force balance

Start with

$$q_s n_s\big(\boldsymbol{E}_\perp + \boldsymbol{u}_{s,\perp}\times\boldsymbol{B}\big) = \boldsymbol{\nabla}_\perp p_s.$$

Step: Eliminate the magnetic cross product

Cross the equation with $\boldsymbol{B}$ from the right. The identity

$$(\boldsymbol{u}\times\boldsymbol{B})\times\boldsymbol{B} = -B^2\boldsymbol{u}_\perp$$

gives

$$q_s n_s \boldsymbol{E}\times\boldsymbol{B} - q_s n_s B^2 \boldsymbol{u}_{s,\perp} = \boldsymbol{\nabla}(p_s)\times\boldsymbol{B}.$$

Step: Separate the two perpendicular drifts

Rearrange and use $\boldsymbol{\nabla}(p)\times\boldsymbol{B} = -\boldsymbol{B}\times\boldsymbol{\nabla}(p)$:

$$\boldsymbol{u}_{s,\perp} = \frac{\boldsymbol{E}\times\boldsymbol{B}}{B^2} + \frac{\boldsymbol{B}\times\boldsymbol{\nabla}(p_s)}{q_s n_s B^2}.$$

The first term is the common electric drift because q_s cancels from the electric force balance. The second is the charge-dependent diamagnetic drift. The spatial ordering requires the pressure-gradient scale $L_\perp$ to be much larger than the species gyroradius $r_{L,s}$, both measured in metres:

$$\frac{r_{L,s}}{L_\perp} \ll 1.$$

This dimensionless inequality supports a local orbit average. Neglecting inertia also requires slow evolution and small convective acceleration relative to the retained forces, as assumed above. Here $r_{L,s}$ denotes orbit size; ρ_s continues to denote mass density.

Rechenbeispiel

Rechenbeispiel 129

Context: in a local Cartesian frame, take $\boldsymbol{B} = 0.0100\,\mathrm{T}\,\boldsymbol{e}_z$, $\boldsymbol{E} = 30.0\,\mathrm{V\,m^{-1}}\boldsymbol{e}_x$, $n_i = n_e = 1.0 \cdot 10^{14}\,\mathrm{m^{-3}}$, and $\boldsymbol{\nabla}(p_i) = \boldsymbol{\nabla}(p_e) = 1.602 \cdot 10^{-5}\,\mathrm{Pa\,m^{-1}}\boldsymbol{e}_x$. Use $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, with $q_i = +e$ and $q_e = -e$.

Assumptions: scalar pressure, locally uniform fields, and the inertia-free perpendicular drift ordering.

Target: report the common electric drift and the ion and electron diamagnetic drift velocities.

Numerical result: $\boldsymbol{u}_{E\times B} = -3.00 \cdot 10^{3}\,\mathrm{m\,s^{-1}}\boldsymbol{e}_y$, $\boldsymbol{u}_{*,i} = +1.00 \cdot 10^{2}\,\mathrm{m\,s^{-1}}\boldsymbol{e}_y$, and $\boldsymbol{u}_{*,e} = -1.00 \cdot 10^{2}\,\mathrm{m\,s^{-1}}\boldsymbol{e}_y$.

Common and species-dependent drifts

Interpretation 130

$\boldsymbol{u}_{E\times B}$ advects both species together, so its current cancels to leading order under quasi-neutrality. For pressure gradients pointing in the same direction, the diamagnetic velocities of ions and electrons point oppositely. These are local fluid velocity moments; they need not represent motion of the orbit centers. Charge-weighting them gives the diamagnetic current discussed in the next section.

Summary

Summary 131

A perpendicular pressure force is balanced locally by the magnetic force. The resulting velocity is the sum of the common $E \times B$ drift and the charge-dependent diamagnetic drift. The expression is valid only when inertia, collisions, and field nonuniformity are subleading on the chosen scale.

Knowledge check

Knowledge check 132

- Which term makes the electric drift common to both species?
- How does reversing the charge affect the diamagnetic drift?
- What scale ordering supports the local drift balance?
- What happens to the pressure-gradient drift when $\boldsymbol{\nabla}(p_s) = \boldsymbol{0}$?

Answers

Which term makes the electric drift common to both species?

The electric force is proportional to q_s, which cancels the $1/q_s$ introduced when the perpendicular Lorentz balance is solved. Thus $\boldsymbol{u}_{E\times B} = \boldsymbol{E} \times \boldsymbol{B}/B^2$.

How does reversing the charge affect the diamagnetic drift?

It reverses $\boldsymbol{u}_{*,s}$ because the drift is proportional to $1/q_s$. The pressure gradient and magnetic-field directions remain unchanged.

What scale ordering supports the local drift balance?

The pressure-gradient scale must be much larger than the species gyroradius, and perpendicular inertia and collisions must be small compared with the Lorentz and pressure forces.

What happens to the pressure-gradient drift when $\boldsymbol{\nabla}(p_s) = \boldsymbol{0}$?

It vanishes. The species can still share a nonzero electric $E \times B$ drift if a perpendicular electric field is present.

8.4 Diamagnetic current

Why can opposite particle drifts produce one current direction? Current is charge-weighted flow. The charge sign that reverses the electron drift is multiplied once more when the electron contribution to the current is formed, so the two species contributions add.

Learning objectives

- sum the species drift currents in a quasi-neutral plasma
- show why the common electric drift contributes little net current
- relate the diamagnetic current to the total pressure gradient
- distinguish current response from bulk mass transport

Quasi-neutral local sum — Assumptions 133

Use $n_e \approx n_i$ only when evaluating the common electric-drift current. Retain separate species pressures and velocities. The magnetic field is locally smooth, and the pressure tensor is represented by scalar pressures for this interpretation.

Insert the perpendicular drift into the current definition:

$$\boldsymbol{j}_\perp = \sum_s q_s n_s \boldsymbol{u}_{s,\perp} = \frac{\rho_q(\boldsymbol{E}\times\boldsymbol{B})}{B^2} + \frac{\sum_s(\boldsymbol{B}\times\boldsymbol{\nabla}(p_s))}{B^2}$$

Under quasi-neutrality, the first term is small and the pressure term is

$$\boldsymbol{j}_* = \frac{\boldsymbol{B}\times\boldsymbol{\nabla}(p_e+p_i)}{B^2}$$

For scalar ideal-gas pressures, $p_s = n_s k_B T_s$. If $n_e \approx n_i \approx n$, then

$$\boldsymbol{\nabla}(p_e+p_i) = k_B((T_e+T_i)\boldsymbol{\nabla}(n) + n\boldsymbol{\nabla}(T_e+T_i))$$

The diamagnetic current is perpendicular to both $\boldsymbol{B}$ and the total pressure gradient. A pressure gradient can therefore drive a current even while the leading-order charge density remains nearly zero.

For example, take uniform $\boldsymbol{B} = B\boldsymbol{e}_z$, no electric field, constant temperature, and a density increasing with x over many gyroradii. More gyro-orbits centered on the dense side cross a given point than do orbits centered on the dilute side. Their opposite local velocities therefore do not cancel: positive ions have a mean velocity along $+\boldsymbol{e}_y$ and electrons along $-\boldsymbol{e}_y$. The guiding centers have no perpendicular drift in this uniform field. The local velocity moment records the imbalance of orbit crossings, not migration of their centers.

Derivation: add the charge-weighted drift responses

Step: Charge-weight one species

For each species, the diamagnetic contribution obeys

$$q_s n_s \boldsymbol{u}_{*,s} = \frac{\boldsymbol{B}\times\boldsymbol{\nabla}(p_s)}{B^2}.$$

The factors $q_s n_s$ cancel the denominator in the species velocity.

Step: Sum the diamagnetic currents

Summing over electrons and ions gives

$$\boldsymbol{j}_* = \frac{\boldsymbol{B} \times [\boldsymbol{\nabla}(p_e) + \boldsymbol{\nabla}(p_i)]}{B^2}.$$

The pressure-drift velocities are opposite for opposite charges, but their charge-weighted currents add.

Step: Check the common electric drift

The common drift instead gives

$$\sum_s q_s n_s \boldsymbol{u}_{E\times B} = \rho_q \boldsymbol{u}_{E\times B}.$$

It vanishes at leading order when $\rho_q \approx 0$. This cancellation is different from the diamagnetic sum because the electric drift is the same for both species.

Step: Interpret the result as a current response

The formula does not by itself assert that the center of mass moves across the pressure gradient. In a stationary, locally ordered configuration, diamagnetic flow can be divergence-free. Whether it transports mass or changes the pressure profile requires the full continuity and momentum equations, boundary conditions, and the terms omitted in the local balance.

Rechenbeispiel

Rechenbeispiel 134

Context: a quasi-neutral hydrogen plasma has $\boldsymbol{B} = 0.0100\,\mathrm{T}\,\boldsymbol{e}_z$ and total pressure gradient $\boldsymbol{\nabla}(p_e + p_i) = 3.204 \cdot 10^{-5}\,\mathrm{Pa\,m^{-1}}\boldsymbol{e}_x$. Neglect the charge-density contribution to the common electric drift current.

Assumptions: local scalar-pressure drift ordering with $\rho_q \approx 0$.

Target: report the diamagnetic current density and the leading-order electric-drift current density.

Numerical result: $\boldsymbol{j}_* = +3.20 \cdot 10^{-3}\,\mathrm{A\,m^{-2}}\boldsymbol{e}_y$, while $\boldsymbol{j}_{E\times B} = 0\,\mathrm{A\,m^{-2}}$.

Current without bulk advection

Interpretation 135

Current weights each species flow by charge; mass flux weights it by mass. They therefore describe different aspects of the same distribution. In the uniform-field example, unequal orbit crossings produce a local diamagnetic current even though the guiding centers do not drift across the field. A local flow moment alone does not establish transport across the pressure profile.

Summary

Summary 136

Charge-weighting converts opposite diamagnetic velocities into additive current contributions. In a quasi-neutral hydrogen plasma,

$$\boldsymbol{j}_* = \frac{\boldsymbol{B} \times \boldsymbol{\nabla}(p_e + p_i)}{B^2},$$

while the common electric drift contributes only through the small charge density. The current is a response to the pressure gradient; its interpretation as transport needs the full fluid balance and boundary conditions.

Exam questions

(b) What is the diamagnetic drift? Derive and interpret it. How does it result in a diamagnetic current?

Verbatim from Plasma Physics Exam.pdf, p. 3

Knowledge check

Knowledge check 137

- Why do the electron and ion diamagnetic currents add?
- What is the quasi-neutral electric-drift current?
- Which pressure enters the two-species diamagnetic current?
- Why should a diamagnetic current not automatically be called a net mass flux?

Answers

Why do the electron and ion diamagnetic currents add?
Their diamagnetic velocities have opposite signs, but each current is multiplied by its own charge. Since

$$q_s n_s \boldsymbol{u}_{*,s} = \frac{\boldsymbol{B} \times \boldsymbol{\nabla}(p_s)}{B^2},$$

both contributions point along the corresponding pressure-gradient cross-field direction.
What is the quasi-neutral electric-drift current?
It is $\boldsymbol{j}_{E\times B} = \rho_q \boldsymbol{u}_{E\times B}$, so it is small when $\rho_q \approx 0$.
Which pressure enters the two-species diamagnetic current?
The total scalar pressure gradient enters: $\boldsymbol{\nabla}(p_e + p_i)$. Separate species pressures add after the charge-weighted drift sum is formed.
Why should a diamagnetic current not automatically be called a net mass flux?
Current is charge-weighted flow, whereas mass flux is $\sum_s \rho_s \boldsymbol{u}_s$. A pressure-driven current can be locally divergence-free and require no comparable center-of-mass transport.

8.5 Parallel force balance and the one-fluid limit

What remains after perpendicular drifts have been separated? Along the magnetic field, the Lorentz force has no magnetic component. Parallel pressure gradients, electric fields, inertia, and collisions therefore determine the species response and provide the bridge to one-fluid models.

Learning objectives

- project the species momentum equation along the magnetic field
- derive the isothermal electron Boltzmann response as a limiting case
- define total mass velocity, pressure, charge, and current variables
- identify the relative-flow stress retained when species equations are summed

Parallel ordering

Assumptions 138

Let $\boldsymbol{b} = \boldsymbol{B}/B$. The magnetic part of the Lorentz force has zero projection along $\boldsymbol{b}$ exactly. Use scalar species pressures in the displayed parallel equation. For the scalar parallel-velocity form, take the field direction as locally fixed on the scale of interest; otherwise retain $\boldsymbol{b} \cdot \partial \boldsymbol{u}_s/\partial t$ and the associated field-line geometry rather than replacing it by a derivative of $u_{\parallel,s}$. Keep inertia and collisions until a further limit is stated. The Boltzmann response below additionally assumes negligible parallel inertia and collisions, scalar isothermal electron pressure, and a connected field line.

Projecting the species momentum equation gives

$$\rho_s\left(\frac{\partial u_{\parallel,s}}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(u_{\parallel,s})\right) = -\boldsymbol{b} \cdot \boldsymbol{\nabla}(p_s) + q_s n_s E_\parallel + R_{\parallel,s}, \quad E_\parallel = \boldsymbol{b} \cdot \boldsymbol{E}$$

For inertialess, collisionless electrons with $q_e = -e$,

$$0 = -\boldsymbol{b} \cdot \boldsymbol{\nabla}(p_e) - e n_e E_\parallel$$

With $p_e = n_e k_B T_e$ and uniform T_e, this becomes

$$E_\parallel = -\left(\frac{k_B T_e}{e}\right)\boldsymbol{b} \cdot \boldsymbol{\nabla}(\ln n_e), \quad E_\parallel = -\boldsymbol{b} \cdot \boldsymbol{\nabla}(\Phi)$$

and integration along a field line gives the electron Boltzmann relation

$$n_e = n_{e,0} \exp\left(\frac{e(\Phi - \Phi_0)}{k_B T_e}\right)$$

Here $n_{e,0}$ is the density at a reference point with potential Φ_0 on the same field line. Parallel balance fixes the density variation along that line; it does not set a common reference density for all field lines.

The exact one-fluid definitions are

$$\rho = \sum_s \rho_s, \quad \rho \boldsymbol{u} = \sum_s \rho_s \boldsymbol{u}_s, \quad \boldsymbol{P}_1 = \sum_s \boldsymbol{P}_s + \sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s$$

$$\rho_q = \sum_s q_s n_s, \quad \boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s, \quad \boldsymbol{V}_s = \boldsymbol{u}_s - \boldsymbol{u}$$

Here $\boldsymbol{P}_1$ includes both random pressure and relative-flow stress. Adding the species momentum equations then yields

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}\boldsymbol{u} + \boldsymbol{P}_1) = \rho_q \boldsymbol{E} + \boldsymbol{j} \times \boldsymbol{B} + \sum_s \boldsymbol{R}_s$$

For isolated elastic collisions, the last term vanishes. Dropping the relative-flow part of $\boldsymbol{P}_1$ is an additional one-fluid closure, not a consequence of adding the equations.

Derivation: parallel balance and species summation

Step: Project one species along the field

Dot the material momentum equation with $\boldsymbol{b}$. Since $\boldsymbol{b} \cdot (\boldsymbol{u}_s \times \boldsymbol{B}) = 0$, the magnetic force disappears. The projected pressure, electric, and collision terms are

$$-\boldsymbol{b} \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s), \quad q_s n_s E_\parallel, \quad R_{\parallel,s} = \boldsymbol{b} \cdot \boldsymbol{R}_s.$$

For scalar pressure, the first term becomes $-\boldsymbol{b} \cdot \boldsymbol{\nabla}(p_s)$.

Step: Derive the electron Boltzmann response

Set electron inertia and collisions to zero and use $q_e = -e$:

$$\boldsymbol{b} \cdot \boldsymbol{\nabla}(p_e) = -e n_e E_\parallel.$$

Isothermal pressure gives $\boldsymbol{b} \cdot \boldsymbol{\nabla}(p_e) = k_B T_e \boldsymbol{b} \cdot \boldsymbol{\nabla}(n_e)$, so

$$E_\parallel = -\left(\frac{k_B T_e}{e}\right) \boldsymbol{b} \cdot \boldsymbol{\nabla}(\ln n_e).$$

With $E_\parallel = -\boldsymbol{b} \cdot \boldsymbol{\nabla}(\Phi)$,

$$\boldsymbol{b} \cdot \boldsymbol{\nabla}\left(\ln n_e - \frac{e\Phi}{k_B T_e}\right) = 0.$$

Along a connected field line,

$$n_e = n_{e,0} \exp\left(\frac{e(\Phi - \Phi_0)}{k_B T_e}\right).$$

Step: Sum the species momentum equations

Write

$$\boldsymbol{u}_s = \boldsymbol{u} + \boldsymbol{V}_s.$$

Then

$$\sum_s \rho_s \boldsymbol{u}_s \boldsymbol{u}_s = \rho \boldsymbol{u}\boldsymbol{u} + \sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s,$$

because

$$\sum_s \rho_s \boldsymbol{V}_s = \sum_s \rho_s \boldsymbol{u}_s - \rho \boldsymbol{u} = \boldsymbol{0}.$$

The field forces become

$$\sum_s q_s n_s \boldsymbol{E} = \rho_q \boldsymbol{E}, \quad \sum_s q_s n_s (\boldsymbol{u}_s \times \boldsymbol{B}) = \boldsymbol{j} \times \boldsymbol{B}.$$

These identities give the summed momentum equation. If interspecies collisions are internal, their momentum sources cancel in the sum.

Rechenbeispiel

Rechenbeispiel 139

Context: an isothermal electron population has reference density $n_{e,0} = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, thermal energy $k_B T_e = 4.806 \cdot 10^{-19}\,\mathrm{J} = 3.00\,\mathrm{eV}$, and a parallel potential increase $\Phi - \Phi_0 = 3.00\,\mathrm{V}$. Use $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$.

Assumptions: connected field line, negligible parallel electron inertia and collisions, electrostatic parallel field, and uniform T_e.

Target: report the Boltzmann density ratio and the resulting electron density.

Numerical result: $n_e/n_{e,0} = 2.72$ (dimensionless) and $n_e = 2.72 \cdot 10^{16}\,\mathrm{m}^{-3}$.

Species differences in the one-fluid limit

Interpretation 140

Summing removes the labels from the total mass balance, but it does not erase current, relative velocity, anisotropic pressure, or collision exchange. A one-fluid model becomes simpler only when an ordering or closure justifies neglecting or modelling those species differences.

Summary

Summary 141

Parallel force balance exposes the electron pressure–electric-field response and, in an isothermal collisionless limit, the Boltzmann density. Summing species equations defines one-fluid mass and momentum variables, but the relative-flow stress and current remain explicit until a further closure, such as the MHD ordering of Chapter 9, is made.

Knowledge check

Knowledge check 142

- Why does the magnetic force disappear from the parallel momentum equation?
- Which assumptions produce the electron Boltzmann relation?
- What is the one-fluid velocity?
- What relative-flow contribution appears in the summed momentum flux?

Answers

Why does the magnetic force disappear from the parallel momentum equation?

The magnetic force is proportional to $\boldsymbol{u}_s \times \boldsymbol{B}$ and is perpendicular to $\boldsymbol{B}$, so its dot product with $\boldsymbol{b}$ is zero.

Which assumptions produce the electron Boltzmann relation?

Negligible parallel electron inertia and collisions, scalar isothermal pressure, electrostatic parallel balance, and a connected field line lead to

$$\frac{n_e}{n_{e,0}} = \exp\left(\frac{e(\Phi - \Phi_0)}{k_B T_e}\right).$$

What is the one-fluid velocity?

It is the mass-weighted velocity defined by $\rho \boldsymbol{u} = \sum_s \rho_s \boldsymbol{u}_s$, with $\rho = \sum_s \rho_s$.

What relative-flow contribution appears in the summed momentum flux?

The tensor $\sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s$, where $\boldsymbol{V}_s = \boldsymbol{u}_s - \boldsymbol{u}$, remains in the total pressure tensor. It can be neglected only under an additional ordering.

9. Single-fluid theory and magnetohydrodynamics

Chapter 8 combined species equations into total-fluid balances. To obtain magnetohydrodynamics (MHD), we also need an equation for the electric field and assumptions that close pressure and heat transport. The resulting model evolves the bulk flow together with the magnetic field. Its ideal and resistive forms apply when charge separation and the omitted species effects are small on the length and time scales of interest.

Steps of the MHD reduction — Key idea 143

Add species equations to obtain mass and momentum. Take an appropriate species difference to obtain generalized Ohm's law. Then state which terms are small before reducing the field equations to ideal or resistive MHD. Each neglected term marks a limit of the resulting model.

9.1 Single-fluid variables and conservation laws

What does a one-fluid variable mean when electrons and ions still move differently? It is a carefully chosen sum. The mass-weighted velocity carries total momentum, while charge-weighted velocity differences still define the current.

Learning objectives

- define the mass, charge, current, pressure, and velocity variables of a single-fluid model
- derive total mass and charge conservation from species balances
- identify the relative-flow stress retained in the summed momentum flux
- separate exact sums from quasi-neutral and one-fluid approximations

Single-fluid variables are weighted sums — Definition 144

Let $\rho = \sum_s \rho_s$ be the total mass density and define the mass velocity by $\rho \boldsymbol{u} = \sum_s \rho_s \boldsymbol{u}_s$. With $\boldsymbol{V}_s = \boldsymbol{u}_s - \boldsymbol{u}$, the total pressure/transport tensor is

$$\boldsymbol{P} = \sum_s \boldsymbol{P}_s + \sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s.$$

The charge density and current remain charge-weighted: $\rho_q = \sum_s q_s n_s$ and $\boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$.

For the hydrogen plasma used here, $q_e = -e$, $q_i = +e$, and the mass velocity is approximately the ion velocity only when $m_i \gg m_e$ and the current correction is small. Quasi-neutrality is the ordering $\rho_q \approx 0$; it does not remove $\boldsymbol{j}$.

The exact summed mass balance is

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}) = 0$$

Charge-weighting the species continuity equations gives

$$\frac{\partial \rho_q}{\partial t} + \boldsymbol{\nabla} \cdot (\boldsymbol{j}) = 0$$

The summed momentum balance from the multiple-fluid chapter is

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}\boldsymbol{u} + \boldsymbol{P}) = \rho_q \boldsymbol{E} + \boldsymbol{j} \times \boldsymbol{B} + \sum_s \boldsymbol{R}_s$$

For isolated elastic interspecies collisions, $\sum_s \boldsymbol{R}_s = \boldsymbol{0}$. In a quasi-neutral, nonrelativistic ordering the electric force density $\rho_q \boldsymbol{E}$ is subleading in the bulk momentum equation, but the magnetic force density remains through the total current.

Derivation: sum species moments without losing relative flow

Step: Sum the mass balances

Start with

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0.$$

Multiply by the constant mass m_s and sum:

$$\sum_s m_s \frac{\partial n_s}{\partial t} + \sum_s m_s \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0.$$

With $\rho = \sum_s \rho_s$ and $\rho \boldsymbol{u} = \sum_s \rho_s \boldsymbol{u}_s$, this becomes the total mass balance.

Step: Sum the charge balances

Multiply the species equations by q_s instead. The two terms become

$$\frac{\partial \sum_s q_s n_s}{\partial t} = \frac{\partial \rho_q}{\partial t}, \quad \boldsymbol{\nabla} \cdot \left(\sum_s q_s n_s \boldsymbol{u}_s \right) = \boldsymbol{\nabla} \cdot (\boldsymbol{j}).$$

This gives charge conservation without using a quasi-neutral approximation.

Step: Decompose the summed momentum flux

Add the conservative species momentum equations. Write $\boldsymbol{u}_s = \boldsymbol{u} + \boldsymbol{V}_s$. Since

$$\sum_s \rho_s \boldsymbol{V}_s = \boldsymbol{0},$$

the flux decomposes as

$$\sum_s \rho_s \boldsymbol{u}_s \boldsymbol{u}_s = \rho \boldsymbol{u}\boldsymbol{u} + \sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s.$$

The last dyadic is the relative-flow stress and remains part of the total pressure tensor.

Step: Sum the forces

The electromagnetic sums are

$$\sum_s q_s n_s \boldsymbol{E} = \rho_q \boldsymbol{E}, \quad \sum_s q_s n_s (\boldsymbol{u}_s \times \boldsymbol{B}) = \boldsymbol{j} \times \boldsymbol{B}.$$

Internal collision sources cancel only when their species sum is zero. These identities yield the displayed total momentum equation.

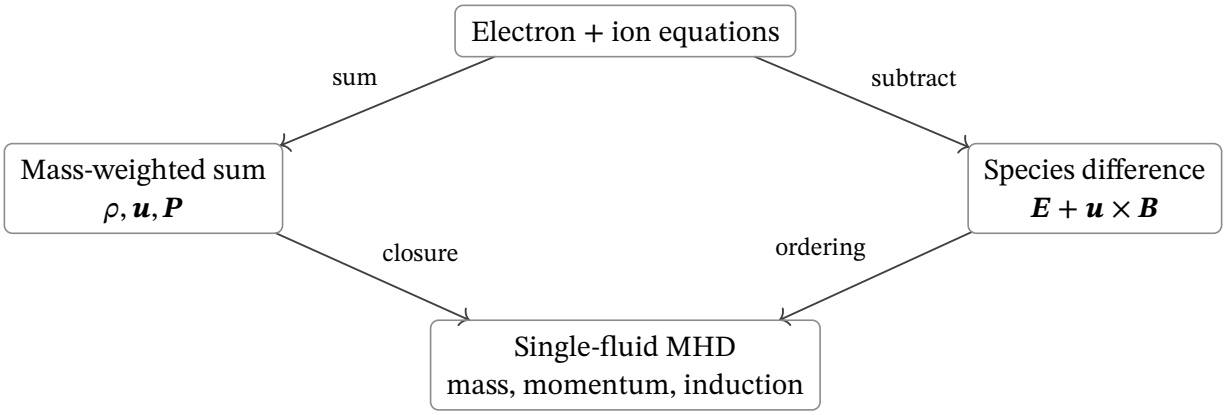


Figure 53: Single-fluid MHD keeps the variables that survive a mass-weighted sum and records the species difference through Ohm's law. Every arrow is an approximation or definition that must be stated.

Rechenbeispiel Rechenbeispiel 145

Context: a quasi-neutral hydrogen plasma has $n_e = n_i = 1.0 \cdot 10^{16}\ \mathrm{m^{-3}}$, $m_e = 9.109 \cdot 10^{-31}\ \mathrm{kg}$, $m_i = 1.673 \cdot 10^{-27}\ \mathrm{kg}$, $u_{i,x} = 2.0 \cdot 10^{5}\ \mathrm{m\,s^{-1}}$, and $u_{e,x} = 1.5 \cdot 10^{5}\ \mathrm{m\,s^{-1}}$. Use $e = 1.602 \cdot 10^{-19}\ \mathrm{C}$.

Assumptions: singly charged species, equal densities, and one-dimensional flows along x.

Target: report the total mass density, mass-weighted velocity, and x-directed current density.

Numerical result: $\rho = 1.674 \cdot 10^{-11}\ \mathrm{kg\,m^{-3}}$, $u_x = 2.00 \cdot 10^{5}\ \mathrm{m\,s^{-1}}$, and $j_x = 80.1\ \mathrm{A\,m^{-2}}$.

One fluid does not mean one velocity for every species Interpretation 146

The mass velocity describes the center-of-mass motion. The current describes a charge-weighted relative motion, and the relative-flow stress stores the corresponding momentum flux. A one-fluid closure becomes accurate only when those additional variables are either small, prescribed, or related to the retained fields by another equation.

Summary Summary 147

Single-fluid variables are exact weighted sums of species moments. The mass balance is obtained by mass-weighting continuity, charge conservation by charge-weighting it, and momentum by adding the species momentum laws. The resulting pressure tensor includes relative-flow stress; quasi-neutrality constrains charge density but does not set current or species velocities to zero.

Exam questions

(a) How are single-fluid mass, charge, current density, pressure and mass velocity defined?

Verbatim from Plasma Physics Exam.pdf, p. 3, section 6(a)

Knowledge check — Knowledge check 148

- What weighting defines the single-fluid velocity?
- Which species difference can remain nonzero in a quasi-neutral plasma?
- What extra term appears in the total pressure tensor?
- What changes in the mass equation if the plasma is quasi-neutral?

Answers

What weighting defines the single-fluid velocity?
Mass weighting defines it: $\rho \boldsymbol{u} = \sum_s \rho_s \boldsymbol{u}_s$ with $\rho = \sum_s \rho_s$. Charge weighting instead defines the current.

Which species difference can remain nonzero in a quasi-neutral plasma?
The velocity difference can remain nonzero. For hydrogen, $\boldsymbol{j} = e n_i \boldsymbol{u}_i - e n_e \boldsymbol{u}_e$ can be finite even when $\rho_q = e(n_i - n_e) \approx 0$.

What extra term appears in the total pressure tensor?
The relative-flow stress $\sum_s \rho_s \boldsymbol{V}_s \boldsymbol{V}_s$ appears in addition to the sum of random pressure tensors. It disappears only when the relative flows are negligible or are closed separately.

What changes in the mass equation if the plasma is quasi-neutral?
Nothing at leading order: mass conservation remains $\partial \rho / \partial t + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}) = 0$. Quasi-neutrality is a charge-density ordering.

9.2 Generalized Ohm's law

How does the current determine the electric field seen by the bulk plasma? The answer comes from the electron momentum equation. Subtracting or rearranging species momentum does not create an arbitrary constitutive law: the Hall, pressure, resistive, and electron-inertia terms each have a definite origin.

Learning objectives

- derive the generalized Ohm-law balance from the electron equation
- identify the Hall, electron-pressure, resistive, and electron-inertia terms
- relate collisional drag to the scalar resistivity and conductivity
- state which ordering produces the ideal electric-field constraint

Unit ledger

A scalar resistivity defined by $\boldsymbol{E} = \eta \boldsymbol{j}$ has units $\Omega\,\mathrm{m}$; the conductivity $\sigma = 1/\eta$ is in $\mathrm{S\,m^{-1}}$. The electron pressure force per charge density is in $\mathrm{V\,m^{-1}}$. The coefficient $m_e/(e^2 n)$ multiplying a current time derivative has the units needed to produce an electric field.

Two-species Ohm-law ordering — Assumptions 149

Use a singly ionized hydrogen plasma with $n_e \approx n_i \approx n$, $m_i \gg m_e$, and $\boldsymbol{j} = en(\boldsymbol{u}_i - \boldsymbol{u}_e)$. Let $\boldsymbol{u}$ be the mass velocity and use $\boldsymbol{u}_e \approx \boldsymbol{u} - \boldsymbol{j}/(en)$ when the ion mass dominates. The generalized balance below keeps scalar electron pressure, collisional drag, and a leading current-inertia term; omitted electron convective-inertia and density-gradient terms must be small under the stated scale ordering.

The electron momentum equation is

$$m_e n\left(\frac{\partial \boldsymbol{u}_e}{\partial t} + \boldsymbol{u}_e \cdot \boldsymbol{\nabla}(\boldsymbol{u}_e)\right) = -en(\boldsymbol{E} + \boldsymbol{u}_e \times \boldsymbol{B}) - \boldsymbol{\nabla}(p_e) + \boldsymbol{R}_e$$

Solving it for the field in the electron frame and replacing $\boldsymbol{u}_e$ by $\boldsymbol{u} - \boldsymbol{j}/(en)$ gives the ordered generalized Ohm law

$$\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = \frac{\boldsymbol{j} \times \boldsymbol{B}}{en} - \frac{\boldsymbol{\nabla}(p_e)}{en} + \eta \boldsymbol{j} + \frac{m_e}{e^2 n}\frac{\partial \boldsymbol{j}}{\partial t}$$

The four right-hand terms are, respectively, the Hall response, the electron-pressure or ambipolar response, collisional resistivity, and the leading electron-inertia response in a slowly varying bulk frame.

The left side is the electric field seen by the moving bulk fluid to nonrelativistic order. The Hall term appears because electrons carry the current relative to that fluid; the pressure term supplies the field needed to balance electron pressure. Resistive drag opposes relative motion, while electron inertia matters when the current changes rapidly.

For a linear electron–ion drag law, the force on electrons is

$$\boldsymbol{R}_e = m_e n \nu_{ei}(\boldsymbol{u}_i - \boldsymbol{u}_e) = \frac{m_e \nu_{ei} \boldsymbol{j}}{e}$$

Therefore

$$\eta = \frac{m_e \nu_{ei}}{ne^2}, \quad \sigma = \frac{1}{\eta} = \frac{ne^2}{m_e \nu_{ei}}$$

Here ν_{ei} is the electron–ion momentum-transfer frequency, and σ is the scalar conductivity in $\mathrm{S\,m^{-1}}$. The frequency itself depends on the collision model and plasma state; the later collisions chapter supplies that kinetic input. Here, resistivity enters as drag per unit current.

Derivation: locate every term in generalized Ohm's law

Step: Isolate the electric field

Start with the electron material momentum equation and move the electric force to the left:

$$\boldsymbol{E} + \boldsymbol{u}_e \times \boldsymbol{B} = -\frac{\boldsymbol{\nabla}(p_e)}{en} + \frac{\boldsymbol{R}_e}{en} - \frac{m_e\left(\frac{\partial \boldsymbol{u}_e}{\partial t} + \boldsymbol{u}_e \cdot \boldsymbol{\nabla}(\boldsymbol{u}_e)\right)}{e}.$$

Step: Expose the Hall term

Since

$$\boldsymbol{u} - \boldsymbol{u}_e \approx \frac{\boldsymbol{j}}{en},$$

adding the difference between bulk and electron magnetic advection gives

$$\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = -\frac{\boldsymbol{\nabla}(p_e)}{en} + \frac{\boldsymbol{R}_e}{en} - \frac{m_e\left(\frac{\partial \boldsymbol{u}_e}{\partial t} + \boldsymbol{u}_e \cdot \boldsymbol{\nabla}(\boldsymbol{u}_e)\right)}{e} + \frac{\boldsymbol{j} \times \boldsymbol{B}}{en}.$$

The displayed Hall contribution is the magnetic force expressed through the electron–bulk relative velocity.

Step: Model pressure and resistive drag

For scalar electron pressure, retain $-\nabla(p_e)/(en)$ as one term; splitting it further is a separate ordering choice. Model interspecies drag as

$$\boldsymbol{R}_e = m_e n \nu_{ei}(\boldsymbol{u}_i - \boldsymbol{u}_e).$$

Since $\boldsymbol{u}_i - \boldsymbol{u}_e = \boldsymbol{j}/(en)$,

$$\frac{\boldsymbol{R}_e}{en} = \frac{m_e \nu_{ei} \boldsymbol{j}}{ne^2} = \eta \boldsymbol{j}.$$

Thus

$$\eta = \frac{m_e \nu_{ei}}{ne^2}, \quad \sigma = \frac{1}{\eta}.$$

Step: Retain electron inertia when needed

Write

$$\boldsymbol{u}_e \approx \boldsymbol{u} - \frac{\boldsymbol{j}}{en}.$$

In the slowly varying-density and slow-bulk-inertia ordering, the current-dependent part of the electron inertia is

$$\frac{m_e \frac{\partial \boldsymbol{j}}{\partial t}}{e^2 n}.$$

If current advection or density variation is not small, replace this approximation by the full electron-inertia operator; the displayed generalized law is then insufficient.

Rechenbeispiel

Rechenbeispiel 150

Context: an electron–ion plasma has $n = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, and $\nu_{ei} = 2.54 \cdot 10^{3}\,\mathrm{s}^{-1}$.

Assumptions: scalar linear electron–ion drag and the Spitzer form $\eta = m_e \nu_{ei}/\left(ne^2\right)$.

Target: report the scalar resistivity and conductivity.

Numerical result: $\eta = 9.01 \cdot 10^{-6}\,\Omega\,\mathrm{m}$ and $\sigma = 1.11 \cdot 10^{5}\,\mathrm{S\,m}^{-1}$.

The ideal field is a limit, not the starting equation

Interpretation 151

If Hall, pressure, resistive, and electron-inertia terms are all small compared with $\boldsymbol{u} \times \boldsymbol{B}$, generalized Ohm's law reduces to $\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} \approx \boldsymbol{0}$. If one correction is not small, the plasma is still a fluid but it is not described by ideal MHD. The ordering must be checked against the length, time, density, and field scales of the problem.

Summary

Summary 152

Generalized Ohm's law is obtained from electron momentum. The Hall term comes from electron–bulk velocity difference, the pressure term from electron pressure, resistivity from collisional drag, and the current term from electron inertia. The ideal constraint is the result of ordering all four corrections below the bulk magnetic-advection term.

Exam questions

(c) Sketch the derivation of Ohm's law. Where does the current term finally come from? What's the simplified form?

(d) Relate collisional drag force to the current to get the Spitzer resistivity in terms of collision frequency.

Verbatim from Plasma Physics Exam.pdf, p. 3, section 6(c)–(d)

Knowledge check

Knowledge check 153

- Which species equation is the most direct source of generalized Ohm's law?
- What physical process gives the scalar resistivity?
- When is the Hall term important?
- What is the ideal-MHD electric-field condition?

Answers

Which species equation is the most direct source of generalized Ohm's law?

The electron momentum equation is the direct source. Solving it for the electric field and replacing the electron–bulk velocity difference with $\boldsymbol{j}/(en)$ produces the Hall and current terms.

What physical process gives the scalar resistivity?

Electron–ion collisional drag gives

$$\boldsymbol{R}_e = \frac{m_e \nu_{ei} \boldsymbol{j}}{e},$$

hence $\eta = m_e \nu_{ei}/\left(ne^2\right)$. More frequent momentum transfer means larger resistivity and smaller conductivity.

When is the Hall term important?

It matters when $\boldsymbol{j} \times \boldsymbol{B}/(en)$ is not small compared with the bulk electric field. Relative to magnetic advection, its magnitude is set by $|\boldsymbol{j}_\perp|/(en|\boldsymbol{u}_\perp|)$: the common factor B cancels. Increasing B alone does not increase this ratio. A scale ordering must justify neglecting the Hall term.

What is the ideal-MHD electric-field condition?

The condition is $\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = \boldsymbol{0}$. It is a reduced generalized Ohm law and therefore requires all retained correction terms to be small.

9.3 Linearized and simplified MHD equations

Which finite set of equations is actually called MHD? After the current response has been ordered, the plasma can be described by mass density, bulk velocity, pressure, and magnetic field. Linearization around a uniform equilibrium then exposes the small-amplitude system that leads to the wave chapters.

Learning objectives

- state the nonlinear ideal-MHD conservation laws and closures
- linearize mass, momentum, induction, and pressure equations about a uniform equilibrium
- define the sound-speed and magnetic-field constraints in the linear system
- distinguish a linear perturbation model from the underlying nonlinear equations

Unit ledger

The adiabatic sound speed $c_s = \sqrt{\gamma p_0/\rho_0}$ is in $\mathrm{m\,s^{-1}}$. Perturbation symbols such as $\delta\rho/\rho_0$, $\delta p/p_0$, and $\delta\boldsymbol{B}/B_0$ are dimensionless ratios.

Simplified MHD ordering — Assumptions 154

Use quasi-neutrality, negligible displacement current, isotropic pressure, negligible Hall and electron-pressure corrections, and a magnetic Reynolds number large enough for ideal induction. Close the pressure with the explicit material-derivative form

$$\frac{\partial p\rho^{-\gamma}}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}(p\rho^{-\gamma}) = 0.$$

The dimensionless adiabatic index γ specifies the pressure response to compression; $\gamma = 5/3$ is the isotropic monatomic choice. For the linearized equations, take a static uniform equilibrium $(\rho_0, p_0, \boldsymbol{B}_0)$ with no equilibrium current or pressure gradient. The displayed pressure-density closure further selects initially isentropic perturbations, meaning no initial entropy perturbation: $\delta p - c_s^2\delta\rho = 0$. General adiabatic perturbations conserve this difference in time but need not set it to zero.

The nonlinear ideal-MHD equations are

$$\frac{\partial\rho}{\partial t} + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0$$

$$\rho\left(\frac{\partial\boldsymbol{u}}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}(\boldsymbol{u})\right) = -\boldsymbol{\nabla}(p) + \frac{\boldsymbol{\nabla}\times(\boldsymbol{B})\times\boldsymbol{B}}{\mu_0}$$

$$\frac{\partial\boldsymbol{B}}{\partial t} = \boldsymbol{\nabla}\times(\boldsymbol{u}\times\boldsymbol{B}), \quad \boldsymbol{\nabla}\cdot(\boldsymbol{B}) = 0$$

$$\frac{\partial p\rho^{-\gamma}}{\partial t} + \boldsymbol{u}\cdot\boldsymbol{\nabla}(p\rho^{-\gamma}) = 0$$

The magnetic force has the coefficient $1/\mu_0$ because the displacement current $\mu_0\varepsilon_0\partial\boldsymbol{E}/\partial t$ has been neglected in Ampere's law (nonrelativistic flows, $u^2 \ll c^2$, and slow time scales):

$$\boldsymbol{j} = \frac{\boldsymbol{\nabla}\times(\boldsymbol{B})}{\mu_0}$$

The electric field is recovered from the ideal constraint, not evolved as an independent MHD variable.

Let $\rho = \rho_0 + \delta\rho$, $p = p_0 + \delta p$, $\boldsymbol{u} = \delta\boldsymbol{u}$, and $\boldsymbol{B} = \boldsymbol{B}_0 + \delta\boldsymbol{B}$, where perturbations are first order. For uniform equilibrium, the linearized equations are

$$\frac{\partial\delta\rho}{\partial t} + \rho_0\boldsymbol{\nabla}\cdot(\delta\boldsymbol{u}) = 0$$

$$\rho_0 \frac{\partial \delta \boldsymbol{u}}{\partial t} = -\boldsymbol{\nabla}(\delta p) + \frac{\boldsymbol{\nabla} \times (\delta \boldsymbol{B}) \times \boldsymbol{B}_0}{\mu_0}$$

$$\frac{\partial \delta \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times (\delta \boldsymbol{u} \times \boldsymbol{B}_0), \quad \boldsymbol{\nabla} \cdot (\delta \boldsymbol{B}) = 0$$

$$\delta p = c_s^2 \delta \rho, \quad c_s^2 = \frac{\gamma p_0}{\rho_0}$$

Here the subscript in c_s labels sound speed, not a particle species. Linearization keeps the response proportional to one perturbation and discards products of perturbations. The pressure relation also uses the initial entropy assumption; it does not follow from small amplitude alone.

Derivation: linearize the ideal-MHD system

Step: Linearize continuity

Insert

$$\rho = \rho_0 + \delta \rho, \quad \boldsymbol{u} = \delta \boldsymbol{u}$$

into

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}) = 0.$$

Since ρ_0 is constant and products of perturbations are second order,

$$\frac{\partial \delta \rho}{\partial t} + \rho_0 \boldsymbol{\nabla} \cdot (\delta \boldsymbol{u}) = 0.$$

Step: Linearize momentum

The equilibrium has $\boldsymbol{u}_0 = \boldsymbol{0}$ and no pressure gradient. The inertial term becomes

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla}(\boldsymbol{u}) \right) = \rho_0 \frac{\partial \delta \boldsymbol{u}}{\partial t} + O(\delta^2).$$

The pressure force is $-\boldsymbol{\nabla}(\delta p)$. Expanding the magnetic force and using $\boldsymbol{\nabla} \times (\boldsymbol{B}_0) = \boldsymbol{0}$ leaves

$$\frac{[\boldsymbol{\nabla} \times (\delta \boldsymbol{B}) \times \boldsymbol{B}_0]}{\mu_0}.$$

Step: Linearize induction and magnetic solenoidality

Keeping one perturbation and one background field in ideal induction gives

$$\frac{\partial \delta \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times (\delta \boldsymbol{u} \times \boldsymbol{B}_0),$$

while divergence-free magnetic fields give

$$\boldsymbol{\nabla} \cdot (\delta \boldsymbol{B}) = 0.$$

Step: Linearize the adiabatic closure

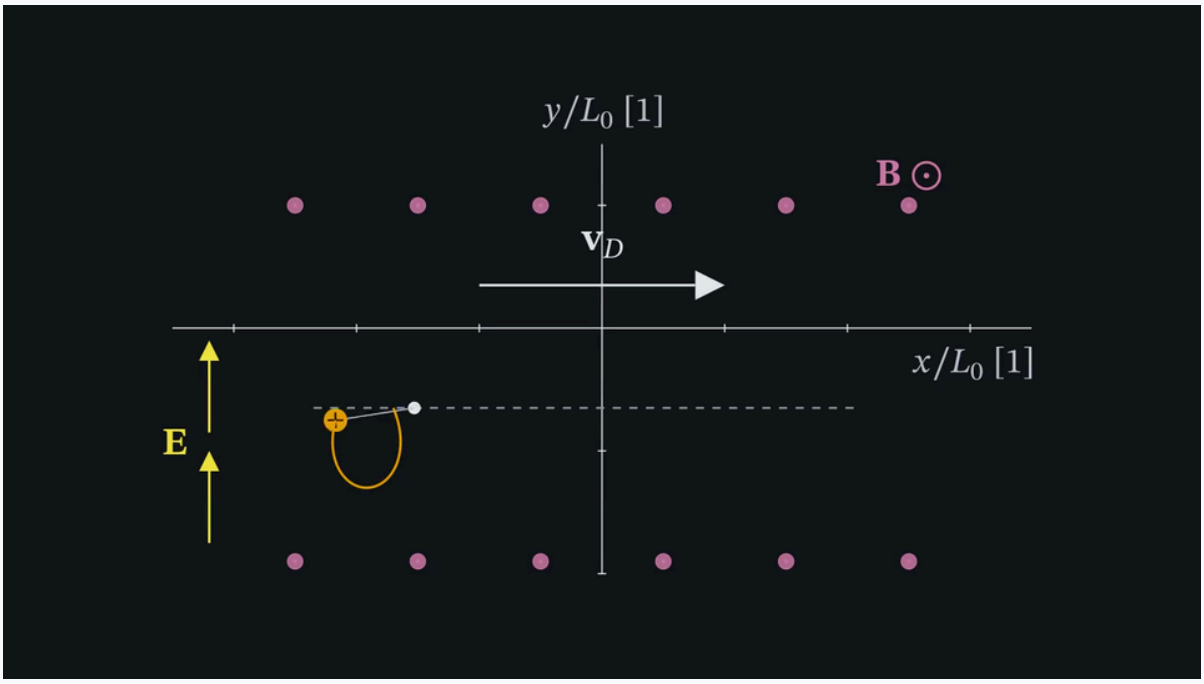
Expand $p\rho^{-\gamma}$ about the static equilibrium. The first-order invariant is proportional to

$$\frac{\delta p}{p_0} - \gamma\left(\frac{\delta \rho}{\rho_0}\right).$$

Its material derivative reduces to a time derivative, so $\partial \delta p - c_s^2 \delta\rho/\partial t = 0$. Initially isentropic perturbations therefore satisfy

$$\frac{\delta p}{p_0} = \gamma\left(\frac{\delta \rho}{\rho_0}\right), \quad \delta p = \left(\frac{\gamma p_0}{\rho_0}\right)\delta\rho = c_s^2 \delta\rho.$$

Every discarded product contains at least two perturbation factors.



Watch animation

Common $E \times B$ advection as a visual bridge to ideal MHD. The reference length L_0 and time t_0 give the gyroradius ratio $r_L/L_0 = 0.65$, $\Omega t_0 = 2$, and $v_D t_0/L_0 = 0.55$, all with unit [1], as in Chapter 5. The animation is a deterministic illustration; it does not show the linearized MHD perturbation equations or measured data.

Figure 54: Common $E \times B$ advection as a visual bridge to ideal MHD. The reference length L_0 and time t_0 give the gyroradius ratio $r_L/L_0 = 0.65$, $\Omega t_0 = 2$, and $v_D t_0/L_0 = 0.55$, all with unit [1], as in Chapter 5. The animation is a deterministic illustration; it does not show the linearized MHD perturbation equations or measured data.

Rechenbeispiel

Rechenbeispiel 155

Context: a uniform equilibrium has $\rho_0 = 1.0 \cdot 10^{-11}\,\mathrm{kg\,m^{-3}}$, $p_0 = 0.10\,\mathrm{Pa}$, $\gamma = 5/3$, and a density perturbation $\delta\rho/\rho_0 = 0.010$.

Assumptions: ideal, adiabatic, small-amplitude MHD perturbations about a static uniform state.

Target: report the adiabatic sound speed and pressure perturbation.

Numerical result: $c_s = 1.29 \cdot 10^5\,\mathrm{m\,s^{-1}}$, $\delta p = 1.67 \cdot 10^{-3}\,\mathrm{Pa}$, and $\delta p/p_0 = 0.0167$ (dimensionless).

Retained and discarded terms in linear MHD — Interpretation 156

The linearized equations preserve pressure restoring forces, magnetic tension and compression, and the divergence-free field constraint. They discard perturbation–perturbation advection and any equilibrium gradients. The resulting waves are small-amplitude limits of nonlinear MHD; they are not valid for shocks, strong reconnection, or perturbations comparable to the background.

Summary — Summary 157

Ideal MHD is a closed reduced system for ρ, $\boldsymbol{u}$, p, and $\boldsymbol{B}$, supplemented by Ampere's reduced law and an adiabatic closure. Linearization around a uniform static state yields four first-order relations for $\delta\rho$, δp, $\delta\boldsymbol{u}$, and $\delta\boldsymbol{B}$. The next wave chapters use this system to derive its characteristic branches.

Exam questions

(b) Write down the linearized single-fluid MHD equations.

Verbatim from Plasma Physics Exam.pdf, p. 3, section 6(b)

Knowledge check — Knowledge check 158

- Which equilibrium assumptions remove the first-order background magnetic force?
- What closure relates pressure and density perturbations in the adiabatic model?
- Why must a magnetic-field perturbation be divergence-free?
- What nonlinear effect is lost on linearization?

Answers

Which equilibrium assumptions remove the first-order background magnetic force?
A uniform, time-independent $\boldsymbol{B}_0$ has $\boldsymbol{\nabla}\times(\boldsymbol{B}_0)=\boldsymbol{0}$, so it carries no equilibrium current in the reduced Ampere law. The background pressure and velocity are also taken uniform and static.
What closure relates pressure and density perturbations in the adiabatic model?
The first-order closure is $\delta p = c_s^2\delta\rho$ with $c_s^2=\gamma p_0/\rho_0$. The sound speed has units $\mathrm{m\,s^{-1}}$, while the perturbation ratios are dimensionless.
Why must a magnetic-field perturbation be divergence-free?
The constraint $\boldsymbol{\nabla}\cdot(\boldsymbol{B})=0$ is part of Maxwell's equations. Splitting $\boldsymbol{B}=\boldsymbol{B}_0+\delta\boldsymbol{B}$ with $\boldsymbol{\nabla}\cdot(\boldsymbol{B}_0)=0$ leaves $\boldsymbol{\nabla}\cdot(\delta\boldsymbol{B})=0$.
What nonlinear effect is lost on linearization?
Products of perturbations, such as $\delta\boldsymbol{u}\cdot\boldsymbol{\nabla}(\delta\boldsymbol{u})$, are discarded. Therefore the linear model cannot represent finite-amplitude steepening, shocks, or perturbation-generated equilibrium changes.

9.4 Frozen-in flux and magnetic diffusion

What does ideal MHD mean geometrically? Its induction equation says that magnetic flux through a surface moving with the fluid is conserved. A finite resistivity adds a competing diffusion process, so the magnetic Reynolds number decides whether field lines are effectively carried by the flow or can slip through it.

Learning objectives

- derive the induction equation from Faraday's law and Ohm's law
- prove conservation of magnetic flux through a material surface
- define magnetic diffusivity and magnetic Reynolds number
- estimate the magnetic-field diffusion timescale and its limiting regime

Unit ledger

The magnetic diffusivity $D_B = \eta/\mu_0$ is in $\mathrm{m^2\,s^{-1}}$. The magnetic Reynolds number $R_m = UL/D_B$ is dimensionless. The advection time $\tau_A = L/U$ and diffusion time $\tau_D = L^2/D_B$ are both in s.

Induction ordering

Assumptions 159

Use Faraday's law, reduced Ampere's law $\boldsymbol{j} = \boldsymbol{\nabla} \times (\boldsymbol{B})/\mu_0$, and a uniform scalar resistivity. For ideal MHD set $\eta = 0$ after the induction equation is derived. Assume $\boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$ when converting the double curl into a Laplacian. A material surface has boundary velocity $\boldsymbol{u}$ and remains smooth while the flux theorem is applied.

Faraday's law and generalized Ohm's law in the resistive MHD limit are

$$\boldsymbol{\nabla} \times (\boldsymbol{E}) = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = \eta \boldsymbol{j}$$

Substitution of reduced Ampere's law gives the induction equation

$$\frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times (\boldsymbol{u} \times \boldsymbol{B}) + D_B \nabla^2 (\boldsymbol{B}), \quad D_B = \frac{\eta}{\mu_0}$$

The first term advects and stretches field; the second diffuses it. The dimensionless competition is

$$R_m = \frac{UL}{D_B} = \frac{\tau_D}{\tau_A}$$

For $R_m \gg 1$, the flow crosses the scale L much sooner than resistivity smooths magnetic structure on that scale. This compares two times; it does not say that diffusion is slow on every observation time. For a characteristic structure of size L, the finite-resistivity diffusion estimate is

$$\tau_D \approx \frac{L^2}{D_B} = \frac{\mu_0 L^2}{\eta}$$

In the ideal limit, follow a loop of fluid elements and a surface spanning that loop. The magnetic flux through the moving surface stays constant even as the surface stretches or bends. The local field strength can still change: frozen flux does not mean a static or spatially uniform field.

Derivation: induction equation and frozen magnetic flux

Step: Insert resistivity into Faraday's law

Begin with

$$\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B} = \eta \boldsymbol{j},$$

so

$$\boldsymbol{E} = -\boldsymbol{u} \times \boldsymbol{B} + \eta \boldsymbol{j}.$$

Faraday's law becomes

$$\frac{\partial \boldsymbol{B}}{\partial t} = -\boldsymbol{\nabla} \times (\boldsymbol{E}) = \boldsymbol{\nabla} \times (\boldsymbol{u} \times \boldsymbol{B}) - \eta \boldsymbol{\nabla} \times (\boldsymbol{j}).$$

Step: Convert the current term to magnetic diffusion

The reduced Ampere law is

$$\boldsymbol{j} = \frac{\boldsymbol{\nabla} \times (\boldsymbol{B})}{\mu_0}.$$

For uniform η,

$$-\eta \boldsymbol{\nabla} \times (\boldsymbol{j}) = -\frac{\eta \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times (\boldsymbol{B}))}{\mu_0}.$$

Use

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times (\boldsymbol{B})) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot (\boldsymbol{B})) - \nabla^2(\boldsymbol{B}).$$

With $\boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$,

$$-\eta \boldsymbol{\nabla} \times (\boldsymbol{j}) = \frac{\eta \nabla^2(\boldsymbol{B})}{\mu_0}.$$

Define $D_B = \eta/\mu_0$ to obtain the displayed induction equation.

Step: Prove frozen magnetic flux in the ideal limit

Let $S(t)$ be a surface whose boundary moves with $\boldsymbol{u}$ and define

$$\Psi_B = \int_{S(t)} \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}.$$

The moving-surface transport theorem gives

$$\frac{\mathrm{d}\Psi_B}{\mathrm{d}t} = \int_{S(t)} \left[\frac{\partial \boldsymbol{B}}{\partial t} - \boldsymbol{\nabla} \times (\boldsymbol{u} \times \boldsymbol{B}) \right] \cdot \mathrm{d}\boldsymbol{S}.$$

Ideal induction sets the integrand to zero, so $\mathrm{d}\Psi_B/\mathrm{d}t = 0$. Stokes' theorem gives the equivalent moving-loop statement. Thus flux through every material surface is constant.

Step: Compare advection and diffusion

The induction terms scale as

$$|\boldsymbol{\nabla} \times (\boldsymbol{u} \times \boldsymbol{B})| \approx \frac{UB}{L}, \quad |D_B \nabla^2(\boldsymbol{B})| \approx \frac{D_B B}{L^2}.$$

Their ratio is

$$\frac{UL}{D_B} = R_m = \frac{\frac{L^2}{D_B}}{\frac{L}{U}} = \frac{\tau_D}{\tau_A}.$$

This magnetic Reynolds number compares advection with resistive diffusion.

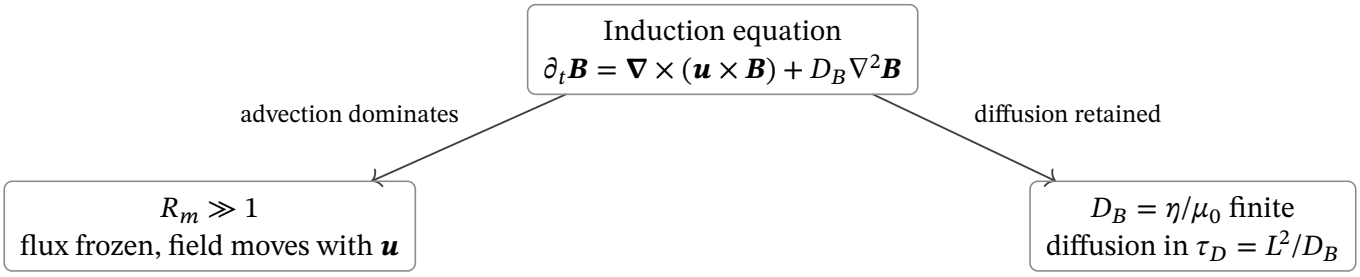


Figure 55: The magnetic Reynolds number R_m compares advection with diffusion in the induction equation. For $R_m \gg 1$ field lines move with the fluid; finite resistivity lets them slip and diffuse.

Rechenbeispiel Rechenbeispiel 160

Context: a magnetic structure has length $L = 10\,\mathrm{m}$ and bulk speed $U = 1.0 \cdot 10^{5}\,\mathrm{m\,s^{-1}}$. Use scalar resistivity $\eta = 9.0 \cdot 10^{-3}\,\Omega\,\mathrm{m}$ and $\mu_0 = 1.257 \cdot 10^{-6}\,\mathrm{H\,m^{-1}}$.

Assumptions: uniform resistivity, divergence-free magnetic field, and the resistive induction ordering stated above.

Target: report the magnetic diffusivity, advection time, diffusion time, and magnetic Reynolds number.

Numerical result: $D_B = 7.16 \cdot 10^{3}\,\mathrm{m^2\,s^{-1}}$, $\tau_A = 1.00 \cdot 10^{-4}\,\mathrm{s}$, $\tau_D = 1.40 \cdot 10^{-2}\,\mathrm{s}$, and $R_m = 1.40 \cdot 10^{2}$ (dimensionless).

Breakdown of frozen flux in thin current layers Interpretation 161

Ideal MHD preserves flux through a surface that moves with $\boldsymbol{u}$, so field-line connectivity is maintained as long as the smooth ideal description remains valid. Finite resistivity permits slippage and, in sufficiently localized regions, changes of connectivity. A large global R_m does not guarantee that diffusion is negligible inside every thin current layer.

Summary Summary 162

Faraday's law plus Ohm's law produces advection and diffusion in the magnetic induction equation. Here $D_B = \eta/\mu_0$, $\tau_D = L^2/D_B$, and $R_m = UL/D_B$. The ideal limit conserves flux through material surfaces; resistivity breaks that material conservation on the diffusion scale.

Exam questions

(e) Sketch the derivation of the frozen-flux theorem.

(f) Derive the diffusion equation for the B field and estimate the timescale.

Verbatim from Plasma Physics Exam.pdf, p. 3, section 6(e)–(f)

Knowledge check

Knowledge check 163

- Which term changes the magnetic-field topology in the resistive induction equation?
- What surface is used in the frozen-flux statement?
- How does the magnetic Reynolds number compare the timescales?
- What assumption is needed to replace the double curl by a Laplacian?

Answers

Which term changes the magnetic-field topology in the resistive induction equation?
The term $D_B \nabla^2(\boldsymbol{B})$ represents magnetic diffusion. It is absent in ideal MHD and becomes important when its scale estimate is comparable with the advection term.

What surface is used in the frozen-flux statement?
It is a material surface whose boundary moves with the bulk velocity $\boldsymbol{u}$. Flux through a fixed laboratory surface is not the same invariant.

How does the magnetic Reynolds number compare the timescales?
$R_m = \tau_D/\tau_A = UL/D_B$. Thus $R_m \gg 1$ means advection is faster than diffusion, while $R_m \ll 1$ means diffusion acts before the flow can transport the field across L.

What assumption is needed to replace the double curl by a Laplacian?
Use $\boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$ and uniform resistivity. Then $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times (\boldsymbol{B})) = -\nabla^2(\boldsymbol{B})$, yielding the diffusion term with coefficient $D_B = \eta/\mu_0$.

9.5 Static MHD equilibrium and magnetic force balance

How can a plasma remain static while its pressure varies across the device? In a magnetized equilibrium, the current creates a magnetic force that balances the pressure gradient. The vector identity for this force separates magnetic pressure from field-line tension and makes cylindrical pinch configurations interpretable.

Learning objectives

- derive the static MHD equilibrium equation from the momentum balance
- separate magnetic-pressure and magnetic-tension contributions
- deduce the restrictions on pressure along field lines and current across them
- explain why the parallel current requires field-line geometry and closure
- interpret theta-pinch, cylindrical pinch, and plasma-beta examples

Unit ledger

Magnetic pressure $B^2/(2\mu_0)$ has the same unit as static pressure, Pa. The plasma beta $\beta = 2\mu_0 p/B^2$ is dimensionless.

Static isotropic equilibrium

Assumptions 164

Set $\partial\rho/\partial t = 0$ and $\boldsymbol{u} = \boldsymbol{0}$, neglect gravity and the bulk electric force under quasi-neutrality, use scalar pressure, and use reduced Ampere's law. The magnetic field remains divergence-free. These assumptions describe magnetohydrostatics.

The static momentum equation is

$$\nabla(p) = \boldsymbol{j} \times \boldsymbol{B} = \frac{\nabla \times (\boldsymbol{B}) \times \boldsymbol{B}}{\mu_0}$$

Using the vector identity $\nabla \times (\boldsymbol{B}) \times \boldsymbol{B} = (\boldsymbol{B} \cdot \nabla)\boldsymbol{B} - \nabla\left(B^2/2\right)$, this becomes

$$\nabla\left(p + \frac{B^2}{2\mu_0}\right) = \frac{1}{\mu_0}(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}$$

The left side is the gradient of gas plus magnetic pressure. The right side is field-line tension and vanishes for a straight, uniform field.

Dotting the force balance with $\boldsymbol{B}$ and with $\boldsymbol{j}$ gives

$$\boldsymbol{B} \cdot \nabla(p) = 0, \quad \boldsymbol{j} \cdot \nabla(p) = 0$$

Thus pressure is constant along each field line, and the current is tangent to pressure surfaces when the scalar-pressure equilibrium is valid. The perpendicular current follows by crossing the equilibrium with $\boldsymbol{B}$:

$$\boldsymbol{j}_\perp = \frac{\boldsymbol{B} \times \nabla(p)}{B^2}$$

The force balance fixes only this perpendicular component. Write the total current as

$$\boldsymbol{j} = j_\parallel\left(\frac{\boldsymbol{B}}{B}\right) + \boldsymbol{j}_\perp$$

Because reduced Ampere's law also implies $\nabla \cdot (\boldsymbol{j}) = 0$, the parallel component obeys the magnetic differential equation

$$\boldsymbol{B} \cdot \nabla\left(\frac{j_\parallel}{B}\right) = -\nabla \cdot (\boldsymbol{j}_\perp)$$

Thus $j_\parallel$ requires field-line geometry and boundary or closure data; it is not determined by the local pressure balance alone.

The ratio of thermal to magnetic pressure is the plasma beta,

$$\beta = \frac{2\mu_0 p}{B^2}$$

Low β means magnetic pressure exceeds thermal pressure; high β means the reverse. This local ratio compares pressures, not their gradients. The forces also depend on spatial variation and field-line curvature, so β alone does not determine an equilibrium or its stability.

Two canonical pinch geometries Definition 165

A θ-pinch uses an externally applied, primarily axial field and an azimuthal current: $\boldsymbol{B} = B_z(r)\boldsymbol{e}_z$ and $\boldsymbol{j} = j_\theta(r)\boldsymbol{e}_\theta$. Straight field lines have no curvature tension, so the radial balance is a total-pressure balance. A cylindrical or z-pinch uses an axial plasma current and a self-generated azimuthal field: $\boldsymbol{j} = j_z(r)\boldsymbol{e}_z$ and $\boldsymbol{B} = B_\theta(r)\boldsymbol{e}_\theta$. The curved field then contributes an inward tension term.

The two radial balances make the distinction quantitative:

For a θ-pinch,

$$\frac{\mathrm{d}p + \frac{B_z^2}{2\mu_0}}{\mathrm{d}r} = 0$$

For a cylindrical or z-pinch,

$$\frac{\mathrm{d}p + \frac{B_\theta^2}{2\mu_0}}{\mathrm{d}r} + \frac{B_\theta^2}{\mu_0 r} = 0$$

The second term is the curvature tension of the azimuthal field. Both relations are special reductions of the same vector equilibrium equation; neither replaces the general force balance.

Derivation: magnetic pressure, tension, and pinch balance

Step: Decompose the magnetic force

Begin with the static momentum equation and reduced Ampere's law:

$$\boldsymbol{0} = -\boldsymbol{\nabla}(p) + \boldsymbol{j} \times \boldsymbol{B}$$

$$\boldsymbol{j} = \frac{\boldsymbol{\nabla} \times (\boldsymbol{B})}{\mu_0}.$$

Therefore,

$$\boldsymbol{\nabla}(p) = \frac{(\boldsymbol{\nabla} \times (\boldsymbol{B})) \times \boldsymbol{B}}{\mu_0}.$$

Use the standard identity

$$\boldsymbol{\nabla}\left(\frac{B^2}{2}\right) = (\boldsymbol{B} \cdot \boldsymbol{\nabla})\boldsymbol{B} + \boldsymbol{B} \times (\boldsymbol{\nabla} \times (\boldsymbol{B})).$$

Since $\boldsymbol{B} \times \boldsymbol{\nabla} \times (\boldsymbol{B}) = -\boldsymbol{\nabla} \times (\boldsymbol{B}) \times \boldsymbol{B}$, rearrangement gives

$$\boldsymbol{\nabla} \times (\boldsymbol{B}) \times \boldsymbol{B} = (\boldsymbol{B} \cdot \boldsymbol{\nabla})\boldsymbol{B} - \boldsymbol{\nabla}\left(\frac{B^2}{2}\right).$$

Substitution separates magnetic pressure from field-line tension:

$$\boldsymbol{\nabla}\left(p + \frac{B^2}{2\mu_0}\right) = \frac{(\boldsymbol{B} \cdot \boldsymbol{\nabla})\boldsymbol{B}}{\mu_0}.$$

The gradient on the left describes compression of the field magnitude; the term on the right describes the tension of curved field lines.

Step: Derive the pressure and current constraints

Dot the force balance with $\boldsymbol{B}$. The right-hand side vanishes, so

$$\boldsymbol{B} \cdot \boldsymbol{\nabla}(p) = 0.$$

Dot it with $\boldsymbol{j}$ instead. Again the right-hand side vanishes:

$$\boldsymbol{j} \cdot \boldsymbol{\nabla}(p) = 0.$$

Next cross the force balance from the left with $\boldsymbol{B}$:

$$\boldsymbol{B} \times \boldsymbol{\nabla}(p) = \boldsymbol{B} \times (\boldsymbol{j} \times \boldsymbol{B}).$$

The triple-product identity gives

$$\boldsymbol{B} \times (\boldsymbol{j} \times \boldsymbol{B}) = B^2 \boldsymbol{j} - \boldsymbol{B}(\boldsymbol{B} \cdot \boldsymbol{j}).$$

Thus the current perpendicular to the field is

$$\boldsymbol{j}_\perp = \frac{\boldsymbol{B} \times \boldsymbol{\nabla}(p)}{B^2}.$$

Decompose the current as

$$\boldsymbol{j} = j_\parallel \left(\frac{\boldsymbol{B}}{B} \right) + \boldsymbol{j}_\perp.$$

Because reduced Ampere's law and $\boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$ imply $\boldsymbol{\nabla} \cdot (\boldsymbol{j}) = 0$, the parallel component obeys

$$0 = \boldsymbol{\nabla} \cdot (\boldsymbol{j}) = \boldsymbol{\nabla} \cdot \left(\frac{j_\parallel \boldsymbol{B}}{B} \right) + \boldsymbol{\nabla} \cdot (\boldsymbol{j}_\perp) = \boldsymbol{B} \cdot \boldsymbol{\nabla} \left(\frac{j_\parallel}{B} \right) + \boldsymbol{\nabla} \cdot (\boldsymbol{j}_\perp).$$

A boundary condition or a separate closure is needed to select the parallel-current solution along each field line.

Step: Reduce the equation to a straight-field pinch

For a straight cylindrical field $\boldsymbol{B} = B_z(r)\boldsymbol{e}_z$, reduced Ampere's law gives

$$\boldsymbol{j} = -\frac{1}{\mu_0} \frac{\mathrm{d}B_z}{\mathrm{d}r} \boldsymbol{e}_\theta.$$

The radial force balance becomes

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{B_z \frac{\mathrm{d}B_z}{\mathrm{d}r}}{\mu_0},$$

so

$$\frac{\mathrm{d}p + \frac{B_z^2}{2\mu_0}}{\mathrm{d}r} = 0.$$

For a θ-pinch with a uniform axial field inside, volume current and magnetic tension can vanish in the interior. The pressure change is then balanced by the magnetic-pressure change across the boundary.

Step: Include field-line curvature in a z-pinch

For a cylindrical or z-pinch, take $\boldsymbol{B} = B_\theta(r)\boldsymbol{e}_\theta$ and $\boldsymbol{j} = j_z(r)\boldsymbol{e}_z$. Cylindrical Ampere's law gives

$$j_z = \frac{1}{\mu_0 r} \frac{\mathrm{d}rB_\theta}{\mathrm{d}r}.$$

Since $\boldsymbol{e}_z \times \boldsymbol{e}_\theta = -\boldsymbol{e}_r$, the radial balance is

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\left(\frac{B_\theta}{\mu_0 r} \right) \frac{\mathrm{d}rB_\theta}{\mathrm{d}r}.$$

Expanding the derivative yields

$$\frac{\mathrm{d}p + \frac{B_\theta^2}{2\mu_0}}{\mathrm{d}r} + \frac{B_\theta^2}{\mu_0 r} = 0.$$

The final term is the inward magnetic tension from curved field lines.

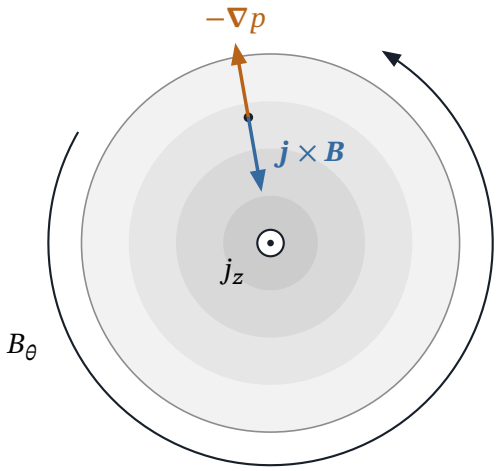


Figure 56: Static force balance $\nabla p = \boldsymbol{j} \times \boldsymbol{B}$ in a z-pinch cross section. Pressure (shading) peaks on the axis, so $-\nabla p$ pushes outward; the axial current j_z and its azimuthal field B_θ give an inward $\boldsymbol{j} \times \boldsymbol{B}$ of equal size.

Rechenbeispiel Rechenbeispiel 166

Context: a straight theta-pinch has constant total pressure across its boundary. Inside, let $p_{\mathrm{in}} = 1.00 \cdot 10^6$ Pa and $B_{z,\mathrm{in}} = 1.00$ T; outside, let $B_{z,\mathrm{out}} = 1.20$ T.

Assumptions: scalar static equilibrium, straight axial field, and no field-line curvature contribution in the pressure balance.

Target: report the outside gas pressure and the inside plasma beta.

Numerical result: $p_{\mathrm{out}} = 8.25 \cdot 10^5$ Pa and $\beta_{\mathrm{in}} = 2.51$ (dimensionless).

Geometric constraints of MHD equilibrium Interpretation 167

The equation $\nabla(p) = \boldsymbol{j} \times \boldsymbol{B}$ constrains the geometry. Pressure cannot vary along a field line, and current-driven force is perpendicular to the field. In a pinch, a pressure profile therefore determines the magnetic-field profile together with boundary conditions; the field is not chosen independently of the plasma pressure.

Summary Summary 168

Static MHD balances pressure force against magnetic force:

$$\nabla(p) = \frac{(\nabla \times (\boldsymbol{B})) \times \boldsymbol{B}}{\mu_0},$$

or equivalently the gradient of gas plus magnetic pressure balances field-line tension. Pressure is constant along field lines, the perpendicular current follows from the pressure gradient, and the parallel current obeys a magnetic differential equation. The dimensionless $\beta = 2\mu_0 p/B^2$ measures the relative strength of thermal and magnetic pressure.

Exam questions

(g) Write down and explain the static MHD equilibrium equation.

Verbatim from Plasma Physics Exam.pdf, p. 3, section 6(g)

Knowledge check

Knowledge check 169

- What is the static MHD force-balance equation?
- What does the magnetic-pressure/tension decomposition show?
- What is fixed locally about the current, and what determines its parallel component?
- What does a large plasma beta indicate?

Answers

What is the static MHD force-balance equation?

With the stated assumptions,

$$\nabla(p) = \boldsymbol{j} \times \boldsymbol{B} = \frac{(\nabla \times (\boldsymbol{B})) \times \boldsymbol{B}}{\mu_0}.$$

What does the magnetic-pressure/tension decomposition show?

It rewrites the force as

$$\nabla\left(p + \frac{B^2}{2\mu_0}\right) = \frac{(\boldsymbol{B} \cdot \nabla)\boldsymbol{B}}{\mu_0}.$$

The gradient term is magnetic pressure and the directional derivative is field-line tension.

What is fixed locally about the current, and what determines its parallel component?

The pressure balance fixes $\boldsymbol{j}_\perp = \boldsymbol{B} \times \nabla(p)/B^2$. The parallel component is constrained by $\boldsymbol{B} \cdot \nabla\left(j_\parallel/B\right) = -\nabla \cdot (\boldsymbol{j}_\perp)$ and therefore requires field-line geometry and boundary or closure data.

What does a large plasma beta indicate?

A large $\beta = 2\mu_0 p/B^2$ means thermal pressure exceeds magnetic pressure locally. Whether magnetic stresses can balance its gradient also depends on the pressure profile, field geometry, and boundary conditions; the ratio alone does not decide force balance.

10. Collisions and plasma conductivity

Gradients and external forces drive net transport; collisions set how quickly momentum relaxes. Inhomogeneity produces diffusion, while an external force produces mobility and electrical current. Collisions alone create no net flux in a homogeneous equilibrium. Resistivity and conductivity models are built on the collision frequencies.

Two routes to a transport coefficient Key idea 170

A collision model supplies a rate at which momentum is redirected or removed. The same bookkeeping supports neutral-collision transport and Coulomb transport. The resulting coefficient is meaningful only together with its collision regime, velocity average, and unit convention.

10.1 Collision bookkeeping and transport mechanisms

What does it mean to say that a particle collides? The useful answer is a rate, a mean free path, and a momentum-transfer rule. A cross section is an effective area; it may depend strongly on relative speed and scattering angle.

Learning objectives

- distinguish collision frequency, collision time, cross section, and mean free path
- derive the collision rate from a path-length probability
- identify diffusion and force-driven mobility as distinct transport routes
- state the Knudsen ordering that separates fluid-like from collisionless motion

Unit ledger

A cross section σ_{ab} is in m^2, mean free path is in m, and collision frequency ν_{ab} is in s^{-1}. A normalized collision count such as $\nu\tau$ and a Knudsen number are dimensionless.

Rate, time, and length scales Definition 171

For a test species a moving through targets b, define the effective collision frequency by $\nu_{ab} = n_b\langle\sigma_{ab}v_{\mathrm{rel}}\rangle$. The brackets denote the prescribed velocity-space average. The collision time is $\tau_{ab} = 1/\nu_{ab}$ and, with the same averaging convention, the mean free path is $\lambda_{ab} = \langle v_{\mathrm{rel}}\rangle/\nu_{ab}$. If the cross section is independent of speed, this becomes $\lambda_{ab} = 1/(n_b\sigma_{ab})$.

With a total scattering cross section, τ_{ab} is a mean waiting time between encounters. With a momentum-transfer cross section, it is a transport relaxation time and λ_{ab} is a relaxation length; many small-angle encounters can occur before directed momentum relaxes.

The two transport mechanisms are separated conceptually. A spatial gradient produces a diffusive particle flux, schematically $\boldsymbol{G}_{\mathrm{diff}} = -D\boldsymbol{\nabla}(n)$, while an external force produces a drift velocity, schematically $\boldsymbol{u}_{\mathrm{mob}} = \mu_{\mathrm{mob}}\boldsymbol{F}$. For an electric force, the charge-weighted drift contributes to the current $\boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$. These are macroscopic summaries; the microscopic coefficient still depends on the collision operator and the ordering of scales.

Here D is a particle diffusivity in $\mathrm{m}^2\,\mathrm{s}^{-1}$, and $\boldsymbol{G}_{\mathrm{diff}}$ is particle flux in $\mathrm{m}^{-2}\,\mathrm{s}^{-1}$. The coefficient μ_{mob} in this force-based definition is mobility, with units $\mathrm{s\,kg}^{-1}$; it is not the magnetic moment of Chapter 5.

An electric-field mobility instead relates velocity directly to the electric field and includes the charge factor from the force.

A dilute binary-collision model Assumptions 172

Treat encounters as independent binary events. The target population is dilute enough that a particle samples one local density during an encounter, and the effective cross section includes the momentum transfer relevant to the observable. A collision frequency is therefore a local model parameter.

Derivation: path probability and the mean free path

Here σ_{ab} is the total event cross section at fixed relative speed. The survival probability counts encounters, not momentum relaxation weighted by scattering angle.

Step: Write the survival equation

During a path segment $\mathrm{d}\ell$, a particle sweeps an effective volume $\sigma_{ab}\,\mathrm{d}\ell$. The expected number of targets is $n_b\sigma_{ab}\,\mathrm{d}\ell$. If $P_0(\ell)$ is the probability of no collision, independent segments give

$$\mathrm{d}P_0 = -n_b\sigma_{ab}P_0\,\mathrm{d}\ell.$$

With $P_0(0) = 1$, integration gives

$$P_0(\ell) = \exp(-n_b\sigma_{ab}\ell).$$

Step: Compute the mean free path and rate

The mean distance is the survival-probability integral

$$\lambda_{ab} = \int_0^\infty P_0(\ell)\,\mathrm{d}\ell = \frac{1}{n_b\sigma_{ab}}.$$

For fixed relative speed, $\mathrm{d}\ell = v_{\mathrm{rel}}\,\mathrm{d}t$, so

$$\nu_{ab} = n_b\sigma_{ab}v_{\mathrm{rel}}.$$

Averaging over the relative-velocity distribution replaces the product by $n_b\langle\sigma_{ab}v_{\mathrm{rel}}\rangle$ and fixes the corresponding convention for the mean free path.

Step: Compare with a macroscopic length

For length L, the Knudsen number is

$$K_{\mathrm{n}} = \frac{\lambda}{L}.$$

The fluid ordering is $K_{\mathrm{n}} \ll 1$, $K_{\mathrm{n}} \approx 1$ marks a transition, and $K_{\mathrm{n}} \gg 1$ requires kinetic or ballistic reasoning.

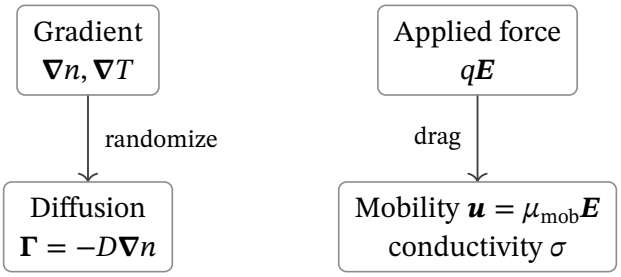


Figure 57: Gradients and forces drive transport; collisions set the response. Collisions alone drive no current in a homogeneous equilibrium.

Rechenbeispiel Rechenbeispiel 173

Consider a test population moving through stationary targets with $n_b = 1.0 \cdot 10^{19}\,\mathrm{m}^{-3}$, $\sigma_{ab} = 2.0 \cdot 10^{-19}\,\mathrm{m}^2$, representative relative speed $v_{\mathrm{rel}} = 1.0 \cdot 10^5\,\mathrm{m\,s}^{-1}$, and macroscopic length $L = 10\,\mathrm{m}$.

Assumptions: independent binary encounters, speed-independent effective cross section, and the fixed-speed convention $\nu_{ab} = n_b \sigma_{ab} v_{\mathrm{rel}}$.

Target: report the collision frequency, collision time, mean free path, and Knudsen number.

Numerical result: $\nu_{ab} = 2.0 \cdot 10^5\,\mathrm{s}^{-1}$, $\tau_{ab} = 5.0 \cdot 10^{-6}\,\mathrm{s}$, $\lambda_{ab} = 0.50\,\mathrm{m}$, and $K_{\mathrm{n}} = 0.050$ (dimensionless).

A collision frequency is an averaged observable Interpretation 174

The same particles have a distribution of relative speeds and scattering angles. The symbol ν is useful only after the averaging and momentum- transfer convention have been stated. Confusing a geometric cross section with a momentum-transfer cross section can change both a transport coefficient and its physical interpretation.

Summary Summary 175

Collision bookkeeping starts with an effective cross section and a target density. It gives $\nu = n\langle \sigma v \rangle$, $\tau = 1/\nu$, and $\lambda = \langle v \rangle / \nu$. Diffusion responds to inhomogeneity, mobility responds to a force, and the Knudsen number states whether a local fluid description is plausible.

Knowledge check — Knowledge check 176

- What is the difference between collision time and collision frequency?
- What are the units of a collision cross section and mean free path?
- Which dimensionless parameter compares a mean free path with a device length?
- What distinguishes diffusion from force-driven mobility?

Answers

What is the difference between collision time and collision frequency?
They are reciprocals under the same averaging convention: $\tau = 1/\nu$. For a total event cross section, these are the encounter rate and mean waiting time. For a momentum-transfer cross section, they describe directed-momentum relaxation, which can require many small-angle encounters.

What are the units of a collision cross section and mean free path?
A cross section is an area in m^2, and a mean free path is a length in m. The collision frequency is in s^{-1}.

Which dimensionless parameter compares a mean free path with a device length?
The Knudsen number $K_\mathrm{n} = \lambda/L$ does so. Small K_n supports a local collisional fluid model, whereas large K_n signals kinetic or ballistic transport.

What distinguishes diffusion from force-driven mobility?
Diffusion is driven by a spatial gradient and has the schematic flux $-D\boldsymbol{\nabla}(n)$. Mobility is driven by an external force and produces a drift proportional to that force. Both coefficients depend on collisions.

10.2 Weakly ionized plasmas: neutral collisions

How does a neutral background impede charged-particle motion? Neutral atoms and molecules act as compact scattering centers. For a weakly ionized plasma, the neutral density is usually much larger than the charged density, so electron–neutral and ion–neutral collisions can dominate even though the plasma remains electrically active.

Learning objectives

- derive the neutral-collision frequency and mean free path
- distinguish total scattering from momentum-transfer cross section
- write the neutral drag force for stationary and moving neutrals
- explain why momentum and energy relaxation need not have the same rate

Weak ionization and stationary neutrals — Assumptions 177

Use $n_\mathrm{n} \gg n_\mathrm{e}$, a neutral temperature and velocity that vary slowly on the collision time, and binary electron–neutral or ion–neutral encounters. For a stationary neutral background $\boldsymbol{u}_\mathrm{n} = \mathbf{0}$, the leading drag is proportional to the charged species drift velocity. Finite neutral mass, excitation, ionization, and charge exchange can require additional collision channels.

The momentum-transfer cross section is the angle-weighted quantity:

$$\sigma_{\mathrm{mt}}(v) = \int (1 - \cos\theta_{\mathrm{sc}}) \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\, \mathrm{d}\Omega$$

Here θ_{sc} is the deflection angle and $\mathrm{d}\Omega$ is a solid-angle element; neither angle carries a dimensional unit. The differential cross section $\mathrm{d}\sigma/(\mathrm{d}\Omega)$ gives scattering area per unit solid angle. The factor $1 - \cos\theta_{\mathrm{sc}}$ weights each direction by its loss of forward momentum. A scattering process dominated by small deflections can therefore have a large total cross section but a small momentum-transfer cross section.

Neutral collision rate and drag — Definition 178

For electrons in a neutral background, $\nu_{\mathrm{en}} = n_{\mathrm{n}} \langle \sigma_{\mathrm{mt,en}} v_{\mathrm{e}} \rangle$ and $\lambda_{\mathrm{en}} = \langle v_{\mathrm{e}} \rangle / \nu_{\mathrm{en}}$. The analogous ion rate is $\nu_{\mathrm{in}} = n_{\mathrm{n}} \langle \sigma_{\mathrm{mt,in}} v_{\mathrm{i}} \rangle$. If neutrals are stationary, the electron drag force density is $\boldsymbol{R}_{\mathrm{en}} = -m_{\mathrm{e}} n_{\mathrm{e}} \nu_{\mathrm{en}} \boldsymbol{u}_{\mathrm{e}}$. For a moving neutral population, replace the drift by $\boldsymbol{u}_{\mathrm{e}} - \boldsymbol{u}_{\mathrm{n}}$.

Derivation: neutral drag from momentum transfer

Step: Count momentum transfer

A collision changes the average directed momentum by an amount that scales as

$$\Delta \boldsymbol{p}_a \text{ scales as } m_a(\boldsymbol{u}_a - \boldsymbol{u}_n).$$

During $\mathrm{d}t$, the number of encounters per particle is $\nu_{an}\, \mathrm{d}t$. Multiplying the momentum change by the charged-particle density gives the force-density scaling

$$\boldsymbol{R}_{an} \text{ scales as } -m_a n_a \nu_{an} (\boldsymbol{u}_a - \boldsymbol{u}_n).$$

Step: Check the stationary-neutral limit

The force is restorative: it vanishes when the charged species and neutral background share a velocity. For stationary neutrals,

$$\boldsymbol{R}_{an} = -m_a n_a \nu_{an} \boldsymbol{u}_a.$$

The rate is a momentum-transfer rate, not necessarily the frequency of every microscopic encounter.

Step: Separate momentum and energy relaxation

Energy transfer has an additional mass-ratio dependence. An electron can reverse direction in an elastic encounter with a heavy neutral while transferring only a small fraction of its kinetic energy. Excitation and ionization are inelastic energy-loss channels; they belong in an energy equation.

Rechenbeispiel — Rechenbeispiel 179

Given a weakly ionized gas with neutral density $n_{\mathrm{n}} = 2.0 \cdot 10^{20}\ \mathrm{m}^{-3}$, momentum-transfer cross section $\sigma_{\mathrm{mt,en}} = 2.0 \cdot 10^{-19}\ \mathrm{m}^2$, and representative electron speed $v_{\mathrm{e}} = 1.0 \cdot 10^{5}\ \mathrm{m\,s}^{-1}$.

Assumptions: stationary neutrals, a fixed representative electron speed, and the momentum-transfer rate $\nu_{\mathrm{en}} = n_{\mathrm{n}} \sigma_{\mathrm{mt,en}} v_{\mathrm{e}}$.

Target: report the effective collision frequency and mean free path.

Numerical result: $\nu_{\mathrm{en}} = 4.0 \cdot 10^{6}\ \mathrm{s}^{-1}$ and $\lambda_{\mathrm{en}} = 2.5 \cdot 10^{-2}\ \mathrm{m}$.

Weak ionization is a collision hierarchy Interpretation 180

The dominant rate is selected by the product of density, cross section, and relative speed. A neutral-rich plasma can therefore be strongly collisional for electron momentum while still having a long energy- relaxation time. The transport model should name the channel being relaxed.

Summary Summary 181

Neutral collisions are described by momentum-transfer cross sections and effective rates such as $\nu_{en} = n_{n\langle\sigma_{mt,en}v_e\rangle}$. Stationary neutrals exert a drag $-mn\nu\boldsymbol{u}$, while moving neutrals replace the velocity by the relative drift. Momentum and energy relaxation are distinct when the neutral is much heavier than the charged particle.

Knowledge check Knowledge check 182

- Why is a momentum-transfer cross section more useful than a total cross section for conductivity?
- How does the neutral density enter the collision frequency?
- What is the drag force when the neutral background moves?
- Why can momentum relaxation be faster than energy relaxation in electron–neutral collisions?

Answers

Why is a momentum-transfer cross section more useful than a total cross section for conductivity?

Conductivity responds to directed momentum loss. The factor $1 - \cos\theta_{sc}$ weights each event by its momentum change, so small-angle scattering is not counted as a full reversal.

How does the neutral density enter the collision frequency?

It enters linearly: $\nu_{en} = n_{n\langle\sigma_{mt,en}v_e\rangle}$. Doubling the target density doubles the rate when the cross section and velocity distribution are unchanged.

What is the drag force when the neutral background moves?

Use the relative velocity: $\boldsymbol{R}_{en} = -m_e n_e \nu_{en}(\boldsymbol{u}_e - \boldsymbol{u}_n)$. It vanishes when the two populations share a velocity.

Why can momentum relaxation be faster than energy relaxation in electron–neutral collisions?

A heavy neutral can redirect an electron efficiently while taking only a small fraction of its kinetic energy. Directional momentum loss and energy exchange therefore have different mass-ratio scalings.

10.3 Fully ionized plasmas: Coulomb collisions

Why is a Coulomb collision not characterized by one hard-sphere radius? The interaction is long range, so most encounters produce small deflections and a few produce large deflections. A controlled collision model must set a lower strong-scattering scale and an upper shielding scale before it can sum the cumulative angular diffusion.

Learning objectives

- define the ninety-degree impact parameter and large-angle collision rate
- show why cumulative small-angle scattering produces a logarithm
- use Debye shielding as the upper impact-parameter cutoff
- state the weak-coupling condition behind a Coulomb collision frequency

Unit ledger

The impact parameter b_{90} and the reduced de Broglie wavelength $\lambda\!\!\!^{-}$ are in m; the Sommerfeld parameter η is dimensionless. The Coulomb cutoff logarithm $\ln\Lambda_{\mathrm{cut}}$ and plasma parameter $\Lambda = n_{\mathrm{e}}\lambda_D^3$ are dimensionless, as is $\ln\Lambda$.

Weakly coupled, fully ionized plasma — Assumptions 183

Use a plasma containing only charged particles on the collision scale, treat a heavy ion as stationary for the electron–ion estimate, and assume $\Lambda = n_{\mathrm{e}}\lambda_D^3 \gg 1$. The interaction is screened beyond approximately λ_D, and many small-angle encounters can be treated statistically. A strongly coupled plasma requires a different kinetic description.

Strong-deflection scale — Definition 184

For charges q_a and q_b with reduced mass m_r and relative speed v_{rel}, define the ninety-degree impact parameter by

$$b_{90} = \frac{|q_a q_b|}{4\pi\varepsilon_0 m_r v_{\mathrm{rel}}^2}.$$

For an electron scattering from a singly charged ion, $b_{90} = e^2/\left(4\pi\varepsilon_0 m_{\mathrm{e}} v_{\mathrm{e}}^2\right)$ in the heavy-ion approximation and $\sigma_{90} = \pi b_{90}^2$. A closest-approach convention based on equating kinetic and Coulomb potential energies differs by an order-one factor; the collision logarithm is insensitive to that convention at leading order.

The large-angle estimate is:

$$\nu_{90} \approx n_{\mathrm{i}}\sigma_{90} v_{\mathrm{e}} = \frac{n_{\mathrm{i}} e^4}{16\pi\varepsilon_0^2 m_{\mathrm{e}}^2 v_{\mathrm{e}}^3}$$

It is not the full electron–ion relaxation rate because the many more distant encounters contribute cumulatively.

Coulomb logarithm and electron–ion rate — Definition 185

For impact parameters b much larger than b_{90}, the deflection is small, $\theta_{\mathrm{sc}}(b) \approx 2b_{90}/b$. The screened upper cutoff is $b_{\max} \approx \lambda_D$. The broad-impact-parameter measure is recorded by $\Lambda_{\mathrm{cut}} = \lambda_D/b_{90}$ and its logarithm $\ln\Lambda_{\mathrm{cut}}$. Separately, define $\Lambda = n_{\mathrm{e}}\lambda_D^3$. The approximate scattering-rate convention of Inan and Golkowski (2011), Eq. (7.5), uses this plasma parameter [1]:

$$\nu_{ei} \approx \frac{\sqrt{2}\omega_{p,e}^4}{64\pi n_{\mathrm{e}}}\left(\frac{k_B T_{\mathrm{e}}}{m_{\mathrm{e}}}\right)^{-\frac{3}{2}} \ln\Lambda,$$

where $\omega_{p,e} = \sqrt{n_{\mathrm{e}} e^2/(\varepsilon_0 m_{\mathrm{e}})}$. The corresponding mean free path is $\lambda_{ei} = \langle v_{\mathrm{e}}\rangle/\nu_{ei}$.

These logarithms are not numerically equal. With the electron Debye length $\lambda_D = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$ and the Maxwellian mean speed used here, $\Lambda_{cut} = 32\Lambda$, so $\ln\Lambda_{cut} = \ln\Lambda + \ln(32)$. Treating the additive constant as negligible is only a leading-log approximation when $\ln\Lambda$ is large compared with it. The rate and numerical tasks below retain $\ln\Lambda$ and the cited approximate prefactor; the impact-parameter integral uses $\ln\Lambda_{cut}$.

The plasma parameter Λ also differs from the Debye-sphere particle count used in Chapter 4: $N_D = (4\pi/3)\Lambda$. Both are dimensionless and express the many-particle screening condition. Keeping these definitions separate avoids introducing an unintended factor into a quoted rate.

Derivation: cumulative small-angle scattering

Step: Accumulate deflection over impact parameters

For $b \gg b_{90}$, Rutherford scattering gives

$$\theta_{sc}(b) \approx \frac{2b_{90}}{b}.$$

During $\mathrm{d}t$, the number of target ions in the annulus from b to $b + \mathrm{d}b$ scales as

$$n_i v_{rel} 2\pi b\,\mathrm{d}b\,\mathrm{d}t.$$

The squared transverse kick scales as $v_{rel}^2\theta_{sc}(b)^2$. Hence the accumulated squared deflection has the scale

$$\frac{\mathrm{d}\langle\Delta v_\perp^2\rangle}{\mathrm{d}t} \propto n_i v_{rel}^3 b_{90}^2 \int_{b_{90}}^{\lambda_D} \frac{\mathrm{d}b}{b}.$$

The integral is

$$\ln\left(\frac{\lambda_D}{b_{90}}\right) = \ln\Lambda_{cut},$$

the Coulomb logarithm. Distant deflections therefore cannot be discarded merely because each individual kick is small.

Step: Insert the Maxwellian velocity scale

For Maxwellian electrons, use

$$\langle v_e\rangle = \sqrt{\frac{8k_B T_e}{\pi m_e}}.$$

Insert this scale and

$$\omega_{p,e}^2 = \frac{n_e e^2}{\varepsilon_0 m_e}.$$

This gives the scaling $\nu_{ei} \propto T_e^{-3/2}\ln\Lambda_{cut}$. It does not fix a kinetic transport prefactor. The displayed textbook estimate uses $\ln\Lambda$ instead, with the leading-log distinction stated above.

Step: State the weak-coupling condition

The ratio of distant to strong-scattering scales is large only when

$$\Lambda = n_e \lambda_D^3 \gg 1.$$

Otherwise independent binary encounters and a weak-coupling collision operator are not self-consistent.

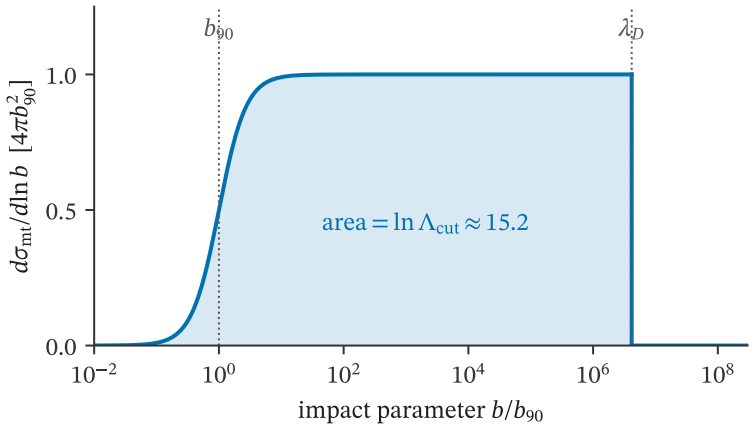


Figure 58: Momentum transfer per logarithmic interval of impact parameter, $2\pi b^2(1-\cos\chi)$ with $\tan(\chi/2) = b_{90}/b$, in units of $4\pi b_{90}^2$. Above b_{90} every decade of b contributes equally until screening cuts the integral at λ_D, so the shaded area is a logarithm, $\ln\Lambda_{cut}$. Here $\lambda_D/b_{90} = 4.2\times10^6$ from the example below: $\ln\Lambda_{cut} \approx 15.2 = \ln\Lambda + \ln 32$.

Rechenbeispiel Rechenbeispiel 186

For a fully ionized hydrogen plasma, use $n_e = 1.0\cdot10^{16}\,\mathrm{m^{-3}}$, $k_B T_e = 10\,\mathrm{eV} = 1.602\cdot10^{-18}\,\mathrm{J}$, $e = 1.602\cdot10^{-19}\,\mathrm{C}$, $\varepsilon_0 = 8.854\cdot10^{-12}\,\mathrm{F\,m^{-1}}$, $m_e = 9.109\cdot10^{-31}\,\mathrm{kg}$, and $\langle v_e\rangle = 2.12\cdot10^6\,\mathrm{m\,s^{-1}}$.

Assumptions: weak coupling $\Lambda \gg 1$, classical Coulomb scattering, a Debye upper cutoff, the heavy-ion approximation, and the Maxwellian mean-speed convention and plasma-parameter logarithm $\ln\Lambda$ stated above.

Target: report λ_D, Λ, $\ln\Lambda$, ν_{ei} using the collision-frequency convention above, and λ_{ei}.

Numerical result: $\lambda_D = 2.35\cdot10^{-4}\,\mathrm{m}$, $\Lambda = 1.30\cdot10^5$, $\ln\Lambda = 11.8$, $\nu_{ei} = 3.60\cdot10^3\,\mathrm{s^{-1}}$, and $\lambda_{ei} = 5.89\cdot10^2\,\mathrm{m}$.

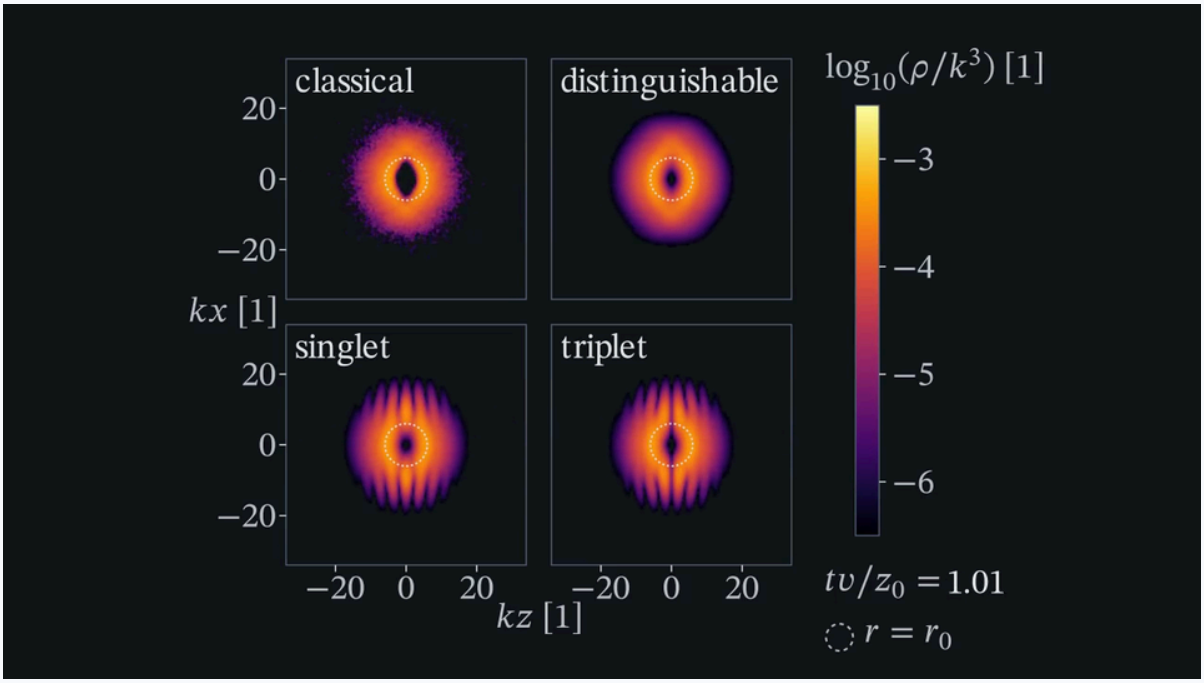
Screening makes the long-range interaction finite Interpretation 187

Without shielding, integrating over arbitrarily large impact parameters would make the cumulative rate ill-defined. Debye shielding supplies the physical upper scale. The lower scale is set by strong deflection or by a quantum correction when classical impact parameters become too small. The Coulomb logarithm is therefore the logarithm of a controlled scale ratio.

The classical deflection $\chi(b)$ assumes that the colliding particles follow trajectories on the scale b_{90}. A quantum particle is localized only to about its reduced de Broglie wavelength $\lambda\!\!\!^{-} = \hbar/(m_r v_{rel})$, where $\hbar$ is the reduced Planck constant. For two unit charges, the ratio of the two lengths is the Sommerfeld parameter

$$\eta = \frac{b_{90}}{\lambda\!\!\!^{-}} = \frac{e^2}{4\pi\varepsilon_0\hbar v_{rel}}.$$

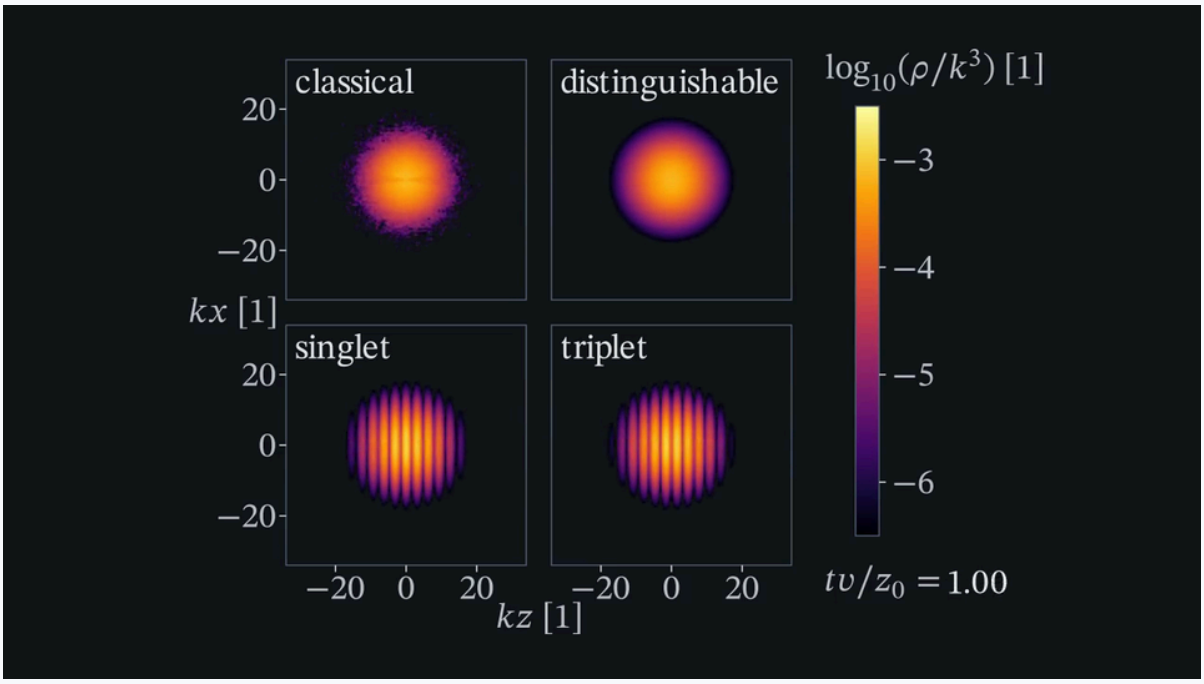
For $\eta \gg 1$ the trajectory picture holds; for $\eta \ll 1$ the wavelength exceeds b_{90} and the lower cutoff must be revised. The two animations follow an encounter of two electrons, $m_r = m_e/2$, in the relative coordinate $\boldsymbol{r} = \boldsymbol{x}_1 - \boldsymbol{x}_2$. A head-on classical pair turns around at $r_0 = 2b_{90} = 2\eta\lambda\!\!\!^{-}$.



Watch animation

Head-on encounter of two electrons at $\eta = 3$ (relative energy 0.76 eV), computed with kin6d. All four panels show the same encounter on one logarithmic colour scale: the density of the relative coordinate in the collision plane, with positions kx, kz and $k = 1/\lambda\!\!\!^{-}$, at time tv_{rel}/z_0, where $z_0 = 20\lambda\!\!\!^{-}$ is the initial separation (all with unit [1]). Top left: an ensemble of classical trajectories with the initial positions and momenta of the quantum packet (Wigner sampling). Top right: the quantum packet of distinguishable particles. Bottom: the singlet (symmetric) and triplet (antisymmetric) spatial states of identical electrons. Because the electrons are identical, each panel also shows the exchanged labelling $\boldsymbol{r} \to -\boldsymbol{r}$. The classical hole $r < r_0$ (dashed circle) is reproduced by the distinguishable quantum panel. Exchange adds interference fringes of opposite sign in singlet and triplet, whose average is the distinguishable panel, and the triplet vanishes on the plane $z = 0$.

Figure 59: Head-on encounter of two electrons at $\eta = 3$ (relative energy 0.76 eV), computed with kin6d. All four panels show the same encounter on one logarithmic colour scale: the density of the relative coordinate in the collision plane, with positions kx, kz and $k = 1/\lambda\!\!\!^{-}$, at time tv_{rel}/z_0, where $z_0 = 20\lambda\!\!\!^{-}$ is the initial separation (all with unit [1]). Top left: an ensemble of classical trajectories with the initial positions and momenta of the quantum packet (Wigner sampling). Top right: the quantum packet of distinguishable particles. Bottom: the singlet (symmetric) and triplet (antisymmetric) spatial states of identical electrons. Because the electrons are identical, each panel also shows the exchanged labelling $\boldsymbol{r} \to -\boldsymbol{r}$. The classical hole $r < r_0$ (dashed circle) is reproduced by the distinguishable quantum panel. Exchange adds interference fringes of opposite sign in singlet and triplet, whose average is the distinguishable panel, and the triplet vanishes on the plane $z = 0$.



Watch animation

The same encounter at $\eta = 0.03$, the relative speed of an electron pair in a 10 keV plasma (relative energy 7.6 keV), on the same axes, clock and colour scale. Now $r_0 = 0.06\lambda\!\!\!^{-}$ lies far inside the packet. The quantum packet passes almost freely: the scattered wave is far below the colour scale. The singlet and triplet fringes are the exchange symmetrization of two overlapping packets; they appear even without interaction. The classical ensemble still keeps a sharp, unresolved hole on the axis, a trajectory feature that the quantum packet does not show.

Figure 60: The same encounter at $\eta = 0.03$, the relative speed of an electron pair in a 10 keV plasma (relative energy 7.6 keV), on the same axes, clock and colour scale. Now $r_0 = 0.06\lambda\!\!\!^{-}$ lies far inside the packet. The quantum packet passes almost freely: the scattered wave is far below the colour scale. The singlet and triplet fringes are the exchange symmetrization of two overlapping packets; they appear even without interaction. The classical ensemble still keeps a sharp, unresolved hole on the axis, a trajectory feature that the quantum packet does not show.

The statistics of the deflections stay Rutherford: the quantum Coulomb cross section of distinguishable particles equals the classical Rutherford formula at every η (identical electrons add an exchange term, the Mott cross section). What changes for $\eta \ll 1$ is the range of impact parameters in which $\chi(b)$ is meaningful, so the lower cutoff moves from b_{90} to $\lambda\!\!\!^{-}$, and $b_{\min} \approx \max(b_{90}, \lambda\!\!\!^{-})$. For an electron–ion pair with $v_{\mathrm{rel}} = \sqrt{k_B T_\mathrm{e}/m_\mathrm{e}}$, $n_\mathrm{e} = 1 \cdot 10^{20}\,\mathrm{m}^{-3}$ and $k_B T_\mathrm{e} = 10\,\mathrm{keV}$: $\lambda_D = 7.4 \cdot 10^{-5}\,\mathrm{m}$, $b_{90} = 1.4 \cdot 10^{-13}\,\mathrm{m}$, $\lambda\!\!\!^{-} = 2.8 \cdot 10^{-12}\,\mathrm{m}$ and $\eta \approx 0.05$. The classical logarithm $\ln(\lambda_D/b_{90}) = 20.1$ drops to $\ln(\lambda_D/\lambda\!\!\!^{-}) = 17.1$, the electron–ion value of the NRL Plasma Formulary [8]. The exact Coulomb phase shifts interpolate between both limits by subtracting Bloch's correction [9],

$$\ln\!\left(\frac{\lambda_D}{b_{90}}\right) - [\ln\eta - \operatorname{Re}\psi(1 + i\eta)],$$

where ψ is the digamma function. The bracket vanishes for $\eta \gg 1$ and tends to $\ln\eta + \gamma$ for $\eta \ll 1$, with Euler's constant $\gamma \approx 0.577$, which gives $\ln(\lambda_D/\lambda\!\!\!^{-}) - \gamma$.

Summary

Summary 188

Fully ionized collisions are dominated in number by small-angle encounters but in cumulative effect by the logarithmic interval between b_{90} and λ_D. The weak-coupling parameter $\Lambda = n_e \lambda_D^3 \gg 1$ justifies the statistical treatment, and the electron–ion rate scales as $T_e^{-3/2} \ln \Lambda$ in the stated convention.

Knowledge check

Knowledge check 189

- Why is the ninety-degree cross section not the complete Coulomb collision rate?
- How does the ninety-degree impact parameter scale with speed?
- What sets the upper impact-parameter cutoff in a plasma?
- What condition makes the weakly coupled Coulomb model self-consistent?

Answers

Why is the ninety-degree cross section not the complete Coulomb collision rate?
Distant encounters are much more numerous. Each gives a small deflection, but their squared kicks add over impact parameters and produce the cutoff logarithm $\ln \Lambda_{\text{cut}}$. The cited rate estimate instead uses the plasma-parameter logarithm $\ln \Lambda$.

How does the ninety-degree impact parameter scale with speed?
For fixed charges and reduced mass,

$$b_{90} = \frac{|q_a q_b|}{4\pi\varepsilon_0 m_r v_{\text{rel}}^2},$$

so faster particles have a smaller strong-deflection scale.

What sets the upper impact-parameter cutoff in a plasma?
Collective Debye shielding sets $b_{\max} \approx \lambda_D$. Encounters at larger scales are screened by the surrounding plasma and cannot be added as independent unscreened Coulomb kicks.

What condition makes the weakly coupled Coulomb model self-consistent?
The plasma parameter must be large: $\Lambda = n_e \lambda_D^3 \gg 1$. It implies many particles occupy a Debye sphere and supports a statistical, weak-coupling description.

10.4 Specific resistivity and Spitzer scaling

How does collisional drag become a resistivity? The current is a measure of relative charge-carrier motion, while drag removes that relative momentum. In a steady scalar response, the electric force balances that drag. Expressing the drag in terms of current then gives the resistivity; the collision model supplies the momentum-transfer frequency.

Learning objectives

- derive scalar resistivity from a drag force and the current
- distinguish neutral-limited and Coulomb-limited resistivity
- obtain the temperature and density scaling of Spitzer resistivity
- connect resistivity to conductivity

Unit ledger
The resistivity η in $\boldsymbol{E} = \eta \boldsymbol{j}$ is in $\Omega\,\mathrm{m}$, and conductivity $\sigma = 1/\eta$ is in $\mathrm{S\,m^{-1}}$. Here σ denotes conductivity; the collision cross sections earlier in this chapter have area units.

Linear, isotropic drag — Assumptions 190

Use a small drift relative to the random thermal speed, so that the collision frequency can be evaluated from the equilibrium distribution. Treat the drag as linear in relative velocity and use a scalar frequency ν for the chosen collision channel. Magnetization and tensor response are postponed to the conductivity section.

Resistivity from momentum transfer — Definition 191

For a hydrogen plasma, the electron–ion drag force density can be written $\boldsymbol{R}_{\mathrm{e\,i}} = m_{\mathrm{e}} n_{\mathrm{e}} \nu_{\mathrm{e\,i}} (\boldsymbol{u}_{\mathrm{i}} - \boldsymbol{u}_{\mathrm{e}})$. Since $\boldsymbol{j} = e n_{\mathrm{e}} (\boldsymbol{u}_{\mathrm{i}} - \boldsymbol{u}_{\mathrm{e}})$, the force per electron charge density is $\eta \boldsymbol{j}$ with $\eta = m_{\mathrm{e}} \nu_{\mathrm{e\,i}} / \left(n_{\mathrm{e}} e^2\right)$. More generally use $\nu = \nu_{\mathrm{en}}$ for a stationary neutral background or the sum of the relevant momentum-transfer frequencies. The scalar conductivity is $\sigma_{\mathrm{dc}} = 1/\eta = n_{\mathrm{e}} e^2 / (m_{\mathrm{e}} \nu)$.

Spitzer scaling in the stated collision convention — Governing law 192

$$\nu_{\mathrm{e\,i}} \approx \frac{\sqrt{2}\omega_{p,e}^4}{64\pi n_{\mathrm{e}}} \left(\frac{k_B T_{\mathrm{e}}}{m_{\mathrm{e}}}\right)^{-\frac{3}{2}} \ln \Lambda$$

gives

$$\eta_{\mathrm{Sp}} \approx \frac{\pi}{2\sqrt{2}} \frac{e^2 \sqrt{m_{\mathrm{e}}}}{(4\pi\varepsilon_0)^2 (k_B T_{\mathrm{e}})^{\frac{3}{2}}} \ln \Lambda.$$

Thus η_{Sp} is approximately independent of density at fixed temperature, apart from the weak density dependence hidden in $\ln \Lambda$. Its dominant temperature scaling is $\eta_{\mathrm{Sp}} \propto T_{\mathrm{e}}^{-3/2}$, while σ_{Sp} scales as

$$\frac{T_{\mathrm{e}}^{\frac{3}{2}}}{\ln \Lambda}.$$

Derivation: drag, current, and Spitzer scaling

Step: Relate friction to current

Start with the electron–ion frictional force density

$$\boldsymbol{R}_{\mathrm{e\,i}} = m_{\mathrm{e}} n_{\mathrm{e}} \nu_{\mathrm{e\,i}} (\boldsymbol{u}_{\mathrm{i}} - \boldsymbol{u}_{\mathrm{e}}).$$

For singly charged hydrogen and quasi-neutrality,

$$\boldsymbol{j} = e n_{\mathrm{e}} (\boldsymbol{u}_{\mathrm{i}} - \boldsymbol{u}_{\mathrm{e}}).$$

Dividing by the charge-density scale $e n_{\mathrm{e}}$ gives

$$\frac{\boldsymbol{R}_{\mathrm{e\,i}}}{e n_{\mathrm{e}}} = \frac{m_{\mathrm{e}} \nu_{\mathrm{e\,i}} \boldsymbol{j}}{n_{\mathrm{e}} e^2}.$$

Step: Identify scalar resistivity and conductivity

The coefficient of $\boldsymbol{j}$ is the resistivity:

$$\eta = \frac{m_e \nu_{e\,i}}{n_e e^2}.$$

Its reciprocal is the DC conductivity:

$$\sigma_{dc} = \frac{n_e e^2}{m_e \nu_{e\,i}}.$$

Step: Insert the Coulomb rate

Use

$$\omega_{p,e}^4 = \left(\frac{n_e e^2}{\varepsilon_0 m_e}\right)^2.$$

Inserting the Coulomb rate gives

$$\eta_{Sp} = \frac{m_e}{n_e e^2}\left[\frac{\sqrt{2}\omega_{p,e}^4}{64\pi n_e}\right]\left(\frac{k_B T_e}{m_e}\right)^{-\frac{3}{2}} \ln\Lambda.$$

Cancelling explicit density factors leaves the displayed temperature scaling and only weak logarithmic density dependence. Density remains physically important because it changes the current for a given drift and changes Λ.

Rechenbeispiel

Rechenbeispiel 193

Using the Coulomb rate from the previous section for $n_e = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$ and $\nu_{e\,i} = 3.60 \cdot 10^3\,\mathrm{s^{-1}}$, calculate the scalar resistivity and DC conductivity with $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$ and $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$.

Assumptions: quasi-neutral singly charged hydrogen, scalar linear electron–ion drag, and no magnetic-field or pressure contribution.

Target: report the scalar resistivity and DC conductivity.

Numerical result: $\eta_{Sp} = 1.28 \cdot 10^{-5}\,\Omega\,\mathrm{m}$ and $\sigma_{dc} = 7.83 \cdot 10^4\,\mathrm{S\,m^{-1}}$.

More carriers do not automatically mean lower resistivity

Interpretation 194

Adding carriers raises the current available at a given drift, but in a fully ionized plasma it also raises the collision rate. At fixed temperature these effects largely cancel in the scalar Spitzer resistivity; the conductivity still tells how much current a specified electric field drives.

Summary

Summary 195

Linear drag converts the collision frequency into $\eta = m_e\nu/(n_e e^2)$ and $\sigma_{dc} = n_e e^2/(m_e\nu)$. Neutral and Coulomb channels supply different ν. With the stated Coulomb convention, Spitzer resistivity scales mainly as $T_e^{-3/2}\ln\Lambda$ and is nearly density-independent at fixed temperature.

Knowledge check

Knowledge check 196

- What physical quantity is divided by the charge density to obtain resistivity?
- How are resistivity and conductivity related?
- What is the dominant temperature scaling of Spitzer resistivity?
- Why must the collision channel be stated before quoting a resistivity?

Answers

What physical quantity is divided by the charge density to obtain resistivity?
The collisional drag force density is divided by the charge density scale and expressed per current. This gives $\eta = m_e\nu/\left(n_e e^2\right)$ in the scalar model.

How are resistivity and conductivity related?
They are reciprocals in the scalar model: $\sigma = 1/\eta$. Resistivity has units $\Omega\,\mathrm{m}$ and conductivity has units $\mathrm{S\,m^{-1}}$ when $\boldsymbol{E} = \eta \boldsymbol{j}$.

What is the dominant temperature scaling of Spitzer resistivity?
At fixed Coulomb logarithm, $\eta_{\mathrm{Sp}} \propto T_e^{-3/2}$. The logarithm varies only weakly with density and temperature in a weakly coupled plasma.

Why must the collision channel be stated before quoting a resistivity?
Because ν may be electron–neutral, electron–ion, or a sum of momentum-transfer channels. Different target populations and cross sections produce different drag and therefore different resistivity.

10.5 Plasma conductivity: DC, AC, and ion motion

How does a magnetic field change the relation between electric field and current? It makes the response anisotropic. Current parallel to the field is not gyroscopically deflected, while perpendicular current is split into a dissipative Pedersen part and a nondissipative Hall part. At finite frequency, the response is generally complex: current can have both an amplitude change and a phase shift relative to the applied field.

Learning objectives

- derive the unmagnetized DC conductivity from the linear momentum equation
- construct the magnetized conductivity tensor and identify its three components
- extend the tensor to harmonic AC response
- include ion motion and identify the limits of an electron-only model

Unit ledger

The conductivity tensor entries are in $\mathrm{S\,m^{-1}}$.

Linearize around a uniform equilibrium with $\boldsymbol{B} = \boldsymbol{B}_0 + \boldsymbol{B}_1$, $n_\mathrm{e} = n_{\mathrm{e}0} + n_{\mathrm{e}1}$, $\boldsymbol{E}_0 = \boldsymbol{0}$, and $\boldsymbol{u}_{\mathrm{e}0} = \boldsymbol{0}$. Neglect pressure gradients in the homogeneous conductivity calculation. First take ions as immobile. For the later independent species tensors, assume drag against a prescribed stationary background, such as neutrals, or take collisionless response. Mutual electron–ion drag with both species mobile requires coupled momentum equations. Use fields proportional to $\exp(-i\omega t)$.

Direct-current (DC) response uses a steady field; alternating-current (AC) response uses a field oscillating at angular frequency ω. With the stated complex convention, the physical field and current are the real parts of their harmonic representations. The factor $-i\omega$ replaces a time derivative.

The linear electron momentum equation becomes

$$m_\mathrm{e}(\nu_\mathrm{e} - i\omega)\boldsymbol{u}_\mathrm{e} = q_\mathrm{e}(\boldsymbol{E} + \boldsymbol{u}_\mathrm{e} \times \boldsymbol{B}_0).$$

For unmagnetized DC response, set $\boldsymbol{B}_0 = \boldsymbol{0}$ and $\omega = 0$, so

$$\boldsymbol{j} = \sigma_\mathrm{dc}\boldsymbol{E}, \quad \sigma_\mathrm{dc} = n_\mathrm{e}q_\mathrm{e}^2/(m_\mathrm{e}\nu_\mathrm{e}).$$

Align the z axis with $\boldsymbol{B}_0$ and define the signed electron cyclotron frequency

$$\Omega_\mathrm{e} = q_\mathrm{e}B_0/m_\mathrm{e}.$$

For DC response, set $\omega = 0$. Write J_x, J_y, and J_z for the Cartesian components of the same current density $\boldsymbol{j}$. The component equations are

$$J_x = \sigma_\mathrm{dc}E_x + \left(\frac{\Omega_\mathrm{e}}{\nu_\mathrm{e}}\right)J_y, \quad J_y = \sigma_\mathrm{dc}E_y - \left(\frac{\Omega_\mathrm{e}}{\nu_\mathrm{e}}\right)J_x, \quad J_z = \sigma_\mathrm{dc}E_z$$

Thus

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix} = \begin{pmatrix} \sigma_\perp & \sigma_\mathrm{H} & 0 \\ -\sigma_\mathrm{H} & \sigma_\perp & 0 \\ 0 & 0 & \sigma_\parallel \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix},$$

with

$$\sigma_\parallel = \sigma_\mathrm{dc}, \quad \sigma_\perp = \frac{\sigma_\mathrm{dc}\nu_\mathrm{e}^2}{\nu_\mathrm{e}^2 + \Omega_\mathrm{e}^2}, \quad \sigma_\mathrm{H} = \frac{\sigma_\mathrm{dc}\nu_\mathrm{e}\Omega_\mathrm{e}}{\nu_\mathrm{e}^2 + \Omega_\mathrm{e}^2}$$

The sign of σ_H follows the signed charge convention. Its magnitude describes current perpendicular to both the applied electric field and the background magnetic field.

In this DC tensor, $\sigma_\perp$ is the Pedersen conductivity: it gives current along the perpendicular electric field. The Hall part gives a current at right angles to it and supplies no electric work because its current is perpendicular to the field.

The harmonic AC tensor follows from the same momentum balance with ν_e replaced by $\nu_\mathrm{e} - i\omega$. Its complex entries encode both response amplitude and phase. The derivation below also states when ion and electron tensors can be added independently.

Derivation: DC and AC conductivity tensor

Step: Write the harmonic current equation

Let $a_\mathrm{e} = \nu_\mathrm{e} - i\omega$. Multiplying the harmonic momentum equation by $n_\mathrm{e}q_\mathrm{e}$ and using $\boldsymbol{j} = n_\mathrm{e}q_\mathrm{e}\boldsymbol{u}_\mathrm{e}$ gives

$$\boldsymbol{j} = \frac{n_\mathrm{e} q_\mathrm{e}^2}{m_\mathrm{e} a_\mathrm{e}} \boldsymbol{E} + \left(\frac{\Omega_\mathrm{e}}{a_\mathrm{e}}\right)(\boldsymbol{j} \times \hat{\boldsymbol{z}}).$$

In components,

$$J_x = \sigma_0 E_x + \left(\frac{\Omega_\mathrm{e}}{a_\mathrm{e}}\right) J_y, \quad J_y = \sigma_0 E_y - \left(\frac{\Omega_\mathrm{e}}{a_\mathrm{e}}\right) J_x,$$

with

$$\sigma_0 = \frac{n_\mathrm{e} q_\mathrm{e}^2}{m_\mathrm{e} a_\mathrm{e}}.$$

Step: Invert the perpendicular response

Solving the two coupled equations gives

$$\sigma_\perp = \frac{n_\mathrm{e} q_\mathrm{e}^2 a_\mathrm{e}}{m_\mathrm{e}(a_\mathrm{e}^2 + \Omega_\mathrm{e}^2)}, \quad \sigma_\mathrm{H} = \frac{n_\mathrm{e} q_\mathrm{e}^2 \Omega_\mathrm{e}}{m_\mathrm{e}(a_\mathrm{e}^2 + \Omega_\mathrm{e}^2)}.$$

The parallel entry is

$$\sigma_\parallel = \frac{n_\mathrm{e} q_\mathrm{e}^2}{m_\mathrm{e} a_\mathrm{e}}.$$

Setting $\omega = 0$ recovers the displayed real DC tensor.

Step: Sum mobile species

For mobile ions with independent stationary-background drag (or collisionless response), define

$$\omega_{p,s}^2 = \frac{n_s q_s^2}{\varepsilon_0 m_s}, \quad \Omega_s = \frac{q_s B_0}{m_s}, \quad a_s = \nu_s - i\omega.$$

The species responses sum to

$$\sigma_\parallel = \varepsilon_0 \sum_s \left(\frac{\omega_{p,s}^2}{a_s}\right),$$

$$\sigma_\perp = \varepsilon_0 \sum_s \frac{\omega_{p,s}^2 a_s}{a_s^2 + \Omega_s^2},$$

and

$$\sigma_\mathrm{H} = \varepsilon_0 \sum_s \frac{\omega_{p,s}^2 \Omega_s}{a_s^2 + \Omega_s^2}.$$

Current always adds as $\boldsymbol{j} = \sum_s n_s q_s \boldsymbol{u}_s$, but these independent Drude tensors also require the stated drag model. For mutual collisions, $\boldsymbol{R}_{ei} = m_e n_e \nu_{ei}(\boldsymbol{u}_i - \boldsymbol{u}_e)$ and $\boldsymbol{R}_{ie} = -\boldsymbol{R}_{ei}$ couple the species velocities; solve that momentum-conserving system before summing its currents. An electron-only model is appropriate only when ions are fixed on the frequency and collision scales; at low frequency, ion motion can change every tensor entry.

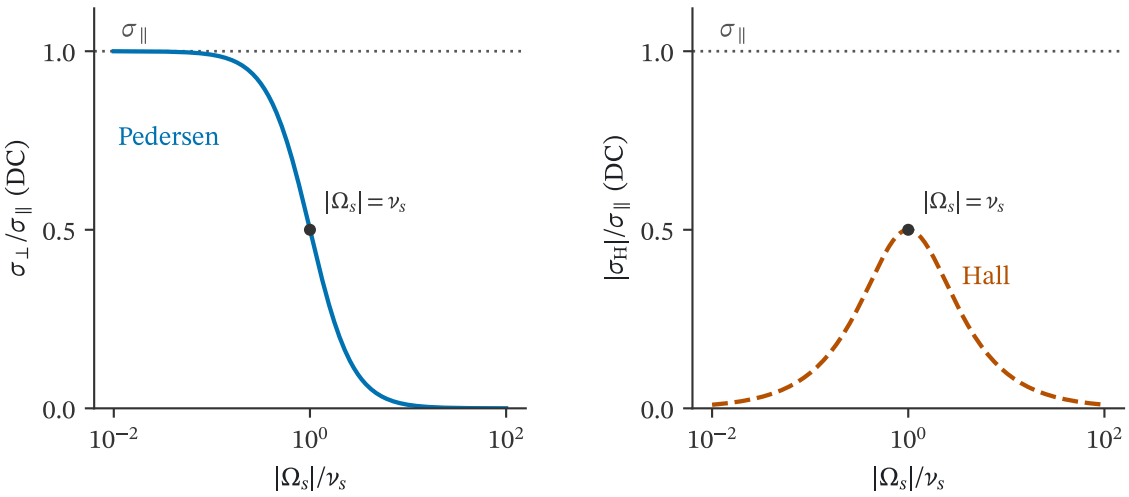


Figure 61: DC conductivity tensor of one species, Pedersen (left) and Hall (right) on the same axes: $\sigma_\perp/\sigma_\parallel = 1/(1+X^2)$ and $|\sigma_H|/\sigma_\parallel = X/(1+X^2)$ with $X = |\Omega_s|/\nu_s$. Weakly magnetized, the current follows $\boldsymbol{E}$; at $|\Omega_s| = \nu_s$ (dot) Pedersen and Hall are equal; strongly magnetized, both vanish while $\sigma_\parallel$ is unchanged. The sign of σ_H follows the sign of q_s.

Rechenbeispiel Rechenbeispiel 198

Consider a homogeneous electron plasma with $n_e = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$, $\nu_e = 2.5 \cdot 10^{3}\,\mathrm{s^{-1}}$, $B_0 = 0.010\,\mathrm{T}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, and $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, with $q_e = -e$.

Assumptions: linear homogeneous DC response, immobile ions, a uniform magnetic field along the z axis, and the signed-charge convention $\Omega_e = q_e B_0/m_e$.

Target: report the signed cyclotron frequency and the three DC tensor entries.

Numerical result: $\Omega_e = -1.76 \cdot 10^{9}\,\mathrm{s^{-1}}$, $\sigma_\parallel = 1.13 \cdot 10^{5}\,\mathrm{S\,m^{-1}}$, $\sigma_\perp = 2.28 \cdot 10^{-7}\,\mathrm{S\,m^{-1}}$, and $\sigma_H = -0.160\,\mathrm{S\,m^{-1}}$.

Parallel, Pedersen, and Hall responses Interpretation 199

The parallel conductivity measures current along $\boldsymbol{B}_0$. The Pedersen conductivity is the dissipative perpendicular response, reduced when $|\Omega_e| \gg \nu_e$. The Hall conductivity rotates the perpendicular current and changes sign when the charge sign changes. AC response replaces the real drag rate by $\nu - i\omega$, revealing phase lag from inertia.

Summary Summary 200

A magnetic field turns scalar Drude conductivity into a tensor with parallel, perpendicular/Pedersen, and Hall entries. In harmonic response use $a_s = \nu_s - i\omega$; with mobile ions, the displayed species tensors add for independent stationary-background drag or collisionless response. Every entry is in $\mathrm{S\,m^{-1}}$ and carries the signed-charge convention through Ω_s.

Knowledge check

- Why is the conductivity tensor anisotropic in a magnetic field?
- What is the physical distinction between Pedersen and Hall current?
- How is DC conductivity generalized to harmonic AC response?
- When must ion motion be included in the conductivity?

Answers

Why is the conductivity tensor anisotropic in a magnetic field?

Gyromotion deflects a perpendicular drift but does not deflect motion parallel to $\boldsymbol{B}_0$. The electric-field direction therefore no longer determines the current direction by a scalar coefficient.

What is the physical distinction between Pedersen and Hall current?

Pedersen current follows the perpendicular electric field and dissipates energy through collisions. Hall current is perpendicular to both the perpendicular electric field and $\boldsymbol{B}_0$ and reverses with the signed charge convention.

How is DC conductivity generalized to harmonic AC response?

For fields proportional to $\exp(-i\omega t)$, replace ν by $a = \nu - i\omega$ in the coupled momentum equations. The resulting conductivities are complex and contain collisional drag and inertia.

When must ion motion be included in the conductivity?

Include it when the frequency is low enough, or the collision and magnetization ordering is such, that ions acquire a significant drift. Then use the additive species current $\boldsymbol{j} = \sum_s n_s q_s \boldsymbol{u}_s$ rather than an electron-only model.

11. Plasma diffusion

Diffusion is the macroscopic signature of many small changes in particle trajectories. Collisions randomize particle directions. Across a density gradient, more particles arrive from the dense side than from the dilute side, producing a net flux toward lower density even when individual steps are symmetric. Magnetic fields do not remove diffusion, but they make the transport direction dependent. This transport hierarchy follows the standard collisional and magnetized plasma description in [3].

Microscopic origin of D — Key idea 202

Which microscopic step length and time set the macroscopic coefficient D? The answer depends on whether particles are unmagnetized, tied together by quasi-neutrality, interrupted by collisions across $\boldsymbol{B}$, or transported by unresolved turbulence.

11.1 Random walks and the diffusion coefficient

A symmetric random walk has no preferred direction, so its mean displacement remains zero. Its variance does not remain zero: independent steps add in quadrature. That distinction is the origin of a diffusion equation.

Learning objectives

- derive the diffusion coefficient from a symmetric random walk
- connect Fick's law to the continuity equation
- interpret the diffusive timescale and the Gaussian spreading solution
- state which assumptions distinguish diffusion from directed advection

Unit ledger

A diffusion coefficient D is in $\mathrm{m^2\,s^{-1}}$, and particle flux $\boldsymbol{\Gamma}$ is in $\mathrm{m^{-2}\,s^{-1}}$. The normalized variables $\xi = x/L_0$ and $\tau = t/\tau_0$ are dimensionless. In the one-dimensional Green function,

$$N_0 = \int_{-\infty}^{\infty} n(x,t)\,\mathrm{d}x$$

is the conserved column density in $\mathrm{m^{-2}}$.

Independent, symmetric steps — Assumptions 203

Use a large number of uncorrelated steps with finite variance, zero mean displacement, and a step time short compared with the macroscopic evolution time. The density varies slowly over one step, so a local gradient expansion is meaningful. Directed drifts are either absent or separated from the diffusive flux.

Diffusion from a random walk Definition 204

For a one-dimensional step of magnitude Δx every Δt, define $D = (\Delta x)^2/(2\Delta t)$. The diffusive particle flux obeys Fick's law $\mathbf{\Gamma}^D = -D\boldsymbol{\nabla}(n)$, and a system of size L has the characteristic diffusion time $\tau_{\mathrm{D}} = L^2/D$. This is a scale estimate; a precise decay time also depends on geometry and boundary conditions. For particles starting at the origin with the same diffusivity in all three directions, the variance relation is $\left\langle |\boldsymbol{r}|^2 \right\rangle = 6Dt$.

Angle brackets denote an ensemble average over possible random walks. A finite collection of walkers need not have exactly zero mean at each time. The ensemble variance describes spreading around the starting point, not a directed displacement of the whole cloud.

Derivation: from step statistics to the diffusion equation

Step: Compute the moments of one random step

Let one step be $+\Delta x$ or $-\Delta x$ with equal probability. Then

$$\langle \Delta x \rangle = \frac{\Delta x}{2} + \frac{-\Delta x}{2} = 0, \quad \left\langle (\Delta x)^2 \right\rangle = \frac{(\Delta x)^2}{2} + \frac{(-\Delta x)^2}{2} = (\Delta x)^2.$$

Step: Accumulate independent steps

After $N = t/(\Delta t)$ statistically independent steps, cross terms in the squared displacement vanish because the individual means are zero. Thus

$$\langle x \rangle = 0, \quad \left\langle x^2 \right\rangle = N(\Delta x)^2 = \left(\frac{(\Delta x)^2 t}{\Delta t} \right) = 2Dt.$$

This defines the one-dimensional coefficient $D = (\Delta x)^2/(2\Delta t)$.

Step: Derive the local diffusive flux

Consider a cell of width Δx. Right-moving and left-moving populations sample densities displaced by one step. Expanding those densities to first order gives

$$\mathbf{\Gamma}^D = -\left(\frac{(\Delta x)^2}{2\Delta t} \right) \boldsymbol{\nabla}(n) = -D\boldsymbol{\nabla}(n).$$

Particle conservation is

$$\frac{\partial n}{\partial t} + \boldsymbol{\nabla} \cdot (\mathbf{\Gamma}) = 0.$$

Step: Obtain the diffusion equation and Green function

Substitution yields

$$\frac{\partial n}{\partial t} = \boldsymbol{\nabla} \cdot (D\boldsymbol{\nabla}(n)).$$

For uniform D, this reduces to $\partial n/\partial t = D\nabla^2(n)$. A point-like initial sheet $n(x,0) = N_0\delta(x)$ has Green-function solution

$$n(x,t) = \frac{N_0}{\sqrt{4\pi D t}} \exp\!\left(-\frac{x^2}{4Dt}\right).$$

Its second moment is $\langle x^2 \rangle = 2Dt$. Independent Cartesian directions add, giving $\langle |\boldsymbol{r}|^2 \rangle = 6Dt$ in three dimensions.

A normalized diffusion equation Governing law 205

Choose reference length L_0 in m, time τ_0 in s, and density n_0 in m^{-3}. Define the reference coefficient $D_0 = L_0^2/\tau_0$ in $\mathrm{m}^2\,\mathrm{s}^{-1}$ and the dimensionless variables $\boldsymbol{\xi} = \boldsymbol{r}/L_0$, $\tau = t/\tau_0$, $n_{\mathrm{norm}} = n/n_0$, and $D_{\mathrm{norm}} = D/D_0$. Spatial derivatives below act on $\boldsymbol{\xi}$. The diffusion equation then has the dimensionless form

$$\frac{\partial n_{\mathrm{norm}}}{\partial \tau} = \boldsymbol{\nabla} \cdot (D_{\mathrm{norm}} \boldsymbol{\nabla}(n_{\mathrm{norm}})).$$

For a system length L with $L_{\mathrm{norm}} = L/L_0$, the normalized diffusion time is $\tau_{\mathrm{D}}/\tau_0 = L_{\mathrm{norm}}^2/D_{\mathrm{norm}}$. Every dimensional result is recovered by restoring the stated reference scales; no unit is hidden in a normalized axis or coefficient.

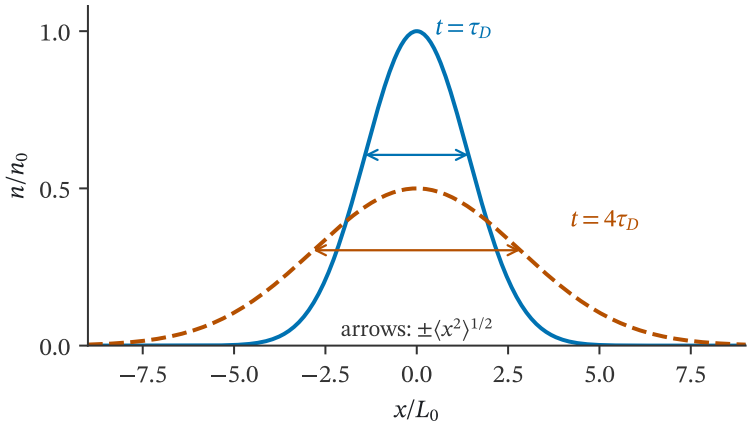
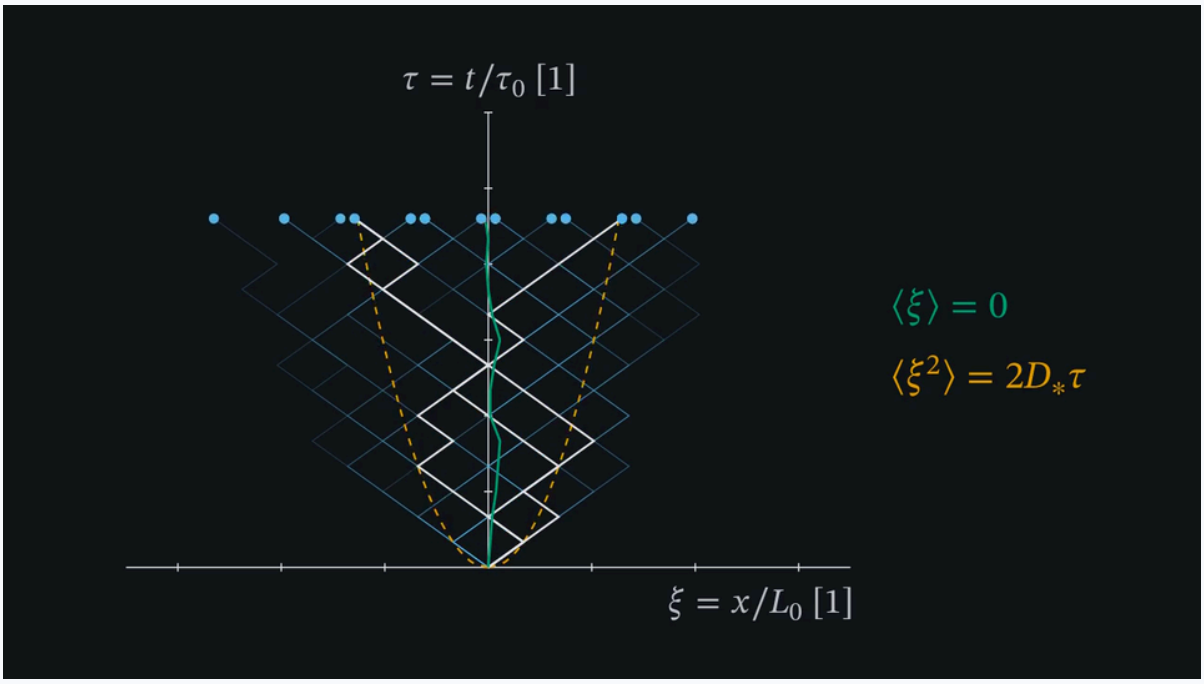


Figure 62: A conserved pulse spreads by diffusion: $n/n_0 = \tau^{-1/2}\exp\!\left(-\xi^2/(4\tau)\right)$ with $\xi = x/L_0$, $\tau = t/\tau_D$, $\tau_D = L_0^2/D$, and $n_0 = N_0/\left(\sqrt{4\pi}L_0\right)$. From $t = \tau_D$ to $t = 4\tau_D$ the rms width $\langle x^2 \rangle^{1/2} = \sqrt{2Dt}$ (arrows) doubles and the peak halves; the area N_0 is conserved.



Watch animation

Random-walk spreading from 36 walkers taking 18 seeded independent symmetric steps. With $\xi = x/L_0$ and $\tau = t/\tau_0$, each step has magnitude $\Delta\xi = 0.34$ [1] and duration $\Delta\tau = 1$ [1]. The coefficient $D_* = D\tau_0/L_0^2 = 0.0578$ [1] is D_{norm} defined above. Restore dimensional steps as $\Delta x = 0.34L_0$ and $\Delta t = \tau_0$. The ensemble mean is zero and its variance is $2D_*\tau$; finite-sample means and variances fluctuate.

Figure 63: Random-walk spreading from 36 walkers taking 18 seeded independent symmetric steps. With $\xi = x/L_0$ and $\tau = t/\tau_0$, each step has magnitude $\Delta\xi = 0.34$ [1] and duration $\Delta\tau = 1$ [1]. The coefficient $D_* = D\tau_0/L_0^2 = 0.0578$ [1] is D_{norm} defined above. Restore dimensional steps as $\Delta x = 0.34L_0$ and $\Delta t = \tau_0$. The ensemble mean is zero and its variance is $2D_*\tau$; finite-sample means and variances fluctuate.

Rechenbeispiel Rechenbeispiel 206

Assume independent symmetric one-dimensional steps in a neutral-collision model, with no directed drift. The model has a step magnitude $\Delta x = 2.0 \cdot 10^{-3}$ m every $\Delta t = 1.0 \cdot 10^{-7}$ s. For a device of length $L = 1.0 \cdot 10^{-1}$ m, determine the diffusion coefficient and the characteristic diffusion time.

Numerical result: $D = 2.0 \cdot 10^{1}$ m^2 s^{-1} and $\tau_D = 5.0 \cdot 10^{-4}$ s.

Diffusion is not advection Interpretation 207

A uniform density has no diffusive flux even when particles move thermally. A density gradient produces a flux down the gradient, whereas an externally imposed velocity produces directed advection. If the collision correlation time is not short compared with the observation time, the random-walk approximation can fail and the transport becomes ballistic or anomalous.

Summary

Summary 208

Independent symmetric steps give $\langle x \rangle = 0$ but $\langle x^2 \rangle = 2Dt$ in one dimension. Fick's law $\boldsymbol{\Gamma}^D = -D\boldsymbol{\nabla}(n)$ and particle conservation produce $\partial n/\partial t = \boldsymbol{\nabla} \cdot (D\boldsymbol{\nabla}(n))$, with the scale $\tau_{\mathrm{D}} = L^2/D$.

Knowledge check

Knowledge check 209

- Why does a symmetric random walk have zero mean displacement but nonzero spreading?
- What physical statement is encoded by Fick's law?
- How does the characteristic diffusion time scale with system size?
- Which assumption separates diffusion from ballistic transport?

Answers

Why does a symmetric random walk have zero mean displacement but nonzero spreading?

The positive and negative steps cancel in the first moment, but the squared steps are positive and add. Thus $\langle x \rangle = 0$ while $\langle x^2 \rangle$ grows linearly with time.

What physical statement is encoded by Fick's law?

The diffusive particle flux points down the density gradient: $\boldsymbol{\Gamma}^D = -D\boldsymbol{\nabla}(n)$. The coefficient D measures the strength of that response.

How does the characteristic diffusion time scale with system size?

For a fixed coefficient, $\tau_{\mathrm{D}} = L^2/D$, so doubling the length makes the diffusive time four times larger.

Which assumption separates diffusion from ballistic transport?

Diffusion requires many weakly correlated steps and a local gradient over the observation time. If a particle retains its direction over the whole experiment, a ballistic description is more appropriate.

11.2 Diffusion in weakly ionized plasmas

In a weakly ionized plasma, collisions with neutrals provide the drag that converts a force or a pressure gradient into a drift. The same momentum equation therefore contains both mobility and diffusion.

Learning objectives

- reduce the collisional momentum equation to a drift-diffusion law
- define signed mobility and positive diffusion coefficients
- use the Einstein relation without losing the charge-sign convention
- identify when the weakly ionized approximation is local and isothermal

Unit ledger

The mobility μ_s has units of velocity divided by electric field, $\mathrm{m^2\,V^{-1}\,s^{-1}}$, and D_s is in $\mathrm{m^2\,s^{-1}}$. This electric-field mobility includes charge, unlike the force-based mobility introduced in Chapter 10.

Local, isothermal neutral drag Assumptions 210

Use a weakly ionized plasma with a prescribed stationary neutral background, a constant momentum-transfer frequency ν_s, small drift relative to the thermal speed, and no magnetic field in this section. Let the species pressure be $p_s = n_s k_B T_s$, with spatially uniform T_s, and consider evolution slow compared with the momentum-relaxation time $1/\nu_s$. Neglect inertia in that regime.

Mobility and diffusion Definition 211

The species momentum equation is

$$m_s n_s\left(\frac{\partial \boldsymbol{u}_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(\boldsymbol{u}_s)\right) = q_s n_s \boldsymbol{E} - \boldsymbol{\nabla}(p_s) - m_s n_s \nu_s \boldsymbol{u}_s.$$

In the steady small-drift limit, $\boldsymbol{u}_s = \mu_s^q \boldsymbol{E} - D_s(\boldsymbol{\nabla}(n_s)/n_s)$, where the signed mobility and diffusion coefficient are $\mu_s^q = q_s/(m_s\nu_s)$ and $D_s = k_B T_s/(m_s\nu_s)$.

Weakly ionized drift-diffusion flux Governing law 212

Multiplying the velocity law by n_s gives $\boldsymbol{\Gamma}_s = n_s\mu_s^q \boldsymbol{E} - D_s\boldsymbol{\nabla}(n_s)$. If a positive mobility is preferred, define $\mu_s = |q_s|/(m_s\nu_s)$. The electric drift is then along the field for positive charges and opposite to it for electrons; multiply the mobility by the charge sign, not by the charge magnitude again. The Einstein relation is $D_s = \mu_s k_B T_s/|q_s|$.

Derivation: collisional force balance

Step: Apply the isothermal pressure law

Start with the species momentum equation and use

$$\boldsymbol{\nabla}(p_s) = k_B T_s \boldsymbol{\nabla}(n_s).$$

Step: Take the long-time force balance

On times longer than ν_s^{-1}, neglect inertia and solve

$$m_s n_s \nu_s \boldsymbol{u}_s = q_s n_s \boldsymbol{E} - k_B T_s \boldsymbol{\nabla}(n_s).$$

Division by $m_s n_s \nu_s$ gives

$$\boldsymbol{u}_s = \frac{q_s}{m_s\nu_s}\boldsymbol{E} - \frac{k_B T_s}{m_s\nu_s}\left(\frac{\boldsymbol{\nabla}(n_s)}{n_s}\right).$$

Step: Identify mobility and diffusion

Define the signed mobility and diffusion coefficient

$$\mu_s^q = \frac{q_s}{m_s\nu_s}, \quad D_s = \frac{k_B T_s}{m_s\nu_s}.$$

Multiplication by n_s produces the particle flux. With the positive mobility magnitude $|\mu_s^q| = |q_s|/(m_s\nu_s)$, the Einstein relation is

$$\frac{D_s}{|\mu_s^q|} = \frac{k_B T_s}{|q_s|}.$$

Step: State the approximation boundary

The derivation assumes local, isothermal, unmagnetized drag. Inertia matters when the forcing varies on a time comparable to ν_s^{-1}, and a scalar coefficient is insufficient when a magnetic field makes the response tensorial.

Rechenbeispiel

Rechenbeispiel 213

Assume local, isothermal, steady, unmagnetized neutral drag with negligible inertia. For electrons in a weakly ionized plasma, use $k_B T_e = 2.0\,\mathrm{eV} = 3.204 \cdot 10^{-19}\,\mathrm{J}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, and $\nu_e = 1.0 \cdot 10^{8}\,\mathrm{s}^{-1}$. Determine the positive mobility magnitude $\mu_e = e/(m_e \nu_e)$ and the diffusion coefficient $D_e = k_B T_e/(m_e \nu_e)$.

Numerical result: $\mu_e = 1.76 \cdot 10^{3}\,\mathrm{m^2\,V^{-1}\,s^{-1}}$ and $D_e = 3.52 \cdot 10^{3}\,\mathrm{m^2\,s^{-1}}$.

The same collision rate controls two responses

Interpretation 214

A larger collision frequency reduces the drift mobility and the unmagnetized diffusion coefficient. This does not mean that collisions always reduce transport: across a magnetic field, collisions are also what interrupt gyromotion and allow a random walk between field lines.

Summary

Summary 215

Neutral drag yields $\mathbf{\Gamma}_s = n_s \mu_s^q \boldsymbol{E} - D_s \boldsymbol{\nabla}(n_s)$ with $\mu_s^q = q_s/(m_s \nu_s)$ and $D_s = k_B T_s/(m_s \nu_s)$. The Einstein relation connects their magnitudes, while the sign of the force response remains set by q_s.

Knowledge check

Knowledge check 216

- Which term in the weakly ionized flux is driven by a density gradient?
- How does increasing the neutral collision frequency affect unmagnetized diffusion?
- Why must the charge sign be kept separate from the positive mobility magnitude?
- When is the scalar Einstein relation insufficient?

Answers

Which term in the weakly ionized flux is driven by a density gradient?
The Fick term $-D_s\nabla(n_s)$ is gradient driven. It points down the density gradient and is distinct from the electric-force term.

How does increasing the neutral collision frequency affect unmagnetized diffusion?
At fixed temperature, $D_s = k_B T_s/(m_s \nu_s)$ decreases inversely with ν_s because the mean free path and drift response are both shortened.

Why must the charge sign be kept separate from the positive mobility magnitude?
Electrons and ions drift in opposite directions under the same electric field. With $\mu_s = |q_s|/(m_s \nu_s)$, the electric drift is $+\mu_s \boldsymbol{E}$ for positive ions and $-\mu_s \boldsymbol{E}$ for electrons. The charge magnitude is already included in μ_s.

When is the scalar Einstein relation insufficient?
It is insufficient when the response is anisotropic, such as in a magnetic field, or when the plasma is non-isothermal or strongly time-dependent.

11.3 Ambipolar diffusion and the ambipolar electric field

Electrons usually diffuse faster than ions because their mass is smaller. The resulting charge separation creates an electric field that couples their transport. With zero-current boundary conditions in the transport direction, the field adjusts until the species carry the same particle flux. This coupled transport is ambipolar diffusion.

Learning objectives

- write the electron and ion fluxes with explicit charge signs
- derive the ambipolar electric field from equal particle fluxes
- obtain the ambipolar diffusion coefficient
- explain the role of quasi-neutrality and zero net current

Unit ledger

The ambipolar diffusion coefficient D_a is in $\mathrm{m}^2\,\mathrm{s}^{-1}$.

Quasi-neutral, singly charged hydrogen

Assumptions 217

Use $n_i \approx n_e = n$, $q_i = +e$, $q_e = -e$, spatially uniform temperatures within each species, and zero current in the transport direction, fixed by the boundary conditions. The temperatures of the two species may differ. The electron and ion drift-diffusion coefficients are evaluated locally, and the ambipolar field adjusts faster than the density profile evolves.

Species fluxes and quasi-neutrality Definition 218

With positive mobility magnitudes, the unmagnetized fluxes are $\boldsymbol{\Gamma}_i = \mu_i n\boldsymbol{E} - D_i\boldsymbol{\nabla}(n)$ and $\boldsymbol{\Gamma}_e = -\mu_e n\boldsymbol{E} - D_e\boldsymbol{\nabla}(n)$. The current is $\boldsymbol{j} = e(\boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_e)$. The zero-current closure used here imposes $\boldsymbol{j} = \boldsymbol{0}$, equivalently $\boldsymbol{\Gamma}_i = \boldsymbol{\Gamma}_e$.

Charge conservation gives $\partial\rho_q/\partial t + \boldsymbol{\nabla}\cdot(\boldsymbol{j}) = 0$. Maintaining quasi-neutrality therefore constrains the leading current divergence; it does not require $\boldsymbol{j} = \boldsymbol{0}$. In one-dimensional steady transport, current is spatially constant, and a zero-current boundary sets that constant to zero. This additional condition gives the equal fluxes used below.

Ambipolar field and diffusion coefficient Governing law 219

Solving the equal-flux condition gives

$$\boldsymbol{E}_a = \left(\frac{D_i - D_e}{\mu_i + \mu_e}\right)\left(\frac{\boldsymbol{\nabla}(n)}{n}\right).$$

Substitution into either species flux gives $\boldsymbol{\Gamma}_a = -D_a\boldsymbol{\nabla}(n)$ with $D_a = (\mu_i D_e + \mu_e D_i)/(\mu_i + \mu_e)$.

Derivation: quasi-neutral flux balance

Step: Write the two species fluxes

For positive ions, the electric force drives a flux in the direction of $\boldsymbol{E}$:

$$\boldsymbol{\Gamma}_i = \mu_i n\boldsymbol{E} - D_i\boldsymbol{\nabla}(n).$$

Electrons have the opposite charge:

$$\boldsymbol{\Gamma}_e = -\mu_e n\boldsymbol{E} - D_e\boldsymbol{\nabla}(n).$$

Step: Impose zero current

The current-free condition is

$$\boldsymbol{0} = \boldsymbol{j} = e(\boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_e).$$

Equating the two fluxes and collecting the field terms gives

$$(\mu_i + \mu_e)n\boldsymbol{E} = (D_i - D_e)\boldsymbol{\nabla}(n).$$

Hence

$$\boldsymbol{E}_a = \left(\frac{D_i - D_e}{\mu_i + \mu_e}\right)\left(\frac{\boldsymbol{\nabla}(n)}{n}\right).$$

Step: Insert the ambipolar field into the flux

The ion flux becomes

$$\boldsymbol{\Gamma}_i = \left(\frac{\mu_i n(D_i - D_e)}{\mu_i + \mu_e}\right)\left(\frac{\boldsymbol{\nabla}(n)}{n}\right) - D_i\boldsymbol{\nabla}(n).$$

Putting both terms over the common denominator produces

$$\boldsymbol{\Gamma}_i = -\frac{\mu_i D_e + \mu_e D_i}{\mu_i + \mu_e} \boldsymbol{\nabla}(n).$$

The same result follows from the electron flux, so define

$$D_a = \frac{\mu_i D_e + \mu_e D_i}{\mu_i + \mu_e}.$$

Step: Interpret the mobility hierarchy

For equal temperatures and comparable collision models, often $\mu_e \gg \mu_i$. Then

$$D_a \approx D_i + \left(\frac{\mu_i}{\mu_e}\right) D_e.$$

Electron diffusion is consequently largely converted into the ambipolar electric field rather than a net current.

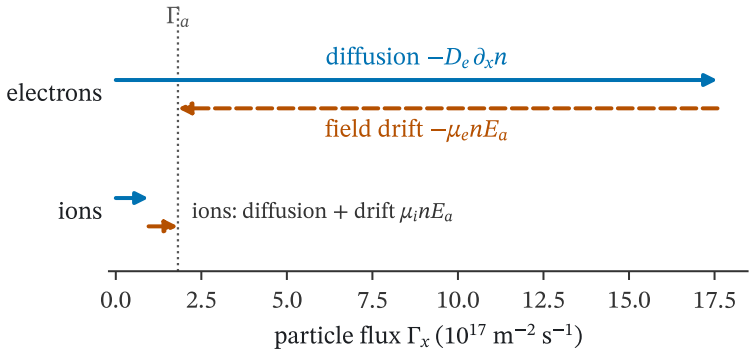


Figure 64: Ambipolar diffusion in the example below. Diffusion alone (solid arrows) would carry electrons about 18 times faster than ions. The ambipolar field $E_a = 0.897\,\mathrm{V\,m^{-1}}$ drifts electrons back and ions forward (dashed arrows) until both fluxes equal $\Gamma_a = D_a|\partial_x n| = 1.82 \times 10^{17}\,\mathrm{m^{-2}\,s^{-1}}$.

Rechenbeispiel Rechenbeispiel 220

Assume local isothermal neutral drag in a weakly ionized, quasi-neutral, singly charged hydrogen plasma with no externally imposed current. Let the ambipolar field adjust rapidly compared with the density evolution. Consider the plasma with $n = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$, $\boldsymbol{\nabla}(n)/n = -1.0\,\mathrm{m^{-1}}$, and $k_B T_i = k_B T_e = 1.0\,\mathrm{eV} = 1.602 \cdot 10^{-19}\,\mathrm{J}$. Use $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $\nu_i = 1.0 \cdot 10^{7}\,\mathrm{s^{-1}}$, and $\nu_e = 1.0 \cdot 10^{9}\,\mathrm{s^{-1}}$. Determine the ambipolar field, coefficient, and particle flux.

Numerical result: $\boldsymbol{E}_a = 0.897\,\mathrm{V\,m^{-1}}$, $D_a = 1.82 \cdot 10^{1}\,\mathrm{m^2\,s^{-1}}$, and $|\boldsymbol{\Gamma}_a| = 1.82 \cdot 10^{17}\,\mathrm{m^{-2}\,s^{-1}}$.

Quasi-neutrality couples otherwise unequal rates Interpretation 221

The ambipolar field is generated by the species' unequal diffusion. A small departure from charge neutrality can support a field strong enough to slow the faster species and accelerate the slower one. Their equal particle fluxes point in the same direction: opposite charges cancel in the current, while both populations continue to spread.

Summary

Summary 222

Equal electron and ion particle fluxes give

$$\boldsymbol{E}_a = \left(\frac{D_i - D_e}{\mu_i + \mu_e}\right)\left(\frac{\boldsymbol{\nabla}(n)}{n}\right)$$

and $\boldsymbol{\Gamma}_a = -D_a \boldsymbol{\nabla}(n)$, where $D_a = (\mu_i D_e + \mu_e D_i)/(\mu_i + \mu_e)$. Ambipolar diffusion here follows from quasi-neutrality together with the zero-current boundary condition.

Knowledge check

Knowledge check 223

- Why does an ambipolar electric field form?
- What condition was used to derive the ambipolar coefficient?
- Can ambipolar diffusion carry particles when the current vanishes?
- What changes if the plasma is not quasi-neutral on the transport scale?

Answers

Why does an ambipolar electric field form?

The more mobile species would otherwise diffuse away faster, creating charge separation. The resulting field opposes that separation until the electron and ion particle fluxes match.

What condition was used to derive the ambipolar coefficient?

For singly charged hydrogen we imposed zero net current, $\boldsymbol{j} = e(\boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_e) = \mathbf{0}$, together with quasi-neutral density.

Can ambipolar diffusion carry particles when the current vanishes?

Yes. Equal oppositely charged particle fluxes cancel in the current but add to the total particle transport.

What changes if the plasma is not quasi-neutral on the transport scale?

The electric field must then be obtained from Poisson's equation and a charge-separation model; the equal-flux ambipolar closure is no longer sufficient.

11.4 Cross-field diffusion in a magnetic field

Along a magnetic field, a colliding particle can random-walk as if it were unmagnetized. Across the field, the guiding center is tied to a field line between collisions. A collision interrupts gyromotion and creates the cross-field step.

Learning objectives

- derive the parallel and perpendicular diffusion coefficients
- identify the magnetization parameter $|\Omega_s|/\nu_s$
- separate ordinary diffusion from Hall-like transverse response
- explain why the collisionless limit suppresses classical cross-field diffusion

Unit ledger

The magnetization parameter $|\Omega_s|/\nu_s$ is dimensionless. Parallel, perpendicular, and Hall diffusion coefficients are in $\mathrm{m^2\,s^{-1}}$.

Uniform magnetic field and isotropic drag Assumptions 224

Use a locally uniform $\boldsymbol{B} = B\hat{\boldsymbol{b}}$, a scalar collision frequency for drag against stationary neutrals, small drift, spatially uniform species temperature, and a time-independent field. Pressure gradients and electric fields vary slowly over one gyration. Curvature, mirrors, finite orbit widths, and turbulence are postponed.

Magnetized diffusion tensor Definition 225

Let $\Omega_s = q_s B/m_s$ and $D_s = k_B T_s/(m_s \nu_s)$. The parallel coefficient is $D_{s,\parallel} = D_s$, while $D_{s,\perp} = D_s/\left(1 + (\Omega_s/\nu_s)^2\right)$. A signed transverse coefficient is

$$D_{s,H} = \frac{D_s\left(\frac{\Omega_s}{\nu_s}\right)}{1 + \left(\frac{\Omega_s}{\nu_s}\right)^2}.$$

The gradient contribution can be written

$$\boldsymbol{\Gamma}_{s,\perp} = -D_{s,\perp}\boldsymbol{\nabla}_\perp(n_s) + D_{s,H}\hat{\boldsymbol{b}} \times \boldsymbol{\nabla}_\perp(n_s),$$

in the chosen orientation convention.

Derivation: solving the perpendicular momentum balance

Step: Write the perpendicular linear system

For $\boldsymbol{B} = B\hat{\boldsymbol{z}}$, arrange the steady perpendicular momentum equation as

$$\begin{pmatrix} \nu_s & -\Omega_s \\ \Omega_s & \nu_s \end{pmatrix}\begin{pmatrix} u_{s,x} \\ u_{s,y} \end{pmatrix} = \left(\frac{q_s}{m_s}\right)\begin{pmatrix} E_x \\ E_y \end{pmatrix} - \frac{k_B T_s}{m_s n_s}\begin{pmatrix} \frac{\partial n_s}{\partial x} \\ \frac{\partial n_s}{\partial y} \end{pmatrix}.$$

Step: Invert the drag–gyro matrix

Its inverse is

$$\frac{1}{\nu_s^2 + \Omega_s^2}\begin{pmatrix} \nu_s & \Omega_s \\ -\Omega_s & \nu_s \end{pmatrix}.$$

Step: Read off the diffusion tensor

The gradient part of the flux has diagonal coefficient

$$\frac{k_B T_s \nu_s}{m_s(\nu_s^2 + \Omega_s^2)} = \frac{D_s \nu_s^2}{\nu_s^2 + \Omega_s^2},$$

and signed off-diagonal coefficient

$$\frac{k_B T_s \Omega_s}{m_s(\nu_s^2 + \Omega_s^2)} = \frac{D_s \nu_s \Omega_s}{\nu_s^2 + \Omega_s^2}.$$

These are the displayed $D_{s,\perp}$ and $D_{s,H}$.

Step: Check the strongly magnetized limit

The electric-force part contains the crossed-field drift

$$\boldsymbol{u}_{E\times B} = \frac{\boldsymbol{E}\times\boldsymbol{B}}{B^2}.$$

When $|\Omega_s| \gg \nu_s$,

$$D_{s,\perp} \approx D_s\left(\frac{\nu_s}{\Omega_s}\right)^2.$$

Thus $\nu_s \to 0$ suppresses classical cross-field diffusion: without interruptions, guiding centers do not make a collisional random walk across field lines.

Step: Relate the coefficient to the thermal gyroradius

With the Chapter 1 convention,

$$v_{\mathrm{th,s}} = \sqrt{\frac{2k_BT_s}{m_s}}, \quad \rho_{\mathrm{th,s}} = \frac{v_{\mathrm{th,s}}}{|\Omega_s|}.$$

Therefore

$$\rho_{\mathrm{th,s}}^2 = \frac{2k_BT_s}{m_s\Omega_s^2}, \quad D_{s,\perp} \approx \left(\frac{\nu_s}{2}\right)\rho_{\mathrm{th,s}}^2.$$

An alternative one-dimensional scale would give $D_{s,\perp} \approx \nu_s\rho_{\mathrm{1D,s}}^2$; it is not the thermal gyroradius convention used elsewhere in this script.

For constant field direction $\boldsymbol{b}$ and constant $D_{s,H}$, the Hall-like gradient flux $D_{s,H}\boldsymbol{b}\times\boldsymbol{\nabla}(n_s)$ has zero divergence, since mixed spatial derivatives commute. It does not by itself broaden a density packet. Spatial variation of the coefficient or field direction, and boundary fluxes, can change that conclusion.

The formal collisionless limit requires care. At fixed nonzero magnetic field, the classical perpendicular coefficient tends to zero. The parallel formula instead grows as $1/\nu_s$; once the mean free path is comparable to the system length, local diffusion along the field no longer applies. Particles then retain memory of their motion and boundaries.

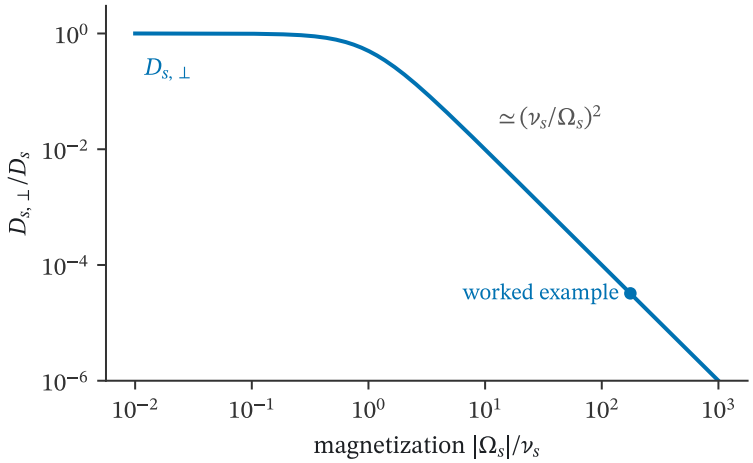


Figure 65: Gyration suppresses diffusion across $\boldsymbol{B}$: $D_{s,\perp}/D_s = 1/\left(1 + (\Omega_s/\nu_s)^2\right)$ with $D_s = k_B T_s/(m_s \nu_s)$, the unmagnetized and parallel coefficient. Beyond $|\Omega_s| = \nu_s$ it falls as $(\nu_s/\Omega_s)^2$. The point is the electron example below, $D_\perp/D_\parallel = 3.23 \times 10^{-5}$.

Rechenbeispiel Rechenbeispiel 226

Assume a local uniform magnetic field, isotropic steady collisional drag and the scalar diffusion model derived above. Ignore field curvature, finite-orbit effects, and turbulence. For electrons at $k_B T_e = 1.0\,\mathrm{eV} = 1.602 \cdot 10^{-19}\,\mathrm{J}$, use $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $B = 1.0 \cdot 10^{-2}\,\mathrm{T}$, and $\nu_e = 1.0 \cdot 10^{7}\,\mathrm{s}^{-1}$. Determine the magnitude of the cyclotron frequency, the parallel coefficient, and the perpendicular coefficient.

Numerical result: $|\Omega_e| = 1.76 \cdot 10^{9}\,\mathrm{s}^{-1}$, $D_{e,\parallel} = 1.76 \cdot 10^{4}\,\mathrm{m}^2\,\mathrm{s}^{-1}$, $D_{e,\perp} = 0.569\,\mathrm{m}^2\,\mathrm{s}^{-1}$, and $D_{e,\perp}/D_{e,\parallel} = 3.23 \cdot 10^{-5}$.

Ambipolarity depends on geometry Interpretation 227

The zero-current closure used in the preceding section does not force transverse electron and ion fluxes to match in every geometry. Rapid parallel electron transport can short-circuit a charge imbalance created by unequal cross-field fluxes. Whether a loss is ambipolar therefore also depends on field-line connection and boundary conditions.

Collisions facilitate cross-field transport Interpretation 228

More collisions shorten the free path, yet a nonzero collision rate is needed to break the indefinite gyromotion. The competition is measured by $|\Omega_s|/\nu_s$: it is small for nearly isotropic transport and large for strong perpendicular suppression.

Summary Summary 229

A magnetic field leaves $D_\parallel = D_s$ but reduces the classical cross-field coefficient to $D_\perp = D_s/\left(1 + (\Omega_s/\nu_s)^2\right)$. The signed Hall-like coefficient describes a transverse response; in the strongly magnetized limit $D_{s,\perp}$ scales as $(\nu_s/2)\rho_{\mathrm{th,s}}^2$ under the thermal-speed convention used in Chapter 1.

Knowledge check

- Why is diffusion along a uniform magnetic field not reduced in the same way as cross-field diffusion?
- What dimensionless parameter measures magnetization?
- What happens to classical cross-field diffusion as collisions vanish?
- What is the physical distinction between $D_\perp$ and D_H?

Answers

Why is diffusion along a uniform magnetic field not reduced in the same way as cross-field diffusion?

Gyromotion constrains only the perpendicular plane. Along the field, particles can continue their collisional random walk, so $D_\parallel = D_s$ in this local model.

What dimensionless parameter measures magnetization?

The ratio $|\Omega_s|/\nu_s$ compares the gyration rate with the collision rate. It is the control parameter in the tensor coefficients.

What happens to classical cross-field diffusion as collisions vanish?

For fixed B it tends to zero as $D_\perp \approx D_s(\nu_s/\Omega_s)^2$. A collisionless particle remains on its guiding-center orbit apart from other drifts.

What is the physical distinction between $D_\perp$ and D_H?

$D_\perp$ multiplies the gradient-aligned dissipative flux, whereas the signed D_H term rotates the response in the perpendicular plane. The latter is Hall-like and changes sign with charge.

11.5 Fully ionized diffusion and characteristic timescales

In a fully ionized plasma, Coulomb collisions couple the species and the electromagnetic force balance must be used consistently. The classical cross-field coefficient follows from pressure balance and conductivity. Experiments can show faster, anomalous transport, often summarized by the empirical Bohm scaling.

Learning objectives

- derive the classical fully ionized cross-field coefficient from force balance
- state the magnetic-field scaling of classical and Bohm-like transport
- use a diffusion coefficient to estimate a loss time
- separate a derived collisional result from an empirical turbulent law

Unit ledger

The classical and Bohm-like diffusion coefficients are in $\mathrm{m^2\,s^{-1}}$.

Quasi-neutral, weakly coupled fluid transport Assumptions 231

Use a fully ionized, quasi-neutral hydrogen plasma with a scalar conductivity, a locally uniform magnetic field, and steady perpendicular force balance. Take each species temperature to be spatially uniform, though the electron and ion temperatures may differ. The classical result neglects turbulent fluctuations and finite-orbit neoclassical geometry. Bohm transport is presented only as an empirical comparison.

Classical fully ionized diffusion Definition 232

The steady force balance and resistive Ohm law are $\mathbf{0} = -\nabla(p) + \boldsymbol{j} \times \boldsymbol{B}$ and $\boldsymbol{j} = \sigma(\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B})$. The perpendicular velocity contains

$$\boldsymbol{u}_\perp = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2} - \frac{\nabla_\perp(p)}{\sigma B^2}.$$

For $p = nk_B(T_e + T_i)$, the pressure-driven flux is $\boldsymbol{\Gamma}_\perp = -D_\perp^{\mathrm{cl}} \nabla_\perp(n)$ with

$$D_\perp^{\mathrm{cl}} = \frac{nk_B(T_e + T_i)}{\sigma B^2} = \frac{\eta n k_B(T_e + T_i)}{B^2}.$$

Classical and Bohm-like scalings Governing law 233

At fixed density, temperatures, and conductivity, $D_\perp^{\mathrm{cl}} \propto B^{-2}$. A commonly used empirical Bohm estimate is $D_\perp^{B} \approx k_B T_e/(16eB)$, so $D_\perp^{B} \propto B^{-1}$. The numerical factor is empirical and should not be mistaken for a derivation from the collisional model.

The coefficient here describes particle transport. It is different from the magnetic diffusivity D_B of Chapter 9, which describes resistive smoothing of the magnetic field. In $D_\perp^B$ above, the superscript labels the Bohm estimate. The identical diffusion units do not make these coefficients interchangeable.

Derivation: pressure-driven classical flux

Step: Combine force balance with Ohm's law

Start from

$$\mathbf{0} = -\nabla(p) + \boldsymbol{j} \times \boldsymbol{B}$$

and substitute

$$\boldsymbol{j} = \sigma(\boldsymbol{E} + \boldsymbol{u} \times \boldsymbol{B}).$$

Using

$$(\boldsymbol{u} \times \boldsymbol{B}) \times \boldsymbol{B} = -B^2 \boldsymbol{u}_\perp$$

gives

$$\mathbf{0} = -\nabla_\perp(p) + \sigma(\boldsymbol{E} \times \boldsymbol{B}) - \sigma B^2 \boldsymbol{u}_\perp.$$

Step: Solve for the perpendicular velocity

Rearranging gives

$$\boldsymbol{u}_\perp = \frac{\boldsymbol{E}\times\boldsymbol{B}}{B^2} - \frac{\boldsymbol{\nabla}_\perp(p)}{\sigma B^2}.$$

Step: Identify the pressure-driven diffusion

Multiply by n. The first term is a common crossed-field drift and does not diffuse a uniform density. For an isothermal two-temperature hydrogen plasma,

$$\boldsymbol{\nabla}_\perp(p) = k_B(T_e + T_i)\boldsymbol{\nabla}_\perp(n).$$

The pressure term in the particle flux is consequently

$$\boldsymbol{\Gamma}_\perp^{\mathrm{diff}} = -\frac{nk_B(T_e + T_i)\boldsymbol{\nabla}_\perp(n)}{\sigma B^2}.$$

Therefore

$$D_\perp^{\mathrm{cl}} = \frac{nk_B(T_e + T_i)}{\sigma B^2} = \frac{\eta n k_B(T_e + T_i)}{B^2}.$$

The factor $1/B^2$ arises because the field enters twice: once through the Lorentz force density $\boldsymbol{j}\times\boldsymbol{B}$ and once through the motional term $\boldsymbol{u}\times\boldsymbol{B}$ in Ohm's law.

Step: State the comparison with anomalous transport

The classical loss time for a macroscopic length L is

$$\tau_{\mathrm{D}} = \frac{L^2}{D_\perp^{\mathrm{cl}}}.$$

The Bohm expression is not obtained by this force-balance derivation; it is a phenomenological comparison motivated by turbulent or anomalous transport. Neoclassical calculations add field geometry, trapped-particle orbits, and finite collisionality.

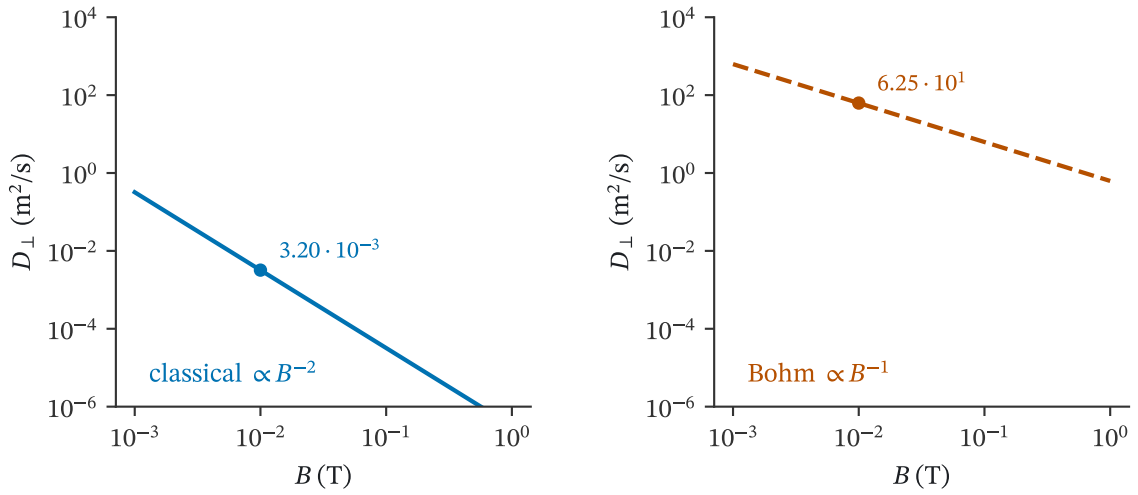


Figure 66: Classical (left) and Bohm (right) cross-field diffusion on the same axes for the plasma of the example below: $n = 10^{16}$ m^{-3}, $k_B T_e = k_B T_i = 10$ eV, $\sigma = 10^5$ S m^{-1}. At $B = 10$ mT (dots) the empirical Bohm estimate exceeds the classical value by 1.95×10^4, and the gap grows in proportion to B.

Rechenbeispiel

Rechenbeispiel 234

Assume a quasi-neutral, fully ionized hydrogen plasma in steady classical perpendicular force balance, with scalar conductivity. Neglect turbulent and finite-orbit corrections; use the Bohm expression only as an empirical comparison. For the plasma, use $n = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$, $k_B T_e = k_B T_i = 10\,\mathrm{eV} = 1.602 \cdot 10^{-18}\,\mathrm{J}$, $\sigma = 1.0 \cdot 10^{5}\,\mathrm{S\,m^{-1}}$, $B = 1.0 \cdot 10^{-2}\,\mathrm{T}$, and $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. Determine the classical coefficient, the Bohm estimate, and their ratio. For a device of length $L = 1.0\,\mathrm{m}$, also estimate the classical diffusion time.

Numerical result: $D_\perp^{\mathrm{cl}} = 3.20 \cdot 10^{-3}\,\mathrm{m^2\,s^{-1}}$, $D_\perp^{B} = 6.25 \cdot 10^{1}\,\mathrm{m^2\,s^{-1}}$, $D_\perp^{B}/D_\perp^{\mathrm{cl}} = 1.95 \cdot 10^{4}$, and $\tau_{\mathrm{D}}^{\mathrm{cl}} = 3.12 \cdot 10^{2}\,\mathrm{s}$.

Classical, neoclassical, and anomalous transport

Interpretation 235

The B^{-2} classical law is a collisional consequence of the local force balance. Neoclassical transport modifies it through magnetic geometry and orbit effects. A Bohm-like B^{-1} law is an empirical anomalous scaling associated with unresolved turbulence; it is not a replacement for stating the collision and geometry model.

Summary

Summary 236

Fully ionized classical transport gives

$$D_\perp^{\mathrm{cl}} = \frac{n k_B (T_e + T_i)}{\sigma B^2}$$

and $\tau_{\mathrm{D}} = L^2/D$. It scales as B^{-2}, while the empirical Bohm estimate scales as B^{-1}. The two rest on different physical assumptions.

Knowledge check

Knowledge check 237

- Why does the classical diffusion coefficient scale as B^{-2} rather than B^{-1}?
- What magnetic-field scaling distinguishes classical from Bohm-like diffusion?
- How is a diffusion loss time estimated from a coefficient?
- Why should a Bohm coefficient be labelled empirical?

Answers

Why does the classical diffusion coefficient scale as B^{-2} rather than B^{-1}?

The field enters through both $\boldsymbol{j} \times \boldsymbol{B}$ in force balance and $\boldsymbol{u} \times \boldsymbol{B}$ in Ohm's law. Solving the perpendicular balance therefore produces the factor $1/(\sigma B^2)$ in $D_\perp^{\mathrm{cl}}$.

What magnetic-field scaling distinguishes classical from Bohm-like diffusion?

The local classical result scales as B^{-2}, whereas the empirical Bohm estimate scales as B^{-1} at fixed temperature.

How is a diffusion loss time estimated from a coefficient?

For a characteristic length L, use $\tau_{\mathrm{D}} = L^2/D$. Geometry changes the numerical factor, but not the basic dimensional scaling.

Why should a Bohm coefficient be labelled empirical?

Its B^{-1} dependence summarizes anomalous transport linked to turbulence; it is not derived from the local collisional force balance used for $D_\perp^{\mathrm{cl}}$.

12. Introduction to waves in plasmas

A plasma wave is a collective perturbation whose restoring force and inertia are distributed among charged particles and fields. This chapter establishes the common language for the later wave chapters: equilibria, linearization, plane-wave response, dispersion, and controlled limits. The chapter follows the standard progression from small-amplitude wave properties to non-magnetized plasma modes in [3].

Selecting a wave branch — Key idea 238

The same plasma can support an electrostatic oscillation, a transverse electromagnetic wave, or an ion-acoustic response. The equilibrium, polarization, frequency, wavelength, and closure select the branch.

12.1 Equilibrium, perturbation, and linearization

How do we turn a nonlinear plasma model into a wave problem? Choose an equilibrium, assign a small parameter to every perturbation, retain terms through first order in that parameter, and then test sinusoidal solutions. The discarded products are precisely where finite-amplitude wave coupling would enter.

Learning objectives

- state the homogeneous equilibrium and its ordering assumptions
- linearize continuity, momentum, and Maxwell equations consistently
- use a plane-wave ansatz to replace derivatives by algebraic operators
- distinguish longitudinal, transverse, electrostatic, and electromagnetic response

Unit ledger

The phase $\boldsymbol{k}\cdot\boldsymbol{r}-\omega t$ and normalized variables such as $\omega/\omega_{p,e}$ are dimensionless.

Uniform, stationary, neutral equilibrium — Assumptions 239

Use $n_s = n_{s,0}$, $\boldsymbol{u}_{s,0} = \boldsymbol{0}$, constant $p_{s,0}$, and $\boldsymbol{E}_0 = \boldsymbol{0}$. The background magnetic field $\boldsymbol{B}_0$ may be retained for the general linear equation, but the unmagnetized examples below set $\boldsymbol{B}_0 = \boldsymbol{0}$. Macroscopic charge neutrality means $\sum_s q_s n_{s,0} = 0$.

Perturbation and plane-wave convention — Definition 240

Introduce a bookkeeping parameter $\varepsilon \ll 1$ and write $n_s = n_{s,0} + \varepsilon n_{s,1}$, $\boldsymbol{u}_s = \varepsilon \boldsymbol{u}_{s,1}$, $\boldsymbol{E} = \varepsilon \boldsymbol{E}_1$, and $\boldsymbol{B} = \boldsymbol{B}_0 + \varepsilon \boldsymbol{B}_1$. A complex amplitude represents the real field through

$$\boldsymbol{A}_1(\boldsymbol{r},t) = \Re\{\tilde{\boldsymbol{A}}_1 \exp(i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t))\}.$$

The wave number $k = |\boldsymbol{k}|$ is in m^{-1} and the angular frequency ω is in s^{-1}.

Polarization describes the direction and time evolution of the electric perturbation. A longitudinal electric field is parallel to $\boldsymbol{k}$; a transverse one is perpendicular to it. For a plane wave at nonzero

frequency, a purely longitudinal field has no magnetic perturbation by Faraday's law and is electrostatic in this sense. A transverse electric wave has an accompanying magnetic perturbation even when the equilibrium field is zero.

Acting on the complex plane wave, a time derivative multiplies its amplitude by $-i\omega$, and a spatial gradient multiplies it by $i\boldsymbol{k}$. These replacements turn the differential equations into algebraic equations for the amplitudes. A dispersion relation specifies which frequencies and wave vectors allow a nonzero solution.

The species equations and Maxwell equations that are linearized are

$$\frac{\partial n_s}{\partial t} + \boldsymbol{\nabla} \cdot (n_s \boldsymbol{u}_s) = 0$$

$$m_s n_s \left(\frac{\partial \boldsymbol{u}_s}{\partial t} + \boldsymbol{u}_s \cdot \boldsymbol{\nabla}(\boldsymbol{u}_s) \right) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) - \boldsymbol{\nabla}(p_s)$$

$$\boldsymbol{\nabla} \times (\boldsymbol{E}) = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \boldsymbol{\nabla} \times (\boldsymbol{B}) = \mu_0 \boldsymbol{j} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{E}) = \frac{\rho_q}{\varepsilon_0}, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{B}) = 0$$

where $\rho_q = \sum_s q_s n_s$ and $\boldsymbol{j} = \sum_s q_s n_s \boldsymbol{u}_s$. For an adiabatic or isothermal closure, the pressure perturbation is written $p_{s,1} = \gamma_s k_B T_s n_{s,1}$, with the appropriate temperature ordering stated explicitly. The dimensionless coefficient γ_s specifies the pressure response; $\gamma_s = 1$ gives an isothermal response. It must be chosen for the process being modeled, not inferred from small amplitude alone.

Derivation: linearized Fourier system

Step: Classify the perturbation order

The equilibrium has no flow. Consequently, products of two first-order quantities are second order and do not appear in the linear problem. In particular, the convective term and the perturbed magnetic force are omitted at first order.

Step: Linearize continuity and momentum

The continuity equation becomes

$$\frac{\partial n_{s,1}}{\partial t} + n_{s,0} \boldsymbol{\nabla} \cdot \left(\boldsymbol{u}_{s,1} \right) = 0.$$

The first-order momentum equation retains the equilibrium magnetic field but not the product of two perturbations:

$$m_s n_{s,0} \frac{\partial \boldsymbol{u}_{s,1}}{\partial t} = q_s n_{s,0} \left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0 \right) - \boldsymbol{\nabla}\left(p_{s,1} \right).$$

Step: Apply the plane-wave replacement

For the convention $\exp(i(\boldsymbol{k} \cdot \boldsymbol{r} - \omega t))$, replace derivatives by algebraic operators:

$$\frac{\partial f}{\partial t} \to -i\omega f, \quad \boldsymbol{\nabla}(f) \to i\boldsymbol{k} f, \quad \boldsymbol{\nabla} \cdot (\boldsymbol{A}) \to i\boldsymbol{k} \cdot \boldsymbol{A}, \quad \boldsymbol{\nabla} \times (\boldsymbol{A}) \to i\boldsymbol{k} \times \boldsymbol{A}.$$

Thus continuity and momentum become

$$-i\omega n_{s,1} + i n_{s,0} \boldsymbol{k} \cdot \boldsymbol{u}_{s,1} = 0$$

and

$$-i\omega m_s n_{s,0} \boldsymbol{u}_{s,1} = q_s n_{s,0} \left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0 \right) - i \boldsymbol{k} p_{s,1}.$$

Step: Form the dispersion condition

Maxwell's equations receive the same algebraic replacement. Equilibrium neutrality removes the zeroth-order charge density. The remaining linear equations form a coefficient matrix for the field and fluid amplitudes. A nonzero wave exists only when this matrix is singular; its determinant is the dispersion relation.

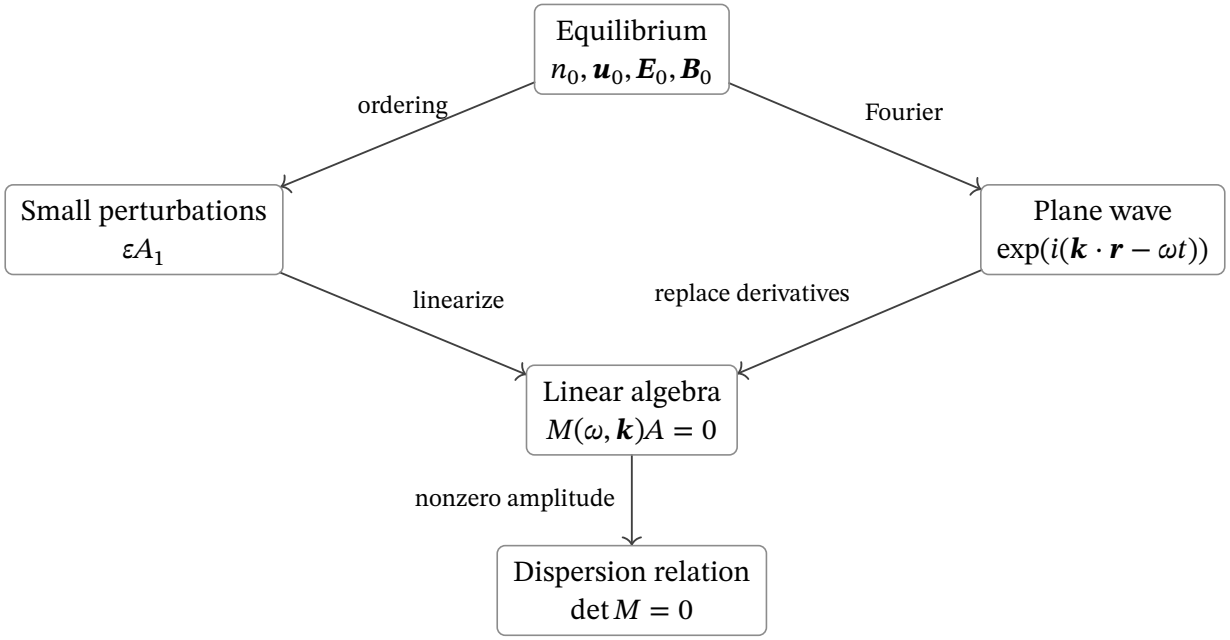


Figure 67: Wave analysis is a sequence of model choices: equilibrium, perturbation ordering, Fourier representation, and a nontrivial-amplitude condition.

Rechenbeispiel Rechenbeispiel 241

A homogeneous hydrogen plasma has equilibrium density $n_{e,0} = n_{i,0} = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, electron temperature energy $k_B T_e = 1.0\,\mathrm{eV} = 1.602 \cdot 10^{-19}\,\mathrm{J}$, and a density perturbation with relative amplitude $\delta n_e / n_{e,0} = 2.0 \cdot 10^{-2}$ [1]. Use $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m^{-1}}$, and a wavelength $\lambda = 0.10\,\mathrm{m}$ at angular frequency $\omega = 1.0 \cdot 10^{10}\,\mathrm{s}^{-1}$, with $k = 2\pi/\lambda$.

Assumptions: cold fixed-ion ordering for the plasma-frequency estimate and small-amplitude perturbation.

Target: report the perturbation parameter, Debye-scale ordering $k\lambda_D$, and frequency ordering $\omega/\omega_{p,e}$.

Numerical result: $\varepsilon = 2.0 \cdot 10^{-2}$ [1], $\lambda_D = 7.43 \cdot 10^{-5}\,\mathrm{m}$, $k\lambda_D = 4.67 \cdot 10^{-3}$ [1], $\omega_{p,e} = 5.64 \cdot 10^{9}\,\mathrm{s}^{-1}$, and $\omega/\omega_{p,e} = 1.77$ [1].

Amplitude ordering and regime ratios Interpretation 242

The small parameter controls amplitudes, while the ratios $\omega/(kv_{\mathrm{th}})$, $\omega/\omega_{p,e}$, and $k\lambda_D$ control the physical regime. A perturbation can be small and still be kinetic, magnetized, or strongly dispersive.

Summary Summary 243

A homogeneous equilibrium plus a first-order expansion turns the fluid and Maxwell system into a linear algebra problem. The plane-wave replacement gives an algebraic response, and the determinant condition selects the allowed wave branches.

Knowledge check Knowledge check 244

- Why must the equilibrium be specified before linearizing a plasma model?
- What does the plane-wave ansatz do to a time derivative and a spatial gradient?
- What is the difference between longitudinal and transverse polarization?
- Why is a dispersion relation a determinant condition?

Answers

Why must the equilibrium be specified before linearizing a plasma model?
The equilibrium determines which zeroth-order forces cancel and which products are first or second order. Without it, terms such as background flow, background magnetic force, and charge density cannot be classified consistently.

What does the plane-wave ansatz do to a time derivative and a spatial gradient?
For the convention $\exp(i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t))$, $\partial f/\partial t$ becomes $-i\omega f$ and $\boldsymbol{\nabla}(f)$ becomes $i\boldsymbol{k}f$ when acting on the amplitude.

What is the difference between longitudinal and transverse polarization?
Longitudinal fields are parallel to $\boldsymbol{k}$ and can create a charge-density perturbation. Transverse fields are perpendicular to $\boldsymbol{k}$ and, in the plane-wave Maxwell system, have vanishing divergence of the electric field.

Why is a dispersion relation a determinant condition?
The linearized equations form a homogeneous matrix equation for the wave amplitudes. A nonzero solution exists only when the matrix is singular, so its determinant must vanish.

12.2 Nonmagnetized response and plasma oscillations

What oscillates when a cold, unmagnetized plasma is displaced? In the high-frequency limit the electrons move against an almost fixed ion background. Charge separation creates an electric restoring force, so the oscillation frequency is set by density and charge, not by a spatial wavelength.

Learning objectives

- derive the electron plasma frequency from continuity, momentum, and Poisson
- identify the restoring force in a cold electrostatic oscillation
- state the correction when both electron and ion inertia are retained
- separate a localized plasma oscillation from a propagating wave

Unit ledger
The plasma frequency $\omega_{p,e}$ is in s^{-1}. The normalized displacement ξ/ξ_0 and time $\omega_{p,e}t$ are dimensionless.

Cold, unmagnetized, collisionless electron response — Assumptions 245

Use $\boldsymbol{B}_0 = \boldsymbol{0}$, negligible pressure, negligible collisions, fixed singly charged ions, and a small longitudinal electric perturbation. The electron equilibrium density equals the ion equilibrium density $n_{e,0} = n_{i,0} = n_0$. Ion motion is added only as a stated extension.

Electron plasma frequency — Definition 246

The electron plasma frequency is $\omega_{p,e} = \sqrt{n_0 e^2/(\varepsilon_0 m_e)}$. It is a local collective frequency. Since the cold longitudinal dispersion relation contains no k, this idealized oscillation has no group propagation.

Cold electrostatic plasma oscillation — Governing law 247

For a longitudinal perturbation with fixed ions, $\omega^2 = \omega_{p,e}^2 = n_0 e^2/(\varepsilon_0 m_e)$. If every species is allowed to move coherently, the restoring frequency becomes $\omega_p^2 = \sum_s \left(n_{s,0} q_s^2/(\varepsilon_0 m_s)\right)$.

Derivation: charge separation supplies the restoring force

Step: Write the electron response

For the electron charge $q_e = -e$, the cold linearized momentum equation is

$$-i\omega m_e \boldsymbol{u}_{e,1} = -e\boldsymbol{E}_1.$$

Longitudinal continuity gives

$$n_{e,1} = \frac{n_0\left(\boldsymbol{k}\cdot\boldsymbol{u}_{e,1}\right)}{\omega}.$$

For a field parallel to $\boldsymbol{k}$, this reduces to

$$n_{e,1} = \frac{n_0 k u_{e,1}}{\omega}.$$

Step: Close the charge–field loop

The fixed ions do not contribute a perturbed charge. Poisson's equation is therefore

$$ikE_1 = \frac{\rho_{q,1}}{\varepsilon_0} = -\frac{e n_{e,1}}{\varepsilon_0}.$$

Solving the momentum equation gives

$$u_{e,1} = \frac{-ieE_1}{m_e\omega}.$$

Inserting this response into continuity and then Poisson gives

$$ikE_1 = -\frac{e}{\varepsilon_0}\left(\frac{-in_0 ekE_1}{m_e\omega^2}\right).$$

Step: Identify the collective frequency

Cancel the nonzero field amplitude and the common factor ik:

$$\omega^2 = \frac{n_0 e^2}{\varepsilon_0 m_e}.$$

The wave number disappears because the cold model has no pressure gradient and hence no spatial restoring scale. If the ions also move, repeat the response calculation for every species and add their charge responses. The result is

$$\omega_p^2 = \sum_s \left(\frac{n_{s,0} q_s^2}{\varepsilon_0 m_s}\right).$$

For hydrogen, the ion contribution is smaller than the electron term by m_e/m_i.

Rechenbeispiel

Rechenbeispiel 248

Assume a homogeneous, cold, collisionless, longitudinal electrostatic perturbation with fixed ions. For a cold hydrogen plasma use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, and $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m^{-1}}$. Determine the electron plasma frequency and its ordinary frequency $f_p = \omega_{p,e}/(2\pi)$.

Numerical result: $\omega_{p,e} = 5.64 \cdot 10^9\,\mathrm{s^{-1}}$ and $f_p = 8.98 \cdot 10^8\,\mathrm{s^{-1}}$.

A plasma oscillation can be local

Interpretation 249

The electron slab can oscillate at every location without transporting information from one location to another in the cold fixed-ion limit. Pressure, magnetic coupling, boundaries, or kinetic phase mixing add the spatial dependence that turns the local oscillation into a propagating or damped mode.

Summary

Summary 250

A displaced cold electron population creates a charge-separation field. This gives $\omega_{p,e}^2 = n_0 e^2/(\varepsilon_0 m_e)$. The frequency is collective and local in the cold fixed-ion limit; ion inertia adds the corresponding ion plasma-frequency contribution.

Knowledge check

Knowledge check 251

- What provides the restoring force for a cold electron plasma oscillation?
- Why does the cold fixed-ion plasma-oscillation frequency not contain the wave number?
- How does allowing the ions to move change the collective frequency?
- When does the cold plasma oscillation cease to be an adequate model?

Answers

What provides the restoring force for a cold electron plasma oscillation?

The electron displacement leaves a small charge imbalance. Poisson's equation turns that imbalance into an electric field that accelerates the electrons back toward neutrality.

Why does the cold fixed-ion plasma-oscillation frequency not contain the wave number?

The cold model has no pressure or other spatial restoring term. The charge-separation balance is local, so the frequency is independent of k in this limit.

How does allowing the ions to move change the collective frequency?

Each mobile species contributes $n_{s,0}q_s^2/(\varepsilon_0 m_s)$ to the squared collective frequency. The ion contribution is usually small because the ion mass is large.

When does the cold plasma oscillation cease to be an adequate model?

Finite pressure, collisions, magnetic fields, finite amplitude, boundaries, or kinetic resonances can alter the response. The relevant ordering must be checked before using the cold local result.

12.3 Phase velocity, group velocity, and electromagnetic dispersion

How does a transverse electromagnetic disturbance differ from a local plasma oscillation? The fields must satisfy both Maxwell's equations and the current response of the electrons. The resulting branch has a cutoff and a wavelength-dependent phase and group velocity.

Learning objectives

- define phase and group velocity for a one-dimensional branch
- derive the cold unmagnetized electromagnetic dispersion relation
- interpret the plasma cutoff and the evanescent regime
- explain why a phase velocity larger than c does not transmit information

Unit ledger

The velocities v_{phi} and v_{g} are in $\mathrm{m\,s^{-1}}$. The dielectric factor ε_r, $kc/\omega_{p,e}$, and $\omega/\omega_{p,e}$ are dimensionless.

Cold transverse response in an unmagnetized plasma

Assumptions 252

Use a homogeneous, collisionless electron fluid with fixed ions, $\boldsymbol{B}_0 = \boldsymbol{0}$, and $\boldsymbol{k} \cdot \boldsymbol{E}_1 = 0$. Retain the displacement current in Maxwell's equations. The current response is linear and isotropic.

Phase and group velocity Definition 253

For a branch $\omega(k)$ with $k > 0$, define $v_{\mathrm{phi}} = \omega/k$ and $v_{\mathrm{g}} = \mathrm{d}\omega/\mathrm{d}k$. Phase velocity tracks a constant phase surface. For a narrow-band packet in a weakly dispersive medium, group velocity describes the motion of its envelope; energy and information transport require the full causal medium and boundary conditions.

Cold electromagnetic branch Governing law 254

The electron current produces $\varepsilon_r(\omega) = 1 - \omega_{p,e}^2/\omega^2$ and $k^2c^2 = \omega^2\varepsilon_r$. Equivalently, $\omega^2 = \omega_{p,e}^2 + c^2k^2$, $v_{\mathrm{phi}} = c\sqrt{1 + \omega_{p,e}^2/(c^2k^2)}$, and $v_{\mathrm{g}} = c^2k/\omega$.

The relative dielectric factor ε_r summarizes how the induced electron current changes the electromagnetic response; vacuum has $\varepsilon_r = 1$. It is distinct from the small perturbation parameter ε used earlier. For a real drive frequency below the plasma frequency, this factor is negative, so k is imaginary. A boundary-driven field then decays into the plasma instead of propagating as a bulk wave.

Derivation: current response and the transverse wave equation

Step: Compute the cold current response

The cold electron momentum equation gives

$$-i\omega m_e \boldsymbol{u}_{e,1} = -e\boldsymbol{E}_1,$$

and hence

$$\boldsymbol{u}_{e,1} = \frac{-ie\boldsymbol{E}_1}{m_e\omega}.$$

The perturbed current is

$$\boldsymbol{j}_1 = -en_0\boldsymbol{u}_{e,1} = \frac{in_0e^2\boldsymbol{E}_1}{m_e\omega}.$$

Step: Eliminate the magnetic amplitude

For a transverse plane wave, Faraday's and Ampere's equations are

$$\boldsymbol{k} \times \boldsymbol{E}_1 = \omega\boldsymbol{B}_1$$

and

$$\boldsymbol{k} \times \boldsymbol{B}_1 = -\frac{\omega\boldsymbol{E}_1}{c^2} - i\mu_0\boldsymbol{j}_1,$$

with $c^2 = 1/(\mu_0\varepsilon_0)$.

Substitute the first relation into the second and use

$$\boldsymbol{k} \times (\boldsymbol{k} \times \boldsymbol{E}_1) = -k^2\boldsymbol{E}_1.$$

The resulting electric-field equation is

$$(\omega^2 - c^2k^2)\boldsymbol{E}_1 = -\frac{i\omega\boldsymbol{j}_1}{\varepsilon_0}.$$

Step: Read off the electromagnetic branch

Insert the current response and define $\omega_{p,e}^2 = n_0 e^2/(\varepsilon_0 m_e)$:

$$\left(\omega^2 - c^2 k^2\right)\boldsymbol{E}_1 = \omega_{p,e}^2 \boldsymbol{E}_1.$$

A nonzero field amplitude therefore requires

$$\omega^2 = \omega_{p,e}^2 + c^2 k^2.$$

Dividing by ω^2 identifies the relative dielectric factor:

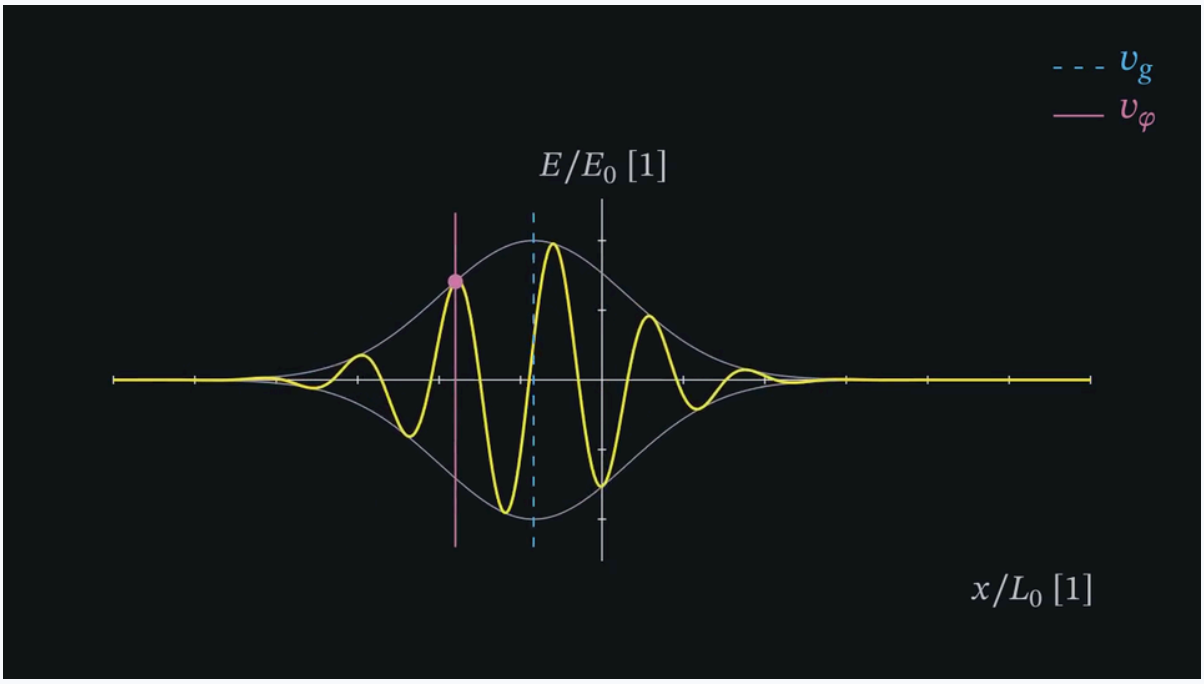
$$\frac{k^2 c^2}{\omega^2} = 1 - \frac{\omega_{p,e}^2}{\omega^2}.$$

Real k requires $\omega > \omega_{p,e}$. Below the cutoff, k is imaginary and the field is evanescent rather than a propagating bulk wave.

Pause and predict

Key idea 255

Before playing the animation, identify which marker should move faster: the carrier phase or the packet envelope. The answer follows from the two velocities in the governing law.



Watch animation

The prescribed carrier and envelope illustrate different phase and group speeds. With $X = x/L_0$, $\tau = t/t_0$, and $A = E/E_0$, the ansatz is $A = \exp\left(-\frac{(X-0.42\tau)^2}{2(1.15)^2}\right)\cos(5.2(X-0.90\tau))$. Here L_0, t_0, and E_0 are arbitrary reference length, time, and field scales; velocities are in units L_0/t_0. Thus $\sigma/L_0 = 1.15$, $kL_0 = 5.2$, $v_g t_0/L_0 = 0.42$, and $v_{phi} t_0/L_0 = 0.90$ are prescribed dimensionless parameters. This ansatz is not an exact solution of the cold-plasma dispersion relation. The phase marker follows a cosine maximum; the varying envelope shifts the maxima of the total field.

Figure 68: The prescribed carrier and envelope illustrate different phase and group speeds. With $X = x/L_0$, $\tau = t/t_0$, and $A = E/E_0$, the ansatz is $A = \exp\left(-\frac{(X-0.42\tau)^2}{2(1.15)^2}\right)\cos(5.2(X-0.90\tau))$. Here L_0, t_0, and E_0 are arbitrary reference length, time, and field scales; velocities are in units L_0/t_0. Thus $\sigma/L_0 = 1.15$, $kL_0 = 5.2$, $v_g t_0/L_0 = 0.42$, and $v_{phi} t_0/L_0 = 0.90$ are prescribed dimensionless parameters. This ansatz is not an exact solution of the cold-plasma dispersion relation. The phase marker follows a cosine maximum; the varying envelope shifts the maxima of the total field.

Rechenbeispiel Rechenbeispiel 256

Assume the homogeneous, cold, collisionless, unmagnetized, fixed-ion transverse electromagnetic model. A cold electromagnetic wave has $\omega/\omega_{p,e} = 2$ [1] in a plasma with $\omega_{p,e} = 5.64 \cdot 10^9\ \mathrm{s}^{-1}$ and $c = 2.998 \cdot 10^8\ \mathrm{m\,s}^{-1}$. Determine k, the phase velocity, the group velocity, and the wavelength.

Numerical result: $kc/\omega_{p,e} = 1.732$ [1], $v_{phi}/c = 1.155$ [1], $v_g/c = 0.866$ [1], $k = 32.6\ \mathrm{m}^{-1}$, and $\lambda = 0.193$ m.

Superluminal phase is not superluminal signalling Interpretation 257

For the cold branch $v_{phi} v_g = c^2$. A phase crest can move faster than c because it is not a localized signal. The packet envelope, causal turn-on, and energy flow carry the physically relevant information and approach the vacuum limit consistently.

Summary

Summary 258

The cold unmagnetized electromagnetic branch obeys $\omega^2 = \omega_{p,e}^2 + c^2k^2$. It has a cutoff at $\omega_{p,e}$, $v_{\mathrm{phi}} > c$, $v_{\mathrm{g}} < c$, and the product $v_{\mathrm{phi}}v_{\mathrm{g}} = c^2$ in the propagating regime.

Knowledge check

Knowledge check 259

- What is the physical meaning of the phase velocity?
- What condition allows the cold electromagnetic branch to propagate in the bulk plasma?
- Why does the cold electromagnetic branch approach the vacuum branch at high frequency?
- Why can the phase velocity exceed the speed of light in this model?

Answers

What is the physical meaning of the phase velocity?

It is $v_{\mathrm{phi}} = \omega/k$, the speed of a constant phase surface. It need not equal the speed of a localized pulse or of information.

What condition allows the cold electromagnetic branch to propagate in the bulk plasma?

The wave number must be real, which requires $\omega > \omega_{p,e}$ in the fixed-ion cold model. Below the cutoff the solution is evanescent.

Why does the cold electromagnetic branch approach the vacuum branch at high frequency?

When $\omega \gg \omega_{p,e}$, electron inertia makes the current response small relative to the displacement current. The dielectric factor approaches one and $\omega \approx ck$.

Why can the phase velocity exceed the speed of light in this model?

The phase surface is not a causal signal. For the branch, $v_{\mathrm{phi}} > c$ is accompanied by $v_{\mathrm{g}} < c$; signal propagation is governed by the causal packet and medium response.

12.4 Warm-fluid wave equations and kinetic limits

What changes when pressure and particle velocity spread are restored? A warm-fluid closure makes the restoring force depend on wavelength. The resulting branches include warm Langmuir and ion-acoustic limits, while kinetic theory adds resonant particles and phase mixing that no finite set of fluid moments can represent exactly.

Learning objectives

- derive the warm electrostatic fluid susceptibility
- identify the warm Langmuir and ion-acoustic limits
- state the ordering behind the quasi-neutral ion-acoustic approximation
- explain what kinetic response adds beyond a fluid pressure closure

Unit ledger

Thermal speed and ion-acoustic speed are in $\mathrm{m\,s^{-1}}$, and $\lambda_D = \sqrt{\varepsilon_0 k_B T_e/(n_0 e^2)}$ is in m. The phase parameters $k\lambda_D$ and $\omega/(kv_{\mathrm{th,s}})$ are dimensionless.

Unmagnetized, electrostatic, warm-fluid closure — Assumptions 260

Use $\boldsymbol{B}_0 = \mathbf{0}$, a longitudinal electric field, collisionless perturbations, and a local pressure closure $p_{s,1} = \gamma_s k_B T_s n_{s,1}$. The equilibrium is quasi-neutral with equal electron and ion densities. Kinetic corrections are discussed separately as a limit comparison.

Warm-fluid scales — Definition 261

Define $c_s^2 = \gamma_s k_B T_s / m_s$ and $\omega_{p,s}^2 = n_{s,0} q_s^2/(\varepsilon_0 m_s)$. Here the index s labels a species: c_e and c_i are its electron and ion pressure-response speeds. Neither is necessarily the phase velocity of a collective branch. The thermal-speed convention used for kinetic comparisons is $v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$.

Plasma-dispersion function convention — Definition 262

For a Maxwellian, the velocity integral in the kinetic response can be expressed using the dimensionless Fried–Conte function

$$Z(\zeta) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{\exp(-x^2)}{x - \zeta}\, \mathrm{d}x,$$

initially defined for $\Im(\zeta) > 0$ and continued causally to real and lower-half-plane frequencies. Here x is a dimensionless integration variable for velocity, not position. The argument ζ compares phase speed with thermal speed, as specified for each species below. For $k > 0$ and $\exp(-i\omega t)$, deform the velocity contour below the pole as it crosses the real axis. The factor of two in the susceptibility below follows from the stated convention $v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$.

Warm electrostatic fluid dispersion — Governing law 263

The longitudinal two-species warm-fluid dispersion relation is $1 - \sum_s \omega_{p,s}^2/(\omega^2 - k^2 c_s^2) = 0$. With fixed ions this gives $\omega^2 = \omega_{p,e}^2 + k^2 c_e^2$. In the low-frequency quasi-neutral hydrogen limit, requiring $\omega \ll \omega_{p,e}$, $k\lambda_D \ll 1$, and $|\omega/(kc_e)| \ll 1$ to neglect electron inertia relative to electron pressure,

$$\omega^2 \approx \frac{k^2(\gamma_e k_B T_e + \gamma_i k_B T_i)}{m_i}.$$

Derivation: warm-fluid susceptibility and ion sound

Step: Obtain the response of one warm species

For one species in one dimension, Fourier continuity gives

$$-i\omega n_{s,1} + ik n_{s,0} u_{s,1} = 0.$$

Therefore

$$n_{s,1} = \frac{k n_{s,0} u_{s,1}}{\omega}.$$

With the pressure closure, the longitudinal momentum equation is

$$-i\omega m_s u_{s,1} = q_s E_1 - \frac{ik\gamma_s k_B T_s n_{s,1}}{n_{s,0}}.$$

Define $c_s^2 = \gamma_s k_B T_s / m_s$. Substitution of continuity and solution for the velocity gives

$$u_{s,1} = \frac{i q_s \omega E_1}{m_s(\omega^2 - k^2 c_s^2)}.$$

The density response is consequently

$$n_{s,1} = \frac{i n_{s,0} q_s k E_1}{m_s(\omega^2 - k^2 c_s^2)}.$$

Step: Sum the species in Poisson's equation

The longitudinal field obeys

$$ikE_1 = \frac{1}{\varepsilon_0} \sum_s q_s n_{s,1}.$$

Cancel ikE_1 and insert the species response:

$$1 = \sum_s \frac{n_{s,0} q_s^2}{\varepsilon_0 m_s(\omega^2 - k^2 c_s^2)}.$$

This is the warm-fluid longitudinal dispersion relation.

Step: Select the two-fluid limits

For a high-frequency electron branch with fixed ions, omit the ion term:

$$\omega^2 = \omega_{p,e}^2 + k^2 c_e^2.$$

For a low-frequency ion-acoustic branch, neglect electron inertia and impose $n_{e,1} = n_{i,1}$ through quasi-neutrality. Electron pressure then supplies the electric field and ion inertia carries the slow wave:

$$\omega^2 = \frac{k^2(\gamma_e k_B T_e + \gamma_i k_B T_i)}{m_i}.$$

This reduction requires $\omega \ll \omega_{p,e}$, $k\lambda_D \ll 1$, and $|\omega/(kc_e)| \ll 1$. The last condition follows by comparing ω^2 with $k^2 c_e^2$ in the electron susceptibility; it controls neglect of electron inertia relative to pressure. The Debye ordering controls charge separation.

Step: Compare with kinetic response

Kinetic theory replaces the fluid closure by a velocity-space response. For isotropic, non-drifting Maxwellian equilibria of each responding species, with $v_{\mathrm{th,s}} = \sqrt{2k_B T_s / m_s}$, the unmagnetized electrostatic form is

$$1 + \sum_s \chi_s(\omega, k) = 0,$$

where

$$\chi_s = \frac{2\omega_{p,s}^2}{k^2 v_{\mathrm{th,s}}^2}(1 + \zeta_s Z(\zeta_s)), \quad \zeta_s = \frac{\omega}{k v_{\mathrm{th,s}}}.$$

The analytically continued plasma-dispersion function Z accounts for resonant particles through causal continuation from $\Im(\omega) > 0$. Other distributions require their own velocity-space integral. The imaginary contribution describes resonant energy exchange: the non-drifting Maxwellian response gives Landau damping, while suitable non-equilibrium distributions can drive growth. A warm-fluid γ_s can reproduce selected long-wavelength real-frequency limits, but it cannot reproduce resonant phase mixing.

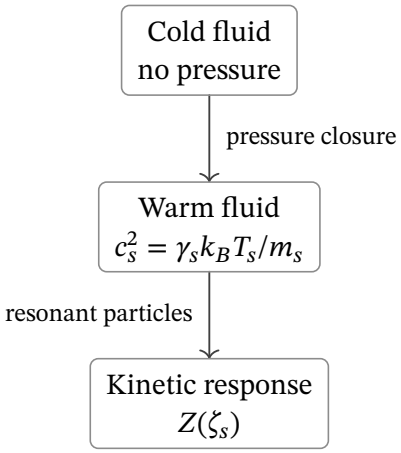


Figure 69: Cold fluid, warm fluid, and kinetic descriptions are nested model choices. Pressure introduces wavelength dependence; kinetic response additionally resolves resonant particles and phase mixing.

In the following example, c_s denotes the collective ion-acoustic speed, as is common in sound-wave notation. It is determined by electron pressure and ion inertia for the stated cold-ion limit, not by the ion pressure-response speed c_i defined above.

Rechenbeispiel Rechenbeispiel 264

Assume an unmagnetized, electrostatic warm-fluid model with cold ions, an isothermal electron response, and quasi-neutral long-wavelength ordering. For cold ions and an isothermal electron response use $k_B T_e = 10\,\mathrm{eV} = 1.602 \cdot 10^{-18}\,\mathrm{J}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, and $k = 1.0\,\mathrm{m}^{-1}$. Determine the ion-acoustic speed, frequency, and the ratio $k\lambda_D$ for $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$. Numerical result: $c_s = 3.09 \cdot 10^4\,\mathrm{m\,s}^{-1}$, $\omega = 3.09 \cdot 10^4\,\mathrm{s}^{-1}$, $\lambda_D = 2.35 \cdot 10^{-4}\,\mathrm{m}$, and $k\lambda_D = 2.35 \cdot 10^{-4}$ [1].

Fluid and kinetic responses Interpretation 265

A warm-fluid dispersion relation is a controlled closure for selected moments and scales. A kinetic response resolves velocity-space structure, including resonant particles and Landau damping. Agreement of real frequencies in one limit does not make the models interchangeable.

Summary Summary 266

Pressure adds k-dependent restoring forces. The warm-fluid longitudinal response obeys $1 - \sum_s \omega_{p,s}^2/(\omega^2 - k^2 c_s^2) = 0$, whose limits include warm Langmuir and ion-acoustic waves. Kinetic response adds resonant velocity-space physics and collisionless damping.

Knowledge check

Knowledge check 267

- What new scale enters the warm-fluid electrostatic dispersion relation?
- Why can an ion-acoustic wave be low frequency even though the electric field is produced by electrons?
- Which ordering justifies the quasi-neutral ion-acoustic limit?
- What physical effect is absent from a finite warm-fluid closure?

Answers

What new scale enters the warm-fluid electrostatic dispersion relation?
The pressure response introduces $k^2c_s^2$, so the frequency depends on wavelength through the species pressure-response speed.

Why can an ion-acoustic wave be low frequency even though the electric field is produced by electrons?
Electrons establish the pressure-supported electric response quickly, while the large ion inertia carries the slow density motion.

Which ordering justifies the quasi-neutral ion-acoustic limit?
The mode must be slow compared with the electron plasma response and long compared with the Debye scale: $\omega \ll \omega_{p,e}$ and $k\lambda_D \ll 1$. Neglecting electron inertia relative to pressure also requires $|\omega/(kc_e)| \ll 1$.

What physical effect is absent from a finite warm-fluid closure?
A finite moment closure does not resolve resonant particles and their phase mixing. Kinetic response is needed for Landau damping or growth and for nonlocal velocity-space structure.

12.5 Reading dispersion diagrams and checking limits

How do we read a dispersion diagram without losing the assumptions? First normalize both axes, then identify real and imaginary wave numbers, slopes, cutoffs, and asymptotic branches.

Learning objectives

- read cutoffs, slopes, and propagating regions from a dispersion plot
- use dimensionless variables to compare plasma and vacuum branches
- check long-wavelength, high-frequency, and weak-plasma limits
- identify when a branch is evanescent or when a fluid model has left its ordering

Unit ledger

The normalized axes use $K = kc/\omega_{p,e}$ and $\hat{\omega} = \omega/\omega_{p,e}$, both dimensionless. The reconstructed wave number is in m^{-1}, frequency in s^{-1}, and velocities in $\mathrm{m\,s}^{-1}$. For warm electrostatic checks, $K_D = k\lambda_D$ is also dimensionless.

Validity range of the plotted branches

Assumptions 268

Read the cold electromagnetic curve only for a homogeneous, unmagnetized, collisionless, fixed-ion plasma. The vacuum line is a reference. If a branch leaves the assumptions, switch model before extrapolating the curve.

Dimensionless dispersion coordinates Definition 269

Use $K = kc/\omega_{p,e}$ and $\hat{\omega} = \omega/\omega_{p,e}$ for the cold electromagnetic branch. Then $\hat{\omega}^2 = 1 + K^2$, $v_{\mathrm{phi}}/c = \hat{\omega}/K$, and $v_{\mathrm{g}}/c = K/\hat{\omega}$. The cutoff is the intercept $\hat{\omega} = 1$ at $K = 0$. The tangent slope $\mathrm{d}\hat{\omega}/\mathrm{d}K$ gives v_{g}/c; the slope of the line from the origin to a point on the branch gives $v_{\mathrm{phi}}/c = \hat{\omega}/K$. These are different geometric measurements of the same curve.

Derivation: limiting-case checks

Step: Long-wavelength limit

Start from the normalized transverse relation

$$\hat{\omega}^2 = 1 + K^2.$$

As $K \to 0$, $\hat{\omega} \to 1$. Hence

$$\omega \to \omega_{p,e}, \quad \frac{v_{\mathrm{g}}}{c} = \frac{K}{\hat{\omega}} \to 0.$$

The phase velocity formally diverges because a nearly spatially uniform oscillation has a finite frequency while its wave number tends to zero.

Step: Short-wavelength and vacuum limits

As $K \to \infty$,

$$\hat{\omega} = \sqrt{1 + K^2} \to K.$$

Therefore

$$\frac{v_{\mathrm{phi}}}{c} = \frac{\hat{\omega}}{K} \to 1, \quad \frac{v_{\mathrm{g}}}{c} = \frac{K}{\hat{\omega}} \to 1.$$

The plasma becomes transparent and the branch approaches the vacuum line. The same result follows dimensionally when the density tends to zero, because $\omega_{p,e} \to 0$ and $\omega = ck$.

Step: Below-cutoff behavior

If $\omega < \omega_{p,e}$, then

$$k^2 = \frac{\omega^2 - \omega_{p,e}^2}{c^2} < 0.$$

Write

$$k = i\alpha, \quad \alpha = \frac{\sqrt{\omega_{p,e}^2 - \omega^2}}{c}.$$

Choosing the decaying boundary solution gives the spatial factor $\exp(-\alpha x)$. This is an evanescent field.

Step: Compare with the cold electrostatic branch

The cold electrostatic fixed-ion branch has $\omega = \omega_{p,e}$ for every k, so its group velocity vanishes. Adding pressure changes the relation to

$$\omega^2 = \omega_{p,e}^2 + k^2 c_e^2.$$

Pressure therefore supplies spatial propagation.

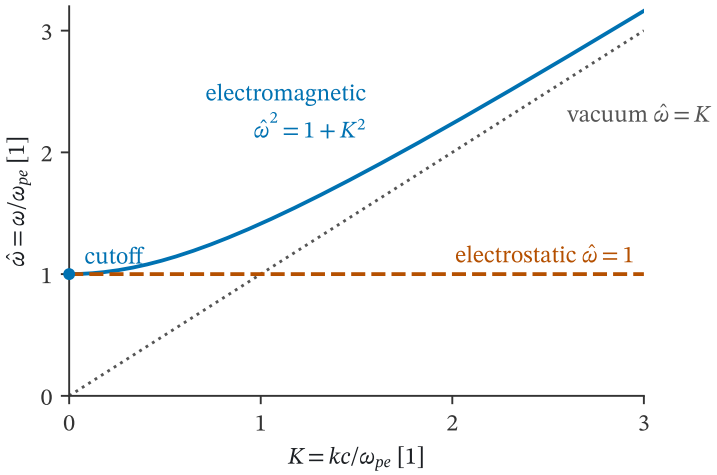


Figure 70: Cold, unmagnetized, fixed-ion branches. The electromagnetic branch $\hat{\omega}^2 = 1 + K^2$ has its cutoff at $\hat{\omega} = 1$ and approaches the vacuum line $\hat{\omega} = K$; the electrostatic oscillation $\hat{\omega} = 1$ has zero group velocity.

Rechenbeispiel Rechenbeispiel 270

Assume a homogeneous, cold, collisionless, unmagnetized, fixed-ion electromagnetic model. A radio source drives a plasma with $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$ and $c = 2.998 \cdot 10^{8}\,\mathrm{m\,s}^{-1}$. Classify a drive at $\omega = 4.0 \cdot 10^{9}\,\mathrm{s}^{-1}$ and another at $\omega = 1.13 \cdot 10^{10}\,\mathrm{s}^{-1}$ as propagating or evanescent. For the propagating drive, report k and the normalized phase and group speeds. Numerical result: $\omega_{p,e} = 5.64 \cdot 10^{9}\,\mathrm{s}^{-1}$; the lower-frequency drive is evanescent with $\alpha = 13.3\,\mathrm{m}^{-1}$. The higher-frequency drive is propagating with $k = 32.7\,\mathrm{m}^{-1}$, $v_{\mathrm{phi}}/c = 1.15$ [1], and $v_{\mathrm{g}}/c = 0.866$ [1].

Cutoff, slope, and asymptote checks Interpretation 271

The intercept identifies the cutoff, the slope gives group velocity, and the asymptote checks the vacuum limit. If a curve violates one of these tests, revisit signs, units, polarization, and the omitted physics before using it for a prediction.

Summary Summary 272

Normalize first, then read the branch: $\hat{\omega}^2 = 1 + K^2$ has a cutoff at $\hat{\omega} = 1$, approaches the vacuum line at large K, and becomes evanescent below the cutoff. Limiting cases expose missing factors, incorrect polarizations, and unjustified model extrapolations.

Knowledge check

- What does the intercept of the cold electromagnetic branch represent?
- How can the slope of a dispersion curve be used physically?
- What is the vacuum limit of the cold electromagnetic branch?
- What is the signature of an evanescent solution in the wave number?

Answers

What does the intercept of the cold electromagnetic branch represent?

The intercept $\hat{\omega} = 1$ at $K = 0$ is the plasma cutoff $\omega = \omega_{p,e}$. Frequencies below it have an imaginary bulk wave number in this model.

How can the slope of a dispersion curve be used physically?

The local slope $\mathrm{d}\omega/\mathrm{d}k$ is the group velocity for a narrow-band packet. It is distinct from the ratio ω/k, which is the phase velocity.

What is the vacuum limit of the cold electromagnetic branch?

As density tends to zero or frequency becomes much larger than the plasma frequency, $\omega \approx ck$, and both normalized phase and group speeds approach the speed of light.

What is the signature of an evanescent solution in the wave number?

A frequency below cutoff gives $k^2 < 0$, so one chooses $k = i\alpha$ and obtains exponential spatial decay rather than a propagating oscillation.

13. Waves in cold magnetized plasmas

A static magnetic field turns the isotropic plasma response into a tensor. This chapter derives that tensor from the cold-fluid momentum equation and uses it to organize parallel, perpendicular, and oblique wave propagation. Propagation direction, polarization, and frequency ordering are carried together throughout, following the standard cold-plasma treatment in [3].

The magnetic field selects a basis — Key idea 274

The magnetic field couples electric-field components perpendicular to $\boldsymbol{B}_0$ and leaves the parallel response distinct. The wave vector $\boldsymbol{k}$ supplies a second direction: a component perpendicular to $\boldsymbol{B}_0$ need not be transverse to the wave. Keep these two geometries separate when identifying polarization.

13.1 Cold magnetized response and the dielectric tensor

How does a uniform magnetic field enter a linear wave calculation? Start with the cold momentum equation for each species, solve the transverse velocity response, and convert the current into a dielectric tensor. Once this tensor is known, Maxwell's equations supply the wave matrix.

Learning objectives

- state the cold, homogeneous, collisionless magnetized ordering
- derive the species current response from the linear momentum equation
- identify the transverse and parallel dielectric coefficients
- write the Maxwell wave matrix and its determinant condition

Unit ledger

The dielectric coefficients and refractive index $N = kc/\omega$ are dimensionless.

Uniform cold magnetized equilibrium — Assumptions 275

Use constant $n_{s,0}$ and $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$, with no equilibrium flow, no collisions, no pressure perturbation, and no spatial gradients of the equilibrium. The ions are fixed for the electron-response formulas below; retaining their inertia means summing the species responses. Use the Fourier convention $\exp(i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t))$.

Signed gyrofrequency and plasma frequency — Definition 276

For species s, define the signed gyrofrequency and the positive plasma frequency by $\Omega_s = q_s B_0/m_s$ and $\omega_{p,s}^2 = n_{s,0} q_s^2/(\varepsilon_0 m_s)$. For electrons it is useful to reserve $\omega_{c,e} = |q_e| B_0/m_e > 0$ for the cyclotron-frequency magnitude. The sign of q_e remains in the transverse polarization convention.

Cold dielectric tensor Definition 277

The current response is represented by $\boldsymbol{\varepsilon}_p \cdot \boldsymbol{E}_1 = \boldsymbol{E}_1 + (i/(\omega\varepsilon_0))\boldsymbol{j}_1$, so that $\boldsymbol{\varepsilon}_p = \boldsymbol{I} + \sum_s \boldsymbol{\chi}_s$ with species susceptibility $\boldsymbol{\chi}_s = \boldsymbol{\sigma}_s/(-i\omega\varepsilon_0)$ for the species conductivity $\boldsymbol{\sigma}_s$ defined by $\boldsymbol{j}_{s,1} = \boldsymbol{\sigma}_s \cdot \boldsymbol{E}_1$, with

$$\boldsymbol{\varepsilon}_p = \begin{pmatrix} \varepsilon_\perp & -i\varepsilon_\times & 0 \\ i\varepsilon_\times & \varepsilon_\perp & 0 \\ 0 & 0 & \varepsilon_\parallel \end{pmatrix}.$$

For several species, $\varepsilon_\perp = 1 - \sum_s \omega_{p,s}^2/(\omega^2 - \Omega_s^2)$,

$$\varepsilon_\times = \sum_s \left(\frac{\Omega_s \omega_{p,s}^2}{\omega(\omega^2 - \Omega_s^2)} \right),$$

and $\varepsilon_\parallel = 1 - \sum_s \omega_{p,s}^2/\omega^2$. The off-diagonal coefficient changes sign with the charge convention.

Each tensor column gives the response to one electric-field component. The coefficients $\varepsilon_\perp$ and $\varepsilon_\parallel$ describe response perpendicular and parallel to the background magnetic field; $\varepsilon_\times$ couples the two perpendicular directions. The symbol $\times$ in its subscript is a label, not an instruction to take another cross product. The identity part is the vacuum response, and the remaining terms describe the induced plasma current.

Magnetized Maxwell wave equation Governing law 278

For a plane wave, Maxwell's equations reduce to

$$\boldsymbol{k} \times (\boldsymbol{k} \times \boldsymbol{E}_1) + \frac{\omega^2}{c^2} \boldsymbol{\varepsilon}_p \cdot \boldsymbol{E}_1 = \boldsymbol{0}.$$

Equivalently,

$$\boldsymbol{k}(\boldsymbol{k} \cdot \boldsymbol{E}_1) - k^2 \boldsymbol{E}_1 + \frac{\omega^2}{c^2} \boldsymbol{\varepsilon}_p \cdot \boldsymbol{E}_1 = \boldsymbol{0}.$$

The dispersion relation is the condition that the associated coefficient matrix has zero determinant.

Derivation: from cold momentum to the dielectric tensor

Step: Resolve the cold momentum equation

The linearized cold momentum equation is

$$-i\omega m_s \boldsymbol{u}_{s,1} = q_s(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0).$$

With $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$, the transverse components form a coupled two-by-two system:

$$-i\omega u_{s,1,x} - \Omega_s u_{s,1,y} = \left(\frac{q_s}{m_s} \right) E_{1,x}$$

and

$$\Omega_s u_{s,1,x} - i\omega u_{s,1,y} = \left(\frac{q_s}{m_s} \right) E_{1,y}.$$

The parallel component is independent:

$$-i\omega u_{s,1,z} = \left(\frac{q_s}{m_s}\right) E_{1,z}.$$

Step: Invert the transverse response

Inverting the transverse system gives

$$u_{s,1,x} = \left(\frac{iq_s\omega}{m_s(\omega^2 - \Omega_s^2)}\right) E_{1,x} - \left(\frac{q_s\Omega_s}{m_s(\omega^2 - \Omega_s^2)}\right) E_{1,y},$$

and

$$u_{s,1,y} = \left(\frac{q_s\Omega_s}{m_s(\omega^2 - \Omega_s^2)}\right) E_{1,x} + \left(\frac{iq_s\omega}{m_s(\omega^2 - \Omega_s^2)}\right) E_{1,y}.$$

Multiply by $q_s n_{s,0}$ and sum over species to obtain $\boldsymbol{j}_1$. The transverse entries are coupled because the Lorentz force rotates the velocity in the $x - y$ plane; the parallel motion has no magnetic force.

Step: Define the dielectric response

Insert the current into

$$\boldsymbol{\varepsilon}_p \cdot \boldsymbol{E}_1 = \boldsymbol{E}_1 + \left(\frac{i}{\omega\varepsilon_0}\right) \boldsymbol{j}_1.$$

Reading off the coefficients gives the cold dielectric tensor displayed above. Its off-diagonal entries encode the rotation of transverse motion.

Step: Insert the response into Maxwell's equations

Fourier-transformed Faraday and Ampere laws are

$$\boldsymbol{k} \times \boldsymbol{E}_1 = \omega \boldsymbol{B}_1$$

and

$$\boldsymbol{k} \times \boldsymbol{B}_1 = -\frac{\omega \boldsymbol{E}_1}{c^2} - i\mu_0 \boldsymbol{j}_1.$$

Substitute the first relation into the second, eliminate $\boldsymbol{j}_1$ by the dielectric definition with $\mu_0\varepsilon_0 = 1/c^2$, and use

$$\boldsymbol{k} \times (\boldsymbol{k} \times \boldsymbol{E}_1) = \boldsymbol{k}(\boldsymbol{k} \cdot \boldsymbol{E}_1) - k^2 \boldsymbol{E}_1.$$

A nonzero field amplitude requires the determinant of the resulting wave matrix to vanish.

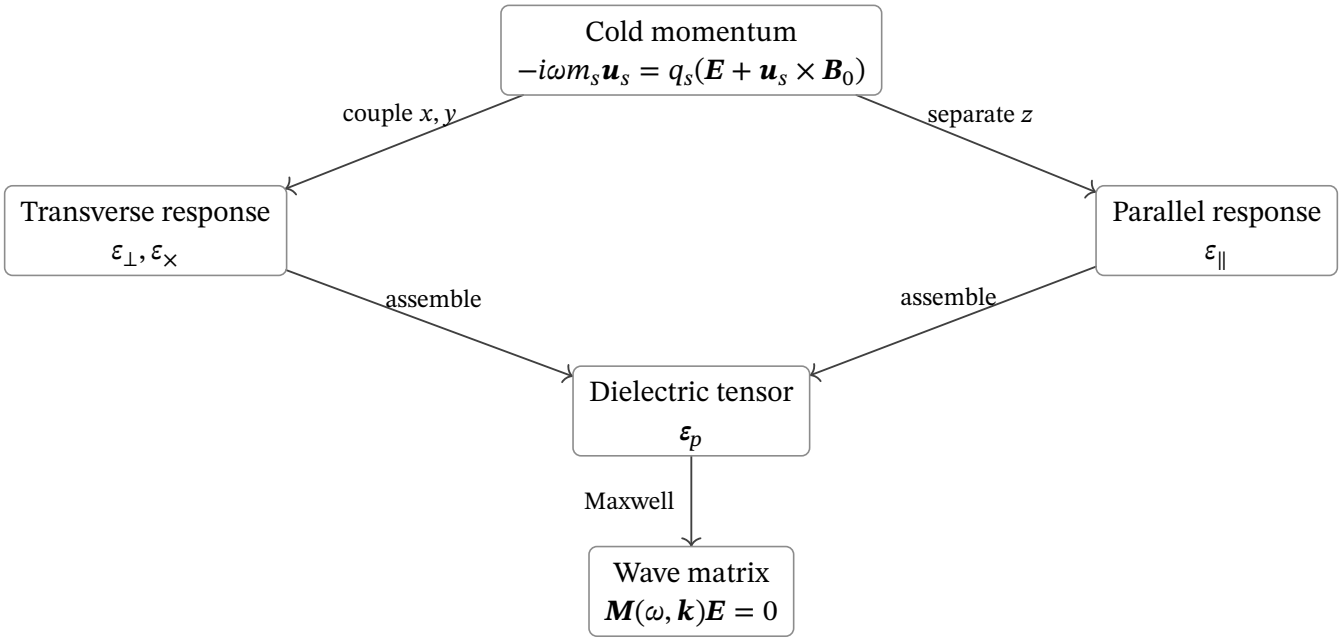


Figure 71: A uniform magnetic field makes the cold plasma dielectric response anisotropic. The transverse pair is coupled through the cyclotron term, while the field component parallel to the background field has its own plasma response.

Rechenbeispiel Rechenbeispiel 279

Assume a homogeneous, cold, collisionless, magnetized electron response with fixed ions and a uniform field $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$. For a hydrogen plasma use $n_0 = 1.0 \cdot 10^{16}\ \mathrm{m}^{-3}$, $B_0 = 1.0 \cdot 10^{-2}\ \mathrm{T}$, $e = 1.602 \cdot 10^{-19}\ \mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\ \mathrm{kg}$, and $\varepsilon_0 = 8.854 \cdot 10^{-12}\ \mathrm{F\,m}^{-1}$. Determine the electron plasma frequency, the electron cyclotron-frequency magnitude, and their ratio.

Numerical result: $\omega_{p,e} = 5.64 \cdot 10^{9}\ \mathrm{s}^{-1}$, $\omega_{c,e} = 1.76 \cdot 10^{9}\ \mathrm{s}^{-1}$, and $\omega_{c,e}/\omega_{p,e} = 0.312$ [1].

Rechenbeispiel Rechenbeispiel 280

Assume a homogeneous, cold, collisionless, fixed-ion electron response with $\boldsymbol{B}_0$ along z. At angular frequency $\omega = 1.00 \cdot 10^{10}\ \mathrm{s}^{-1}$, use $\omega_{p,e} = 5.64 \cdot 10^{9}\ \mathrm{s}^{-1}$, $\omega_{c,e} = 1.76 \cdot 10^{9}\ \mathrm{s}^{-1}$, and the signed convention $\Omega_e = -\omega_{c,e}$. Evaluate the cold dielectric coefficients $\varepsilon_\perp$, $\varepsilon_\times$, and $\varepsilon_\parallel$.

Target: report the three dimensionless entries of the dielectric tensor.

Numerical result: $\varepsilon_\perp = 0.672$ [1], $\varepsilon_\times = -5.78 \cdot 10^{-2}$ [1], and $\varepsilon_\parallel = 0.682$ [1].

Anisotropy of a homogeneous plasma Interpretation 281

The tensor is anisotropic even though the equilibrium is homogeneous. Homogeneity lets us use plane waves; the magnetic field breaks rotational symmetry and determines which field components can exchange phase and energy.

Summary Summary 282

A cold uniform magnetic field makes the plasma response tensorial. Solving the transverse Lorentz-force response gives $\varepsilon_\perp$ and $\varepsilon_\times$, while parallel motion gives $\varepsilon_\parallel$. Maxwell's wave matrix and its determinant then determine the modes.

Knowledge check

- Which assumption makes the cold-fluid momentum equation algebraic in Fourier space?
- Why are the two transverse electric components coupled?
- Why is the parallel dielectric coefficient different from the transverse coefficient?
- What mathematical condition selects a wave mode?

Answers

Which assumption makes the cold-fluid momentum equation algebraic in Fourier space?

A homogeneous equilibrium together with the plane-wave ansatz replaces derivatives by multiplication by $-i\omega$ and $i\boldsymbol{k}$. The cold ordering removes pressure gradients and the collisionless ordering removes drag terms.

Why are the two transverse electric components coupled?

The transverse velocity crossed with $\boldsymbol{B}_0$ points into the other transverse direction. The Lorentz force therefore couples the x and y momentum equations and creates the off-diagonal dielectric coefficient $\varepsilon_\times$.

Why is the parallel dielectric coefficient different from the transverse coefficient?

The magnetic force vanishes for motion parallel to $\boldsymbol{B}_0$. Parallel motion therefore responds with the unmagnetized factor $1 - \sum_s \omega_{p,s}^2/\omega^2$, while transverse motion contains the gyrofrequency denominators.

What mathematical condition selects a wave mode?

The wave amplitudes satisfy a homogeneous matrix equation. A nonzero electric-field vector exists only when the matrix is singular, so its determinant must vanish.

13.2 Parallel propagation and circular polarization

When the wave travels along the magnetic field, the longitudinal and transverse problems separate. The transverse block is diagonalized by circular polarization: the two rotation senses see different effective dielectric coefficients, while the longitudinal branch remains a local plasma oscillation in the cold limit.

Learning objectives

- factor the parallel wave matrix into longitudinal and circular branches
- derive the two circular-mode dispersion relations
- calculate the cutoff frequencies and identify the cyclotron resonance
- explain how different circular wave numbers rotate linear polarization

Unit ledger

The refractive index $N = kc/\omega$, normalized frequency is $\hat{\omega} = \omega/\omega_{p,e}$, and magnetization is $Y = \omega_{c,e}/\omega_{p,e}$; all three are dimensionless. The dimensional k is in m^{-1}, ω and $\omega_{c,e}$ are in s^{-1}, and phase or group velocities are in $\mathrm{m\,s}^{-1}$.

Parallel fixed-ion propagation Assumptions 284

Take $\boldsymbol{k} = k\boldsymbol{e}_z$ parallel to $\boldsymbol{B}_0$, retain the cold collisionless electron response, and treat ions as stationary. The positive magnitude $\omega_{c,e}$ is used below. The labels + and − refer to the chosen Fourier and field-basis convention; reversing the time convention swaps the handedness names but not the two eigenvalues.

Circular eigenmodes Definition 285

For an electron plasma with fixed ions, define the circular eigenmode label $s = +1$ or $s = -1$ by $E_{1,y} = -isE_{1,x}$. The corresponding refractive indices are

$$N_s^2 = \varepsilon_s = \varepsilon_\perp - s\varepsilon_\times = 1 - \frac{\omega_{p,e}^2}{\omega\left(\omega + s\omega_{c,e}\right)}.$$

Thus the $s = +1$ branch has the lower positive-frequency cutoff, while the $s = -1$ branch has a cyclotron-sensitive denominator at $\omega = \omega_{c,e}$.

In these circular-mode formulas, s labels polarization, not particle species. The factor $-is$ specifies equal amplitudes with a quarter-cycle phase difference between the two electric components. This algebraic convention fixes the rotation sense without relying on an unstated viewing direction.

Parallel dispersion and cutoffs Governing law 286

The longitudinal branch is $\varepsilon_\parallel = 0 \quad \Rightarrow \quad \omega = \omega_{p,e}$. The two transverse circular branches are

$$N_s^2 = \varepsilon_s = 1 - \frac{\omega_{p,e}^2}{\omega\left(\omega + s\omega_{c,e}\right)}.$$

Their positive-frequency cutoffs satisfy $N_s = 0$ and are

$$\omega_{\mathrm{cut,s}} = \frac{\sqrt{\omega_{c,e}^2 + 4\omega_{p,e}^2} - s\omega_{c,e}}{2}.$$

A real N denotes bulk propagation in this idealized model; $N^2 < 0$ denotes an evanescent branch.

Derivation: circular factorization and Faraday rotation

Step: Factor the parallel wave matrix

For parallel propagation, divide the wave matrix by ω^2/c^2:

$$\begin{pmatrix} \varepsilon_\perp - N^2 & -i\varepsilon_\times & 0 \\ i\varepsilon_\times & \varepsilon_\perp - N^2 & 0 \\ 0 & 0 & \varepsilon_\parallel \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The longitudinal factor gives $\varepsilon_\parallel = 0$. The transverse determinant is

$$\left(\varepsilon_\perp - N^2\right)^2 - \varepsilon_\times^2 = 0.$$

Thus the two transverse eigenvalues are

$$N^2 = \varepsilon_\perp + \varepsilon_\times, \quad N^2 = \varepsilon_\perp - \varepsilon_\times.$$

Step: Express the circular branches

For an electron, the signed gyrofrequency is negative when B_0 points along positive z. Use its positive magnitude $\omega_{c,e}$ and the circular basis

$$E_{1,y} = -isE_{1,x}.$$

The two indices become

$$N_s^2 = \varepsilon_\perp - s\varepsilon_\times = 1 - \frac{\omega_{p,e}^2}{\omega\left(\omega + s\omega_{c,e}\right)}.$$

Setting $N_s = 0$ gives

$$\omega^2 + s\omega_{c,e}\omega - \omega_{p,e}^2 = 0.$$

The positive root is the cutoff stated above. For $s = -1$, the denominator vanishes at the electron cyclotron frequency. The cold response is then singular, so finite temperature or collisions are required before that limit is interpreted.

Step: Accumulate the relative phase

A linearly polarized wave is the equal-amplitude sum of the two circular eigenmodes. After a distance L, their relative phase is

$$(k_+ - k_-)L.$$

The major axis rotates by half this phase:

$$\theta_F = \frac{(k_+ - k_-)L}{2}.$$

If the plasma varies slowly along the ray, use the local wave-number difference instead:

$$\theta_F = \frac{\int (k_+(z) - k_-(z))\,\mathrm{d}z}{2}.$$

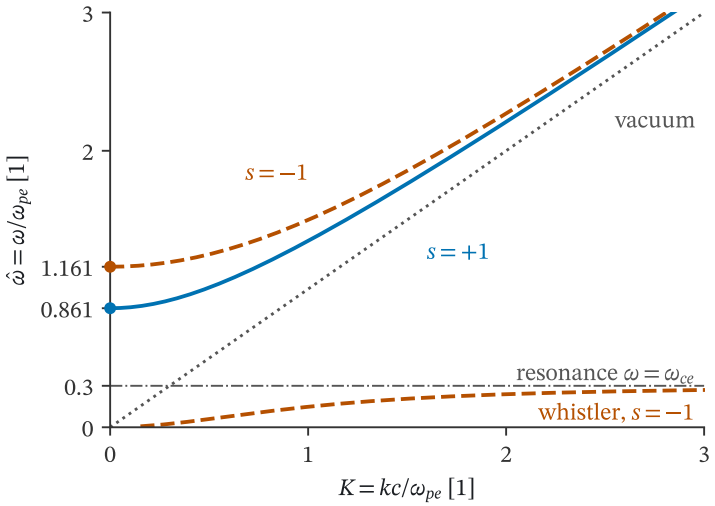


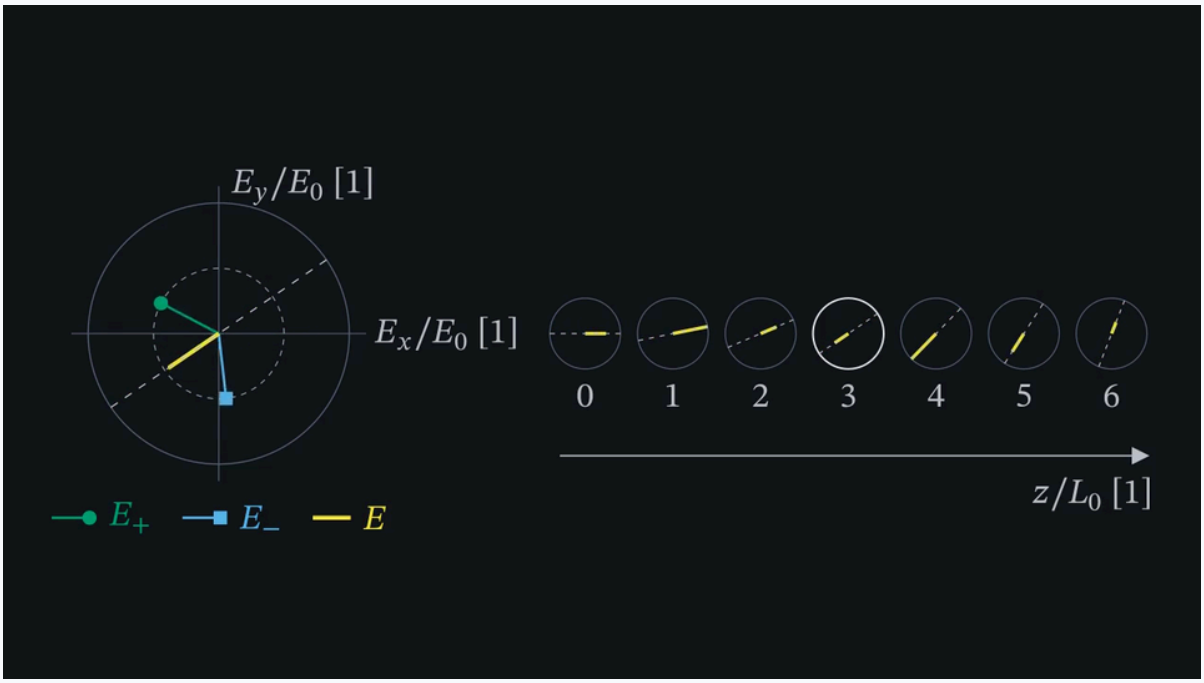
Figure 72: Cold fixed-ion parallel propagation at $Y = \omega_{c,e}/\omega_{p,e} = 0.3$ [1], from $N_s^2 = 1 - \frac{1}{\hat{\omega}(\hat{\omega}+sY)}$ with $K = \hat{\omega}N_s$. The two circular modes have different cutoffs $W_{\text{cut,s}} = \frac{\sqrt{Y^2+4}-sY}{2}$ (dots); only $s = -1$ has the cyclotron resonance $\hat{\omega} = Y$, below which it propagates as the whistler.

Faraday rotation compares two circular modes at the same frequency. When both propagate without attenuation and have equal amplitudes, their superposition remains linearly polarized. Different wave numbers rotate its polarization axis with distance; at a fixed position that axis remains fixed as the electric vector oscillates. If one component is evanescent or attenuated differently, the equal-amplitude linear-polarization picture no longer applies.

Pause and predict

Key idea 287

A linearly polarized wave enters the plasma with its electric field along $\boldsymbol{e}_x$. Before opening the animation, predict whether the output polarization can remain fixed when $k_+ \neq k_-$. It cannot: the two circular components accumulate different phases, so their superposition rotates. The viewing direction and Fourier convention determine the sign of the rotation; its magnitude follows from $k_+ - k_-$.



Watch animation

Faraday rotation: a linearly polarized wave can be decomposed into two circular eigenmodes. Because the magnetic field gives them different wave numbers, their relative phase changes along the path and the polarization plane rotates. The prescribed illustration uses $k_+L_0 = 1.2$, $k_-L_0 = 0.8$, and $\omega t_0 = 1$, with each component of amplitude $E_0/2$. Here L_0, t_0, and E_0 are reference length, time, and electric-field scales. The transverse panel shows the highlighted position $z = 3L_0$. At fixed position the polarization axis is stationary in time; the wave numbers are prescribed.

Figure 73: Faraday rotation: a linearly polarized wave can be decomposed into two circular eigenmodes. Because the magnetic field gives them different wave numbers, their relative phase changes along the path and the polarization plane rotates. The prescribed illustration uses $k_+L_0 = 1.2$, $k_-L_0 = 0.8$, and $\omega t_0 = 1$, with each component of amplitude $E_0/2$. Here L_0, t_0, and E_0 are reference length, time, and electric-field scales. The transverse panel shows the highlighted position $z = 3L_0$. At fixed position the polarization axis is stationary in time; the wave numbers are prescribed.

Rechenbeispiel Rechenbeispiel 288

Assume a homogeneous, cold, collisionless, fixed-ion electron response with uniform $\boldsymbol{B}_0$ and parallel propagation. For the fixed-ion plasma with $\omega_{p,e} = 5.64 \cdot 10^9\,\mathrm{s}^{-1}$, $\omega_{c,e} = 1.76 \cdot 10^9\,\mathrm{s}^{-1}$, and a parallel wave with $\omega = 2.00 \cdot 10^{10}\,\mathrm{s}^{-1}$ crossing a path of $L = 0.10\,\mathrm{m}$, determine the two refractive indices and the Faraday-rotation angle

$$\theta_F = \frac{(k_+ - k_-)L}{2}.$$

Numerical result: $N_+ = 0.963$ [1], $N_- = 0.955$ [1], and $\theta_F = 2.45 \cdot 10^{-2}\,\mathrm{rad} = 1.41°$.

The resonant branch is not an ordinary cutoff Interpretation 289

At a cutoff, N goes to zero and the wavelength becomes large. At the cyclotron resonance, the cold susceptibility becomes singular and N can grow without bound. The former is a propagation boundary; the latter is a warning that neglected orbit-scale, thermal, collisional, or kinetic physics may control the response.

Summary Summary 290

Parallel propagation separates a longitudinal plasma oscillation from two circularly polarized transverse modes. The magnetic field splits their cutoffs and creates a cyclotron-sensitive resonance. A linear polarization rotates because its two circular components accumulate different phases.

Knowledge check Knowledge check 291

- Why are circular polarizations the natural eigenvectors for parallel propagation?
- What is the physical meaning of a circular-mode cutoff?
- Which cold-plasma feature identifies the cyclotron-sensitive branch?
- Why does a linearly polarized wave rotate in a magnetized plasma?

Answers

Why are circular polarizations the natural eigenvectors for parallel propagation?
The transverse dielectric block has equal diagonal entries and antisymmetric off-diagonal entries. Its eigenvectors are the two circular combinations of E_x and E_y, which remove the coupling.

What is the physical meaning of a circular-mode cutoff?
It is the frequency at which the corresponding refractive index vanishes. The wave number is then zero, so the branch meets the zero-wave-number boundary between propagation and non-propagation.

Which cold-plasma feature identifies the cyclotron-sensitive branch?
Its denominator contains $\omega(\omega - \omega_{c,e})$ under the chosen convention, so the response becomes singular as ω approaches the electron cyclotron frequency.

Why does a linearly polarized wave rotate in a magnetized plasma?
A linear wave is a sum of two circular eigenmodes. Since the modes have different wave numbers, their relative phase changes with distance and the direction of the linear polarization changes.

13.3 Perpendicular propagation: ordinary and extraordinary modes

Turning the wave vector perpendicular to the magnetic field changes which field component is decoupled. The ordinary mode has its electric field parallel to the background field and therefore sees the unmagnetized plasma response. The extraordinary mode has electric components in the plane perpendicular to the background field, including a component along the wave vector, and contains an upper-hybrid resonance.

Learning objectives

- factor the perpendicular wave matrix into ordinary and extraordinary modes
- derive the ordinary and extraordinary refractive indices
- identify the upper-hybrid resonance and the extraordinary polarization
- distinguish a cutoff from a resonant stopband

Unit ledger

The angle θ is dimensionless and measured in radians. The refractive index $N = kc/\omega$, $\hat{\omega} = \omega/\omega_{p,e}$, and $Y = \omega_{c,e}/\omega_{p,e}$ are dimensionless. Dimensional k is in m^{-1}, ω in s^{-1}, and wavelengths in m.

Perpendicular fixed-ion propagation — Assumptions 292

Take $\boldsymbol{k} = k\boldsymbol{e}_x$ and $\boldsymbol{B}_0 = B_0\boldsymbol{e}_z$. Retain a cold collisionless electron fluid with fixed ions. The ordinary and extraordinary names refer to this geometry; they should not be transferred to an oblique mode without checking its polarization.

Ordinary and extraordinary indices — Definition 293

For the ordinary mode, $\boldsymbol{E}_1$ is parallel to $\boldsymbol{B}_0$ and $N_O^2 = \varepsilon_\parallel = 1 - \omega_{p,e}^2/\omega^2$. The extraordinary mode is polarized in the $x - y$ plane and has

$$N_X^2 = \frac{\varepsilon_\perp^2 - \varepsilon_\times^2}{\varepsilon_\perp}.$$

Define the upper-hybrid frequency by $\omega_{\mathrm{UH}} = \sqrt{\omega_{p,e}^2 + \omega_{c,e}^2}$; then

$$N_X^2 = 1 - \frac{\omega_{p,e}^2\left(\omega^2 - \omega_{p,e}^2\right)}{\omega^2\left(\omega^2 - \omega_{\mathrm{UH}}^2\right)}.$$

The extraordinary polarization ratio is $E_x/E_y = i\varepsilon_\times/\varepsilon_\perp$, away from zeros of the denominator.

Ordinary and extraordinary branches — Governing law 294

The ordinary branch obeys $\omega^2 = \omega_{p,e}^2 + c^2k^2$ and has the same cutoff as the cold unmagnetized transverse wave. The extraordinary branch obeys $N_X^2 = \varepsilon_+\varepsilon_-/\varepsilon_\perp$, with circular factors ε_s from the parallel problem. Its denominator vanishes at $\omega = \omega_{\mathrm{UH}}$, so the cold wave number diverges at the upper-hybrid resonance.

Derivation: perpendicular factorization

Step: Separate the ordinary mode

For $\boldsymbol{k} = k\boldsymbol{e}_x$, the wave equation is

$$\begin{pmatrix} \varepsilon_\perp & -i\varepsilon_\times & 0 \\ i\varepsilon_\times & \varepsilon_\perp - N^2 & 0 \\ 0 & 0 & \varepsilon_\parallel - N^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The z equation is independent of the $x - y$ block. A nonzero E_z therefore requires

$$N^2 = \varepsilon_{\parallel}.$$

This is the ordinary mode. Its electric field is parallel to $\boldsymbol{B}_0$ but transverse to $\boldsymbol{k}$, so the Lorentz force does not alter this component.

Step: Factor the extraordinary block

The determinant of the $x - y$ block is

$$\varepsilon_{\perp}\left(\varepsilon_{\perp} - N^2\right) - \varepsilon_{\times}^2 = 0.$$

Solving for the index gives

$$N_X^2 = \frac{\varepsilon_{\perp}^2 - \varepsilon_{\times}^2}{\varepsilon_{\perp}}.$$

For one electron species, define

$$\varepsilon_{\perp} = 1 - \frac{\omega_{p,e}^2}{\omega^2 - \omega_{c,e}^2}$$

and

$$\varepsilon_{\times} = -\frac{\omega_{c,e}\omega_{p,e}^2}{\omega\left(\omega^2 - \omega_{c,e}^2\right)}.$$

Step: Expose the upper-hybrid denominator

Let $d = \omega^2 - \omega_{c,e}^2$ and $p = \omega_{p,e}^2$. Then

$$N_X^2 = \frac{(d-p)^2 - \left(\frac{\omega_{c,e}^2 p^2}{\omega^2}\right)}{d(d-p)}.$$

Multiplying the two circular factors and simplifying yields

$$N_X^2 = 1 - \frac{\omega_{p,e}^2\left(\omega^2 - \omega_{p,e}^2\right)}{\omega^2\left(\omega^2 - \omega_{p,e}^2 - \omega_{c,e}^2\right)}.$$

The denominator vanishes at

$$\omega_{\mathrm{UH}}^2 = \omega_{p,e}^2 + \omega_{c,e}^2,$$

which is the upper-hybrid resonance.

Step: Read the extraordinary polarization

The first row of the transverse block is

$$\varepsilon_{\perp}E_x - i\varepsilon_{\times}E_y = 0.$$

Hence

$$\frac{E_x}{E_y} = \frac{i\varepsilon_{\times}}{\varepsilon_{\perp}},$$

which is generally elliptical rather than purely linear polarization.

Rechenbeispiel

Rechenbeispiel 295

Assume a homogeneous, cold, collisionless, fixed-ion electron response with perpendicular propagation and uniform $\boldsymbol{B}_0$. For $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$ and $B_0 = 1.0 \cdot 10^{-2}\,\mathrm{T}$, use $\omega_{p,e} = 5.64 \cdot 10^{9}\,\mathrm{s}^{-1}$, $\omega_{c,e} = 1.76 \cdot 10^{9}\,\mathrm{s}^{-1}$, and $\omega = 5.50 \cdot 10^{9}\,\mathrm{s}^{-1}$. Classify the ordinary and extraordinary branches and report the wave number for the propagating branch.

Numerical result: the ordinary branch is evanescent with $\alpha_O = 4.17\,\mathrm{m}^{-1}$; the extraordinary branch propagates with $N_X = 0.805$ [1], $k_X = 14.8\,\mathrm{m}^{-1}$, and $\lambda_X = 0.425\,\mathrm{m}$.

Ordinary does not mean unmagnetized everywhere

Interpretation 296

The ordinary mode is unaffected by the magnetic field only because its electric field is parallel to $\boldsymbol{B}_0$ in perpendicular propagation. Changing the angle mixes the field components and restores the full anisotropic determinant.

Summary

Summary 297

Perpendicular propagation splits into an ordinary branch with $N_O^2 = \varepsilon_\parallel$ and an extraordinary branch with $N_X^2 = \varepsilon_+\varepsilon_-/\varepsilon_\perp$. The latter is elliptically polarized in general and has an upper-hybrid resonance.

Knowledge check

Knowledge check 298

- Why is the ordinary mode unaffected by the magnetic field in perpendicular propagation?
- What distinguishes the extraordinary mode from the ordinary mode?
- What is the upper-hybrid frequency in the cold fixed-ion model?
- How can a branch be evanescent even when the plasma is collisionless?

Answers

Why is the ordinary mode unaffected by the magnetic field in perpendicular propagation?

Its electric field is parallel to $\boldsymbol{B}_0$, so the perturbed velocity is parallel to the field and the magnetic part of the Lorentz force vanishes.

What distinguishes the extraordinary mode from the ordinary mode?

Its electric field lies in the plane perpendicular to $\boldsymbol{B}_0$, with components both along and across $\boldsymbol{k}$. Its index contains both circular factors and its polarization is generally elliptical.

What is the upper-hybrid frequency in the cold fixed-ion model?

It is $\omega_{\mathrm{UH}} = \sqrt{\omega_{p,e}^2 + \omega_{c,e}^2}$. It marks the zero of the extraordinary-mode denominator and the associated cold resonance.

How can a branch be evanescent even when the plasma is collisionless?

A collisionless response can still give $N^2 < 0$. Then k is imaginary and the spatial solution decays rather than propagates; this is a stopband, not collisional damping.

13.4 Oblique propagation and mode coupling

At an arbitrary angle the clean parallel and perpendicular factorizations no longer apply. The wave vector has components both along and across the

magnetic field, so the two electromagnetic branches are generally elliptically polarized and coupled through the full three-component wave matrix.

Learning objectives

- set up the oblique coordinate system and its angle convention
- write the full cold magnetized wave matrix
- identify the two electromagnetic roots of the Appleton–Hartree form
- recover the parallel and perpendicular limits from the oblique model

Unit ledger

The propagation angle θ is dimensionless. Use $N = kc/\omega$, $X_\omega = \omega_{p,e}^2/\omega^2$, and $Y_\omega = \omega_{c,e}/\omega$, all dimensionless. The dimensional wave number k is in m^{-1} and frequency ω in s^{-1}.

Oblique cold-fluid ordering — Assumptions 299

Let $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$ and choose $\boldsymbol{k} = k(\sin\theta \boldsymbol{e}_x + \cos\theta \boldsymbol{e}_z)$, with no y component. Keep the same homogeneous, cold, collisionless, fixed-ion ordering as before. The mode labels are branch labels, not fixed polarization labels at every angle.

Oblique wave matrix and Appleton–Hartree roots — Governing law 300

With $S = \varepsilon_\perp$, $D = \varepsilon_\times$, and $P = \varepsilon_\parallel$, the normalized wave matrix is

$$\begin{pmatrix} S - N^2\cos^2\theta & -iD & N^2\sin\theta\cos\theta \\ iD & S - N^2 & 0 \\ N^2\sin\theta\cos\theta & 0 & P - N^2\sin^2\theta \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Its determinant gives two electromagnetic roots. For one cold electron species they can be written in the Appleton–Hartree form

$$N_\pm^2 = 1 - \frac{X_\omega}{1 - \frac{Y_\omega^2\sin^2\theta}{2(1-X_\omega)} \pm \sqrt{\left(\frac{Y_\omega^2\sin^2\theta}{2(1-X_\omega)}\right)^2 + Y_\omega^2\cos^2\theta}}.$$

The upper sign defines N_+ and the lower sign defines N_-; their polarization and longitudinal content vary continuously with θ.

Here S, D, and P are dimensionless dielectric abbreviations; D is not a diffusion coefficient and P is not pressure. The parameters X_ω and Y_ω use the wave frequency as their reference, whereas the earlier $\hat{\omega}$ and Y use the plasma frequency. Their relations are $X_\omega = 1/\hat{\omega}^2$ and $Y_\omega = Y/\hat{\omega}$.

The two roots give possible wave numbers at a specified frequency and angle. For each root, the matrix also determines the relative electric-field components. Following those components is necessary to identify a physical branch.

Derivation: oblique matrix and limiting geometry

Step: Insert the oblique wave vector

Use

$$\boldsymbol{k} = k(\sin\theta \boldsymbol{e}_x + \cos\theta \boldsymbol{e}_z)$$

in

$$\boldsymbol{k}(\boldsymbol{k}\cdot\boldsymbol{E}_1) - k^2\boldsymbol{E}_1 + \frac{\omega^2}{c^2}\boldsymbol{\varepsilon}_p\cdot\boldsymbol{E}_1 = \boldsymbol{0}.$$

After division by ω^2/c^2, the geometric term contributes

$$-N^2\cos^2\theta E_x + N^2\sin\theta\cos\theta E_z$$

to the x component, $-N^2E_y$ to the y component, and

$$N^2\sin\theta\cos\theta E_x - N^2\sin^2\theta E_z$$

to the z component. Adding the dielectric tensor gives the displayed oblique matrix.

Step: Reduce the determinant to a quadratic

Set $Z = N^2$, $a = \sin\theta$, and $b = \cos\theta$. Expanding the determinant gives

$$\det(M) = \big(P - Za^2\big)\big(\big(S - Zb^2\big)(S - Z) - D^2\big) - Z^2a^2b^2(S - Z).$$

Collect terms as

$$AZ^2 - BZ + C = 0,$$

where

$$A = Sa^2 + Pb^2, \quad B = \big(S^2 - D^2\big)a^2 + PS\big(1 + b^2\big), \quad C = P\big(S^2 - D^2\big).$$

The two roots before specialization are

$$Z_{\pm} = \frac{B \pm \sqrt{B^2 - 4AC}}{2A}.$$

Step: Specialize to one cold electron species

Insert

$$S = 1 - \frac{X_\omega}{1 - Y_\omega^2}, \quad P = 1 - X_\omega,$$

with D of magnitude

$$\frac{Y_\omega X_\omega}{1 - Y_\omega^2}.$$

Since $a^2 + b^2 = 1$, useful identities are

$$S^2 - D^2 = \frac{(1 - X_\omega)^2 - Y_\omega^2}{1 - Y_\omega^2}, \quad A = Sa^2 + Pb^2.$$

Substitution into the quadratic formula, extraction of the common factor $1 - X_\omega$, and completion of the square in the angle-dependent term give the displayed Appleton–Hartree roots. The square-root term measures the splitting produced by the field component along the propagation direction together with transverse coupling.

Step: Check the parallel and perpendicular endpoints

For $\theta = 0$, the matrix separates into the longitudinal factor P and circular transverse factors $S \pm D$. For $\theta = \pi/2$, the z component decouples as the ordinary mode and the $x - y$ block produces the extraordinary mode. Without its endpoint polarization, an algebraic root can be assigned to the wrong physical branch.

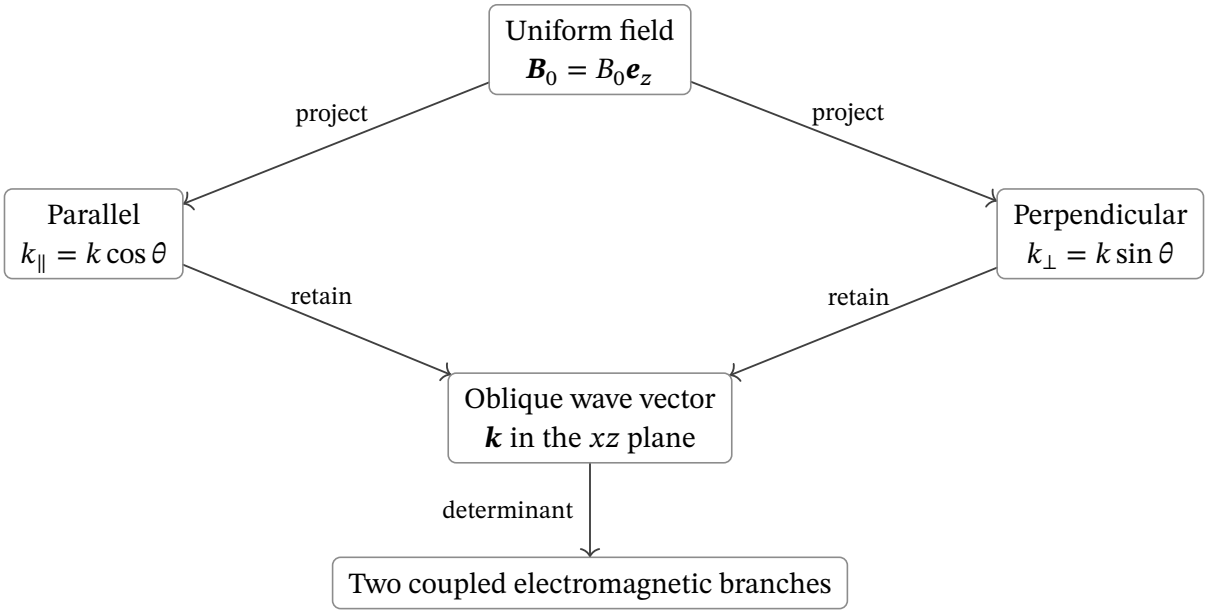


Figure 74: Oblique propagation is represented by $\boldsymbol{k} = k(\sin\theta \boldsymbol{e}_x + \cos\theta \boldsymbol{e}_z)$, with $\boldsymbol{B}_0$ along z. The angle is a model parameter: $\theta = 0$ is parallel propagation and $\theta = \pi/2$ is perpendicular propagation.

Rechenbeispiel Rechenbeispiel 301

Assume a homogeneous, cold, collisionless, fixed-ion electron response with uniform $\boldsymbol{B}_0$ and oblique propagation. Use the normalized parameters $Y = \omega_{c,e}/\omega_{p,e} = 0.30$ [1], $\hat{\omega} = \omega/\omega_{p,e} = 1.50$ [1], and $\theta = \pi/4$ radians. Evaluate the two Appleton–Hartree refractive indices.

Numerical result: $X_\omega = 0.444$ [1], $Y_\omega = 0.200$ [1], and the two roots are $N_+ = 0.778$ [1] and $N_- = 0.686$ [1], where the signs follow the displayed denominator signs.

Oblique modes exchange character Interpretation 302

An oblique branch can be mostly transverse at one frequency and mostly longitudinal near a resonance or slow-wave region. Naming a mode only by its endpoint label hides this conversion; track N, polarization, and charge response together.

Summary Summary 303

Oblique propagation requires the full three-component determinant. The Appleton–Hartree roots are the two electromagnetic branches of the cold anisotropic response, and the parallel or perpendicular modes appear as controlled endpoint limits.

Knowledge check

Knowledge check 304

- Why can the oblique wave matrix not be reduced to the parallel transverse block?
- How many electromagnetic refractive-index roots does the cold oblique determinant provide?
- What happens to the oblique roots as the angle approaches zero?
- Why should a mode label be accompanied by its propagation angle?

Answers

Why can the oblique wave matrix not be reduced to the parallel transverse block?
Both $k_{\parallel}$ and $k_{\perp}$ are nonzero. The geometric part of Maxwell's equation then couples E_x and E_z in addition to the dielectric coupling of E_x and E_y.

How many electromagnetic refractive-index roots does the cold oblique determinant provide?
The determinant is quadratic in N^2, so it provides two electromagnetic roots. Their polarization vectors, not only their scalar indices, identify how they connect to the principal modes.

What happens to the oblique roots as the angle approaches zero?
The wave vector becomes parallel to the magnetic field, the longitudinal factor separates, and the two electromagnetic roots approach the circularly polarized parallel branches.

Why should a mode label be accompanied by its propagation angle?
The same branch can change from mostly transverse to partly longitudinal as the angle changes. The polarization and cutoff or resonance locations are angle-dependent even when some principal cutoffs are shared.

13.5 Cutoffs, resonances, and controlled limits

A cutoff, a resonance, an evanescent interval, and a high-frequency vacuum limit are different statements about the same refractive-index equation. Each holds only within the model that produced the diagram.

Learning objectives

- distinguish $N = 0$ cutoffs from divergent resonances
- classify propagation using the sign of N^2
- recover the unmagnetized and vacuum limits
- state when warm, collisional, or kinetic corrections are required

Unit ledger

The normalized frequency $\hat{\omega} = \omega/\omega_{p,e}$ and magnetization $Y = \omega_{c,e}/\omega_{p,e}$ are dimensionless. Define the normalized wave number $K = kc/\omega_{p,e} = \hat{\omega}N$. The refractive index N and angle θ are dimensionless; dimensional k is in m^{-1}, ω in s^{-1}, and wavelengths in m.

Validity of the cold model near resonances

Assumptions 305

The diagnostic uses a homogeneous, collisionless, cold, fixed-ion plasma with a uniform magnetic field. Near a cyclotron or upper-hybrid resonance, the wavelength can approach the particle orbit scale and the cold fluid approximation loses its ordering. Collisions, finite temperature, boundaries, and kinetic resonances must then be added explicitly.

Cutoff, resonance, and propagation test Definition 306

A cutoff is a point where $N^2 = 0$, so $k = 0$. A resonance is a singular response where the cold dispersion relation drives N^2 to infinity or a dielectric denominator to zero. A pole in an individual dielectric entry need not survive in every mode: check the complete dispersion relation for cancellations. For a real frequency, $N^2 > 0$ gives a real propagating wave number, whereas $N^2 < 0$ gives an imaginary wave number and spatial evanescence. The sign test is necessary but not sufficient: one must also check polarization and the approximation's scale ordering.

Normalized landmarks Governing law 307

For $Y = \omega_{c,e}/\omega_{p,e}$, the parallel circular cutoffs are

$$\hat{\omega}_{\mathrm{cut,s}} = \frac{\sqrt{Y^2 + 4} - sY}{2}.$$

The perpendicular extraordinary resonance is $\hat{\omega}_{\mathrm{UH}} = \sqrt{1 + Y^2}$, while the ordinary cutoff is $\hat{\omega} = 1$. When $Y \to 0$, both circular modes merge into the unmagnetized transverse branch $\hat{\omega}^2 = 1 + K^2$ from the previous chapter, where $K = \hat{\omega}N$. At frequencies much larger than both plasma and cyclotron frequencies, the propagating electromagnetic branches approach $N \to 1$, or equivalently $\hat{\omega} \approx K$.

Derivation: limiting-case checks

Step: Remove the magnetic field

Set $Y = 0$ in the circular response. Both eigenvalues become

$$N^2 = 1 - \frac{1}{\hat{\omega}^2},$$

so multiplying by $\hat{\omega}^2$ and using $K = \hat{\omega}N$ gives the cold unmagnetized electromagnetic branch

$$\hat{\omega}^2 = 1 + K^2.$$

The longitudinal branch remains $\hat{\omega} = 1$ in the cold fixed-ion limit, so its zero group velocity is recovered as well.

Step: Check high-frequency, cutoff, and resonance limits

As $\hat{\omega} \to \infty$, every susceptibility term scales as $\hat{\omega}^{-2}$ or faster away from a resonance. Therefore

$$N^2 \to 1, \quad \omega \to ck.$$

The plasma becomes transparent. A cutoff is found from $N^2 = 0$ and is a finite-frequency, zero-wave-number endpoint. At a resonance, a denominator such as $\omega(\omega - \omega_{c,e})$ or $\omega^2 - \omega_{\mathrm{UH}}^2$ vanishes. Then N^2 becomes large and the wave length becomes small. The cold response is singular because it has no thermal spread, collisions, finite orbit width, or nonlinear saturation.

Step: Classify propagation intervals

For the extraordinary perpendicular branch, the two circular cutoffs and the upper-hybrid resonance divide the frequency axis into propagation and stopband intervals. Determine the correct interval by evaluating the sign of N_X^2 between adjacent landmarks. If density or magnetic field

varies along a path, a local cutoff can reflect or mode-convert a wave. Approximating the wave locally by a plane wave additionally requires the background scale length to be long compared with the local wavelength; this condition fails near a cutoff where the wavelength grows without bound.

Step: Return to the physical ordering

If $k\lambda_D$ is not small, pressure and kinetic dispersion matter. If ω is comparable to a collision frequency, the dielectric coefficients become complex. If $\omega - k v_\parallel$ is near zero for a significant particle population, kinetic susceptibility and resonant damping or growth replace the cold-fluid response.

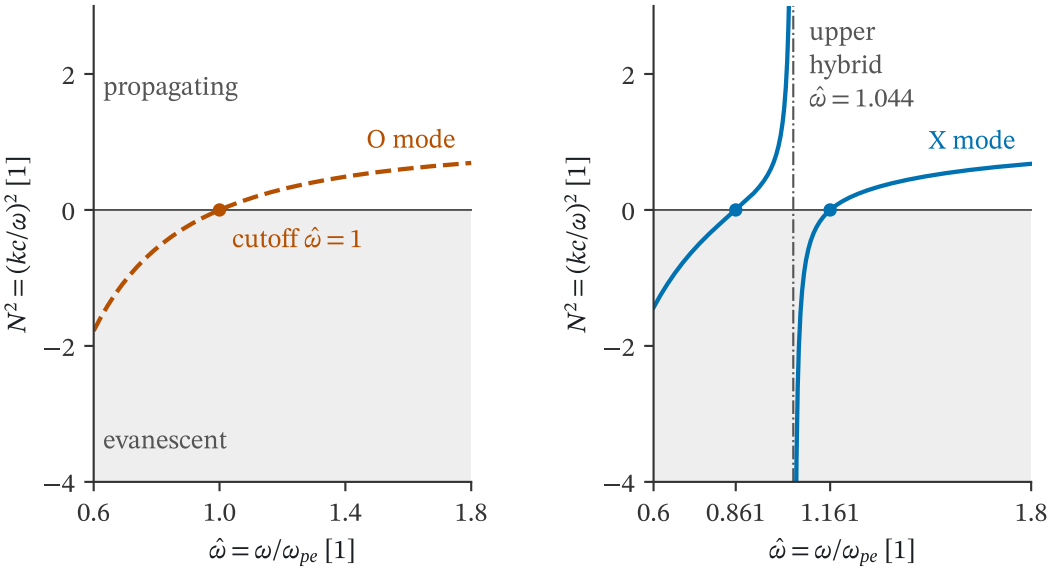


Figure 75: Perpendicular cold modes, ordinary (left) and extraordinary (right), at $Y = \omega_{c,e}/\omega_{p,e} = 0.3$ [1], the values of the following example. Cutoffs are zeros of N^2 (dots), the upper-hybrid resonance is a pole of N_X^2; between a resonance and the next cutoff the extraordinary mode is evanescent (shaded, $N^2 < 0$).

Rechenbeispiel

Rechenbeispiel 308

Assume a homogeneous, cold, collisionless, fixed-ion electron response with uniform $\boldsymbol{B}_0$. Let $Y = \omega_{c,e}/\omega_{p,e} = 0.30$ [1]. Determine the normalized circular cutoffs and upper-hybrid resonance, then classify the ordinary and extraordinary modes at $\hat{\omega} = 0.90$ [1], $\hat{\omega} = 1.10$ [1], and $\hat{\omega} = 1.30$ [1].

Numerical result: $\hat{\omega}_{\mathrm{cut},+} = 0.861$ [1], $\hat{\omega}_{\mathrm{cut},-} = 1.161$ [1], and $\hat{\omega}_{\mathrm{UH}} = 1.044$ [1]. At $\hat{\omega} = 0.90$ [1], ordinary is evanescent and extraordinary propagates with $N_X = 0.403$ [1] at $\hat{\omega} = 1.10$ [1], ordinary propagates with $N_O = 0.417$ [1] and extraordinary is evanescent; at $\hat{\omega} = 1.30$ [1], both propagate with $N_O = 0.639$ [1] and $N_X = 0.565$ [1].

Unmagnetized, vacuum, and sign checks

Interpretation 309

The unmagnetized limit checks the tensor algebra, the vacuum limit checks the current response, and the sign of N^2 checks propagation. Failure of any one of these tests usually indicates a convention, polarization, or omitted-physics error before it indicates an exotic new mode.

Summary

Summary 310

Cutoffs are zero-wave-number boundaries; resonances are singular short- wavelength responses. Normalizing by $\omega_{p,e}$ and tracking Y makes the parallel, perpendicular, unmagnetized, and vacuum limits comparable. Near a resonance, the cold model must hand off to warm, collisional, or kinetic theory.

Knowledge check

Knowledge check 311

- What equation identifies a cutoff?
- What is the vacuum limit of the cold magnetized response?
- Why can the sign of N^2 be used to classify propagation?
- Which corrections should be considered near a cold-plasma resonance?

Answers

What equation identifies a cutoff?

A cutoff is identified by $N^2 = 0$. The wave number vanishes at the cutoff, unlike a resonance where the refractive index becomes singular.

What is the vacuum limit of the cold magnetized response?

For frequencies much larger than the plasma and cyclotron frequencies, the dielectric response approaches the identity, N^2 approaches one, and the dispersion relation approaches $\omega = ck$.

Why can the sign of N^2 be used to classify propagation?

For a real prescribed frequency and a homogeneous lossless cold model, $N^2 > 0$ gives real k and oscillatory spatial dependence, while $N^2 < 0$ gives imaginary k and exponential spatial dependence.

Which corrections should be considered near a cold-plasma resonance?

Finite temperature, collisions, particle-orbit effects, kinetic resonances, boundaries, and nonlinear effects can all regularize or replace the singular cold response. The relevant correction is selected by the ordering of the physical scales.

14. Collisions, ions, and finite-temperature effects on magnetized waves

The cold magnetized response is an organizing limit. Collisions make the response complex, ion inertia opens low-frequency branches, and pressure gives longitudinal waves a spatial dispersion. This chapter adds those effects in a controlled order and connects the resulting two-fluid branches to warm MHD, following [3].

Relax one ordering at a time — Key idea 312

A reliable extension of a cold-plasma result starts by naming the small parameter that is being released. Compare ν/ω, $\omega/\omega_{c,i}$, and $k\lambda_D$ before choosing a collisional, two-fluid, warm-fluid, or kinetic description.

14.1 Collisional response and wave attenuation

What changes when particles lose momentum between wave cycles? With a linear drag term, the cold susceptibility becomes complex, the refractive index becomes complex, and a propagating wave acquires spatial attenuation.

Learning objectives

- insert a linear collision frequency into the time-harmonic momentum equation
- interpret the collision term as a complex effective mass
- identify the real and imaginary parts of a collisional wave number
- distinguish the exact drag substitution from a weak-collision expansion

Unit ledger

The damping rate is in s^{-1}, and wave numbers k_r and k_i are in m^{-1}. The effective mass has units kg, while N and m_{eff}/m are dimensionless.

Linear drag as a collisional model — Assumptions 313

Use a homogeneous, cold, magnetized equilibrium and the Fourier convention $\exp(i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t))$. Represent collisions by a velocity-independent drag $-m_s\nu_s\boldsymbol{u}_1$ against a stationary background. If the collision partner also moves, its velocity must enter the relative drag. In this momentum-transfer model, energy exchange, velocity-dependent Coulomb operators, and boundary collisions require a more detailed kinetic treatment.

Complex effective mass Definition 314

With the drag term included, write the time-harmonic momentum equation as

$$-i\omega m_s \boldsymbol{u}_{s,1} = q_s\left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0\right) - m_s \nu_s \boldsymbol{u}_{s,1}.$$

Moving the drag to the left gives

$$-i\omega m_{\mathrm{eff},s} \boldsymbol{u}_{s,1} = q_s\left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0\right)$$

with $m_{\mathrm{eff},s} = m_s(1 + i\nu_s/\omega)$. The sign of the imaginary part follows the stated Fourier convention. For nonzero ω, this is an exact algebraic rewrite of the stated constant-drag equation, including near cyclotron resonance. The complex mass is a way to combine inertia and drag; the physical particle mass remains m_s.

Complex refractive index Governing law 315

The cold tensor formulas can be reused with m_s replaced by $m_{\mathrm{eff},s}$ when the drag model is valid. For the cyclotron-sensitive electron circular branch, one convenient convention is

$$N_{\mathrm{RH}}^2 = 1 - \frac{\omega_{p,e}^2}{\omega\left(\omega - \omega_{c,e} + i\nu_e\right)}.$$

For a wave driven at real positive ω in a homogeneous medium, write $N = N_r + iN_i$ and $k = \omega N/c = k_r + ik_i$. The field factor is

$$\exp(ikz - i\omega t) = \exp(ik_r z - i\omega t)\exp(-k_i z),$$

so $k_i > 0$ is amplitude attenuation and $1/(2k_i)$ is the intensity attenuation length along positive z. This describes spatial decay; temporal damping instead uses a complex frequency at real wave number.

Derivation: drag, complex susceptibility, and attenuation

Step: Introduce the effective mass

Start from the linearized momentum equation with drag:

$$-i\omega m_s \boldsymbol{u}_{s,1} = q_s\left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0\right) - m_s \nu_s \boldsymbol{u}_{s,1}.$$

Move the drag term to the left:

$$(-i\omega m_s + m_s \nu_s)\boldsymbol{u}_{s,1} = q_s\left(\boldsymbol{E}_1 + \boldsymbol{u}_{s,1} \times \boldsymbol{B}_0\right).$$

Factor the coefficient:

$$-i\omega m_s\left(1 + \frac{i\nu_s}{\omega}\right) = -i\omega m_{\mathrm{eff},s},$$

where

$$m_{\mathrm{eff},s} = m_s\left(1 + \frac{i\nu_s}{\omega}\right).$$

Step: Insert the effective parameters into the cold response

The species plasma-frequency factor becomes $n_{s,0}q_s^2/(\varepsilon_0 m_{\mathrm{eff},s})$, and the signed gyrofrequency becomes $q_s B_0/m_{\mathrm{eff},s}$. Insert both into the cold transverse response. For the electron branch whose collisionless denominator is $\omega(\omega-\omega_{c,e})$, the two substitutions combine to

$$\frac{\left[\frac{\omega_{p,e}^2}{1+\frac{i\nu_e}{\omega}}\right]}{\left[\omega\left(\omega-\frac{\omega_{c,e}}{1+\frac{i\nu_e}{\omega}}\right)\right]}=\frac{\omega_{p,e}^2}{\left[\omega(\omega-\omega_{c,e}+i\nu_e)\right]}.$$

This gives the stated collisional circular response.

Step: Extract weak attenuation

Away from resonances and for $\nu_e/\omega \ll 1$, the unmagnetized susceptibility is

$$\frac{\omega_{p,e}^2}{\omega(\omega+i\nu_e)}\approx\frac{\omega_{p,e}^2}{\omega^2}\left(1-\frac{i\nu_e}{\omega}\right).$$

Therefore N^2 has a positive imaginary part in the chosen convention. Taking the square root gives $N=N_r+iN_i$ with $N_i>0$. Since $k=\omega N/c$,

$$\exp(i(k_r+ik_i)z-i\omega t)=\exp(ik_r z-i\omega t)\exp(-k_i z).$$

The amplitude falls by e^{-1} after $1/k_i$, while intensity, which is proportional to amplitude squared, falls by e^{-1} after $1/(2k_i)$.

Step: State the validity boundary

The effective-mass substitution remains exact for constant linear drag. An expansion of the circular denominator instead requires $\nu_e \ll |\omega-\omega_{c,e}|$, not merely $\nu_e \ll \omega$. Near resonance retain the full complex denominator. Near a cutoff, expanding its square root also fails if the imaginary correction is comparable to the real N^2. A direct solution of the same drag tensor gives the same result as the effective-mass substitution. A more realistic collision operator is needed only when the physical model must include velocity dependence, energy exchange, or coupled species momentum.

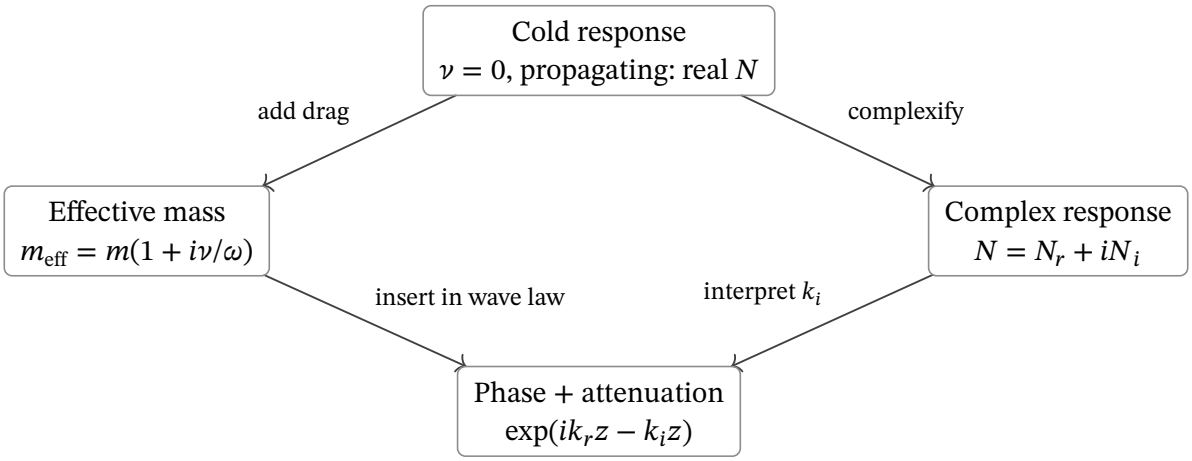


Figure 76: For a propagating cold branch, collisions introduce absorption into a lossless response. The lossless tensor is Hermitian away from poles; its entries need not all be real. Stopbands can have imaginary wave number even without collisions. The real wave number controls phase advance; the imaginary part controls attenuation with the stated Fourier convention.

Rechenbeispiel

Rechenbeispiel 316

Assume a homogeneous, cold, unmagnetized, weakly collisional, fixed-ion electron plasma with linear momentum drag and the stated Fourier convention. Use $\omega_{p,e} = 5.64 \cdot 10^9\,\mathrm{s}^{-1}$, $\omega = 2.00 \cdot 10^{10}\,\mathrm{s}^{-1}$, and a constant $\nu_e = 1.00 \cdot 10^9\,\mathrm{s}^{-1}$. In the high-frequency weak-collision approximation, estimate the effective-mass ratio $m_{\mathrm{eff},e}/m_e$, k_r, k_i, and the amplitude attenuation length for the unmagnetized branch.

Numerical result: $m_{\mathrm{eff},e}/m_e = 1.00 + i5.00 \cdot 10^{-2}$ [1], $N \approx 0.959 + i2.07 \cdot 10^{-3}$ [1], $k_r \approx 64.0\,\mathrm{m}^{-1}$, $k_i \approx 1.38 \cdot 10^{-1}\,\mathrm{m}^{-1}$, and the amplitude attenuation length is approximately 7.23 m.

Damping is part of the response

Interpretation 317

A complex dielectric coefficient changes phase velocity, polarization, cutoff structure, and energy absorption together. The sign of the imaginary part must always be read with the declared Fourier convention.

Summary

Summary 318

Linear drag can be represented by $m_{\mathrm{eff}} = m(1 + i\nu/\omega)$ in a time-harmonic cold response. The resulting N and k are complex: k_r controls phase advance and k_i controls spatial attenuation. Near resonances retain the full drag response; a weak-collision expansion can fail even though the effective-mass rewrite remains exact. The drag model's physical adequacy must be assessed separately.

Knowledge check

Knowledge check 319

- Why does a collision frequency make the refractive index complex?
- What determines whether a complex wave number attenuates or grows in space?
- Why are collisions especially important near a cyclotron resonance?
- What information is missing from a constant-drag model?

Answers

Why does a collision frequency make the refractive index complex?

The drag term changes the coefficient of the time-harmonic velocity from $-i\omega m$ to $-i\omega m_{\mathrm{eff}}$ with complex m_{eff}. The susceptibility and therefore the dielectric coefficient acquire an imaginary part.

What determines whether a complex wave number attenuates or grows in space?

With $\exp(ikz - i\omega t)$ and $k = k_r + ik_i$, the spatial factor is $\exp(-k_i z)$. Positive k_i attenuates the forward wave; the sign must be interpreted together with the chosen propagation and Fourier conventions.

Why are collisions especially important near a cyclotron resonance?

The collisionless susceptibility denominator becomes small there. The collisional term then competes directly with the small detuning and can set both the phase and absorption instead of being a small correction.

What information is missing from a constant-drag model?

A constant drag does not describe velocity-dependent scattering, energy exchange, conservation constraints, or kinetic resonances. Those effects require an appropriate collision operator and often a kinetic dielectric calculation.

14.2 Ion inertia and low-frequency branches

Fixed ions hide the low-frequency part of the spectrum. Once ions are allowed to move, their opposite charge and much larger mass add a second cyclotron scale. The two circular branches then connect the cold electromagnetic modes to shear-Alfvén, whistler-like, and ion-cyclotron behavior.

Learning objectives

- add ion inertia to the magnetized dielectric response
- write the two circular parallel branches for an electron–ion plasma
- derive the low-frequency Alfvén limit
- explain how ion-cyclotron and whistler continuations arise

Unit ledger

The Alfvén, sound, and wave phase speeds are in $\mathrm{m\,s^{-1}}$. All refractive indices are dimensionless.

Two-fluid electron–ion response — Assumptions 320

Use a neutral, homogeneous hydrogen plasma with $n_{e,0} = n_{i,0} = n_0$, no pressure and no collisions for the first branch calculation. Retain both electron and ion inertia, use $m_e/m_i \ll 1$, and keep $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$. The labels RH and LH follow the present Fourier and viewing convention; they denote circular eigenvalues.

Two-fluid dielectric coefficients — Definition 321

The species sum gives

$$\varepsilon_\perp = 1 - \left[\frac{\omega_{p,e}^2}{\omega^2 - \omega_{c,e}^2} + \frac{\omega_{p,i}^2}{\omega^2 - \omega_{c,i}^2}\right],$$

$$\varepsilon_\times = -\frac{\omega_{c,e}\omega_{p,e}^2}{\omega\left(\omega^2 - \omega_{c,e}^2\right)} + \frac{\omega_{c,i}\omega_{p,i}^2}{\omega\left(\omega^2 - \omega_{c,i}^2\right)},$$

and

$$\varepsilon_\parallel = 1 - \frac{\left[\omega_{p,e}^2 + \omega_{p,i}^2\right]}{\omega^2}.$$

This is the signed-Ω_s convention from the previous chapter: for $\boldsymbol{B}_0$ along positive z, the electron contribution is negative and the ion contribution is positive. Charge neutrality and singly charged species imply

$$\frac{\omega_{p,i}^2}{\omega_{p,e}^2} = \frac{m_e}{m_i} = \frac{\omega_{c,i}}{\omega_{c,e}}.$$

Circular branches and the Alfvén limit

Governing law 322

For parallel propagation, the two circular branches can be written, to leading order in m_e/m_i, as

$$N_{\mathrm{RH}}^2 \approx 1 - \frac{\omega_{p,e}^2}{(\omega + \omega_{c,i})(\omega - \omega_{c,e})}$$

and

$$N_{\mathrm{LH}}^2 \approx 1 - \frac{\omega_{p,e}^2}{(\omega - \omega_{c,i})(\omega + \omega_{c,e})}.$$

For $\omega \ll \omega_{c,i}$ both approach

$$N^2 \approx 1 + \frac{\omega_{p,e}^2}{\omega_{c,e}\omega_{c,i}} \approx \frac{c^2}{v_A^2},$$

where

$$v_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}} = \frac{c\sqrt{\omega_{c,e}\omega_{c,i}}}{\omega_{p,e}}$$

and $\rho_0 \approx n_0 m_i$. The last approximation also requires $v_A \ll c$, so the added unity is negligible. In this nonrelativistic limit, $\omega \approx k v_A$ at low frequency.

Derivation: species sum, circular factors, and low frequency

Step: Sum the species responses

Insert the cold transverse velocity response from the previous chapter into

$$\boldsymbol{j}_1 = \sum_s q_s n_{s,0} \boldsymbol{u}_{s,1}.$$

Use the positive magnitudes $\omega_{c,e}$ and $\omega_{c,i} > 0$ for electrons and ions. The transverse current sum defines $\varepsilon_\perp$ and $\varepsilon_\times$; the two species carry opposite signs in the off-diagonal coefficient. Parallel motion feels no magnetic force, so its response is the sum of the unmagnetized susceptibilities.

Step: Factor the parallel-propagation determinant

For parallel propagation, the transverse determinant factors as

$$N^2 = \varepsilon_\perp + \varepsilon_\times$$

$$N^2 = \varepsilon_\perp - \varepsilon_\times.$$

Consider the factor with an electron denominator $(\omega - \omega_{c,e})$. Its exact species-sum structure is

$$1 - \frac{\omega_{p,e}^2}{\omega(\omega - \omega_{c,e})} - \frac{\omega_{p,i}^2}{\omega(\omega + \omega_{c,i})}.$$

Use

$$\frac{\omega_{p,i}^2}{\omega_{p,e}^2} = \frac{\omega_{c,i}}{\omega_{c,e}},$$

and put the terms over a common denominator. The numerator is

$$\omega_{p,e}^2\left(1 + \frac{\omega_{c,i}}{\omega_{c,e}}\right).$$

Since $\omega_{c,i}/\omega_{c,e} = m_e/m_i$ is small, retain the ion factors in the denominator but drop this small numerator correction:

$$N_{\mathrm{RH}}^2 \approx 1 - \frac{\omega_{p,e}^2}{(\omega + \omega_{c,i})(\omega - \omega_{c,e})}.$$

Interchanging the two circular senses gives the LH expression.

Step: Take the low-frequency limit

When $\omega \ll \omega_{c,i} \ll \omega_{c,e}$, both denominators have the leading product $-\omega_{c,e}\omega_{c,i}$. Therefore

$$N^2 \approx 1 + \frac{\omega_{p,e}^2}{\omega_{c,e}\omega_{c,i}}.$$

In a dense nonrelativistic plasma this term is much larger than one. Use

$$\frac{\omega_{c,e}\omega_{c,i}}{\omega_{p,e}^2} = \frac{\varepsilon_0 B_0^2}{n_0 m_i} = \frac{B_0^2}{\mu_0 n_0 m_i c^2}.$$

It follows that

$$N^2 \approx \frac{c^2}{\left[\frac{B_0^2}{\mu_0 n_0 m_i}\right]} = \frac{c^2}{v_A^2}.$$

Since $N = kc/\omega$, the low-frequency phase speed is $\omega/k \approx v_A$.

Step: Interpret the two circular branches

Under this convention, the RH branch is regular through the positive ion-cyclotron frequency and develops a whistler-like continuation as the frequency rises. The LH branch has a denominator that vanishes at $\omega = \omega_{c,i}$ and therefore supports an ion-cyclotron-sensitive continuation. Far above both cyclotron scales, the ion correction becomes negligible and the fixed-ion electron branches are recovered.

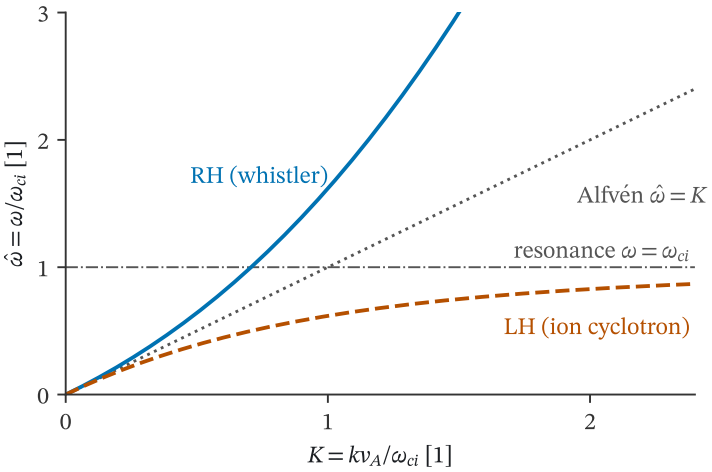


Figure 77: Cold parallel two-fluid branches for $\omega \ll \omega_{c,e}$ and $v_A \ll c$, with $K = kv_A/\omega_{c,i}$ and $\hat{\omega} = \omega/\omega_{c,i}$: $W_{\mathrm{RH}} = \frac{K^2+\sqrt{K^4+4K^2}}{2}$ and $W_{\mathrm{LH}} = \frac{\sqrt{K^4+4K^2}-K^2}{2}$. Both start as Alfvén waves; LH stops at the ion cyclotron resonance, RH continues as the whistler.

Rechenbeispiel Rechenbeispiel 323

Assume a homogeneous, cold, collisionless, two-fluid hydrogen plasma with both species mobile and parallel low-frequency propagation. For the neutral hydrogen plasma use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $B_0 = 1.0 \cdot 10^{-2}\,\mathrm{T}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, and $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. Consider a parallel low-frequency wave at $\omega/\omega_{c,i} = 0.10$ [1]. Determine $\omega_{c,i}$, v_A, $k \approx \omega/v_A$, and the wavelength.

Numerical result: $\omega_{c,i} = 9.58 \cdot 10^5\,\mathrm{s}^{-1}$, $v_A = 2.18 \cdot 10^6\,\mathrm{m\,s}^{-1}$, $k = 4.39 \cdot 10^{-2}\,\mathrm{m}^{-1}$, and $\lambda = 1.43 \cdot 10^2\,\mathrm{m}$.

Low-frequency circular modes can become linear Interpretation 324

The two circular eigenvalues remain the convenient algebraic basis, but their low-frequency indices become nearly equal. Equal counter-rotating components can therefore combine into a linearly polarized shear-Alfvén perturbation. Ion inertia is the physics that makes this low-frequency branch possible.

Summary Summary 325

Mobile ions add $\omega_{c,i}$ and $\omega_{p,i}$ to the dielectric response. The parallel circular branches approach $\omega = kv_A$ at low frequency, then separate into whistler-like and ion-cyclotron continuations. The fixed-ion cold result is recovered only above the ion-inertia scale.

Knowledge check

- Why do ions matter at frequencies far below the electron cyclotron frequency?
- How does the Alfvén speed arise from the parallel circular dispersion?
- What distinguishes the ion-cyclotron and whistler-like continuations?
- Why can the RH and LH labels not be used without a convention?

Answers

Why do ions matter at frequencies far below the electron cyclotron frequency?
Their response is controlled by the much smaller ion gyrofrequency. At frequencies comparable to or below $\omega_{c,i}$, ion inertia and ion current are no longer negligible in the transverse dielectric tensor.

How does the Alfvén speed arise from the parallel circular dispersion?
In the low-frequency limit the circular refractive index becomes $N^2 \approx c^2/v_A^2$. Combining this with $N = kc/\omega$ gives the nondispersive relation $\omega = kv_A$.

What distinguishes the ion-cyclotron and whistler-like continuations?
Their circular denominators contain opposite ion and electron cyclotron factors. One branch is sensitive to the positive ion-cyclotron resonance, while the other continues smoothly into a whistler-like branch under the chosen convention.

Why can the RH and LH labels not be used without a convention?
Handedness depends on the Fourier sign, propagation direction, and viewing direction. The physical eigenvalues are invariant, but the names assigned to the two circular polarizations can exchange.

14.3 Finite-temperature longitudinal waves

Pressure gradients are invisible to a perfectly cold plasma. Restoring a finite temperature makes density perturbations communicate through the fluid, so plasma oscillations acquire a k-dependent frequency and a low-frequency ion-acoustic branch appears when both species move.

Learning objectives

- insert an adiabatic or isothermal pressure closure into the linear response
- derive the warm longitudinal dielectric coefficient
- recover the warm plasma-oscillation dispersion relation
- derive the low-frequency ion-acoustic speed from the two-fluid response

Unit ledger

The normalized combination $k\lambda_D$ and the ratio ω/ω_p are dimensionless.

Warm-fluid pressure closure Assumptions 327

Use a homogeneous unmagnetized longitudinal perturbation, or one with both $\boldsymbol{k}$ and $\boldsymbol{E}_1$ parallel to $\boldsymbol{B}_0$, with $p_{s,1} = \gamma_s k_B T_s n_{s,1}$. Define $c_s^2 = \gamma_s k_B T_s / m_s$. Here s labels a species; below, $c_{s,e}$ and $c_{s,i}$ denote the electron and ion pressure-response speeds. They are distinct from the collective ion-acoustic speed. The closure is local and fluid-like; it requires $k\lambda_D$ to remain small enough that kinetic phase mixing is not the leading correction. Small $k\lambda_D$ alone does not choose γ_s: the closure must match the thermal response. For collisionless Maxwellian Langmuir waves the leading warm correction has $\gamma_e = 3$; an isothermal closure is a different model. Perpendicular longitudinal response retains the cyclotron term derived in the next section.

Warm longitudinal response Definition 328

For a longitudinal perturbation, the pressure-corrected dielectric coefficient is $\varepsilon_\parallel = 1 - \sum_s \omega_{p,s}^2/(\omega^2 - k^2 c_s^2)$. For fixed ions and warm electrons this gives

$$\omega^2 = \omega_{p,e}^2 + k^2 c_{s,e}^2 = \omega_{p,e}^2\left(1 + \gamma_e k^2 \lambda_{D,e}^2\right),$$

with $\lambda_{D,e}^2 = \varepsilon_0 k_B T_e/\left(n_0 e^2\right)$ for an isothermal electron reference.

Warm plasma oscillation and ion acoustic wave Governing law 329

The fixed-ion warm branch is $\omega^2 = \omega_{p,e}^2 + k^2 c_{s,e}^2$. When both species move, the longitudinal condition is

$$1 - \frac{\omega_{p,e}^2}{\omega^2 - k^2 c_{s,e}^2} - \frac{\omega_{p,i}^2}{\omega^2 - k^2 c_{s,i}^2} = 0.$$

Its low-frequency root has

$$\omega^2 \approx \frac{k^2\left(\omega_{p,e}^2 c_{s,i}^2 + \omega_{p,i}^2 c_{s,e}^2\right)}{\omega_{p,e}^2 + \omega_{p,i}^2}.$$

For a hydrogen plasma this is approximately

$$\omega^2 = \frac{k^2 \gamma k_B (T_e + T_i)}{m_i}$$

when the same adiabatic convention is used for both species.

Derivation: pressure response and the two longitudinal roots

Step: Obtain the species response

The linearized continuity equation is

$$\frac{\partial n_{s,1}}{\partial t} + n_{s,0} \boldsymbol{\nabla} \cdot \left(\boldsymbol{u}_{s,1}\right) = 0.$$

For a longitudinal plane wave it becomes

$$-i\omega n_{s,1} + i n_{s,0} k u_{s,1} = 0,$$

hence

$$\frac{n_{s,1}}{n_{s,0}} = \frac{k u_{s,1}}{\omega}.$$

The pressure perturbation is

$$p_{s,1} = \gamma_s k_B T_s n_{s,1}.$$

The longitudinal momentum equation is therefore

$$-i\omega m_s u_{s,1} = q_s E_1 - \frac{ik\gamma_s k_B T_s n_{s,1}}{n_{s,0}}.$$

Use continuity and define $c_s^2 = \gamma_s k_B T_s / m_s$:

$$-i\omega m_s u_{s,1} = q_s E_1 - i\frac{k^2 m_s c_s^2 u_{s,1}}{\omega}.$$

After moving the pressure term to the left and multiplying by i,

$$m_s\left(\omega - \frac{k^2 c_s^2}{\omega}\right)u_{s,1} = i q_s E_1.$$

Thus

$$u_{s,1} = \frac{i q_s \omega E_1}{m_s(\omega^2 - k^2 c_s^2)}.$$

Step: Construct the longitudinal susceptibility

The current or charge response is proportional to $q_s n_{s,0} u_{s,1}$. Insert it into

$$\boldsymbol{\varepsilon} \cdot \boldsymbol{E} = \boldsymbol{E} + \left(\frac{i}{\varepsilon_0 \omega}\right)\boldsymbol{j}.$$

The contribution of species s is the susceptibility

$$-\frac{\omega_{p,s}^2}{\omega^2 - k^2 c_s^2}.$$

Summing over species gives the displayed longitudinal dielectric response $\varepsilon_{\parallel}$.

Step: Identify the fixed-ion warm branch

For fixed ions, set the electron-only coefficient to zero:

$$1 - \frac{\omega_{p,e}^2}{\omega^2 - k^2 c_{s,e}^2} = 0.$$

Multiplication by the denominator gives

$$\omega^2 - k^2 c_{s,e}^2 - \omega_{p,e}^2 = 0.$$

This is the warm plasma-oscillation branch. The definitions

$$\lambda_{D,e}^2 = \frac{\varepsilon_0 k_B T_e}{n_0 e^2}$$

$$c_{s,e}^2 = \frac{\gamma_e k_B T_e}{m_e}$$

imply

$$\frac{c_{s,e}^2}{\omega_{p,e}^2} = \gamma_e \lambda_{D,e}^2.$$

Step: Separate the two-species roots

For two mobile species, multiply the longitudinal condition by both denominators:

$$(\omega^2 - k^2 c_{s,e}^2)(\omega^2 - k^2 c_{s,i}^2) - \omega_{p,e}^2(\omega^2 - k^2 c_{s,i}^2) - \omega_{p,i}^2(\omega^2 - k^2 c_{s,e}^2) = 0.$$

This is quadratic in ω^2. For the low-frequency root, terms of order $k^4 c_{s,e}^2 c_{s,i}^2$ are smaller than the plasma-frequency terms. Retaining the terms of order ω^2 and k^2 gives

$$-\left(\omega_{p,e}^2 + \omega_{p,i}^2\right)\omega^2 + k^2\left[\omega_{p,e}^2 c_{s,i}^2 + \omega_{p,i}^2 c_{s,e}^2\right] \approx 0.$$

Solving gives the displayed ion-acoustic speed. Charge neutrality implies

$$\frac{\omega_{p,i}^2}{\omega_{p,e}^2} = \frac{m_e}{m_i},$$

so

$$c_{\mathrm{ia}}^2 \approx c_{s,i}^2 + \left(\frac{m_e}{m_i}\right) c_{s,e}^2 = \frac{\gamma_i k_B T_i}{m_i} + \frac{\gamma_e k_B T_e}{m_i}.$$

The high-frequency root approaches the electron plasma-oscillation branch.

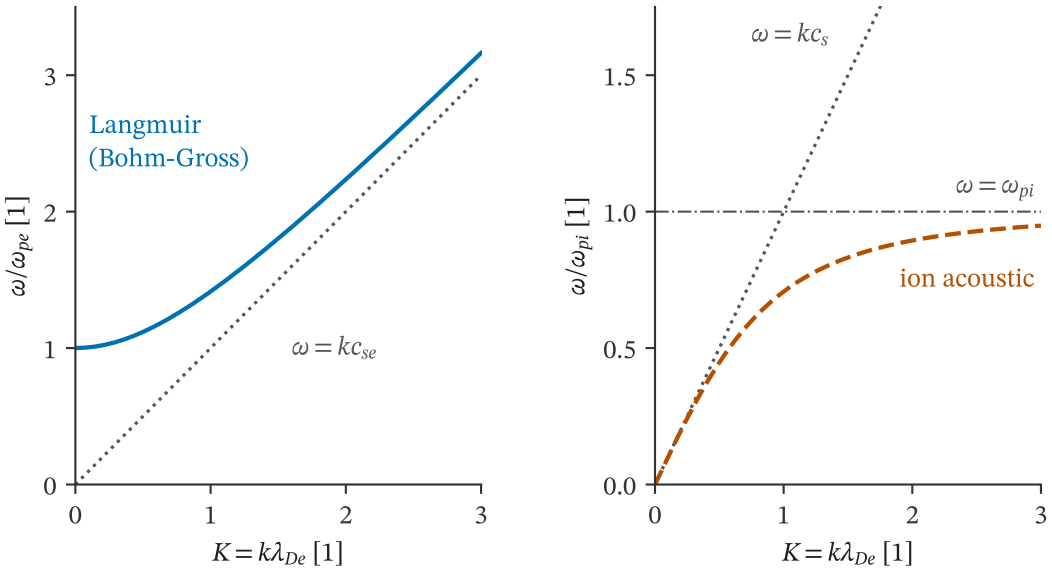


Figure 78: The two roots of the warm two-species longitudinal relation for isothermal electrons ($\gamma_e = 1$), cold ions and $m_i/m_e = 1836$, with $K = k\lambda_{D,e}$. Left, the electron plasma wave in units of $\omega_{p,e}$ follows $\omega^2 \approx \omega_{p,e}^2(1 + K^2)$. Right, the ion-acoustic branch in units of $\omega_{p,i}$ rises as $\omega \approx kc_s$ and saturates at $\omega_{p,i}$ for $K \gg 1$, where the fluid closure is no longer controlled. The two frequency scales differ by $\sqrt{m_i/m_e} \approx$

Rechenbeispiel

Rechenbeispiel 330

Assume a homogeneous, unmagnetized, collisionless, fixed-ion warm-fluid electron plasma with an isothermal closure. Use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $k_B T_e = 1.602 \cdot 10^{-18}\,\mathrm{J}$ (the same energy as 10 eV), $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, and $\gamma_e = 1$. This is a formal isothermal-closure calculation outside the controlled small-$k\lambda_D$ regime; its result requires kinetic comparison before interpretation as a collisionless plasma wave. At $k\lambda_{D,e} = 0.80$ [1], determine $\lambda_{D,e}$, k, and the normalized warm plasma-oscillation frequency.

Numerical result: $\lambda_{D,e} = 2.35 \cdot 10^{-4}\,\mathrm{m}$, $k = 3.40 \cdot 10^{3}\,\mathrm{m}^{-1}$, and $\omega/\omega_{p,e} = 1.28$ [1].

Pressure couples neighboring density perturbations

Interpretation 331

Without pressure, each cold-fluid density displacement oscillates at the same plasma frequency. The pressure gradient couples neighboring density perturbations and introduces wave-number dependence. Its relative contribution is $\gamma_e k^2 \lambda_{D,e}^2$ on the fixed-ion branch. Once $k\lambda_D$ is not small, the fluid branch must be compared with kinetic phase mixing.

Summary

Summary 332

A pressure closure changes the longitudinal response from $\omega = \omega_p$ to $\omega^2 = \omega_p^2 + k^2 c_s^2$. With mobile ions, the same response contains a low-frequency ion-acoustic root. The relevant ordering is measured by $k\lambda_D$ and by the ratio of electron and ion plasma frequencies.

Knowledge check

Knowledge check 333

- Why does a cold plasma oscillation have no wave-number dependence?
- Which term gives the warm plasma-oscillation branch its dispersion?
- Why does the two-fluid warm system have an ion-acoustic root?
- When should the warm-fluid longitudinal branch be checked against kinetic theory?

Answers

Why does a cold plasma oscillation have no wave-number dependence?
The cold momentum equation contains no pressure gradient. Without a spatial restoring term, the longitudinal charge displacement oscillates locally at the plasma frequency.

Which term gives the warm plasma-oscillation branch its dispersion?
The pressure perturbation $-\nabla(p_{s,1})$ couples density variation to velocity through continuity, producing the $k^2 c_s^2$ term in the denominator.

Why does the two-fluid warm system have an ion-acoustic root?
Electrons and ions can move together nearly quasineutrally at low frequency. Their pressure forces provide the restoring force while the ion mass supplies most of the inertia.

When should the warm-fluid longitudinal branch be checked against kinetic theory?
When $k\lambda_D$ is no longer small or when the phase velocity approaches a substantial particle population. Then velocity-space resonance and phase mixing can be as important as fluid pressure.

14.4 Warm transverse waves and magnetosonic motion

Pressure also modifies waves propagating across the magnetic field, but propagation direction and electric-field polarization must be distinguished. Near the

upper-hybrid frequency it adds a warm spatial correction to the extraordinary branch. At much lower frequency, mobile ions and nearly ideal conductivity reorganize the wave into MHD motion: magnetic tension gives shear-Alfvén waves, while pressure and magnetic compression combine into magnetosonic waves.

Learning objectives

- identify the warm upper-hybrid correction in perpendicular propagation
- derive shear-Alfvén and compressional restoring forces from warm MHD
- distinguish field-line tension from pressure compression
- connect the wave speeds to the single-fluid variables

Unit ledger

The speeds v_A, v_s, and v_m are in $\mathrm{m\,s^{-1}}$. The ratios $\omega/\omega_{c,i}$ and $k\lambda_D$ are dimensionless.

Warm-fluid and ideal-MHD limits — Assumptions 334

Use the warm-fluid pressure closure for the upper-hybrid correction, with fixed ions, $\boldsymbol{k} \perp \boldsymbol{B}_0$, and an electrostatic perturbation $\boldsymbol{E}_1 \parallel \boldsymbol{k}$. This is the longitudinal approximation near the extraordinary branch's upper-hybrid resonance. For the low-frequency MHD derivation assume quasineutrality, $\omega \ll \omega_{c,i}, k\lambda_D \ll 1$, a single bulk velocity, and sufficiently large conductivity that the ideal induction law applies. The bulk velocity is mass-weighted. The relative drift remains through $\boldsymbol{j} = en(\boldsymbol{u}_i - \boldsymbol{u}_e)$ and Ampere's law; Hall and relative inertial corrections are ordered small, rather than setting the current to zero. Use an adiabatic closure $p_1 = v_s^2 \rho_1$ with $v_s^2 = \gamma p_0/\rho_0$.

Alfvén, sound, and magnetosonic speeds — Definition 335

Define $v_A = B_0/\sqrt{\mu_0 \rho_0}$, $v_s = \sqrt{\gamma p_0/\rho_0}$, and $v_m = \sqrt{v_A^2 + v_s^2}$ for perpendicular compressional motion. The square v_A^2 is magnetic tension per unit mass density, v_s is the pressure-wave speed, and v_m is their warm perpendicular combination.

Warm upper-hybrid and MHD branches — Governing law 336

In the perpendicular electrostatic approximation near the upper-hybrid branch, the warm-fluid correction gives $\omega^2 = \omega_{\mathrm{UH}}^2 + k^2 c_{s,e}^2$ in the simple electron pressure ordering. At low frequency, ideal MHD gives the shear-Alfvén branch $\omega^2 = k_\parallel^2 v_A^2$ and the perpendicular compressional branch $\omega^2 = k_\perp^2 (v_A^2 + v_s^2)$. The latter is often called the magnetosonic or magnetosonic-Alfvén branch.

Derivation: warm upper hybrid and magnetosonic restoring forces

Step: Perpendicular warm-electron response

Choose $\boldsymbol{k} = k\boldsymbol{e}_x$, $\boldsymbol{E}_1 = E_{1,x}\boldsymbol{e}_x$, and $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$. Continuity gives

$$-i\omega n_{e,1} + ikn_0 u_{e,1,x} = 0$$

and therefore

$$\frac{n_{e,1}}{n_0} = \frac{k u_{e,1,x}}{\omega}.$$

With the signed electron gyrofrequency $\Omega_e = -\omega_{c,e}$, the transverse momentum equations are

$$-i\omega u_{e,1,x} - \Omega_e u_{e,1,y} = \left(\frac{q_e}{m_e}\right) E_{1,x} - \frac{ikc_{s,e}^2 n_{e,1}}{n_0}$$

and

$$\Omega_e u_{e,1,x} - i\omega u_{e,1,y} = 0.$$

The second equation gives

$$u_{e,1,y} = -i\left(\frac{\Omega_e}{\omega}\right) u_{e,1,x}.$$

Insert this relation and the continuity result into the first equation. Collect the velocity terms and multiply by i:

$$\left(\frac{\omega^2 - \Omega_e^2 - k^2 c_{s,e}^2}{\omega}\right) u_{e,1,x} = \left(\frac{iq_e}{m_e}\right) E_{1,x}.$$

Hence

$$u_{e,1,x} = \left(\frac{iq_e\omega}{m_e\left(\omega^2 - \omega_{c,e}^2 - k^2 c_{s,e}^2\right)}\right) E_{1,x},$$

and continuity gives

$$n_{e,1} = \left(\frac{in_0 q_e k}{m_e\left(\omega^2 - \omega_{c,e}^2 - k^2 c_{s,e}^2\right)}\right) E_{1,x}.$$

Step: Close the electrostatic response

For an electrostatic wave, Gauss's law is

$$ikE_{1,x} = \frac{q_e n_{e,1}}{\varepsilon_0}.$$

Substitute the density response and cancel the nonzero factor $ikE_{1,x}$:

$$1 = \left(\frac{n_0 q_e^2}{\varepsilon_0 m_e\left(\omega^2 - \omega_{c,e}^2 - k^2 c_{s,e}^2\right)}\right) = \frac{\omega_{p,e}^2}{\omega^2 - \omega_{c,e}^2 - k^2 c_{s,e}^2}.$$

The warm upper-hybrid branch is therefore

$$\omega^2 = \omega_{p,e}^2 + \omega_{c,e}^2 + k^2 c_{s,e}^2 = \omega_{\mathrm{UH}}^2 + k^2 c_{s,e}^2.$$

This is a local warm-fluid approximation near the upper-hybrid branch. The full electromagnetic warm tensor contains additional polarization terms.

Step: Recover the perpendicular MHD branch

For the low-frequency one-fluid limit, use continuity, $\partial\rho/\partial t + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0$, the ideal induction equation, $\partial\boldsymbol{B}/\partial t = \boldsymbol{\nabla}\times(\boldsymbol{u}\times\boldsymbol{B})$, and the MHD momentum equation

$$\rho\frac{\partial\boldsymbol{u}}{\partial t} = -\boldsymbol{\nabla}(p) + \frac{[\boldsymbol{\nabla}\times(\boldsymbol{B})\times\boldsymbol{B}]}{\mu_0}.$$

This momentum equation is written to first order about a static equilibrium; the advective acceleration is second order in the perturbations. Linearize about $\rho = \rho_0$, $p = p_0$, and $\boldsymbol{B} = B_0\boldsymbol{e}_z$. For $\boldsymbol{k} = k\boldsymbol{e}_x$ and $\boldsymbol{u}_1 = u_x\boldsymbol{e}_x$, continuity gives

$$-i\omega\rho_1 + ik\rho_0 u_x = 0, \quad \rho_1 = \frac{\rho_0 k u_x}{\omega}.$$

The pressure closure is $p_1 = v_s^2\rho_1$. The induction equation gives

$$-i\omega B_{1,z} = -ikB_0 u_x, \quad B_{1,z} = \frac{B_0 k u_x}{\omega}.$$

The x component of momentum is

$$-i\omega\rho_0 u_x = -ikp_1 - i\frac{kB_0 B_{1,z}}{\mu_0}.$$

Substitute the density, pressure, and magnetic perturbations, then divide by the nonzero factor $\rho_0 u_x/\omega$:

$$\omega^2 = k^2\left[v_s^2 + \frac{B_0^2}{\mu_0\rho_0}\right] = k^2\left(v_s^2 + v_A^2\right).$$

Thus the perpendicular compressional speed is

$$v_m = \sqrt{v_A^2 + v_s^2}.$$

Step: Recover the shear-Alfvén branch

For a shear perturbation with $\boldsymbol{k} = k_\parallel\boldsymbol{e}_z$, the pressure and density perturbations vanish to first order. Induction and transverse momentum retain only magnetic tension:

$$\frac{\partial^2\boldsymbol{u}_\perp}{\partial t^2} = v_A^2\frac{\partial^2\boldsymbol{u}_\perp}{\partial z^2}.$$

A plane wave then satisfies

$$\omega^2 = k_\parallel^2 v_A^2.$$

Tension bends field lines, whereas perpendicular compression changes both density and magnetic-field strength.

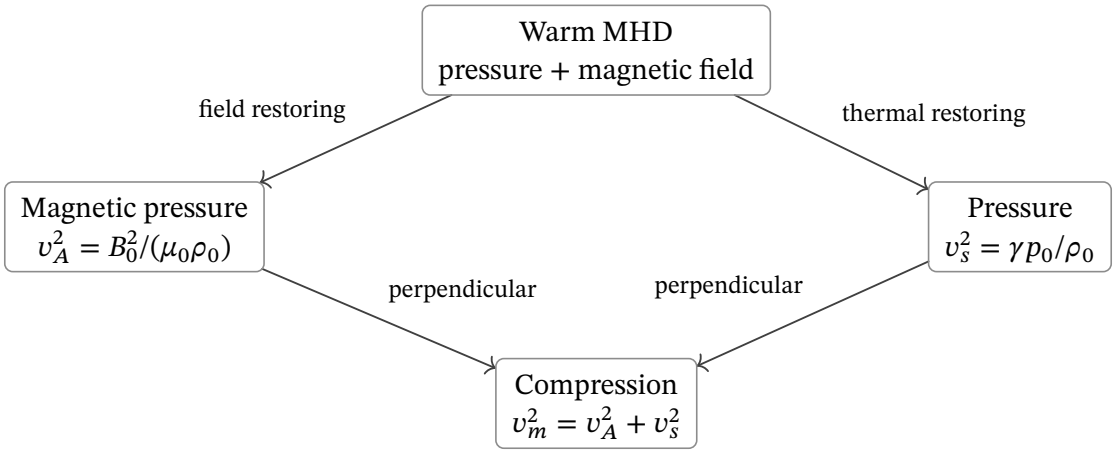


Figure 79: In the warm MHD limit, magnetic tension restores parallel shear-Alfvén waves. For perpendicular compression, magnetic pressure and thermal pressure combine to give $v_m = \sqrt{v_A^2 + v_s^2}$.



Watch animation

Warm magnetized-wave patterns: pressure compression, shear-Alfvén field displacement, and perpendicular compressional magnetosonic motion. All patterns use $X = x/L_0$ and $\tau = t/t_0$, with reference speed L_0/t_0. Their dimensionless speeds are $v_s t_0/L_0 = 0.6$, $v_A t_0/L_0 = 1$, and $v_m t_0/L_0 = \sqrt{1.36}$; the common wave number is $kL_0 = 1.25$. The prescribed linear patterns illustrate $v_m^2 = v_A^2 + v_s^2$; L_0 and t_0 are arbitrary reference length (m) and time (s).

Figure 80: Warm magnetized-wave patterns: pressure compression, shear-Alfvén field displacement, and perpendicular compressional magnetosonic motion. All patterns use $X = x/L_0$ and $\tau = t/t_0$, with reference speed L_0/t_0. Their dimensionless speeds are $v_s t_0/L_0 = 0.6$, $v_A t_0/L_0 = 1$, and $v_m t_0/L_0 = \sqrt{1.36}$; the common wave number is $kL_0 = 1.25$. The prescribed linear patterns illustrate $v_m^2 = v_A^2 + v_s^2$; L_0 and t_0 are arbitrary reference length (m) and time (s).

Rechenbeispiel

Rechenbeispiel 337

Assume a homogeneous, quasineutral hydrogen plasma in the low-frequency, strongly conducting ideal-MHD limit, with perpendicular and parallel propagation compared. Use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $B_0 = 1.0 \cdot 10^{-2}\,\mathrm{T}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, $k_B T_e = k_B T_i = 1.602 \cdot 10^{-18}\,\mathrm{J}$ (10 eV), and isothermal $\gamma_e = \gamma_i = 1$, determine v_A, the total-pressure sound speed v_s, and v_m. For $k = 1.0 \cdot 10^{-3}\,\mathrm{m}^{-1}$, report the parallel shear-Alfvén and perpendicular magnetosonic frequencies.

Numerical result: $v_A = 2.18 \cdot 10^{6}\,\mathrm{m\,s}^{-1}$, $v_s = 4.38 \cdot 10^{4}\,\mathrm{m\,s}^{-1}$, and $v_m = 2.18 \cdot 10^{6}\,\mathrm{m\,s}^{-1}$ to the shown precision. The two example frequencies are $\omega_A = 2.18 \cdot 10^{3}\,\mathrm{s}^{-1}$ and $\omega_m = 2.18 \cdot 10^{3}\,\mathrm{s}^{-1}$.

The same field can support different restoring forces

Interpretation 338

Shear-Alfvén motion bends field lines with little compression. Magnetosonic motion compresses the field and the fluid, so thermal pressure adds to magnetic pressure. The appropriate branch is selected by propagation angle and frequency ordering.

Summary

Summary 339

Warm pressure shifts the upper-hybrid branch and supplies the sound speed. In the low-frequency ideal-MHD limit, magnetic tension gives shear-Alfvén waves and pressure plus magnetic compression gives $v_m = \sqrt{v_A^2 + v_s^2}$. These are controlled limits of the same multi-species response.

Knowledge check

Knowledge check 340

- Why does finite temperature affect the upper-hybrid branch?
- Why is the shear-Alfvén wave not strongly modified by pressure in the ideal low-frequency limit?
- What is added when the magnetosonic speed is formed from v_A and v_s?
- Which assumptions are needed to use the ideal-MHD wave equations?

Answers

Why does finite temperature affect the upper-hybrid branch?

The perpendicular density perturbation produces a pressure gradient. Continuity converts it into a k-dependent restoring term, shifting the cold upper-hybrid frequency by a warm-fluid contribution.

Why is the shear-Alfvén wave not strongly modified by pressure in the ideal low-frequency limit?

Its transverse field-line displacement is nearly incompressible, so the first-order density and pressure perturbations vanish. Magnetic tension supplies the leading restoring force.

What is added when the magnetosonic speed is formed from v_A and v_s?

Magnetic compression contributes v_A^2 and thermal pressure contributes v_s^2 to the squared speed. For perpendicular compression their restoring forces add.

Which assumptions are needed to use the ideal-MHD wave equations?

Use a quasineutral, strongly conducting plasma with a mass-weighted bulk velocity, frequencies below the ion-inertia scale, and fluid length scales. Retain the relative species drift as the current in Ampere's law, while ordering Hall and relative inertial corrections small. Exact equality of electron and ion velocities would incorrectly remove that current.

14.5 Ordering map: from cold waves to warm MHD

The collisional, two-fluid, warm-fluid, and MHD responses of the preceding sections are branches of one model-selection map. Four dimensionless orderings decide whether a cold, collisional, two-fluid, warm-fluid, MHD, or kinetic calculation is justified.

Learning objectives

- define the collision, ion-inertia, thermal, and mass-ratio orderings
- recover the cold fixed-ion result as a controlled limit
- identify when a two-fluid or MHD description is preferable
- recognize the scale overlap that requires kinetic theory

Unit ledger

The ordering parameters $C = \nu/\omega$, $I = \omega/\omega_{c,i}$, $K = k\lambda_D$, and $M = m_e/m_i$ are dimensionless. Dimensional ν, ω, and $\omega_{c,i}$ are in s^{-1}; k is in m^{-1}; and λ_D and the pressure-response gyroradius $\rho_s = c_s/\omega_{c,s}$ are in m. Particle and wave speeds are in $\mathrm{m\,s}^{-1}$.

One model boundary at a time — Assumptions 341

Use the cold magnetized response as the reference model. Release one ordering while holding the others explicit: collisions through C, ion motion through I, pressure through K, and species separation through M. If more than one parameter is order unity, retain all corresponding terms before making a branch identification.

Model-selection ordering — Definition 342

The collisionless cold fixed-ion model requires $C \ll 1$, $I \gg 1$, and $K \ll 1$ for the mode under study. If C is not small, use a complex collisional response. If I is not large, retain ion inertia and use a two-fluid or MHD reduction. If K is not small, retain pressure and test the warm-fluid result against kinetic scales. The small mass ratio $M \ll 1$ justifies electron-dominated plasma frequencies but does not justify removing ion motion at low frequency.

Controlled limits — Governing law 343

The branches must satisfy four checks: (1) $\nu \to 0$ recovers the lossless cold tensor away from poles; N is real on propagating branches but imaginary in stopbands; (2) $m_i \to \infty$ or $I \to \infty$ recovers the fixed-ion branches; (3) $T_s \to 0$ or $K \to 0$ removes the pressure correction; and (4) $\omega \ll \omega_{c,i}$ with high conductivity reduces the two-fluid system to MHD, with v_A and v_m as the leading speeds. If $k\rho_s$ is not small, $\omega - k_\parallel v_\parallel$ approaches zero, or the distribution is strongly non-Maxwellian, a kinetic susceptibility is required even when the fluid equations look closed.

Derivation: recovering the neighboring models

Step: Recover the collisionless model

Begin with

$$m_{\text{eff}} = m\left(1 + \frac{i\nu}{\omega}\right).$$

Let $\nu \to 0$ at real frequency away from poles. Then $m_{\text{eff}} \to m$ and the cold dielectric tensor is Hermitian: its diagonal coefficients are real and its gyrotropic off-diagonal entries remain imaginary complex conjugates. Absorption vanishes, but a stopband with $N^2 < 0$ still has imaginary k and spatial evanescence. Real N is recovered only on propagating branches.

Step: Take the fixed-ion limit

For a singly charged electron–ion plasma, define

$$\frac{\omega_{c,i}}{\omega_{c,e}} = \frac{m_e}{m_i} = M, \quad \frac{\omega_{p,i}^2}{\omega_{p,e}^2} = M.$$

At fixed ω with $\omega \gg \omega_{c,i}$, the ion denominators are regular and the ion susceptibility is smaller by M than the electron susceptibility. Sending $M \to 0$ removes the ion contribution and recovers the fixed-ion tensor. This argument fails when ω is also reduced to the ion-cyclotron scale, because the ion denominator is then no longer an order-one factor.

Step: Recover the warm and cold pressure limits

The pressure-corrected species susceptibility contains

$$\omega^2 - k^2 c_s^2$$

instead of ω^2. Taking $T_s \to 0$ sends $c_s \to 0$ and returns the cold denominator. Conversely, when $k\lambda_D$ becomes order unity, the pressure term changes the longitudinal root by an order-one amount and cannot be treated as a perturbation.

Step: Recover the MHD endpoint

Take $\omega \ll \omega_{c,i}$ while maintaining high conductivity and quasineutrality. The weighted momentum sum gives

$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} = -\boldsymbol{\nabla}(p) + \frac{[\boldsymbol{\nabla} \times (\boldsymbol{B}) \times \boldsymbol{B}]}{\mu_0}.$$

The induction equation becomes

$$\frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times (\boldsymbol{u} \times \boldsymbol{B}).$$

Linearizing these equations gives the shear-Alfvén and magnetosonic dispersions derived in the previous section.

Step: Check the scale-ordering boundaries

A fluid closure fails when a wave samples unresolved particle scales. The independent checks are

$$k\lambda_D \ll 1, \quad k_\perp \rho_s \ll 1, \quad \left|\frac{\omega}{k_\parallel}\right| \gg \left|v_\parallel\right|$$

for cold quasineutral motion, finite-Larmor-radius effects, and resonant kinetic corrections respectively, with $v_\parallel$ a thermal parallel speed. The three conditions are independent.

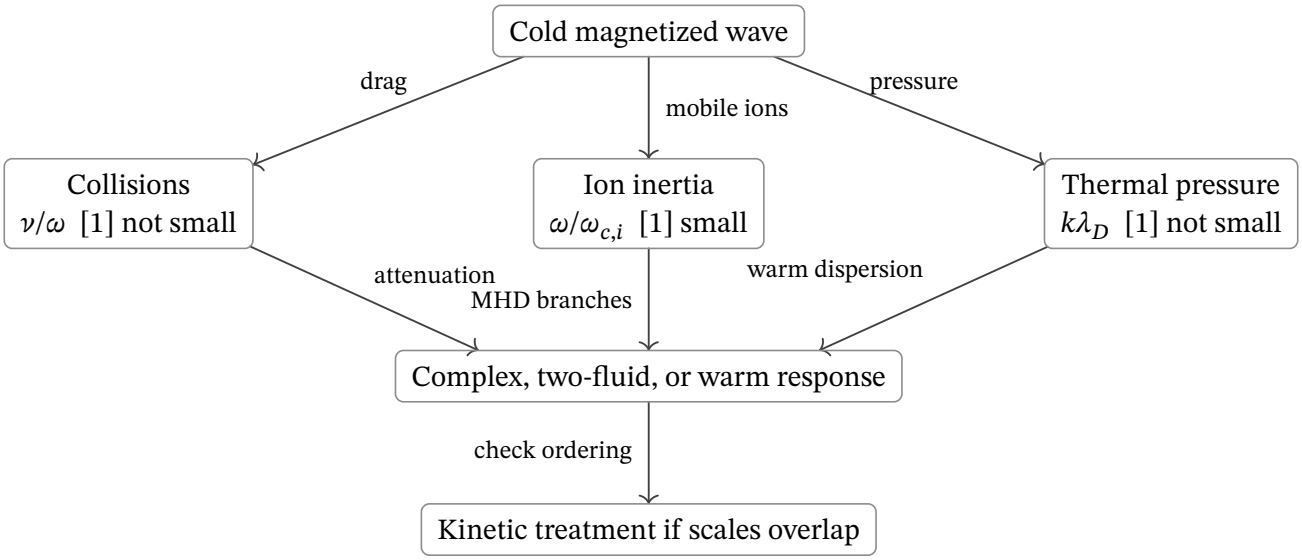


Figure 81: Model selection is controlled by normalized orderings with unit [1]. Compare ν/ω, $\omega/\omega_{c,i}$, and $k\lambda_D$ before interpreting a cold-plasma branch.

Rechenbeispiel Rechenbeispiel 344

Classify two model orderings. Case A has $C = 0.02$ [1], $I = 50$ [1], $K = 0.05$ [1], and $M = 5.45 \cdot 10^{-4}$ [1]. Case B has $C = 0.30$ [1], $I = 0.40$ [1], $K = 0.80$ [1], and the same $M = 5.45 \cdot 10^{-4}$ [1]. State the simplest justified description and the first correction to add in each case.

Numerical result: Case A is collisionless cold fixed-ion to leading order; the first corrections are collisional damping, ion inertia, and warm dispersion in the displayed order. Case B requires a collisional two-fluid warm model; a kinetic check is recommended because all three primary orderings are no longer asymptotically small.

Limits as consistency checks Interpretation 345

Every reduction should be reversible in a stated asymptotic limit. If a proposed branch does not recover its cold, fixed-ion, or MHD endpoint, the first suspects are a sign convention, an omitted current, or an inconsistent scale ordering.

Summary Summary 346

Collisions, ions, and temperature extend the cold magnetized spectrum in different directions. The dimensionless orderings C, I, and K make those extensions explicit. The correct endpoint is selected by scale separation; overlapping scales require a kinetic response.

Knowledge check

- Which limit removes collisional attenuation?
- Why does the fixed-ion limit fail at low frequency?
- What does $k\lambda_D$ measure in the model hierarchy?
- What is the first sign that a fluid branch needs a kinetic replacement?

Answers

Which limit removes collisional attenuation?

Taking $\nu \to 0$ at fixed real frequency away from poles removes collisional absorption and recovers the Hermitian cold tensor. Propagating branches have real N, while lossless stopbands can retain imaginary N and evanescence.

Why does the fixed-ion limit fail at low frequency?

Reducing the frequency to the ion-cyclotron scale makes the ion response resonant or order one even though m_e/m_i is small. The small mass ratio alone does not remove ion inertia.

What does $k\lambda_D$ measure in the model hierarchy?

It compares the perturbation wavelength with the Debye shielding length. Small $k\lambda_D$ supports a cold or quasineutral fluid ordering; order-one values make pressure and kinetic dispersion important.

What is the first sign that a fluid branch needs a kinetic replacement?

A wave-particle or cyclotron resonance, finite-Larmor-radius scale, strong non-Maxwellian distribution, or unresolved Debye-scale structure invalidates the assumed fluid closure. The relevant condition must be checked explicitly.

15. Waves in hot plasmas

A fluid closure retains only a few velocity moments. A hot plasma wave can depend on the particles that move nearly with the wave, so the full distribution function becomes part of the response. This chapter builds that kinetic response from the Vlasov equation, identifies resonant particles, derives collisionless damping and two-stream growth, and then extends the resonance picture to a magnetized plasma, following the hot- plasma treatment in [3]. Here "hot" means that the velocity distribution matters for the wave response; it is not an absolute temperature threshold.

Distribution slope at the phase velocity — Key idea 348

A hot-plasma dispersion relation is not specified by density and temperature alone. The slope of the velocity distribution projected along the wave helps decide whether resonant particles absorb energy from it or supply energy to it.

15.1 Hot isotropic plasma dispersion

A kinetic wave calculation starts with an equilibrium distribution. Perturb it weakly and solve the Vlasov equation along the unperturbed particle trajectories. The resulting susceptibility contains a resonant denominator that a finite set of fluid moments cannot reproduce.

Learning objectives

- write the collisionless Vlasov equation and its linear perturbation
- derive the electrostatic kinetic dispersion relation from Poisson's equation
- recover the cold plasma frequency and warm long-wave correction as limits
- define the plasma dispersion function and state its contour prescription

Unit ledger

The distribution f_s is a phase-space density with units $\mathrm{m}^{-6}\,\mathrm{s}^3$. The plasma dispersion function, $k\lambda_D$, and $\omega/\omega_{p,e}$ are dimensionless.

Collisionless isotropic equilibrium — Assumptions 349

Use a homogeneous unmagnetized electron plasma with fixed neutralizing ions, $\boldsymbol{B}_0 = \boldsymbol{0}$, and a smooth isotropic equilibrium $f_{e,0}(\boldsymbol{v})$. Let the perturbation be small and use the Fourier convention $\exp(i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t))$. The linear calculation describes normal modes; it does not describe nonlinear trapping or a finite-amplitude rearrangement of the distribution.

Kinetic perturbation — Definition 350

The collisionless kinetic equation for species s is

$$\frac{\partial f_s}{\partial t} + \boldsymbol{v}\cdot\nabla(f_s) + \left(\frac{q_s}{m_s}\right)(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B})\cdot\frac{\partial f_s}{\partial \boldsymbol{v}} = 0.$$

Expand $f_s = f_{s,0}(\boldsymbol{v}) + \varepsilon f_{s,1}(\boldsymbol{r},\boldsymbol{v},t)$, $\boldsymbol{E} = \varepsilon\boldsymbol{E}_1$, and use

$$f_{s,1} = \tilde{f}_{s,1}(\boldsymbol{v})\exp(i(\boldsymbol{k}\cdot\boldsymbol{r} - \omega t)).$$

The dimensionless parameter ε orders perturbation amplitudes. It is distinct from the dielectric function ε_L below.

Electrostatic kinetic dispersion — Governing law 351

For a longitudinal perturbation with $\boldsymbol{k} \parallel \boldsymbol{E}_1$, the allowed modes satisfy

$$\varepsilon_L(\omega,\boldsymbol{k}) = 1 + \sum_s \frac{q_s^2}{\varepsilon_0 m_s k^2}\int_{-\infty}^{\infty}\frac{\left[\boldsymbol{k}\cdot\frac{\partial f_{s,0}(\boldsymbol{v})}{\partial \boldsymbol{v}}\right]}{[\omega - \boldsymbol{k}\cdot\boldsymbol{v}]}\mathrm{d}^3\boldsymbol{v} = 0.$$

The species sum includes mobile species; fixed ions supply equilibrium neutrality but no perturbed response. The integral uses causal analytic continuation from $\Im(\omega) > 0$, with the Landau contour passing below the moving pole for $k > 0$. For an electron Maxwellian,

$$f_{e,0}(\boldsymbol{v}) = \frac{n_0}{\pi^{\frac{3}{2}} v_{\mathrm{te}}^3}\exp\left(-\frac{v^2}{v_{\mathrm{te}}^2}\right),$$

$v_{\mathrm{te}} = \sqrt{2k_B T_e/m_e}$, and $\varepsilon_L = 1 + \left(1/\left(k^2\lambda_D^2\right)\right)[1 + \zeta Z(\zeta)]$ with $\zeta = \omega/(kv_{\mathrm{te}})$ and $\lambda_D^2 = \varepsilon_0 k_B T_e/\left(n_0 e^2\right)$. Here

$$Z(\zeta) = \frac{1}{\sqrt{\pi}}\int\frac{\exp(-x^2)}{x-\zeta}\,\mathrm{d}x$$

is the plasma dispersion function on that contour from $-\infty$ to ∞; $x = v_\parallel/v_{\mathrm{te}}$ is a dimensionless integration variable.

Derivation: Vlasov response, Poisson, and the Maxwellian limit

Step: Linearize the Vlasov equation

Insert the perturbation expansion and retain first-order terms. With $\boldsymbol{B}_0 = \boldsymbol{0}$, the linear equation is

$$\frac{\partial f_{s,1}}{\partial t} + \boldsymbol{v}\cdot\nabla\left(f_{s,1}\right) + \frac{q_s}{m_s}\boldsymbol{E}_1\cdot\frac{\partial f_{s,0}}{\partial \boldsymbol{v}} = 0.$$

For the Fourier convention used here,

$$\frac{\partial f}{\partial t} \to -i\omega f, \quad \nabla(f) \to i\boldsymbol{k}f.$$

Therefore

$$i(\boldsymbol{k}\cdot\boldsymbol{v}-\omega)\tilde{f}_{s,1}+\frac{q_s}{m_s}\boldsymbol{E}_1\cdot\frac{\partial f_{s,0}}{\partial \boldsymbol{v}}=0,$$

and

$$\tilde{f}_{s,1}=\frac{-iq_s}{m_s}\frac{\boldsymbol{E}_1\cdot\frac{\partial f_{s,0}}{\partial \boldsymbol{v}}}{\omega-\boldsymbol{k}\cdot\boldsymbol{v}}.$$

Step: Close the electrostatic response with Poisson's equation

For an electrostatic wave, set

$$\boldsymbol{E}_1=-i\boldsymbol{k}\Phi_1.$$

The perturbed charge density is

$$\rho_1=\sum_s q_s\int \tilde{f}_{s,1}\mathrm{d}^3\boldsymbol{v}.$$

Inserting the distribution response gives

$$\rho_1=-\sum_s\frac{q_s^2\Phi_1}{m_s}\int\frac{\boldsymbol{k}\cdot\frac{\partial f_{s,0}}{\partial \boldsymbol{v}}}{\omega-\boldsymbol{k}\cdot\boldsymbol{v}}\mathrm{d}^3\boldsymbol{v}.$$

Poisson's equation is

$$\boldsymbol{\nabla}\cdot(\boldsymbol{E}_1)=\frac{\rho_1}{\varepsilon_0}.$$

After the Fourier substitution, $k^2\Phi_1=\rho_1/\varepsilon_0$. A nonzero potential therefore requires the displayed condition $\varepsilon_L=0$.

Step: Evaluate an isotropic Maxwellian

Choose the z axis along $\boldsymbol{k}$. The perpendicular velocity integrals reduce the response to the one-dimensional plasma-dispersion function

$$Z(\zeta)=\frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{\exp(-x^2)}{x-\zeta}\,\mathrm{d}x.$$

Define the integral first for $\Im(\omega)>0$ and $k>0$, then continue causally, keeping the contour below the pole as it crosses the real axis. For one isotropic electron Maxwellian, the result is

$$\varepsilon_L=1+\frac{1}{k^2\lambda_D^2}[1+\zeta Z(\zeta)].$$

Step: Recover cold and warm limits

For a cold distribution, the integration-by-parts form reduces to

$$\varepsilon_L=1-\frac{\omega_{p,e}^2}{\omega^2},$$

so $\omega=\omega_{p,e}$. For $|\zeta|\gg 1$,

$$Z(\zeta) \approx -\frac{1}{\zeta} - \frac{1}{2\zeta^3} - \frac{3}{4\zeta^5}.$$

Substitution into $\varepsilon_L = 0$ gives

$$\omega^2 \approx \omega_{p,e}^2 + \frac{3k^2 k_B T_e}{m_e}.$$

The warm-fluid coefficient is the leading real kinetic correction, while the contour contribution contains collisionless damping omitted by the moment closure.

Rechenbeispiel Rechenbeispiel 352

Use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $k_B T_e = 1.602 \cdot 10^{-18}\,\mathrm{J}$ (that is, $10\,\mathrm{eV}$), $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, and $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m}^{-1}$. Take $k\lambda_D = 0.20$ [1] and use the long-wavelength warm approximation $\omega_r^2/\omega_{p,e}^2 \approx 1 + 3(k\lambda_D)^2$ [1]. Determine λ_D, k, and ω_r.

Numerical result: $\lambda_D \approx 2.35 \cdot 10^{-4}\,\mathrm{m}$, $k \approx 851\,\mathrm{m}^{-1}$, and $\omega_r/\omega_{p,e} \approx 1.06$ [1] or $\omega_r \approx 5.97 \cdot 10^9\,\mathrm{s}^{-1}$.

The kinetic correction has two parts Interpretation 353

The real part of the kinetic susceptibility shifts the wave frequency, while its contour contribution supplies a small imaginary part when resonant particles are present. The warm-fluid result can reproduce the long-wave real shift, but it cannot decide the sign of wave-particle energy exchange.

Summary Summary 354

The Vlasov equation plus Poisson's equation produces a dielectric function containing the distribution function and a resonant denominator. A cold delta distribution recovers $\omega = \omega_{p,e}$; a Maxwellian gives the plasma-dispersion function and the warm long-wave shift $\omega^2 \approx \omega_{p,e}^2 + 3k^2 k_B T_e/m_e$.

Knowledge check

- Why can a finite set of fluid moments miss a hot-plasma wave effect?
- What role does the Landau contour play in the electrostatic dielectric function?
- Which limit recovers the cold electron plasma frequency?
- What does the coefficient three in the warm Langmuir correction represent?

Answers

Why can a finite set of fluid moments miss a hot-plasma wave effect?
The moment hierarchy compresses velocity space. A resonant denominator samples the distribution at a particular velocity, so the local slope and contour contribution are not determined by a finite density, momentum, and pressure set alone.

What role does the Landau contour play in the electrostatic dielectric function?
It specifies how the velocity integral bypasses the pole at the wave phase velocity. That analytic prescription supplies the imaginary contribution associated with resonant particles.

Which limit recovers the cold electron plasma frequency?
Taking the equilibrium distribution toward a cold velocity-space delta function, or taking the long-wave thermal correction to zero, gives $\varepsilon_L = 1 - \omega_{p,e}^2/\omega^2$ and $\omega = \omega_{p,e}$.

What does the coefficient three in the warm Langmuir correction represent?
For an isotropic Maxwellian and a long-wavelength expansion, the second velocity moment enters the real susceptibility and gives the coefficient $3k_B T_e/m_e$. A different closure or distribution must be stated before changing it.

15.2 Resonant particles and velocity-space response

The strongest resonant response comes from particles near the phase velocity $v_{\mathrm{phi}} = \omega_r/k$, which selects a narrow region of velocity space. The denominator $\omega - kv$ becomes small there, so the slope of the equilibrium distribution near v_{phi} controls the direction of the resonant energy exchange.

Learning objectives

- define the phase velocity and resonant velocity for a longitudinal wave
- locate the pole in the velocity integral and explain its contour treatment
- relate the sign of the velocity-space slope to damping or growth
- distinguish a smooth resonance from nonlinear particle trapping

Unit ledger

The phase velocity v_{phi} and thermal speed v_{te} are in $\mathrm{m\,s^{-1}}$. The resonant velocity v_{res} is in $\mathrm{m\,s^{-1}}$, ω_r and the weak rate γ are in $\mathrm{s^{-1}}$, and k is in $\mathrm{m^{-1}}$. The one-dimensional marginal $F_{s,0}$ is in $\mathrm{m^{-4}\,s}$, and its slope $\mathrm{d}F_{s,0}/\mathrm{d}v$ is in $\mathrm{m^{-5}\,s^2}$. The ratios $v_{\mathrm{phi}}/v_{\mathrm{te}}$ and $\gamma/\omega_{p,e}$ are dimensionless.

Weak, smooth, collisionless resonance

Assume a smooth equilibrium and a small complex correction to a real wave root, $\omega = \omega_r + i\gamma$ with $|\gamma| \ll \omega_r$. The perturbation remains linear, so resonant trapping width and nonlinear plateau formation are outside this section.

Velocity marginal along the wave Definition 357

For $\boldsymbol{k} = k\boldsymbol{e}_z$ with $k > 0$, write $v = v_z$ and integrate out the perpendicular velocities:

$$F_{s,0}(v) = \int_{\mathbb{R}^2} f_{s,0}(v_x, v_y, v)\,\mathrm{d}v_x\,\mathrm{d}v_y,$$

so

$$\int_{-\infty}^{\infty} F_{s,0}(v)\,\mathrm{d}v = n_{s,0}.$$

This is a marginal distribution, not a slice at $v_x = v_y = 0$. For the isotropic Maxwellian,

$$F_{e,0}(v) = \left(\frac{n_0}{\sqrt{\pi}v_{\mathrm{te}}}\right)\exp\left(-\frac{v^2}{v_{\mathrm{te}}^2}\right).$$

Phase and resonant velocity Definition 358

For a forward longitudinal wave, define $v_{\mathrm{phi}} = \omega_r/k$ and $v_{\mathrm{res}} = v_{\mathrm{phi}}$. The pole of the response is at $\omega_r - kv_{\mathrm{res}} = 0$ in the real-frequency limit. For a marginal that varies slowly across the resonant layer, only its local derivative $\mathrm{d}F_{s,0}/\mathrm{d}v$ evaluated at $v = v_{\mathrm{res}}$ is needed for the leading sign of the collisionless energy exchange.

Slope criterion Governing law 359

For a weakly damped or growing root,

$$\gamma = -\frac{\Im(\varepsilon_L(\omega_r, k))}{\frac{\partial \Re(\varepsilon_L(\omega,k))}{\partial\omega}}.$$

The derivative in the denominator is evaluated at $\omega = \omega_r$. With the present Fourier convention, $\gamma < 0$ means temporal damping and $\gamma > 0$ means temporal growth. For the usual positive-frequency branch, with positive $\partial\Re(\varepsilon_L)/\partial\omega$, a Maxwellian marginal slope $\mathrm{d}F_{e,0}/\mathrm{d}v < 0$ gives damping, while a positive bump-on-tail slope can give growth. The sign is a physical statement only after the Fourier convention and propagation direction are fixed.

Derivation: pole, contour, and local slope

Step: Expose the resonant denominator

Choose $\boldsymbol{k} = k\boldsymbol{e}_z$ with $k > 0$. Integrate the three-dimensional response over v_x, v_y before integrating by parts in v:

$$\varepsilon_L = 1 - \sum_s \frac{q_s^2}{\varepsilon_0 m_s k^2}\int_{-\infty}^{\infty} \frac{\frac{\mathrm{d}F_{s,0}}{\mathrm{d}v}}{v - \frac{\omega}{k}}\,\mathrm{d}v.$$

The boundary term $F_{s,0}v/(v - \omega/k)$ vanishes at infinity. Since

$$\frac{\mathrm{d}\frac{1}{v-\frac{\omega}{k}}}{\mathrm{d}v} = -\frac{1}{\left(v-\frac{\omega}{k}\right)^2},$$

this becomes

$$\varepsilon_L = 1 - \sum_s \frac{q_s^2}{\varepsilon_0 m_s k^2} \int_{-\infty}^{\infty} F_{s,0} \frac{v}{\left(v-\frac{\omega}{k}\right)^2}\,\mathrm{d}v.$$

The contour prescription is unchanged. The equivalent first-derivative form is useful because the pole contribution is directly proportional to $\mathrm{d}F_{s,0}/\mathrm{d}v$ at $v = \omega_r/k$.

Step: Take the causal boundary value

Start in $\Im(\omega) > 0$ and continue toward the real axis. For a smooth test function $g(v)$, the boundary value is

$$\int \frac{g(v)}{v - v_{\mathrm{res}} - i0}\,\mathrm{d}v = \mathrm{PV}\int \frac{g(v)}{v - v_{\mathrm{res}}}\,\mathrm{d}v + i\pi g(v_{\mathrm{res}}).$$

Here PV denotes the Cauchy principal value. Consequently,

$$\Im(\varepsilon_L(\omega_r, k)) = -\sum_s \frac{\pi q_s^2}{\varepsilon_0 m_s k^2} \frac{\mathrm{d}F_{s,0}}{\mathrm{d}v}.$$

Each derivative on the right is evaluated at $v = \omega_r/k$.

A negative electron marginal slope therefore gives positive $\Im(\varepsilon_L)$. On the Langmuir branch the real-frequency derivative is positive, so the weak-rate relation gives damping. The following section derives that relation and its Maxwellian asymptote.

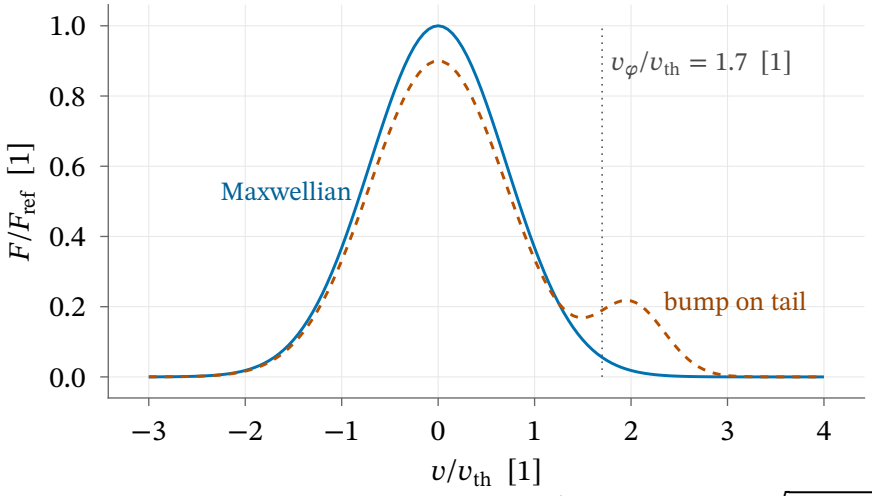


Figure 82: One-dimensional velocity marginals, with $\xi = v/v_{\mathrm{th}}$, $v_{\mathrm{th}} = \sqrt{2k_B T_e/m_e}$ and $F_{\mathrm{ref}} = n_0/\left(\sqrt{\pi}v_{\mathrm{th}}\right)$. The Maxwellian is $F/F_{\mathrm{ref}} = \exp\left(-\xi^2\right)$; the equal-density mixture is $0.9\exp\left(-\xi^2\right) + 0.2\exp\left(-4(\xi-2)^2\right)$. At the marked $v_\varphi/v_{\mathrm{th}} = 1.7$, near the steepest positive slope of the mixture, their slopes have opposite signs. A positive slope can supply growth; a full dispersion calculation is still needed. Both axes use unit [1].

Rechenbeispiel

Rechenbeispiel 360

For $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$ and $k_B T_e = 1.602 \cdot 10^{-18}\,\mathrm{J}$, use $\lambda_D = 2.35 \cdot 10^{-4}\,\mathrm{m}$ and $\omega_{p,e} = 5.64 \cdot 10^{9}\,\mathrm{s}^{-1}$. At $k\lambda_D = 0.30$ [1], estimate $\omega_r/\omega_{p,e}$ [1] from the long-wave real root, and locate the resonance using $v_{\mathrm{te}} = \sqrt{2}\omega_{p,e}\lambda_D$.

Numerical result: $\omega_r/\omega_{p,e} \approx 1.13$ [1], $v_{\mathrm{res}}/v_{\mathrm{te}} \approx 2.66$ [1]. The Maxwellian marginal slope there is negative.

Resonance is not the same as trapping

Interpretation 361

The linear theory identifies the particles that exchange energy with the wave. At larger amplitude those particles can become trapped in the wave potential, flatten the distribution near v_{phi}, and change the rate. The slope criterion is therefore a linear diagnostic.

Summary

Summary 362

The resonant velocity is the phase velocity for an unmagnetized longitudinal wave. The Landau contour converts the local slope of $F_{e,0}(v)$ into an imaginary frequency correction: negative slope damps the positive-frequency Maxwellian branch, while a positive slope can drive it.

Knowledge check

Knowledge check 363

- Which particles are selected by a longitudinal Landau resonance?
- Why does the slope of $F_{e,0}$ matter more than its value at the resonance?
- What does $\gamma < 0$ mean with the convention $\exp(i(kx - \omega t))$?
- When should the weak-damping estimate be replaced by a complex root calculation?

Answers

Which particles are selected by a longitudinal Landau resonance?
Particles with velocity near the wave phase velocity satisfy $\omega_r - kv \approx 0$. They are the particles for which the wave and particle remain in phase long enough for a coherent energy exchange.

Why does the slope of $F_{e,0}$ matter more than its value at the resonance?
The contour contribution compares particles just below and just above the resonant velocity. Their imbalance is represented by the local derivative, which determines the net direction of energy transfer.

What does $\gamma < 0$ mean with the convention $\exp(i(kx - \omega t))$?
Since $\omega = \omega_r + i\gamma$, the time factor contains $\exp(\gamma t)$. Therefore $\gamma < 0$ is temporal damping and $\gamma > 0$ is temporal growth.

When should the weak-damping estimate be replaced by a complex root calculation?
Solve the full complex dispersion relation when the rate is not small compared with the real frequency or when coupled resonances defeat the weak-rate expansion. A nonsmooth distribution needs its own integral treatment. Nonlinear trapping requires a nonlinear model, not just a more accurate linear root.

15.3 Landau damping and wave growth

The resonant-particle argument becomes quantitative when the complex root is separated into an oscillation frequency and a rate. Landau damping is collision-

less attenuation caused by phase-correlated particles, whereas growth occurs when a non-equilibrium distribution has a free-energy slope that reverses the exchange.

Learning objectives

- separate the real frequency and temporal rate of a complex root
- derive the weak-rate formula and follow energy into velocity-space structure
- interpret collisionless damping as wave–particle energy exchange
- convert a temporal rate to a spatial attenuation scale when appropriate

Unit ledger

The real frequency ω_r, temporal rate γ, and plasma frequency $\omega_{p,e}$ are in s^{-1}. The wave number k is in m^{-1}, the group velocity is in $\mathrm{m\,s}^{-1}$, and the temporal and spatial e-folding scales are in s and m. Rates normalized by $\omega_{p,e}$ and wave numbers normalized by λ_D are dimensionless.

Weak complex-frequency correction — Assumptions 364

Let the real part of the kinetic dispersion relation have a simple root $\omega_r(k)$. Assume $|\gamma| \ll \omega_r$, a collisionless plasma, and a contour prescription that is causal for the chosen initial-value problem. The mode is followed before nonlinear saturation or strong collisional broadening occurs.

Temporal and spatial damping scales — Definition 365

With $\omega = \omega_r + i\gamma$, the time factor is $\exp(-i\omega t) = \exp(-i\omega_r t)\exp(\gamma t)$. A damping time is $\tau_\mathrm{d} = 1/|\gamma|$ for $\gamma < 0$. For a forward packet with group velocity $v_\mathrm{g} > 0$, the corresponding spatial rate is $k_i \approx -\gamma/v_\mathrm{g}$ and the amplitude attenuation length is $L_\mathrm{amp} = 1/k_i$. These are distinct from the intensity e-folding length, which is $1/(2k_i)$.

Landau damping estimate — Governing law 366

The general weak-rate relation is

$$\gamma = -\frac{\Im(\varepsilon_L(\omega_r, k))}{\frac{\partial \Re(\varepsilon_L(\omega,k))}{\partial \omega}}.$$

For a Maxwellian Langmuir branch at $k\lambda_D \ll 1$,

$$\gamma \approx -\sqrt{\frac{\pi}{8}}\omega_{p,e}\exp\left(-\frac{3}{2}\right)(k\lambda_D)^{-3}\exp\left[-\frac{1}{2(k\lambda_D)^2}\right].$$

This is collisionless damping: no collision frequency appears. A positive velocity-space slope changes the sign of the resonant contribution and can turn damping into growth.

Derivation: complex root and energy exchange

Step: Expand the complex dispersion relation

Write

$$\varepsilon_L(\omega, k) = \varepsilon_r(\omega, k) + i\varepsilon_i(\omega, k),$$

and let $\omega = \omega_r + i\gamma$. To first order in γ,

$$0 \approx \varepsilon_r(\omega_r, k) + i\varepsilon_i(\omega_r, k) + i\gamma \frac{\partial \varepsilon_r(\omega, k)}{\partial \omega}.$$

Separating real and imaginary parts gives

$$\varepsilon_r(\omega_r, k) = 0, \quad \gamma = -\varepsilon_i \frac{\omega_r, k}{\frac{\partial \varepsilon_r(\omega,k)}{\partial \omega}}.$$

Step: Evaluate the Maxwellian contour contribution

Causal continuation gives

$$\Im(Z(\zeta)) = \sqrt{\pi} \exp(-\zeta^2)$$

for real positive ζ, while

$$\Re(Z(\zeta)) \approx -\frac{1}{\zeta} - \frac{1}{2\zeta^3} - \ldots.$$

The imaginary contribution near a positive phase velocity is therefore proportional to $\zeta \exp(-\zeta^2)$. The real expansion supplies

$$\omega_r^2 \approx \omega_{p,e}^2 \left[1 + 3(k\lambda_D)^2\right].$$

Put $a = k\lambda_D$. Then $\zeta = \omega_r / \left(\sqrt{2} a \omega_{p,e}\right)$,

$$\varepsilon_i = \frac{\sqrt{\pi}\zeta \exp(-\zeta^2)}{a^2},$$

and $\partial\varepsilon_r/\partial\omega \approx 2/\omega_{p,e}$ at leading order. Use $\zeta \approx 1/\left(\sqrt{2}a\right)$ in the prefactor but retain $\zeta^2 \approx 1/\left(2a^2\right) + 3/2$ in the exponential. The weak-rate formula then gives

$$\frac{\gamma}{\omega_{p,e}} \approx -\sqrt{\frac{\pi}{8}} a^{-3} \exp\left(-\frac{1}{2a^2} - \frac{3}{2}\right).$$

Step: Interpret the sign by energy exchange

The field establishes correlations between particle phase and velocity. An ensemble with negative marginal slope at positive v_{phi} absorbs net wave energy on this branch. The familiar slower/faster particle picture describes that phase-correlated ensemble exchange: an individual particle's instantaneous gain or loss also depends on its wave phase. A bump-on-tail can reverse the net exchange. Collisionless streaming transfers the perturbation into progressively finer velocity-space structure, allowing the coherent field to decay without collisional loss of the total particle-plus-field energy.

Step: Relate temporal and spatial descriptions

Expand a real-frequency dispersion relation around a temporal root:

$$\omega(k_r + ik_i) \approx \omega_r + i\gamma + ik_i v_{\mathrm{g}}.$$

Holding ω real requires $\gamma + k_i v_g = 0$, hence

$$k_i = -\frac{\gamma}{v_g}.$$

This relation applies to a simple branch with small damping and nonzero group velocity.

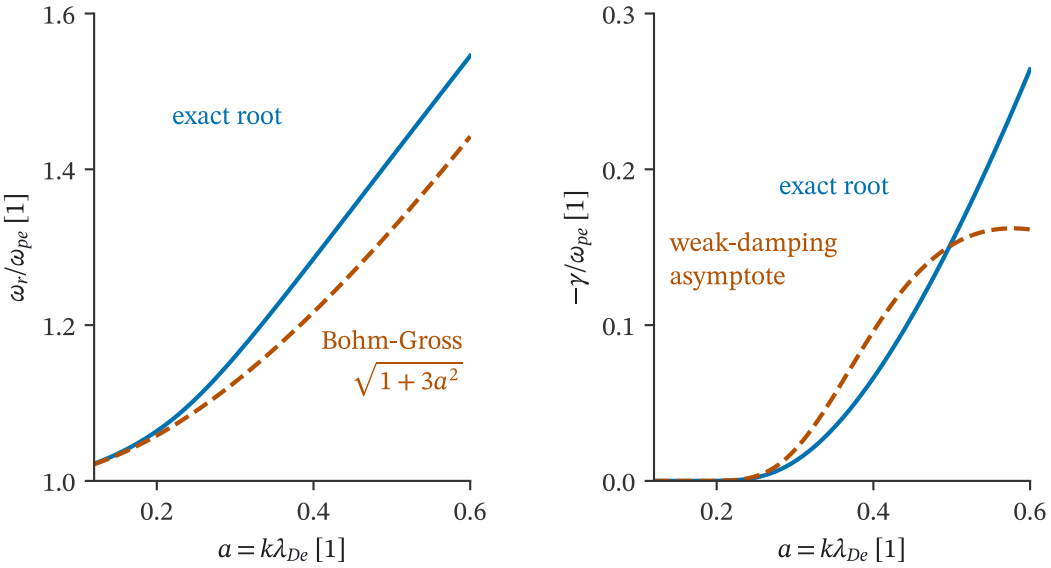
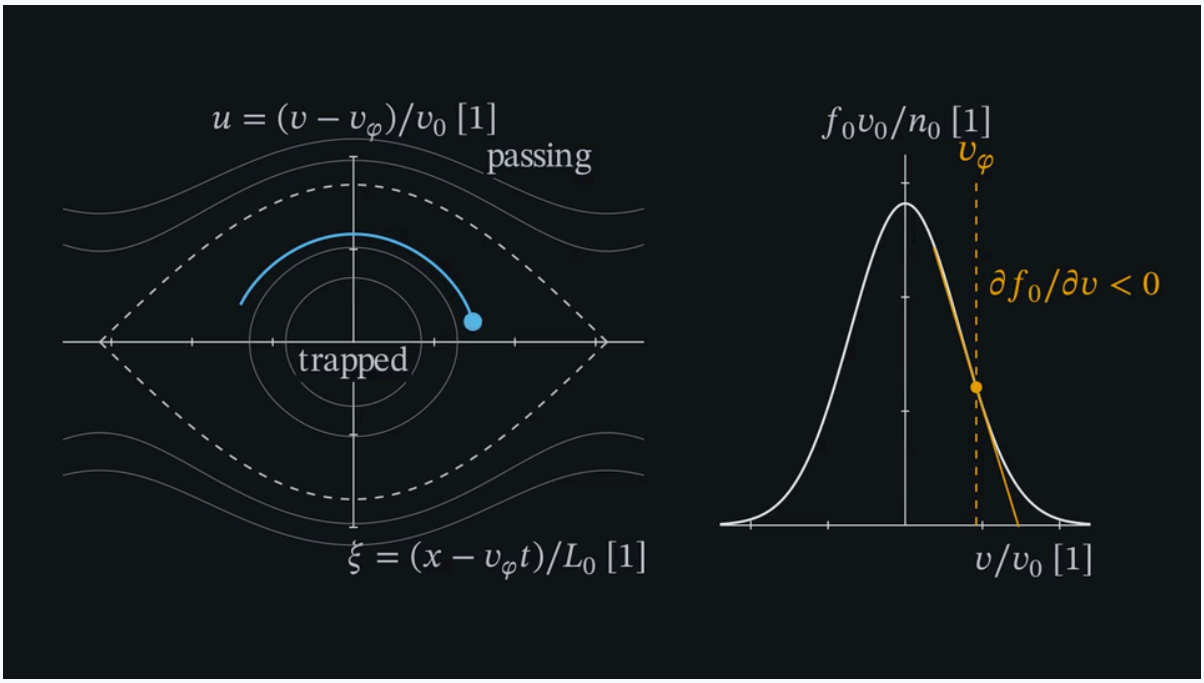


Figure 83: Exact complex root $\omega = \omega_r + i\gamma$ (real part left, damping rate right) of $\varepsilon_L = 1 + (1 + \zeta Z(\zeta))/a^2 = 0$ for one electron Maxwellian, $a = k\lambda_{D,e}$, against the asymptotes $\omega_r/\omega_{p,e} = \sqrt{1 + 3a^2}$ and $\frac{\gamma}{\omega_{p,e}} = -\sqrt{\pi/8}\, a^{-3} \exp\!\left(-1/\left(2a^2\right) - 3/2\right)$. The asymptotes hold for $a \lesssim 0.25$; beyond, the wave is strongly damped and only the full root is meaningful.



Watch animation

Landau damping is a phase-correlated exchange between a wave and particles near its phase velocity. The orbit panel instead follows one finite-amplitude test electron without wave feedback, using $X = x/L_0$, $V = v/v_0$, $\tau = tv_0/L_0$, $\mathrm{d}X/\mathrm{d}\tau = V$, and $\mathrm{d}V/\mathrm{d}\tau = -0.18\sin(X - 0.92\tau)$. It starts at $X = -1.4$, $V = 1.12$ and displays $X - 0.92\tau$ and $V - 0.92$. The field scale is $E_0 = m_e v_0^2/(eL_0)$; L_0 and v_0 are reference length and speed. In the distribution panel $v_0 = v_{\mathrm{te}}$; the plotted f_0 is the marginal $F_{e,0}$ defined above, with $F_{e,0}v_0/n_0 = \exp(-V^2)/\sqrt{\pi}$.

Figure 84: Landau damping is a phase-correlated exchange between a wave and particles near its phase velocity. The orbit panel instead follows one finite-amplitude test electron without wave feedback, using $X = x/L_0$, $V = v/v_0$, $\tau = tv_0/L_0$, $\mathrm{d}X/\mathrm{d}\tau = V$, and $\mathrm{d}V/\mathrm{d}\tau = -0.18\sin(X - 0.92\tau)$. It starts at $X = -1.4$, $V = 1.12$ and displays $X - 0.92\tau$ and $V - 0.92$. The field scale is $E_0 = m_e v_0^2/(eL_0)$; L_0 and v_0 are reference length and speed. In the distribution panel $v_0 = v_{\mathrm{te}}$; the plotted f_0 is the marginal $F_{e,0}$ defined above, with $F_{e,0}v_0/n_0 = \exp(-V^2)/\sqrt{\pi}$.

Rechenbeispiel Rechenbeispiel 367

For a forward packet use a temporal rate $\gamma = -1.13 \cdot 10^8\,\mathrm{s}^{-1}$ and group velocity $v_g = 1.00 \cdot 10^6\,\mathrm{m\,s}^{-1}$. Use the weak-rate conversion $k_i \approx -\gamma/v_g$, with $L_{\mathrm{amp}} = 1/k_i$ and $\tau_d = 1/|\gamma|$. Determine the spatial attenuation rate, amplitude attenuation length, and temporal damping time.

Numerical result: $k_i \approx 113\,\mathrm{m}^{-1}$, $L_{\mathrm{amp}} \approx 8.85 \cdot 10^{-3}\,\mathrm{m}$, and $\tau_d \approx 8.85 \cdot 10^{-9}\,\mathrm{s}$.

Damping is a collective collisionless process Interpretation 368

Landau damping does not require binary collisions or entropy production through a collisional operator. The phase mixing of a reversible Vlasov system moves perturbation energy into fine velocity-space structure. A coarse-grained field measurement sees attenuation even though the underlying collisionless equation is Hamiltonian.

Summary

A complex kinetic root has a real oscillation frequency and an imaginary rate. On the declared convention $\gamma < 0$ damps the wave. The rate is fixed by the imaginary resonant response divided by the slope of the real dispersion function; for a forward Maxwellian Langmuir wave this is the familiar Landau-damping asymptote.

Knowledge check

- Why is Landau damping called collisionless damping?
- How are temporal and spatial attenuation related for a forward packet?
- What must be true for the weak-rate expansion to be reliable?
- How can a collisionless wave grow instead of damp?

Answers

Why is Landau damping called collisionless damping?

Its rate comes from the resonant contour contribution of the Vlasov response, not from a binary-collision frequency. Phase mixing transfers coherent wave energy into fine velocity-space structure.

How are temporal and spatial attenuation related for a forward packet?

For small rates and group velocity $v_g > 0$, the fixed-frequency spatial rate is $k_i \approx -\gamma/v_g$. Thus temporal damping with $\gamma < 0$ gives positive spatial attenuation.

What must be true for the weak-rate expansion to be reliable?

The complex correction must be small compared with the real frequency, the real root must be simple, and the distribution and contour contribution must vary smoothly enough for a first-order expansion.

How can a collisionless wave grow instead of damp?

A non-equilibrium distribution can have a positive slope at the resonant velocity. Then resonant particles lose net energy to the wave, making $\gamma > 0$ on the present convention.

15.4 Transverse kinetic waves

The kinetic response is not restricted to electrostatic oscillations. For a transverse electromagnetic wave, the magnetic part of the wave force acts on an anisotropic equilibrium distribution even when the equilibrium field is zero. For isotropic $f_{s,0}$ its velocity gradient is parallel to $\boldsymbol{v}$, so the magnetic force dotted into that gradient vanishes. The resulting dispersion relation recovers the cold plasma cutoff, while an anisotropic distribution can turn the transverse branch unstable.

Learning objectives

- derive the transverse kinetic response from the Vlasov and Maxwell equations
- recover the cold unmagnetized electromagnetic dispersion relation
- identify the pole that can produce transverse wave damping or growth
- show how a perpendicular velocity-space anisotropy supplies free energy

Unit ledger
The phase-space density $f_{s,0}$ is in $\mathrm{m}^{-6}\,\mathrm{s}^3$. The ratios $kc/\omega_{p,s}$, $\omega/\omega_{p,s}$, and v_x^2/c^2 are dimensionless.

Unmagnetized transverse plane wave — Assumptions 371

Use a homogeneous collisionless plasma with $\boldsymbol{B}_0 = \boldsymbol{0}$ and a small wave with $\boldsymbol{k} = k\boldsymbol{e}_z$ and $\boldsymbol{E}_1 = E_{1,x}\boldsymbol{e}_x$. Then $\boldsymbol{k}\cdot\boldsymbol{E}_1 = 0$ and the wave magnetic field is determined by Faraday's law. The equilibrium can be isotropic for the ordinary branch or anisotropic when testing a free-energy-driven instability. Require reflection symmetry separately in v_x and v_y with these axes as principal directions. Then the response driven by $E_{1,x}$ is odd in v_x and even in v_y, so its density, $J_{1,y}$, and $J_{1,z}$ vanish. This decouples the scalar transverse dispersion; a general anisotropic equilibrium requires the full dielectric matrix.

Transverse polarization — Definition 372

A transverse mode has its electric field perpendicular to the propagation direction. With the stated geometry, Faraday's law gives $\boldsymbol{B}_1 = (k/\omega)E_{1,x}\boldsymbol{e}_y$. The perturbed current is obtained from the first velocity moment of $f_{s,1}$, so the kinetic dielectric response depends on the full equilibrium distribution rather than on density and temperature alone.

Transverse kinetic dispersion relation — Governing law 373

For a species equilibrium $f_{s,0}(\boldsymbol{v})$, the transverse normal modes satisfy

$$k^2c^2 = \omega^2 + \sum_s \frac{q_s^2}{\varepsilon_0 m_s}\int_{\mathbb{R}^3}\left[v_x\frac{\partial f_{s,0}(\boldsymbol{v})}{\partial v_x} + \frac{kv_x^2}{\omega - kv_z}\frac{\partial f_{s,0}(\boldsymbol{v})}{\partial v_z}\right]\mathrm{d}^3\boldsymbol{v}.$$

If the distribution decays at velocity-space infinity, integration by parts gives the equivalent form

$$k^2c^2 = \omega^2 - \sum_s \omega_{p,s}^2 - \sum_s \frac{q_s^2k^2}{\varepsilon_0 m_s}\int_{\mathbb{R}^3}\left[\frac{v_x^2 f_{s,0}(\boldsymbol{v})}{(\omega - kv_z)^2}\right]\mathrm{d}^3\boldsymbol{v},$$

where $\omega_{p,s}^2 = n_s q_s^2/(\varepsilon_0 m_s)$. The velocity integral uses the causal contour when the pole $\omega - kv_z = 0$ lies on the integration path.

Derivation: transverse current and cold limit

Step: Linearize and solve the transverse Vlasov equation

For the Fourier convention $\exp(i(kz - \omega t))$, the linearized Vlasov equation for the transverse geometry is

$$-i(\omega - kv_z)f_{s,1} + \left(\frac{q_s}{m_s}\right)E_{1,x}\left[\left(1 - \frac{kv_z}{\omega}\right)\frac{\partial f_{s,0}}{\partial v_x} + \frac{kv_x}{\omega}\frac{\partial f_{s,0}}{\partial v_z}\right] = 0.$$

Solving for the distribution perturbation gives

$$f_{s,1} = \frac{-iq_sE_{1,x}}{m_s\omega}\frac{\left[(\omega - kv_z)\frac{\partial f_{s,0}}{\partial v_x} + kv_x\frac{\partial f_{s,0}}{\partial v_z}\right]}{\omega - kv_z}.$$

Step: Insert the distribution into the current

The transverse current is

$$J_{1,x} = \sum_s q_s \int_{\mathbb{R}^3} v_x f_{s,1} \mathrm{d}^3\boldsymbol{v}.$$

Substitution into the transverse component of Ampere's law gives

$$k^2c^2 = \omega^2 + \sum_s \frac{q_s^2}{\varepsilon_0 m_s} \int_{\mathbb{R}^3} \left[v_x \frac{\partial f_{s,0}}{\partial v_x} + \frac{k v_x^2}{\omega - k v_z} \frac{\partial f_{s,0}}{\partial v_z} \right] \mathrm{d}^3\boldsymbol{v}.$$

Step: Integrate the velocity derivative by parts

To integrate the second term by parts, use

$$\frac{\partial \frac{1}{\omega - k v_z}}{\partial v_z} = \frac{k}{(\omega - k v_z)^2}.$$

The boundary terms vanish for a decaying distribution. The two needed identities are

$$\int_{\mathbb{R}^3} v_x \frac{\partial f_{s,0}}{\partial v_x} \mathrm{d}^3\boldsymbol{v} = -n_s$$

and

$$\int_{\mathbb{R}^3} \frac{k v_x^2}{\omega - k v_z} \frac{\partial f_{s,0}}{\partial v_z} \mathrm{d}^3\boldsymbol{v} = -\int_{\mathbb{R}^3} \left[\frac{k^2 v_x^2 f_{s,0}}{(\omega - k v_z)^2} \right] \mathrm{d}^3\boldsymbol{v}.$$

These identities produce the equivalent governing law

$$k^2c^2 = \omega^2 - \sum_s \omega_{p,s}^2 - \sum_s \frac{q_s^2 k^2}{\varepsilon_0 m_s} \int_{\mathbb{R}^3} \left[\frac{v_x^2 f_{s,0}}{(\omega - k v_z)^2} \right] \mathrm{d}^3\boldsymbol{v}.$$

Step: Recover the cold electromagnetic branch

For a cold equilibrium

$$f_{s,0} = n_s \delta(v_x) \delta(v_y) \delta(v_z),$$

the resonant integral containing v_x^2 vanishes in the distributional limit. Therefore

$$k^2c^2 = \omega^2 - \sum_s \omega_{p,s}^2,$$

or

$$\omega^2 = k^2c^2 + \sum_s \omega_{p,s}^2.$$

This is the cold electromagnetic cutoff already obtained from the fluid model.

For an electron population with fixed neutralizing ions, take an equilibrium of the form $f_0(\boldsymbol{v}) = \delta(v_z)F(v_x, v_y)$ with

$$n_0 = \int_{\mathbb{R}^2} F(v_x, v_y)\,\mathrm{d}v_x\,\mathrm{d}v_y \quad \text{and} \quad \langle v_x^2 \rangle = \left(\frac{1}{n_0}\right)\int_{\mathbb{R}^2} v_x^2 F(v_x, v_y)\,\mathrm{d}v_x\,\mathrm{d}v_y.$$

The transverse relation becomes $k^2c^2 = \omega^2 - \omega_p^2\left[1 + k^2\langle v_x^2\rangle/\omega^2\right]$. Equivalently, $\omega^4 - \left(k^2c^2 + \omega_p^2\right)\omega^2 - k^2\omega_p^2\langle v_x^2\rangle = 0$. The lower value of ω^2 is negative whenever $k \neq 0$ and $\langle v_x^2\rangle > 0$, so this idealized anisotropy contains a growing transverse mode. The instability is powered by the anisotropic velocity-space free energy.

Derivation: the negative transverse branch

Step: Evaluate the kinetic correction

For the stated equilibrium, the pole denominator is independent of the nonzero perpendicular velocities because $v_z = 0$. The integrated form of the dispersion relation therefore gives

$$\int_{\mathbb{R}^3}\left[\frac{v_x^2 f_0(\boldsymbol{v})}{(\omega - kv_z)^2}\right]\mathrm{d}^3\boldsymbol{v} = n_0\frac{\langle v_x^2\rangle}{\omega^2}.$$

Inserting this result together with

$$\omega_p^2 = \frac{n_0 e^2}{\varepsilon_0 m_e}$$

yields the quartic governing the transverse branch.

Step: Solve the quartic as a quadratic in frequency squared

Set $Y = \omega^2$. The two roots are

$$Y_\pm = \frac{\left[\left(k^2c^2 + \omega_p^2\right) \pm \sqrt{\left(k^2c^2 + \omega_p^2\right)^2 + 4k^2\omega_p^2\langle v_x^2\rangle}\right]}{2}.$$

Since the square root is larger than $k^2c^2 + \omega_p^2$ when $k \neq 0$ and $\langle v_x^2\rangle > 0$, the lower root satisfies $Y_- < 0$.

Step: Convert the negative root to a growth rate

Writing $\omega = i\gamma$ on the growing member gives

$$\gamma^2 = \frac{\left[\sqrt{\left(k^2c^2 + \omega_p^2\right)^2 + 4k^2\omega_p^2\langle v_x^2\rangle} - \left(k^2c^2 + \omega_p^2\right)\right]}{2}.$$

Thus the idealized anisotropy produces exponential growth. A finite parallel temperature, collisions, magnetic field, or nonlinear redistribution changes this branch and must be included before applying the formula to a particular plasma.

Rechenbeispiel Rechenbeispiel 375

Use an electron plasma with $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m}^{-1}$, and $c = 2.998 \cdot 10^{8}\,\mathrm{m\,s}^{-1}$. Let $kc/\omega_p = 0.50$ [1] and $\langle v_x^2 \rangle / c^2 = 1.00 \cdot 10^{-2}$ [1]. Determine k, the wavelength, and the positive growth rate of the anisotropic transverse branch.

Numerical result: $\omega_p \approx 5.64 \cdot 10^{9}\,\mathrm{s}^{-1}$, $k \approx 9.41\,\mathrm{m}^{-1}$, $\lambda \approx 0.668\,\mathrm{m}$, $\gamma/\omega_p \approx 4.47 \cdot 10^{-2}$ [1], and $\gamma \approx 2.52 \cdot 10^{8}\,\mathrm{s}^{-1}$.

Velocity spread in the transverse response Interpretation 376

The cold cutoff depends only on the total plasma frequency. The kinetic correction samples perpendicular velocity spread and the pole at the parallel phase velocity. An isotropic distribution usually leaves the ordinary high-frequency branch stable in the nonrelativistic model, while anisotropy can release free energy into a transverse perturbation.

Summary Summary 377

A transverse kinetic wave follows from the Vlasov response together with Maxwell's equations. Its cold limit is $\omega^2 = k^2c^2 + \sum_s \omega_{p,s}^2$; finite velocity spread adds a resonant integral. An anisotropic distribution can make the lower transverse branch have $\omega^2 < 0$, producing growth from velocity-space free energy.

Knowledge check

- What distinguishes a transverse kinetic wave from the longitudinal calculation?
- Which limit gives the cold electromagnetic cutoff?
- Why does the transverse kinetic response contain a squared resonant denominator after integration by parts?
- What supplies the free energy in the anisotropic transverse example?

Answers

What distinguishes a transverse kinetic wave from the longitudinal calculation?
The electric field is perpendicular to $\boldsymbol{k}$, so the wave magnetic field contributes to the Lorentz force and the current must be coupled to Maxwell's transverse wave equation rather than only to Poisson's equation.

Which limit gives the cold electromagnetic cutoff?
Taking each equilibrium toward a cold velocity-space delta distribution removes the finite-velocity integral and gives $\omega^2 = k^2c^2 + \sum_s \omega_{p,s}^2$.

Why does the transverse kinetic response contain a squared resonant denominator after integration by parts?
Integration by parts moves the velocity derivative from the distribution to $1/(\omega - kv_z)$, producing

$$\frac{k}{(\omega - kv_z)^2}.$$

This is the same response in an equivalent form; it retains the original causal contour prescription.

What supplies the free energy in the anisotropic transverse example?
The unequal velocity-space spread, represented here by a finite perpendicular second moment and a cold parallel direction, stores free energy that can be transferred to the growing transverse field.

15.5 Two-stream instability

A distribution with two separated beams is unstable to electrostatic perturbations. The electrostatic field couples the streams, and for a range of wavelengths the two real oscillatory branches merge into a complex pair. The negative value of ω^2 on the lower branch is the algebraic signal of exponential growth.

Learning objectives

- construct the cold symmetric two-stream distribution
- derive its longitudinal dispersion relation and solve for ω^2
- identify the unstable wave-number interval
- locate the maximum normalized growth rate

Unit ledger

The total density n_0 is in m^{-3}, beam speed v_0 and the light speed c are in $\mathrm{m\,s}^{-1}$, k is in m^{-1}, and ω,

γ, and ω_p are in s^{-1}. The wavelength is in m. The normalized variables kv_0/ω_p, ω/ω_p, and γ/ω_p are dimensionless.

Cold symmetric electron beams — Assumptions 379

Use two equal electron beams with densities $n_0/2$ and velocities $+v_0$ and $-v_0$ along $\boldsymbol{e}_z$. Ions form a fixed neutralizing background, the plasma is unmagnetized and collisionless, and the wave is one-dimensional and electrostatic. The total electron plasma frequency is $\omega_p^2 = n_0 e^2/(\varepsilon_0 m_e)$.

Two-stream equilibrium — Definition 380

The one-dimensional equilibrium can be written $F_{e,0}(v) = (n_0/2)[\delta(v - v_0) + \delta(v + v_0)]$, the marginal along the beams, with units $\mathrm{m}^{-4}\,\mathrm{s}$ and

$$\int F_{e,0}(v)\,\mathrm{d}v = n_0.$$

The delta functions idealize cold beams. A finite beam temperature replaces them by narrow distributions and introduces thermal spreading, resonant damping, and additional kinetic structure.

Cold two-stream dispersion — Governing law 381

With $x = \left(kv_0/\omega_p\right)^2$, the dispersion relation is

$$D(\omega, k) = 1 - \frac{\omega_p^2}{2(\omega - kv_0)^2} - \frac{\omega_p^2}{2(\omega + kv_0)^2} = 0.$$

Its two values of ω^2 are

$$\frac{\omega^2}{\omega_p^2} = x + \frac{1}{2} \pm \frac{\sqrt{1 + 8x}}{2}.$$

The lower branch is negative when $0 < x < 1$, so

$$\frac{\gamma}{\omega_p} = \left[\frac{\sqrt{1 + 8x}}{2} - x - \frac{1}{2}\right]^{\frac{1}{2}}$$

in that interval. The maximum occurs at $x = 3/8$ and has $\gamma_{\max}/\omega_p = 1/\left(2\sqrt{2}\right)$.

Derivation: symmetric beams and the unstable branch

Step: Write the response of the two cold beams

For the two delta-function beams, insert the equilibrium into the first-derivative form of the electrostatic susceptibility. Equivalently, each cold beam responds as a cold fluid whose equilibrium drift shifts the frequency to $\omega - kv_0$ or $\omega + kv_0$. The total response is

$$D(\omega, k) = 1 - \frac{\omega_p^2}{2(\omega - kv_0)^2} - \frac{\omega_p^2}{2(\omega + kv_0)^2}.$$

Step: Clear the beam denominators

Multiply by $\left(\omega^2 - k^2 v_0^2\right)^2$ and collect powers of ω^2:

$$\omega^4 - \left(2k^2 v_0^2 + \omega_p^2\right)\omega^2 + k^2 v_0^2\left(k^2 v_0^2 - \omega_p^2\right) = 0.$$

Define the dimensionless variables

$$x = \frac{k^2 v_0^2}{\omega_p^2}, \quad y = \frac{\omega^2}{\omega_p^2}.$$

Dividing by ω_p^4 gives

$$y^2 - (2x + 1)y + x(x - 1) = 0.$$

The quadratic formula yields

$$y_\pm = x + \frac{1}{2} \pm \frac{\sqrt{1 + 8x}}{2}.$$

Step: Identify the unstable interval

The upper branch is positive for all $x \geq 0$. For the lower branch, $y_- < 0$ precisely when

$$\sqrt{1 + 8x} > 1 + 2x.$$

Both sides are nonnegative, so squaring gives

$$1 + 8x > 1 + 4x + 4x^2,$$

or $0 < x < 1$. In that interval write $\omega = i\gamma$ and obtain

$$\frac{\gamma}{\omega_p} = \sqrt{-y_-}.$$

Step: Locate the maximum growth rate

To locate the maximum, differentiate

$$\frac{\gamma^2}{\omega_p^2} = \frac{\sqrt{1 + 8x}}{2} - x - \frac{1}{2}.$$

The derivative vanishes when

$$\frac{2}{\sqrt{1 + 8x}} = 1,$$

so $x = 3/8$. Substitution gives

$$\frac{\gamma^2}{\omega_p^2} = \frac{1}{8},$$

and therefore

$$\frac{\gamma_{\max}}{\omega_p} = \frac{1}{2\sqrt{2}}.$$

The instability draws on the relative drift energy. Beam temperature, collisions, boundaries, and nonlinear trapping all modify this ideal result.

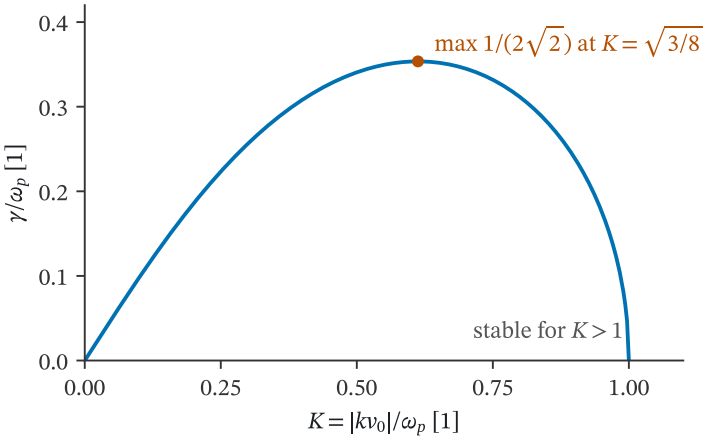
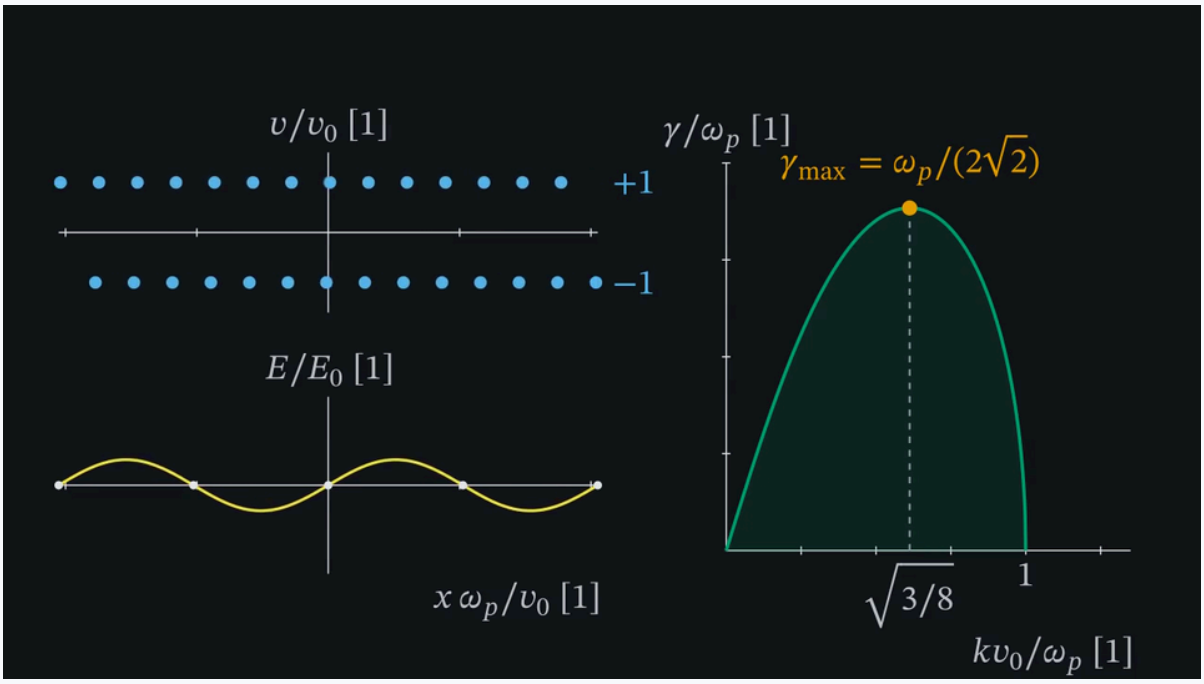


Figure 85: Cold symmetric equal-density electron beams with immobile ions; ω_p uses the total electron density and $K = |kv_0|/\omega_p$. The unstable root gives $\frac{\gamma}{\omega_p} = \sqrt{\frac{\sqrt{1+8K^2}-1-2K^2}{2}}$ for $0 < K < 1$, with the maximum $1/\left(2\sqrt{2}\right)$ at $K = \sqrt{3/8}$.



Watch animation

Counter-streaming beams can convert directed kinetic energy into an electrostatic wave. The growth curve is the cold symmetric two-stream result in normalized variables. The field uses $X = x\omega_p/v_0$, $\tau = \omega_p t$, and $E_0 = m_e v_0 \omega_p / e$, with ω_p based on total beam density. At $K = kv_0/\omega_p = \sqrt{3/8}$ it is $\frac{E}{E_0} = 0.01 \exp\left(\frac{\tau}{2\sqrt{2}}\right) \sin(KX)$: its real frequency is zero.

Figure 86: Counter-streaming beams can convert directed kinetic energy into an electrostatic wave. The growth curve is the cold symmetric two-stream result in normalized variables. The field uses $X = x\omega_p/v_0$, $\tau = \omega_p t$, and $E_0 = m_e v_0 \omega_p / e$, with ω_p based on total beam density. At $K = kv_0/\omega_p = \sqrt{3/8}$ it is $\frac{E}{E_0} = 0.01 \exp\left(\frac{\tau}{2\sqrt{2}}\right) \sin(KX)$: its real frequency is zero.

Rechenbeispiel Rechenbeispiel 382

Let $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$, $v_0/c = 0.10$ [1] with $c = 2.998 \cdot 10^8\,\mathrm{m\,s^{-1}}$, and choose $kv_0/\omega_p = 0.50$ [1]. Use $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$, $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, and $\varepsilon_0 = 8.854 \cdot 10^{-12}\,\mathrm{F\,m^{-1}}$. Determine k, the wavelength, and the positive growth rate.

Numerical result: $\omega_p \approx 5.64 \cdot 10^9\,\mathrm{s^{-1}}$, $k \approx 94.1\,\mathrm{m^{-1}}$, $\lambda \approx 6.68 \cdot 10^{-2}\,\mathrm{m}$, $\gamma/\omega_p \approx 0.341$ [1], and $\gamma \approx 1.92 \cdot 10^9\,\mathrm{s^{-1}}$.

A negative squared frequency gives growth and decay Interpretation 383

At short wavelengths the two beams support ordinary oscillatory roots. In the unstable band, the lower root crosses through $\omega = 0$ and becomes a purely imaginary pair. One member grows and the other decays; the physical initial-value problem selects their combination from the initial perturbation.

Summary

Summary 384

The cold symmetric two-stream model has a finite unstable band $0 < |kv_0|/\omega_p < 1$. The maximum growth rate is $\omega_p/\left(2\sqrt{2}\right)$, reached at $|kv_0|/\omega_p = \sqrt{3/8}$. Finite temperature and kinetic phase mixing determine how this ideal beam instability is modified.

Knowledge check

Knowledge check 385

- What supplies the free energy in the two-stream instability?
- Why is the unstable interval finite in wave number?
- What does a positive imaginary frequency mean here?
- Which idealization is most directly relaxed by giving the beams a finite temperature?

Answers

What supplies the free energy in the two-stream instability?

The relative directed kinetic energy of the counter-streaming beams supplies free energy. The electrostatic perturbation extracts it when the beam response makes the lower branch have negative ω^2.

Why is the unstable interval finite in wave number?

For $|kv_0|/\omega_p > 1$, both squared-frequency branches are positive. The lower branch reaches zero at the boundary and becomes negative only for $0 < |kv_0|/\omega_p < 1$. Thus the boundary itself is marginal, not oscillatory.

What does a positive imaginary frequency mean here?

With $\exp(-i\omega t)$, a root $\omega = i\gamma$ with $\gamma > 0$ gives $\exp(\gamma t)$, so its amplitude grows exponentially in time.

Which idealization is most directly relaxed by giving the beams a finite temperature?

The delta-function velocity distributions become narrow but smooth profiles. Thermal spread then introduces phase mixing and resonant corrections that can reduce, shift, or reshape the cold growth band.

15.6 Hot magnetized waves and branch interpretation

A magnetic field adds another way for particles to stay in phase with a perturbation. They stream along the field while gyrating across it, so the resonance condition contains both a Doppler shift and an integer cyclotron harmonic. Cold-fluid branches provide reference limits for interpreting the kinetic spectrum, while Landau and cyclotron resonances explain how velocity spread changes the response.

Learning objectives

- state the equilibrium condition for a gyrotropic distribution
- expand the perturbed distribution into gyroangle harmonics
- derive the Doppler-shifted cyclotron resonance condition
- connect hot magnetized branch changes to velocity-space gradients

Unit ledger
For a characteristic perpendicular speed $v_{\perp,s}$, define the thermal gyroradius $\rho_s = v_{\perp,s}/|\Omega_s|$, which is in m. Harmonic number n, $k_\perp \rho_s$, $\omega/|\Omega_s|$, and the resonance condition after division by a frequency are dimensionless. The individual orbit radius $r_{L,s} = v_\perp/|\Omega_s|$ is in m and varies across the velocity integral; it differs from the thermal scale ρ_s.

Homogeneous gyrotropic magnetized equilibrium — Assumptions 386

Use a collisionless homogeneous plasma with $\boldsymbol{B}_0 = B_0 \boldsymbol{e}_z$ and an equilibrium $f_{s,0}(v_\parallel, v_\perp)$ independent of gyroangle. Begin with parallel propagation to expose the harmonic denominators; finite $k_\perp \rho_s$ introduces Bessel factors and finite-Larmor-radius effects. For thermal estimates, take $v_{\perp,s}$ to be the characteristic perpendicular thermal speed of species s.

Gyroangle harmonic response — Definition 387

Define the signed gyrofrequency $\Omega_s = q_s B_0/m_s$. Choose $v_x = v_\perp \cos(\theta)$ and $v_y = -v_\perp \sin(\theta)$, so increasing θ rotates from positive x toward negative y and $\mathrm{d}\theta/\mathrm{d}t = \Omega_s$. Thus $\theta = -\theta_{\mathrm{std}}$, where the glossary's usual counterclockwise angle has $v_y = v_\perp \sin(\theta_{\mathrm{std}})$ and $\mathrm{d}\theta_{\mathrm{std}}/\mathrm{d}t = -\Omega_s$. These are two coordinates for the same physical orbit. Write the perturbed distribution as

$$f_{s,1} = \sum_{n=-\infty}^{\infty} f_{s,1,n}(v_\parallel, v_\perp) \exp(in\theta).$$

The integer n counts the number of gyroangle phase windings sampled by the perturbation. The harmonic $n = 0$ is the parallel Landau resonance; nonzero n are cyclotron harmonics.

Doppler-shifted cyclotron resonance — Governing law 388

The hot magnetized response contains denominators of the form $\omega - k_\parallel v_\parallel - n\Omega_s$. Resonant particles satisfy $\omega_r - k_\parallel v_\parallel - n\Omega_s = 0$, so $v_{\parallel,\mathrm{res}} = (\omega_r - n\Omega_s)/k_\parallel$ when $k_\parallel \neq 0$. The harmonic weights depend on the equilibrium velocity-space gradients and, for oblique propagation, on the signed orbit argument $a_s = k_\perp v_\perp/\Omega_s$ through Bessel functions J_n of the first kind, with $|a_s| = k_\perp r_{L,s}$. These weights describe how strongly an orbit couples to each harmonic. A branch can therefore be damped, amplified, or strongly refracted near a resonance even when its cold-fluid counterpart is smooth.

Derivation: gyroangle harmonics and resonant denominators

Step: Characterize the unperturbed gyro-orbit

The unperturbed characteristic equations in a uniform field are

$$\frac{\mathrm{d}v_\parallel}{\mathrm{d}t} = 0, \quad \frac{\mathrm{d}v_\perp}{\mathrm{d}t} = 0, \quad \frac{\mathrm{d}\theta}{\mathrm{d}t} = \Omega_s.$$

Indeed, differentiating $v_x = v_\perp \cos(\theta)$ and $v_y = -v_\perp \sin(\theta)$ gives $\mathrm{d}v_x/\mathrm{d}t = \Omega_s v_y$ and $\mathrm{d}v_y/\mathrm{d}t = -\Omega_s v_x$, the Lorentz-force components $\mathrm{d}\boldsymbol{v}/\mathrm{d}t = (q_s/m_s)\boldsymbol{v} \times \boldsymbol{B}_0$.

A stationary gyrotropic equilibrium therefore satisfies

$$\Omega_s \frac{\partial f_{s,0}}{\partial \theta} = 0,$$

and can depend on $v_\parallel$ and $v_\perp$ but not on θ.

Step: Expand the perturbed distribution in harmonics

Linearize the Vlasov equation about this equilibrium and write the perturbed distribution as a Fourier harmonic in time, space, and gyroangle. The streaming and gyroangle pieces of the operator become

$$\frac{\partial f}{\partial t} + v_\parallel \frac{\partial f}{\partial z} + \Omega_s \frac{\partial f}{\partial \theta} \to -i\big(\omega - k_\parallel v_\parallel - n\Omega_s\big) f.$$

Define $S_{s,n}$ as the harmonic of the negative perturbed-force term, so $-i\big(\omega - k_\parallel v_\parallel - n\Omega_s\big) f_{s,1,n} = S_{s,n}$. Its response is

$$f_{s,1,n} = \frac{iS_{s,n}}{\big[\omega - k_\parallel v_\parallel - n\Omega_s\big]}.$$

Step: Identify the parallel and cyclotron harmonics

For parallel propagation, the transverse electric field decomposes into circular polarizations. The $n = +1$ and $n = -1$ terms are selected by the corresponding sense of gyromotion, while $n = 0$ describes parallel motion. In chapter 13's circular basis $E_y = -i\sigma E_x$ with $\sigma = \pm 1$, $\boldsymbol{E}_\perp \cdot \boldsymbol{v}_\perp = E_x v_\perp \exp(i\sigma\theta)$. Thus that circular forcing selects $n = \sigma$; for electrons the positive-frequency cyclotron pole has $n = -1$ because $\Omega_e < 0$. The resonance condition is therefore

$$\omega - k_\parallel v_\parallel - n\Omega_s = 0.$$

Step: Add finite-Larmor-radius weights

For oblique propagation take $\boldsymbol{k}_\perp = k_\perp \boldsymbol{e}_x$. Integrating $\mathrm{d}x/\mathrm{d}t = v_\perp \cos(\theta)$ with $\mathrm{d}\theta/\mathrm{d}t = \Omega_s$ gives $x = X + (v_\perp/\Omega_s)\sin(\theta)$, where X is the guiding-center coordinate. The perpendicular phase factor along this individual orbit therefore expands as

$$\exp(ia_s \sin(\theta)) = \sum_n J_n(a_s) \exp(in\theta), \quad a_s = \frac{k_\perp v_\perp}{\Omega_s}.$$

Finite-Larmor-radius physics thus supplies Bessel weights but leaves the same Doppler-shifted cyclotron denominator. These weights remain inside the velocity integral. The thermal $k_\perp \rho_s$ estimates their importance but cannot replace the individual velocity-dependent argument.

Step: Interpret the contour contribution

The contour prescription around each pole gives an imaginary contribution proportional to the appropriate derivative of the equilibrium. In an isotropic Maxwellian the parallel $n = 0$ resonance gives Landau damping on the usual positive-frequency branch. An anisotropic distribution can have a gradient of the opposite sign, and cyclotron harmonics can then drive growth or produce strong absorption. Sending $B_0 \to 0$ removes the harmonic spacing and recovers the unmagnetized velocity-space response.

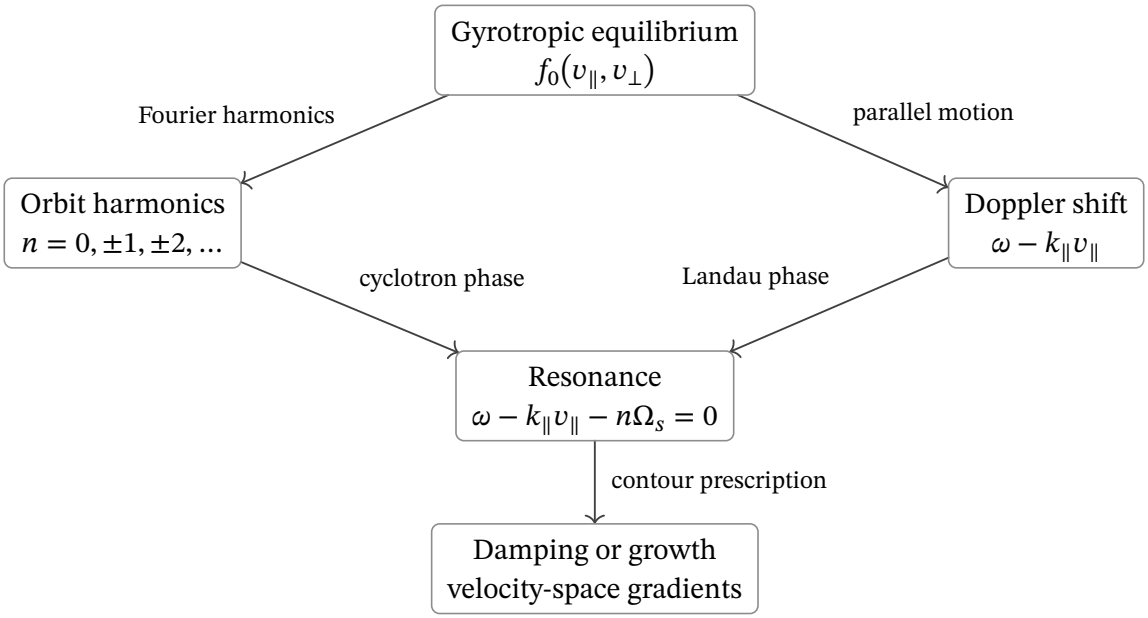


Figure 87: In a hot magnetized plasma, particle orbits create a family of resonances: parallel Landau resonance for harmonic zero and cyclotron resonances for nonzero harmonic number. The branch can be damped or driven by the local velocity-space gradients.

Rechenbeispiel Rechenbeispiel 389

For electrons in a $B_0 = 1.0 \cdot 10^{-2}\,\mathrm{T}$ field use $\omega_{ce} = 1.76 \cdot 10^{9}\,\mathrm{s}^{-1}$ as the positive electron gyrofrequency magnitude and $\Omega_e = -\omega_{ce}$. Let $\omega/\omega_{ce} = 0.80$ [1] and $k_\parallel v_{te}/\omega_{ce} = 1.50$ [1], with $v_{te} = 1.88 \cdot 10^{6}\,\mathrm{m\,s}^{-1}$. Report the resonant parallel velocities for $n = 0$ and $n = -1$, normalized by v_{te}, and give $k_\parallel$ and ω.

Numerical result: for $n = 0$, $v_{\parallel,\mathrm{res}}/v_{te} = 0.533$ [1] for $n = -1$, $v_{\parallel,\mathrm{res}}/v_{te} = -0.133$ [1] $k_\parallel \approx 1.40 \cdot 10^{3}\,\mathrm{m}^{-1}$ and $\omega \approx 1.41 \cdot 10^{9}\,\mathrm{s}^{-1}$.

Cold branches as reference limits Interpretation 390

The cold dielectric tensor identifies the principal wave branches and their polarizations. The hot tensor adds orbit averaging, Doppler shifts, harmonic resonances, and distribution gradients. A cold cutoff or resonance is therefore a location at which kinetic corrections should be checked first.

Summary Summary 391

Hot magnetized waves inherit their cold branch structure but acquire resonant denominators $\omega - k_\parallel v_\parallel - n\Omega_s$. The $n = 0$ harmonic is Landau-like, nonzero harmonics are cyclotron-like, and the sign of the velocity-space gradient determines damping or growth. Oblique propagation additionally weights the harmonics through the individual orbit argument $k_\perp v_\perp/\Omega_s$; the thermal $k_\perp \rho_s$ is only an ordering scale.

Knowledge check

- What is the difference between a signed gyrofrequency and its positive magnitude?
- Which harmonic corresponds to the parallel Landau resonance?
- Why do Bessel functions appear for oblique hot-plasma waves?
- What can make a cold magnetized resonance physically finite?

Answers

What is the difference between a signed gyrofrequency and its positive magnitude?

The positive magnitude $\omega_c = |q|B/m$ is a rate, while the signed $\Omega = qB/m$ retains the charge-dependent sense of gyromotion and enters the harmonic resonance with its sign.

Which harmonic corresponds to the parallel Landau resonance?

The $n = 0$ harmonic removes the cyclotron shift and gives $\omega - k_\parallel v_\parallel = 0$. Nonzero n add gyrofrequency multiples and are cyclotron resonances.

Why do Bessel functions appear for oblique hot-plasma waves?

Each perpendicular orbit samples the oblique phase periodically. Its harmonic expansion gives Bessel weights with signed argument $k_\perp v_\perp/\Omega_s$, whose magnitude is $k_\perp r_{L,s}$. The thermal radius only estimates the ordering; the orbit radius varies over velocity space.

What can make a cold magnetized resonance physically finite?

Finite temperature, orbit averaging, collisions, and the analytic contour contribution spread or regularize the ideal cold response. The correct mechanism is selected by the relevant ordering and distribution.

16. Plasma sheaths and Langmuir probes

Electrons and ions arrive at a material surface with different thermal speeds, so the surface charges until an electric field filters the particle fluxes. This chapter derives the planar collisionless sheath model, obtains the Bohm entry condition and floating potential, and turns the same flux picture into a Langmuir-probe diagnostic, following the boundary treatment in [3].

A sheath is a self-consistent boundary condition — Key idea 393

At a wall, the surface potential, the directed ion flow, the electron distribution, and Poisson's equation must agree. A probe measures the resulting current–voltage response, not a local density or temperature in isolation.

16.1 Particle flux and the need for a sheath

A surface collects the particles moving toward it, so the incoming flux is a one-sided velocity moment. At comparable electron and ion temperatures, the electron thermal speed is much larger, and an initially uncharged wall receives a much larger electron current. The resulting negative charging creates the barrier that defines a sheath.

Learning objectives

- define directed particle flux as a half-space velocity moment
- derive the isotropic Maxwellian flux to a planar surface
- compare electron and ion flux scales without confusing flux and density
- explain why a wall potential must form before steady current balance

Unit ledger

The directed particle flux Γ_s is in $\mathrm{m^{-2}\,s^{-1}}$, so a current density $J_s = q_s\Gamma_s$ is in $\mathrm{A\,m^{-2}}$. Ratios such as Γ_e/Γ_i and $e\Phi/(k_B T_e)$ are dimensionless.

Isotropic distribution next to a planar surface — Assumptions 394

Take a locally planar boundary with normal $\boldsymbol{e}_n$ directed from the plasma into the material and a distribution that is isotropic on the plasma side. The surface is collisionless over the collection distance, absorbs particles that reach it, and has no secondary emission in the first flux estimate. This estimate is a boundary flux.

Directed particle flux Definition 395

The number of particles of species s crossing a unit area toward the surface is

$$\Gamma_s = \int_{v_n>0} v_n f_s(\boldsymbol{v}) \mathrm{d}^3\boldsymbol{v},$$

where $v_n = \boldsymbol{v} \cdot \boldsymbol{e}_n$. For an isotropic distribution, the tangential velocity integrals can be performed first, leaving the positive half of the normal-velocity distribution.

Maxwellian half-space flux Governing law 396

For a three-dimensional Maxwellian with density n_s and temperature T_s, the unretarded flux toward a plane is

$$\Gamma_{s,0} = n_s \sqrt{\frac{k_B T_s}{2\pi m_s}} = \frac{n_s v_{\mathrm{th,s}}}{2\sqrt{\pi}} = \frac{n_s v_{\mathrm{mean,s}}}{4},$$

where $v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$ and $v_{\mathrm{mean,s}} = \sqrt{8k_B T_s/(\pi m_s)}$. At equal densities and temperatures, $\Gamma_{e,0}/\Gamma_{i,0} = \sqrt{m_i/m_e}$. Equal particle densities therefore do not imply equal arrival rates.

Derivation: the one-sided Maxwellian flux

Step: Write the isotropic distribution

Choose the surface normal as the z direction. The three-dimensional Maxwellian is

$$f_s(\boldsymbol{v}) = n_s \left(\frac{m_s}{2\pi k_B T_s}\right)^{\frac{3}{2}} \exp\left(-\frac{m_s\left(v_x^2 + v_y^2 + v_z^2\right)}{2k_B T_s}\right).$$

Insert it into the half-space flux definition:

$$\Gamma_{s,0} = \int_0^\infty \int_{-\infty}^\infty \int_{-\infty}^\infty v_z f_s(\boldsymbol{v})\, \mathrm{d}v_x\, \mathrm{d}v_y\, \mathrm{d}v_z.$$

Step: Evaluate the tangential integrals

Each tangential Gaussian integral is

$$\int_{-\infty}^\infty \exp\left(-\frac{m_s v_x^2}{2k_B T_s}\right) \mathrm{d}v_x = \sqrt{\frac{2\pi k_B T_s}{m_s}}.$$

Evaluating both tangential integrals leaves

$$\Gamma_{s,0} = n_s \sqrt{\frac{m_s}{2\pi k_B T_s}} \int_0^\infty v_z \exp\left(-\frac{m_s v_z^2}{2k_B T_s}\right) \mathrm{d}v_z.$$

Step: Evaluate the normal half-space integral

Use

$$\int_0^\infty v \exp(-a v^2)\,\mathrm{d}v = \frac{1}{2a}, \quad a = \frac{m_s}{2k_B T_s}.$$

The normal integral is $k_B T_s/m_s$, so

$$\Gamma_{s,0} = n_s \sqrt{\frac{k_B T_s}{2\pi m_s}}.$$

With $v_{\mathrm{th,s}} = \sqrt{2k_B T_s/m_s}$, this becomes

$$\Gamma_{s,0} = \frac{n_s v_{\mathrm{th,s}}}{2\sqrt{\pi}}.$$

The mean speed of the same Maxwellian is

$$v_{\mathrm{mean,s}} = \sqrt{\frac{8k_B T_s}{\pi m_s}},$$

which gives the equivalent form

$$\Gamma_{s,0} = \frac{n_s v_{\mathrm{mean,s}}}{4}.$$

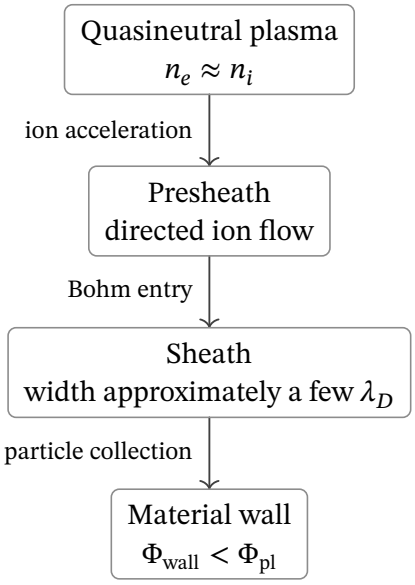


Figure 88: A planar boundary is organized into a quasineutral plasma, a presheath that supplies directed ion flow, a charge-separated sheath of order a few Debye lengths, and the material wall. The labels describe model regions.

Rechenbeispiel Rechenbeispiel 397

Consider a hydrogen plasma with equal electron and ion temperatures and $n_e = n_i$. Let $m_i/m_e = 1836$ [1]. Determine the ratio of the unretarded electron and ion fluxes to the same planar surface.

Numerical result: $\dfrac{\Gamma_{e,0}}{\Gamma_{i,0}} = \sqrt{1836} \approx 42.8$ [1].

Flux imbalance charges the boundary Interpretation 398

At equal density and temperature, electrons arrive faster by the square root of the ion-to-electron mass ratio. Electrons lost by the plasma are gained by the wall, leaving a negative wall charge. Its potential repels further electrons and attracts ions until the net current is compatible with the electrical boundary condition. The density equality in the plasma therefore does not imply equal wall fluxes.

Summary Summary 399

A surface samples a half-space moment, not the total thermal density. An isotropic Maxwellian supplies $\Gamma_{s,0} = n_s\sqrt{k_B T_s/(2\pi m_s)}$. The much larger electron flux charges an initially neutral wall negatively, creating the electrostatic sheath required for steady current balance.

Knowledge check Knowledge check 400

- Why is the flux to a plane not $n_s v_{\mathrm{mean,s}}$?
- Why can equal electron and ion densities still produce unequal wall currents?
- What physical process starts the formation of a floating sheath?
- Which assumption makes the half-space flux formula a local estimate rather than a sheath solution?

Answers

Why is the flux to a plane not $n_s v_{\mathrm{mean,s}}$?
Only the velocity component normal to the plane contributes, and only particles with positive normal velocity cross the surface. The angular average and half-space restriction together give $\Gamma_{s,0} = n_s v_{\mathrm{mean,s}}/4$.

Why can equal electron and ion densities still produce unequal wall currents?
The flux contains the thermal speed, which scales as $m_s^{-1/2}$ at equal temperature. Electrons therefore arrive much more rapidly than ions even when the volume densities are equal.

What physical process starts the formation of a floating sheath?
The initial electron flux is larger, so the wall gains electrons and its potential becomes negative relative to the plasma. This potential filters electrons and accelerates ions.

Which assumption makes the half-space flux formula a local estimate rather than a sheath solution?
The distribution is taken to be isotropic on the plasma side of the boundary. Inside a sheath the electric field makes the distribution anisotropic, so the collected flux must be obtained from the orbit and boundary conditions.

16.2 Sheath structure and the Bohm criterion

The flux argument says that a boundary layer must form, but not how its potential varies in space. A one-dimensional model makes the coupling explicit: Boltzmann electrons respond to the potential, cold ions accelerate through it, and Poisson's equation determines the curvature. The requirement that a monotone sheath can attach to a quasineutral plasma is the Bohm criterion.

Learning objectives

- set a consistent wall-to-plasma coordinate and potential reference
- combine Boltzmann electrons with cold-ion continuity and energy
- nondimensionalize Poisson's equation with the electron Debye length
- derive the sonic lower bound on ion speed at the sheath edge

Unit ledger

The coordinate x and Debye length λ_D are in m, potential Φ is in V, electric field is in $\mathrm{V\,m^{-1}}$, and densities are in $\mathrm{m^{-3}}$. Ion speed and the cold-ion sound speed are in $\mathrm{m\,s^{-1}}$. The normalized coordinate $\xi = x/\lambda_D$, potential barrier $\eta = -e\Phi/(k_B T_e)$, and Mach number $M = u_s/c_s$ are dimensionless. The electron Debye length is $\lambda_D^2 = \varepsilon_0 k_B T_e/(n_0 e^2)$, with ε_0 in $\mathrm{F\,m^{-1}}$.

Steady planar collisionless sheath

Assumptions 401

Let the wall be at $x = 0$ and the plasma occupy $x > 0$, with $\Phi(\infty) = 0$ and a negative wall potential. Use cold ions with charge $+e$, Boltzmann electrons with temperature T_e, no magnetic field, no collisions inside the sheath, and a steady ion flux. The sheath edge is an asymptotic matching region. The limit $x \to \infty$ represents the sheath-edge reservoir in this local model; the presheath that joins it to the bulk plasma is not included.

Normalized sheath variables

Definition 402

Define the positive electron barrier and distance from the wall by $\eta(x) = -e\Phi(x)/(k_B T_e) \geq 0$ and $\xi = x/\lambda_D$. The electron density is $n_e/n_0 = \exp(-\eta)$. Let u_s and n_s be the ion speed and density at the sheath edge, define $c_s = \sqrt{k_B T_e/m_i}$ and $M = u_s/c_s$, and take $\eta = 0$ at the matching edge, with $n_s = n_0$. Throughout this sheath model $u_i > 0$ is the inward ion speed: the signed velocity is $v_{i,x} = -u_i$ because x increases away from the wall. The barrier $-e\Phi$ is positive electron potential energy, although Φ itself is negative. Here the subscript s on u_s and n_s means "sheath edge," not a species index.

Planar sheath equation

Governing law 403

Ion continuity and energy give

$$\frac{n_i}{n_s} = \frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}}.$$

Poisson's equation, written with $\boldsymbol{E} = -\boldsymbol{\nabla}(\Phi)$ and

$$\boldsymbol{\nabla} \cdot (\boldsymbol{E}) = \frac{e(n_i - n_e)}{\varepsilon_0},$$

becomes

$$\frac{\mathrm{d}^2\eta}{\mathrm{d}\xi^2} = \frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}} - \exp(-\eta).$$

Expanding at the quasineutral edge gives $\mathrm{d}^2\eta/\mathrm{d}\xi^2 \approx (1 - M^{-2})\eta$. A monotone solution that leaves the edge therefore requires $M \geq 1$, or $u_s \geq c_s = \sqrt{k_B T_e/m_i}$. This is the cold-ion Bohm criterion.

Derivation: Boltzmann response, ion flow, and the sheath edge

Step: Write the Boltzmann electron response

For a species in a static electrostatic potential, the stationary Maxwell–Boltzmann response is

$$n_s(\Phi) = n_{s,\infty} \exp\left(-\frac{q_s \Phi}{k_B T_s}\right).$$

For electrons $q_e = -e$ and with $\Phi(\infty) = 0$ as the plasma reference,

$$n_e = n_0 \exp\left(\frac{e\Phi}{k_B T_e}\right) = n_0 \exp(-\eta).$$

Step: Use ion continuity and energy

For cold steady ions,

$$\frac{\mathrm{d}n_i u_i}{\mathrm{d}x} = 0, \quad n_i u_i = n_s u_s = \Gamma_i.$$

The ion momentum equation is

$$m_i u_i \frac{\mathrm{d}u_i}{\mathrm{d}x} = -e\frac{\mathrm{d}\Phi}{\mathrm{d}x}.$$

Since $u_i \mathrm{d}u_i/\mathrm{d}x = \mathrm{d}u_i^2/2/\mathrm{d}x$, direct integration gives the conserved energy:

$$\frac{\mathrm{d}\frac{m_i u_i^2}{2} + e\Phi}{\mathrm{d}x} = 0.$$

At the edge, where $\eta = 0$,

$$\frac{m_i u_i^2}{2} + e\Phi = \frac{m_i u_s^2}{2}.$$

Since $e\Phi = -k_B T_e \eta$,

$$u_i^2 = u_s^2 + \frac{2k_B T_e \eta}{m_i} = c_s^2\left(M^2 + 2\eta\right).$$

Continuity then gives

$$\frac{n_i}{n_s} = \frac{u_s}{u_i} = \frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}}.$$

Step: Nondimensionalize Poisson's equation

Gauss's law reads

$$\boldsymbol{\nabla} \cdot (\boldsymbol{E}) = \frac{\rho_q}{\varepsilon_0} = \frac{e(n_i - n_e)}{\varepsilon_0}.$$

In one dimension, $E_x = -\mathrm{d}\Phi/\mathrm{d}x$, so

$$-\frac{\mathrm{d}^2\Phi}{\mathrm{d}x^2} = \frac{e(n_i - n_e)}{\varepsilon_0}.$$

Set $\Phi = -(k_B T_e/e)\eta$ and $x = \lambda_D \xi$. The left side becomes

$$\left(\frac{k_B T_e}{e\lambda_D^2}\right)\frac{\mathrm{d}^2\eta}{\mathrm{d}\xi^2}.$$

Using $\lambda_D^2 = \varepsilon_0 k_B T_e/\left(n_0 e^2\right)$ and $n_s = n_0$ at the ideal matching edge gives

$$\frac{\mathrm{d}^2\eta}{\mathrm{d}\xi^2} = \frac{n_i}{n_0} - \frac{n_e}{n_0} = \frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}} - \exp(-\eta).$$

Step: Expand at the sheath edge

For small η,

$$\frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}} = \left(1 + \frac{2\eta}{M^2}\right)^{-\frac{1}{2}} \approx 1 - \frac{\eta}{M^2},$$

while

$$\exp(-\eta) \approx 1 - \eta.$$

Their difference is $\left(1 - M^{-2}\right)\eta$. If $M < 1$, the coefficient is negative and the edge curvature has the wrong sign for a monotone positive barrier attached to the plasma. Therefore $M \geq 1$. Equality is the sonic threshold; ion temperature and kinetic effects modify the sound speed and the generalized criterion.

Derivation: electron-free planar edge and Child–Langmuir scaling

Step: Relate ion speed and density to the potential drop

Near a sufficiently negative wall, take the electron density to be negligible and write $V = -\Phi > 0$ for the potential drop from the plasma reference. Let x increase from the wall toward the sheath edge, and let the electron-free layer have thickness d.

The large-drop limit neglects the Bohm entry kinetic energy. Extending this approximation to $V = 0$ imposes zero injection speed there; this is the idealized Child–Langmuir boundary, not an exact match to the Bohm sheath, whose entry speed is finite. In this idealization ion energy gives

$$u_i \approx \sqrt{\frac{2eV}{m_i}}.$$

Constant ion flux $\Gamma_i = n_i u_i$ then gives

$$n_i = \Gamma_i \sqrt{\frac{m_i}{2eV}}.$$

Step: Reduce Poisson's equation

For $\Phi = -V$, Poisson's equation is

$$\frac{\mathrm{d}^2 V}{\mathrm{d}x^2} = \frac{e n_i}{\varepsilon_0} = C V^{-\frac{1}{2}},$$

where $C = (\Gamma_i/\varepsilon_0)\sqrt{em_i/2}$. Introduce the inward distance $s = d - x$ measured from the electron-free edge toward the wall. The equation becomes

$$\frac{\mathrm{d}^2V}{\mathrm{d}s^2} = CV^{-\frac{1}{2}}.$$

Step: Integrate from the sheath edge

Multiply by $\mathrm{d}V/\mathrm{d}s$ and integrate once:

$$\frac{\left(\frac{\mathrm{d}V}{\mathrm{d}s}\right)^2}{2} = 2CV^{\frac{1}{2}} + C_1.$$

At the idealized injection boundary, $s = 0$, $V = 0$, and the separately imposed zero-field condition gives $\mathrm{d}V/\mathrm{d}s = 0$, so $C_1 = 0$. Taking the positive branch gives

$$V^{-\frac{1}{4}}\frac{\mathrm{d}V}{\mathrm{d}s} = 2\sqrt{C}.$$

Integrating from the edge,

$$\frac{4}{3}V^{\frac{3}{4}} = 2\sqrt{C}s,$$

and therefore

$$V(s) = \left(\frac{9C}{4}\right)^{\frac{2}{3}} s^{\frac{4}{3}}.$$

Step: Evaluate the space-charge current

At the wall, $s = d$ and $V = V_w$:

$$V_w^{\frac{3}{2}} = \left(\frac{9C}{4}\right)d^2.$$

Solving for the ion flux and multiplying by e gives the planar Child–Langmuir law:

$$J_i = e\Gamma_i = \frac{4\varepsilon_0}{9}\sqrt{\frac{2e}{m_i}}\frac{V_w^{\frac{3}{2}}}{d^2}.$$

This is a near-wall space-charge result. It does not replace the Boltzmann-electron sheath equation across the complete boundary layer.

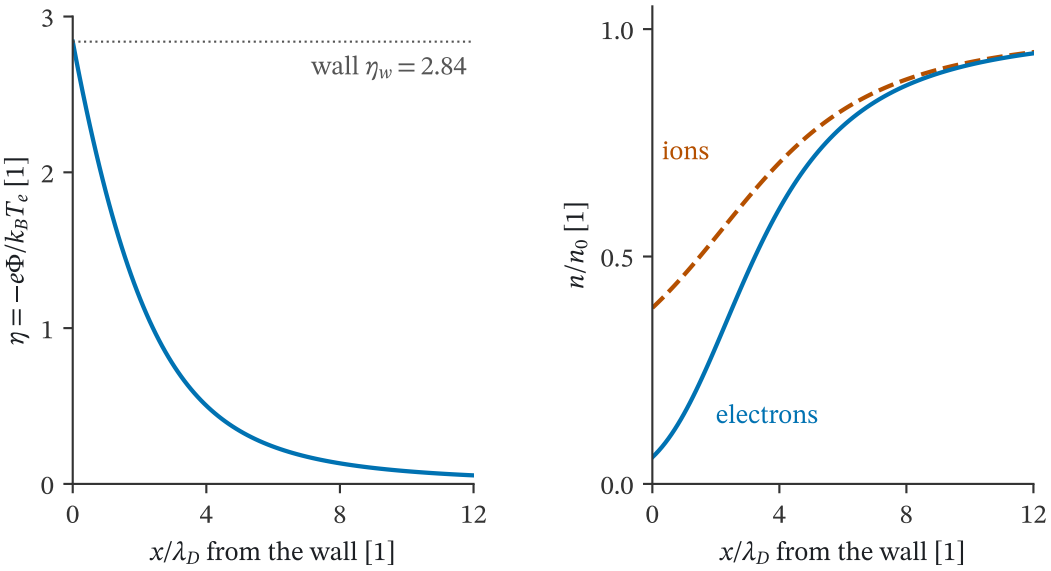


Figure 89: Potential (left) and densities (right) from the Poisson solution of the sheath equation $\eta'' = \frac{M}{\sqrt{M^2+2\eta}} - \exp(-\eta)$ for ions entering at the Bohm speed ($M = 1$), integrated from a hydrogen wall at the floating potential $\eta_w = -\ln\sqrt{2\pi m_e/m_i} = 2.84$, with $\eta = -e\Phi/(k_B T_e)$ and x measured from the wall. Boltzmann electrons $n_e/n_0 = \exp(-\eta)$ are depleted faster than the cold ions $\frac{n_i}{n_0} = \frac{M}{\sqrt{M^2+2\eta}}$, so the sheath carries positive space charge.

Rechenbeispiel Rechenbeispiel 404

For a hydrogen plasma use $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m}^{-3}$, $k_B T_e = 10.0\,\mathrm{eV}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, and $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. Let the normalized sheath-edge speed be $M = 1.50$ [1]. Determine the electron Debye length, cold-ion sound speed, sheath-edge ion speed, and ion particle flux $\Gamma_i = n_0 M c_s$.

Numerical result: $\lambda_D \approx 2.35 \cdot 10^{-4}\,\mathrm{m}$, $c_s \approx 3.09 \cdot 10^{4}\,\mathrm{m\,s^{-1}}$, $u_s \approx 4.64 \cdot 10^{4}\,\mathrm{m\,s^{-1}}$, and $\Gamma_i \approx 4.64 \cdot 10^{20}\,\mathrm{m^{-2}\,s^{-1}}$. The chosen $M = 1.50 > 1$ [1] is above the cold-ion Bohm threshold.

The Bohm condition is an attachment condition Interpretation 405

The criterion does not say that every ion in the plasma moves at exactly the sound speed. It says that the directed ion flow arriving at the sheath edge must be at least sonic in the cold-ion model. The presheath, a nearly quasineutral region connecting the bulk plasma to the sheath, accelerates ions into this flow. The Debye-scale sheath then supplies the remaining potential drop and particle filtering.

Summary

Boltzmann electrons, cold-ion continuity, ion energy, and Poisson's equation combine into

$$\frac{\mathrm{d}^2\eta}{\mathrm{d}\xi^2} = \frac{M}{(M^2 + 2\eta)^{\frac{1}{2}}} - \exp(-\eta).$$

Its small-η expansion requires $M \geq 1$: ions must enter the sheath with at least $c_s = \sqrt{k_B T_e/m_i}$. A strongly electron-free near-wall region has the separate Child–Langmuir scaling J_i scales as

$$\frac{V_w^{\frac{3}{2}}}{d^2}.$$

Knowledge check

- Why is the electron density Boltzmann-suppressed in a negative sheath?
- Where does ion acceleration enter the cold sheath model?
- What mathematical sign produces the Bohm criterion?
- Why is the Child–Langmuir profile not the full sheath solution?

Answers

Why is the electron density Boltzmann-suppressed in a negative sheath?

For electrons $q_e = -e$, the equilibrium factor is

$$\exp\left(-\frac{q_e\Phi}{k_B T_e}\right) = \exp\left(\frac{e\Phi}{k_B T_e}\right).$$

A negative potential makes this factor smaller than one, so electrons are repelled.

Where does ion acceleration enter the cold sheath model?

The ion energy integral gives $u_i^2 = u_s^2 + 2k_B T_e\eta/m_i$. As the potential becomes more negative, eta grows and the ion speed increases; continuity then changes the ion density.

What mathematical sign produces the Bohm criterion?

Near the quasineutral edge the normalized Poisson equation becomes $\mathrm{d}^2\eta/\mathrm{d}\xi^2 \approx (1 - M^{-2})\eta$. A monotone positive barrier can attach to the edge only when this coefficient is nonnegative, giving $M \geq 1$.

Why is the Child–Langmuir profile not the full sheath solution?

It neglects electrons and entry kinetic energy, and imposes zero injection speed and field at zero drop. These idealized boundaries produce space-charge scaling; a Bohm sheath instead has finite ion entry speed and matches to a plasma with both species present.

16.3 Floating potential and current balance

A conducting surface can be electrically isolated, externally biased, or held at a prescribed potential. An isolated surface settles at the floating potential, for which the net conventional current vanishes. The mass asymmetry of the incoming flux then enters through one logarithm.

Learning objectives

- write electron and ion currents with an explicit sign convention
- derive the exponential electron collection factor for a negative wall
- obtain the floating potential from zero net current
- separate the ideal cold-ion result from presheath and surface corrections

Unit ledger

The floating potential Φ_f is in V relative to the sheath-edge plasma potential, and $k_B T_e/e$ is the corresponding potential scale. Fluxes are in $\mathrm{m^{-2}\,s^{-1}}$, current densities in $\mathrm{A\,m^{-2}}$, and probe area is in $\mathrm{m^2}$. The normalized wall bias $u_f = e\Phi_f/(k_B T_e)$ and all flux ratios are dimensionless.

Floating planar wall with cold-ion Bohm entry — Assumptions 408

Use a singly ionized hydrogen plasma, a Maxwellian electron population at the sheath edge, cold ions entering at the sonic speed $c_s = \sqrt{k_B T_e/m_i}$, and an absorbing surface without secondary emission. Take conventional current into the wall as positive for ions: $J = +e\Gamma_i - e\Gamma_e$. Set the sheath-edge potential to zero and let n_0 be the density there. A potential quoted relative to the bulk plasma must additionally include the presheath potential drop.

Floating potential — Definition 409

The floating potential Φ_f is the surface potential for which $J(\Phi_f) = 0$. For a negative surface bias, the electron flux is the unretarded Maxwellian flux multiplied by the Boltzmann transmission factor: $\Gamma_e(\Phi_f) = \Gamma_{e,0}\exp(e\Phi_f/(k_B T_e))$. The ion flux is set primarily by the presheath and sheath-edge entry condition, so in the simplest model $\Gamma_i = n_0 c_s$.

Zero-current balance — Governing law 410

Set $0 = J_f = e\Gamma_i - e\Gamma_{e,0}\exp(e\Phi_f/(k_B T_e))$. Solving gives $\Phi_f = (k_B T_e/e)\ln(\Gamma_i/\Gamma_{e,0})$. With $\Gamma_i = n_0\sqrt{k_B T_e/m_i}$ and $\Gamma_{e,0} = n_0\sqrt{k_B T_e/(2\pi m_e)}$, $\Phi_f = (k_B T_e/(2e))\ln(2\pi m_e/m_i)$. For hydrogen this ideal cold-ion value is approximately $\Phi_f \approx -2.84 k_B T_e/e$.

Derivation: electron transmission and the floating logarithm

Step: Compute the transmitted electron flux

Let the sheath-edge potential be zero and the wall potential be $\Phi_w < 0$. An electron has potential energy $q_e\Phi = -e\Phi$. It reaches the wall only if its normal kinetic energy exceeds the barrier $e|\Phi_w|$. Integrating the Maxwellian over the transmitted velocities multiplies the unretarded flux by a Boltzmann factor:

$$\Gamma_e(\Phi_w) = \Gamma_{e,0}\exp\left(-\frac{e|\Phi_w|}{k_B T_e}\right) = \Gamma_{e,0}\exp\left(\frac{e\Phi_w}{k_B T_e}\right).$$

The final equality uses $\Phi_w < 0$.

Step: Write the net current

Take ion current into the wall as positive and electron charge transport as negative:

$$J_i = +e\Gamma_i, \quad J_e = -e\Gamma_e.$$

Thus

$$J(\Phi_w) = e\Gamma_i - e\Gamma_{e,0}\exp\left(\frac{e\Phi_w}{k_B T_e}\right).$$

A floating surface has no external current, so $J(\Phi_f) = 0$ and

$$\Gamma_{e,0}\exp\left(\frac{e\Phi_f}{k_B T_e}\right) = \Gamma_i.$$

Taking the logarithm gives

$$\Phi_f = \left(\frac{k_B T_e}{e}\right)\ln\left(\frac{\Gamma_i}{\Gamma_{e,0}}\right).$$

Step: Insert the ideal edge fluxes

The cold-ion Bohm flux and the unretarded thermal electron flux are

$$\Gamma_i = n_0\sqrt{\frac{k_B T_e}{m_i}}, \quad \Gamma_{e,0} = n_0\sqrt{\frac{k_B T_e}{2\pi m_e}}.$$

Their ratio is $\sqrt{2\pi m_e/m_i}$. Therefore

$$\Phi_f = \left(\frac{k_B T_e}{2e}\right)\ln\left(\frac{2\pi m_e}{m_i}\right).$$

For hydrogen, $2\pi m_e/m_i \approx 0.00342$, whose logarithm is approximately -5.68. The ideal cold-ion floating potential is thus approximately $-2.84 k_B T_e/e$. The numerical example evaluates these stated definitions without an additional probe-geometry factor.

Rechenbeispiel Rechenbeispiel 411

Use a hydrogen plasma with $n_0 = 1.0 \cdot 10^{16}\,\mathrm{m^{-3}}$ and $k_B T_e = 3.00\,\mathrm{eV}$. Take $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$, $m_i = 1.673 \cdot 10^{-27}\,\mathrm{kg}$, and $e = 1.602 \cdot 10^{-19}\,\mathrm{C}$. Use the cold-ion Bohm flux and a planar area of $A = 1.0 \cdot 10^{-4}\,\mathrm{m^2}$. Determine λ_D, the electron and ion edge fluxes, the floating potential, and the ion current collected at the floating surface.

Numerical result: $\lambda_D \approx 1.29 \cdot 10^{-4}\,\mathrm{m}$, $\Gamma_{e,0} \approx 2.90 \cdot 10^{21}\,\mathrm{m^{-2}\,s^{-1}}$, $\Gamma_i \approx 1.70 \cdot 10^{20}\,\mathrm{m^{-2}\,s^{-1}}$, $e\Phi_f/(k_B T_e) \approx -2.84$ [1], $\Phi_f \approx -8.52\,\mathrm{V}$, and $e\Gamma_i A \approx 2.72\,\mathrm{mA}$.

Floating does not mean field-free Interpretation 412

Zero net current is a global electrical condition. The sheath field can be strong while the integrated ion and electron charge currents cancel. Changing ion temperature, presheath losses, secondary emission, magnetic incidence, or surface geometry changes the fluxes and therefore the floating potential.

Summary

For a negative absorbing wall, electron collection is exponentially reduced: $\Gamma_e = \Gamma_{e,0}\exp(e\Phi_w/(k_B T_e))$. The ideal cold-ion floating condition is

$$\Phi_f = \left(\frac{k_B T_e}{e}\right)\ln\left(\frac{\Gamma_i}{\Gamma_{e,0}}\right) \approx -\frac{2.84 k_B T_e}{e}$$

for hydrogen. The coefficient is a model result.

Knowledge check

- Why does the electron current contain an exponential wall-potential factor?
- What current sign convention is used for a floating surface?
- Why does the floating potential scale with electron temperature?
- Why should the coefficient -2.84 not be transferred unchanged to every experiment?

Answers

Why does the electron current contain an exponential wall-potential factor?
A negative wall imposes a kinetic-energy barrier on electrons. Integrating the Maxwellian tail above that barrier multiplies the unretarded flux by $\exp(e\Phi_w/(k_B T_e))$.

What current sign convention is used for a floating surface?
Conventional current into the surface is positive for ions and negative for electrons: $J = e\Gamma_i - e\Gamma_e$. Floating means this signed sum is zero.

Why does the floating potential scale with electron temperature?
The electron barrier is measured by the dimensionless ratio $e\Phi_f/(k_B T_e)$. A hotter electron population requires a larger potential-energy barrier, so the potential drop scales with $k_B T_e/e$.

Why should the coefficient -2.84 not be transferred unchanged to every experiment?
It assumes cold ions at the Bohm speed, planar collection, Maxwellian electrons, no secondary emission, and no magnetic or collisional correction. Changing any of these changes the ion-to-electron flux ratio.

16.4 Langmuir-probe current–voltage diagnostics

A Langmuir probe turns the floating-surface balance into a controlled measurement. By sweeping the probe potential relative to the plasma and separating ion, retarding-electron, and electron-saturation regions, one can infer electron temperature, density, and an estimate of the plasma potential. The fit is only meaningful when the collection model and surface physics are stated.

Learning objectives

- define probe bias and conventional probe current consistently
- identify ion-saturation, electron-retardation, and electron-saturation regions
- derive the temperature from the semilog slope of the electron current
- infer density from the electron saturation scale and list major corrections

Unit ledger
Probe bias $\Phi_p - \Phi_{\mathrm{pl}}$ is in V, current I_p is in A (often reported in mA), and collection area A is in m^2. The normalized bias $u = e\big(\Phi_p - \Phi_{\mathrm{pl}}\big)/(k_B T_e)$ and normalized current $I_p/\big(e\Gamma_{e,0}A\big)$ are dimensionless. When temperature is quoted in electron-volts, $k_B T_e$ is an energy and $e\Phi/(k_B T_e)$ is evaluated with the same energy unit.

Ideal planar probe characteristic — Assumptions 415

Use an absorbing probe that is small enough not to change the bulk plasma, but whose collecting face is large compared with the sheath thickness so that local planar collection applies. Take stationary Maxwellian electrons and the stated cold-ion entry model. In the negative-bias branch, take the ion current as approximately saturated and the electron current as Boltzmann-retarded. Neglect magnetic-orbit effects, collisions in the sheath, secondary emission, photoemission, RF fluctuations, and probe perturbation of the bulk plasma. These assumptions define a diagnostic baseline.

Probe bias and current regions — Definition 416

Let Φ_p be the probe potential and Φ_{pl} the plasma potential. Define

$$u = \frac{e\big(\Phi_p - \Phi_{\mathrm{pl}}\big)}{k_B T_e}.$$

With conventional current into the probe, $I_p = A(e\Gamma_i - e\Gamma_e)$. For strongly negative u, ions form an ion-saturation branch. As u increases, the electron current grows exponentially in the electron-retardation region. At still more positive bias, the electron collection becomes geometry- and sheath-limited, producing an electron saturation region.

Idealized probe characteristic — Governing law 417

On the negative-bias side, write $I_p(u) = A\big[e\Gamma_i - e\Gamma_{e,0}\exp(u)\big]$. The floating point is the zero of this curve. If $I_e = -eA\Gamma_e$ denotes the electron current alone and $I_{e,0} = eA\Gamma_{e,0} > 0$ its unretarded magnitude, then the electron-retarding branch satisfies $\ln\big(|I_e|/I_{e,0}\big) = u$. Its slope is therefore

$$\frac{\mathrm{d}\ln\left(\frac{|I_e|}{I_{e,0}}\right)}{\mathrm{d}\Phi_p} = \frac{e}{k_B T_e}.$$

Obtain I_e by subtracting the estimated ion contribution $I_i = eA\Gamma_i$ from the measured signed current: $I_e = I_p - I_i$. The logarithm is taken of the positive, dimensionless current ratio. The electron saturation scale gives

$$n_e = \frac{|I_{e,0}|}{\left[eA\sqrt{\frac{k_B T_e}{2\pi m_e}}\right]}$$

for the ideal planar collection model.

Derivation: semilog temperature and density inversion

Step: Linearize the retarding electron current

The negative-bias electron flux is

$$\Gamma_e(\Phi_p) = \Gamma_{e,0} \exp\left(\frac{e(\Phi_p - \Phi_{\mathrm{pl}})}{k_B T_e}\right).$$

With $I_e = -eA\Gamma_e$, its magnitude is

$$|I_e| = eA\Gamma_{e,0} \exp\left(\frac{e(\Phi_p - \Phi_{\mathrm{pl}})}{k_B T_e}\right).$$

Divide by the fixed reference $I_{e,0} = eA\Gamma_{e,0}$ and take the natural logarithm to obtain a straight line:

$$\ln\left(\frac{|I_e|}{I_{e,0}}\right) = \frac{e(\Phi_p - \Phi_{\mathrm{pl}})}{k_B T_e}.$$

Step: Read temperature from the semilog slope

Differentiate with respect to probe potential:

$$\frac{\mathrm{d}\ln\left(\frac{|I_e|}{I_{e,0}}\right)}{\mathrm{d}\Phi_p} = \frac{e}{k_B T_e}.$$

Inverting the slope gives

$$k_B T_e = e\left(\frac{\mathrm{d}\ln\left(\frac{|I_e|}{I_{e,0}}\right)}{\mathrm{d}\Phi_p}\right)^{-1}.$$

If the voltage unit is volts and the energy unit is electron-volts, the numerical value of $k_B T_e$ in eV is the inverse slope in V^{-1}.

Step: Read density from electron saturation

At the extrapolated electron-saturation reference,

$$|I_{e,0}| = eA\Gamma_{e,0}.$$

The Maxwellian half-space flux is

$$\Gamma_{e,0} = n_e \sqrt{\frac{k_B T_e}{2\pi m_e}}.$$

Solving for density gives

$$n_e = \frac{|I_{e,0}|}{eA\sqrt{\frac{k_B T_e}{2\pi m_e}}}.$$

The measured current must be in amperes and the area in square metres for this expression to return n_e in m^{-3}.

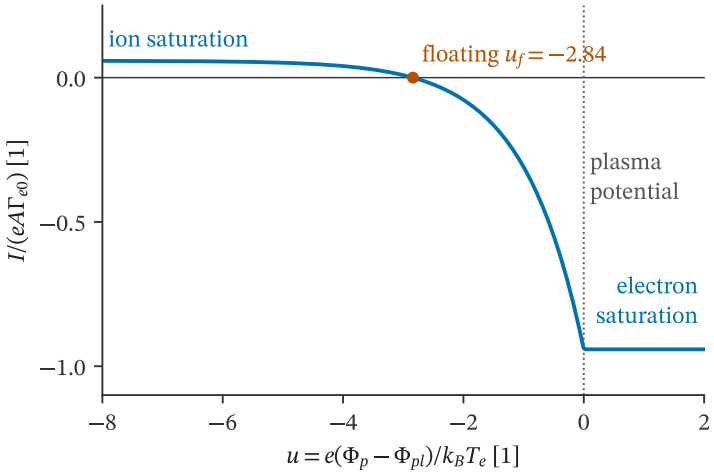
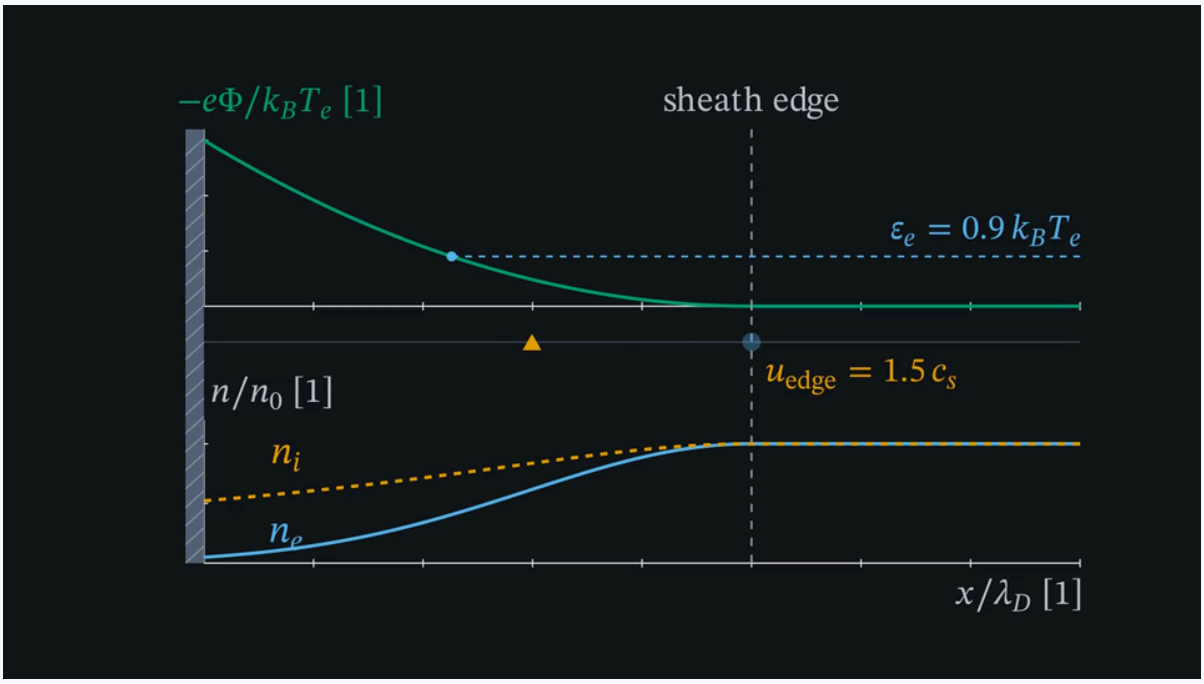


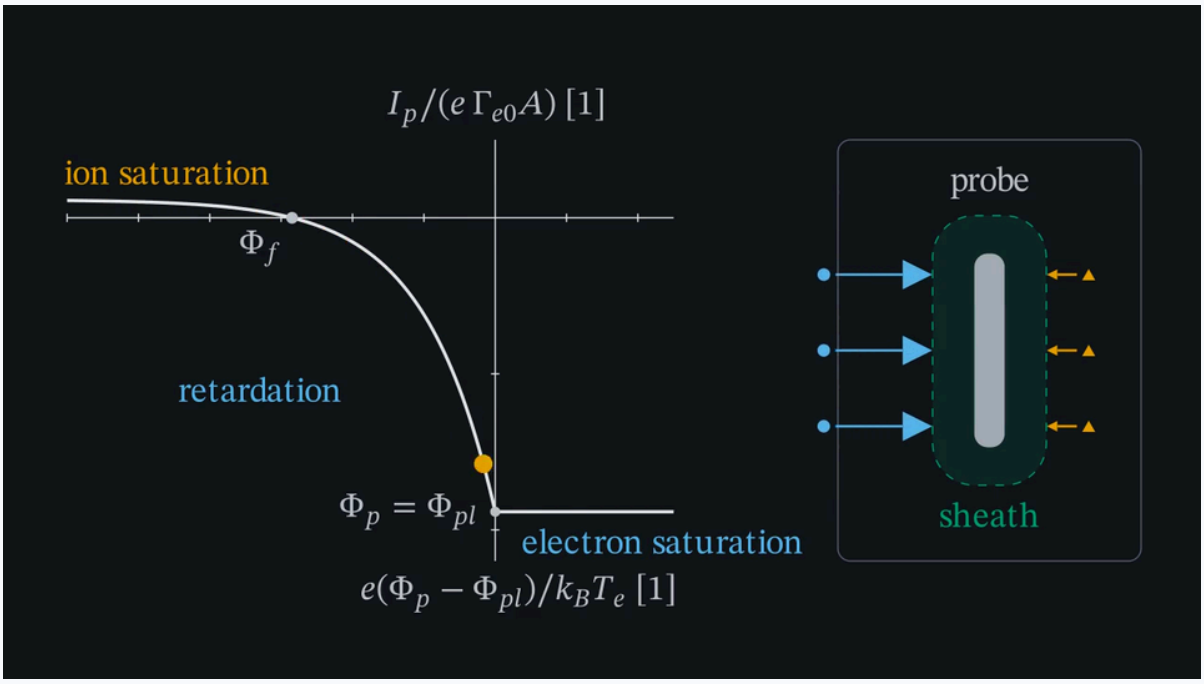
Figure 90: Idealized planar probe, ion current positive: $\frac{I}{e\Gamma_{e,0}A} = \frac{\Gamma_i}{\Gamma_{e,0}} - \exp(u)$ for $u \le 0$ and electron saturation for $u > 0$, with $u = \frac{e(\Phi_p - \Phi_{\mathrm{pl}})}{k_B T_e}$. For hydrogen $\frac{\Gamma_i}{\Gamma_{e,0}} = \sqrt{\frac{2\pi m_e}{m_i}} = 0.0585$, so the current vanishes at $u_f = -2.84$. The exponential retardation branch carries the temperature diagnostic; real saturation depends on geometry.



Watch animation

The prescribed profile uses $X = x/\lambda_D$, edge $X = 5$, $\eta = 0.12(5 - X)^2$ inside the sheath, and $\eta = 0$ outside. Densities follow $n_e/n_0 = \exp(-\eta)$ and $\frac{n_i}{n_0} = \frac{1.5}{\sqrt{1.5^2+2\eta}}$; Poisson's equation is not solved. The electron's entry normal energy is $0.9k_BT_e$. The electron transit is played in units of $\lambda_D/\sqrt{k_BT_e/m_e}$ and the ion transit in units of λ_D/c_s, so their playback speeds are not comparable. Physically the electron barrier is $-e\Phi > 0$: reflection requires normal kinetic energy below this barrier, while sufficiently energetic electrons reach the wall. The equations above use distance increasing away from the wall and positive inward ion speed.

Figure 91: The prescribed profile uses $X = x/\lambda_D$, edge $X = 5$, $\eta = 0.12(5 - X)^2$ inside the sheath, and $\eta = 0$ outside. Densities follow $n_e/n_0 = \exp(-\eta)$ and $\frac{n_i}{n_0} = \frac{1.5}{\sqrt{1.5^2+2\eta}}$; Poisson's equation is not solved. The electron's entry normal energy is $0.9k_BT_e$. The electron transit is played in units of $\lambda_D/\sqrt{k_BT_e/m_e}$ and the ion transit in units of λ_D/c_s, so their playback speeds are not comparable. Physically the electron barrier is $-e\Phi > 0$: reflection requires normal kinetic energy below this barrier, while sufficiently energetic electrons reach the wall. The equations above use distance increasing away from the wall and positive inward ion speed.



Watch animation

Probe bias moves the operating point across collection regimes. The retarding branch carries the temperature information in its logarithmic slope, while the saturation scale carries the density information under the stated collection model. This ideal planar illustration uses $I_p/(e\Gamma_{e,0}A) = 0.058 - \exp(\min(u, 0))$: it is continuous at zero bias and caps the electron contribution at its planar saturation scale. Geometry-dependent rising collection branches require a different model.

Figure 92: Probe bias moves the operating point across collection regimes. The retarding branch carries the temperature information in its logarithmic slope, while the saturation scale carries the density information under the stated collection model. This ideal planar illustration uses $I_p/(e\Gamma_{e,0}A) = 0.058 - \exp(\min(u, 0))$: it is continuous at zero bias and caps the electron contribution at its planar saturation scale. Geometry-dependent rising collection branches require a different model.

Rechenbeispiel Rechenbeispiel 418

A planar probe has area $A = 1.0 \cdot 10^{-5}\,\mathrm{m}^2$. In its electron-retarding region, a fit gives

$$\frac{\mathrm{d}\ln\left(\frac{|I_e|}{I_{e,0}}\right)}{\mathrm{d}\Phi_p} = 0.40\,\mathrm{V}^{-1}.$$

The extrapolated zero-bias electron saturation current magnitude is $|I_{e,0}| = 4.0\,\mathrm{mA}$. Use the ideal planar model and determine the electron temperature energy $k_B T_e$ in eV and density in m^{-3}. Numerical result: $k_B T_e/e \approx 2.50\,\mathrm{V}$, so $k_B T_e \approx 2.50\,\mathrm{eV}$, and $n_e \approx 9.44 \cdot 10^{15}\,\mathrm{m}^{-3}$.

A probe measures a model-dependent collection response Interpretation 419

The semilog slope uses only the retarding branch, but the density scale depends on area and collection geometry. Cylindrical or spherical probes, magnetized plasmas, collisional sheaths, ion drift, secondary emission, photoemission, RF fluctuations, and contamination can all alter the characteristic. A reliable analysis reports the fit interval and the corrections used.

Summary

Summary 420

A Langmuir probe sweeps

$$u = \frac{e\left(\Phi_p - \Phi_{\mathrm{pl}}\right)}{k_B T_e}$$

through ion saturation, electron retardation, and electron saturation. On the ideal retarding branch,

$$\frac{\mathrm{d}\ln\left(\frac{|I_e|}{I_{e,0}}\right)}{\mathrm{d}\Phi_p} = \frac{e}{k_B T_e};$$

the extrapolated saturation scale then gives density after the collection area and model are fixed.

Knowledge check

Knowledge check 421

- Which part of an ideal probe characteristic carries the electron-temperature information?
- What does the zero-current point represent on a floating probe?
- Why can the electron saturation current be used to estimate density?
- Why is an ideal Langmuir-probe fit not automatically a local plasma measurement?

Answers

Which part of an ideal probe characteristic carries the electron-temperature information?

After subtracting the ion contribution, the slope of $\ln\left(|I_e|/I_{e,0}\right)$ against probe potential is $e/(k_B T_e)$. Fit only the electron-retarding region, excluding the ion and electron saturation regions.

What does the zero-current point represent on a floating probe?

It is the bias at which the signed ion and electron currents cancel. The point is the probe analogue of the floating wall condition.

Why can the electron saturation current be used to estimate density?

Under the planar Maxwellian model it is $eAn_e\sqrt{k_B T_e/(2\pi m_e)}$. Once temperature and collection area are known, the current scale is proportional to density.

Why is an ideal Langmuir-probe fit not automatically a local plasma measurement?

The probe creates a sheath and samples an orbit-selected collection region. Geometry, magnetic field, collisions, emission, and RF effects can change the current before it is interpreted as a bulk parameter.

Appendix: Mathematical toolkit

The course repeatedly moves between kinetic, fluid, and field descriptions. This appendix collects the energy-weighted second moment of the kinetic equation and the differential operators used in Cartesian, cylindrical, and spherical coordinates, with collapsed derivations for reconstructing each result.

Checklist for compact plasma equations — Key idea 422

Before manipulating a compact plasma equation, identify the species, the integration measure, the active unit convention, the coordinate scale factors, and the boundary terms. Most apparent sign or missing-factor errors can be found at one of those five checkpoints.

Energy-weighted second moment

How does the kinetic equation produce an energy equation? Weighting by the single-particle kinetic energy and integrating over velocity converts velocity-space transport into electromagnetic work, spatial transport, and collisional energy exchange. The result is the bridge between a distribution function and the fluid energy equations used later in the script.

Learning objectives

- define the bulk, internal, and heat-flux contributions to a species energy moment
- carry out the velocity-space integration by parts without dropping the boundary condition
- identify why the magnetic part of the Lorentz force does no work
- separate the total-energy equation from the internal-energy equation

Unit ledger

The species distribution is defined by $dN_s = f_s \mathrm{d}^3\boldsymbol{r}\mathrm{d}^3\boldsymbol{v}$. The energy density and energy flux below are in $\mathrm{J\,m^{-3}}$ and $\mathrm{W\,m^{-2}}$, respectively.

Kinetic equation and velocity-space boundary — Assumptions 423

For species s, use a smooth kinetic equation with acceleration $\boldsymbol{a}_s = (q_s/m_s)(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$. The distribution and the velocity-space flux $f_s\boldsymbol{a}_s$ vanish sufficiently rapidly as $|\boldsymbol{v}| \to \infty$, so surface terms at infinite speed are zero. Require species-number-conserving collisions,

$$\int C_s[f]\mathrm{d}^3\boldsymbol{v} = 0,$$

so $\partial\rho_s/\partial t + \boldsymbol{\nabla}\cdot(\rho_s\boldsymbol{u}_s) = 0$. Ionization and recombination are excluded from the internal-energy balance below. Collisions may exchange energy between species, but their weighted integral is retained as a source.

Bulk and thermal variables Definition 424

The species flow velocity and random velocity are

$$\boldsymbol{u}_s = \frac{\int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}}{n_s}$$

and $\boldsymbol{c}_s = \boldsymbol{v} - \boldsymbol{u}_s$. Define the bulk kinetic-energy density, internal-energy density, pressure tensor, and heat-flux vector by $K_s = \rho_s|\boldsymbol{u}_s|^2/2$, $U_s = \operatorname{tr}\boldsymbol{P}_s/2$,

$$\boldsymbol{P}_s = m_s \int \boldsymbol{c}_s\boldsymbol{c}_s f_s \mathrm{d}^3\boldsymbol{v}, \quad \text{and} \quad \boldsymbol{q}_s = \frac{m_s \int |\boldsymbol{c}_s|^2 \boldsymbol{c}_s f_s \mathrm{d}^3\boldsymbol{v}}{2}.$$

Here $\boldsymbol{c}_s\boldsymbol{c}_s$ is a dyadic product. The total second moment is

$$W_s = K_s + U_s = \frac{m_s \int |\boldsymbol{v}|^2 f_s \mathrm{d}^3\boldsymbol{v}}{2}.$$

For a collision operator $C_s[f]$, define the collisional momentum transfer and kinetic-energy transfer by

$$\boldsymbol{R}_s = m_s \int \boldsymbol{v} C_s[f] \mathrm{d}^3\boldsymbol{v} \quad \text{and} \quad Q_s = \frac{m_s \int |\boldsymbol{v}|^2 C_s[f] \mathrm{d}^3\boldsymbol{v}}{2}.$$

The source $\boldsymbol{R}_s$ is in $\mathrm{N\,m^{-3}}$ and Q_s is in $\mathrm{W\,m^{-3}}$.

Species total-energy equation Governing law 425

The energy-weighted kinetic equation gives

$$\frac{\partial W_s}{\partial t} + \boldsymbol{\nabla} \cdot ((W_s)\boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_s) = q_s n_s \boldsymbol{E} \cdot \boldsymbol{u}_s + Q_s,$$

where Q_s is the collisional rate of kinetic-energy transfer to species s in $\mathrm{W\,m^{-3}}$. The magnetic force is absent from the power term because $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0$.

The three terms in the energy flux have distinct meanings: advection of total particle kinetic energy $W_s\boldsymbol{u}_s$ (bulk plus internal), pressure work transport $\boldsymbol{P}_s \cdot \boldsymbol{u}_s$, and random-energy transport $\boldsymbol{q}_s$. They must not be merged into a scalar heat flux unless isotropy and a closure have been stated.

Derivation route Key idea 426

Multiply the kinetic equation by $m_s|\boldsymbol{v}|^2/2$, integrate over all velocity space, commute time and space derivatives with the integral, integrate the acceleration term by parts in velocity, then decompose $\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{c}_s$. The collapsed derivation gives each step and the boundary terms.

Derivation: total energy from the kinetic equation

Step: Weight the kinetic equation

Start from the non-conservative form

$$\frac{\partial f_s}{\partial t} + \boldsymbol{v} \cdot \boldsymbol{\nabla}(f_s) + \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} = C_s[f].$$

Since $\boldsymbol{v}$ is independent of $\boldsymbol{r}$, the streaming term is also $\boldsymbol{\nabla} \cdot (f_s \boldsymbol{v})$. Define the single-particle kinetic energy

$$\varepsilon_s(\boldsymbol{v}) = \frac{m_s |\boldsymbol{v}|^2}{2}.$$

Multiply the kinetic equation by this energy and integrate over all velocity space:

$$\int \varepsilon_s \frac{\partial f_s}{\partial t} \mathrm{d}^3\boldsymbol{v} + \int \varepsilon_s \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) \mathrm{d}^3\boldsymbol{v} + \int \varepsilon_s \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} \mathrm{d}^3\boldsymbol{v} = \int \varepsilon_s C_s[f] \mathrm{d}^3\boldsymbol{v}.$$

Step: Evaluate the temporal and streaming terms

The temporal term is

$$\int \varepsilon_s \frac{\partial f_s}{\partial t} \mathrm{d}^3\boldsymbol{v} = \frac{\partial \int \varepsilon_s f_s \mathrm{d}^3\boldsymbol{v}}{\partial t} = \frac{\partial W_s}{\partial t}.$$

The streaming term commutes with the velocity integral:

$$\int \varepsilon_s \boldsymbol{\nabla} \cdot (f_s \boldsymbol{v}) \mathrm{d}^3\boldsymbol{v} = \boldsymbol{\nabla} \cdot \left(\frac{m_s \int |\boldsymbol{v}|^2 \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}}{2} \right).$$

Step: Integrate the acceleration term by parts

Write the contracted velocity divergence explicitly as

$$\boldsymbol{\nabla} \cdot_{\boldsymbol{v}} (\boldsymbol{A}) = \sum_j \frac{\partial A_j}{\partial v_j}.$$

The velocity-space product rule is

$$\varepsilon_s \boldsymbol{a}_s \cdot \frac{\partial f_s}{\partial \boldsymbol{v}} = \boldsymbol{\nabla} \cdot_{\boldsymbol{v}} (\varepsilon_s f_s \boldsymbol{a}_s) - f_s \boldsymbol{a}_s \cdot \frac{\partial \varepsilon_s}{\partial \boldsymbol{v}} - \varepsilon_s f_s \boldsymbol{\nabla} \cdot_{\boldsymbol{v}} (\boldsymbol{a}_s).$$

For the Lorentz acceleration, $\boldsymbol{\nabla} \cdot_{\boldsymbol{v}} (\boldsymbol{a}_s) = 0$: the electric field is independent of velocity and the magnetic linear map is antisymmetric, hence has zero trace. The last term therefore vanishes.

The first term becomes the surface integral

$$\int_{\text{boundary V_v}} \varepsilon_s f_s \boldsymbol{a}_s \cdot d\boldsymbol{S}_v,$$

which vanishes at infinite speed by the boundary assumption. Because $\partial \varepsilon_s / \partial \boldsymbol{v} = m_s \boldsymbol{v}$, the remaining term is

$$-m_s \int f_s \boldsymbol{a}_s \cdot \boldsymbol{v} \mathrm{d}^3\boldsymbol{v}.$$

Insert the Lorentz acceleration. The magnetic contribution vanishes pointwise, leaving

$$-q_s \boldsymbol{E} \cdot \int \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v} = -q_s n_s \boldsymbol{E} \cdot \boldsymbol{u}_s.$$

Define the collision power

$$Q_s = \int \varepsilon_s C_s[f] \mathrm{d}^3\boldsymbol{v}.$$

Moving the electric term to the right gives the conservative energy equation

$$\frac{\partial W_s}{\partial t} + \boldsymbol{\nabla} \cdot \left(\frac{m_s \int |\boldsymbol{v}|^2 \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}}{2} \right) = q_s n_s \boldsymbol{E} \cdot \boldsymbol{u}_s + Q_s.$$

Step: Decompose the energy flux

Substitute

$$\boldsymbol{v} = \boldsymbol{u}_s + \boldsymbol{c}_s$$

$$|\boldsymbol{v}|^2 = |\boldsymbol{u}_s|^2 + 2\boldsymbol{u}_s \cdot \boldsymbol{c}_s + |\boldsymbol{c}_s|^2.$$

The first-order random moment vanishes:

$$\int \boldsymbol{c}_s f_s \mathrm{d}^3\boldsymbol{v} = \boldsymbol{0}.$$

The surviving flux terms are

$$\frac{m_s \int |\boldsymbol{v}|^2 \boldsymbol{v} f_s \mathrm{d}^3\boldsymbol{v}}{2} = K_s \boldsymbol{u}_s + U_s \boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_s.$$

Since $W_s = K_s + U_s$, this becomes

$$W_s \boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_s.$$

Substitution gives the stated total-energy equation.

Internal-energy equation

Governing law 427

Subtracting the bulk kinetic-energy equation obtained by dotting the species momentum equation with $\boldsymbol{u}_s$ gives

$$\frac{\partial U_s}{\partial t} + \boldsymbol{\nabla} \cdot (U_s \boldsymbol{u}_s + \boldsymbol{q}_s) = -\boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) + Q_s - \boldsymbol{u}_s \cdot \boldsymbol{R}_s.$$

The double contraction is

$$\boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) = \sum_{i,j} (P_s)_{ij} \frac{\partial (u_s)_i}{\partial r_j}.$$

For an isotropic pressure tensor $\boldsymbol{P}_s = p_s \boldsymbol{I}$, the pressure work reduces to $p_s \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s)$ and $U_s = 3p_s/2$ in three spatial dimensions.

Derivation: subtract bulk energy and identify pressure work

Step: Start from the momentum balance

The first moment of the kinetic equation is

$$\frac{\partial \rho_s \boldsymbol{u}_s}{\partial t} + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s \boldsymbol{u}_s + \boldsymbol{P}_s) = q_s n_s (\boldsymbol{E} + \boldsymbol{u}_s \times \boldsymbol{B}) + \boldsymbol{R}_s.$$

Dot this equation with $\boldsymbol{u}_s$. The number-conserving assumption gives $\partial \rho_s / \partial t + \boldsymbol{\nabla} \cdot (\rho_s \boldsymbol{u}_s) = 0$, so the product rule yields

$$\frac{\partial K_s}{\partial t} + \boldsymbol{\nabla} \cdot (K_s \boldsymbol{u}_s) = q_s n_s \boldsymbol{E} \cdot \boldsymbol{u}_s - \boldsymbol{u}_s \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) + \boldsymbol{u}_s \cdot \boldsymbol{R}_s.$$

Step: Subtract bulk energy from total energy

The total-energy equation contains the same electromagnetic power. Insert $W_s = K_s + U_s$ and subtract the bulk identity:

$$\frac{\partial U_s}{\partial t} + \boldsymbol{\nabla} \cdot (U_s \boldsymbol{u}_s + \boldsymbol{P}_s \cdot \boldsymbol{u}_s + \boldsymbol{q}_s) = \boldsymbol{u}_s \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) + Q_s - \boldsymbol{u}_s \cdot \boldsymbol{R}_s.$$

Step: Rewrite pressure work

Apply

$$\boldsymbol{\nabla} \cdot (\boldsymbol{P}_s \cdot \boldsymbol{u}_s) = \boldsymbol{u}_s \cdot \boldsymbol{\nabla} \cdot (\boldsymbol{P}_s) + \boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s).$$

The first term cancels, leaving the internal-energy equation with pressure work and heat-flux transport. For $\boldsymbol{P}_s = p_s \boldsymbol{I}$,

$$\boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) = p_s \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s).$$

This is the isotropic limit.

Content of the second-moment equations Interpretation 428

The total-energy equation tracks directed and random kinetic energy together. The internal form isolates compressional work, anisotropic shear work, and heat-flux transport. A fluid model closes this equation only after choosing how $\boldsymbol{P}_s$, $\boldsymbol{q}_s$, and Q_s are represented. When $C_s[f] = 0$, both $Q_s = 0$ and $\boldsymbol{R}_s = \boldsymbol{0}$ for each species. Energy exchange with the fields enters through electric work, not a collision source. For collisional species exchange, the internal source is $Q_s - \boldsymbol{u}_s \cdot \boldsymbol{R}_s$: total collisional power and bulk momentum transfer must be separated.

Summary Summary 429

Weighting the kinetic equation by $m_s |\boldsymbol{v}|^2/2$ gives a total energy balance. Velocity-space integration by parts turns the Lorentz force into electric work; the magnetic part does no work. Decomposing velocity into bulk plus random motion separates total-energy advection, pressure work, and heat-flux transport. Representing $\boldsymbol{P}_s$ and $\boldsymbol{q}_s$ in terms of lower moments is a closure choice.

Knowledge check

- Why does the magnetic part of the Lorentz force not appear in the single-species energy source?
- Which term transports random kinetic energy relative to the species flow?
- What assumption removes the velocity-space surface term in the energy derivation?
- How does isotropic pressure simplify the internal-energy equation?

Answers

Why does the magnetic part of the Lorentz force not appear in the single-species energy source?

Its power is proportional to $\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B})$, which is zero for every particle. The electric part contributes $q_s n_s \boldsymbol{E} \cdot \boldsymbol{u}_s$ in $\mathrm{W\,m^{-3}}$.

Which term transports random kinetic energy relative to the species flow?

The heat-flux vector

$$\boldsymbol{q}_s = \frac{m_s \int |\boldsymbol{c}_s|^2 \boldsymbol{c}_s f_s \mathrm{d}^3\boldsymbol{v}}{2}$$

transports random energy. It is a third central velocity moment and needs a closure in a truncated fluid model.

What assumption removes the velocity-space surface term in the energy derivation?

The distribution and velocity-space flux must decay fast enough that the surface integral at $|\boldsymbol{v}| \to \infty$ vanishes. If a high-energy boundary or source is present, that flux must be retained.

How does isotropic pressure simplify the internal-energy equation?

With $\boldsymbol{P}_s = p_s \boldsymbol{I}$, the tensor contraction becomes $\boldsymbol{P}_s : \boldsymbol{\nabla}(\boldsymbol{u}_s) = p_s \boldsymbol{\nabla} \cdot (\boldsymbol{u}_s)$ and the internal energy is $U_s = 3p_s/2$ in three dimensions.

Vector operators and coordinate geometry

Plasma equations are often simplest in the coordinates selected by the geometry: Cartesian coordinates for a local wave, cylindrical coordinates for a column or pinch, and spherical coordinates for a nearly isotropic expansion. The physical vector is unchanged by the coordinate choice, but the basis vectors and area elements vary. Where do the extra factors of r and $\sin(\theta)$ come from? They are metric scale factors, not new physics.

Learning objectives

- state the gradient, divergence, curl, and scalar Laplacian in orthogonal coordinates
- specialize the formulas to Cartesian, cylindrical, and spherical systems
- use a divergence or curl identity as a dimensional and limiting-case check
- choose coordinates that expose the symmetry of a plasma configuration

Unit ledger

Coordinates x, y, z, r are lengths in m, angles φ and θ are dimensionless, and a scalar field ψ and vector field $\boldsymbol{A}$ retain whatever physical units the model assigns. Thus $\boldsymbol{\nabla}(\psi)$ has units of ψ per m, $\boldsymbol{\nabla} \cdot (\boldsymbol{A})$ has units of $\boldsymbol{A}$ per m, and $\boldsymbol{\nabla} \times (\boldsymbol{A})$ has the same units. The formulas below are dimensional coordinate identities; no normalized radius is implied.

Orthogonal curvilinear coordinates

Assumptions 431

Let (q_1, q_2, q_3) be right-handed orthogonal coordinates with unit basis vectors $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$. Define scale factors by $d\boldsymbol{r} = \sum_i h_i dq_i \boldsymbol{e}_i$. The basis is orthonormal at each point, but it may vary with position. Components A_i are physical components along the local unit vectors, not covariant components.

Metric scale factors

Definition 432

The scale factors h_i convert coordinate increments to physical line elements. The three systems used most often here are

$$\text{Cartesian: } (q_1, q_2, q_3) = (x, y, z), (h_1, h_2, h_3) = (1, 1, 1),$$

$$\text{cylindrical: } (q_1, q_2, q_3) = (r, \varphi, z), (h_1, h_2, h_3) = (1, r, 1),$$

and

$$\text{spherical: } (q_1, q_2, q_3) = (r, \theta, \varphi), (h_1, h_2, h_3) = (1, r, r\sin(\theta)).$$

In spherical coordinates θ is the polar angle from positive z, and φ is the azimuth measured from positive x toward positive y. The volume element is $\mathrm{d}V = h_1 h_2 h_3 \, \mathrm{d}q_1 \, \mathrm{d}q_2 \, \mathrm{d}q_3$.

Orthogonal-coordinate operators

Governing law 433

For a scalar ψ and vector $\boldsymbol{A} = \sum_i A_i \boldsymbol{e}_i$, the operators are

$$\boldsymbol{\nabla}(\psi) = \sum_i \left(\frac{\boldsymbol{e}_i}{h_i}\right) \frac{\partial \psi}{\partial q_i},$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{A}) = \frac{\sum_i \frac{\partial (h_j h_k A_i)}{\partial q_i}}{h_1 h_2 h_3},$$

where (i, j, k) cycles through $(1, 2, 3)$, and

$$\boldsymbol{\nabla} \times (\boldsymbol{A}) = \sum_i \left(\frac{\boldsymbol{e}_i}{h_j h_k}\right) \left(\frac{\partial h_k A_k}{\partial q_j} - \frac{\partial h_j A_j}{\partial q_k}\right).$$

The scalar Laplacian is

$$\nabla^2(\psi) = \frac{\sum_i \frac{\partial \left(\frac{h_j h_k}{h_i}\right) \frac{\partial \psi}{\partial q_i}}{\partial q_i}}{h_1 h_2 h_3}.$$

Each derivative is with respect to the displayed coordinate, while the scale factors account for physical distance and area.

In the cyclic formulas, for the component with index i, the remaining indices (j, k) are taken in cyclic order. For example, the first curl component is

$$\frac{\frac{\partial h_3 A_3}{\partial q_2} - \frac{\partial h_2 A_2}{\partial q_3}}{h_2 h_3}.$$

Writing the scale factors explicitly is safer than importing a Cartesian formula and inserting r by intuition.

Coordinate choice is a symmetry choice Key idea 434

A straight, locally uniform wave naturally uses Cartesian components. A cylindrical pinch makes A_φ and B_z natural components, while a spherically expanding plasma makes the area factor $4\pi r^2$ visible in a divergence. The operator does not change the physical conservation law; only its coordinate representation changes.

Derivation: origin of the scale factors

Step: Derive the gradient components

The directional differential of a scalar field is

$$d\psi = \boldsymbol{\nabla}(\psi) \cdot d\boldsymbol{r} = \sum_i (\boldsymbol{\nabla}(\psi))_i h_i dq_i.$$

The chain rule gives $d\psi = \sum_i \partial\psi/\partial q_i dq_i$. Comparing coefficients yields

$$(\boldsymbol{\nabla}(\psi))_i = \frac{\frac{\partial \psi}{\partial q_i}}{h_i}.$$

Step: Derive the divergence

A small coordinate cell has physical volume

$$\mathrm{d}V = h_1 h_2 h_3 \, \mathrm{d}q_1 \, \mathrm{d}q_2 \, \mathrm{d}q_3.$$

The outward flux through the pair of faces normal to $\boldsymbol{e}_1$ is the difference of $h_2 h_3 A_1$ at the two faces, multiplied by $\mathrm{d}q_2 \, \mathrm{d}q_3$. Cyclic permutations give the other faces. Divide the total flux by $\mathrm{d}V$:

$$\boldsymbol{\nabla} \cdot (\boldsymbol{A}) = \frac{\frac{\partial h_2 h_3 A_1}{\partial q_1} + \frac{\partial h_3 h_1 A_2}{\partial q_2} + \frac{\partial h_1 h_2 A_3}{\partial q_3}}{h_1 h_2 h_3}.$$

Step: Derive curl and the Laplacian

Circulation around the face normal to $\boldsymbol{e}_1$ gives

$$(\boldsymbol{\nabla} \times (\boldsymbol{A}))_1 = \frac{\frac{\partial h_3 A_3}{\partial q_2} - \frac{\partial h_2 A_2}{\partial q_3}}{h_2 h_3}.$$

Cyclic permutation gives the other components. Finally insert $\nabla^2(\psi) = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla}(\psi))$:

$$\nabla^2(\psi) = \frac{\sum_i \frac{\partial\left(\frac{h_j h_k}{h_i}\right)\frac{\partial \psi}{\partial q_i}}{\partial q_i}}{h_1 h_2 h_3}.$$

The scale factors must be differentiated whenever they vary with position; otherwise physical flux and circulation are misrepresented.

Cartesian specialization Definition 435

For (x, y, z), all scale factors are one:

$$\boldsymbol{\nabla}(\psi) = \boldsymbol{e}_x \frac{\partial \psi}{\partial x} + \boldsymbol{e}_y \frac{\partial \psi}{\partial y} + \boldsymbol{e}_z \frac{\partial \psi}{\partial z},$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{A}) = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z},$$

and

$$\nabla^2(\psi) = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2}.$$

The Cartesian curl is

$$\boldsymbol{\nabla} \times (\boldsymbol{A}) = \boldsymbol{e}_x \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] + \boldsymbol{e}_y \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] + \boldsymbol{e}_z \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right].$$

These are the local forms used in most linear wave derivations.

Cylindrical specialization Definition 436

For (r, φ, z), the physical line element is $d\boldsymbol{r} = dr\boldsymbol{e}_r + rd\varphi \boldsymbol{e}_\varphi + dz\boldsymbol{e}_z$. Therefore

$$\boldsymbol{\nabla}(\psi) = \boldsymbol{e}_r \frac{\partial \psi}{\partial r} + \left(\frac{\boldsymbol{e}_\varphi}{r} \right) \frac{\partial \psi}{\partial \varphi} + \boldsymbol{e}_z \frac{\partial \psi}{\partial z},$$

$$\boldsymbol{\nabla} \cdot (\boldsymbol{A}) = \frac{\frac{\partial r A_r}{\partial r}}{r} + \frac{\frac{\partial A_\varphi}{\partial \varphi}}{r} + \frac{\partial A_z}{\partial z},$$

and

$$\nabla^2(\psi) = \frac{\frac{\partial r \frac{\partial \psi}{\partial r}}{\partial r}}{r} + \frac{\frac{\partial^2 \psi}{\partial \varphi^2}}{r^2} + \frac{\partial^2 \psi}{\partial z^2}.$$

In an axisymmetric state $\partial\psi/\partial\varphi = 0$, so the cylindrical area factor remains in the radial divergence even though the azimuthal derivative disappears.

Spherical specialization

Definition 437

For (r, θ, φ), the physical line element is

$$d\boldsymbol{r} = dr\boldsymbol{e}_r + rd\theta\boldsymbol{e}_\theta + r\sin(\theta)d\varphi\boldsymbol{e}_\varphi.$$

The gradient and divergence are

$$\boldsymbol{\nabla}(\psi) = \boldsymbol{e}_r\frac{\partial\psi}{\partial r} + \left(\frac{\boldsymbol{e}_\theta}{r}\right)\frac{\partial\psi}{\partial\theta} + \left(\frac{\boldsymbol{e}_\varphi}{r\sin(\theta)}\right)\frac{\partial\psi}{\partial\varphi},$$

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}) = \frac{\frac{\partial r^2 A_r}{\partial r}}{r^2} + \frac{\frac{\partial \sin(\theta)A_\theta}{\partial\theta}}{r\sin(\theta)} + \frac{\frac{\partial A_\varphi}{\partial\varphi}}{r\sin(\theta)},$$

and the scalar Laplacian is

$$\nabla^2(\psi) = \frac{\frac{\partial r^2\frac{\partial\psi}{\partial r}}{\partial r}}{r^2} + \frac{\frac{\partial \sin(\theta)\frac{\partial\psi}{\partial\theta}}{\partial\theta}}{r^2\sin(\theta)} + \frac{\frac{\partial^2\psi}{\partial\varphi^2}}{r^2\sin(\theta)^2}.$$

The factors r^2 and $\sin(\theta)$ express the changing physical area of spherical coordinate surfaces.

Fast identity checks

Governing law 438

The coordinate formulas must satisfy $\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times(\boldsymbol{A})) = 0$ and $\boldsymbol{\nabla}\times(\boldsymbol{\nabla}(\psi)) = \boldsymbol{0}$ whenever the fields are sufficiently smooth. A purely radial inverse-square flux in spherical geometry has $A_r = C/r^2$ and zero divergence away from the origin, because $r^2A_r = C$. A constant scalar has zero gradient and zero Laplacian in every coordinate system. These checks catch missing metric factors before a plasma equilibrium or wave equation is trusted.

Derivation: cylindrical radial conservation check

Step: Choose the radial flux

Take an axisymmetric purely radial vector

$$\boldsymbol{A} = A_r(r)\boldsymbol{e}_r.$$

Step: Compute the cylindrical surface flux

The cylindrical divergence is

$$\boldsymbol{\nabla}\cdot(\boldsymbol{A}) = \frac{\frac{\partial rA_r}{\partial r}}{r}.$$

The flux through a cylindrical surface of radius r and length L is

$$\Phi(r) = 2\pi rLA_r(r).$$

Between r and $r + \mathrm{d}r$,

$$\mathrm{d}\Phi = 2\pi L\frac{\partial rA_r}{\partial r}\,\mathrm{d}r,$$

while the shell volume is

$$\mathrm{d}V = 2\pi r L\,\mathrm{d}r.$$

Step: Compare flux change with volume

Dividing gives

$$\frac{\mathrm{d}\Phi}{\mathrm{d}V} = \frac{\frac{\partial r A_r}{\partial r}}{r},$$

exactly the cylindrical divergence. The factor $1/r$ converts radial flux change into flux per physical volume.

Rechenbeispiel

Rechenbeispiel 439

In an axisymmetric cylindrical plasma column, let the radial particle-flux vector be $\boldsymbol{\Gamma} = \Gamma_r(r)\boldsymbol{e}_r$ with $\Gamma_r(r) = C/r$. Use $C = 2.00 \cdot 10^{12}\,\mathrm{m^{-1}\,s^{-1}}$, and evaluate the flux at $r_1 = 0.100\,\mathrm{m}$ and $r_2 = 0.200\,\mathrm{m}$. Determine both radial fluxes and $\boldsymbol{\nabla} \cdot (\boldsymbol{\Gamma})$ for $r > 0$.

Numerical result: $\Gamma_r(r_1) = 2.00 \cdot 10^{13}\,\mathrm{m^{-2}\,s^{-1}}$, $\Gamma_r(r_2) = 1.00 \cdot 10^{13}\,\mathrm{m^{-2}\,s^{-1}}$, and $\boldsymbol{\nabla} \cdot (\boldsymbol{\Gamma}) = 0\,\mathrm{m^{-3}\,s^{-1}}$.

Read operators geometrically

Interpretation 440

A gradient points in the direction of greatest scalar increase. A divergence measures net outward flux density from a small physical volume. A curl measures circulation per physical area. The scale factors ensure that these meanings survive a coordinate change. In a normalized model, one may later set $\xi = r/L$ [1] and obtain

$$\frac{\partial \psi}{\partial r} = \frac{\frac{\partial \psi}{\partial \xi}}{L},$$

but the reference length L in m must be stated explicitly.

Summary

Summary 441

Orthogonal-coordinate formulas are generated by physical line elements, face areas, and volumes. Cartesian coordinates have unit scale factors; cylindrical geometry contributes r; spherical geometry contributes r and $r\sin(\theta)$. Check formulas with $\boldsymbol{\nabla} \times (\boldsymbol{\nabla}(\psi)) = 0$, $\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times (\boldsymbol{A})) = 0$, and a symmetric flux before using them in a plasma derivation.

- Why does the cylindrical divergence contain $1/r$?
- What does the factor $r\sin(\theta)$ represent in spherical coordinates?
- Which coordinate system is usually the cleanest starting point for a locally plane plasma wave?
- How would a normalized radial coordinate change the gradient?

Answers

Why does the cylindrical divergence contain $1/r$?

A radial shell has volume proportional to $r\,\mathrm{d}r$, while its outward surface flux is proportional to rA_r. Their ratio gives

$$\frac{\frac{\partial rA_r}{\partial r}}{r},$$

including the changing circumference.

What does the factor $r\sin(\theta)$ represent in spherical coordinates?

It is the physical line scale for an azimuthal angle increment: $dl_\varphi = r\sin(\theta)d\varphi$. It also enters the face areas and therefore the divergence, curl, and Laplacian.

Which coordinate system is usually the cleanest starting point for a locally plane plasma wave?

Cartesian coordinates, because the basis vectors and scale factors are constant. Cylindrical or spherical coordinates become preferable when the equilibrium or boundary has that symmetry.

How would a normalized radial coordinate change the gradient?

If $\xi = r/L$ [1] with stated reference length L in m, then

$$\frac{\partial \psi}{\partial r} = \frac{\frac{\partial \psi}{\partial \xi}}{L}.$$

Each physical gradient component therefore carries the inverse length scale, which must be recorded before a normalized operator is used.

Appendix: Gaussian CGS translation

The script uses SI throughout; much of the older plasma literature uses Gaussian CGS, and the table below translates the equations and scales used in this script from one system to the other.

Quantity	SI	Gaussian CGS
Coulomb force	$F = q_1 q_2/(4\pi\varepsilon_0 r^2)$	$F = q_1 q_2/r^2$
Poisson equation	$\nabla^2\Phi = -\rho_q/\varepsilon_0$	$\nabla^2\Phi = -4\pi\rho_q$
Lorentz force	$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B})$	$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B}/c)$
Gauss's law	$\boldsymbol{\nabla}\cdot\boldsymbol{E} = \rho_q/\varepsilon_0$	$\boldsymbol{\nabla}\cdot\boldsymbol{E} = 4\pi\rho_q$
Ampère–Maxwell law	$\boldsymbol{\nabla}\times\boldsymbol{B} = \mu_0\boldsymbol{j} + \mu_0\varepsilon_0\partial\boldsymbol{E}/\partial t$	$\boldsymbol{\nabla}\times\boldsymbol{B} = \frac{4\pi}{c}\boldsymbol{j} + \frac{1}{c}\frac{\partial\boldsymbol{E}}{\partial t}$
Faraday's law	$\boldsymbol{\nabla}\times\boldsymbol{E} = -\partial\boldsymbol{B}/\partial t$	$\boldsymbol{\nabla}\times\boldsymbol{E} = -(1/c)\partial\boldsymbol{B}/\partial t$
No magnetic monopoles	$\boldsymbol{\nabla}\cdot\boldsymbol{B} = 0$	$\boldsymbol{\nabla}\cdot\boldsymbol{B} = 0$
Debye length	$\lambda_D = \sqrt{\varepsilon_0 k_B T_e/(ne^2)}$	$\lambda_D = \sqrt{k_B T_e/(4\pi ne^2)}$
Electron plasma frequency	$\omega_{pe} = \sqrt{ne^2/(\varepsilon_0 m_e)}$	$\omega_{pe} = \sqrt{4\pi ne^2/m_e}$
Signed gyrofrequency	$\Omega = qB/m$	$\Omega = qB/(mc)$
Magnetic pressure	$B^2/(2\mu_0)$	$B^2/(8\pi)$
Plasma beta	$\beta = 2\mu_0 p/B^2$	$\beta = 8\pi p/B^2$
$\boldsymbol{E}\times\boldsymbol{B}$ drift	$\boldsymbol{v}_E = \boldsymbol{E}\times\boldsymbol{B}/B^2$	$\boldsymbol{v}_E = c\boldsymbol{E}\times\boldsymbol{B}/B^2$
Magnetic moment of a current loop	$\mu_{\mathrm{mag}} = IS$	$\mu_{\mathrm{mag}} = IS/c$
MHD force density	$\boldsymbol{j}\times\boldsymbol{B}$	$\boldsymbol{j}\times\boldsymbol{B}/c$
Magnetic field	1 T	10^4 G
Energy	1 J	10^7 erg
Number density	$1\ \mathrm{m}^{-3}$	$10^{-6}\ \mathrm{cm}^{-3}$
Charge	1 C	$\approx 2.998\cdot 10^9$ statC
Electric potential	1 V	$\approx (1/299.8)$ statV

Purely geometric factors, such as the Debye number $N_D = (4\pi/3)n\lambda_D^3$, are identical in both systems. Temperatures quoted as $k_B T$ in electronvolts carry the same number in both systems; only the conversion to energy differs,

$$1\ \mathrm{eV} = 1.602176634\cdot 10^{-19}\ \mathrm{J} = 1.602176634\cdot 10^{-12}\ \mathrm{erg}\,.$$

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